



FRIEDRICH-SCHILLER-  
UNIVERSITÄT  
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# **Pronominal clitics in West Slavic:**

## **An exploratory corpus study of spoken varieties**

### **Dissertation**

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## Motto

Back to “Go”!

## Clitimeric: A clitic limerick

There was a clitic that longed to be free  
So, it climbed a syntactic tree  
But on the top, it found  
That it still was bound  
And would not ever be free



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# Abstract

Clitic phenomena in Slavic have received a great deal of attention since the 1990s. Generally, clitic auxiliaries and pronouns in West and South Slavic are classified as special clitics exhibiting intricate grammatical properties. In contrast, clitics in Polish, which is a West Slavic language, have been assigned a distinct typological status, since they lack recognised traits of special-clitic-hood, with Wackernagel (or second-position) placement, clitic clustering, and fixed word order among clitics being most decisive here. However, phenomena considered prototypical for Polish clitics are attested in other Slavic languages, too, which is why it is argued here that the typological classification of Polish is not well-grounded. The present study identifies the root cause of such discrepancies in the selective and subjective databases usually employed in grammar research, as a result of which portions of linguistic variation have gone unnoticed. The thesis presents an exploratory corpus study of the clitic systems of spoken Polish and cognate West Slavic lects, i.e. spoken Czech, Goralish, Slovak, Silesian, and Kashubian. In particular, it examines the distribution of formally short pronouns and reflexives (= constant clitics) in the clause quantitatively. The analysis shows that Polish and other Lechitic lects align with languages like Czech and Slovak in being second-position clitic languages. A revision of the typological classification of the pronominal and reflexive clitic systems in West Slavic is thus suggested, and second-position cliticisation is argued to be best captured in statistical terms of likelihood of occurrence rather than being grammatically inevitable. In addition, the study unveils novel data evidencing pronominal clitics licensed by non-verbal predicates. The study further provides the first systematic account of register variation in Czech clitic positioning by comparing frequencies of second-position cliticisation with frequencies of clitic placement in first and third position in edited Czech writing and spontaneous Czech speech. Finally, delayed clitic placement in Polish is investigated and demonstrated to be contingent upon notions of information structure. The thesis highlights the importance of linguistic variation in cliticisation and the need to consider this variation in research. From a methodological perspective, the study contributes to the debate on the appropriateness of introspectively constructed examples vs. empirically obtained data in supporting the importance of empirical data for scientific progress in linguistic research.

# Abbreviations, glosses, and marks

In this study, the Leipzig Glossing Rules (LGR) are applied for glossing linguistic examples with some minor adjustments.<sup>1</sup> Glottonyms are abbreviated by two-, three-, or four-letter codes, but do not necessarily follow any specific system (e.g. ISO). Below, a comprehensive list of abbreviations, glosses, and marks used in this thesis is given.

|      |                                                                                                                                                    |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| *    | ungrammaticality / unacceptability judgement                                                                                                       |
| ?    | unclear status / uncertain with respect to grammaticality / acceptability                                                                          |
| <    | linear precedence relation (what is on the left precedes that what is on the right)                                                                |
| 1    | first person                                                                                                                                       |
| 1P   | first position                                                                                                                                     |
| 2    | second person                                                                                                                                      |
| 2D   | second daughter = second position / Wackernagel position (with respect to syntactic constituency)                                                  |
| 2P   | second position / Wackernagel position (indeterminant with respect to word level or Constituency level)                                            |
| 2W   | second word = second position / Wackernagel position (with respect to individual syntactic or morphological or phonological or orthographic words) |
| 3    | third person                                                                                                                                       |
| 3P   | third position                                                                                                                                     |
| 4P   | fourth position                                                                                                                                    |
| 5P   | fifth position                                                                                                                                     |
| 6P   | sixth position                                                                                                                                     |
| 7P   | seventh position                                                                                                                                   |
| 10P  | tenth position                                                                                                                                     |
| ACC  | accusative                                                                                                                                         |
| ACI  | accusativus cum infinitivo / accusative and infinitive (cf. also ECM)                                                                              |
| ADJ  | adjective                                                                                                                                          |
| AdjP | adjective phrase                                                                                                                                   |
| ADV  | adverb                                                                                                                                             |
| AdvP | adverb phrase                                                                                                                                      |
| AGr  | Ancient Greek                                                                                                                                      |

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<sup>1</sup> <https://www.eva.mpg.de/lingua/pdf/Glossing-Rules.pdf>

|                       |                                                    |
|-----------------------|----------------------------------------------------|
| ANI                   | animate                                            |
| AOR                   | aorist                                             |
| AUX                   | auxiliary                                          |
| BCMS                  | Bosnian-Croatian-Montenegrin-Serbian               |
| BCr                   | Burgenland Croatian                                |
| Bg                    | Bulgarian                                          |
| Bos                   | Bosnian                                            |
| Ča                    | Čakavian                                           |
| CC                    | clitic climbing                                    |
| CL                    | clitic                                             |
| COMP / C <sup>0</sup> | complementizer head                                |
| COND                  | conditional (mood / marker / element)              |
| COP                   | copula                                             |
| CP                    | complementizer phrase                              |
| Csb                   | Kashubian (Cassubian)                              |
| Cz                    | Czech                                              |
| DAT                   | dative                                             |
| DEF                   | definite                                           |
| DEIX                  | deixis / deictic                                   |
| DERV                  | derivation / derivational (affix)                  |
| DetP                  | determiner phrase                                  |
| DIM                   | diminutive                                         |
| DP                    | Delayed Placement (third to <i>n</i> -th position) |
| ECM                   | exceptional case marking (cf. also ACI)            |
| EMPH                  | emphatic                                           |
| Eng                   | English                                            |
| ETH                   | ethical (pronoun / element)                        |
| F                     | feminine gender                                    |
| FOC                   | Focus                                              |
| Fra                   | French                                             |
| FUT                   | future tense                                       |
| GB                    | Government and Binding (theory)                    |
| Ger                   | German                                             |
| GEN                   | genitive                                           |
| Gor                   | Goralish                                           |



|      |                                                                                                                                   |
|------|-----------------------------------------------------------------------------------------------------------------------------------|
| HPSG | Head-driven Phrase Structure Grammar                                                                                              |
| Hrv  | Croatian                                                                                                                          |
| IMP  | imperative (mood / form)                                                                                                          |
| INA  | inanimate                                                                                                                         |
| INDF | indefinite                                                                                                                        |
| INF  | infinitive                                                                                                                        |
| INFL | inflection / inflectional (affix)                                                                                                 |
| INS  | instrumental                                                                                                                      |
| INT  | interjection                                                                                                                      |
| IP   | inflectional phrase                                                                                                               |
| IPFV | imperfective aspect                                                                                                               |
| Ita  | Italian                                                                                                                           |
| ıP   | intonation Phrase                                                                                                                 |
| Kaj  | Kajkavian                                                                                                                         |
| LFG  | Lexical Functional Grammar                                                                                                        |
| LOC  | locative                                                                                                                          |
| M    | masculine gender                                                                                                                  |
| Mac  | Macedonian                                                                                                                        |
| N    | neuter gender                                                                                                                     |
| NN   | noun                                                                                                                              |
| NOM  | nominative                                                                                                                        |
| NP   | noun phrase                                                                                                                       |
| NUM  | quantificational expression: numeral proper or quantifier                                                                         |
| NVIR | non-virile gender: non-masculine-animate-personal (Goralish, Kashubian, Polish, Silesian) / non-masculine-animate (Czech, Slovak) |
| OBJ  | object                                                                                                                            |
| OCS  | Old Church Slavonic                                                                                                               |
| OCz  | Old Czech                                                                                                                         |
| OPol | Old Polish                                                                                                                        |
| ORu  | Old Russian                                                                                                                       |
| PA   | predicate-adjacent                                                                                                                |
| PART | particle                                                                                                                          |
| PASS | passive                                                                                                                           |
| PFV  | perfective aspect                                                                                                                 |
| PIE  | Proto-Indo-European                                                                                                               |

|      |                                                                                                                                                                                                                                                                   |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PL   | plural                                                                                                                                                                                                                                                            |
| Pol  | Polish                                                                                                                                                                                                                                                            |
| PP   | prepositional phrase                                                                                                                                                                                                                                              |
| PRED | predicative                                                                                                                                                                                                                                                       |
| PRF  | perfect                                                                                                                                                                                                                                                           |
| PRN  | pronoun, pronominal                                                                                                                                                                                                                                               |
| PRO  | opaque pronoun (big PRO)                                                                                                                                                                                                                                          |
| PRS  | present tense                                                                                                                                                                                                                                                     |
| PST  | past tense                                                                                                                                                                                                                                                        |
| PTCP | participle                                                                                                                                                                                                                                                        |
| Q    | question particle / marker                                                                                                                                                                                                                                        |
| REFL | reflexive marker / pronoun / element                                                                                                                                                                                                                              |
| Rue  | (Carpatho-)Rusyn                                                                                                                                                                                                                                                  |
| Rus  | Russian                                                                                                                                                                                                                                                           |
| S    | sentence / clause                                                                                                                                                                                                                                                 |
| SBJ  | subject                                                                                                                                                                                                                                                           |
| SG   | singular                                                                                                                                                                                                                                                          |
| Sil  | Silesian                                                                                                                                                                                                                                                          |
| Skr  | Sanskrit                                                                                                                                                                                                                                                          |
| Slk  | Slovak                                                                                                                                                                                                                                                            |
| Slv  | Slovene                                                                                                                                                                                                                                                           |
| Spa  | Spanish (indifferent between European and American Spanish)                                                                                                                                                                                                       |
| Srp  | Serbian                                                                                                                                                                                                                                                           |
| SUB  | subordinator (incl. complementizer, subordinating conjunction/adverb, relative pronoun/adverb)                                                                                                                                                                    |
| TO   | syntactic-pragmatic particle <i>to</i> , formally identical to neuter demonstrative pronoun in nominative/accusative; gloss is used if (i) <i>to</i> ≠ demonstrative, (ii) <i>to</i> = ambiguous between multiple functions, (iii) status of <i>to</i> is unclear |
| TOP  | TOPIC                                                                                                                                                                                                                                                             |
| TRAN | transgressive                                                                                                                                                                                                                                                     |
| Ukr  | Ukrainian                                                                                                                                                                                                                                                         |
| V    | verb                                                                                                                                                                                                                                                              |
| VA   | verb-adjacent                                                                                                                                                                                                                                                     |
| VIR  | virile gender: masculine animate personal (Goralish, Kashubian, Polish, Silesian) or masculine animate (Czech, Slovak)                                                                                                                                            |

|                |                                                                                                |
|----------------|------------------------------------------------------------------------------------------------|
| VOC            | vocative                                                                                       |
| VP             | verb(al) phrase                                                                                |
| vP             | small v(erb(al)) phrase                                                                        |
| W2             | Wackernagel position / second positions                                                        |
| WH             | interrogative expression (so-called <i>wh</i> -expression)                                     |
| ω              | phonological word                                                                              |
| X <sup>0</sup> | head of arbitrary syntactic category <i>x</i> / any syntactic head                             |
| XP             | phrase/maximal projection of arbitrary syntactic category <i>x</i> / any syntactic constituent |

# Transliteration

In the thesis the Latin script is used consequently. Examples from languages, which are written in a script different from Latin, are always transliterated.

For Cyrillic, I follow the international scientific transliteration as it is exemplified in Comrie & Corbett (1993: xii–xiii).<sup>2</sup>

Old Church Slavonic examples are transliterated according to Table 2.2 from Cubberley (1993: 24).

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<sup>2</sup> Cf. also the exhaustive transliteration sheet by Bunčić (2020), which is freely available under the Creative Commons licence.

## **PART I: FOUNDATIONS**

# 1 Introduction

## 1.1 Clitics (in Slavic) as an object of study

Until recently, clitic phenomena have been intriguing generations of linguists as confirmed by numerous monographs (Anderson 2005; Franks & King 2000; Klavans 1995; Kolaković et al. 2022; Leeuw 1997; Milićević 2023; Nevis et al. 1994; Spencer & Luís 2012a; Zwicky 1977), anthologies (Beukema & Dikken 2000; Borer 1986; Gerlach & Grijzenhout 2000; Halpern & Zwicky 1996; Riemsdijk 1999; Salvesen & Helland 2013), and individual research articles (see, e.g., the references cited throughout the thesis).<sup>3</sup> This arises from the many idiosyncratic properties observed in clitics. It is generally accepted that these properties affect all traditionally distinguished levels of grammatical description and the interfaces between them: Phonology, morphology, syntax, and semantics.<sup>4</sup> Clitics are traditionally distinguished by phonology in that they lack inherent word stress. From this perspective, they are prosodically dependent on another word, the so-called HOST, with which they need to constitute a prosodic unit.<sup>5</sup> Morphologically, particular clitic + clitic or word + clitic combinations may yield alternations in form (= shape of a morph) and substance (= specification of phonetic/phonological features), i.e. fusions, dissimilations, or haplology (cf. Zwicky 1977: 16). A sequence of contiguous clitics is called CLITIC CLUSTER, while the host + clitic(s) sequence is referred to as CLITIC GROUP.<sup>6</sup> Typologically common ordering patterns found within clitic clusters have been attributed to either morphological or syntactic principles. Especially in generative syntax, clitic distribution and ordering have been attributed to an underlying cascade of functional heads ( $C^0$ , AgrS<sup>0</sup>, T<sup>0</sup> etc.).<sup>7</sup> Within the domain of semantics, it has been argued that clitic and orthotonic

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<sup>3</sup> In 2023, the collaborative research project *The Behavior of Czech and Slovenian Clitics* between the Masaryk University (Brno, Czech Republic) and the University of Nova Gorica (Republic of Slovenia) began. The project is led by Petr Karlík (Brno) and Franc Marušič (Nova Gorica) and two teams of prominent experts in clitic research (<https://www.phil.muni.cz/vyzkum/resene-projekty/70161>; <https://www.ung.si/en/research/center-for-cognitive-science-of-language/projects/n6-0314/>; accessed 20 Oct 2023).

<sup>4</sup> Recently, great interest sparked in clitics in first language acquisition, cf., e.g. Radeva-Bork (2012), Varlokosta et al. (2016), Mykhaylyk & Sopata (2016), or *Lingua*'s special issue on the *Acquisition of Clitics* edited by Grohmann & Neokleous (2015).

<sup>5</sup> Alternatively, the term *anchor* is used sometimes synonymously. However, Haspelmath (2023: 14–15) distinguishes a syntactic notion of *anchor* from a prosodic notion of *host*, both of which may but not need to be congruent in a given example (cf. also Fried 1994; Klavans 1995). I do not adopt this distinction for two reasons: First, the anchor/host distinction does not seem to apply to the object languages examined here (i.e. West Slavic). Second, I do not investigate prosody (and I do not have access to a sufficient amount of acoustic data), such that distinguishing those concepts is not applicable here, at all.

<sup>6</sup> The clitic cluster is called *clitic chain* by Aguado & Dogil (1989), while Zwicky (1977) indirectly refers to it as *clitic string*: “I know of no rules affecting whole groups, although there are some affecting entire strings of clitics (without the host)” (Zwicky 1977: 15).

<sup>7</sup> Throughout the thesis I will stick to the still widely common terminology of transformational syntax, as I will use the notation  $X^0$  for a head of the category X and XP for the maximal projection or phrasal constituent of X instead of the Minimalist proper notions  $X^{\min}$  and  $X^{\max}$  respectively.

pronouns have different referential properties. To date there is no consensus on how to analyse clitics properly. Phonologists, morphologists, and syntacticians argue about which domain of grammar is the appropriate level of representation for clitic phenomena.

There are two major points on the issue of lacking consensus that I will briefly address: First, investigations into clitics usually have been selective with respect to data. It depends heavily on the individual scholar and the scientific framework utilised, which data are considered relevant or marginal. Second, the lack of consensus in terms of a missing unified formal analysis for all phenomena called “clitic” might be considered problematic only in case one assumes or demands all superficially similar phenomena to be cross-linguistically identical in principle. A representative of this very point is Klavans’s (1995: 94) hypothesis “that cliticization is a unitary phenomenon” (her HYPOTHESIS A). Still, one could take a perspective which acknowledges that different approaches might be suitable for accounting for clitic phenomena across the globe. In other words, perhaps linguists should allow different components of grammar to be responsible to different degrees for a given cross-linguistic observation. Suppose two languages  $L_A$  and  $L_B$ , such that cliticisation is more dependent on principles of morphology in  $L_A$ , while cliticisation is mainly driven by principles of syntax in  $L_B$ . There is no sound a priori reasoning why linguists should abandon this idea entirely.<sup>8</sup> Therefore, the present thesis is fully in accordance with Diesing (2010), who notes that

“it is not sufficient to merely judge grammaticality of sentences, but their appropriateness in a particular context must be assessed. If semantic or pragmatic factors play a role in clitic placement, then the problem needs to be approached from a broader view of the interfaces involved. Current research has also relied heavily on a relatively small database of native speaker judgements, much of it harvested from previously published work and based on the judgements of native speaker linguists. While this is not uncommon in the field of theoretical linguistics, there is clearly something to be gained from expanding both the database and the methodology – by corpus sources and experimental investigations.” (Diesing 2010: 104–105)

Furthermore, the present study has several parallels to the research project on BCMS by Kolaković et al. (2022), who scrutinised a rich and diverse data set of auxiliary/verbal, pronominal, and reflexive clitics. One of the traits shared between their project and the present study is its variationist perspective. But while my work does neither rely on Diesing (2010) nor on Kolaković et al. (2022)

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<sup>8</sup> A similar assessment was already made by Zwicky (1977: 24): “Whatever the appropriate descriptive device – *and this might well be different for different languages* – there is a rich collection of phenomena that deserve careful study and comparison, with an attempt to draw some generalizations from these data.” [my emphasis] In a similar vein, Kolaković et al. (2022: 6–7) “agree with Rosen & Hana (2017: 4) that in many respects, CL[itic]s in different languages are similar with regard to inventory, positioning, and internal order (CL[itic] clusters), but the mix of properties in each language may be unique.”

and has evolved independently, it appears that it arises from the same suspicion about what is sometimes regarded as an anti-empirical attitude in certain linguistic research traditions.

The present study has arisen against the background of various claims and hypotheses about cliticisation in Slavic in general and Polish in particular. Readers will find a primarily empirical investigation into the surface-syntactic distribution of West Slavic pronominal and reflexive clitics, which utilises corpus linguistics as its basic research method or research paradigm.<sup>9</sup> The investigation will be exploratory in nature and provide data for both a language-particular description and a comparative evaluation for a micro-typology of genetically closely-related languages. Thus, the thesis represents an approach, which fits the phrase *back to “Go”* that serves as the dissertation’s motto and refers to a talk by Götz Keydana in November 2011.<sup>10</sup>

## 1.2 Objectives of the thesis

Clitics are linguistic entities that not only received a great deal of scientific attention from scholars of diverse subfields, but that also became core elements in linguistic theorising associated with a diverse set of phenomena. Therefore, they have been considered to represent interface material par excellence (cf. Spencer & Luís 2012a: 3).

The doctoral thesis arose out of the original question of whether Polish clitics are in fact typologically peculiar within the Slavic language family due to their high degree of distributional freedom, as has been argued by several linguists. This issue has two sides: On the one hand, one might wonder whether virtually “anything goes” in Polish and whether Polish clitics lack properties typically associated with clitic-hood in other Slavic languages. On the other hand, it is a contentious issue, whether other Slavic clitic systems are as well-behaved as they are thought to be. The question arises, whether one or more of Polish’s cognate languages – especially closely-related Czech – may exhibit properties reminiscent of the putative peculiarities.

In order to approach these questions, a number of related issues needs clarification. As it turns out, some of the most fundamental issues in cliticisation – not only in Slavic – are rather unclear or controversial. However, linguistic analyses operate with diverse notions and concepts, as if they were self-evident. This is why the reader will come across some fundamental considerations and

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<sup>9</sup> I will not take any position in the controversy of whether corpus linguistics is only a method of linguistic research or whether it is a linguistic discipline on its own.

<sup>10</sup> On 17 November 2011, Götz Keydana gave an invited plenary talk under the title *Zurück auf los!* ‘Back to “Go”!’ during the 50. *Studentische Tagung Sprachwissenschaft* (StuTS, ‘50th students’ conference on linguistics’) organised by the dissertation’s author and his peer students. The talk provided a critical review of the state of the art in contemporary grammar research.



discussions in the thesis. It is necessary to provide a detailed discussion of the very notion of *clitic* as well as of the status of clitics in (West) Slavic. Furthermore, I will provide a discussion of methodological considerations of grammar research and empiricism, which remain a field of – sometimes bitter – controversy. The corpus linguistic portion of the thesis is essentially motivated by the conviction that clitics in West Slavic have not been adequately described, yet, due to a rather fragmented set of data. On the one hand, Polish exhibits the predominant tendency of clitic second position placement, which has gone unnoticed. On the other hand, Czech is prone to a higher degree of positional variation than usually assumed for a Wackernagel clitic language. The study is data-oriented and pursues a holistic approach in that it takes linguistic variation and statistical distributions into account, which, representing linguistic facts, need to be taken into consideration when modelling linguistic phenomena. Thus, it is hoped that typological assessments of Polish versus Slavic can be refined and, if necessary, corrected. Furthermore, the approach concurs with Van Valin & LaPolla’s (1997: 3) argument that an adequate explanation presupposes an adequate description. Most analyses of clitic phenomena remain primarily descriptive despite their explanatory aims, and it has yet to be clarified what the driving forces behind phenomena such as second-position placement, delayed clitic placement, and clitic climbing are.

There is a rich literature on clitics in general. An impressive body of research has accrued, particularly with respect to clitics in Romance languages. As Russi (2008: 6) puts it:

“It is not surprising, then, that Romance pronominal clitics represent one of the most actively and thoroughly investigated topics in modern linguistics, especially within the generative tradition. In fact, it would not be an exaggeration to say that any aspect and property of Italian clitic pronouns has been *meticulously (and typically very satisfactorily) scrutinized* already.” (Russi 2008: 6; my emphasis in italics)

Particular attention has been paid to Slavic clitics since the 1990s. Most recently, Kolaković et al. (2022) published a thorough empirical study on clitics in BCMS, while Milićević (2023) presents a new, dependency-grammatical treatment of Serbian clitics.<sup>11</sup> One might wonder, therefore, why we need a further investigation into Slavic clitics. In the following, I would like to provide a concise answer to this question and justify the motivation of the thesis.

First and foremost, clitic phenomena in Slavic cannot be considered “meticulously and typically very satisfactorily scrutinized”, as Russi (2008: 6) claims for Romance. There is a lot of

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<sup>11</sup> I will consequently use Bosnian-Croatian-Montenegrin-Serbian (BCMS) to refer to the pluricentric language formerly known as Serbo-Croatian (SC) and Croatia-Serbian (CS) respectively. In *BCMS*, the linguistic denotations are simply arranged in alphabetical order and are not meant to imply any hierarchy among the national languages. Examples cited within the thesis will refer to BCMS, when the source indistinctively labels it SC, CS, or BCS. I will cite the respective national language (Bosnian [Bos]; Croatian [Hrv]; Serbian [Srp]), when provenance of the data is indisputably certain.

controversy and – taking the perspective of variationist linguistics – a yet too small set of data that most analyses rely on. As will become clear in the course of the thesis, a relevant set of data containing significant variation has mostly gone unnoticed. Kolaković et al. (2022) set a new benchmark for investigations into clitic phenomena, since they draw upon a rich empirical database. However, a survey of this scale is quite unique and there is still a lot of work to do. As for theoretical linguistics, a new set of BCMS data has come up that needs to be considered in linguistic theorising. Taking a closer look at West Slavic, there has not been any research that would in the least resemble Kolaković et al.’s (2022) study in terms of methodological thoroughness and empirical coverage.<sup>12</sup>

Clitics – or, put more carefully, elements that are considered to exhibit “typical” clitic behaviour – occur cross-linguistically in genetically unrelated and typologically divergent languages. On the one hand, investigations in cliticisation contribute to the study of language universals and general properties of human language.<sup>13</sup> On the other hand, a careful consideration of clitic phenomena in diverse languages might reveal typological variation of phenomena that are usually lumped together. Clitic phenomena are of particular interest for theoretical linguistics, because clitics seem to challenge the modularity of language, which is widely assumed among linguists of diverse theoretical schools. Phonological, morphological, and syntactic effects are interspersed in a non-trivial way.<sup>14</sup> The study of clitics, thus, deepens our understanding of how the components of grammar interact. Put differently, findings on cliticisation contribute to the study of grammatical interfaces.

However, I suspect that Slavic linguistics is still not at the point that we can be confident about having adequate descriptions of clitic phenomena. There is still a lack of empirical coverage, which partly arises from an atomistic approach to the study of language favouring small grammar fragments at the expense of comprehensiveness. The present study is meant to be a descriptively more adequate and empirically more exhaustive contribution to the study of clitics, but I do not claim that it is the most adequate or most exhaustive account possible.

Studying pronominal clitics in West Slavic in particular, I hope to achieve several goals: First, I would like to provide an adequate (or, at least, more adequate) description of parts of these languages’ clitic systems. Doing so, I hope to contribute to the clarification of some basic properties that, as I contend, are still not well understood. This is why some portions of Part III will appear to be rather documentary in nature. Second, I hope to bring data to the reader’s attention, which have been mostly neglected. Third, based on a discussion of these data, I would like to reconsider the

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<sup>12</sup> I depart from Kolaković et al.’s (2022: 4) assessment that clitics “are particularly well described in [...] Czech”.

<sup>13</sup> I use *language universal* in the broadest sense covering absolute, implicational, and statistical universals.

<sup>14</sup> This problem arises only, if one presupposes the modular division of language into components like phonology, morphology, and syntax. I do not wish to reconsider the concept of modularity here but would like to point out that it is not taken for granted by all linguists or within all frameworks. I refer the reader to Hilbert (2008 and references therein) for discussion.

typology of Slavic clitics and suggest some refinements. The present thesis thus attempts to meet the desideratum identified in I. Kulik (2023b: 226), i.e. “to revisit and refine the micro-typology of Slavic cliticization on a sound empirical basis.” Fourth, from a methodological perspective, I would like to exemplify that empirical studies – i.e. a corpus study in the present case – have some advantages over the usually encountered example corpora constructed by native speaker linguists.<sup>15</sup> I think that the empirical approach adopted here increases descriptive adequacy, i.e. empirical coverage. Fifth, explanatory adequacy presupposes an adequate description (cf. Van Valin & LaPolla 1997: 3). I use the term *explanation* in the traditional sense referring to the actuation or causation of a phenomenon, i.e. in terms of a cause-effect relation. Thus, I do not refer to Chomsky’s (1965) idiosyncratic concept of **explanatory adequacy**, which presupposes the concept of Universal Grammar. Instead, I refer to Weinreich et al.’s (1968) notion of the **actuation principle**, which was originally coined against the background of historical linguistics and sound change. I adopt the concept here for synchronic linguistics and grammar research.

To date, no attempts have been made to scrutinise the positional capacity of clitics in contemporary Slavic, even though there is a number of empirical studies focusing on clitic placement properties. However, these have not taken numerical values of clausal positions into consideration, which is a novelty of the present thesis’s (but cf. I. Kulik 2019 for a first application). It has remained unclear, therefore, which positions a clitic can be expected to occupy at all, i.e. on a purely phenomenological level. Another question concerns frequency, which is hardly ever taken into consideration (but cf. Kosek 2009, 2011, 2015, 2017 for Old and Middle Czech; Kolaković et al. 2022; Reinkowski 2001 for BCMS). Linguistic variation cannot be reduced to chance but exhibits quantifiable patterns that allow us to account for common vs. rare phenomena.

### 1.3 Meta-theoretical approach

The present study attempts to bridge a gap by connecting theoretical perspectives and methodological approaches of different kinds. Theoretical and formal linguistics must be recognised as the study’s point of departure, but I do not wish to favour any approach to the formal study of natural language’s syntax over another. For one thing, the majority of research cited in the thesis is nested within Generative Grammar, i.e. the Principle-and-Parameters approach instantiated in the Government and Binding framework (GB) and the subsequent Minimalist Program (MP). The study of clitics has been

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<sup>15</sup> Scholars using such databases are sometimes called “armchair linguists”, which Fillmore (1992) uses (auto-) ironically. Published reference to this notion is also found, e.g. in Juola (2018), Phillips (2009), and Tajsner (2017). I will avoid this term.

the most productive within these frameworks. However, I will try to incorporate work done in other frameworks, too, e.g. Head-driven Phrase Structure Grammar and other schools of thought like Dependency Grammar.

The methodological cornerstone of the thesis is computer-assisted corpus linguistics, as it has been evolving since the beginning of this millennium. Thus, I utilise data that are stored in large electronic corpora and are retrievable via standardised queries. This kind of approach allows for two things: First, it provides our field with a broad empirical basis, while second, it allows us to test hypotheses in terms of Karl Popper's falsifiability criterion. Science is dependent upon critical assessments and reassessments of current theorising and methodology to make progress. Consequently, the *modus operandi* will be in essence empirical, utilising corpus data and quantitative exploration in order to objectify the data discussion. Unfortunately, parts of the scientific discourse in our discipline are characterised by mutual disregard rather than perception and constructive discussion.

Furthermore, I acknowledge the perspectives common in both language typology and variationist linguistics, which are of significance for the present thesis, too. Functional considerations play a major role within both fields. Although the three cornerstones appear to be highly disparate and even conflicting, a synthesis is very fruitful, as distinct approaches might complement each other. This is why the thesis has an agnostic attitude towards the various theoretical frameworks currently available. In other words, the present study is theory-neutral or meta-theoretical and does not favour one theory over another *a priori*.<sup>16</sup> Nevertheless, theory-neutral does not mean pre-theoretical, un-theoretical, or anti-theoretical. On the contrary, the reader will recognise notions common to linguistic theorising throughout the thesis, especially from the domain of syntax. Concepts that are sometimes treated as mutually exclusive, will occur next to one another to enrich the description of natural language. In fact, many notions of one framework may be translated into another framework's ones and I will rely on assumptions that are shared by all – or, at least, most – of the theories cited. For one thing, the syntax I assume is a phrase structure grammar (PSG) in essence, but I also attempt to integrate insights and notions from dependency grammar (DG), whenever possible. One may proceed this way, because syntactic structures in terms of PSG may be transposed into syntactic structures of DG and vice versa. For instance, syntactic constituents might be identified as complete subtrees and thus utilised for further analysis in DGs, while the PSG principle of endocentricity basically reflects

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<sup>16</sup> The closeness to the term *meta-analysis* is intended. While systematic reviews and meta-analyses have become a standard procedure in various disciplines, there has been little effort to adopt the approach in linguistics. Of course, numerous studies in linguistic typology operate on a large scale in comparing dozens or hundreds of individual genetically and areally unrelated languages and dialects with respect to linguistic structures. However, these works do not compare studies and theoretical approaches. The only work that comes closest to a systematic review I am aware of is Stefan Müller's (2023) monograph *Grammatical Theory*, whose major objective is the comparison of syntactic theories.

a DG notion. Thus, I hope to provide a study that is compatible with various frameworks, while it attempts to omit most of their idiosyncrasies, which might sometimes lead to confusion. Beyond that, I aim to reduce bias by using theory-specific constructs, which can sometimes lead to tunnel-type vision on the part of the researcher. This can be the case when researchers are more engaged with a particular formal theory rather than the language data or the linguistic phenomena they would like to examine. It would be the greatest success of the present contribution to scientific discourse if linguists from diverse fields would talk to each other and mutually benefit from each other.<sup>17</sup>

## 1.4 Structure of the thesis

The thesis consists of three parts each of which serves a different purpose. Part I sets the ground for the study in introducing the research topic, object languages, and pertinent issues. Part II is devoted to the methodological domain and provides some fundamental reflections. Furthermore, I present the database and explain the procedures I executed. Part III is the empirical core of the thesis and presents the corpus-based study. The thesis is structured as follows:

Chapter 2 introduces the object languages, which are all spoken varieties. Chapter 3 gives a critical discussion of the notion of clitic. Chapter 4 provides a state-of-the-art report of cliticisation from a cross-Slavic perspective. Chapter 5 presents an overview of Part II. Chapter 6 discusses several methodological issues, which are basically linked to the ongoing debate of introspective vs. empirical data acquisition. Chapter 7 presents the utilised corpora and data sources. Chapter 8 describes the method underlying the present study step by step. Chapter 9 opens Part III of the thesis with an overview. Chapter 10 presents the first case study on the empirically attested distribution patterns in the object languages. Chapter 11 is devoted to clitics subcategorised for by non-verbal heads. Chapter 12 zooms into Czech and explores the frequency distributions of second position cliticisation versus clitic placement in third and first position by contrasting written and spoken Czech. Chapter 13 attempts to provide an analysis of delayed clitic placement in Polish with particular reference to information structure. Chapter 14 concludes the thesis and discusses future prospects.

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<sup>17</sup> I intentionally refer to Martin Haspelmath's (2000) review "Why can't we talk to each other?" of Newmeyer's (1998) monograph.

## 2 Object area: Languages and clitic inventories

### 2.1 Introductory remarks

The study investigates the clitic distribution in West Slavic spoken varieties. A reasoning for the choice of spoken lects will be provided in Chapter 6. At this point, it is sufficient to delimit and briefly describe the object area. The entire database consists of six object languages. Three of them build the empirical core, i.e. the respective data samples are of considerable size: Czech, Polish, and the Spisz variety of Goralish. Three more lects serve as a supplementary means only. The respective data samples successively decrease in size: Slovak, Silesian, and Kashubian. Thus, the object area covers the Lechitic and Czech-Slovak branches of West Slavic. Sorbian is not covered, at all.<sup>18,19</sup> While some of the object languages are comparatively well-studied, others might be hardly known even by a professional audience. This is why some of the smaller and lesser known lects will be introduced to the reader in more detail in the next sections. The order of presentation will be kept mostly throughout the thesis whenever these lects will be discussed individually, i.e. Czech, Polish, Goralish, Slovak, Silesian, and finally Kashubian.

### 2.2 Czech

The spoken colloquial variety of Czech (Cz) is usually referred to as *obecná čeština* ‘Common Czech’. While the term itself suggests a spoken colloquial variety used all over the Czech Republic,

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<sup>18</sup> Corpus data for both Upper and Lower Sorbian exist including spoken language. For Upper Sorbian, a corpus of spoken data (corpus of Colloquial Upper Sorbian) was assembled within the French-German collaborative project *EuroSlav 2010* (cf. Breu & Adamou 2011, Breu et al. 2023) by Lenka Scholze and Walter Breu. The corpus of Colloquial Upper Sorbian (Catholic vernacular SWR) is based on speech recordings from 8 native speaker informants from 2012 and can be accessed under the URL:

[https://pangloss.cnrs.fr/corpus/Sorabe%20sup%C3%A9rieur%20\(courant\)?lang=en&seeMore=true](https://pangloss.cnrs.fr/corpus/Sorabe%20sup%C3%A9rieur%20(courant)?lang=en&seeMore=true) (accessed 8 Apr 2024). For spoken Lower Sorbian, the *Korpus GENIE – GESprochenes NIEdersorbisch* was compiled under the supervision of Roland Marti and is available under the URL: <https://genie.coli.uni-saarland.de/> (accessed 8 Apr 2024). The GENIE corpus covers broadcast and dialectal field recordings from the 1950 to 2006. The corpus is described in Marti et al. (2012).

<sup>19</sup> There is a number of structural differences falling under the cover term of “archaic features” between Sorbian on the one hand and the remainder of West Slavic on the other hand, which are relevant for clitic research in Slavic. Generally speaking – and abstracting from microtypological differences between Upper and Lower Sorbian – Sorbian retained e.g. the dual number, aorist and imperfect, and a complete paradigm of past tense auxiliary forms incl. 3SG/PL (cf. Franks & King 2000: 162–186 [Ch. 3.4]; Stone 1993: 614, 633, 635). However, Scholze’s (2007) study of colloquial Upper Sorbian indicates that the contemporary spoken vernacular of Upper Sorbian significantly deviates from the codified standard, e.g. by a decline of the dual (cf. Scholze 2007: §IV.4, 109–132), loss of the aorist/imperfect, which comes about with the generalisation of the *l*-participle as sole past tense form (cf. Scholze 2007: 182, 210–211), and the omission of 3SG/PL and even 1SG/PL past tense auxiliaries in the past tense formation (cf. Scholze 2007: 108). These phenomena render colloquial Upper Sorbian more like most other West Slavic languages.

the exact denotation of Common Czech is the spoken colloquial language of Bohemia with its centre Prague.<sup>20</sup> Common Czech may be described as an interdialect that arose from levelling of the local Bohemian dialects. It represents the primary form of everyday communication in Bohemia and significantly differs from the Czech standard language *spisovná čeština* ‘Written Czech’. Differences between both varieties are located on all linguistic levels giving rise to what has been described as diglossia. Common Czech isoglosses may be traced back to the Baroque era before the Battle of White Mountain in 1620: some linguistic features are attested as early as the emergence of the Kralice Bible in the late 16th century. The sociolinguistic situation of the Czech Republic becomes more complex, when taking Moravia (incl. the Czech part of Silesia) into consideration. While Common Czech is the dominant code in Bohemia, Moravia is still dominated by the traditional autochthonous dialects, which are roofed by Written Czech. Thus, the sociolinguistic situation in Moravia is essentially diaglossic.<sup>21</sup> However, Common Czech features have been reported to occur in the colloquial language of Moravian cities suggesting that Common Czech is spreading from Bohemia to Moravia (cf. E. Eckert 1993: 14; Nekvapil 2005: 13).<sup>22</sup> The data in the present study come from the ORAL conversational subcorpus, which covers the whole Czech Republic. Irrespective of the localisation of the data points, the data may be considered to represent contemporary spoken or colloquial Czech, but it surely does not only involve Common Czech in the strict sense.

## 2.3 Polish

Spoken Polish (Pol) or *polszczyzna mówiana* is primarily represented by spoken colloquial Polish, although a few data points from television and radio broadcasts are included, too, which might represent a more sophisticated kind of speech sometimes. Colloquial Polish is the code of informal communication and geographically distributed all around Poland. It is generally accepted that the traditional Polish dialects rapidly diminished during the second half of the 20th century. There are

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<sup>20</sup> The term *Česko* originally refers to Bohemia only and did not involve Moravia and Silesia in the first place.

<sup>21</sup> The terms *diglossy* and *diaglossy* refer to different sociolinguistic notions and should not be confused. Diglossy refers to complementary bilingualism, whereby two linguistic codes are used in mutually exclusive communication situations. Here, one code is a native language proper, i.e. a first language naturally acquired in childhood, while another code is a second language learnt in an institutional context via instruction. (A similar situation in the Germanophone world can be found in Switzerland, where people speak their local Swiss German dialect as native language and learn the Swiss variant of standard German only in school. According to anecdotal reports, Swiss people perceive standard German as a foreign language.) Diaglossy refers to a language in terms of a diasystem, i.e. a complex linguistic system composed of multiple varieties, which may be determined along various axes (areal: diatopic, social: diastratic, register: diaphasic). Different varieties of the respective axes are not complementary, but compose a complex, possibly layered set of features of a particular native speaker’s linguistic competence.

<sup>22</sup> Jiří Muryc (p.c., 5 Dec 2024) informed me that it is not the case that there is linguistic levelling between Bohemia and Moravia. On the contrary, the sociolinguistic differences between both areas are still persistent.

several reasons for the decline of the traditional rural dialects. First, the forced migrations accompanying Poland's so-called "westward shift" after World War II led to a dissolution of the Polish linguistic landscape in the Kresy. As a result, a bundle of "new mixed dialects" emerged in the so-called Recovered Territories in the west and north. Second, the communist regime neglected the internal linguistic differentiation to a large extent for political reasons. Third, as a general trend all over Europe, dialect levelling increased during the second half of the 20th century due to the spread of new media, a rise of the education level and drop of the illiteracy rate, and urban migration. Consequently, one would expect contemporary colloquial Polish to rather not exhibit dialectal interferences. Still, one can occasionally encounter dialectal features in the spoken data, which can be treated as artefacts. Nevertheless, bear in mind that there is still a relevant degree of diatopic variation in Polish. On the one hand, autochthonous lects are better preserved in regions of the periphery (Silesian, Goralish, but also Kashubian, Lemko). On the other hand, features of the traditional macrodialects are retained in urban dialects (e.g. in Kraków) and regiolects. Some former dialectal contrasts are still preserved in Polish orthoepy, which recognises two prescriptive norms. The so-called "Kraków-Poznań pronunciation" exhibits regressive voicing assimilation across word boundaries, the voiced realisation of /v/ after voiceless obstruents in the onset, and the positional assimilation of /n/ to a following velar stop after morpheme concatenation. In contrast, the nowadays dominant "Warsaw pronunciation" does not apply regressive voice assimilation, devoices /v/ after voiceless onset consonants, and retains /n/ before velar stops across morpheme boundaries.

## 2.4 Goralish (Spisz dialect)

The term *Goralish* (Gor) (Pol. *góralszczyzna*) or Goralish dialect (Pol. *gwara góralaska*, Cz. *goralská nářečí*, Slk. *goralské nárečia*) denotes linguistic varieties autochthonous in the Western Carpathians and western parts of the Eastern Carpathians, i.e. the Beskid and Tatra mountains. Goralish varieties are spoken in the south of contemporary Poland in the Lesser Poland Voivodship (Pol. *województwo małopolskie*), the Subcarpathian Voivodeship (Pol. *województwo podkarpackie*), and, to a lesser extent, the Silesian Voivodship (Pol. *województwo śląskie*). Outside Poland, Goralish is autochthonous to minor parts of the Czech and especially Slovak Republic. Genetically, it is a part of the Lesser Polish dialect cluster. Goralish varieties on Slovak and Czech territory have been traditionally regarded as Polish lects by Polish linguists (cf. Dudášová-Kriššáková 1995: 92 for a critical perspective, but Dudášová-Kriššáková 2016: 7 agrees). On the other hand, Dudášová-Kriššáková (1995: 92) remarks that Slovak dialectology has neglected Goralish. Though sharing a historical spark with other Lesser Polish varieties, Goralish possesses several distinctive features and,



thus, comprises a group of lects on its own within the Lesser Polish macro group according to Sowa & Sowa (2002: 135). Furthermore, Goralish exhibits transitional features from the Lesser Polish macrodialect (Pol. *dialekt małopolski*) to Slovak, particularly Eastern Slovak.<sup>23</sup> Goralish also exhibits an areal convergence with other linguistic varieties of the Carpathians, e.g. Rusyn.<sup>24</sup> Goralish has received considerable attention in Polish dialectology and it is widely known across Poland in popular culture, whereby the diatopic variety of Podhale represents the iconic model of Goralish for most Polish people.<sup>25</sup>

The thesis particularly draws from the diatopic variety of Spisz (*gwara spiska*). Spisz/Spiš (Ger. *Zips*, Hu. *Szepes*) belongs geologically to the Outer Western Carpathian Mountains. The area largely lies within the borders of the Slovak Republic (henceforth: Slovakia) and to a lesser extent within the Republic of Poland (henceforth: Poland). It does not have an official administrative status, but is the name of a historic region, which nowadays is integrated into several administrative units. The main part of Spisz/Spiš is situated in Eastern Slovakia (Slk. *Východné Slovensko*) and ranges over the western parts of the Prešov and Košice Regions (Slk. *Prešovský kraj* and *Košický kraj* respectively). Its Polish portion is in the High Tatra Mountains in the south of the Lesser Poland Voivodship (Pol. *województwo małopolskie*) and belongs to the three municipalities (Pol. *gminy*) of Łapsze Niżne (comprising the biggest portion), Bukowina Tatrzańska, and Nowy Targ. Polish Spisz is roughly bounded by the Polish-Slovak state border in the south and east, by the Białka river (Slk. *Biela voda*) in the west and north, and by the Dunajec river, which is staunches to the reservoirs of the Czorsztyn lake (Pol. *jezioro czorszyńskie*) and Sromowce lake (Pol. *jezioro sromowieckie*), in the north-east.<sup>26</sup>

Following Grochola-Szczepanek (2012: 33–34), I will give a brief overview of the history of the region, as it was ethnically heterogeneous during most part of its history. In the High Middle Ages (10th–15th c.), the area of Spisz/Spiš was initially populated by Poles and Slovaks coming from the north and south respectively. Germans from the west began their residence towards the end of the High Middle Ages (12th c.). In the Late Middle Ages (15th c.), Aromanian and Rusyn shepherds coming from the east settled in this area. During the same period, Gypsy nomads arrived through the

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<sup>23</sup> Eastern Slovak has a number of features setting it apart from standard Slovak, which is based on Central Slovak dialects. It shows convergence with its immediate Slavic neighbours Polish and Rusyn. For one thing, East Slovak dialects do not possess the quantitative opposition between long and short vowels and have penult word stress (cf. Short 1993b: 589).

<sup>24</sup> See Danylenko (2020a) for an overview of the so-called Carpathian Convergence Area. But note that the article is particularly focused on Ukrainian and relevant Balkan languages, whereas it does not provide information on Slovak, Polish, or related lects.

<sup>25</sup> The two largest settlements of Podhale are Nowy Targ and Zakopane. While Nowy Targ is the former capital town of the historical region of Podhale and its economical and logistical centre to date, Zakopane is famous for winter sports and as an important destination for tourists.

<sup>26</sup> The area delimited in this way is missing the village of Nowa Biała on the other side of the Białka river. This area belongs to Polish Spisz in the narrow sense, too.

Balkans. The first Jewish population arises during the Early Modern Period (18th c.). Being part of the Hungarian county of Spiš (Hu. *Szepes vármegye*), present day Polish Spisz belonged – either directly or indirectly – to the Hungarian crown during the Middle Ages and was part of the Habsburg Monarchy later. During the Partitions of Poland, the Białka river was the geographical and political border between the Habsburg crownland of Galicia (parts of which comprise present-day Poland around Kraków and present-day Ukraine around Lviv) and Hungary (which incorporated the whole of present-day Slovakia).<sup>27</sup> After World War I, there was dispute between Poland and Czechoslovakia due to conflicting territorial claims. In 1920, Poland obtained an area of roundabout 195.5 km<sup>2</sup> with over 8200 inhabitants living in 14 municipalities following a decision of the Allies.<sup>28</sup> After 1920, the Białka river has formed the inner-Polish border between Podhale and Spisz, which is still valid today (cf. Dudášová-Kriššáková 2016: 21–22).

Dudášová-Kriššáková (2016: 20–23) divides the varieties of the entire Spisz/Spiš into a northern part in Zamagurí, which lays partly in Poland and partly in Slovakia, and a southern part in *porieči Popradu*. The variety of Polish Spisz belongs to the northern group, which, though neighbouring with Slovak for a long time, was influenced less by Slovak than the southern group. Dudášová-Kriššáková (2016: 23) regards the northern group as transition varieties between Slovak and Polish.

The relevance of the Spisz dialect is grounded in its character of being a contact or transitional variety between Polish and Slovak, the latter of which is regarded as an uncontroversial Wackernagel or 2P clitic language (cf. Franks & King 2000: 139; Short 1993b: 566–567). It seems therefore plausible to assume that it stands somewhere in the middle between Polish and Slovak. This could be either instantiated by a strong Wackernagel pattern, such that Goralish would mirror Slovak, or by a weak Wackernagel pattern, whereby Goralish would exhibit more Wackernagel cliticisation than Polish. Parallels with or interferences from Slovak are evident on diverse linguistic areas, the most obvious being the lexicon (incl. phraseology), cf. (2.1)–(2.4).<sup>29</sup>

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<sup>27</sup> I am simplifying the historical course of events here and take the situation after the Third Partition of Poland as a point of departure.

<sup>28</sup> Ciągwa (1995) gives an area of 170 km<sup>2</sup>.

<sup>29</sup> Examples are numbered for each chapter separately. The figure before the dot corresponds to the chapter, while the figure following it represents the consecutive examples' numbering. As far as possible, examples are cited with their source of origin. In case of data from the research literature, quotations adhere to the author-year-page scheme. Empirically obtained data are provided with the exact source. Corpora are followed by specifications of subcorpora, if applicable. Data elicited from the World Wide Web by informal internet search engine queries are provided with a short code under which the respective source is bibliographically listed including the exact URL the data was taken from. Finally, examples constructed by the author of the present thesis are marked as such. Sometimes, examples are provided with a two-layered layout involving a normalized/standardised or latinised line next to a dialectal or Cyrillic line between angle brackets <>. The procedure is motivated by the thesis's corpus linguistic foundation in order to ensure replicability.

- (2.1) a. (matka jej umarła wcas) bo ona  
 <(matka jej umarła wczas) bo ona  
 mother.NOM.SG her.DAT die.PST.SG.F early because she.NOM  
 miała kielo rokók?  
 miała kielo roków?>  
 have.PST.SG.F how\_much year.GEN.PL  
 ‘(her mother died early) because how old was she?’ (Gor; KoSpi)
- b. koľko rokov si mal?  
 how\_much year.GEN.PL AUX.3SG have.PST.SG.M  
 ‘how old were you?’ (Slk)
- (2.2) a. (baca przigrozoł palcem  
 <(baca przygrażał palcem  
 elder\_shepherd.NOM.SG threaten.PST.SG.M finger.INS.SG  
 juhasowi i zawołał) ino daj pozór  
 juhasowi i zawołał) ino daj pozór>  
 younger\_shepherd.DAT.SG and shout.PST.SG.M only give.IMP.2SG attention.ACC.SG  
 ‘(the [elder] shepherd wagged his finger at the [younger] shepherd and shouted) just pay  
 attention’ (Gor; KoSpi)
- b. dávat’ pozor (Slk)
- c. dávat pozor (Cz)  
 give.INF attention.ACC.SG  
 ‘(to) pay attention’
- (2.3) a. kumoter (Gor; KoSpi)
- b. kmotor (Slk)
- c. kmotr (Cz)  
 ‘godfather’
- (2.4) a. zaczek / zocek  
 ‘acolyte.DIM’ (Gor; KoSpi)
- b. žiak (Slk)
- c. žák (Cz)  
 ‘pupil; student’

The Spisz dialect is the only Goralish variety an electronic corpus is currently available for (cf. Grochola-Szczepanek et al. 2019).<sup>30</sup> There have been several publications dealing with the Goralish variety of Polish Spisz (cf. Dudášová-Křiššáková 1995, 2016; Grochola-Szczepanek 2006, 2008, 2009, 2010, 2012; Sowa 1976, 1994, 2006; Sowa & Sowa 2002, with further references). These publications address phonology, morphology, and lexis as well as their variation with respect to social and demographic factors rather than syntax. Especially, there has not been any survey of this variety's clitic system or a subcomponent thereof to date.<sup>31,32</sup> It seems fair to say that the present thesis pioneers in this respect and introduces Goralish data into the big picture of cliticisation in Slavic. Since knowledge about Goralish is not widespread and there is no widely accessible overview article yet, I will summarise the most important features relevant for cliticisation in this study (cf. Grochola-Szczepanek 2012 for a more comprehensive overview):

1. The past tense person markers are fully mobile, i.e. they have clitic status.
2. The conditional marker *-by-* 'COND' is freely mobile.
3. Retention of the old aorist ending [x] *-ch* as either [k] *-k* (common across Lesser Polish) or [f] *-f* (particularly Spisz Goralish) in the 1SG conditional (*byk*, *byf* 'COND.1SG'), which has become generalised to the past tense (*bylek*, *bylef* 'be.PST.1SG').
4. Grammatical agreement drop of PST.1SG/PL with simultaneous occurrences of overtly realised subject pronouns.
5. First person plural past tense marker/clitic and the corresponding subject pronoun *my* 'we' are homophonous.<sup>33</sup>

<sup>30</sup> Language samples of Goralish varieties incl. Spisz (though dated) can be found in traditional dialectological treatises, cf. e.g. the classical surveys by Nitsch (1957) or Urbańczyk (1976), as well as individual surveys (cf. e.g. the respective references in this section). Usually, these publications are available only in print and have a low circulation; it is difficult to access and process them electronically. Some of the data dates back to the mid-20th century or before. An electronic resource for Polish dialectology, which, however, does not comprise an electronic corpus in the narrow sense, is the online compendium *Dialekty i gwary polskie. Kompendium internetowe* edited by Warsaw-based dialectologist Halina Karaś (<http://www.dialektologia.uw.edu.pl/index.php>).

<sup>31</sup> Cf. Grochola-Szczepanek (2012: 45–51) for a summary of main research on the linguistic varieties of Polish Spisz.

<sup>32</sup> Admittedly, the present thesis cannot fill this gap, because of a different scope. A survey of the clitic system of Spisz Goralish will have to take both auxiliary and pronominal clitics into consideration. An exhaustive account shall not restrict the set of pronominal clitics to constant clitics but also incorporate inconstant clitics. I leave this enterprise for future research.

<sup>33</sup> Example (i) nicely illustrates that both superficially similar occurrences of *my* need to be distinguished. However, it is sometimes difficult or even impossible to determine, which instance one encounters in a given example. As a consequence, scholars are facing some degree of ambiguity in the data.

(i) no to **my** se postavili **my**  
 < no to **my** sobie postavili **śmy** >  
 well TO we.NOM REFL.DAT place.PST.PL.VIR AUX.1PL  
 'well, so we raised [a house] by/for ourselves'

(Gor; KoSpi)

## 2.5 Slovak

To begin with, Slovak (Slk) is underrepresented as an object language within Slavic studies and may be regarded as understudied especially with respect to its clitic system.<sup>34</sup> In fact, I am aware only of six scholarly publications of different range and scope relevant for the issues of Slavic cliticisation with particular reference to Slovak, i.e. Chidambaram (2013), Franks & King (2000), Ivanová & Bashir Géčová (2021), Ivanová et al. (2021), Janečka (2020), Short (1993b), and Szczegielniak (1997).<sup>35</sup> Fortunately, Slovak spoken data is available and accessible through the Slovak National Corpus.

Generally, Slovak appears only *en passant* in most work on Slavic 2P clitics. It is consistently considered a Wackernagel clitic language but not scrutinised individually in more detail. One may only speculate, whether Czech is taken as a general substitution model for Slovak given the high degree of linguistic convergence between both languages. They are closely related and form a branch of West Slavic sharing an original dialect continuum (via Moravian and West Slovak).<sup>36</sup> During the longest portion of the 20th century, Czech and Slovak were spoken in the common state of Czechoslovakia, which had an impact on the languages, as they were tied with a joint political system, economy, society, educational system, and media landscape. To date, Czech and Slovak are mutually intelligible (cf. Golubović & Gooskens 2015; Nábělková 2007). However, Slovak can and should be studied in its own right. It is an independent ethnic, national, or standard language, roofing a distinct set of various dialects, having its own diachrony, and exhibiting areal-linguistic interrelations distinct from those ones of the Czech language.<sup>37</sup>

Despite these arguments, I can treat Slovak only as supplementary material due to limitations of time and workforce. Still, a look at Slovak clitics *in vivo* is a gain for the present study and may serve as the basis for future research, where Slovak should be given more space. Finally, bear in mind that the codification of present-day standard Slovak goes back to the so-called *Štúrovci*, named after their leading figure Ľudovít Štúr, on the basis of a central Slovak dialect.<sup>38</sup>

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<sup>34</sup> This situation was also confirmed by Martina Ivanová (p.c., 11 Apr 2023). Remarkably, this matter of facts leads to the situation that Ivanová et al.'s (2021) research article on clitic placement in Slovak as a foreign language predominantly relies on research literature on Czech.

<sup>35</sup> Slovak is also one of Rivero's (1991) main object languages, but she is not concerned with cliticisation, even though all her examples contain clitics.

<sup>36</sup> As noted in Section 2.4 on Goralish, the Eastern Slovak dialect cluster shares features with Polish.

<sup>37</sup> While Czech was in contact with Austrian German as its superstrat language, Slovak was linguistically dominated by Hungarian. Furthermore, Slovak has inner-Slavic areal transitions to Polish (particularly, to Lesser Polish and Goralish) and East Slavic (Rusyn, Ukrainian).

<sup>38</sup> Earlier attempts to codify a Slovak linguistic norm on the basis of a western Slovak dialect did not become prevalent.

## 2.6 Silesian

Silesian (Sil) is a Lechitic variety autochthonous in south-western Poland. It is traditionally treated as a dialect of Polish. Silesian, whose status has been subject to a long-standing controversy (cf. Kocyba 2015), consists of three areal groups that are distributed over the Opole voivodship (Pol. *województwo opolskie*) and the Silesian voivodship (Pol. *województwo śląskie*). The northern group is located in the Opole voivodship by and large. The central group is distributed across the area of the Upper Silesian Industrial Region with Gliwice and the voivodship's capital city Katowice.<sup>39</sup> The southern group in the area around the city of Cieszyn (Cz. *Těšín*) has a somewhat different status. First, the Cieszyn portion of Silesia remained a part of the Habsburg Empire after the Silesian Wars (around the middle of the 18<sup>th</sup> century) and was never incorporated into Prussia (later Germany). This situation is mirrored by isoglosses distinguishing southern Silesian from the central and northern varieties (but shared with the neighbouring Goralish varieties in the east).<sup>40</sup> Second, Cieszyn Silesia is split between present-day Republic of Poland and the Czech Republic. The Czech part is called *Zaolzie* 'Trans-Olza' by Poles and *České Těšínsko* 'Czech Cieszyn Silesia' by Czechs.<sup>41</sup> Accordingly, there are two very different sociolinguistic situations for southern Silesian: The Poland-based portion is roofed by Polish, but it also exhibits some degree of convergence with northern Silesian varieties (cf. Greń 2001). On the other hand, the Czechia-based portion of Silesian is roofed by Czech, which is why Silesian speakers are bi- or multilingual with Silesian, Czech, and (Standard) Polish (through schooling) here.<sup>42</sup> Accordingly, different kinds of language mixing of the various linguistic codes can be found, e.g., interlocutor-dependent code-switching from one lect to the other or nonce borrowings from Czech in predominantly Silesian utterances (cf. Bogoczová & Bortliczek 2014; Bogoczová & Muryc 2021).

During its colourful history, the region of Silesia belonged to different neighbouring powers. For centuries, Silesia was the bone of contention between the Polish and Czech crown, followed by

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<sup>39</sup> Note that Zagłębie Dąbrowskie belongs to the contemporary Upper Silesian Industrial Region but is not part of Silesia proper in terms of a linguistic area or cultural heritage. Historically, Zagłębie Dąbrowskie was a part of Lesser Poland and annexed by Prussia during the partitions of the Polish-Lithuanian Commonwealth. However, after the Congress of Vienna in 1815, Zagłębie Dąbrowskie belonged to the so-called Kingdom of Poland (= Congress Poland), a client state of the Russian Empire. Silesia proper remained uninterruptedly part of Prussia (later, the German Empire) after the First and Second Silesian War in the 1740s (the Third Silesian War in the 1760s did not involve geopolitical changes). The Cieszyn area remained a part of Austria after the Silesian Wars between Prussia and Austria.

<sup>40</sup> For example, Cieszyn Silesian (CSil) and Goralish share some inflectional paradigms, cf. PRS.1SG -*ym* in *myśl<sup>ym</sup>* 'I think', *chod<sup>ym</sup>* 'I walk' vs. Polish *myślę*, *chodzę* and central Silesian *myśla*, *chodza*, or the morphological form of indefinite pronouns (CSil. *cośi*, *cośika* 'something', Gor. *cosi*, *cosik* 'something').

<sup>41</sup> I thank Jiří Muryc (p.c.) for making this distinction clear to me. Kai Witzlack-Makarevich (p.c.) pointed out to me that the term *Zaolzie* is controversial insofar as it represents a purely Polish perspective on the area (meaning 'behind the Olza river'), which today is predominantly inhabited by ethnic Czechs.

<sup>42</sup> This situation is reminiscent of what Rabus (2019) calls Border effects, i.e. the internal diversification of a lect due to different sociolinguistic embedding of its regional subcomponents.

Austrian and, finally, Prussian dominance. Therefore, Silesian is strongly influenced by the German language due to sociolinguistic roofing and bilingualism, which led to an increase of German loans in the Silesian lexicon in the first place, but transferences exist in phraseology and grammar, too (cf. Tambor 2013, 2014). Being a part of the West Slavic (Lechitic-Czechoslovak) dialect continuum, Silesian shares areal features with neighbouring varieties. On the one hand, there are commonalities with Greater Polish in the north and Lesser Polish in the east. Notably, the eastern area of the central group around present-day Katowice was formerly regarded as a – political as well as linguistic – part of Lesser Poland (cf. S. Bąk 1974: 14; Urbańczyk 1976: 13, 29, 81).<sup>43</sup> Originally, the eastern varieties were distinguished by two isoglosses from the core Silesian area around Gliwice, i.e. the so-called *mazurzenie* ‘mazuration’ and the Lesser Polish fortition of word-final velar fricative *-ch* [x] > *-k* (cf. also Urbańczyk 1976: map 1 „Granice zjawisk fonetycznych”, map 3 „Schematyczny podział dialektów polskich”, map 5 „Granice ważniejszych zjawisk w Małopolsce”).<sup>44</sup> Contemporarily, the eastern central group is treated as a Silesian-Lesser Polish transitional dialect (Pol. *gwary pogranicza śląsko-małopolskiego*; cf. S. Bąk 1974, map 6 „Nowa mapa dialektów śląskich”). On the other hand, Silesian shares commonalities with Czech and Slovak in the south, some of which are archaisms, e.g. the retention of the close front vowel [i] after etymologically soft *r'* > *rz* [ʒ], e.g. *trzi* ‘three’.<sup>45</sup> Others are contact-induced transitions. From a historical point of view, it should also be noted that, while German became the dominant language of legislation and administration in Lower Silesia, Czech became the dominant Chancery language in Upper Silesia roughly in the mid-15th century and kept its status as an administrative language until the 17th century (cf. Jurek 2003). One of the most iconic phonological isoglosses is the so-called “open articulation” of the former front nasal vowel *ę* > *a* (most consistently applied at the right word margin), which gave rise to a complete nominative-accusative syncretism in the female declension. Another phonological feature is the raising of etymologically long vowels *ē* > *y*, *ā* > *o*, e.g. *chlyb* ‘bread.NOM/ACC.SG’, *piynkno* ‘beautiful.NOM/ACC.SG.F’ (cf. I. Kulik 2021), which is not restricted to Silesian, however.

Silesian is a predominantly oral variety, although there have been attempts to develop a standardised written norm for Silesian in recent years (cf. Henzelmann 2016, 2019: 16–19, and references therein). Therefore, it has been sometimes regarded as a “(literary) microlanguage” following Duličenko’s (1981) framework (cf. Czesak 2006, 2012; Henzelmann 2016, 2019: 16–19). The considerable amount of text production by activists covers a wide range of genres and styles.

<sup>43</sup> I am thankful to Robert Hammel (p.c.) for making me aware of the relevant literature.

<sup>44</sup> The notion of *mazurzenie* ‘mazuration’ denotes the merger of the postalveolar fricatives and affricates with the corresponding alveolar ones: [ʃ] <sz> > [s] <s>, [ʒ] <ż> > [z] <z>, [tʃ] <cz> > [ts] <c>, [dʒ] <dż> > [dz] <d>. This phonological neutralisation has equivalents in other Slavic idioms like the so-called *cakavizam* ‘cakavism’ attested in some Čakavian varieties – <š> > <s>, <ž> > <z>, <č> > <c> (cf. Brozović 1998: 223) – or *tsokanye* known from Northern Russian dialects (cf. Sussex & Cubberly 2006: 523).

<sup>45</sup> By the same token, [i] is retained in Goralish, which is referred to as *Podhalań archaism* in Polish linguistics, but, as far as I am aware, there has not been any link with the similar diachronic fact in Silesian.

This is fortunate insofar as it provides corpus linguists with an increasing amount of electronically processable data. However, no spoken corpus of Silesian exists yet. Speech data could be crawled from diverse sources in the World Wide Web in principle, but this would involve high amounts of work and data processing before a usable corpus would be built.<sup>46</sup> An exception is the data collected by dialectologists, as it solely represents spoken language. The most prominent data collections by Nitsch (1957) and Urbańczyk (1976) are dated, however. A more recent and electronically retrievable data sample has been edited by Karaś (2010).<sup>47</sup>

Although being regarded as a traditional dialect of Polish, Silesian is exceptional in several respects. It is the most vital code among the diatopic varieties or minority languages in present-day Poland. According to the 2021 census, approximately 467,000 speakers (= 1.2% of Poland's population) use Silesian as a means of communication at home.<sup>48</sup> Approximately 55,000 respondents thereof declared to use Silesian exclusively in this domain.<sup>49</sup> Furthermore, Silesian is not a solely rural dialect restricted to the so-called immobile residents of the countryside, but it is also an urban variety (cf. Hentschel 2003: 68; Tambor 2013: 8). Among others, it is spoken by inhabitants of the agglomeration of the Upper Silesian Industrial Region around its two biggest cities Katowice and Gliwice (cf. Hentschel 2003: 59).<sup>50</sup> Furthermore, it is not restricted to a particular social class definable by economic or educational parameters (cf. Bogoczová & Bortliczek 2014: 34–35/94–95;

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<sup>46</sup> The creation of a corpus does not belong to the goals of the present study.

<sup>47</sup> An older version of Karaś's (2010) compendium is the dialectological guide (Karaś 2009), which presents a subset of the compendium's material.

<sup>48</sup> The 2021 census's report on national or ethnic identity and language usage has not been published, yet. The statistical data is available online: <https://stat.gov.pl/spisy-powszechnie/nsp-2021/nsp-2021-wyniki-ostateczne/> (accessed 26 Apr 2024).

<sup>49</sup> There is a significant decrease in the numbers between the 2021 census and the previous one from 2011. According to the 2011 census (Gudaszewski 2015: 70), approximately 529,000 respondents (= 1.4% of Poland's population) used Silesian at home, more than 126,000 of which declared to use it exclusively. For comparison: In 2021, ca. 89,000 respondents provided to use Kashubian at home. German was declared to be used by approximately 216,000 respondents. (The number for German requires discussion, as it dramatically changed in the previous decade: In 2011, it was roundabout 96,000 questionees, who provided German as a language of private use. Perhaps, the excessive increase within these 10 years could be attributed to a growing number of returnees, who have spent a considerable time in German-speaking countries but returned to Poland with their families eventually. This seems to be corroborated by data for English presumably telling something about returnees from the UK and Ireland (and, perhaps, the US): In 2011, ca. 103,000 respondents provided English as a language at home, whereas the number increased to ca. 737,000 individuals in 2021.)

<sup>50</sup> This is geographically somewhat imprecise, since the municipalities of Zagłębie Dąbrowskie 'Dąbrowa Basin' belong to the very same industrial region and the metropolitan area associated with it. However, Zagłębie Dąbrowskie has not been part of Silesia in the narrow sense, but belonged to Lesser Poland instead. During the Polish partitions, it was under Russian annexion, while Silesia was part of the Prussia/Germany and – to a smaller extent – Austro-Hungary. These historical facts are reflected linguistically to date, as the Dąbrowa Basin does not belong to the Silesian dialect continuum.



Tambor: 2013: 4).<sup>51,52</sup> Tambor (2013: 19–20) informs that Silesian features have made it into the colloquial Polish speech of inhabitants of Silesia who have never been acquainted with Silesian.

Despite recent efforts to codify and standardise Silesian, there is still no consensus among scholars as to the linguistic status of this code. Jaroszewicz (2020) is perhaps the first scholar to conclude that Silesian literary production has achieved the stylistic diversity and orthographic stability that is usually expected from a written standard language. Still, Silesian is a predominantly oral variety; literature and journalism have been niches for language enthusiasts until recently.<sup>53</sup> Thus, it seems fair to still follow Hentschel's (2003: 68) assessment that the sociolinguistic situation in Upper Silesia (covering the *opolskie* and *śląskie* voivodships by and large) may be characterised as some sort of diglossia between Polish and Silesian. Furthermore, no comprehensive standard grammar or standard dictionary of Silesian exists.<sup>54</sup> On the other hand, it needs to be acknowledged that Jaroszewicz (2022) presents an orthographical dictionary for Silesian. It is explicitly normative in nature and contains the sketch of a grammar, which, however, is confined to inflectional morphology.<sup>55</sup>

As with Goralish, cliticisation has never been purposefully studied in Silesian to date, such that the thesis breaks new ground. In the following I will provide those isoglosses of Silesian, which are most important for cliticisation. This is why I will delimit the scope to (morpho-)syntactic features

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<sup>51</sup> This might be a general trend in 21st century Europe, where there is a comparably high degree of social permeability, and speakers of traditional dialects and minority languages have, in principle, equal access to institutions of higher education with corresponding repercussions for their social and economic advancement.

<sup>52</sup> In their Czech-Polish study, Bogoczová & Bortliczek (2014) focus on the South Silesian variety spoken in the Czech Republic, i.e. in Trans-Olza region, but their statements agree with what can be observed on the Polish side of Upper Silesia.

<sup>53</sup> However, the political change after the parliamentary election for the 10th Sejm on 15 October 2023 initiated political developments, which could have a significant impact on Silesian's status in the near future:

- i) The Civic Platform's top candidate Donald Tusk promised in March 2023 to recognise Silesian as a regional language (source: <https://www.pap.pl/aktualnosci/news%2C1550466%2Ctusk-jezyk-slaski-bedzie-uznany-za-jezyk-regionalny.html>, accessed 24 Apr 2024).
- ii) In December 2023, the Silesian Language Council (Sil. *Rada Ślōnskigo Jynzyka*) was founded by Silesian scholars, writers and translators, artists, editors, and politicians (source: <https://wiadomosci.onet.pl/slask/powstaje-rada-jezyka-slaskiego-jej-czlonkowie-chca-oficjalnego-uznania-slazakow-za/bdr7519>, accessed 24 Apr 2024).
- iii) In January 2024, Tusk's government started to work on a reform of the *Law concerning national and ethnic minorities and regional language* (source: <https://tvs.pl/informacje/lekcje-slaskiego-w-szkole-taka-mozliwosc-daje-uznanie-slaskiego-za-jezyk-regionalny/>, accessed 24 Apr 2024).
- iv) In March 2024, the draft bill was forwarded to the Polish parliament's *Committee of National and Ethnic Minorities*, after the right-wing opposition parties attempted to reject the bill in the first reading (source: [https://www.rmfm24.pl/regiony/slaskie/news-jezyk-slaski-jezykiem-regionalnym-projekt-ustawy-trafil-do-k.nId.7402075#crp\\_state=1](https://www.rmfm24.pl/regiony/slaskie/news-jezyk-slaski-jezykiem-regionalnym-projekt-ustawy-trafil-do-k.nId.7402075#crp_state=1), accessed 24 Apr 2024).
- v) On 26 April 2024, the Sejm accepted an amendment to the law to recognise Silesian as a regional language, which has not been ratified, yet. President Andrzej Duda vetoed against the bill on 29 May 2024.

<sup>54</sup> There are various dictionaries of different size, which are ranging from serious descriptive dictionaries to collections of curiosities. There is, however, not a single work, which were universally accepted or whose provisions were binding by the time this thesis is written.

<sup>55</sup> This normative orthography is a revised version of the spelling of the *Gōrnoślōnski ślabikōrz* 'Upper Silesian primer' published in 2010 by the association Pro Loquela Silesiana, which may be seen as the most important agent of the Silesian linguistic emancipation movement.

and ignore the levels of lexis, phonology, morphology, semantics, and pragmatics (cf. I. Kulik 2021; Tambor 2013, 2014, with references, for an overview).

1. Retained mobility of the past tense person markers, i.e. clitic status, including the present tense forms of the copula *być* ‘be’.
2. Full mobility of the conditional marker *by* ‘COND’.
3. Retention of the old AOR.1SG formative *-ch* [x] in the conditional (*bych*) and its generalisation for PST.1SG (absence of corresponding Polish *-m*)
4. So-called agreement drop of PST.1SG/PL is grammatical under overt realisation of the subject pronoun.
5. Grammaticality of the accusative with infinitive construction (ACI)
6. The past tense person marker/clitic ‘AUX.1PL’ is homophonous with the 1PL subject pronoun *my* ‘we’ (absence of the form *-śmy*)

## 2.7 Kashubian

Kashubian (Csb) is a West Slavic language belonging to the Pomeranian branch of the Lechitic subgroup. It is genetically closely related to extinct Slovincian and contemporary Polish. Three macro-dialects are recognised with stress assignment as the main classificatory isogloss. The North Kashubian dialects retained free and mobile stress, which is called “partially mobile” by some scholars, because paradigmatic mobility occurs only in a small set of inherited lexemes. Central and South Kashubian exhibit fixed, i.e. immobile stress. Central Kashubian shows free stress, i.e. any syllable may bear stress, which, however, does not change its position through inflection. South Kashubian has initial stress like Czech, Slovak, and Goralish. Today, Kashubian is sociolinguistically roofed by Polish, but it came into contact with German in the Middle Ages (cf. Zieniukowa 2015: 45), which was roofing Kashubian until the end of World War II. Therefore, the language exhibits a considerable amount of Germanisms (cf. Lubaś 2002: 265–266; Stone 1993: 790). Expectedly, borrowings from German prevail in the lexicon, but there are also instances of loan grammar. For one thing, the Kashubian past tense formation *miec* ‘have’ + past passive particle is commonly known for being modelled after the German perfect tense form (cf. Bartelik 2018; Menzel 2013; Tommola 2000, esp. 469–470).<sup>56</sup>

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<sup>56</sup> The data in (i)–(iii) suggest that not only the perfect tense formation, but also auxiliary choice is modelled after German, i.e. Csb. *běc*, Ger. *sein* ‘(to) be’ with unaccusative verbs (incl. verbs of motion proper), Csb. *miec*, Ger. *haben* ‘(to) have’ otherwise. I thank Hagen Pitsch for providing me with (iii). Compare the parallels with the German

Until recently, Kashubian has been considered a dialect in traditional Polish dialectology, although assessments differed outside of Poland. There is a growing number of Polish linguists who regard Kashubian as an independent language or ethnolect, but there is still no overall consensus, and the matter remains subject to controversy. Nevertheless, Kashubian was recognised as a regional language according to the Act on national and ethnic minorities and on the regional languages in 2005, which liberalised its usage in municipal institutions (the European Charter for Regional or Minority Languages came into effect in Poland in 2009). Furthermore, the law grants the possibility of voluntary Kashubian language classes in schooling, public financial support for language maintenance, and bilingual street signs in municipalities with at least 20% of the relevant population among the inhabitants. The first official grammar of standard written Kashubian was published by Makurat-Snuzik in 2016 (cf. Makúrôt 2016).

There is a considerable amount of text production with different styles and genres, including the press, as well as radio broadcasting (Radio Kaszëbë).<sup>57</sup> However, Makurat-Snuzik (2020) tells that various translations of English and Russian literature to Kashubian are not based on the original text, but on a Polish translation. Consequently, interferences of Polish occur in the Kashubian target text. Importantly, Kashubians are generally Kashubian-Polish bilingual, as pointed out by Makurat-Snuzik (Makurat 2007: *passim*).<sup>58</sup> She identifies four groups of speakers distinguished by age with different degrees of Kashubian-Polish bilingualism in terms of active (speaking, writing) and passive (listening-comprehension, reading-comprehension) linguistic skills, cf. Table 2.1.

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translations in (iv)–(vi) (n.b. the distribution of the Kashubian demonstratives mirrors German finite determiners to a large extent).

- |       |                                              |               |              |               |             |               |
|-------|----------------------------------------------|---------------|--------------|---------------|-------------|---------------|
| (i)   | To                                           | aùto          | mô           | rozjachoné    | kùrà.       |               |
|       | this.NOM.SG.N                                | car.NOM       | have.PRS.3SG | run_over.PTCP | hen.ACC.SG  |               |
|       | ‘This car run over the hen.’                 |               |              |               |             |               |
| (ii)  | Mój                                          | woejc         | mô           | ten           | čòłn        | zbudovõné.    |
|       | My.NOM.SG.M                                  | father.NOM.SG | have.PRS.3SG | this.ACC.SG.M | boat.ACC    | build.PTCP    |
|       | ‘My father built this boat.’                 |               |              |               |             |               |
|       | (Csb; Pitsch 2023a: 147)                     |               |              |               |             |               |
| (iii) | Ta                                           | białka        | je           | precz         | jidzenô.    |               |
|       | this.NOM.SG.F                                | woman.NOM.SG  | be.PRS.3SG   | away          | go.PTCP     |               |
|       | ‘This woman went away.’                      |               |              |               |             |               |
|       | (Csb; Stone 1993: 777)                       |               |              |               |             |               |
| (iv)  | Das                                          | Auto          | hat          | das           | Huhn        | überfahren.   |
|       | this.NOM.SG.N                                | car.NOM.SG    | have.PRS.3SG | the.ACC.SG.N  | hen.ACC.SG  | run_over.PTCP |
|       | ‘The car run over the hen.’                  |               |              |               |             |               |
| (v)   | Mein                                         | Vater         | hat          | das           | Boot        | gebaut.       |
|       | my.NOM.SG.M                                  | father.NOM.SG | have.PRS.3SG | the.ACC.SG.N  | boat.ACC.SG | build.PTCP    |
|       | ‘My father built the boat.’                  |               |              |               |             |               |
|       | (Ger; constructed, based on Pitsch 2023a)    |               |              |               |             |               |
| (vi)  | Die                                          | Frau          | ist          | weg-gegangen. |             |               |
|       | the.NOM.SG.F                                 | woman.NOM.SG  | be.PRS.3SG   | away-go.PTCP  |             |               |
|       | ‘The woman went away.’                       |               |              |               |             |               |
|       | (Ger; constructed, based on Stone 1993: 777) |               |              |               |             |               |

<sup>57</sup> There was also a television broadcast *CSB TV (Cassubia TV)* from 2010–2012, which has been cancelled (source: Csb: [https://csb.wikipedia.org/wiki/CSB\\_TV](https://csb.wikipedia.org/wiki/CSB_TV), Pol: [https://pl.wikipedia.org/wiki/CSB\\_TV](https://pl.wikipedia.org/wiki/CSB_TV), accessed 23 Aug 2023).

<sup>58</sup> Sometimes, Kashubian-Polish-German trilingualism can be found which is limited to the oldest generations (cf. Zieniukowa 2015: 49). Nomachi (2019: 454) considers even quadrilingualism between Kashubian, Polish, Standard High German, and Low German.

The emerging picture is that of a decline of bilingualism, to the disadvantage of Kashubian (cf. Makurat 2007: 102). Makurat-Snuzik concludes that, instead of substituting the Kashubian minority language by the Polish majority language, Kashubian-Polish bilingualism gives rise to a mixed code (Makurat 2007: 100–102):

“Obecnie sytuację językową na Kaszubach należy określić jako sytuację przejściową między etapem stosunkowo równomiernej znajomości języków a etapem kształtowania się polsko-kaszubskiego dialektu mieszanego.”

‘Currently, the linguistic situation in Kashubia needs to be classified as an intermediate stage between a relatively even proficiency in both languages and a phase of an emerging mixed Polish-Kashubian dialect.’ (Makurat 2007: 102)

Table 2.1: Degrees of bilingualism across different groups of Kashubian speakers (Makurat 2007: 98).

|                                            |                                                                     |
|--------------------------------------------|---------------------------------------------------------------------|
| Young generation      ( $\leq 30$ years):  | predominantly passive bilingualism (passive knowledge of Kashubian) |
| Middle generation      (31–50 years):      | predominantly active bilingualism                                   |
| Older generation      (51–70 years):       | predominantly active bilingualism                                   |
| Oldest generation      ( $\geq 71$ years): | predominantly passive bilingualism (passive knowledge of Polish)    |

This situation is reminiscent of other Slavic-Slavic mixed lects, i.e. the so-called *Trasyanka* (Belarusian-Russian mixed lect) in Belarus and *Surżyk* (Ukrainian-Russian mixed lect) in Ukraine.<sup>59</sup> Zieniukowa (2015: 50) rates Kashubian as an endangered language. In fact, the figures dropped significantly between the two last censuses in 2011 and 2021.<sup>60</sup>

I am not aware of any study that would examine Kashubian clitics (or what one would expect to be clitics). Therefore, there are no statements on pronominal positioning in Kashubian in the Slavicist literature on clitics. Franks & King’s (2000) handbook does not cover Kashubian, at all. Stone’s (1993) overview of Kashubian does not provide information on pronominal placement. To phrase a tentative observation, in those examples cited by Stone (1993) and Nomachi (2019), which contain a short pronoun (or object pronoun), these pronouns always occur contiguously to the verb. Instances of simultaneous second position placement is attested, too. The present thesis can be

<sup>59</sup> A comparable situation of Slavic-Slavic lect mixing is described for Trans-Olza by Bogoczová & Bortliczek (2014), who inform that the code, which they refer to by the label *po naszymu* ‘our way [to speak]’, has a Silesian-Czech basis.

<sup>60</sup> Compare the figures of both censuses: In 2011, almost 110,000 respondents declared the use of Kashubian at home, approximately 4,000 of which told to use it as the only means of communication. In 2021, ca. 89,000 respondents told to use Kashubian at home (–19%), whereby less than 1,800 (–55%) questionees confirmed its exclusive usage.

regarded as involving the first humble attempt to shed some light on the issue under discussion in Kashubian.

## 2.8 Clitic paradigms

Now that the object languages have been sufficiently described, I shall continue to delimit the scope of the study and determine the exact set of elements under scrutiny. The study is devoted to pronominal clitics, but the term is broad both in its common sense and its use in the study's title. Reflecting the denotation of *pronominal clitic* in the thesis' title first, I subsumed the reflexive clitics under the pronominal label (cf. Franks & King 2000 for a similar strategy). As to common sense, it is frequently acknowledged that pronouns come in two species in West Slavic. The first one is ambiguous between clitic and non-clitic, and it represents the majority of pronoun types. The other one formally distinguishes between clitic and non-clitic pronoun. In other words, two sets of clitics are commonly distinguished, whereby the first set comprises elements which are ambiguous between clitic and non-clitic, while the second set consists of only-clitic units morphologically distinct from never-clitic units. Several terminologies have been proposed to distinguish the former from the latter, which I summarise in Table 2.2.

Table 2.2: Terminologies for ambiguous vs. unambiguous clitic pronouns.

| Ambiguous clitic/non-clitic                      | Unambiguous clitic                         | Source                                         |
|--------------------------------------------------|--------------------------------------------|------------------------------------------------|
| long(er) pronoun/form                            | short(er) pronoun/form                     | traditional Slavistics                         |
| nestálá příklonka<br>'inconstant clitic'         | stálá příklonka<br>'constant clitic'       | Trávníček (1959a);<br>Rosen (2001)             |
| semi-clitic                                      | pure clitic                                | Avgustinova & Oliva (1997)                     |
| phonologisches Klitikon<br>'phonological clitic' | lexikalisches Klitikon<br>'lexical clitic' | Junghanns (2002a)                              |
| volatile (en)clitic                              | (en)clitic tantum                          | Ivanová et al. (2021 [citing<br>Mistrík 1966]) |

It is precisely the unambiguous clitic forms, which are in the present study's scope. I will adopt the conceptual pair CONSTANT CLITIC vs. INCONSTANT CLITIC from Czech *stálá příklonka* and *nestálá příklonka*, respectively, which are common technical terms in indigenous Czech linguistics (cf. Karlík et al. 1995; Petkevič 2009; Trávníček 1959a, 1959b) and have been introduced to a broader audience

translated to English by Rosen (2001, 2014) and Hana (2007).<sup>61</sup> The usefulness of the distinction becomes apparent once we examine the paradigms in Table 2.3. The table summarises the paradigms of the object pronouns in the accusative/genitive and dative in the six object languages.<sup>62</sup> Constant clitics are highlighted in bold face. They represent only a very small portion of the paradigm. In fact, it is only five unambiguous forms per lect – with the exception of Slovak (six forms) and Kashubian (four forms).

Table 2.3: Object pronouns (ACC/GEN, DAT) in West Slavic (Cz, Pol, Gor, Slk, Sil, Csb).

|                  | Czech                                                      | Polish                                                    | Goralish                                                  | Slovak                                | Silesian                                                  | Kashubian                             |
|------------------|------------------------------------------------------------|-----------------------------------------------------------|-----------------------------------------------------------|---------------------------------------|-----------------------------------------------------------|---------------------------------------|
| ACC/GEN.1SG      | mě   (mne)                                                 | (mię)   mnie                                              | (mię)   mnie                                              | <b>ma</b>   mňa                       | (mie)   mnie                                              | miã <sub>ACC</sub>   mie              |
| ACC/GEN.2SG      | tě   tebe                                                  | cię   ciebie                                              | <b>cie</b> <cię>  <br>ciebie                              | <b>t'a</b>   teba                     | <b>cie/czie</b>  <br>ciebie                               | <b>cã<sub>ACC</sub>/ce</b>  <br>cebie |
| ACC/GEN.3SG.M    | <b>ho</b>   jeho                                           | <b>go</b>   jego                                          | <b>go</b>   jego                                          | <b>ho</b>   jeho                      |                                                           | <b>go</b>   jego                      |
| ACC/GEN.3SG.F    | jì <sub>ACC</sub> /jí <sub>GEN</sub>                       | jã <sub>ACC</sub> /jej <sub>GEN</sub>                     | jã <sub>ACC</sub> /jej <sub>GEN</sub>                     | jù <sub>ACC</sub> /jej <sub>GEN</sub> | jom <sub>ACC</sub> /ji/jej <sub>GEN</sub>                 | jã <sub>ACC</sub> /jí <sub>GEN</sub>  |
| ACC/GEN.3SG.N    | jě <sub>ACC</sub> / <b>ho</b> <sub>GEN</sub>  <br>jej/jeho | jě <sub>ACC</sub> / <b>go</b> <sub>GEN</sub>  <br>je/jego | jě <sub>ACC</sub> / <b>go</b> <sub>GEN</sub>  <br>je/jego | <b>ho</b>   jeho                      | jě <sub>ACC</sub> / <b>go</b> <sub>GEN</sub>  <br>je/jego | jego                                  |
| ACC/GEN.1PL      | nás                                                        | nas                                                       | nos <nas>                                                 | nás                                   | nos                                                       | nas/naju                              |
| ACC/GEN.2PL      | vás                                                        | was                                                       | wos <was>                                                 | vás                                   | wos                                                       | was/waju                              |
| ACC/GEN.3PL.VIR  | jich                                                       | ich                                                       | ich                                                       | ich                                   | ich/jich                                                  | jich                                  |
| ACC/GEN.3PL.NVIR | je                                                         | je                                                        | je                                                        |                                       | je                                                        | je                                    |
| DAT.1SG          | <b>mí</b>   mně                                            | <b>mí</b>   mnie                                          | <b>mí</b>   jemu                                          | <b>mí</b>   mne                       | <b>mí</b>   mnie                                          | mie                                   |
| DAT.2SG          | <b>tí</b>   tobě                                           | <b>ci</b>   tobie                                         | <b>ci</b>   tobie                                         | <b>tí</b>   tebe                      | <b>ci</b>   tobie                                         | <b>cě</b>  <br>tobie(/cebie)          |
| DAT.3SG.M/N      | <b>mu</b>   jemu                                           | <b>mu</b>   jemu                                          | <b>mu</b>   jemu                                          | <b>mu</b>   jemu                      | <b>mu</b>   jemu                                          | <b>mu</b>   jemu                      |
| DAT.3SG.F        | jí                                                         | jej                                                       | ji(j)/jej                                                 | jej                                   | ji(j)/jej                                                 | ji                                    |
| DAT.1PL          | nám                                                        | nam                                                       | nom <nam>                                                 | nám                                   | nom                                                       | nama                                  |
| DAT.2PL          | vám                                                        | wam                                                       | wom <wam>                                                 | vám                                   | wom                                                       | wama                                  |
| DAT.3PL          | jim                                                        | im                                                        | im                                                        | im                                    | im                                                        | jima                                  |

The reflexive pronouns in Table 2.4 below all manifest the formal dichotomy between orthotonic and clitic form, such that reflexive clitics are constant clitics. Czech and Slovak employ only accusative reflexive clitics and do not exhibit genitive reflexives. This is because there are no verbs subcategorised for genitive objects and, because the genitive of negation is obsolete. Note that I

<sup>61</sup> I. Kulik (2023: 213, n. 3) erroneously attributed the term *contant clitic* to a Czech notion of *konstantní příklonka*, which does not exist, in fact.

<sup>62</sup> Instead of providing the paradigm for each language individually in Sections 2.2–2.7, I preferably chose to compile them in a single table for ease of comparison. I accept a certain degree of imprecision because of broad coverage, and I refrain from pointing out phonetic/phonological peculiarities or making usage notes for the sake of conciseness. The columns for Czech, Polish, and Slovak in Table 2.3 basically follow Franks & King (2000), while Kashubian is drawn upon Makurat-Snuzik (2016) – with a partly simplified orthography – and, partly, Stone (1993). Goralish and Silesian forms are based on observational data from utilised corpus data.

explicitly take the Polish clitic dative reflexive *se* into consideration. Usually, this form is neglected, since it is not part of Polish written standard norm and considered to occur in rural dialects only (cf. Urbańczyk 1976: 58).<sup>63</sup> However, *se* occurs in contemporary spoken Polish in both urban and rural domains (cf. also Franks & King 2000: 150; Swan 2002: 158). Totalled up, there are seven constant clitics per lect (eight in Slovak, six in Kashubian).

Table 2.4: Reflexive pronouns (ACC/GEN, DAT) in West Slavic (Cz, Pol, Gor, Slk, Sil, Csb).

| Gloss          | Czech                           | Polish              | Goralish                     | Slovak                          | Silesian                           | Kashubian                    |
|----------------|---------------------------------|---------------------|------------------------------|---------------------------------|------------------------------------|------------------------------|
| REFL.ACC(/GEN) | <b>se</b> <sub>ACC</sub>   sebe | <b>się</b>   siebie | <b>sie</b> <się>  <br>siebie | <b>sa</b> <sub>ACC</sub>   seba | <b>się/sie/še/szie</b><br>  siebie | <b>se, sã, so</b>  <br>sebie |
| REFL.DAT       | <b>si</b>   sobě                | <b>se</b>   sobie   | <b>se</b>   sobie            | <b>si</b>   sebe                | <b>se</b>   sobie                  | <b>se</b>   sobie            |

It is common knowledge in Slavic linguistics that the reflexive is polyfunctional (cf. Golden 2003: 216; Hana 2007: 79–80; Kupść 2000b: 87–90; Wiemer 2007). First, it can serve as an argument proper in genuinely reflexive and reciprocal constructions. Second, in impersonal and anti-causative constructions it is a grammatical marker. Third, there are semantically empty reflexives that occur as lexically mandatory element of certain verbs, which have been traditionally called *reflexive tantum* or, more recently, *inherently reflexive*. Fourth, there are seemingly pragmatically loaded reflexives, e.g., the occurrence of the dative reflexive pronoun with so-called diminutive-iterative verbs in Polish (cf. Wyroślak 2022 for an analysis of the datival reflexive in Polish).<sup>64</sup> Only in the first and fourth case, orthotonic as well as clitic forms are licit. The other two cases occur only with the clitic form and do not permit orthotonic forms. As I cannot do justice to the extensive research literature on reflexives in Slavic and, furthermore, will not go into the details of reflexive forms, I will disregard the different classification of reflexives for practical reasons (cf. Fehrmann et al. 2010, 2014; Hudousková 2016; Janic 2023; Kolaković et al. 2022: 41–49; Wiemer 2007 for some relevant discussion from diverse perspectives). Therefore, the reader will see that the reflexive clitic

<sup>63</sup> As a notable exception, Aguado & Dogil (1989) explicitly take Polish *se* into consideration.

<sup>64</sup> I am not aware, whether these types of construction have gained considerable attention, yet. I provide the Polish examples in (i)–(ii) for the sake of illustration. Examples (iii)–(iv) show that the occurrence of the pragmatically loaded reflexive is not confined to diminutive-iterative verbs. Similar constructions can be found in Czech.

- (i) Po-chodziłem po domu godzinę i z powrotem do łóżka.  
DIM-go.PST.SG.M around house.LOC.SG hour.DIM.ACC.SG and with return.INS.SG to bed.GEN.SG  
'I walked around the house for an hour and [went] back to bed.'
- (ii) Po-chodziłem **sobie** po Ustroniu, spacerkiem pod górami.  
DIM-go.PST.SG.M REFL.DAT around Ustroń.LOC walk.DIM.INS.SG beneath mountain.INS.SG  
'I walked around Ustroń, a promenade near the foot of the mountains.'
- (iii) A idź, ty kretynie!  
ah go.IMP.2SG you.SG.VOC idiot.VOC  
'Ah, go away, you idiot!'
- (iv) Idź **sobie**, ty świnio!  
go.IMP.2SG REFL.DAT you.SG.VOC pig.VOC  
'Go away, you pig!'

(Pol; NKJP full pelcra)

*se/się/sie/sa* will almost always be glossed ‘REFL.ACC(/GEN)’ irrespective of its function. This is meant as purely morphological information.<sup>65</sup>

Pronominal and reflexive clitics are consistently reported to linearise with clitics of other morphosyntactic categories in a contiguous string known as CLITIC CLUSTER. Auxiliary and conditional clitics constitute a second major class in this respect.<sup>66</sup> I shall present the respective paradigms for the sake of completeness. Table 2.5 provides the paradigms of the auxiliary clitics vis-à-vis the copula verb.

Table 2.5: Auxiliary clitics (and putative counterparts) in West Slavic (Cz, Pol, Gor, Slk, Sil, Csb).

| Gloss   | Czech                      | Polish               | Goralish             | Slovak    | Silesian             | Kashubian          |
|---------|----------------------------|----------------------|----------------------|-----------|----------------------|--------------------|
| AUX.1SG | jsem / sem<br>(jsem)       | -m (jestem)          | -f/-k (jestem)       | som (som) | -ch (jest<br>[ż]ech) | —/jem (jem)        |
| AUX.2SG | jśi / si, s (jśi /<br>seś) | -ś (jesteś)          | -ś (jesteś)          | si (si)   | -ś (jesteś)          | —/jes (jes)        |
| AUX.3SG | — (je)                     | — (jest)             | — (jest)             | — (je)    | — (jest)             | —/je (je)          |
| AUX.1PL | jsme / sme<br>(jsme)       | -śmy<br>(jesteśmy)   | -my<br>(jesteśmy)    | sme (sme) | -my (somy)           | —/jesmë<br>(jesmë) |
| AUX.2PL | jste / ste (jste)          | -ście<br>(jesteście) | -ście<br>(jesteście) | ste (ste) | -ście<br>(jesteście) | —/jesta<br>(jesta) |
| AUX.3PL | — (jsou)                   | — (są)               | — (są)               | — (sú)    | — (som)              | —/są (są)          |
| be.INF  | být                        | być                  | być                  | byť       | być                  | bëc                |

The copula is the clitics’ diachronic source, but it cannot be considered to represent an orthotonic allomorph in synchrony (cf. Toman 1980 for Czech). Note that in West Slavic, clitic auxiliary verbs are employed only for past tense formation. This is a different situation than in South Slavic, where clitic auxiliaries also take part in the future tense formation. Still, a few remarks are necessary for the sake of accuracy. First, there are some significant differences between standard written Czech and colloquial Czech, which are tentatively represented in the respective columns. Second, the status of Polish auxiliary clitics is controversial, since they have been argued to undergo grammaticalisation to fully-fledged inflectional affixes (cf. Abramowicz 2008; Franks 2010: 133; Franks & Bański 1999; Franks & King 2000: 144; Rappaport 1988: 320; Ventayol Boada 2014). It is common to refer to these elements as *mobile inflection* (cf. Embick 1995). Third, Goralish and Silesian differ from Polish

<sup>65</sup> The only exception will be *se/się/sie/sa* co-occurring with a real accusative argument, whereby the reflexive cannot bear genuine accusative case. The gloss will be reduced to ‘REFL’ then.

<sup>66</sup> Auxiliary and conditional clitics often are subsumed under the label of *verbal clitics*, which can be explained by etymology (originating from the copula verb ‘be’), selectional restrictions (roughly speaking, co-occurring with the *l*-participles), and semantics (being associated with tense and mood respectively). I omit this usage throughout the thesis in order not to confuse it with the homonymic notion referring to VA clitic languages like Bulgarian and Macedonian.



with respect to both auxiliary cliticisation and copula formation in relevant aspects, which cannot be discussed here. The abundance of variation in these lects is not represented in Table 2.5.<sup>67</sup> Fourth, several past tense formations compete in Kashubian, which are likewise not well represented in Table 2.5.

Finally, Table 2.6 summarises the conditional clitics. Conditional clitics usually occur adjacent to first- and second-person auxiliary clitics in the relevant environments. The standard written Czech conditional clitic is idiosyncratic in the first persons, as it represents a portmanteau morph that codes the conditionals’ inherent semantics with person agreement. In contrast, the paradigm is transparent in colloquial Czech. Similarly, there is a clear split between written Kashubian, which realises a plain conditional particle without overt person marking in terms of inflection or cliticisation, and colloquial speech, which highly prefers to “inflect” conditionals with person endings.

Table 2.6: Conditional clitics in West Slavic (Cz, Pol, Gor, Slk, Sil, Csb).

| Gloss    | Czech             | Polish | Goralish     | Slovak   | Silesian         | Kashubian  |
|----------|-------------------|--------|--------------|----------|------------------|------------|
| COND.1SG | bych / bysem      | bym    | bym/byf/byk  | by [som] | bych             | bě / bēm   |
| COND.2SG | bys / bysi        | byś    | byś          | by [si]  | byś              | bě / bēs   |
| COND.3SG | by                | by     | by           | by       | by               | bě         |
| COND.1PL | bychom /<br>bysme | byśmy  | bymy / byśmy | by [sme] | by my /<br>byśmy | bě / bēsmě |
| COND.2PL | byste             | byście | byście       | by [ste] | byście           | bě / bēsta |
| COND.3PL | by                | by     | by           | by       | by               | bě / bē    |

Auxiliary and conditional clitics will be considered whenever necessary, but pronominal and reflexive constant clitics will be the primary focus of investigation.

## 2.9 Excursus: Elements not treated as clitics

Besides the largely parallel sets of pronominal, reflexive, auxiliary, and conditional clitics in West Slavic, there are language-particular elements, which are sometimes regarded as clitics, too. Czech has adverbial and pragmatic clitics like *tu* ‘here’, *už* ‘already’, *však* ‘however’, *prý* ‘allegedly’ etc. They cluster with the other clitics but enjoy more positional flexibility than auxiliary and pronominal

<sup>67</sup> I will refer to past tense auxiliary solely by the gloss AUX. Future tense auxiliaries will be specified by the gloss FUT. Furthermore, I will maintain the distinction between auxiliary clitics and the homophonous (clitic) copula by glossing the latter ‘be’ (possibly accompanied by the gloss CL).

clitics (cf. Franks & King 2000: 103–104; Hana 2007: 89–91; Rosen 2001: 210–213). Therefore, their status is not as clear as in the case of the core set of clitics. In fact, elements like *tu*, *už*, *však*, or *prý* may be freely distributed across an utterance from first and last position, whereby they do not necessarily team up with other clitics to build a cluster. Furthermore, they may perfectly occur in isolation or be coordinated, cf. (2.5). While auxiliary and pronominal clitics possess – or are parts of – inflectional paradigms, the modal particles are inflectionally invariant. The same holds for interrogative *-li*, but it occurs always in clausal second position even in Russian, where it follows the first prosodic word according to Franks & King (2000: 188–190). The modal particles are not dependents of a verbal predicate, but they modify a constituent, e.g. the clause. Furthermore, this set of alleged clitics is language-specific, while the auxiliary and pronominal clitics comprise a set of shared cognates. Given their rather untypical behaviour, I will treat these and similar elements not as clitics.

Speaker B: Vrátil jsem z Prahy.  
return.PST.SG.M AUX.1SG from Prague.GEN  
'I've returned from Prague.'

b. Speaker A: Mámo,      kde      jsi?  
mom.VOC    where    be.PRS.2SG  
‘Mom, where are you?’

(Cz; constructed)

reference, topicalisation, and clause linking.<sup>68</sup> However, it must be said the ACC reflexive clitic *se/siĕ* is notorious for its polyfunctionality and quite a textbook example of a clitic. Second, *to* may appear in diverse positions, some of which are generally not acknowledged to be clitic positions including isolation. Furthermore, *to* occurs as complement in prepositional constructions including syntactically complex relativizers. While this position is highly restricted for pronominal clitics, as discussed in Section 3.10, it is also not available for orthotonic *j*-initial forms, too. Instead, the *n*-initial allomorph must be realised. Evidence against a genuine clitic status of *to* is provided in example (2.6). The demonstrative co-occurs with a conditional, a reflexive, and a pronominal clitic in (2.6a) and apparently forms a cluster with them. However, it cannot be an object clitic, for it is not an object, at all. Instead, it is the subject of the utterance. Czech does not possess pronominal subject clitics. Moreover, *to* enjoys more distributional flexibility than its fellow clitics. It freely occupies clausal first position like in (2.6b).<sup>69</sup> The contrast between (2.6c) and (2.6d) corroborates that *to* cannot be a clitic. Generally, clitics are unselective with respect to their host in 2P languages and it is irrelevant, what syntactic category precedes the clitic cluster. However, if the verbal predicate hosts the enclitics, then only utterance (2.6c) with cluster-final *to* is eligible, whereas (2.6d) is unacceptable. It is not possible to elicit such sequences from the corpus. This can be explained by two things: First, clitics cannot be positioned further to the right than immediately following their verbal predicate.<sup>70</sup> Second, the demonstrative *to* is not a clitic, such that the clitic cluster is disjoint from its verbal head by a non-clitic expression.

- (2.6) a. **taky by se mi to** líbilo  
 also COND REFL.ACC me.DAT this.NOM.SG.N like.PST.SG.N  
 ‘I would like it, too.’ [lit.: ‘It would please me, too.’]
- b. **no to by se mi** líbilo  
 well this.NOM.SG.N COND REFL.ACC me.DAT like.PST.SG.N  
 ‘Well, I would like it.’ (Cz; ČNK oral\_v1)
- c. Jsem hrozný bídák, líbilo **by se mi to**.  
 be.PRS.1SG terrible scoundrel like.PST.SG.N COND REFL.ACC me.DAT this.NOM.SG.N  
 ‘I am a terrible scoundrel; I would like it.’ (Cz; ČNK syn\_v11)
- d. \*líbilo **to se mi** moc.  
 Intended: ‘I liked it very much.’ (Cz; constructed)

<sup>68</sup> The status of *to* will be briefly picked up in Section 13.4 again.

<sup>69</sup> Note that *no* ‘well’ is an interjection and parenthetical therefore, such that it is not part of the syntactic structure and does not count for the purpose of clitic placement.

<sup>70</sup> See Section 4.8 for the rightwards positional restriction of clitic placement.

The arguments presented in this brief section are the rationale for counting only the set of pronouns, reflexives, past tense auxiliaries, and conditional markers as clitics *sensu stricto* and disregarding all other putative clitic elements for the purpose of the dissertation (except for interrogative *-li*, where it still occurs as a productive morpheme). I treat the demonstrative *to* like all other demonstratives irrespective of its syntactic function (subject/object) and morphological case (nominative/accusative). This approach is generally applied to all object languages surveyed in the present study. Consequently, Polish emphatic morphs *-no*, *-że* are not treated as clitics, too. Usually, they are considered (en)clitics and a part of the clitic cluster occupying cluster-initial position. While a clitic analysis might have been appropriate for older stages of Polish, I argue that this is not adequate for contemporary Polish. Now, the morphs are best analysed as emphatic suffixes, for their distribution is highly restricted. They only occur right-adjoined to the imperative verb. Hence, we observe categorical selectivity not only with respect to the syntactic category (part of speech) of the preceding element, but also with respect to grammatical mood. This is atypical for other clitic elements in Polish. Note that it is not possible to draw parallels to Slavic interrogative *-li* or Romance clitics, because *-li* does not show such a high degree of selectivity in other Slavic languages (e.g. based on mood), whereas Romance clitics are category-sensitive, but not limited to post-verbal placement. Therefore, emphatic morphs *-no*, *-że* will not be assigned clitic status in the present thesis.

## 3 Approaching the clitic

### 3.1 Introductory remarks

The following discussion will raise the problem of the notion *clitic*, which usually is taken for granted in linguistic surveys. Large parts of Chapter 3 review the concept of clitic-hood on a general level before turning to Slavic clitics in particular. This discussion will necessarily cover various testing criteria for identifying clitics. In practice, testing for clitic-hood means to tell apart clitics from affixes and words. Some level of detail is needed to point to basic problems in approaching the object under scrutiny, which usually are not addressed, but have severe repercussions for the way clitics in Slavic are approached.

### 3.2 The clitic: A cover term

It is a common observation that various notions have been used to describe the very properties of clitics, e.g. “conjunct”, “conjoined”, “conjunctive”, “weak”, “bound”, “atonic” or “dependent” forms, since they must appear in an embedding environment. In this manner they are contrasted with “disjunct”, “disjoint”, “disjunctive” “free”, “strong”, “independent”, “tonic” or “full” words that may be stressed or appear in isolation (cf. Klavans 1995: 24; Zwicky 1977: 3). Coming from traditional classical grammar and being coined from Greek ἐγκλίνειν/*enklínein* ‘to lean on, to bend’, the term *clitic* originally denotes linguistic signs that lack inherent stress and, hence, need prosodic support from stress-bearing words (cf. Klavans 1995: 24, 129). The traditional notion is still reminiscent in contemporary linguistics. Phonological approaches particularly focus on prosodic deficiency. Another common view of clitics is syntax-based and relies on the observation that clitics obey strict distributional restrictions in the clause, such that they are bound or fixed with respect to a specific syntactic position or category in the clause, e.g. the so-called Wackernagel position. Morphological approaches treat clitics as some kind of affix or affixoid adjoined to some sort of morphological or syntactic constituent by means of a morphological operation. Nevertheless, clitics are typically identified in terms of prosodic deficiency and/or syntactic distribution as pointed out by Anderson (2005: 1–3).<sup>71</sup> Still, one might wonder, whether this analytical richness in terms of different

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<sup>71</sup> In contrast, Zwicky (1987: 133) suggests prosodic dependency not to be a sufficient condition for clitic-hood. Thus, the set of requirements for identifying clitics need to be more comprehensive than referring to prosodic deficiency and the phonologically motivated “leaning” on a host. His discussion also suggests that clitics are not derived by

components of grammar could have originated from properties divergent across individual languages or even across individual types of clitics.

Zwicky (1994: xiii) explicitly states that the notion *clitic* “is an umbrella term, not a genuine category in grammatical theory.” Thus, it does not denote a single class of elements, but is used for different linguistic phenomena, and diverse linguistic expressions have been subsumed under a common term: (a.) auxiliaries of periphrastic tenses and modal constructions, (b.) pronouns, (c.) determiners, (d.) expletive and pleonastic elements, (e.) adpositions, (f.) conjunctions and complementizers, and (g.) diverse adverbs and adverbials, including negation and discourse particles (cf. Franks & King 2000: 8–12; Klavans 1995: 30; Zwicky 1977: 10). Unsurprisingly, these elements show a wide range of properties in terms of phonology, morphology, syntax, and semantics, such that the question arises whether one could find generalisations over such a heterogeneous range of categories (cf. Zwicky 1987: 133). Even the superficially homogeneous set of pronominal and reflexive clitics manifests a wide range of different functions on different levels of linguistic description. Among others, they realise pronominal arguments (direct object, indirect object, negated direct object, subcategorised genitive argument), free datives (cf. Rubadeau 1996: 122), ethic datives, dative of interest (cf. Rubadeau 1996: 123), dative of possession, resumptive pronouns after an invariant subordinator (cf. Rubadeau 1996: 130–131), reflexive and reciprocal pronoun, passive/medium marker, pleonastic morphemes of inherently reflexive verbs.

In a similar manner, Schwarze (2001: 284–285) points out through the example of French that clitics are ambiguous in multiple respects. He distinguishes seven types of ambiguity, four of which are reproducible in Slavic: First, categorical ambiguity refers to the double nature of several pronominal forms being orthotonic as well as clitic, which is well-known for Slavic clitics, e.g. Cz. *jí* ‘she.DAT’, *vás* ‘you.ACC/GEN.PL’, *jsme* ‘1PL’, *jsi* ‘2SG’. Second, semantic ambiguity relies on the opposition between semantically contentful and semantically empty expressions. It is based on the observation that French *il* may serve both as a referential pronoun and as an expletive. Although there are hardly any expletive expressions in Slavic (e.g. Cz. *ono* ‘it.NOM’), the criterion of semantic ambiguity may be readily addressed in reference to Slavic polyfunctional clitics with the reflexive BCMS/Bg./Cz./Slv. *se*, Pol. *się*, Slk. *sa* being the most prominent representative.<sup>72</sup> It can be a true reflexive or reciprocal pronoun, a purely grammatical/functional marker of the passive voice, medium, or an impersonal construction, and, finally, a semantically empty expression obligatorily co-occurring with a set of verbs traditionally called *reflexiva tantum* (pleonastic REFL) and a small subset of derived perfective verbs (e.g. *napít se* ‘(to) drink’). Third, some clitics exhibit an ambiguity

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synchronic phonological rules but should be listed in the mental lexicon. He adds that their distribution “must be described, at least in part, by rules of syntax and morphology” (Zwicky 1987: 133).

<sup>72</sup> Another example are the polyfunctional dative clitics, especially in the singular, which may be an indirect object proper, a possessive marker, *dativus* (in-)commodi, or an ethical dative.

of syntactic function, which is harder to reproduce for Slavic due to the morphological distinctions across cases. Still, Cz. *ji* ‘she.DAT+GEN’, Pol. *jej* ‘she.DAT+GEN’ reflect this ambiguity, as they may serve either as indirect object or as direct object of verbs that subcategorise for a genitival object.<sup>73</sup> Fourth, just as the French clitic *en* can either be an oblique argument or a partitive element modifying the direct object, genitival clitics in Czech (*jich* ‘them.GEN/ACC’) and Polish (*ich* ‘them.GEN/ACC’) occur as partitives in quantified constructions which they can be extracted from (cf. Toman 1986 for Czech incl. a comparison with Romance).

While the entire set of what is usually labelled *clitic* displays a multitude of categories and functions, the attempt of narrowing it down to a seemingly homogeneous class of “clitics” yields an as yet insufficient reduction of heterogeneity. In fact, Franks (2016: 91) is suspicious about achieving unifying generalisations and asserts

“that it strikes me as highly unlikely that all the things linguists refer to as clitics can (or should) be analyzed in the same way. This much bandied-about term instead subsumes a variety of canonical properties, there being no general agreement among scholars as to which to take as definitive.” (Franks 2016: 91)

Rubadeau (1996: 132) states that clitic systems exhibit “variations and innovations that are language peculiar” and concludes that “no matter which definition of clitics one chooses to employ, each definition is still insufficient to describe thoroughly all clitics in all languages”. Consider also Klavans’s (1995: 5) observation that analyses of clitic phenomena within different syntactic frameworks arrive at completely disparate conclusions:

“Observe, for example, the fact that clitics provide evidence for a lexicalist analysis and for base-generation within the Generalized Phrase Structure Grammar (GPSG) formalism (Miller 1992), for movement in Principles and Parameters theory (Kayne 1991) or for providing evidence to support the claim that affixes within a word can satisfy argument structure requirements in Lexical Functional Grammar (LFG) (Bresnan and Mchombo 1987) [...]” (Klavans 1995: 5)

To conclude, the notion of clitic is a cover term, which involves varying phenomena even in cases, where the object area is restricted to a rather small and homogeneous set of expressions. This situation might be one of the very reasons for persistent confusion.

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<sup>73</sup> I am abstracting away from the fact that Cz. *ji* is also the non-prepositional instrumental. Note, furthermore, that direct objects occur more frequently in the genitive case in Polish due to the productivity of the genitive of negation.

### 3.3 An essential dichotomy: Simple clitics and special clitics

Zwicky's (1977) seminal work on clitics is regarded as the first attempt to frame a general typology of clitics (cf. Klavans 1995: 48). He introduced the widely received distinction between SIMPLE CLITICS and SPECIAL CLITICS, which is still prominent in the field (cf. Anderson 2005; Franks & King 2000; Luraghi 2013; Spencer & Luis 2012a). Primarily, simple clitics are regarded as phonologically reduced variants of unstressed morphemes subordinate to an adjacent word. They show "ordinary" phonology and obey the same synchronic rules of assimilation, reduction, and elision as other unstressed syllables of the object language (Spencer 1991: 376–377; Zwicky 1977: 5–6, 26).<sup>74</sup> Furthermore, simple clitics are positioned due to canonical syntax, i.e. they appear in the same position as their non-clitic counterparts (cf. Zwicky 1977: 6, 1985: 295). This type of cliticisation is located in casual speech. The German examples in (3.1), provided with a broad phonetic transcription of colloquial pronunciation, are prototypical representatives of simple clitic occurrences.<sup>75</sup> The clitic expletive *'s* < *es* 'it' is phonologically reduced and occurs in the same position as its non-reduced counterpart, such that 'it' procliticises in (1a)<sup>76</sup> and encliticises in (1b).<sup>77</sup>

- (3.1) a. *'s*=      gibt                    nichts            in der            Mensa.<sup>78</sup>            [sɡɪpt̪ nɪks ʔɪnə mɛnza]  
           it.NOM give.PRS.3SG nothing.ACC in the.DAT.F cafeteria.DAT  
           'There isn't anything [tasty] in the cafeteria.'

<sup>74</sup> The phonologically transparent derivation of simple clitics from orthotonic forms is abandoned by Zwicky & Pullum (1983) without further discussion: "we do not require that simple clitics derive synchronically from full forms by processes associated with casual or fast speech." (Zwicky & Pullum 1983: 505, n. 3) It seems to be linked to a point made by Spencer (cf. 1991: 377, 383) that English contracted auxiliaries are not output forms of phonological rules, but lexically specified alterations, i.e. allomorphs (cf. Spencer 1991: 376, 383). For one thing, the auxiliary verbs *is* and *has* contract to clitic *'s*, but no such alleged reduction is observed with *was* (cf. Spencer 1991: 377). Similarly, Spencer (1991: 383) argues that "there are no regular phrase phonology rules which could produce [...] /d/ from /had/". However, Spencer (1991: 376) argues that instances like English infinitival *to* reduce regularly [tu] > [tə] because of informal or allegro speech. (Contrarywise, Zwicky 1985: 292 concludes from syntactic and phonological considerations that infinitival "*to* is not a clitic".) Note also the less trivial reduction exemplified in (3.2.c). I will maintain the phonological reasoning and continue to refer to it, since it is relevant for a set of elements that falls under simple clitics.

<sup>75</sup> Note that, for constructed examples, *German* refers to a linguistic variety distributed in the northern half of Germany. The data reflect the speech of the Hannover–Braunschweig–Göttingen–Wolfsburg Metropolitan Region (Ger. *Metropolregion Hannover-Braunschweig-Göttingen-Wolfsburg*). I will not provide if the example belongs to a more sophisticated or more colloquial register. I consider registers to be a matter of social interaction and, therefore, relevant for sociolinguistics, but not for grammar research dealing with the grammatical competence that native speakers acquire in first language acquisition.

<sup>76</sup> I do not consider the case of so-called TOPIC DROP, whereby the expletive pronoun *es* 'it' can be omitted in colloquial German: *Gibt nichts in der Mensa*. 'There isn't anything [tasty] in the cafeteria.' Cf. Fries (1988) for Topic Drop in German and Trutkowski (2016) for a recent analysis.

<sup>77</sup> Clitics are typographically highlighted in the examples by italics throughout the thesis.

<sup>78</sup> Each example will be provided with language abbreviation and source. Examples constructed by the author of the study will be declared as such.



- b. Gibt =’s nichts in der Mensa? [gɪpt’s nɪks ʔɪnə mɛnza]  
 give.PRS.3SG it.NOM nothing.ACC in the.DAT.F cafeteria.DAT  
 ‘Isn’t there anything [tasty] in the cafeteria?’ (Ger; constructed)

With respect to “ordinary” phonological processes applying to simple clitics in the same way as to other unaccented syllables, I provide a few illustrative examples for the reduction and elision of vowels in colloquial German in (3.2), where a careful (tendentially hypercorrective) orthoepic pronunciation is followed by successively reduced realisations of the same expressions. Such reduced phonetic forms typify colloquial speech and commonly occur in everyday communication. We encounter the same phonological processes both in word level phonology and in simple cliticisation. Utterances (3.2a–b) exemplify multiple degrees of reduction in cliticising pronouns as well as their hosts, which is a subject pronoun in (3.2a) and a temporal auxiliary verb in (3.2b). The reduction of *habe* ‘I have’ applies irrespective of whether it is an auxiliary or a full lexical verb, however. Several variants of shwa-elision in the environment of liquids are found in (3.2c–d), the latter also showing *r*-vocalisation. Example (3.2e) provides an instance of vowel loss of unstressed [i]. There is variation, which could be due to speech rate or speaker-dependent, but it does not harm the argument.

- (3.2) a. Hat sie es gemacht? → Hat sie=’s gemacht?  
 have.PRS.3SG she.NOM it.ACC make.PTCP  
 [zi: ʔɛs] ~ [zi: ʔɛs] ~ [zi: ʔəs] ~ [zi:s] ~ [zɪs] ~ [zəs]  
 ‘Did she do it?’
- b. (Das) habe ich gemacht. → (Das) hab=’ich gemacht.  
 that.ACC.SG.N have.PRS.1SG I.NOM make.PTCP  
 [’ha:bə ’ʔɪç] ~ [’ha:p ’ʔɪç] ~ [’ha:bɪç] ~ [’habɪç]  
 ‘I did that.’
- c. (ich) handele [’handələ] ~ [’handlə] ~ [’handəl] ~ [’handl]  
 ‘(I) act; (I) trade’
- d. (ich) hadere [’ha:dəʁə] ~ [’ha:dʁə] ~ [’ha:dɐ]<sup>79</sup>  
 ‘(I) quarrel’
- e. Original / original [ʔɔʁɪgi’na:l] ~ [ʔɔʁgi’na:l]<sup>80</sup>  
 ‘original (noun / adjective)’ (Ger; constructed)

<sup>79</sup> Note that the sequence [əʁ]/[ɐʁ] is not licensed/eligible in German.

<sup>80</sup> The pronunciation variant [ʔɔʁgi’na:l] is an actual auditory attestation (dated between 2018–2019).

Special clitics do not have a direct phonological relation to their full counterparts by means of synchronic rules of grammar. The formal relation between both members of such a pair is opaque (cf. Zwicky 1977: 4, 29). Besides, they show a property Zwicky (1977: 4) calls “special syntax”:<sup>81</sup> they obligatorily appear in a position in the syntactic string different from that of the full form. That is, the pronominal clitics in French (Fra) (3.3) and Spanish (Spa) (3.4) are in complementary distribution with the corresponding orthotonic expressions.

- (3.3) a. Je        vois        Jean.  
           I.NOM see.PRS.1SG Jean.ACC  
 b. \*Je Jean vois.  
           ‘I see Jean.’  
 c. Je **le**        vois.  
           I him.ACC see.PRS.1SG  
 d. \*Je vois **le**.  
           ‘I see him.’

(Fra; Zwicky 1977: 4–5)

- (3.4) a. Veo        el libro.  
           see.PRS.1SG the book  
 b. \*El libro veo.  
           ‘I see the book.’  
 c. **Lo**        veo.  
           him.ACC see.PRS.1SG  
 d. \*Veo **lo**.  
           ‘I see it.’

(Spa; Klavans 1995: 24–25)

Zwicky (1977: 23) informs that another property of special clitics is the formation of a cluster. Within the cluster, clitics appear in a more or less rigid order with respect to one another; no language allows all logically possible orders. The clitics may be ordered with respect to their morphosyntactic categories (AUX < REFL < DAT < ACC) or due to their phonological structure (monosyllabic < disyllabic) (cf. Zwicky 1977: 24).

Zwicky (1977: 6) determines BOUND WORDS as a third class. It is an element without a strong counterpart, such that only clitic forms exist. In subsequent work, Zwicky & Pullum (1983: 510) refine Zwicky’s (1977) original three-way typology by merging bound words with special clitics.

<sup>81</sup> This is reminiscent of Wackernagel (1892: 378), who flags the second positioning of clitics as “spezielle[n] Stellungsregel der Enklitika” ‘special positioning rule of the enclitics’.

Thus, special clitics are characterised by one of two parameters: either a clitic occupies a syntactic position different from a corresponding full form or it does not have an orthotonic counterpart, at all.

From a diachronic perspective Zwicky (1977: 6) suggests that special clitics develop from simple clitics. The idea implies that primordially independent words become deaccented and cliticise onto their neighbouring word, while they may undergo phonological reduction characteristic of allegro speech. These simple clitics then further develop into special clitics. Accordingly, they lose the originally transparent phonological relation to their strong counterpart and develop new distributional properties, i.e. “special syntax”. The whole life cycle from word to ultimately inflectional or derivational affix drawn by Zwicky (1977: 6) may be sketched along the lines of (3.5) adopted from Klavans (1995: 51).

(3.5) word > simple clitic > special clitic > affix

Nübling (1998: 274) proposes the grammaticalisation path from word to affix in (3.6).

(3.6) Syntax → Allegroverschmelzung → einfache Klise → spezielle Klise → Flexion  
‘syntax’ → ‘allegro speech merger’ → ‘simple clisis’ → ‘special clisis’ → ‘inflection’

Nübling (2005) interprets Zwicky’s (1977) dichotomy in terms of different degrees of grammaticalisation. She investigates the coalescence of definitive articles with prepositions in German (e.g. *in* ‘in’ + *dem* ‘the.M/N.DAT.SG’ > *im* ‘in the’). Her distinction of simple clitics and special clitics is based on considerations about grammaticalisation (cf. Nübling 2005: 112–113): Special clitics are more grammaticalised than simple clitics. On the grammaticalisation path from word through clitic to affix, special clitics represent the stage directly preceding affix-hood. In her view, special clitics are not exchangeable with their orthotonic counterparts, which is in sharp contrast to simple clitics. Substitution of a special clitic with a full form either results in ungrammaticality (3.7a) or yields different semantics (3.7b). Simple clitics can be freely substituted by full forms (3.7c). Furthermore, Nübling (2005: 113) argues that the special clitic article bears only semantic definiteness, whereas the full form bears pragmatic definiteness.

- (3.7) a. vom / \* von dem Rauchen  
          ‘because of smoking [cigarettes]’  
      b. im Garten ≠ in dem Garten  
          ‘in the garden’

c. *gegens / fürs* = *gegen das / für das Rauchen*

‘against / for smoking [cigarettes]’

(Ger; Nübling 2005: 112)

Therefore, semantics is affected by the simple/special dichotomy in Nübling’s (2005) account. Moreover, Nübling (2005: 113) provides the quantitative observation that the occurrence of special clitic forms of preposition-article-complexes significantly outranks full forms by number. While special clitics occur in > 90% of all cases, the definite article’s full form appears correspondingly in < 10% of all observed cases. In contrast, simple clitic occurrences of the definite article are generally outranked by the full form (although different statistical values for certain preposition-article combinations occur). Nübling’s (2005: 114, Figure 2) quantitative data suggests that the preference of special clitic forms is statistically very robust and, hence, categorical. On the other hand, the ratio between simple clitic and orthotonic forms is gradual in principle but significantly favours full forms over simple clitics.

### 3.4 Distinguishing clitics from affixes

Clitics have been repeatedly regarded as linguistic expressions intermediate between word and affix, because they share a subset of properties of either category (cf. Klavans 1995: 16, 94–95; Zwicky 1977: 1, 1994: xii). That is why there have been efforts to distinguish clitics from independent words on the one hand and from affixes on the other hand. However, taking into consideration the long tradition of the term *clitic* – namely as *enclitic* and *proclitic* – scholars rather late began to strive for something like a definition of clitic-hood in order to distinguish it from other linguistic categories in a principled way. To my knowledge, Zwicky (1977) was the first to attempt classifying clitics in a principled way on the basis of cross-linguistic data. However, it was pretheoretical in nature, as Zwicky (1985: 284) himself remarks. At that very time, the term *clitic* itself was not meant to have any theoretical value. For instance, it did not denote an individual class of entities in a model of grammar. Zwicky (1985: 283–284) suggests that, if the notion *clitic* is meant to be of theoretical significance, tests to distinguish clitics should be applied. Accordingly, Zwicky & Pullum (1983) were the first to account for the distinction of clitics from inflectional affixes. Both categories of morphemes share the property of being bound in some sense (cf. Zwicky & Pullum 1983: 502). Analysing English auxiliary verbs and the negation *not*, they use six criteria (enumerated A–F) to discriminate the clitics from affixes:

**A) Degree of selection:** Clitics are considered to be not very selective with respect to their host in contrast to affixes, which are restricted to a specific set of roots and stems belonging to a

particular syntactic category.<sup>82</sup> For one thing, inflectional affixes coding case, tense, or the comparative/superlative are limited to nouns, verbs, and adjectives respectively.<sup>83</sup> English auxiliary simple clitics like *'s < is*, *'s < has*, on the other hand, can attach on any syntactic category. Zwicky & Pullum (1983: 504) provide the examples in (3.8a–d), where the clitic attaches to a preposition, a verb, an adjective, and an adverb respectively. For the sake of completeness, I add examples (3.8e–f) with auxiliaries cliticising to a pronoun and a full noun from the British National Corpus (BNC).

- (3.8) a. The person I was walking to's going to be angry with me.  
 b. The ball you hit's just broken my dining room window.  
 c. Any answer not entirely right's going to be marked as an error.  
 d. The drive home tonight's been really easy. (Eng; Zwicky & Pullum 1983: 504)  
 e. He's going to be an expert on dinosaurs.  
 f. (As a matter of fact) your father's been in touch with us. (Eng; BNC)

**B) Arbitrary lexical gaps:** Clitics are considered not to exhibit gaps with respect to attested or grammatically possible combinations with their host. So, a clitic can in principle attach to any lexical element eligible as host. In other words, any lexical element fulfilling the structural conditions for hosting clitics in general can combine with a clitic. In contrast, lexical gaps are significant for affixes, which may or may not occur with certain stems. Zwicky & Pullum (1983: 505) provide the example of the English verb *stride*, which they consider lacking a past participle form, which would be an arbitrary lexical gap in the verb's paradigm.<sup>84</sup>

**C) Morphophonological idiosyncrasies:** Idiosyncrasies of the morphophonological shape of clitic-host combinations are generally absent. While the hosts' forms are not influenced by attaching clitics at all, clitic allomorphy can be traced back to "general rules referring to the phonological and morphological properties of the hosts." (Zwicky & Pullum 1983: 505) This is very different with stem-affix combinations, where morphophonological idiosyncrasies are quite common, e.g. irregular inflection forms or suppletion. Zwicky & Pullum (1983: 505) provide *dice (< dice)*<sup>85</sup>, *oxen (< ox)*,

<sup>82</sup> This criterion corresponds to Zwicky's (1987: 136) and Kupść's (2000b) term of "promiscuity".

<sup>83</sup> This global statement might be valid only for Indo-European languages. Some refinements would be at hand, but for the purpose of the present study, it is sufficient to provide such examples. Nevertheless, note that the Japanese so-called *i*-adjectives have verb properties in that they may be both attributes and predicates inflecting for past tense, negation, and obeying politeness distinctions, among others, e.g. *taka-i* 'high', *taka-katta* 'was high (informal)', *taka-katta desu* 'was high (formal)', *taka-ku nai* '(is) not high (informal)', *taka-ku nai desu* '(is) not high (formal)'.  
<sup>84</sup> The issue with *stride* seems to be somewhat controversial, but I will not go into details. The main point, however, is that the past participle of *stride* is rare to such an extent that native speakers have doubts about the actual form and even try to avoid constructions that would license it.

<sup>85</sup> The form *dice* is both singular and plural according to the *Oxford Dictionary of English* (Stevenson 2010). However, some controversy remains, as *dice* still can function as plural of the singular form *die* for some speakers. Note also that the *Oxford Dictionary of English* provides the singular form *die* and the plural form *dice* remarking that the form *dice* is more frequently used for both singular and plural. Nevertheless, the mere recognition of differences in

*feet* (< *foot*) for idiosyncratic plural forms, *slept* (< *sleep*), *thought* (< *think*), *went* (< *go*) for idiosyncratic past tense forms, and *best* (< *good*), *worst* (< *bad*) for idiosyncratic superlative forms.

**C2) Morphophonological processes:** Somewhat different from the idiosyncrasies just cited, which can be viewed as peculiarities of a small set of individual lexical items, Zwicky (1985: 300) brings up a new criterion missed by Zwicky and Pullum (1983), in order to distinguish affixes from clitics. I would like to posit it as a variant of Zwicky and Pullum's (1983) criterion (C), as it refers to morphonology as well. Zwicky (1985: 300) tells that "morphophonological processes like ablaut, umlaut, consonant changes, reduplication, accent shifts, and tone alterations are fairly common as the phonological exponents of inflectional or derivational formations in morphology." This can be proved true, when considering, for instance, ablaut in German strong verbs' past and perfect forms (*singen* – *sang* – *gesungen* 'sing – sang – sung', *laufen* – *lief* – *gelaufen* 'run – ran – run'), umlaut as a marker of plurality in a set of German nouns, occurring either alone or in combination with a plural suffix (*Mutter* – *Mütter* 'mother – mothers', *Haus* – *Häuser* 'house – houses'), consonantal and vowel alterations in Polish noun declension (*noga* – *nodze* 'leg.NOM.SG – leg.LOC.SG', *mąż* – *męża* 'man.NOM.SG – man.GEN.SG), consonantal and quantitative vowel alteration in Czech derivation (*kniha* – *knížka* 'book – book.DIM', *muka* – *mučit* 'anguish – (to) agonize'), or paradigmatic stress shift in some Russian declensions (*móre/mópe* – *morjá/morjá* 'sea.NOM.SG – sea.NOM.PL'). These processes are linked to affixation or, generally speaking, the domain of morphology and are more common in the respective languages' morphologies. They do not occur, for instance, with Czech or Polish auxiliary or pronominal clitics.

**D) Semantic idiosyncrasies:** Semantic idiosyncrasies are meant as deviations from compositional semantics, such that the meaning of the whole cannot be derived from its immediate parts. Semantic idiosyncrasies are not observed with host-clitic combinations. Therefore, the examples in (3.8) above do not differ semantically in any respect from the corresponding utterances containing orthotonic auxiliaries. In contrast, semantic idiosyncrasies may appear with stem-affix combinations. Zwicky & Pullum (1983: 505) exemplify this point by *last* (< *late*), which in the syntagma *last words* does not share the temporal semantics of its etymological positive form. Another example is *most* (< *many*, *more*), which took the meaning of *best* in the 1950s substandard English (3.9) and does not exhibit the meaning of measurable or countable quantification anymore.

(3.9) Franckie Avalon is the most.

'Franckie Avalon is the best.'

(Eng; Zwicky & Pullum 1983: 505)

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frequency does not imply categorical linguistic distinctions, yet. It seems reasonable to infer a sort of lexical diffusion for this particular case, which should not be understood as the incomplete spread of some phonological or morphological feature within a monolithically conceived lexicon, but as gradual spread across a speech community.

So far, clitics and affixes can be distinguished by the criteria (A) to (D). The comparison of the inflectional examples in the respective paragraphs with the sentences in (3.8) confirm that clitics and affixes behave differently with respect to the proposed diagnostics. Zwicky & Pullum (1983: 505) point out that the distinctions would become more significant, when contrasting clitics with derivational affixes rather than inflectional affixes, as the former are much more prone to selectivity, lexical gaps, and idiosyncrasies in the domains of morphophonology and semantics.

**E) Syntactic operations affecting free-word-bound-morpheme-combinations (non-constituency):** Zwicky & Pullum (1983: 504) assume that “[a]ll cliticization [...] follows syntax”. Therefore, syntactic rules are deemed to operate on inflected words, but not on clitics. Though Zwicky and Pullum (1983) do not suggest a label, I refer to their diagnostic (E) by the term *non-constituency*. They elaborate that “no syntactic operations exist which treat a word combined with one of the clitics [...] as a unit. [...] But inflected nouns, verbs, adjectives, and adverbs are of course regularly treated as units by syntactic operations.” (Zwicky & Pullum 1983: 505–506) Therefore, I conclude that diagnostic (E) should be viewed in terms of constituency.

**F) Combinability restrictions (uncombinability):** Clitics can combine with other clitics, but affixes cannot combine with clitics or suffix onto a clitic group consisting of a host and clitics attached to it (Zwicky & Pullum 1983: 504, 507). Thus, in (3.10a) the clitic form of *have* can attach to the clitic form of *would*, cf. (3.10b). On the other hand, arguing that the contracted form *-n’t* of the negation *not* is an affix rather than a cliticised form, Zwicky and Pullum (1983: 507) show that *-n’t* cannot attach to the cliticised form of the modal *would* (3.10c–e). For the sake of completeness, I would like to remind the reader of the obvious observation that affixes can combine with other affixes, even when they belong to different classes, namely inflectional and derivational affixes.<sup>86</sup> Some sort

<sup>86</sup> I provide a small, non-exhaustive set of German examples in (i)–(iv) for illustrative purposes. In (i) an adjectival stem (ADJ) combines with the derivational suffix (DERV) *-heit* that establishes abstract nouns. In (ii) a nominal stem (NN) combines with the derivational suffix *-al* that establishes adjectives and further is concatenated with the derivational suffix *-ität* that gives rise to a noun again. In (iii) a verbal stem (V) is combined with the derivational suffix *-ung* that generates deverbal nouns. Finally, all the derived nouns are combined with the allomorph *-en* of the plural number morpheme, which is an inflectional suffix (INFL). Note that inflectional suffixes are not necessarily the last morphemes in line, as indicated by (iv), where the derivational suffixes *-chen* and *-lein* forming the diminutive follow the inflectional plural allomorph *-er*. Note that the examples in (iv) belong to a rather small set of lexemes (mostly neuter nouns with *-er* plural, but cf. (iv.f)), which allow for this highly restricted paradigm. Usually, the diminutive suffix is inherently ambiguous between singular and plural, cf. (v).

- (i) *Schönheiten* < {schön}<sub>ADJ</sub> + {-heit}<sub>DERV</sub> + {en}<sub>INFL</sub>  
‘beauties’
- (ii) *Nationalitäten* < {Nation}<sub>NN</sub> + {-al}<sub>DERV</sub> + {-ität}<sub>DERV</sub> + {en}<sub>INFL</sub>  
‘nationalities’
- (iii) *Meinungen* < {meinen}<sub>V</sub> + {-ung}<sub>DERV</sub> + {en}<sub>INFL</sub>  
‘opinions’
- (iv) a. *Kinderchen* < {Kind}<sub>NN</sub> + {-er}<sub>INFL</sub> + {-chen}<sub>DERV</sub>  
b. *Kinderlein* < {Kind}<sub>NN</sub> + {-er}<sub>INFL</sub> + {-lein}<sub>DERV</sub>  
‘little children’  
c. *Eierchen* < {Ei}<sub>NN</sub> + {-er}<sub>INFL</sub> + {-chen}<sub>DERV</sub>  
‘small eggs’

of a category distinction between clitics and affixes seem to exist in terms of combinability, such that clitics and affixes cannot combine.

- (3.10) a. I'd've done it if you'd asked me.  
 b. I would have done it if you had asked me. (Eng; Zwicky & Pullum 1983: 506)  
 c. I'd not be doing this until I had to.  
 d. I wouldn't be doing this until I had to.  
 e. \*I'dn't be doing this until I had to. (Eng; Zwicky & Pullum 1983: 507)

### 3.5 Distinguishing clitics from independent words

Taking the perspective from the other side of the scale, Zwicky (1985) elaborates on the distinction between clitics and independent words. Importantly, he proposes a list of tests like the diagnostics by Zwicky & Pullum (1983). The criteria proposed by Zwicky (1985) outnumber Zwicky & Pullum's (1983). Zwicky (1985: 284–285) reviews the notion of *test* in linguistics critically and warns that it does not provide definitive conditions but is rather a differential diagnosis (similar to medical diagnosis) flagging characteristic traits. Tests point to overall tendencies and are rather statements of probability than absolute (cf. Zwicky 1985: 285). From this it follows that a linguistic test yields evidence for most clear-cut cases, but it can still fail, especially with borderline cases.

Zwicky (1985: 286–290) divides the tests he proposes for clitic-hood – in contrast to word-hood – into six groups, some of which consist of a single test only: 1) phonological tests, 2) an accentual test, 3) tests utilising the resemblance between clitics and affixes, 4) syntactic tests, 5) a test based on assumptions about interfaces between modules of grammar, and 6) a metaconsideration.

**1.1) Internal/external sandhi:** Host-clitic combinations are regarded as phonological words in contrast to word-word combinations, which are then treated as phonological phrases. Consequently, internal sandhi rules are deemed to apply within phonological words, whereas external sandhi rules are applying within the domain of the phonological phrase. In other words: clitics can be

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- d. Kleiderchen < {Kleid}<sub>NN</sub> + {-er}<sub>INFL</sub> + {-chen}<sub>DERV</sub>  
 'small dresses'  
 e. Rinderchen < {Rind}<sub>NN</sub> + {-er}<sub>INFL</sub> + {-chen}<sub>DERV</sub>  
 'small cattle'  
 f. Männerchen < {Mann}<sub>NN</sub> + {-er}<sub>INFL</sub> + {-chen}<sub>DERV</sub>  
 'small men'  
 (v) a. ein Häuschen<sub>SG</sub> / Häuslein<sub>SG</sub>  
 'a small house'  
 b. viele Häuschen<sub>PL</sub> / Häuslein<sub>PL</sub> (alongside *Häuserchen*)  
 'many small houses'

(Ger; personal knowledge)



prone to internal sandhi phenomena observed in an object language, but not to external sandhi phenomena that affect independent words only.

**1.2) Word/phrase domains in prosodic phonology:** Prosodic structure (stress, length, tone) that is associated particularly with the domain of either the prosodic word or the prosodic phrase can serve to distinguish clitics and words. For instance, if stress assignment operates on the level of the phonological word, then clitics are expected to be part of the stress domain. Otherwise, clitics should not be affected by stress assignment.

**1.3) Word/phrase domains in segmental phonology:** Rules that manipulate features of phonological segments in either domain can be used as a diagnostic, too. Zwicky (1985: 286) gives vowel harmony as an example. If a rule of vowel harmony applies on the phonological word level, an element prone to such vowel harmony is a clitic. If vowel harmony is a phonological phrase level operation than the affected element is not clitic, but an independent word.

**2) An accentual test:** This test bears on the observation that clitics lack word-level accent, hence, they are considered inherently stressless. Zwicky (1985: 287) mentions two flaws of this test: first, clitics are stressed in some languages, indeed, whereby they receive stress through general stress assignment rules, emphasis, or contrast. However, stress assignment by phonological rules can be subsumed under test 1.2 on prosodic phonology. As for emphasis and contrasting stress, we will see later that these are considered as properties precluding clitic status by other approaches. Second, some elements that are considered independent words are realised without stress. Zwicky (1985: 287) gives prepositions, determiners, and auxiliary verbs in English as examples. Again, depending on the particular approach, such elements – at least prepositions and determiners – could be deemed clitics. For one thing, prepositions in Czech and Polish are considered proclitics (cf. Kraska-Szlenk 1995: 62, 149; Skibicki 2016: 18–19 for Polish; Janda & Townsend 2000: 8 for Czech; more generally Haspelmath 2023: 10). In sum, test 2 is quite controversial, so it is no wonder that Zwicky (1985: 287) considers it “most unreliable”.

**3.1) Binding:** Elements that are notoriously bound and cannot appear in isolation are clitics (or affixes). Independent words, on the other hand, can appear in isolation and form one-word utterance.

**3.2) Closure:** Elements, which block further attachment of clitics (or affixes), are clitics. The idea of this test is that some phonotactic, morphological or syntagmatic string cannot be prone to further structure building operations on a certain level of description. In a sense, clitics terminate the cycle of structure building. As argued by Zwicky & Pullum (1983), inflectional affixation cannot apply to strings containing clitics. Considering the termination of clitic sequence building, this test presupposes the assumption of a “final clitic” that eventually prevents further clitic attachment.

**3.3) Construction:** Zwicky (1985: 287) suggests that clitics appear in combination with single words, whereas independent words appear with phrases (potentially) consisting of multiple words. However, he admits the controversial status of this test, since the criteria used by Klavans (1985 [cited after Zwicky 1985: 287])<sup>87</sup> to distinguish clitics differ from his own: in her approach, clitics combine with multi-word phrases in contrast to affixes, which appear with single words. Zwicky (1985: 288) summarises that “construction with phrases is not a reliable test for words as opposed to clitics.” I shall, however, briefly exemplify the test used by Zwicky (1985: 291–292) to argue that the so-called particles of English have the status of words not of clitics or affixes. In (3.11) the adverbial particles *not*, *only*, *even* combine with different phrasal categories. Perhaps, Zwicky does not suggest that the alleged particles combine with the next adjacent word, but with the whole phrase, due to the fact that the particle has scope over the phrase and not only over parts of it, e.g. the phrasal head.

- (3.11) a. not / only / even [<sub>S</sub> because I asked you ]  
 b. not / only / even [<sub>PP</sub> with a pick-axe ]  
 c. not / only / even [<sub>NP</sub> the bravest of us ] (Eng; after Zwicky 1985: 292)

**3.4) Ordering:** Being similar to affixes, clitics obey a rather strict order with respect to their host, i.e. independent words. On the other hand, words generally may have alternative orders with respect to other words. However, some variation in the ordering of clitics with respect to each other seems to be allowed – or has to be accounted for (cf. Zwicky 1985: 288). Zwicky (1985: 292) exemplifies the ordering alternatives of independent words with *even* and *only* in (3.12) both of which are either positioned in front of the VP or within it.

- (3.12) a. even [<sub>VP</sub> saw Adeline ]  
 b. [<sub>VP</sub> saw even Adeline ]  
 c. only [<sub>VP</sub> took a drop ]  
 d. [<sub>VP</sub> took only a drop ] (Eng; Zwicky 1985: 292)

**3.5) Distribution:** Zwicky (1985: 288) refers to “a single principle governing [the] distribution” of affixes and clitics. He cites derivational *-ness* limited to adjectival stems and inflectional *-ing* restricted to verbal stems as examples for the principle governing affix distribution. For clitics, he cites principles like the attachment to the verbal head of a clause, attachment to either first word or

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<sup>87</sup> Klavans’s (1985) paper is a revised version of the fourth chapter from her 1980’s PhD thesis. It later circulated and gained attention as the so-called Indiana University Linguistics Club version, i.e. Klavans (1982).

first phrase of a clause, and attachment to an NP. In contrast, independent words have far more complex distributions.

**3.6) Complexity:** Clitics (and affixes) are morphologically simple units, whereas independent words are morphologically complex consisting of a combination of morphemes.

**4.1) Deletion:** Independent words and syntactic phrases may undergo “deletion under identity”, whereas parts of words, i.e. affixes, and clitics cannot. Zwicky (1985: 292) provides the following example for deletion: *I urged him to (have the penguin stuffed)*. Here, the material in parentheses is recoverable from the (hypothetical) preceding context, which presupposes *have the penguin stuffed*. For instance, one could imagine the following question prior to Zwicky’s example sentence, where the deleted material is placed between square brackets: *Did he [have the penguin stuffed]*? Such deletion analyses are not eligible for the affixes in (3.13).

- (3.13) a. Where are my sons and your daughters?  
b. # The boy and girls are on the playground.  
c. # The boys and girl are on the playground.  
d. # The boy and girl are on the playground.  
e. The boys and girls are on the playground. (Eng; constructed)

Although the previous context (3.13a) presupposes the plurality of either *boys* and *girls*, (3.13b–d) are inappropriate replies (marked with a superscribed rhomb #), because the plural affix *-s* cannot be deleted.

**4.2) Replacement:** Independent words may be replaced by a pro-form, which generally can substitute a single word, a syntactic phrase, or a partial or complete clause. Clitics on the other hand cannot be substituted by pro-forms. Zwicky (1985: 292) gives *I urged him to do so*. as an example with *so* being the pro-form in question substituting the phrase *have the penguin stuffed*.

**4.3) Movement:** Independent words are prone to movement via movement rules, but clitics are not.

**5) Test derived from interface assumptions:** As cliticisation is assumed to apply after syntax, only independent words are subject to or conditions for syntactic rules, whereas clitics or clitic groups are not (cf. also Zwicky & Pullum 1983). This test applies only in case of special clitics, as simple clitics, being reduced forms of orthotonic words, may be prone to syntactic rules in the orthotonic shape.<sup>88</sup>

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<sup>88</sup> This and other tests either explicitly or implicitly presuppose a derivational account of grammar, which is not a necessity. For one thing, instead of presupposing that simple clitics are synchronically derived from orthotonic forms via grammatical rules, one could also assume a direct insertion of a simple clitic into the syntactic structure, whereby the relation between an orthotonic lexeme and its variant (or variants) could be specified in the mental lexicon.

**6) Metaconsideration:** In unclear cases – or if evidence is missing – an analysis as an independent word or – next – as an affix is preferable to an assessment as clitic. Zwicky (1985: 290) admits that criterion 6 is speculative in nature and builds upon the assumption that cliticisation is generally marked in comparison to word-hood and affixation.

Although Zwicky (1985) proposes a comprehensive and fine-grained list of tests, I see several problems and unresolved questions with respect to the word-clitic distinction. The criterion of Word/Phrase Domains refers to three disparate sets of phonology, namely sandhi phenomena (1.1), suprasegmental or prosodic phonology (1.2), and segmental phonology (1.3) with particular reference to phonological, i.e. distinctive features. As for internal/external sandhi, one must acknowledge that the test is vacuous. Consider the regressive voicing assimilation of obstruents in Slavic, which applies both within words and across word boundaries in various morphological and syntactic contexts. Obstruents assimilate in single words, in morphologically derived, complex words, and between syntagmatically adjacent words. Similar can be observed with the assimilation of other phonological features, most notably place of articulation and palatalisation. Regressive assimilations are insensitive to the word-phrase distinction in Slavic. The same applies to the word-phrase distinction in relation to segmental phonology. Slavic does not exhibit phenomena like vowel harmony or umlaut, and I am not aware of any comparable process in Slavic. Possibly one could consider the occurrence of so-called floating vowels, i.e. the vowel-zero alternations that emerged with the deletion of weak and the vocalisation of strong yers. These occur word-internally in a set of non-derived lexemes and complex, morphologically derived words as well as in some syntactic contexts with prepositions. With respect to prosody, I first would like to address the fact that the assumption of suprasegmental domains like prosodic word, prosodic phrase, and intonation phrase are theoretical in nature.<sup>89</sup> Another problem is linked with stress domains in Slavic. Whereas some stress systems are lexically-morphologically determined, as in Russian, Bulgarian, or the 2P-clitic language BCMS, others are metrically determined, e.g. word-initial stress in Czech and penult stress in Polish. As for prepositions, it is generally acknowledged that these are proclitics in Slavic (Kraska-Szlenk 1995: 62, 149; Skibicki 2016: 18–19 for Polish; Janda & Townsend 2000: 8 for Czech). Such an assessment has some issues, as prepositions in Slavic (like their counterparts in English or German) can receive emphatic or contrastive stress and appear in isolation, cf. (3.14).<sup>90</sup>

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<sup>89</sup> For the sake of completeness, I would like to point out that even the prosodic unit of the syllable, which most people share a pre-theoretic but common intuition of, is not uncontroversial in phonological theory. See Steriade (1999), who proposed a purely linear account of consonantal phonotactics that accounts for the parsing of consonant clusters into syllable onsets and codas without reference to the notion of the syllable itself. Syllabification is accounted for not in terms of an abstract model of the syllable but as analogically derived effect by reference to word-internal and word-final consonant clusters. I owe this reference to Esther Kunay (p.c.), whom I thank for discussion.

<sup>90</sup> Stressed syllables are highlighted by capital letters throughout the thesis.

- (3.14) a. Lej DO szklanki a nie POZA (szklankę)!  
 pour.IMP.2SG to drinking\_glass.GEN.SG and NEG beyond drinking\_glass.ACC.SG  
 ‘Pour INTO the glass and not BEYOND (the glass)!’
- b. A: Gdzie go mogę znaleźć? Pod półką?  
 where him.ACC can.PRS.1SG find.INF under shelf.INS.SG  
 ‘Where can I find it? Under the shelf?’
- B: NAD!  
 above  
 ‘ABOVE [the shelf]!’ (Pol; constructed)

However, prepositions are relevant for metrical stress assignment in Czech and Polish. The stress position shifts from the noun to the preposition in a P<sup>0</sup> NP syntagma in Czech (e.g. *ŠKOla* ‘school’ – *DO školy* ‘to school’). Thus, prepositions can be regarded as belonging to the domain of the phonological word, which is in accord with Zwicky’s prosodic criterion. However, taking into consideration the criterion of Binding (occurrence in isolation), one would have to conclude that Slavic prepositions are ambiguous between words and clitics. Returning to pronominal and auxiliary clitics, one has to acknowledge that the test is not helpful for Czech, as the clitic auxiliaries and pronouns usually follow their hosts as enclitics. As stress is determined from the left word edge, there is little to say about enclitics in relation to the prosodic word. A solution to this problem would be the construction of environments, in which auxiliary and pronominal clitics are forced to procliticise to the word following them, e.g. after an intonational break. Turning to Polish penult stress there is little to say indeed, as neither pronominal nor auxiliary clitics participate in metrical stress assignment (contrary to prepositions in Polish, cf. e.g. *wiDZImy* ‘we see’ – *wiDZImy go* ‘we see him’ – \**widziMY go* vs. *bez CIEbie* ‘without you’ – *BEZ nich* ‘without them’ – *beZE mnie* ‘without me’). In sum, the phonological tests proposed by Zwicky (1985) are not suitable or do not yield unequivocal results when applied to two different sets of presumed clitics.

The Binding test (3.1) can be considered a safe candidate for a test of clitic- and affix-hood, as both types of entities require to be adjoined to some other element, which is the host in case of clitics and a stem/root in case of affixes.

The Closure criterion (3.2) is significant only for the clitic-affix distinction but is not helpful for the word-clitic dichotomy. Nor does it have significance for determining clitic clusters. One element, which could be assigned a closure function is the Bg/BCMS auxiliary clitic (*j*)e, which as the only auxiliary clitic follows the pronominal clitics in the clitic cluster, where it is the last element. Therefore, the placement of *je* would posit some sort of structural closure and no other clitics (or affix) could be attached. However, the closure criterion is controversial at best. First, clitic clusters

are not immune to syntactic or syntagmatic manipulations. Two instances pose a serious problem for the test. First, what would be considered a clitic cluster might be discontinuous, as will be shown in the later course of discussion. That is, multiple clitics can occur separated from one another by other syntactic material (words and phrases). Second, clitics may occur in an order different from what is generally believed to be “fixed order”. This is especially significant for West Slavic, where an idiosyncratic instance of auxiliary clitic placement of the BCMS type does not exist. Therefore, the accusative or genitive pronominal clitics are the cluster-final clitics. However, taking into consideration that the order  $DAT < ACC$  may be inverted (hence,  $ACC < DAT$ ), the notion of *closure* is at odds with what one would expect from the theoretical consideration presupposed by the notion itself.

In its strongest version, the Ordering criterion (3.4) requires clitics to appear in a fixed order with respect to one another and with respect to their host. This requirement might be suitable in one way or another for verb-adjacent clitic language, but must necessarily fail, if one takes Wackernagel clitic languages into consideration. Czech and BCMS auxiliary and pronominal clitics may attach to whatever category they find to their left. Still, if one argues that the host is an underlying syntactic position, the problem arises that Wackernagel clitics may also occupy positions other than the second position of the clause, while Bošković (2001) and Marušič (2008) argue that clitics do not occupy a single syntactic position in BCMS and Slovene respectively. A softer version of the Ordering criterion refers only to the linearisation within the clitic cluster but excludes the notion of the host. Still, serious problems arise, if variation comes into play and clitics are positioned more freely with respect to each other than expected. As we will see in the course of the thesis, some degree of variation may be found with clitic orders and Zwicky himself admits that variation needs to be accounted for. If we must weaken criteria to account for certain phenomena, then it is questionable, if they are valid criteria, at all.

Problems for the Distribution criterion (3.5) are twofold. First, although clitic distribution can be described in terms of simple principles in most cases, it turns out that clitic placement does not obey only a single principle, but several principles interact, sometimes in a subtle way. As we will see, information structure interferes with the basic placement rules for clitics in a way that is far from trivial. This state of affairs becomes especially clear, if one considers both clitic distribution in root clauses and in complex clauses, where so-called clitic climbing can occur. Second, a problem from the domain of independent words is posed by verb second languages, particularly by German. The placement of the finite verb in German main and embedded clauses may be reduced to the two simple principles in (i) and (ii).

- (i) Finite verbs are positioned in the second position of the main clause. (Alternatively: There may be only one constituent before the finite verb in assertive main clauses.)
- (ii) Finite verbs are positioned clause-finally in embedded clauses.<sup>91</sup>

German verbs are independent words and generally incapable of cliticisation, but their distribution is captured by simple statements, which would give preference to a clitic interpretation. Note also that the Distribution criterion can only apply to special clitics and does not have anything to say about simple clitics, like German reduced pronouns, which are distributed like their full counterparts.

The Complexity (3.6) criterion is problematic for several reasons. A significant number of independent words are morphologically simple. For instance, a large number of English singular nouns are bare stems/roots except in the genitive (e.g. *dog*, *house*, *lord*). An analogous observation applies, among others, to the NOM.SG of the West Slavic masculine declensions and NOM/ACC.SG of inanimate masculines (e.g. Cz. *muž* ‘man.NOM.SG’, *hrad* ‘castle.NOM/ACC.SG’, Pol. *pan* ‘mister.NOM.SG’, *dom* ‘house.NOM/ACC.SG’), the GEN.PL of the feminine *a*-declension (Cz. *žen* ‘woman.GEN.PL’, Pol. *kobiet* ‘woman.GEN.PL’), and the NOM/ACC.SG of other female declensions (e.g. Cz. *kost* ‘bone.NOM/ACC.SG’, Pol. *kość* ‘bone.NOM/ACC.SG’). They represent simple forms in terms of morphological substance, as they cannot be partitioned into smaller morphological units. On the other hand, both affixes and clitics are morpho-semantically complex in Indo-European languages, since they often represent portmanteau morphs coding multiple information. For one thing, the Polish verbal inflectional affix *-cie* in *widzicie* ‘see.PRS.2PL’ codes person and number, while the Cz./Pol. pronominal clitic *mu* ‘3SG.M/N.DAT’ codes person, number, gender, and case. Therefore, complexity is an inaccurate criterion and does not help to determine the status of a relevant portion of the object languages’ lexicon.

The Deletion test (4.1) fails with Slavic clitics. The clitic can be omitted in (3.15a–f).<sup>92</sup>

- (3.15) a. *Widziała-m samochód i kupiła-m go.*  
           see.PST.1SG.F-1SG car.ACC.SG and buy.PST.1SG.F-1SG him.ACC  
           ‘[I] saw a car and [I] bought it.’
- b. *Widziałam samochód i kupiłam.*  
           ‘[I] saw a car and bought [it].’

<sup>91</sup> For the sake of convenience, I am abstracting away from the topological model of the German clause.

<sup>92</sup> The gloss PST for past will be generally used for all *l*-participles in past tense examples for the sake of convenience. It is the generalised past tense form in most Slavic languages. The gloss PTCP will be retained for the *l*-participle (i) in the periphrastic future in Polish and Slovene, (ii) in Old Czech and Old Polish, when several past tense forms were still distinguished, and (iii) in some rare citations of pluperfect forms.

- c. Widziałam **go** i kupiłam **go**.  
 ‘[I] saw it and [I] bought it.’
- d. Widziałam **go** i kupiłam.  
 ‘[I] saw it and [I] bought [it].’
- e. Widziała-ś { ten samochód / **go** } na wystawie?  
 see.PST.SG.F-2SG this.ACC car.ACC him.ACC on exhibition.LOC.SG  
 ‘Did you see { the car / him } at the exhibition?’
- f. Tak, widziałam i kupiłam.  
 ‘Yes, [I] saw [it] and [I] bought [it].’ (Pol; constructed)

The Replacement test (4.2) is vacuous with respect to pronominal clitics taking into consideration that they are pro-forms themselves.

The Movement test (4.3) is built upon the theory-dependent (transformational) notion of movement. Of course, it is possible to translate transformational notions into the vocabulary of another framework, as Zwicky (1985: 288) says. The general notion would be one of dislocation or unbound dependency. However, it remains unclear what these notions’ significance is with respect to cliticisation. If one thinks of movement as the derivation of a surface position from a presumed underlying (logical, semantic, “deep”, or base generate) position, then it is a theoretical decision again, if pronominal clitics are taken as base-generated in their surface position or if they reach it after several transformational procedures (cf. Toman 1999 for the former account). It is questionable if a theory-dependent criterion is suitable. Even if the Movement test refers to the fact of unbound dependencies, then it is still insignificant for the word-clitic dichotomy. First, it is unclear with respect to which element a potential unbound dependency should be tested. Consider the Czech example in (3.17a).<sup>93</sup> If it were the *l*-participle, which licenses the dative clitic and co-occurs with the auxiliary clitic as a morphosyntactic complex of a periphrastic tense, then the test turns out to be negative. Neither clitic needs to be adjacent to the verbal predicate. The clitic climbing construction in (3.17b) presents another challenge, as the reflexive clitic *se* is licensed by the deeply embedded infinitive *učit* ‘(to) teach/learn’ but realised in the finite matrix clause, such that three non-governing verbs intervene between the clitic and its verbal head. So, the clitic is clearly dislocated from its verbal head.

<sup>93</sup> The examples in (3.17) are taken from the SYN verze 8 corpus as indicated by the specification *syn\_v8* (cf. Hnátková et al. 2014, Křen et al. 2019). The SYN v8 subcorpus is unbalanced but significantly large, such that it is potentially broader in empirical coverage than balanced corpora, which are necessarily restricted in various ways.



- (3.17) a. Už včera **jsme mu** výhru převedli na  
already yesterday AUX.1PL him.DAT win.ACC.SG transfer.PST.PL.VIR on  
účet.  
account.ACC.SG  
'We transferred the win to his [banking] account already yesterday.'
- b. (Na gymplu najednou zjistil, že má  
on grammar\_school.LOC.SG suddenly find\_out.PST.SG.M that have.PRS.3SG  
čtyřky) a asi **se<sub>4</sub>** bude<sub>1</sub> [ muset<sub>2</sub> [ začít<sub>3</sub> [ učit<sub>4</sub> ]]].  
D-grades and maybe REFL.ACC be.FUT.3SG must.INF begin.INF teach.INF  
'(In secondary school he suddenly found out that he had D grades) and that maybe he  
would have to start learning.'  
(Cz; ČNK syn v8)

Turning to Zwicky's (1985: 289) Interface assumptions (5), I would like to remind the reader that interface assumptions are only at hand, if one assumes a modular conception of grammar, such that the notion of interfaces could be employed in a meaningful way. In a fully parallel architecture of grammar with no division between individual grammatical domains, test (5) becomes obsolete. As some other criteria, Interface is theory-dependent, because it relies on the assumption of a stratified mental grammar, to which a (radical) connectionist model of grammar (or language) would be a diametral alternative.

Finally, Zwicky's (1985: 289–290) Metaconsideration (6) has another side, which is not necessarily bound with the linguistic notion of markedness, nor with linguistics in particular. Taking a somewhat different perspective on the matter of preference of one type of analysis over another one, the metaconsideration can be seen as a paraphrase or application of Occam's Razor. As a heuristic principle utilised in science, Occam's Razor favours explanations or theories, which presuppose less entities or rely on less assumptions. An analysis of a linguistic expression as either word or affix (but not clitic) is in accordance with Occam's Razor, if the tests applied are indecisive – especially, if the ultimate result would be the abandonment of the category of clitics. However, it would still be an application of Occam's Razor to assume a smaller than bigger set of clitics, in case there is some controversy on the matter. Being a sort of general scientific heuristics, Occam's Razor (or Zwicky's Metaconsideration) is clearly a criterion applicable for the word-clitic-affix distinction in its entirety.

In sum, some of the criteria proposed by Zwicky (1985) are inconclusive as to the word-clitic distinction on conceptual or empirical grounds, while others seem to rely on theory-dependent assumptions and, thus, are not eligible for a largely theory-neutral classification. Yet other criteria rely on tests that might be applicable in one set of languages, but not in another, as the relevant

phenomena do not occur, such that the respective tests cannot be applied to the object languages. Despite the problems, recall that Zwicky (1985) does not demand all criteria to apply at once, nor does he claim that these criteria would automatically license discrete category distinctions. As linguistic tests, they refer to tendencies and are based on probabilities and I agree with this line of reasoning. The successful application of some tests to a defined set of linguistic expressions may lend plausibility to clitic-hood. Nevertheless, most criteria are problematic and inapplicable. In fact, only the Binding criterion (3.1) and the Metaconsideration (6) appear to be uncontroversial, while the Distribution criterion (3.5) needs further refinement.

## 3.6 Successive deficiency: Strong, weak, and clitic pronouns

### 3.6.1 The tripartition of pronouns: Basic properties

Based on a cross-linguistic comparison, Cardinaletti & Starke (1999a) propose a three-way classification of strong ( $z_s$ ), weak ( $y_D$ ), and clitic pronouns ( $x_D$ ), which received considerable attention.<sup>94</sup> The tripartition is accounted for by a structural deficiency, according to which pronouns possess successively less structure, i.e. functional heads, the more deficient they become. The effects of this gradual loss of functional structure apply to all grammatical levels, i.e. phonology, morphology, syntax, and semantics. Weak and clitic pronouns constitute the class of deficient pronouns, as opposed to strong pronouns. However, the opposition is not strictly binary (“ $\{x_D, y_D\}$  vs.  $z_s$ ”), but hierarchically ordered (“ $x_D < y_D < z_s$ ”) (Cardinaletti & Starke 1999a: 167). Weak pronouns take an intermediate position between strong and clitic pronouns sharing some, but not all, properties with either group (cf. Cardinaletti & Starke 1999a: 167). Assuming an underlying structure for pronouns which mirrors the major divisions of clausal structure in Generative Grammar, Cardinaletti & Starke (1999a: 194–195, 214) assume the syntactic representations as in (3.18).<sup>95</sup>

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<sup>94</sup> I will not summarise or discuss any of the peer comments on Cardinaletti & Starke’s (1999a) proposal in Riemsdijk (1999: 235–271) with one exception. I will briefly refer to Dobrovie-Sorin (1999: 249–255) after my discussion of Cardinaletti & Starke (1999a). In summary, all commentators agree with the typological tripartition.

<sup>95</sup> Cardinaletti & Starke (1999a: 194–195) adopt Laka’s (1990) assumption of a functional projection  $\Sigma P$  posited between CP and IP, which was developed on the basis of Basque.

- (3.18) a.  $[_{CP} C^0 [_{\Sigma P} \Sigma^0 [_{IP} I^0 [_{LP} L^0 ]]]]$  strong pronouns<sup>96</sup>  
 b.  $[_{\Sigma P} \Sigma^0 [_{IP} I^0 [_{LP} L^0 ]]]$  weak pronouns  
 c.  $[_{IP} I^0 [_{LP} L^0 ]]$  clitic pronouns
- } deficient pronouns

Each functional layer is the locus of distinct features. The CP-layer, which they identify as the locus of (functional) case features, is missing in deficient pronouns (Cardinaletti & Starke 1999a: 185–186). Thus, deficient pronouns must occur in a particular surface position associated with the assignment of case.<sup>97</sup> The same is taken to apply on the level of semantics, whereby strong pronouns receive [+human] referential features by the C-head (Cardinaletti & Starke 1999a: 182–183). The  $\Sigma P$ -shell is responsible for polarity and Focus interpretation and is the locus of prosodic features. Clitics are mere IPs and lack other projections. This is seen as the driving force behind clitic second-position placement, which corresponds to  $\Sigma P$  on the sentence-level. Clitics move to  $\Sigma P$  to become prosodically licensed.<sup>98</sup> In summary, while strong pronouns do not obey any particular restrictions, weak pronouns move only to a case-associated position (identified with an agreement phrase [AgrP]), but clitic pronouns must license both their case and prosodic features. In fact, Cardinaletti & Starke (1999a) identify a comprehensive list of distinguishing criteria for the three classes of pronouns, which they pose in terms of empirical observations. Table 3.1 summarises the criteria for the respective domains and pronominal classes and provides page references.

Table 3.1: Summary of Cardinaletti & Starke's (1999a) typology.

| Domain              | Strong pronouns                                                                                           | Deficient pronouns                                                             | Page(s) |
|---------------------|-----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|---------|
| Phonology & Prosody | are not subject to phonological processes, application of phonological rules would yield ungrammaticality | are prone to phonological processes like sandhi rules (e.g. liaison in French) | 159     |
|                     | build a prosodic unit of their own, cannot build a common unit with another neighbouring element          | build a single prosodic unit with a neighbouring lexical element               | 159–160 |

<sup>96</sup>  $L^0$  may be any lexical head, e.g. nominal  $N^0$ . The specification of the other functional heads will be addressed in the course of the section.

<sup>97</sup> In Cardinaletti & Starke's (1999a: 186–187) account, it is the specifier position of an agreement phrase (AgrP).

<sup>98</sup> Cardinaletti & Starke (1999a: 196) argue that prosodic features are present in  $\Sigma P$  and necessarily in  $LP$ , which is the lexical head projecting the entire structure. As such, it must contain all structural information projected in syntax including prosodic features. Thus, they derive the distinction between 2P- and VA-cliticisation – the former positing clitics relative to  $\Sigma^0$ , the latter positing clitics relative to  $L^0 = V^0$ .

|                     |                                                                                                                         |                                                                                                                             |                 |
|---------------------|-------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|-----------------|
| Phonology & Prosody | cannot be subject of reduction phenomena                                                                                | undergo reduction phenomena (e.g. destressing, contraction)                                                                 | 160, 221, n. 23 |
| Morphology          | not reduced                                                                                                             | reduced compared with strong pronouns: represent a subset in terms of morphological shape                                   | 149             |
| Syntax              | strong and deficient pronouns occur in different positions: they are in complementary distribution                      |                                                                                                                             | 150             |
|                     |                                                                                                                         | cannot appear in what is taken as the base/theta position                                                                   | 150–151         |
|                     |                                                                                                                         | cannot occupy peripheral positions, i.e. cleft sentences, left dislocation (= topicalisation), right dislocation, isolation | 151             |
|                     | may be modified by adverbs, which have scope over the entire NP (c-modification) and may be coordinated with another NP | cannot be modified and coordinated at all.                                                                                  | 151–152         |
|                     |                                                                                                                         |                                                                                                                             |                 |
| Semantics           | can refer to entities previously absent in the discourse (but also to present entities)                                 | must refer to entities already present in the discourse                                                                     | 153–154         |
|                     | cannot serve as (quasi) expletive expressions                                                                           | can serve as expletive subjects                                                                                             | 154             |
|                     | cannot serve as subject in impersonal constructions                                                                     | can serve as subject in impersonal constructions                                                                            | 154–155         |
|                     | cannot serve as non-referential dative                                                                                  | can serve as non-referential (= ethical) dative                                                                             | 155             |
|                     | can only refer to [+human] referents, cannot refer to [–human] referents                                                | can refer to both [+human] and [–human] referents                                                                           | 145–147, 155    |

The internal differentiation of deficient pronouns is less systematic and does not refer to all modules of grammar. It is grounded on a mixed set of observational statements and theoretical considerations.

Cardinaletti & Starke (1999a: 168–172) distinguish weak and clitic pronouns by the following six criteria:

- 1) **Head-phrase distinction:** Clitic pronouns are heads  $X^0$ , while weak pronouns are phrases  $XP$  (p. 168).
- 2) **Doubling:** Clitic pronouns can double another (non-clitic) NP, but weak pronouns cannot (p. 169).
- 3) **Clustering:** Clitic pronouns build a cluster, which is the domain of morpho-phonological processes, but weak pronouns do not (p. 169).
- 4) **Co-occurrence restrictions (Person Case Constraint):** Co-occurrence of 1st and 2nd person accusative with 3rd person dative clitic pronouns is ruled out (so-called Person Case Constraint, PCC). The PCC does not hold for weak pronouns (p. 170).
- 5) **Inherent word-level stress:** Clitic pronouns do not have word stress, only weak pronouns possess word stress (pp. 172–173).
- 6) **Overall preference:** As long as no conflicting conditions apply, more deficient pronouns are preferred over less deficient ones: clitic > weak > strong (pp. 173–175).

Cardinaletti & Starke (1999a: 166) provide Slovak examples as representatives of Slavic, cf. (3.19), suggesting that Slavic pronominal systems exhibit the tripartition. The difference between the three classes is exemplified by the strong pronoun *jemu* ‘him.DAT’, weak *ono* ‘it.NOM’, and clitic *mu* ‘him.DAT’. Only the strong and weak pronouns are licit clause-initially (3.19a–b), while the clitic is banned from this position (3.19c). On the other hand, it is only strong *jemu*, which can be coordinated (3.19d), while deficient pronouns, weak *ono* and clitic *mu*, cannot occur as a conjunct (3.19e–f).

- (3.19) a. *Jemu to bude pomáhať.*  
           him.DAT this.NOM.SG.N be.FUT.3SG help.INF
- b. *Ono mu to bude pomáhať.*  
           it.NOM.SG.N him.DAT this.NOM.SG.N be.FUT.3SG help.INF
- c. \**Mu to bude pomáhať.*  
           him.DAT.CL this.NOM.SG.N be.FUT.3SG help.INF  
           ‘It will help him.’
- d. *Jemu a Milanovi to bude pomáhať.*  
           him.DAT and Milan.DAT this.NOM.SG.N be.FUT.3SG help.INF  
           ‘It will help him and Milan.’

- e. \*Ono            a        to                    druhe [sic]   **mu**        budu        pomáhat'. [sic]  
                          it.NOM.SG.N   and this.NOM.SG.N   second.NOM   him.DAT   be.FUT.3PL   help.INF  
                          'It and the other will help him.'
- f. \***Mu**            a        jej        to                    bude        pomáhat'.  
                          him.DAT.CL   and her.DAT   this.NOM.SG.N   be.FUT.3SG   help.INF  
                          'It will help him and her.'
- (Slk; Cardinaletti & Starke 1999a: 166)

Cardinaletti & Starke (1999a) declare their tripartition as a universal linguistic typology, but they also point out that not every language utilises a tripartite pronominal system. Still, every pronoun can be assigned to one of the categories (cf. Cardinaletti & Starke 1999b: 273). Thus, a language  $L_x$  can possess a bipartite pronominal system, but both classes of pronouns should be still analysable in terms of the proposed typology, i.e. e.g. strong pronouns vis-à-vis weak or clitic pronouns.<sup>99</sup> The proposal has become highly influential in the study of clitics and makes strong predictions. For instance, based on the discussion in Cetnarowska (2004), Jung & Migdalski (2022) propose to extend the tripartite typology to a four-way classification. However, being a product of significant abstraction, Cardinaletti & Starke's (1999a) tripartite typology is prone to a far too high degree of idealisation, which might give rise to inconsistencies. In the following subsections, I will attempt to critically review selected issues addressing each linguistic domain step by step. Since Cardinaletti & Starke's (1999a) proposal has become highly influential, it deserves a more careful and detailed discussion, which is why each grammatical level will be provided a separate subsection.

### 3.6.2 Phonology

Most of Cardinaletti & Starke's (1999a: 159–160) phonological criteria are unsubstantiated. First, the criterion of inherent stress is challenged by Talić's (2018) observation that clitics affect stress of neighbouring elements in Bosnian. The prosodic patterns are observed for a small set of elements only but lead her to the conclusion that the respective clitics are lexically specified for stress. Second, deficient – especially clitic – pronouns are not necessarily subject to phonological processes like sandhi phenomena and reductions. No such phenomena are observed with Slovak or Polish clitics, and they are also not restricted to “weak” elements. Strong elements may exhibit sandhi phenomena and reductive phonological processes, too. For one thing, Slavic languages like Russian, Czech, and Polish have regressive voicing assimilation both within and across words, the latter of which

<sup>99</sup> In principle, the typology predicts that languages with unary pronominal systems would possess either strong pronouns, or weak pronouns, or clitic pronouns exclusively. It is unclear to me whether this prediction is borne out.

represents word-external sandhi. The word-internal voicing assimilation in (3.20a–b) is not contingent on notions of strong vs. deficient elements. The examples in (3.20c–g) unequivocally show that external sandhis are not restricted to deficient elements.<sup>100</sup> Admittedly, prepositions like *bez* ‘without’ in (3.20c–d) are proclitics in Slavic. For instance, they are integrated to the phonological word of their host, where they count for word-level stress assignment in Czech and Slovak. However, not only can they be contrastively stressed, but also occur in isolation and be coordinated. Thus, prepositions in Slavic do not fit any particular category of the proposed tripartite classification. On the other hand, the voicing assimilation targets the coda consonants of nouns and verbs in (3.20e–g). The noun *závod* ‘competition.NOM.SG’ in (3.20e) is the head of a complex NP and governs the genitive case of the embedded noun *aut* ‘car.GEN.PL’.<sup>101</sup> In (3.20f), the noun *krev* ‘blood.NOM.SG’ is the subject of the verb or clause. Finally, the infinitive *napsat* ‘write.PFV.INF’ is the verbal predicate in (3.20g).<sup>102</sup> It is not justified to regard any of these elements as deficient (neither weak nor clitic).<sup>103</sup>

- (3.20) a. *prosit* [prosiːt] ‘(to) ask for, beg’  
 b. *prosba* [prozba] ‘plea’  
 c. *bez dechu* [bɛz dɛxu] ‘out of breath’  
 d. *bez tebe* [bɛs tɛbɛ] ‘without you’ (Cz)  
 e. *závod aut* [za:vot ʔaut] / [za:vod aut] ‘car race’  
 f. *krev teče* [krɛf tɛtʃɛ] ‘blood flows’ (Cz; Šimáčková et al. 2012: 227)  
 g. *napsat větu* [napsad vjetu] ‘(to) write a sentence’ (Cz; Šimáčková et al. 2012: 228)

Next, consider Cardinaletti & Starke’s (1999a: 159–160) claim that strong pronouns generate a prosodic unit on their own and do not share a common unit with neighbouring elements. This

<sup>100</sup> Cf. also the German examples in Section 3.3, which corroborate the argument made here through the example of Czech. In German, hosts of simple clitics are also subject to phonological reductions (vowel shortening) and sandhi phenomena (voicing over word boundaries, resyllabification).

<sup>101</sup> Devoiced [za:vot] is Bohemian Czech, voiced [za:vod] is Moravian Czech. Šimáčková et al. (2012: 227) discuss this minimal pair in the context of diatopic differences with respect to word-final devoicing before another vowel-initial word. According to their analysis, Bohemian [za:vot] is an instance of plain word-final devoicing. In contrast, the Moravian variant is analysed as resyllabification into [za:vo.daut]. However, an alternative analysis is possible: Note that in the Bohemian syntagma, the subordinate noun *aut* starts with a glottal stop. Nothing prevents an analysis along the lines of a regressive voicing assimilation of /d/ to /ʔ/ yielding [t]. Issues are more complicated with the voiced variant [za:vod aut]. The regressive voicing assimilation applies in consonant clusters, and basically obstruents are trigger of as well as subject to assimilation (cf. Short 1993a: 458; Šimáčková et al. 2012: 227–228). Therefore, the Moravian example does not fit the traditional view of regressive voicing, because the trigger is a vowel then. I must leave this issue open.

<sup>102</sup> The voicing triggered by [v] is a feature specific for Moravian Czech, where it is mandatory, while in Bohemian Czech [v] triggers voicing only irregularly (cf. Šimáčková et al. 2012: 228).

<sup>103</sup> In any case of doubt if the consideration of linguistic categories other than pronouns is legitimate, note that Cardinaletti & Starke (1999a: 207) themselves suggest that the tripartition could possibly be extended to other syntactic categories. Furthermore, their arguments are not exclusively limited to pronouns, such that they refer to more general concepts and categories themselves.

argument is incorrect in two respects: First, any strong element serving as host for a clitic shares a common prosodic unit with this clitic if we stick to the common observation that clitics are incorporated into their host’s prosodic word or intonation phrase. Second, “strong” elements share prosodic units with each other, too. Bartnicka et al. (2004: 80) inform that underlyingly orthotonic elements may behave like clitics in allegro speech.

“Beim schnellen Sprechen können orthotonische Ausdrücke ihre Eigenständigkeit verlieren und sich wie Klitika verhalten d.h. eine Akzenteinheit eingehen.” (Bartnicka et al. 2004: 80)

‘In fast speech, orthotonic expressions may lose their independence and behave like clitics, i.e. strike up an accentual unit [= share an accentual unit].’

They cite the Polish examples in (3.21), which I provide with a broad phonetic transcription next to the standard orthographic representation.

- (3.21) a. Czemu milczysz? [tʃɛmu\_ˈmiltʃɨʃ]  
           why be\_silent.PRS.2SG  
           ‘Why are you silent?’
- b. Która godzina? [ktura\_gɔˈdʒina]  
           which.SG.F hour.NOM  
           ‘What time [is it]?’
- c. Czy wiesz? [tʃɨ\_ˈvʲɛʃ]  
           Q know.PRS.2SG  
           ‘Do you know?’
- d. Co to wszystko znaczy? [tsɔ\_tɔ\_ˈfʲɨʂtkɔ\_ˈznatʃɨ]  
           what this.NOM.SG.N everything.NOM.SG.N mean.PRS.3SG  
           ‘What does all that mean?’ (Pol; Bartnicka et al. 2004: 80)

### 3.6.3 Morphology

In terms of morphology, Cardinaletti & Starke (1999a) compare the additional morpheme of Italian strong pronouns (which they call a “dummy marker”) as in *a loro* ‘to them’ with complex prepositions (e.g. English *instead of*) and complex complementizers (e.g. French *avant que* ‘before that’). They infer a nominal structure parallel to the architecture of verbal projections, i.e. CP-IP-like layers above



the nominal head (= split DP-hypothesis) in (3.22b), which mirrors the verbal superstructure in (3.22a) (cf. Cardinaletti & Starke 1999a: 183–185).<sup>104</sup>

- (3.22) a. [CP [IP [VP V<sup>0</sup> ]]]  
 b. [CP [IP [NP N<sup>0</sup> ]]] < [DP [NP N<sup>0</sup> ]]] (Cardinaletti & Starke 1999a: 184)

The scholars distinguish morphemes like *je-* in Slovak *jeho, jemu* vs. *ho, mu* or BCMS *nje-* in *njega* vs. *ga* from the case-related dummy morphemes mentioned above and consider them “semantically empty ‘dummy’ morpheme[s]” serving as “syntactic support” for the clitic proper (Cardinaletti & Starke 1999a: 193). They identify the same morpheme in Slavic auxiliary clitics, e.g. BCMS copula *je-sam, je-si, je-st, je-smo, je-ste, je-su* vs. auxiliary clitics *sam, si, je, smo, ste, su*. Therefore, Cardinaletti & Starke (1999a: 178) assume an underlyingly syntactic, but morphologically realised (i.e. visible) structural condition for the morphological relationship between forms like *jemu* and *mu* and the asymmetries these pronouns exhibit. This account is grounded on the generative assumption that morphological formants represent functional syntactic heads.<sup>105</sup> The strong pronoun *jemu* is taken to possess more morphemes than the clitic *mu*. It means that the traditionally not further analysable syllable *je-* is taken to realise a morpheme with an abstract syntactic function, which is absent in the case of *mu*.

This pursuit of the strong structuralist hypothesis that identical form implies identical function is problematic, however. The form-function-identity claim is explicated in connection with the morphological analysis of Slavic auxiliary and pronominal clitics vs. full copula and strong pronouns through the example of BCMS (cf. Cardinaletti & Starke 1999a: 193–194). Accordingly, the phonotactic sequence /je/ in BCMS strong lexemes like *jesam* ‘be.PRS.1SG’ and *jemu* ‘him.DAT.SG.M/N’ but absent in clitic *sam* ‘AUX.1SG’ and *mu* ‘him.DAT.SG.M/N’ represents an identical piece of underlying structure, i.e. a morpheme {je}, which “is part of a tripartite system of support prefixes.” (Cardinaletti & Starke 1999a: 194) Thus, {je} is the overt realisation of the functional head C<sup>0</sup>.<sup>106</sup> The argument overlooks an important diachronic fact about linguistic form, namely the emergence of homonymy through sound change. For one thing, the German lexemes *Kiefer* ‘pine’ and *Kiefer* ‘jaw’ are homophones but have different etymological sources with originally distinct

<sup>104</sup> The nominal C-head is a “complementizer-like preposition” that “closes off the extended projection of the noun”, e.g. Italian *a* ‘in; to; at’ in *a loro* ‘them.3PL.DAT’ (Cardinaletti & Starke 1999a: 184).

<sup>105</sup> By (*morphological*) *formant* I mean ‘bound morpheme; affixe’ and not refer to formants in phonetics, i.e. bands of acoustic frequencies.

<sup>106</sup> This type of reasoning is already a precursor of the later upcoming framework of Nanosyntax, which Michal Starke is a primary representative of (cf. Franks 2016: 124, n. 53 for a similar assessment). Cf. Starke (2009) and Taraldsen (2019) for brief introductions to Nanosyntax.

morphonological forms.<sup>107</sup> Consider also the homonymy of the pronouns and non-pronominal forms in (3.23).

- (3.23) a. er [ʔe:ɐ̯] ‘he’ ~ Ehr’ [ʔe:ɐ̯] (with apocope) ‘honour’  
 b. sie/Sie [zi:] ‘she; they/you.PL’ ~ sieh! [zi:] ‘look.IMP.2SG’  
 c. es [ʔɛs] ‘it’ ~ ess! [ʔɛs] ‘eat.IMP.2SG (non-standard)’  
 ~ ess’ [ʔɛs] (with apocope) ‘eat.PRS.1SG (non-standard)’  
 (Ger; personal knowledge)

In a similar manner, the putative morpheme {je} in *jesam* and *jemu* have different etymological sources. The syllable *je-* of the copula’s strong forms is diachronically derived from Proto-Slavic (or Balto-Slavic) *\*es-* < PIE *\*h<sub>1</sub>es-* (cf. Derksen 2008: 146).<sup>108</sup> On the other hand, the occurrence of the syllable *je-* in Slavic pronouns trace back to the Proto-Slavic pronoun *\*jī* < PIE *\*io-* (cf. Derksen 2008: 208). Whereas the syllable in question is the etymological reflex of a verbal root in one case, it continues a pronoun in the other case. The formal similarity of an additional syllable or (alleged) morpheme *je-* is the accidental outcome of sound change.<sup>109</sup> Still, one must take the possibility of reanalysis into consideration, whereby the *je*-syllable received a unified status in the auxiliary and pronominal paradigms. If this were the case, then one would expect the tendency to increase formal regularity and transparency. In fact, the BCMS copula paradigm shows a formally regular distinction between strong and clitic forms, but this is not true in case of the pronominal paradigm (cf. Franks & King 2000: 19, 24).

Furthermore, Czech is not in accordance with Cardinaletti & Starke’s (1999a) reasoning. First, consider the morphological form of the copula inflection (taken to represent the strong form) vs. the clitic auxiliary paradigm in Table 3.2.

<sup>107</sup> As I am focusing on the form, I may ignore that (*die*) *Kiefer* ‘(the) pine’ has feminine and (*der*) *Kiefer* ‘(the) jaw’ has masculine gender (neuter gender in Austrian German: (*das*) *Kiefer* ‘(the) jaw’). Likewise, I may ignore the fact that both lexemes have distinct plural forms ‘(*die*) *Kiefer-n* ‘(the) pines’ and (*die*) *Kiefer-Ø* ‘(the) jaws’ for the sake of argumentation. The logic of the argument is not affected.

<sup>108</sup> The PIE root *\*h<sub>1</sub>es-* is also the source of several inflectional forms of the copula verb *be* in other branches of Indo-European, e.g. Skr. *ásmi* ‘be.1SG’, *ásti* ‘be.3SG’, AGr. εἰμί ‘be.1SG’, ἐστί ‘be.3SG’, or German *ist* ‘be.PRS.3SG’ (cf. Derksen 2008: 146).

<sup>109</sup> Additionally, one must consider a formal reanalysis in the copula paradigm, for the formal relation between the BCMS clitic *sam* and the strong form *jesam* does not mirror the diachronically underlying morphological structure, which would be PIE *\*{h<sub>1</sub>es}{mi}* vs. BCMS *{je}{sam}* (cf. Derksen 2008: 146). This is comparable to the case of the suffix *-burger* originating from *{{Hamburg}{er}}* reanalysed as *{{Ham}{burger}}*, which gave rise to the formation of *cheese-burger*, *chicken-burger*, *veggie-burger* etc. However, if one takes a closer look at the BCMS or Czech pronominal paradigms, one will find a rather low degree of regularity and transparency.

Table 3.2: Paradigms of copula *být* ‘be’ (present tense) and auxiliary clitic in Czech.

| Copula <i>být</i> | Auxiliary clitic | Gloss                   |
|-------------------|------------------|-------------------------|
| jsem [sɛm]        | jsem [sɛm]       | ‘(I) am’ / ‘AUX.1SG’    |
| jsi [si]          | jsi [si]         | ‘(you) are’ / ‘AUX.2SG’ |
| je [jɛ]           | —                | ‘(he/she/it) is’ / —    |
| jsme [smɛ]        | jsme [smɛ]       | ‘(we) are’ / ‘AUX.1PL’  |
| jste [stɛ]        | jste [stɛ]       | ‘(you) are’ / ‘AUX.2PL’ |
| jsou [sou]        | —                | ‘(they) are’ / —        |

There is no formal morphological difference among first persons and second persons. This is not striking, as it is commonly known that most Czech strong and clitic pronouns are homophones. However, there is a significant difference for the third person forms, which do not realise clitic forms. The fact that the 3SG copula is *je*, while the clitic is phonetically null appears striking at first glance, for it fits Cardinaletti & Starke’s (1999a: 193) BCMS examples. However, the 3PL form does not have a *je-* syllable but possesses a distinct form. Therefore, it is impossible to relate the pronominal forms to the verbal ones in any meaningful way for Czech, cf. e.g. pronominal *jeho* ~ *ho*, *jemu* ~ *mu* vs. verbal/auxiliary *je* ~  $\emptyset$ , *jsou* ~  $\emptyset$ .

Second, a serious problem is brought about by the morpho-phonological forms of the strong vs. clitic pronouns themselves. Recall the paradigms of object pronouns in Section 2.8, which do not display a regular pattern for deriving clitic pronouns from strong ones. Consider Table 3.3 for the sake of argument, which provides the unambiguous clitic pronouns of Czech, Slovak, and Polish vis-à-vis their strong counterparts. Obviously, the orthotonic pronouns are more complex in that they comprise more phonological segments, but only the pairs *jeho/jego* ~ *ho/go* and *jemu* ~ *mu* exhibit the formal regularity Cardinaletti & Starke (1999a) refer to. In all other cases, the pronouns differ by other means. There is no correspondence among the hypothetical morphemes *te-/cie-*, *je-*, *to-/te-*, *se-/sie-*, *so-/se-* and it is impossible to derive the clitic by simply *detaching* the first syllable of *tobě/tebe/tobie* ~ *ti/ci* and *tebe/teba/ciebie* ~ *tě/t’a/cię* (the same holds for BCMS *tebi* ~ *ti* ‘you.DAT’, *tebe* ~ *te* ‘you.SG.ACC’). The 1SG.DAT strong and clitic pronouns are both monosyllabic *mně/mne/mnie* ~ *mi*, such that there is not even an additional syllable (but differently for BCMS *meni* ~ *mi* ‘me.DAT’). Slovak shows a genuine distinction between strong *mňa* vs. clitic *ma* only in terms of consonant cluster vs. mono-segmental onset.<sup>110</sup> It is therefore doubtful that *je-* represents a morpheme which is shared by the disparate morphosyntactic categories of auxiliary and pronominal clitics and is lacking

<sup>110</sup> Czech clitic *mě* ‘me.ACC’ is homophonous with orthotonic *mně* ‘me.DAT’. Polish *mię* ‘me.ACC’ is rather obsolete as a distinct form. On the other hand, the strong form *mnie* ‘me.ACC/DAT’ tends to be pronounced as [mʲɛ] in contemporary Polish, thus, blurring the original formal distinction between strong and clitic pronoun.

in the respective deficient counterparts. Thus, the argument that strong and clitic pronouns may alternate by an individual morpheme, which, in turn, is the phonetic realisation of an underlying functional head, is empirically untenable and not convincing. This reasoning does not challenge the claim that strong(er) forms are more complex. It is an argument against the structural identity of parts of lexemes, which accidentally appear to be phonetically similar.

Table 3.3: Czech, Slovak, and Polish strong pronouns and their corresponding clitic forms.

| Czech  |        | Slovak |        | Polish |        | Gloss            |
|--------|--------|--------|--------|--------|--------|------------------|
| strong | clitic | strong | clitic | strong | clitic |                  |
| mne    | (mě)   | mňa    | ma     | mnie   | (mię)  | ‘me.ACC(/GEN)’   |
| tebe   | tě     | teba   | ťa     | ciebie | cię    | ‘you.SG.ACC’     |
| jeho   | ho     | jeho   | ho     | jego   | go     | ‘he.ACC/GEN’     |
| mně    | mi     | mne    | mi     | mnie   | mi     | ‘me.DAT’         |
| tobě   | ti     | tebe   | ti     | tobie  | ci     | ‘you.DAT’        |
| jemu   | mu     | jemu   | mu     | jemu   | mu     | ‘he.DAT’         |
| sebe   | se     | seba   | sa     | siebie | się    | ‘REFL.ACC(/GEN)’ |
| sobě   | si     | sebe   | si     | sobie  | se     | ‘REFL.DAT’       |

Furthermore, most strong and clitic pronouns are homophonous in Czech and Polish.<sup>111</sup> Since the majority of pronominal clitics represents unstressed strong pronouns, a purely phonological account would suffice to derive deaccented forms. This raises the question of the value of the tripartition. Rather than assuming a globally applying typology, a simple prosodic rule of deaccentuation would satisfy Occam’s razor. Keeping the typology comes with the problem of how to deal with homophony. Do we assume two distinct sets of morphologically identical pronouns, one consisting of accented full phrases and the other of the deficient kind of bare heads? Or should we assume a single set of pronouns, whereby structure becomes attenuated under cliticisation? The former conflicts with Occam’s razor, while the latter implies that the decrease in underlying syntactic structure entails a loss of stress in the first place but misses the significant morphological and syntactic predictions. Especially the hypothetical morpheme – or some other sort of corresponding structure – is visible only in the smallest portion of the paradigm, which yields the impression of a largely arbitrary theoretical apparatus.

<sup>111</sup> Cf. Table 2.3 in Section 2.8 for an overview of the respective pronominal forms.

### 3.6.4 Syntax

Cardinaletti & Starke (1999a: 168) derive the head-phrase distinction between clitic and weak pronouns by the observation that 2P clitics like Slovak *mu* ‘him.DAT’ “strongly amalgamate with the verb when enclitic, and pattern together with ‘clitic’ verbs, particles etc. themselves clear heads.” On the other hand, *ono* may occur in sentence-initial position, which is “only available to topicalized and subject XPs (except for the special case of verb-inversion)” (Cardinaletti & Starke 1999a: 169). Furthermore, the grammatical placement of *ono* between a complementizer and the clitic cluster in embedded clauses, suggests phrasal status for *ono*, cf. (3.24). This position can be filled exactly by one XP, as Cardinaletti & Starke (1999a: 169) argue. Consequently, clitics will end up in a functional head position, while weak pronouns target a functional specifier position in a generative syntactic framework (cf. Cardinaletti & Starke 1999a: 171).

- (3.24) a.  $C^0$  XP CL  
b.  $C^0$  *ono* CL  
c. \* $C^0$  XP XP CL  
d. \* $C^0$  XP *ono* CL (Slk; Cardinaletti & Starke 1999a: 169)

However, both the discrimination of weak and clitic pronouns on the one hand and deficient and strong pronouns on the other hand is problematic, when considering the modification and coordination ban, which is explained by deficient pronouns’ lack of CP (cf. Cardinaletti & Starke 1999a: 187). Veselovská (1995: §9.3.1, n. 23) points out that the ban does not follow from the syntactic status of the deficient pronouns.<sup>112</sup> Weak pronouns are syntactic phrases XP in Cardinaletti & Starke’s (1999a) approach, similar to strong pronouns and full NPs. On the other hand, clitic pronouns are heads  $X^0$ . Veselovská (1995: §9.3.1, n. 23) remarks that not only XPs may serve as conjuncts in coordinate structures but  $X^0$ s may too.<sup>113</sup> Thus, it is impossible to derive the different

<sup>112</sup> Veselovská (1995) refers to Cardinaletti & Starke (1994). Unlike other references, I do not cite Veselovská (1995) with page numbers but follow Hana (2007) in providing chapter and section numbers, because there is no coherent pagination in Veselovská’s (1995) doctoral dissertation.

<sup>113</sup> Structures like (ii)–(iii) show that modification and coordination are equally available for what in GB-terms are a CP and an IP. Example (iii) shows that both CP and IP may be equally modified by the restrictive particle *nur* ‘only’ in German. Cardinaletti & Starke (1999a: 187) base their argument on (i) telling that “modifiers always modify a full clause, nominal or verbal, and never a subpart of the clause”. However, if we assume that IP – or any equivalent syntactic structure – is subordinate to CP, then it is a subpart of CP by definition. Example (iv) shows that both CP and IP can be conjuncts in the same coordinate construction. It is generally known that conjuncts may belong to disparate syntactic categories, cf. (iv). Veselovská (1995: §9.3.1, n. 23) cites (v) showing coordinated heads. Thus, it is questionable to account for restrictions on modification and coordination in terms of a distinction between phrasal categories.

(i) a. [ only that it is so cold down there ] bothers me  
b. \* [ that only it is so cold down there ] bothers me (Eng; Cardinaletti & Starke 1999a: 187)

behaviour of strong vs. weak vs. clitic pronouns with respect to coordination and modification based on the CP/IP or the head/phrase distinction.

The clustering criterion, too, cannot serve as a test or classificatory criterion, for it does not account for the linguistic data. Cardinaletti & Starke (1999a: 169) tell that “a co-occurrence of clitics leads to the formation of a clitic cluster with characteristic morpho-phonemic processes applying.” First, it has been observed that clitics may occur disjoint from one another, e.g. in Czech (cf. Hana 2007; I. Kulik 2023b; Lenertová 2004).<sup>114</sup> Second, cluster-internal morpho-phonological processes hardly occur in Slavic. Only BCMS and Czech have been reported to show some relevant processes, all of which apply in connection with auxiliary clitics. There are two commonly known processes in BCMS, one of which is the replacement of the pronominal clitic form *je* ‘her.ACC’ by *ju* when immediately followed by homophonic *je* ‘AUX.3SG’, while the other entails the omission of *je* ‘AUX.3SG’, if it would be preceded by *se* ‘REFL.ACC’.<sup>115</sup> Czech is known for the fusion of *se* ‘REFL.ACC’ and *si* ‘REFL.DAT’ with *jsi* ‘AUX.2SG’ giving rise to the fused forms *ses* and *sis* respectively. However, these processes are not mandatory and there is variation as to whether these processes apply or not, cf. the examples from Czech in (3.25) (cf. Kolaković et al. 2022: 21, 94–95, 103–106 for BCMS).<sup>116</sup>

- (3.25) a. To **ses** ne-bála? Vážně?  
 TO REFL.ACC.AUX.2SG NEG-fear.PST.SG.F seriously  
 ‘So, you weren’t afraid? Seriously?’

- 
- (ii) Er hat den Fall übernommen,  
 he.NOM have.PRS.3SG the.ACC.SG.M case.ACC.SG take\_care\_of.PTCP  
 a. [CP nur [CP damit er gut vor dem Chef da-steht ]].  
 only so\_that he.NOM well before the.DAT.SG.M boss.DAT.SG there-stand.PRS.3SG  
 b. [IP nur [IP um gut vor dem Chef da-zustehen ]].  
 only in\_order\_to well before the.DAT.SG.M boss.DAT.SG there-stand.INF  
 ‘He took the case, so that he would just have a good standing in the boss’s eye.’  
 (iii) Er nimmt eine längere Auszeit, [CP damit er sich  
 he.NOM take.PRS.3SG a.ACC.SG.F longer.ACC.SG.F break.ACC.SG in\_order\_that he.NOM REFL  
 regenerieren kann ] und [IP um über sein Leben nach-zu-denken ].  
 regenerate.INF can.PRS.3SG and in\_order\_to over his.ACC.SG.N life.ACC.SG over-to-think.INF  
 ‘He takes a longer break, so that he can recover and reflect about his life.’  
 (iv) a. Sie fahren mit der Regionalbahn [AdvP der Umwelt  
 they.NOM go.PRS.3PL with the.DAT.SG.F regional\_train.DAT.SG the.GEN.SG.F environment.GEN.SG  
 zuliebe ] und [IP um Geld zu sparen ].  
 for\_the\_sake\_of and in\_order\_to money.ACC.SG to save.INF  
 ‘They take the regional train for the sake of the environment and to save money.’  
 b. Er ist [NP ein sensibler Junge ] und [AdjP sehr schüchtern ].  
 he.NOM be.PRS.3SG a.NOM.SG.M sensitive.NOM.SG.M boy.NOM.SG and very shy  
 ‘He is a sensitive boy and very shy.’ (Ger; constructed)  
 (v) John [ does and can ] swim. (Eng; Veselovská 1995: §9.3.1, n. 23)

<sup>114</sup> See Section 4.5 on clitic clustering.

<sup>115</sup> See Section 4.9 on co-occurrence restrictions in the clitic cluster.

<sup>116</sup> CQL query: (i) [word="jsi"] [word="se|si"], (ii) [word="ses"], (iii) [word="sis"]

- b. (Myslela jsem,) že už **jsi** se mě nabažil.  
 think.PST.SG.F AUX.1SG that already AUX.2SG REFL me.ACC have\_enough.PST.SG.M  
 ‘(I thought) that you’re already fed up with me.’
- c. Všechno **sis** vymyslel!  
 everything REFL.DAT.AUX.2SG make\_up.PST.SG.M  
 ‘You made it all up!’
- d. Chtěl **jsi** si hrát na hrdinu (a dobře  
 want.PST.SG.M AUX.2SG REFL.DAT play.INF on hero.ACC.SG and good.ADV  
 víš, že jsem tě varovala,) ale ty  
 know.PRS.2SG that AUX.1SG you.SG.ACC warn.PST.SG.F but you.NOM.SG  
**jsi** si říct ne-dal.  
 AUX.2SG REFL.DAT tell.INF NEG-give.PST.SG.M  
 ‘You wanted to act the hero (but you [are] well aware that I warned you,) but you didn’t  
 listen to reason.’ (Cz; ČNK syn2020)

Finally, the co-occurrence restriction of the Person Case Constraint (PCC) does not apply to Slavic contrary to Cardinaletti & Starke’s (1999a: 170) claim that “[t]he sharp ungrammaticality of such examples [...] is constant across Romance and Slavic languages, but also in many different language groups”. Various linguists have demonstrated that the PCC does not apply in Slavic, cf. Hana (2007: 132), Lenertová (2004), for Czech; Cetnarowska (2003) for Polish; Stegovec (2020) for Slovene; and Migdalski (2013: 144–145) for Slovene and especially BCMS.<sup>117</sup>

### 3.6.5 Semantics

From Table 3.1 we read that Cardinaletti & Starke (1999a: 145–146) make the generalisation that strong pronouns cannot refer to non-human entities. They exemplify their point with the German and Slovak examples in (3.26) among others. Witkoś (1998: 153) and Kupść (2000b: 34 [citing Witkoś 1998: 153]) confirm the prediction for Polish, although it must be pointed out that Witkoś (1998: 153) remains unclear about which semantic feature is relevant. While he uses [ $\pm$ human] in reference to Cardinaletti & Starke (1994/1999a), at first, Witkoś continues to make a stronger claim, in fact, when utilising the feature [ $\pm$ animate] throughout his argumentation.<sup>118</sup>

<sup>117</sup> Migdalski (2006: 198) points out that the PCC does not hold for BCMS, Czech, and Polish.

<sup>118</sup> The claim is insofar stronger, as the [ $\pm$ human] and [ $\pm$ animate] refer to properties of sets of referents, whereby [ $\pm$ human] is a subset of [ $\pm$ animate]. Thus, [ $\pm$ animate] is far more restricted than [ $\pm$ human] in set-theoretic terms.

- (3.26)
- |    |                                               |                |          |                 |              | [+human]                                | [-human] |   |
|----|-----------------------------------------------|----------------|----------|-----------------|--------------|-----------------------------------------|----------|---|
| a. | Sie                                           | sind           | groß.    |                 |              | √                                       | √        |   |
|    | they.NOM                                      | be.PRS.3PL     | tall/big |                 |              |                                         |          |   |
|    | 'They are tall/big.'                          |                |          |                 |              |                                         |          |   |
| b. | Sie                                           | und die        | daneben  | sind            | groß.        | √                                       | *        |   |
|    | they.NOM                                      | and the.NOM.PL | besides  | be.PRS.3PL      | tall/big     |                                         |          |   |
|    | 'They and those [ones] next [to it] are big.' |                |          |                 |              | (Ger; Cardinaletti & Starke 1999a: 146) |          |   |
|    |                                               |                |          |                 |              | [+human]                                | [-human] |   |
| c. | Vidiel                                        | som            | ich.     |                 |              | √                                       | √        |   |
|    | see.PST.SG.M                                  | AUX.1SG        | them.ACC |                 |              |                                         |          |   |
| d. | Vidiel                                        | som            | ich      | a               | tých         | druhých.                                | √        | * |
|    | see.PST.SG.M                                  | AUX.1SG        | them.ACC | and this.ACC.PL | other.ACC.PL |                                         |          |   |
|    | 'They and those [ones] next [to it] are big.' |                |          |                 |              | (Slk; Cardinaletti & Starke 1999a: 147) |          |   |

Nevertheless, it is not correct that German strong pronouns, identified by means of coordination, cannot refer to non-human entities. On the contrary, they are capable of referring to inanimate referents, in fact, cf. (3.27) from an article description of a winter garden.

- (3.27) Verbringen Sie mehr Zeit 'draußen' im Garten<sub>j</sub>, wo Sie  
 spend.IMP.2PL you.PL more time outdoors in.DAT.SG garden.DAT.SG where you.PL  
**ihn<sub>j</sub>** und andere Dinge genießen können, (ohne sich Gedanken um  
 him.ACC and other things enjoy.INF can.PRS.2PL without REFL thought.ACC.PL about  
 das Wetter machen zu müssen.)  
 the.ACC.SG.N weather.ACC.SG make.INF to must.INF  
 'Spend more time "outdoors" in the garden, where you can enjoy it [= the garden] and other  
 things (without bothering about the weather).' (Ger; S1)

Particularly compelling counterevidence is provided in (3.32)–(3.34) with pronouns being complements of prepositions. Clitics are generally known for being unable to occur as complements of prepositional heads (cf. Veselovská 1995: §11.3.3 for Czech; Eichler 1997: 71; Progovac 1996: 415–416; Radanović-Kocić 1996: 430 for BCMS; Kupść 2000b: 38 for Polish).<sup>119,120</sup>

<sup>119</sup> See Section 3.10 for contradicting structures.

<sup>120</sup> In a similar fashion, Milićev & Bešlin (2019) argue that there are no instrumental clitics in BCMS – which is the case across all Slavic languages, in fact – because the instrumental is always underlyingly a PP. Bare instrumentals are analysed as PPs, whereby the comitative preposition is phonetically null.



Abels (2003: 443) captures the generalisation by the declarative statement in (3.28).<sup>121</sup> He provides example (3.29), which shows that the generalisation holds in German, too.

(3.28) \* [P clitic] [= prohibition of a clitic as complement of a P-head] (Abels 2003: 443)

(3.29) a. Ich habe mich auf ihn verlassen.

I.NOM have.PRS.1SG me.ACC on him.DAT rely.PTCP

b. \*Ich habe mich auf 'n verlassen.

'I have relied on him.'

(Ger; Abels 2003: 443)

In Slavic, only orthotonic pronouns can be complements of a P-head, whereby a form with a glide-onset [j] surfaces as an allomorph with a nasal onset [ɲ].<sup>122</sup> Already Veselovská (1995) observes that pronominal complements of a preposition, i.e. strong pronouns, can refer to [-human] antecedents in Slavic contra Cardinaletti & Starke (1999a) and provides the Czech examples in (3.30). The clitic pronoun *ho* 'him.ACC' in (3.30a) can have both human and non-human referents as expected, such that the translation provides two possibilities here, i.e. *him/it*. Example (3.30b) complies with the predictions of the tripartite model, since the complement of the P-head *za* 'behind' is the strong pronoun *něho* 'him.ACC', which does not refer to a non-human entity, as shown in the translation line. Interestingly, example (3.30c) provides that *něho* may have human as well as non-human reference, indeed. Since no context with a relevant antecedent is provided, we are forced to solely rely on the English translation, which is somewhat unfortunate.

(3.30) a. Včera jsem ho viděl.

yesterday AUX.1SG him.ACC see.PST.SG.M

'Yesterday I saw him/it.'

(Cz; Veselovská 1995: §9.3.4, ex. [30a])

<sup>121</sup> Abels's (2003) exact analysis, which links the generalisation to P-stranding, is not important here. It is noteworthy that the generalisation, though worded imprecisely, applies cross-linguistically, as Abels (2003: 443) demonstrates on the basis of a small set of different languages. The generalisation in (3.28) is imprecise insofar as it is a statement over a general class of clitics, although Abels (2003: 443) himself specifies that it "is a ban against using a clitic *pronoun* as the complement of a preposition (P)." [my emphasis] This is an important detail, as the difference in grammaticality between Abels's German example (3.29) and (i) below cannot be captured otherwise. Note that the indefinite article *einen* 'a.ACC.M' has two clitic allomorphs in colloquial German, 'nen and 'n.

(i) a. Wir gehen auf einen / den Ball.  
we.NOM go.PRS.1PL on a.ACC.M / the.ACC.M ball.ACC

b. Wir gehen auf 'n Ball.

'We (will) go to a/the ball.'

(Ger, colloquial; constructed)

<sup>122</sup> I will not take the fossilised complexes of preposition + consonantal pronoun into consideration like, e.g., Cz. *naň* 'onto him/it', Pol. *dlań* 'for him/it', Slk. *zaň* 'for him/it, behind him/it'. They are archaic and do not represent productive structures.

b. Postav se [PP Za **něho** ]!  
 stand.IMP.2SG REFL.ACC behind him.ACC  
 ‘Stand behind him/\*it!’

c. [PP Bez **něho** ] se ne-vracej!  
 without him.GEN REFL.ACC NEG-return.IMP.2SG  
 ‘Don’t come back without him/it.’ (Cz; Veselovská 1995: §9.3.4, n. 32, ex. [i])

Kupść (2000b: 93) tells for Polish that, in contrast to pronominal clitics, reflexives do not exhibit differences in semantic reference to [±human]. Example (3.31a) provides that both clitic *się* and orthotonic *siebie* refer to the subject NP *psy* ‘dogs’. But Polish orthotonic pronouns are perfectly capable of non-human reference in Polish as evidenced by examples (3.31b–c), where the antecedents of *jego* and *jemu* are inanimate, in fact. Obviously, this referential possibility is not confined to reflexives. This contradicts the argument that “[t]he clitic pronoun can freely refer to inanimate objects, while the full form is not used in this role.” (Witkoś 1998: 153)<sup>123</sup>

(3.31) a. Gdy psy<sub>j</sub> zobaczyły **się** / **siebie**<sub>j</sub> nawzajem,  
 when dog.NOM.PL see.PST.PL.NVIR REFL.CL.ACC REFL.ACC each-other  
 zaczęły szczekać.  
 start.PST.PL.NVIR bark.INF  
 ‘When the dogs saw each other, they started barking.’ (Pol; Kupść 2000b: 93)

<sup>123</sup> It follows that long pronouns can refer to non-human animate entities. In (i)–(ii), the full forms *jemu* and *jego* refer to animals, which is ruled out in Cardinaletti & Starke’s (1999a) typology.

- (i) Żółw<sub>j</sub> klika krotnie w ciągu dnia przerzucił całą  
 turtle.NOM.SG several times in course.LOC.SG day.GEN.SG knock over.PST.SG.M full.ACC.SG.F  
 zawartość miski niczego nie jedząc. Popołudniu wymieniłem **jemu**<sub>j</sub>  
 content.ACC.SG bowl.GEN.SG nothing.GEN.SG NEG eat.PTCT afternoon exchange.PST.1SG.M him.DAT  
 pokarm na świeżą porcję, ale nadal nic nie je.  
 food.ACC.SG on fresh.ACC.SG.F portion.ACC.SG but still nothing.ACC NEG eat.PRS.3SG  
 ‘Several times during the day, the turtle<sub>j</sub> knocked over the bowl’s entire charge not eating anything [at all].  
 In the afternoon, I changed his<sub>j</sub> fodder to a new portion, but he still did not eat anything.’ (Pol; S4)
- (ii) Kot<sub>j</sub> dziwnie się zachowywał. (Nagle złapał mysz i ją  
 cat.NOM.SG strangely REFL behave.PST.SG.M suddenly catch.PST.SG.M mouse.ACC.SG and her.ACC  
 zjadł. Podszedłem do niego pomimo, ten mógł na mnie skoczyć,  
 PFV-eat.PST.SG.M go.PST.1SG.M to him.GEN.SG despite this.NOM.SG.M can.PST.SG.M on me.ACC jump.INF  
 ale wbrew przekonaniu dał mi się pogłaskać.) Od tamtego  
 but contrary to belief.DAT.SG give.PST.SG.M me.DAT REFL stroke.INF from that.GEN.SG.M  
 czasu karmiłem **jego**<sub>j</sub> i często do mnie przychodził.  
 time.GEN.SG feed.PST.1SG.M him.ACC and often to me.GEN come.PST.SG.M  
 ‘The cat<sub>j</sub> behaved strangely. (Suddenly, he caught a mouse and ate it [up]. I went to him despite [the fact that] he  
 could jump onto me, but contrary [to my] concerns, he let me stroke him.) From this time on I fed him<sub>j</sub> and he  
 came to me often.’ (Pol; S5)

- b. Uszkodzeniu uległ magnes<sub>j</sub> a czytałem że jego<sub>j</sub>  
 damage.DAT.SG undergo.PST.SG.M magnet.NOM.SG and read.PST.1SG.M that him.ACC  
 się nie da naprawić [...]  
 REFL NEG give.PRS.3SG repair.INF  
 ‘The magnet broke and I read that it’s impossible to repair it ...’ (Pol; S2)
- c. Ja posiadam Kena staruszkę TS-830S i  
 I.NOM possess.PRS.1SG Ken[wood].ACC.SG old\_guy.ACC.SG TS-830S and  
 jestem bardzo zadowolony z tego sprzętu,  
 be.PRS.1SG very satisfied.NOM.SG.M from this.GEN.SG device.GEN.SG  
 wymieniałem jemu<sub>j</sub> nowe wszystkie 3 lampy  
 exchange.PST.1SG.M him.DAT new.ACC.PL all.ACC three valve.ACC.PL  
 (przegląd maleńki i przepięknie dalej pracuje.)  
 inspection.NOM.SG small.DIM.NOM.SG.M and beautifully further work.PRS.3SG  
 ‘I have an old Ken[wood] TS-830S and I am very satisfied with this device. I exchanged  
 all its three valves [to] new [ones] – a small inspection and it works beautifully again.’  
 (Pol; S3)

Likewise, the Czech and Polish locative plural pronouns *nim* and *něm* in (3.33)–(3.34) are strong pronouns by definition. They occur as first conjunct in a coordination structure headed by a preposition. Nevertheless, these pronouns are coreferential with an antecedent that is [–human, –animate]. The same applies to German (3.32). Note that all examples in (3.32)–(3.34) are empirically attested.

- (3.32) Am 21.11.1964 wurde das [NP Dekret über den Ökumenismus]<sub>j</sub> mit  
 on 21.11.1964 was the decree about the ecumenism with  
 überwältigender Mehrheit von 2137 Ja- zu 11 Nein-Stimmen verabschiedet.  
 overwhelming majority of 2,137 yes- to 11 no-votes adopted  
 [PP In ihm<sub>j</sub> und anderen Konzilsdokumenten] öffnet die katholische Kirche  
 in it.DAT and other documents\_of\_the\_council opens the Catholic church  
 sich ökumenischen Denkweisen [...]  
 REFL ecumenical ways\_of\_thinking  
 ‘On 11 Nov. 1964, the [NP decree Restoration of unity]<sub>j</sub> was adopted by an overwhelming  
 majority of 2,137 votes in favour to 11 votes against. Within it<sub>j</sub> and other documents of the  
 council, the Catholic church opens up for ecumenical ways of thinking ...’ (Ger; S6)

- (3.33) Homesharing vznikl v [NP Irsku ]j a úspěšně [PP v něm] a dalších  
 homesharing originated in Ireland and successfully in it.LOC and other  
 západních zemích ] funguje už 30 let.  
 Western countries works already 30 years  
 ‘Homesharing originated in Irelandj and [it] successfully works therej [= in itj] and in other  
 Western countries [for] already 30 years.’ (Cz; S7)
- (3.34) Mówiono o jej naukowej współpracy z [NP warszawskim  
 discussed about her scientific cooperation with Warsaw.ADJ.INS.SG  
 Instytutem “Pomnik – Centrum Zdrowia Dziecka” ]j oraz możliwości  
 institute.INS.SG “Pomnik – Centrum Zdrowia Dziecka” and possibility  
 odbywania [PP W nim] i innych zakładach opieki zdrowotnej  
 spending in it.LOC and other institutions care.GEN.SG health.ADJ.GEN.SG  
 w naszym kraju staży przez ukraińskich lekarzy i pielęgniarki.  
 in our country internships by Ukrainian physicians and nurses  
 ‘Her scientific cooperation with the Warsaw-based Institute “Pomnik – Centrum Zdrowia  
 Dziecka” [= lit.: Monument – Centrum of Child’s Health] was discussed as well as the  
 possibility of internships for Ukrainian physicians and nurses in it and in other healthcare  
 facilities in our country.’ (Pol; S8)

Consequently, Cardinaletti & Starke’s (1999b: 283) affirmation that “[s]trong pronouns loose [sic] the ability to refer to non-human entities” is falsified. Furthermore, their claim that “[t]he ‘exceptions’ are in fact no exceptions, but demonstratives” is incorrect (Cardinaletti & Starke 1999b: 283). None of the strong pronouns in (3.32)–(3.34) is a demonstrative itself or resembles one in terms of morphology.<sup>124</sup> Therefore, one cannot recognise an (absolute) universal in the semantic behaviour of strong vs. deficient pronouns contra Cardinaletti & Starke (1999a: 146).

<sup>124</sup> From this it follows that English should not be treated as peculiar with respect to [–human] reference of coordinate personal pronouns, as done by Cardinaletti & Starke (1999a: 218, n. 2). They marginalise instances of non-human reference of the English personal pronouns *they*, *them* by reference to morphological resemblance to demonstratives (in terms of a putative morpheme *th-*). I conclude therefore that English does provide a genuine counterexample to their claim, just like German, Polish, and Czech do. Accordingly, there is reasonable doubt that the Slovak examples cited by Cardinaletti & Starke (1999a: 147, 149) in fact support their claim. They merely do not provide negative evidence and thus can be considered vacuous with respect to Karl Popper’s requirement of falsifiability.

### 3.6.6 Interim conclusion

From a meta-theoretical perspective, Cardinaletti & Starke's (1999a: 179) approach of clitic-hood in terms of a deficiency of functional structure has great appeal within the generative framework, where a diverse set of functional heads is assumed.<sup>125</sup> However, the account is not tenable within theoretical frameworks that do not recognise functional heads or that have a different stance on the head/phrase distinction. This may cast doubt on whether Cardinaletti & Starke (1999a) propose a linguistic generalisation proper, applicable in other theoretical settings, or a theory-dependent generalisation, whose validity is limited to the framework of Generative Grammar.<sup>126</sup> At best, the generative formulation of functional deficiency can be reproduced in the formal vocabulary of another theory or in a theory-neutral way. In the worst case, it has only a meaning in Generative Grammar and cannot be translated into or emulated by another theory or theory-neutral notions. If the latter were true, the question arises whether such a generalisation is of any significance at all.<sup>127</sup> I will not go into further detail here, but I conclude that the conceptual foundations of structural deficiency are likely to be subject to controversy.

Several scholars point out that Cardinaletti & Starke's (1999a) tripartite pronominal typology is not applicable to Slavic. Kupść (2000b: 34) notes that Polish atonic pronouns do not exhibit a clear distinction between weak and clitic, thus leaving only the strong/deficient dichotomy.<sup>128</sup> Kosta & Zimmerling (2014: 450, n. 6) do not adopt Cardinaletti & Starke's (1999a) proposal and argue that Slavic has a bi-partite system, at least with respect to pronouns. Werkmann (2007) states that there are no traces of a tripartite system in Russian, which exhibits a monolithic pronominal system. She attributes differences in distribution and (referential) semantics by pragmatic conditions. Based on Romanian data, Somesfalean (2007: 5–11) argues that a tripartition does not hold for Romanian as well, where only a bipartition of the pronominal system in strong vs. deficient pronouns can be observed. Dobrovie-Sorin (1999) argues that the tripartition by successive loss of functional structure is not fully mirrored in syntax proper and rejects the assumption of a  $\Sigma P$ . Instead, she argues in favour

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<sup>125</sup> There is no consensus among generative grammarians about the exact number and categories of functional heads.

<sup>126</sup> I would like to point out that scope is an important matter here. A linguistic generalisation proper is a generalisation about language, linguistic structure, or a linguistic phenomenon, which can (but does not need to) be stated within the (formal) vocabulary of a particular theory. However, the generalisation is not dependent on the vocabulary or the theory, and it may be restated in another formal vocabulary or theory. The validity of the generalisation is not affected by an individual choice of analytical tools. On the other hand, by a theory-dependent generalisation I mean a generalisation, which is not only stated within the formal vocabulary of a particular theory, but also has meaning only within that particular theory. Thus, a theory-dependent generalisation cannot be restated in any other (formal) vocabulary or theory.

<sup>127</sup> It would be, if a unique property  $G_P$  of Generative Grammar, absent in other approaches to language and grammar, covered an essential property  $L_P$  of natural human language. Then any theory lacking property  $G_P$  were inadequate with respect to property  $L_P$ .

<sup>128</sup> Kupść (2000b) refers to Cardinaletti & Starke (1994), which is an earlier version of Cardinaletti & Starke (1999a). I will refer to the more recent article.

for a bipartite distinction of strong vs. deficient pronouns on the syntactic level. The internal differentiation among deficient elements, to which she adds inflectional affixes, is a matter of morphology and phonology ultimately.

The objection that Slavic languages do not exhibit a three-class pronominal system does not pose a serious challenge in Cardinaletti & Starke's (1999b: 273) opinion, as they make clear that languages do not need to exploit all three classes simultaneously. A different line of argumentation is employed in Cardinaletti & Starke's (1999b: 282) response to Dobrovie-Sorin (1999). They point out that her account is "losing the fact that the tripartition embodies a (universal) cross-modular generalization", which they consider "the most fascinating aspect of the tripartition". In their opinion, attributing the distinctions between different types of pronouns to different components of grammar would blur their strong generalisation.

In sum, Cardinaletti & Starke's (1999a) proposal involves a tripartite classification of pronouns (and supposedly other categories, too) into strong, weak, and clitic. The tripartition results from a successive structural deficiency in terms of missing functional projections. It has been shown in the previous sections that the proposal suffers from empirical inconsistencies, which might be the byproduct of a too high degree of idealisation. This casts the validity of the approach into doubt. What is left from Cardinaletti & Starke's (1999a) proposal is a number of tests in the sense of Zwicky (1985) for the strong vs. deficient (and perhaps the weak vs. clitic) distinction. These are based on a small number of negatively defined syntactic idiosyncrasies identified for deficient elements. That is, deficient pronouns cannot be modified or coordinated, and they cannot appear in peripheral positions (cleft sentences, topicalisation, right dislocation), or in isolation. The claimed typological value of Cardinaletti & Starke's (1999a) proposal is doubtful. Besides the conceptual problems and empirical falsifications, there is also a methodological problem with respect to the database used to establish the alleged universal typology. Within their paper, Cardinaletti & Starke (1999a) refer to 15 languages and dialects from 8 language families or 5 macro families respectively.<sup>129</sup> Romance data, especially French and Italian, outnumber the examples from other language families and languages. The question arises if the proposal is perhaps biased towards a specifically Romance picture of pronounhood. At least, it is built upon an insufficiently small data set, as the data points have a high density in Europe, especially the Indo-European macro family in general and the Romance branch in

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<sup>129</sup> The authors provide examples from:

- (i) Indo-European: (a) Romance: Italian, French, Trentino dialect, Senigallia dialect, Spanish (b) Germanic: German, English, South Bavarian, (c) Slavic: Slovak, BCMS, (d) Greek: Greek
- (ii) Uralic: Finno-Ugric: Hungarian
- (iii) Afroasiatic: Semitic: Hebrew
- (iv) Niger-Congo: Volta-Niger: Gun
- (v) isolated: Basque

particular.<sup>130</sup> It is therefore questionable to claim a universal linguistic typology, which is supposed to cover languages of different genetic origin and diverse typological make-up from all around the globe.

Given the numerous challenges discussed so far, the proposal stands on weak foundations. I have demonstrated that problematic cases and severe contradictions come from “core languages” like German or Slavic (represented by BCMS, Czech, Polish, and Slovak). The tripartition is not a typological classification proper and neither qualifies as an absolute nor as statistical typology. Since Cardinaletti & Starke’s (1999a) account has been shown to be inconsistent, it is problematic to adopt the approach for a treatment of clitics. Therefore, there is little justification for pursuing such a proposal or extending it as Jung & Migdalski (2022) do, who propose a quadripartite typology of pronouns.

### 3.7 Clitics as minimal signs

Franks (2016) proposes that clitics are deficient elements in multiple levels. At first sight, this is reminiscent of Cardinaletti & Starke’s (1999a) tripartite classification. However, instead of a strictly monotone approach of increasing deficiency, he proposes that clitics are deficient with respect to three main criteria, not all of which must be met simultaneously. According to his approach, clitics are more or less “perfect” in being deficient. Franks (2016: 91) regards clitics as being defined exclusively negatively, since “they cannot do something other, ‘normal’ words can do.” In short, clitics are deficient in three domains: phonology, semantics, and syntax.

Following Anderson (2005), Franks argues that the clitics’ phonological deficiency lies in their lack of inherent prosodic structure above the syllable. That is, clitics are prosodically deficient in that they do not parse feet on their own and, hence, are not capable of being accentuated like lexical items. Instead, clitics may receive stress by means of regular stress assignment in phonology. Furthermore, Franks notes that clitics are unable to occur initially, whereby initiality may be defined with respect to different domains, i.e. intonation phrase, prosodic word, or utterance. The domain, which the ban applies in, is a language-particular specification. Therefore, clitics are prohibited in tP-initial position in BCMS, but not in Slovene. Clitics cannot be placed at the beginning of an utterance in Bulgarian, such that the Tobler-Musafia-law applies, whereas this is not an issue in Macedonian. There are also cases of item-particular constraints, as Franks argues. For instance, the Slavic question particle *-li/-nu* cannot appear initially in a prosodic word.

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<sup>130</sup> Another possibility, which should be considered, is the areal convergence of the majority of the cited languages in the sprachbund of Average Standard European (cf. Haspelmath 2001).

On the level of semantics, clitics are always function words without lexical meaning and exclusively code grammatical information. In other words, clitics necessarily are synsemantic, but never autosemantic lexemes. On the other hand, autosemantica can never be clitic, while synsemantica may (but do not need to) be clitic. From the generative point of view advocated by Franks, clitics are functional heads, which makes them similar to a “grammatical morpheme, akin to inflection (although of course syntactically independent [...]).” (Franks 2016: 107–108)

Finally, the syntactic deficiency of clitics is instantiated by bare phrase structure. Clitics lack syntactic complexity in that they are non-branching elements. Thus, they are ambiguous between syntactic heads  $X^0$  and syntactic phrases  $XP$  showing properties of both at the same time. This property also implies that clitics are capable of moving as ambiguous elements, i.e. either as heads or phrases. Essentially, Franks assumes that clitics move as heads in the first place. However, they might have the ability to move as phrases, which he discusses for the facts of Slovene clitics showing  $ACC < DAT$  order or discontinuous clitics (cf. Franks 2016: 118–119). Furthermore, being synsemantica, clitics are bundles of formal features (cf. Franks 2016: 114–116). These features are always moving in syntax to the highest possible licensing head. Full NPs, on the other hand, contain formal as well as conceptual features. Their formal features, too, move as high as possible (or necessary) in the first place. The overt dislocation of an NP is thought of as pied-piping the conceptual features to the position, where its formal features have already been licensed. However, if pied-piping does not occur, formal and conceptual features are licensed in different structural positions, which yields the respective autosemantic unpronounceable. As a result, the respective content word/NP is realised in a lower position, where both formal and conceptual features are present. In this way, 2P cliticisation is understood as the pronunciation of a clitic’s formal features in the top-most syntactic position. As clitics do not contain conceptual content, there is no pied-piping and the clitic is always realised in the structurally highest position available. From this it also follows that clitics notoriously appear in positions higher than the corresponding NPs.

In this way, Franks derives the difference between an accusative pronominal clitic like BCMS *ga* ‘him.ACC’ and an accusative noun like *prijatelj* ‘friend.ACC’ in terms of syntactic deficiency, given that *prijatelj* is a non-branching noun. Like the clitic, *prijatelj* would lack internal structure, but it still would have conceptual content and, therefore, conceptual features. On the other hand, both the pronominal clitic *ga* and the orthotonic pronoun *njega* ‘him.ACC’ consist of formal features solely. They are still distinguished by their internal syntax, as *njega* is a branching element and has internal structure. In Franks’s (2016: 121) words: “clitics can be better or worse instantiations of what it means to be a clitic, depending on how minimal they are.” However, the semantic deficiency is the basic one for Franks (2016: 121), since “no vocabulary item can be a clitic if it has lexical (substantive or contentful) meaning.” Variation thus occurs with regard to prosodic and syntactic deficiency. On the



one hand, Franks (2016: 121) argues that weak pronouns could be analysed in such a way that they are both semantically and prosodically deficient, but not syntactically. On the other hand, it is possible that a clitic is semantically and syntactically deficient but still has full prosodic structure. Franks (2016: 121) provides the Italian pronominal *loro* ‘them.DAT’ as an example for such a clitic type. Still, “[t]he perfect clitic is [...] the most minimal word” according to Franks (2016: 121), i.e. a lexeme which is deficient on all three levels.

In sum, Franks (2016) identifies clitics as minimal linguistic units, which he calls “vocabulary items”. Their minimality is manifested negatively by being unable to behave like regular words do. Franks’s (2016) concept of minimality is essentially a deficiency on three levels, cf. (3.35):

- (3.35) a. **The Prosodic Deficiency:** Clitics *cannot* project prosodic feet.  
 b. **The Semantic Deficiency:** Clitics *cannot* instantiate lexico-conceptual features.  
 c. **The Syntactic Deficiency:** Clitics *cannot* express syntactic complexity.

(Franks 2016: 121)

It was indicated at the beginning of the section that Franks’s (2016) proposal of understanding clitics as deficient elements resembles Cardinaletti & Starke’s (1999a) tripartition. However, there are some essential differences between both approaches, which are briefly discussed by Franks (2016: 123–124): First, Franks does not agree with the tripartition but argues for a greater degree of variation than envisaged in Cardinaletti & Starke’s (1999a) monotonic deficiency hierarchy. Franks (2016) proposes that pronominal clitics are kase heads ( $K^0$ ) heading a kase phrase (KP). As such, they are capable of overtly realising case values. If the head does not project, we receive a non-branching, i.e. head-only element, which is the smallest possible structure. However, the pronominal’s internal structure may be more complex. For instance, the K-head may have an NP-complement allowing to account for the fact that some pronouns can be realised as either clitic or orthotonic expression. As a result, an entire KP is projected, and the pronoun moves as phrase in syntax. The KP may involve more phrasal categories like, e.g., determiner phrase (DetP), which is language-specific (DetP occurs in Bulgarian but not in BCMS). Second, Franks takes issue with the circumstance that, while Cardinaletti & Starke (1999a) elaborate on different classes of pronouns, they do not consider the more fundamental question of what clitic-hood essentially is. Third, Cardinaletti & Starke (1999a) regard clitics strictly as heads, while Franks takes the point of view from Bare Phrase Structure, which renders clitics as non-branching elements ambiguous between head and phrase.

In fact, it is a widespread view that clitics are somehow deficient in comparison to non-clitic elements. This account is manifest e.g. in the lack of inherent prosodic structure, in the (sometimes) reduced morpho-phonological shape of clitics (cf. Czech/Slovak *ho* vs. *jeho*), or defective paradigms

(as in the case of the Czech/Slovak auxiliary clitics, which do not have 3rd person forms). Nevertheless, Franks' criteria are essentially descriptive statements about clitics. These statements are phrased in the formal vocabulary of phonological theory and Minimalist syntax. As a matter of fact, the structural correlates of the generalisations are theory-dependent. It is surely undisputed in the phonological literature that words parse feet, and that accent is parsed upon feet. It is also commonly known that clitics invariably belong to synsemantic categories. However, his employment of lexico-conceptual features for autosemantica is a theory-dependent assumption. It is furthermore a particularly generative approach to long-distance dependencies to presume a mechanism like movement or copy-and-delete, not to mention the feature-based ontology of their very existence. Finally, the notion of syntactic complexity adopted by Franks is specifically minimalist and not necessarily shared by other frameworks, (cf. Kolaković et al. 2020: 389–407 [Ch. 16] for an alternative approach to complexity phenomena). In this respect, Franks (2016) and Cardinaletti & Starke (1999a) are parallel, as both approaches take syntactic complexity to be some sort of internal complexity of a linguistic sign. Therefore, weak or orthotonic pronouns possess internal complexity in the form of internal phrasal structure, while clitic pronouns are internally plain (non-branching) signs. Towards this background, it is surprising that Franks's argumentation allows for syntactically complex orthotonic pronouns, e.g. *njega*, next to syntactically plain appellative nouns, e.g., *prijatelj*. The implication following from it that a full pronoun is internally more complex than an appellative noun, is neither justified by arguments or evidence, nor is it convincing. Note furthermore that the perspective of bare phrase structure is not shared across all frameworks, such that it may be subject to controversy. Its highly theory-dependent background renders Franks's (2016) conception of clitic-hood rather unsuitable for the theory-neutral/meta-theoretical perspective taken in the present study. The three levels of deficiency are too general and too abstract to serve as tests for clitic-hood. The proposal also does not qualify as a definition proper. Finally, the negatively stated criteria render clitics a left-over category, which casts doubt on the sufficiency of both the criteria themselves and the category they are meant to delimit.<sup>131</sup>

### 3.8 A test set for Czech

Hana (2007: 74–95) develops an exclusively syntactic or syntagmatic test battery of clitic-hood for Czech, basically following Zwicky and Pullum's work cited in the previous sections. Hana considers most of the criteria proposed in the literature either unreliable in general or not applicable to Czech

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<sup>131</sup> Left-over categories usually are sets of disregarded or unsolved problems.

in particular. His remarks on the clitic-affix distinction are straightforward and brief. The main criterion for telling clitics from affixes is the clitics' ability to be hosted by any syntactic category, whereas affixes are highly selective with respect to what stem they can attach to (cf. Hana 2007: 75). Another point made by Hana (2007: 75) is the high degree of mobility that pronominal clitics exhibit. Unlike affixes, pronominal clitics are capable of clitic climbing. As for the clitic-word distinction, the fulfilment of one of the following three properties suffices for an element to qualify as a clitic in Czech, according to Hana (2007: 75–76).

- 1) [*\*Alone*] Clitics cannot appear in isolation and cannot act as utterances on their own, e.g. as answers to previously posed questions.
- 2) [*\*Final*] Clitics cannot be placed in sentence-final position unless this position is identical with the second clausal position.
- 3) A member of a clitic cluster is itself a clitic. This criterion instantiates two particular test cases:
  - a) [*1P-Cl*] An element qualifies as a clitic in the position between the first clausal position and a(nother) clitic. Hana (2007: 76) points out that the reliability of this test relies on the element in question not being part of the first position but following it actually. Ideally, the first position should be filled with a pragmatically unmarked proper name, since one has also to ensure that the first position of the clause is not occupied by multiple constituents due to a special information structure configuration. Furthermore, the test is unsuitable to identify elements as clitics, which are notoriously placed at the right margin of the clitic cluster.
  - b) [*Cl-Cl*] An element situated between two clitics which are not or cannot be separated by prosodic boundaries qualifies as a clitic. As a result, the element in question belongs to the same clitic cluster as its two neighbours.

Hana (2007) utilises a very small set of tests for clitic-hood. The freedom in host-selection with respect to category is uncontroversial and one of few clear instances of the clitic/affix distinction. It works particularly well for 2P languages like Czech or BCMS.<sup>132</sup> The mobility criterion applies to pronominal and reflexive clitics only, because auxiliary clitics are not subject to clitic climbing. From the clitic/word distinguishing criteria, only [*\*Alone*] is a safe test. The [*\*Final*] condition is stipulated, in fact. It follows prescriptive grammaticography, but conflicts with the empirical data. Example (3.36) shows clause-final auxiliary and reflexive clitics following the verbal predicate in subordinate

<sup>132</sup> Hana (2007: 75) remarks that Czech *-li* 'if' and *-s* 'AUX.2SG' are the only exceptions to this criterion. More generally, the criterion would not be useful for VA clitic languages like Bulgarian, Macedonian or Romance languages.

clauses introduced by *že* ‘that’.<sup>133</sup> Note that clause-final position may coincide with sentence-final position, cf. (3.36a–b).

- (3.36) a. [...] *vtom si uvědomuji, [s že zaspal jsem].*  
 suddenly REFL.DAT realise.PRS.1SG that oversleep.PST.SG.M AUX.1SG  
 ‘... suddenly I realise that I overslept.’
- b. [...] *[s že po-povídají si].*  
 that PFV-talk.PRS.3SG REFL.DAT  
 ‘... that they have a chat.’
- c. ([...] *a moc omlouvám, [s že ne-věděl jsem], (jak*  
 and much apologise.PRS.1SG that NEG-know.PST.SG.M AUX.1SG how  
*vám tu říkají.)*  
 you.PL.DAT here say.PRS.3SG  
 ‘(... and I very much apologize) that I did not know (how they call you here.)’
- d. (*Však myslím, [s že ne-sluší se], (abych Vás*  
 however think.PRS.1SG that NEG-suit.PRS.3SG REFL.ACC that.COND.1SG you.PL.ACC  
*podobným povídáním obtěžoval.)*  
 similar.INS.SG story.INS.SG annoy.PST.3SG.M  
 ‘However, I think that it is inappropriate to bother you with such kind of chatter.’
- (Cz; ČNK syn\_v8)

The third criterion with its two variants suffers from circularity, as it identifies an element as clitic, if it occurs in the environment of clitics. The crucial point is that one cannot presuppose an ideal scenario, when the task is to identify clitics in an arbitrary data set. Hana (2007: 76) admits that further restrictions are necessary for the [1P-Cl] test in order to work appropriately. In fact, these restrictions boil down to pragmatically marked single or multiple constituents occupying the first clausal position. If information about the pragmatic status of the clause-initial expressions is not available, it is hardly possible to decide, whether an object pronoun occupying the superficially third position in Hana’s scheme is a clitic. Bear in mind that Hana’s tests represent a declarative constraint that ought to mechanically identify clitics. Consider a yet unknown, but pragmatically marked test element  $T_x$  positioned between some arbitrary syntactic entity in the first position and a safe clitic in the third position. In this scenario, the [1P-Cl] set-up would always favour a clitic analysis, which ultimately yields an incorrect classification.

<sup>133</sup> CQL query: [word="že"] [tag="V[pB].\*"]  
 [word="jsem|jsi|jsme|jste|mu|ho|mi|tě|se|si"]{1,2} [tag="Z:.\*"]

Returning to the problem of circularity, the [1P-Cl] test relies on the establishment of the right-hand environment. None of Hana's tests can determine the status of a clitic-cluster-final element. Hana (2007: 76) admits that "this test is not able to identify clitics that are [...] required to be on the end right of the cluster". However, it is not able to identify any arbitrary element, which appears at the right edge of the clitic cluster, as clitic, strictly speaking. Suppose a string [  $W_1$   $Prn_{CL}$   $Prn_{NON-CL}$   $W_2$  ], whereby  $W_1$ ,  $W_2$  are arbitrary orthotonic words,  $Prn_{CL}$  is a clitic pronoun, and  $Prn_{NON-CL}$  is a non-clitic, i.e. strong pronoun. Let  $Prn_{NON-CL}$  be an inconstant clitic, ambiguous between clitic and orthotonic realisation, then it is impossible to distinguish between a clitic and an orthotonic occurrence of  $Prn_{NON-CL}$ . The [1P-Cl] criterion falls short irrespective of the exact positional requirements of  $Prn_{NON-CL}$  in the clitic cluster, i.e. whether it invariably closes off the cluster or appears in cluster-medial position. For the sake of illustration, consider colloquial Czech (3.37). In both (3.37a) and (3.37b), the inconstant clitic *jim* 'them.DAT' occurs before the constant clitic *ho* 'him.ACC'. The status of *jim* is uncontroversial in (3.37a): it is classified as clitic, for it is surrounded by two constant clitics (*sem* 'AUX.1SG' and *ho*) and thus passes the [Cl-Cl] test. Things become complicated in (3.37b), where *jim* occurs in clausal second position between the subject pronoun *já* and the pronominal clitic *ho*. If one mechanically applies the [1P-Cl] criterion, then *jim* is a plain clitic. However, it is possible to provide a different analysis for *jim*, namely that of a discourse prominent element bearing Contrastive Focus (3.37c). As a focused constituent, *jim* cannot be but an orthotonic dative pronoun. Neither can clitics bear discourse functions of Topic and Focus (cf. Dotlačil 2007) nor be the locus of a corresponding stress.

The [1P-Cl] test bears on the common idealisation of the second position as the clitic slot par excellence. But clitic and orthotonic pronouns do not target mutually exclusive positions. Morphologically distinct orthotonic pronouns occupy clausal second position in main and subordinate clauses in (3.38) and (3.39) respectively. Notably, these occurrences result in clitic third position placement in (3.38a) and (3.39). From this it follows that [1P-Cl] is not a suitable test condition. Therefore, one cannot deduce clitic-hood for Czech particles like *však* 'however' and *prý* 'supposedly' by application of the test contrary to Hana (2007: 91).

- (3.37) a. ne to ne ale ale já sem jim ho ne-votrhala  
 no this no but but I.NOM AUX.1SG them.DAT him.ACC NEG-pick.PST.SG.F  
 'no, that not, but I didn't pick it from them'
- b. já jim ho dělat ne-budu  
 I.NOM them.DAT him.ACC do.INF NEG-be.FUT.1SG  
 'I will not do it for/to them'

c. [ JÁ ]<sub>TOP</sub> [ JIM ]<sub>FOC</sub> **ho** dělat [ NEbudu ]<sub>FOC</sub>

‘For/to them, I will not do it.’

(Cz; ČNK oral\_v1)

(3.38) a. Také<sup>1</sup> **jemu**<sup>2</sup> **ho**<sup>3</sup> Městský úřad v Nymburce  
also him.DAT him.CL.ACC city.ADJ.NOM.SG office.NOM.SG in Nymburk.LOC  
říjnu 2010 odebral.

October.LOC 2010 take\_away.PST.SG.M

‘The municipal authority in Nymburk took it [= driver’s licence] away from him in October 2010, too.’

b. Děkuju, já<sup>1</sup> **tobě**<sup>2</sup> přeju taky všechno nejlepší.  
thank.PRS.1SG I.NOM you.DAT wish.PRS.1SG also all best

‘Thank you, I wish all the best to you, too.’

c. Klaus<sup>1</sup> **jemu**<sup>2</sup> zase udělil Řád bílého  
Klaus.NOM him.DAT again award.PST.SG.M order.ACC.SG white.GEN.SG.M  
lva.

lion.GEN.SG

‘[President] Klaus invested him with the Order of the White Lion, in turn.’

d. [...] já<sup>1</sup> **jemu**<sup>2</sup> dala kufr s čistým  
I.NOM him.DAT give.PST.SG.F suitcase.ACC.SG with clean.INS.SG.N  
prádlem a on mně ten s použitým.  
laundry.INS.SG and he.NOM me.DAT this.ACC.SG.M.INA with used.INS.SG.N

‘... I gave a suitcase with fresh clothes to him and he [gave] the one with [the] used [ones] to me.’

e. Ale já<sup>1</sup> **tebe**<sup>2</sup> vlastně taky ne-znám.  
but I.NOM you.SG.ACC actually also NEG-know.PRS.1SG

‘But actually I don’t know you, too.’

f. (Novináři mi to ještě nikdy ne-řekli)  
journalist.NOM.PL me.CL.DAT this still never NEG-tell.PST.PL.VIR  
a já<sup>1</sup> **sebe**<sup>2</sup> ne-můžu hodnotit.  
and I.NOM REFL.ACC NEG-can.PRS.1SG evaluate.INF

‘(Journalists never told me that) and I cannot assess myself.’

(Cz; ČNK syn\_v8)

- (3.39) a. (Pak si řekneš, že to výtvarník udělal  
 then REFL.CL.DAT say.PST.2SG that this.ACC artist.NOM.SG do.PST.SG.M  
 zajímavě,) protože<sup>1</sup> **tebe**<sup>2</sup> **by**<sup>3</sup> napadlo toho  
 interesting.ADV because you.ACC COND cross\_one\_'s\_mind.PST.SG.N this.ACC.SG  
 čerta „jen“ namalovat.  
 devil.ACC.SG only paint.INF  
 ‘(Then you say to yourself that the artist did it in an interesting way,) because you would  
 have thought of “only” painting this devil.’
- b. (Myslíš,) že<sup>1</sup> **tebe**<sup>2</sup> **se**<sup>3</sup> to ne-týká?  
 think.PRS.2SG that you.ACC REFL.CL.ACC this.NOM.SG NEG-concern.PRS.3SG  
 ‘(Do you think) that it doesn’t concern you?’
- c. (Kdo si myslí,) že<sup>1</sup> **jemu**<sup>2</sup> **se**<sup>3</sup> to nikdy  
 who.NOM REFL.CL.DAT think.PRS.3SG that him.DAT REFL.CL.ACC this.NOM.SG never  
 ne-může stát, (je na velkém omylu.)  
 NEG-can.PRS.3SG happen.INF be.PRS.3SG on huge.LOC.SG mistake.LOC.SG  
 ‘(Whoever thinks) that it cannot happen to him (is gravely mistaken.)’
- d. ([...] já ne-vím,) jestli<sup>1</sup> **jemu**<sup>2</sup> **se**<sup>3</sup> to  
 I.NOM NEG-know.PRS.1SG if him.DAT REFL.CL.ACC this.NOM.SG  
 nelíbí [...]   
 NEG-like.PRS.3SG  
 ‘(... I don’t know,) if he likes it ...’
- e. (Druhá věc je,) že<sup>1</sup> **sebe**<sup>2</sup> **bych**<sup>3</sup> do  
 second.NOM.SG.F thing.NOM.SG be.PRS.3SG that REFL.ACC COND.1SG to  
 nominace za letošní výkon  
 nomination.GEN.SG for this\_year's.ACC.SG.M.INA performance.ACC.SG  
 ne-dal.  
 NEG-give.PST.SG.M  
 ‘(The second point is) that I wouldn’t have let myself being nominated for this year’s  
 performance.’

f. ([...] ale nic ne-může zvrátit můj  
 but nothing.NOM NEG-can.PRS.3SG reverse.INF my.ACC.SG.M.INA  
 žal,) protože<sup>1</sup> tobě<sup>2</sup> se<sup>3</sup> nic ne-vyrovná [...]   
 sorrow.ACC.SG because you.DAT REFL.CL.ACC nothing.NOM NEG-equal.PRS.3SG  
 ‘(... but nothing can take away my sorrow,) because nothing compares to you ...’<sup>134</sup>  
 (Cz; ČNK syn\_v8)

In sum, Hana’s (2007) test set faces several problems, as only the “Promiscuity” and [\*Alone] tests turned out to be reliable. The “Mobility” condition referring to clitic climbing is restricted to pronominal and reflexive clitics. The remaining tests proved insufficient, for they rest upon false premises and circular reasoning. Crucially, it has been shown that clitics are not in complementary distribution with orthotonic expressions as to the available syntactic positions.

### 3.9 Typology of clitics in Slavic

Franks (2010) and Franks & King (2000) follow Zwicky (1977) in that special clitics are characterised by special syntax.<sup>135</sup> Towards the background of the comparatively free constituent order in Slavic, special syntax manifests as a restricted distribution of clitics in the utterance, i.e. a special syntactic position (cf. Franks & King 2000: 6). Thus, clitics are either placed with respect to the left periphery of the sentence in case of the WACKERNAGEL or SECOND POSITION clitics (2P) or with respect to the verb in the case of so-called VERB-ADJACENT clitics (VA).<sup>136</sup> The distinction between 2P cliticisation and VA cliticisation is typologically important for Slavic. The BCMS example (3.40) shows that the dative clitic *mi* obligatorily occupies the second position in BCMS irrespective of which morphosyntactic category it is leaning on. Deviations from the second position yield the utterance ungrammatical as exemplified by (3.40e–g).

<sup>134</sup> The translation of (3.39f) is based on the Czech example, which itself is a comparatively free translation of a line of Irish singer Sinéad O’Connor’s interpretation of “Nothing Compares 2 U” (from 1990), which was originally written by US musician Prince for his side project *The Family* (published in 1985) (cf. Adelman 2021). The English original words are provided in (i) for the sake of transparency.

(i) but nothing can take away these blues / ’cause nothing compares to you

<sup>135</sup> Franks (2010) is a revised version of a position paper that previously circulated as Franks (1998). The position paper sets the ground for the publication of Franks and King (2000). Throughout the thesis, I will refer to Franks (2010).

<sup>136</sup> Cardinaletti & Starke (1999a: 196) refer to the respective clitic types as *C2 clitics* (occupying the second clausal position) and *ad-verbal clitics* (occurring on the verb). Klavans (1995) distinguishes *Second Position Enclitics* from *Verbal Clitics* (the latter being verb-adjacent). Bošković (2001: 97) refers to Bulgarian and Macedonian clitics by the term *verbal clitics*, too. Bennett (2011: 66) distinguishes *verb-clitics* from 2P clitics. Note that Klavans’s and Bošković’s notion of *verbal clitics* has a different denotation than the one by Franks & King (2000), Kolaković et al. (2022), Thorpe (1991), and Veselovská (1995), who mean by this auxiliary clitics (and possibly the clitic form of the copula) by this token.



- (3.40) a. Zoran<sup>1</sup>      **mi**<sup>2</sup>      stalno   kupuje      knjige.  
                  Zoran.NOM   me.DAT   always   buy.PRS.3SG   book.ACC.PL
- b. Stalno<sup>1</sup> **mi**<sup>2</sup> kupuje knjige Zoran.
- c. Knjige<sup>1</sup> **mi**<sup>2</sup> Zoran stalno kupuje.
- d. Kupuje<sup>1</sup> **mi**<sup>2</sup> stalno knjige Zoran.
- e. \*Zoran<sup>1</sup> stalno<sup>2</sup> **mi**<sup>3</sup> kupuje knjige.
- f. \*Zoran<sup>1</sup> stalno<sup>2</sup> kupuje<sup>3</sup> **mi**<sup>4</sup> knjige.
- g. \***Mi**<sup>1</sup> Zoran stalno kupuje knjige.
- ‘Zoran always buys me books.’ (BCMS; Franks 2009a: 728)

In contrast, the Macedonian example (3.41) exemplifies VA cliticisation. Every clitic position is grammatical as long as the clitics are contiguous with the verbal head of the clause. The sentence turns ungrammatical, if a non-clitic element disrupts the clitic cluster from the verb, even if the clitics occupy the second position as in (3.41d).

- (3.41) a. Vera<sup>1</sup>      **mi**<sup>2</sup>      **go**<sup>2</sup>      [v dade ]      včera.  
                  Vera.NOM   me.DAT   him.ACC   give.AOR.3SG   yesterday
- b. Včera<sup>1</sup> Vera<sup>2</sup> **mi**<sup>3</sup> **go**<sup>3</sup> [v dade ].
- c. **Mi**<sup>1</sup> **go**<sup>1</sup> [v dade ] Vera včera.
- d. \*Vera<sup>1</sup> **mi**<sup>2</sup> **go**<sup>2</sup> [AdvP včera ] [v dade ].
- ‘Vera gave him/it to me yesterday.’ (Mac; Franks 2010: 78–79)

Note that verb-adjacency is not required in 2P clitic languages, such that any arbitrary number of elements may intervene between the verb and its clitics. The adverb *stalno* ‘always’ intervenes between the pronominal clitic *mi* and the finite verb *kupuje* ‘buy.PRS.3SG’ in (3.40a), while it is the plain subject NP *Zoran* and the adverb in (3.40c). On the contrary, verb-adjacency in (3.40e–f) does not save the grammaticality of the utterances. However, Klavans (1995: 35) argues on the basis of Uto-Aztecan languages that the 2P/VA dichotomy is not as clear as usually thought and that both types are diachronically linked. The same diachronic linkage has been observed in Slavic (cf. Bennett 2006; Pancheva 2005).

Furthermore, Franks (2010: 5) and Franks & King (2000: 187, n. 2) use paradigmatic inflection as a distinguishing criterion: only special clitics possess inflectional paradigms. In contrast, simple clitics neither have special syntax nor do they exhibit inflection. The lack of prosodic structure is their only idiosyncrasy, therefore. This point of view persists until recently:

“simple clitics have no syntactic needs, whereas special clitics do. Simple clitics, being fully specified, stay put. But, because special clitics have paradigms and inflect, they have unvalued features which leads them to move, so that they appear higher than they might otherwise be expected to.” (Franks 2016: 114–115)

Typologically, Franks (2010) and Franks & King (2000) draw a first line of division within Slavic between East Slavic on the one hand as simple clitic languages and West/South Slavic on the other hand as special clitic languages. The classification is based on Jakobson’s (1971) discussion of the loss of short pronouns and present/past tense auxiliaries in East Slavic. The reflexivizer *-sja/-sja* is an exception and formally continues the ACC/GEN reflexive clitic *сѧ/se*, but it does not have the status of a syntactic word anymore. Instead, *-sja* is considered to be a bare morphological element, i.e. a suffix or postfix.<sup>137</sup> It is, therefore, generally accepted that East Slavic does not have special clitics (cf. Franks & King 2000: 187–188, 193). Nevertheless, the absence of auxiliary and pronominal clitics is a feature of the standard languages only. Franks & King (2000: 187, 193–194) cite that some Ukrainian dialects exhibit a system of auxiliary and pronominal clitics.<sup>138</sup> The clitic typology of West and South Slavic does not follow the traditional distinction of the Slavic branches: Czech, Slovak, and Sorbian (West Slavic) as well as BCMS and Slovene (South Slavic) are 2P languages.<sup>139</sup> The remaining South Slavic languages Bulgarian and Macedonian are VA languages.<sup>140</sup> Notably, Polish

<sup>137</sup> This putative affix still resembles its cognates in other Slavic languages in many respects in bearing multiple functions, e.g. it realises a genuine argument as a true reflexive or reciprocal element, it is a grammatical marker for passive and middle voice, and it is a lexicalised but semantically empty companion of inherently reflexive verbs like *смеяться/smejat'sja* ‘(to) laugh’ and *бояться/bojat'sja* ‘(to) fear’ etc. As an argument, the postfix *-sja* is in complementary distribution with the reflexive pronoun *себя/sebja*, whereby the latter is used under stress, i.e. Contrastive Focus. Furthermore, *-sja* occurs with a small set of perfectivised verbs, whose direct objects exhibit accusative-genitive alternation between perfective and imperfective aspect, e.g. *numь<sub>IPFV</sub> vody<sub>ACC</sub>/pit'<sub>IPFV</sub> vodu<sub>ACC</sub> – napit'sja<sub>PFV</sub> vody<sub>GEN</sub>/napit'sja<sub>PFV</sub> vody<sub>GEN</sub>* ‘(to) drink water’. This pattern is observable across Slavic, cf. e.g. Cz. *pit<sub>IPFV</sub> vodu<sub>ACC</sub> – napit<sub>PFV</sub> se vody<sub>GEN</sub>*, BCMS *piti<sub>IPFV</sub> vodu<sub>ACC</sub> – napiti<sub>PFV</sub> se vode<sub>GEN</sub>*. According to Uwe Junghanns (p.c.), an analysis of Russian *-sja* other than suffix is not precluded, in principle. In fact, Basilico (2017) exclusively refers to it by the notion *clitic*.

<sup>138</sup> See Section 3.10 for a discussion of East Slavic pronominal (clitic) data.

<sup>139</sup> Molise Slavic, an originally Croatian (predominantly Štokavian-Ikavian) idiom spoken in the province of Campobasso in Southern Italy, changes into a VA clitics language due to language contact with Italian (cf. Breu 2019).

<sup>140</sup> In contrast, Kosta & Zimmerling (2014: 462–463) argue that Bulgarian and Macedonian clitics appear in fixed clausal positions. Bulgarian obeys 2P placement, while Macedonian is 1P/2P. The VA placement is an additional requirement then. Zimmerling & Kosta (2013: 184, n. 3) claim that the Wackernagel status of Bulgarian can be shown by restrictions on the number of constituents that may precede the clitic cluster. The data they provide seem to suggest that at most one constituent is allowed in front of the clitic or clitic cluster. In contrast, Spencer (1991: 358) states that “Macedonian clitics don’t respect Wackernagel’s Law” and Werkmann (2003: 161–162) tells that Bulgarian clitics can be placed far to the right in a clause as long as they are contiguous with the verb. In her example (i), the accusative clitic *go* occurs in fourth position left-adjoined to the verb *izbracha* ‘elect.AOR.3PL’. Franks’ (2016: 103) example (ii) with the clitic cluster in fifth position corroborates a view in favour of VA rather than Wackernagel cliticisation for Bulgarian.

(i) [AdvP Včera]<sup>1</sup> [pp na zasedanie-to na narodno-to sâbranie]<sup>2</sup> [pp za predsedatel na komisija-ta  
yesterday on session-DEF on national-DEF gathering as chairman on committee-DEF  
po prava-ta na čoveka]<sup>3</sup> **go**<sup>4</sup> izbracha edva sled povtorno glasuvane.  
for rights-DEF on human him.ACC elect.AOR.3PL only after repeated voting  
‘Yesterday, on the parliamentary session, they elected him as chair of the human rights committee only after the second voting.’  
(Bg; Werkmann 2003: 162)

(West Slavic) is assigned a typologically peculiar status, according to which it does not have special clitics. While special clitics in Slavic are thought to be syntactic heads, clitics in Polish are regarded to have phrasal status instead.

There are some inconsistencies in Franks' (2010) and Franks & King's (2000) typology: First, although being premised on Zwicky's (1977) simple/special dichotomy, their distinction of simple and special clitics does not fully concur with Zwicky's (1977). Franks (2010) and Franks & King (2000) do not refer to the phonological relatedness between clitic and full forms. The derivation of simple clitics from orthotonic lexemes disappeared with Zwicky & Pullum (1983), in fact, but the parallel of clitic vs. orthotonic forms has been kept. None of this is discussed in Franks (2010) or Franks & King (2000). Second, the simple clitic status of interrogative *-li* is debatable, provided its structural position is identified with  $C^0$  by Franks (2010: 115, 121) and Franks & King (2000: 350–351, 355–356). Since *-li* targets a fixed structural position in the syntactic tree, it is reasonable to take this as an instance of special syntax as against the otherwise flexible constituent order in Slavic. Alternatively, one may account for clitic-hood in the widely accepted dichotomy of head  $X^0$  vs. maximal projection  $XP$ . If one simply takes all clitics to be heads, then the simple/special distinction is lost. If one accounts for simple/special dichotomy by treating simple clitics as  $XP$  and special clitics as  $X^0$ , then we are left with a special-clitic interpretation of *-li*, again. From a different angle, special clitic status of *-li* follows from the fact that it does not have an orthotonic counterpart. Note that already Zwicky (1977: 13) regards Russian *-li* as special clitic, and he suggests that it is in COMP (= complementizer position). Third, the criterion of paradigmatic inflection is problematic. The truncated copula verb forms in English (3.42) and the reduced personal pronouns in colloquial German (3.43) unequivocally inflect.<sup>141</sup> On the other hand, it is a typical property of pronominal elements in Slavic to inflect for case, number, and gender. The same applies to short pronouns and reflexives in Polish, of course (cf. Rubadeau 1996: 4).

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(ii) [<sub>AdvP</sub> Včera ]<sup>1</sup> [<sub>PP</sub> v gradina-ta ]<sup>2</sup> [<sub>NP</sub> Mila ]<sup>3</sup> [<sub>VP</sub> sigurno<sup>4</sup> **mu**<sup>5</sup> **gi**<sup>5</sup> dade ].  
 yesterday in garden-DEF Mila surely him.DAT them.ACC gave

‘Yesterday, in the garden, Mila surely gave them to him.’ (Bg; Franks 2016: 103)

<sup>141</sup> The phonetic transcription shows numerous phonological processes the simple clitics and their hosts undergo, which provides counterevidence to Cardinaletti & Starke's (1999a) phonological reasoning discussed in Section 3.6.2:

- (i) lack of final devoicing of the host's coda consonant in (3.43a);
- (ii) elision of the pronoun's word-initial glottal stops in (3.43a, 3.43c);
- (iii) resyllabification of the host's coda, which is parsed in the onset of the following syllable and, thus, is shared with the clitic in (3.43a, 3.43c), less trivially in (3.43b) (cf. next point);
- (iv) phonological merger of host and clitic in (3.43b), with the consequence that the formal boundary between two originally distinct morphological words disappears;
- (v) phonological reduction in terms of quantity (*hab(e)*: [ha:bə] or apocopic [ha:p] > [hab]/[hap]; *habt*: [ha:pt] > [hapt]) and quality (*du*: [du:] > [(t)ə]; *ihr*: [ʔi:ɐ̯] > [ɐ̯]).

- (3.42) a. I'm a celebrity... Get me out of here!<sup>142</sup>  
 b. You're the one that I want.<sup>143</sup>  
 c. She's the man.<sup>144</sup> (Eng)

- (3.43) a. Habe **ich** schon gemacht. [ha(:)bɪç ʃo:n gəmaχt]  
 have.PRS.1SG I.NOM already make.PTCP  
 'I've already done it.'  
 b. Hast **du** mal 'ne Mark? [hastə ma: nə maŋk]  
 have.PRS.2SG you.SG once a.CL.ACC.SG.F German\_mark.ACC.SG  
 'Do you got some change?'  
 c. Habt **ihr** mal '(ne)n Lappen? [hap'tʊ ma: (nə)n lap'm]  
 have.PRS.2PL you.PL once a.CL.ACC.SG.M rag.ACC.SG  
 'Do you have a (cleaning) rag?' (Ger; constructed)

Second, utilising Zwicky's (1977) original criterion of transparent vs. opaque phonology for the simple/special distinction, there is no synchronic phonological rule in Czech or Polish that derives short *mu*, *ho/go* from *jemu*, *jeho/jego*. On the one hand, it is impossible to derive the b-examples in (3.44) and (3.45) from the a-examples; they simply represent two unrelated sets of lexemes. On the other hand, as I argued in Section 3.6 contra Cardinaletti & Starke's (1999a) morphological considerations, there is no morphological or phonological process which could operate on the Czech pronominal forms *jeho* ~ *ho*, *tobě* ~ *ti*, *tebe* ~ *tě*, *sobě* ~ *si* etc.

- (3.44) a. *jelen* 'deer'  
 b. *len* 'flax' (Cz)

- (3.45) a. *jelen* 'deer'  
 b. *leń* 'deadbeat' (Pol)

<sup>142</sup> Title of a reality TV show that originated in the UK in 2002.

<sup>143</sup> Title of a 1978 song performed by John Travolta and Olivia Newton-John for the motion picture *Grease*.

<sup>144</sup> Title of a 2006 US movie by director Andy Fickman.

### 3.10 Re-introducing East Slavic

It was mentioned in the previous Section 3.9 that an unspecified number of Ukrainian dialects exhibit clitics. However, their exact sociolinguistic status is subject to controversy. Disagreement arises over the question of whether the respective varieties, e.g. Hucul, are dialects of Ukrainian (Ukr) or belong to an independent branch of (Carpatho-)Rusyn (Rue). Further problems arise from the fact that the respective dialect continuum is geographically distributed across the tri-border region between Poland, Slovakia, and Ukraine. I am not concerned with these sociolinguistic issues here, but the reader should not be confused by the seemingly interchangeable use of the glottonyms *Ukrainian* and *Rusyn* in this section. It is a byproduct of sticking to an original reference's usage.

Shevelov (1993: 996) informs that south-western Ukrainian dialects retained auxiliary and pronominal clitics.<sup>145</sup> Similarly, Schweier (2002: 537) mentions enclitic pronouns in the south-western Ukrainian macrodialect and lists the Hutsul and Lemko lects, among others, as subdialects. Franks & King (2000: 196–202) examine Hutsul data as a representative of dialectal Ukrainian. Coyne (2014: 166–184) devotes a chapter to Hutsul clitics in her dissertation on language change in Hutsul by language contact with contemporary standard Ukrainian.<sup>146</sup> Another study of language contact of south-western Ukrainian with particular reference to auxiliary clitics is Fellerer's (2022) article. He analyses the alternation in the past tense formation between an overt subject pronoun accompanied by a bare *l*-participle and a pro-dropped auxiliary clitic construction in the first and second persons. He continues to compare the Ukrainian dialectal facts with similar patterns occurring in “South-Eastern ‘Borderland’ Polish”, i.e. the Kresy dialects in contemporary western Ukraine. Danylenko (2012) examines the grammaticalisation of auxiliary clitics in the past and future tense in southwestern Ukrainian dialects. Danylenko's (2020b) encyclopaedic article on Carpatho-Rusyn addresses, among other things, the contrast of orthotonic vs. clitic pronouns (*mene* – *mja/nja/mna* ‘me.ACC’) and the past tense formation with a mobile auxiliary (*čyťal(a) jem* ‘I read’, *čyťal(a) jes* ‘you<sub>SG</sub> read’, *čyťaly sme* ‘we read’, *čyťaly ste* ‘you<sub>PL</sub> read’). Rabus (2019) provides a brief overview

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<sup>145</sup> On a different note, Urbańczyk (1976: 58) notes that in the Polish dialects at the eastern borders and in Mazovia (and in parts of Cassubia), the enclitic pronominal forms *go* ‘him.ACC’, *mu* ‘him.DAT’ do not exist. Perhaps, this could be either a structural convergence to the neighbouring East Slavic languages or a contact-induced interference from Russian during the Polish partitions (cf. Witzlack-Makarevich's 2020: 224–228, inter alia, summary of purists' critique of Russian grammatical interferences in Polish during the interwar period). I refer the reader to Rabus (2013: 248–316 [Ch. 5.6], with references) for East Slavic influence on Polish due to language contact (whereby the prestigious Kresy varieties seemed to play an important role). Note that Kashubian maintains a formal distinction between long and short pronouns and reflexives (cf. Makūrôt 2016: 81–83; Lorentz 1925: 156).

<sup>146</sup> Coyne (2014: 184) concludes that “[w]hile clitics were once a fairly salient feature of Hutsul morphosyntax, evidence indicates that they are undergoing a process of attrition under the increased influence of CSU [= contemporary standard Ukrainian].”

of three formally distinct types of the past tense formation in Rusyn.<sup>147</sup> First, there is a “Slovak type” consisting of *l*-participle and a clitic auxiliary for first and second persons, which is syllabic in the singular and plural. Second, the “Polish type” differs from the previously mentioned in that it has non-syllabic auxiliary clitics in the singular. Third, the “Ukrainian type” does not employ auxiliary clitics at all, and follows the common East Slavic past tense formation with obligatorily overt subject pronoun and *l*-participle. All past tense formations are attested in each diatopic variety of Rusyn, i.e. in Slovakia, Poland, and Ukraine, but they tend to converge with the past tense forms of the roofing (standard) language (which reflects so-called *border effects*). I cite the examples in (3.46) from the Corpus of Spoken Rusyn (CSR; Rabus & Schimon 2015) for illustrative purposes.<sup>148</sup> The clitics occur in clausal second position either following a subordinate conjunction (3.46a,c) or the clause-initial verb (3.46b, 3.46d). Verb-adjacent clitic placement is not obligatory in Rusyn (3.46a). Multiple clitics cluster (3.46b–c).

- (3.46) a. [...], že jak **jem** [NP vodu] nesla, [...]  
 <[...], же як **єм** [NP воду] несла, [...]>  
 that when AUX.1SG water.ACC.SG carry.PST.SG.F  
 ‘... that, when I carried water, ...’
- b. povil **jes mi**, že po-lamleš šafy [...]  
 <повіл **єс мі**, же по-ламлєш шафы [...]>  
 tell.PST.SG.M AUX.2SG me.DAT that PFV-break.2SG wardrobe.ACC.PL  
 ‘you told me that you will tear the wardrobes to pieces ...’ (Rue, Lemko; CSR)

<sup>147</sup> Rabus’s (2019) main objective is the statistical analysis of the areal distribution of the different patterns. As a result, Rusyn exhibits border effects, i.e. the convergence of a linguistic variety or a dialect continuum, which is distributed across multiple national states and roofed by different standard languages, with the respective standard language. Consequently, the linguistic variety or dialect continuum in question diverges internally and the respective diatopic varieties become more distinct.

<sup>148</sup> The data points are provided with the respective diatopic subvariety: *Lemko* refers to Rusyn in Poland, *Prešov* to Slovakia, and *Transcarpathian* to Ukraine. Note that the term *Lemko* in (3.46a) is essentially ambiguous. According to Vaňko (no date) “a number of linguistic factors suggest it may be preferable to divide Carpatho-Rusyn dialects into two basic groups, which will be referred to here as Western, or Lemko Rusyn dialects, and Eastern, or Subcarpathian Rusyn dialects.” Under this perspective, the Prešov lects cover the southern varieties of the Lemko dialect group.

- c. (Včily      **z'me**      **sa**      ukrajinčinu.  
 <(Вчили      **зьме**      **са**      українчину.  
 learn.PST.PL AUX.1PL REFL.ACC Ukrainian.ACC.SG  
 Včily      **zme**      **sa**      rus'kŭj      jazŭk.)  
 Вчили      **зме**      **са**      руський      язык.)  
 learn.PST.PL AUX.1PL REFL.ACC Rusyn.ADJ.ACC.SG.N language.ACC.SG  
 Ne nadavu,      že **jem**      **sa**      **ho**      navčila,      lebo  
 Не надаву,      же **єм**      **са**      **го**      навчила,      лебо  
 NEG give.PRS.1SG that AUX.1SG REFL.ACC him.ACC learn.PST.SG.F because  
 navčila      **sa**      **ho**,      znavu.  
 навчила      **са**      **го**,      знаву.>  
 learn.PST.SG.F REFL him.ACC know.PRS.1SG  
 '(We were learning Ukrainian. We were learning the Rusyn language.) I am not just  
 saying that I learned it, because I learned it, I know [it].'  
 (Rue, Prešov; CSR)
- d. Bŷly      **z'me**      na vŷpusknomu      vyčori,  
 <Были      **зьме**      на выпускному      вичорі,  
 be.PST.PL AUX.1PL on graduation.ADJ.M.LOC.SG evening.LOC.SG  
 u muzyčnij      školi.  
 у музичній      школі.>  
 at musical.F.LOC.SG school.LOC.SG  
 'We were on a ball at the music school.'  
 (Rue, Transcarpathian; CSR)

Given the limited data set, Rusyn seems to resemble Slavic second position clitic languages like neighbouring Slovak. In contrast, Franks & King (2000: 196–201) come to a different conclusion on the basis of Hucul data and report similarities with Polish. Boudovskaia (2017: 3–4), in turn, informs that Transcarpathian Rusyn clitics occupy the position after the first stressed word and occur in a cluster with fixed word order. Like Fellerer (2022), she analyses the complementary distribution of overt subject pronouns and auxiliary clitics in the past tense formation of the Rusyn variety of the village of Novoselytsia (Zakarpattia Oblast, contemporary Khust Raion, formerly Mizhhiria Raion). Her extensive data discussion also provides occurrences of reflexive and – to a lesser degree – pronominal clitics. A clitic cluster in second position is cited in example (3.47). Example (3.48) shows that the relevant level for clitic positioning is not the word, but the constituent. In (3.49), the clitic cluster occurs after the clause-initial verbal predicate. While clitics occur contiguously to the verb in most examples, verb-adjacency is not mandatory, cf. (3.50). However, the clitics can occur beyond clausal second position as in (3.51). We encounter a clitic in third position in a root clause in (3.51a)

and a clitic in fourth position in a subordinate (relative) clause in (3.51b). Finally, the clitic cluster may be split up in Rusyn, cf. (3.52). While the linguistic facts seem to contradict a Wackernagel treatment of Rusyn clitics, we shall see later that such patterns recur across languages some of which are undisputed 2P languages.<sup>149</sup>

- (3.47) ta [AdvP fúrt ] **ys'** **my** **s'a** snylá  
 and all\_the\_time AUX.2SG me.DAT REFL.ACC appear\_in\_dream.PST.SG.F  
 'and I dreamed of you all the time' (Rue; Boudovskaia 2017: 4)

- (3.48) a. a [PP W dêwjatnác'c'it' ] **ym** **s'a** viddála  
 and in nineteen.ACC AUX.1SG REFL.ACC give\_away.PST.SG.F  
 'and at [the age of] nineteen I got married.' (Rue; Boudovskaia 2017: 14, 24)

- b. [PP W zymí ] **s'me** **s'a** pibrály  
 in winter.LOC.SG AUX.1PL REFL.ACC take.PST.PL  
 'in the winter we got married' (Rue; Boudovskaia 2017: 12, 13)

- (3.49) a. [V poženyw ] **ym** **s'a**  
 marry.PST.SG.M AUX.1SG REFL.ACC  
 '[I] got married,' (Rue; Boudovskaia 2017: 16)

- b. ta [V poklály ] **s'te** **ho** u čěl'usty tam  
 and put.PST.PL AUX.2PL it.ACC in front\_part\_of\_oven.ACC.SG there  
 'and you put it in the front part of the oven,' (Rue; Boudovskaia 2017: 50)

- (3.50) taj [AdvP málo ] **mu** [ hy ] pittrúčovaly tak id hóri  
 and a\_little him.DAT as\_if push.PST.PL so from above.GEN.SG  
 'and they, so to speak, pushed it up a little to him' (Rue; Boudovskaia 2017: 46)

- (3.51) a. a [AdvP túj ]<sup>1</sup> [V pryjšów ]<sup>2</sup> **ym**<sup>3</sup> u prýjmý  
 and here come.PST.SG.M AUX.1SG in husband\_in\_wife's\_house.ACC.PL  
 'and here, I came as a husband.' (Rue; Boudovskaia 2017: 16)

- b. u nas tam [S de<sup>1</sup> [NP. ja ]<sup>2</sup> [V ródyła ]<sup>3</sup> **s'a**<sup>4</sup> hôr.í  
 by us.GEN there where I.NOM be\_born.PST.SG.F REFL.ACC uphill  
 'at our place, where I was born, up there,' (Rue; Boudovskaia 2017: 20)

<sup>149</sup> See Section 4.4.2 for delayed clitic placement and Section 4.5 for non-clustering clitics.



- (3.52) užé      jak **ym**      pidn'áv      **s'a**      na nóhý      býw  
 already as AUX.1SG raise.PST.SG.M REFL.ACC on foot.ACC.PL be.PST.SG.M  
 'already as I became self-supporting,' (Rue; Boudovskaia 2017: 16)

Another East Slavic variety that has not been considered in Slavic clitic research yet is colloquial Russian (*rususkaja razgovornaja reč'*). It seems to have gone unnoticed that short pronominal forms do occur in this lect, e.g. *мѧ/mja* 'I.ACC/GEN', *мѧ/tja* 'you.ACC/GEN', *me/te* 'you.DAT'.<sup>150</sup> Such forms are commonly used by speakers of the standard language (cf. Zemskaja 1973) and may be encountered in contemporary informal communication.<sup>151</sup> Indeed, Dimitrova-Vulchanova (1999: 83) mentions Russian "weak forms of the 2P,SG pronominal and the Dative reflexive", but does not specify, which forms are intended. I illustrate the short pronominal occurrences in (3.53).<sup>152</sup> The short form *мѧ/tja* in (3.53a) and (3.53c) corresponds to the standard Russian genitive pronoun *тебѧ/tebjja*. It is the complement of the prepositional head *y/u* 'at, by' in (3.53a) and occurs sentence-initially following a coordinating conjunction in (3.53c), where it is focused and presumably bears stress. The form *me/te* in (3.53b) and (3.53d) is the short counterpart of the standard dative pronoun *тебе/tebe*. It appears in clausal second position following the subject pronoun *я/ja* 'I.NOM'. The examples in (3.54) are taken from Werkmann (2007) and involve accusative *мѧ/tja* and dative *me/te* again. The accusative form *мѧ/tja* occurs in third position in (3.54a) following the subject pronoun in a subordinate clause. The dative form occupies the second position after the subject pronoun in (3.54b). Werkmann (2007: 402) states that the short forms are integrated into the prosodic word of the preceding subject pronoun *я/ja* and occupy the same structural position as the corresponding long form. Although she does not make it fully explicit, Werkmann (2007: 402) seems to suggest that both cliticisation and morpho-phonological reduction of the short pronouns are synchronic phonological processes. This implies that the short forms were derived from the long forms, but Werkmann does not provide a reasoning for this assumption, nor does she cite a reference.

<sup>150</sup> The question whether these are continuants of older short pronouns or innovations, cannot be addressed here. The forms *мѧ/mja* 'I.ACC/GEN', *мѧ/tja* 'you.ACC/GEN' correspond to Old Slavic *\*mę*, *\*tę* in terms of Neogrammarian sound change (*\*ę > ja*, cf. e.g. OCS *ѣмъ/sę* vs. Rus. *-ся/-sja* 'REFL', OCS *пѣтъ/pęti* vs. Rus. *пѣтъ/pjat'* 'five'). Therefore, a potential research question was whether such alleged old short pronouns were neglected during the process of standardisation. On the other hand, the dative form *me/te* does not correspond to older *\*ti*, cf. OCS *ти/ti*. It would be speculative to draw conclusions without a detailed analysis. Such an endeavour would have to take both diachronic data and variational data from dialects and colloquial speech into consideration. (This complex of questions about retention vs. innovation was independently posed by Vladimir Aleksandrovič Panov [Institute of Linguistics, Russian Academy of Sciences] on 10 June 2023 on the Mosling mailing list. I thank Ruprecht von Waldenfels, p.c., for sharing the posting with me.)

<sup>151</sup> I owe this observation and the relevant reference to Eugen Klein (p.c.).

<sup>152</sup> I am thankful to Genia Böhnisch (p.c., 16 Jun 2016) for providing me with the data points in (i)–(ii) from Russian web forums. Examples (iii)–(iv) are cited from Böhnisch's (2018) dissertation. Examples are provided in the first line in transliteration, while the Cyrillic original spelling is given in the second line framed by angle brackets. The third and fourth line contain the interlinear glosses and translation respectively.

- (3.53) a. Temperans, superskaja u **tja** avatarka  
 <Темперанс, суперская у **тя** аватарка>  
 Temperans super.NOM.SG.F at you.GEN.SG avatar  
 ‘Temperans, you got an awesome avatar.’
- b. Doča, (a zaodno i prosto vse sanktpeterburžcy i  
 <Доча, (а заодно и просто все санктпетербуржцы и  
 daughter.VOC and simultaneously and simply all Petersburg.M.NOM.PL and  
 sanktpeterburžicy) ja **te** napominaju, [...]>  
 санктпетербуржцы) я **те** напоминаю, [...]>  
 Petersburg.F.NOM.PL I.NOM you.DAT remind.PRS.1SG  
 ‘Dear, (and, simultaneously, simply all Petersburgers and Petersburgines) I’m reminding  
 you ...’ (Rus, non-standard; Böhnisch, p.c.)
- c. User A: Urja! Vot i prišel novyj, 2010 god!  
 <Уря! Вот и пришел новый, 2010 год!  
 hooray here and come.PST.M new.NOM.SG.M 2010 year.NOM.SG  
 Vsex pozdravljaju!  
 Всех поздравляю!>  
 all congratulate.PRS.1SG  
 ‘Hooray! The new year 2010 has arrived! My best wishes to all [of you]!’
- User B: I **tja** tož s Novym godom!  
 <И **тя** тож с Новым годом!>  
 and you.GEN also with new.INS.SG.M year.INS.SG  
 ‘And a Happy New Year to you, too!’  
 (Rus, non-standard; Böhnisch 2018: 267)
- d. Mozgovitaja, pozdravljaju, ja **te** grila, što vse  
 <Мозговитая, поздравляю, я **те** грила, что все  
 brainy congratulate.PRS.1SG I.NOM.SG you.DAT.SG tell.PST.1SG.F that all  
 budet per’jevym pučkom  
 будет перьевым пучком>  
 be.FUT.3SG feather.ADJ.INS.SG bunch.INS.SG  
 ‘Congratulations, smart one! I told you that everything would work out great.’  
 (Rus, non-standard; Böhnisch 2018: 253)

- (3.54) a. Šuruj                    otsjuda,    i    čtob ja    **tja**            tut    bol'se    ne  
 <Шуруй                    отсюда,    и    чтоб я    **тя**            тут    больше не  
 get\_lost.IMP.2SG from\_here and that I.NOM you.ACC here more NEG  
 videl.  
 видел.>  
 see.PST.SG.M  
 'Get lost, and don't let me see you [around] here again.'
- b. Ja            **te**            pokažu,            kak jabloki            vorovat'!  
 <Я            **те**            покажу,            как яблоки            воровать!>  
 I.NOM you.DAT show.PFV.PRS.1SG how apple.ACC.PL steal.INF  
 'I'll show you [what happens] when [you] steal apples!' (Rus; Werkmann 2007: 402)

More research is needed to unveil the properties of Russian short(ened) pronouns, such that this issue represents a clear desideratum. The respective forms are attested in the *Nacional'nyj Korpus Russkogo Jazyka* 'Russian National Corpus' (NKRJ) and may be obtained from both the main (3.55) and the spoken (3.56) subcorpus.

- (3.55) a. My            **tja**            ne    boim=sja,            ty, maneken! (Ty vnutri  
 <Мы            **тя**            не    боим=ся,            ты, манекен! (Ты внутри  
 we.NOM you.GEN NEG fear.PRS.1PL=REFL you mannequin you internally  
 pustaja!)  
 пустая!>  
 empty.NOM.SG.F  
 'We're not afraid of you, you mannequin! (You are empty inside!)
- b. (Olja,    daj            svoj            domašnij,)            ja            **te**  
 <(Оля,    дай            свой            домашний,)            я            **те**  
 Olja.VOC give.IMP.2SG POSS.REFL home.ADJ.ACC.SG.M.INA I.NOM you.DAT  
 večerkom            po-zvonju.  
 вечером            по-звоню.>  
 evening.DIM.INS.SG PFV-call.PRS.1SG  
 'Olya, give [me] your private [number], I'll call you in the evening.' (Rus; NKRJ, main)

- (3.56) a. Sědnja u **mja** pervyj rabočij den' byl!  
 <Седня у **мя** первый рабочий день был!>  
 today at me.GEN first work.ADJ.NOM.SG day.NOM.SG be.PST.SG.M  
 'Today, I had my first workday!'
- b. "A kak rabotat' s mečom?" / a on skazal /  
 <"А как работать с мечом?" / а он сказал /  
 and how work.INF with sword.INS.SG and he.NOM tell.PST.SG.M  
 "Voz'měš meč – on **tja** sam povedět".  
 "Возьмёшь меч – он **тя** сам поведёт".>  
 take.PFV.PRS.2SG sword.ACC.SG he.NOM you.ACC alone lead.PFV.PRS.3SG  
 "A how to work with a sword?" And he said: "When you'll take the sword, it'll lead you  
 itself."
- c. (Ona fotografiruet=sja postojanno / vezde prosto /)  
 <(Она фотографирует=ся постоянно / везде просто /)  
 she.NOM take\_photographs.PRS.3SG=REFL constantly everywhere simply  
 ja **te** govorju / (ona delaet èto celymi dnjami.)  
 я **те** говорю / (она делает это целыми днями.) >  
 I.NOM you.DAT tell.PRS.1SG she.NOM do.PRS.3SG this whole.INS.PL day.INS.PL  
 '(She is constantly taking photos of herself, simply everywhere) I'm telling you (she is  
 doing it all day long.)' (Rus; NKRJ, spoken)

There is reason to believe that Slavic clitic research will benefit from examining short pronouns in colloquial Russian. Consider the prepositional phrases *u tja* 'your.SG' in (3.53a) and *u mja* 'my' in (3.56a). One may regard them as compelling evidence that Russian short pronouns are contracted ordinary pronouns in Russian, since clitics cannot be complements of prepositions as mentioned in Section 3.6.5. There is, however, an important caveat. Dimitrova-Vulchanova (1999: 106) informs that accusative clitics can complement prepositions in BCMS, cf. (3.57), even though this option is highly restricted (cf. also Browne 1974: 40, 50 n. 4 for BCMS; Ćavar & Wilder 1999: 445 for Croatian [Hrv]). Strikingly, Czech resembles BCMS in this respect. Smetánka (1914: 47) tells that accusative clitics may exceptionally occur as complements of prepositions in Czech (3.58a):

"Nach Präpositionen stehen in der Regel die volleren, ausnahmsweise auch die kürzeren Formen."  
 (Smetánka 1914: 47)

'As a rule, the fuller [forms] are placed after prepositions, exceptionally also the shorter forms [may occur after prepositions].'

Despite the age of Smetánka's publication, this observation still seems to hold for contemporary Czech. Example (3.58b) is from Jan Němec's novel *Dějiny světa* published in 2013 by the publishing house Host in Brno. Utterance (3.58c) is from a journalistic source from 2013 but represents colloquial Czech. Comparable attestations occur in several varieties from present-day Poland. Example (3.59) provides Polish data from the NKJP, while (3.60) presents Spisz Goralish, and (3.61) written Silesian.<sup>153</sup> Coming back to East Slavic, Franks & King (2000) cite (3.62) from Hucul.

- (3.57) a. Majka je ovo [PP za **te**] spremila  
 mother has this for you.ACC.CL prepared  
 'Mom prepared this for you.' (BCMS; Dimitrova-Vulchanova 1999: 106)
- b. ... da je Ivan računao [PP na **me**].  
 that AUX.3SG Ivan.NOM count.PST.SG.M on me.ACC.CL  
 '... that Ivan counted on me.' (Hrv; Čavar & Wilder 1999: 445)

- (3.58) a. čekám [PP na **tě**]  
 waiting.PRS.1SG for you.ACC.SG.CL  
 'I am waiting for you' (Cz; Smetánka 1914: 47)
- b. A já přitom myslel [PP na **Tě**] ...  
 and I.NOM at\_that\_time think.PST.SG.M on you.ACC.SG.CL  
 'And I was thinking about you at the same time ...'
- c. (Leošu, na rovinu, ty=s byl takovej  
 Leoš.VOC on plain.ACC.SG you.NOM.SG=AUX.2SG be.PST.SG.M such.NOM.SG  
 vůl), že mě [PP za **tě**] hanba fackovala [...]  
 ox.NOM.SG that me.ACC for you.ACC.SG.CL shame.NOM.SG slap.PST.SG.F  
 '(Leoš, [I tell you] straight, you were such an idiot) that I felt embarrassed for you ...'  
 (Cz; ČNK syn\_v8)

- (3.59) a. ... i tapeta na kompie mówi sama  
 and wallpaper.NOM.SG on computer.LOC.SG say.PRS.3SG alone  
 [PP za **się**].  
 for REFL.ACC.CL  
 '... and the wallpaper on the computer speaks for itself.'

<sup>153</sup> Written Silesian data are drawn from Ruda et al.'s (2023a) corpus *Teksty śląskie* 'Silesian texts' (SIL).

- b. ([...] że ktoś odebrał wydruk z faksu,  
 that someone.NOM.SG take.PST.SG.M printout.ACC.SG from fax.GEN.SG  
 który czeka [PP na cię].  
 which.NOM.SG.M wait.PRS.3SG on you.ACC.SG.CL  
 ‘(... that someone took a printout from the fax,) which is waiting for you.’

(Pol; NKJP full poliqarp)

- (3.60) a. (dobrze ci we młynie) robi [PP na cię] woda<sup>154</sup>  
 well you.DAT in mill.LOC.SG work.PRS.3SG on you.ACC.CL water.NOM.SG  
 ‘(you are fine in the mill) the water is working for you’
- b. [PP na się] nie da podskakuwać (bo bo inaczej  
 <[PP na się] nie da podskakiwać (bo bo inaczej  
 on REFL.ACC.CL NEG give.PRS.3SG jump.INF because because otherwise  
 cię zawrejóm do hereśtu)  
 cię zawrą do aresztu>  
 you.ACC close.PRS.3PL to arrest.GEN.SG  
 ‘he doesn’t allow to mess with him, because otherwise they’ll put you to jail’ (Gor; KoSpi)

- (3.61) a. Pacza [PP przed sia]. Pacza [PP za sia].<sup>155</sup>  
 look.PRS.1SG before REFL.ACC.CL look.PRS.1SG behind REFL.ACC.CL  
 (I widza, co mi sie jedyn prziglōndo.)  
 and see.PRS.1SG what me.DAT REFL.ACC one.NOM.SG watch.PRS.3SG  
 ‘I look in before me. I look behind me. (And I see that someone’s watching me.)’
- b. [...] a ciōngła [PP do sia] nazod. Była fest  
 and pull.PST.SG.F to REFL.GEN.CL back be.PST.SG.F very  
 zło [PP na sia] [...]  
 angry.NOM.SG.F on REFL.ACC.CL  
 ‘... and she pulled him back towards herself. She was very upset with herself ...’

<sup>154</sup> Most likely, this attestation is part of a traditional folk song, hence a rhythmic genre.

<sup>155</sup> The unusual *a*-vocalism of the PP-internal reflexive has parallels in attested spoken data, cf. (i).

(i) [PP Dlo **cia**] fajnie, bo=ś stary i w klotce zrodzony [...]  
 for you.GEN fine.ADV because=be.PRS.2SG old.NOM.SG.M and in cage.LOC.SG given\_birth.NOM.SG.M  
 ‘For you, [it’s] fine, ’cause [you]’re old and were born in the cage ...’ (Sil; GiD)

c. [...]že=by Goście s Podhala czuli sie tak  
 that=COND guest.NOM.PL from Podhale.GEN feel.PST.PL.VIR REFL.ACC so  
 jak [pp u sia ].  
 how at REFL.GEN.CL  
 ‘... that the guests from Podhale felt like home.’ (Sil; SIL)

(3.62) Mama mene [pp za tja ] ne dast’.  
 mother.NOM me.ACC for you.ACC.CL NEG give.PFV.PRS.3SG  
 ‘Mom won’t let me marry you.’ [lit.: ‘Mom will not give me to you.’]  
 (Ukr, Hucul; Franks & King 2000: 199)

Presumably, these occurrences do not represent a productive linguistic structure, but some sort of fossilised syntagm.<sup>156</sup> This conjecture seems to be corroborated by Old Russian (ORu) (3.63) with the attested PP *na mę*. Prepositional constructions of this kind motivate Jung & Migdalski’s (2022) proposal of a quadripartite pronominal typology. Parallel structures are attested in Old Czech (Ocz) (3.64) and Old Polish (OPol) (3.65) (cf. Kosek 2009: 128–129 for Old Czech).<sup>157</sup>

(3.63) postrъčъtъ užь [pp na mę ] i na moe deti.  
 provoke.3SG already against me.ACC and against my child.PL.ACC  
 ‘[He] then provokes against me and against my children.’ (ORu; Jung & Migdalski 2022: 2)

(3.64) a. [pp Na tě ], milý Jezukriste, spadly sú mé viny, [...]  
 on you.ACC.CL kind Jesus\_Christ fall.PTCP.PL.NVIR AUX.3PL my guilt.NOM.PL  
 ‘Upon you, dear Jesus Christ, my sins have fallen, ...’  
 (Ocz; Milíčovský sborník modliteb)

b. V tu dobu Ježíš pozvav svých apostołów  
 in this time Jesus invite.AOR.3SG POSS.REFL.ACC.PL.VIR Apostle.ACC.PL  
 [pp před sě ] [...]  
 before REFL.ACC.CL  
 ‘[And when day came], he called his disciples [to him] ...’ (Ocz; Život Krista Pána)

<sup>156</sup> Perhaps, such structures could be conserved in a (peripheral) diatopic variety or a traditional body of folk songs, gospels, or prayers.

<sup>157</sup> The Old Czech examples were elicited from ČNK’s diachronic subcomponent, Diakorp v6 (Kučera et al. 2015). The Old Polish examples have been obtained from the PDF-version of the *Korpus tekstów staropolskich* ‘Corpus of Old Polish texts’ (<https://ijp.pan.pl/publikacje-i-materialy/zasoby/korpus-tekstow-staropolskich/>, accessed 19 Jun 2023), by aid of the corpus analysis software *AntConc* (Anthony 2022).

- c. [...] na zemi ho [PPpod **sě**] udeřivši [...]  
 on ground him.ACC under REFL.ACC.CL beat.TRAN.PST.SG.F  
 ‘... [she] threw him [down] to the ground [that was] beneath her ...’

(OCz; Pasionál muzejní)

- (3.65) a. Niechaj będzie wzgląd moj [PPna **cię**], Pana mego, [...]  
 may be.FUT.3SG view my on you.ACC.CL Lord.ACC.SG my.ACC.SG  
 ‘My eyes are ever toward the Lord, ...’ (OPol; Modlitwy Wacława)
- b. [...] gdy niekto albo niekto przysięgał  
 when somebody.NOM.SG.M or somebody.NOM.SG.F swear.PTCP.SG.M  
 albo przysięgała tako [PPprzez **się**] [...]  
 or swear.PTCP.SG.F like\_this by REFL.ACC.CL  
 ‘... whether she has sworn [an oath] herself ...’ (OPol; Kodeks Świątosławów)
- c. Wiele, już nastoja [PPna **mię**] i mąca mię; [...]  
 many already harass.PRS.3PL on me.ACC.CL and torture.PRS.3PL me.ACC.CL  
 ‘Many are they that persecute me, and afflict me; ...’ (OPol; Psalterz floriański)

More research is needed, but as a first descriptive generalisation, note that examples (3.57)–(3.65) involve locative or (al)lative prepositional heads complemented either by a 1/2SG.ACC/GEN clitic pronoun or the reflexive clitic.<sup>158</sup> This tentative generalisation fits an independent observation by Madariaga (2023: 5, n. 5), who informs that “accusative clitics, and only accusative clitics, are regularly selected by prepositions, namely in the earliest O[ld] R[ussian] texts.”<sup>159</sup> Their restricted occurrence notwithstanding, the attested [PP P CL] structures pose a problem for the general ban on clitics after P-heads in face of their wide spatial and temporal distribution.<sup>160</sup> Summing up, East Slavic

<sup>158</sup> These structures must be distinguished from the meta-communicatively or meta-linguistically motivated occurrences of clitics following prepositional heads, which may occur rather freely. I cite a passage from Trávníček (1959b: 130) for illustrative purposes in (i).

(i) Např. v našem příkladě [VP stojí [CL **jsem**] [PP před [CL **se**]].  
 e.g. in our.LOC.SG examples.LOC.SG stand.PRS.3SG AUX.1SG before REFL.ACC  
 ‘E.g., *jsem* stands before *se* in our example.’ (Cz; Trávníček 1959b: 130)

<sup>159</sup> Madariaga (2023: 5, n. 5) hypothesises that the class of Old Russian pronominal clitics was heterogenous, in fact, with accusative clitics enjoying more distributional flexibility, while dative clitics were more restricted.

<sup>160</sup> One could hypothesise that the putative clitics in the preposition’s complement position are strong pronouns in fact. Such an assumption faces several problems, however. First, clitics are in a paradigmatic relation with their strong counterparts, such that strong pronouns substitute clitic pronouns in those environments, where clitics are not licit. Why would this pattern not be effective in the prepositional domain anymore? Second, if the pronouns in a preposition’s complement position are indeed strong, why are the clitics not substituted by the corresponding orthotonic form as expected but seem to have become strengthened? Such a behaviour is nowhere else observable in the grammar of the languages under discussion – at least, as far as West Slavic is concerned. Third, it is only the accusative (and genitive) constant clitics, which show up in this position, but no datives. This makes the productivity of the pattern by some sort of grammatical process (e.g. a rule, derivation) unlikely. This is in line with the assumption mentioned in the main text that these occurrences could represent fossilised syntagms. Anyway, such data remain a



is of considerable relevance for research in Slavic clitics, once we leave the limitations of the national standard languages behind. On the one hand, full paradigms of auxiliary and pronominal clitic manifest in south-western Ukrainian Dialects and/or Carpatho-Rusyn. On the other hand, short pronouns in colloquial Russian reveal parallels with clitic constructions in other Slavic languages. One can suspect that the cross-linguistic occurrence of short pronominal complements of prepositions has implications for both linguistic typology and theory.

### 3.11 The status of clitics: Heads or phrases?

The “special” syntactic properties that clitics exhibit motivate the frequent utilisation of the head/phrase distinction in the literature. For instance, Czech clitics are treated as textbook examples of Wackernagel clitics and prototypical representatives of special-clitic-hood. Polish clitics, again, are considered to disobey any property of special-clitic-hood. This division has been translated into various approaches, which have been reviewed in the previous sections. Some of them make explicit reference to the head/phrase dichotomy. Table 3.4 summarises selected clitic typologies.

Table 3.4: Status of Czech and Polish clitics in selected clitic typologies.

|                                                                 | Czech                                                                       | Polish                                           |
|-----------------------------------------------------------------|-----------------------------------------------------------------------------|--------------------------------------------------|
| special/simple dichotomy<br>(Zwicky 1977)                       | special clitic (special syntax)                                             | simple clitic (ordinary syntax)                  |
| structural deficiency<br>(Cardinaletti & Starke 1999a)          | clitic pronoun ( $X^0$ )                                                    | weak pronoun ( $XP \approx IP$ )                 |
| head/phrase distinction<br>(Franks 2010;<br>Franks & King 2000) | head $X^0$                                                                  | phrase $XP$                                      |
| clitics = minimal<br>(Franks 2016)                              | prosodically deficient<br>semantically deficient<br>syntactically deficient | prosodically deficient<br>semantically deficient |

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severe problem for Cardinaletti & Starke’s (1999a) tripartition of pronouns. The generalisation that clitics are globally banned from a preposition’s complement position is either wrong, or the morphological reasoning is disproved according to which the additional *je*-syllable of orthotonic pronouns (*mu* vs. *je-mu*) in Slavic exhibits a morpheme, which spells out the functional head distinctive of strong pronouns vs. weak pronouns.

We read from Table 3.4 that Czech special clitics are true clitic pronouns, which are non-projecting syntactic heads  $X^0$ . Polish simple clitics are weak pronouns, which are phrasal elements XP (cf. Franks 2010: 35, 135–136). In terms of minimality, Czech clitics are more minimal than Polish ones, the latter not being syntactically deficient, indeed. It will be argued later on that this perspective is simplistic and is not represented in empirical data.

Returning to the issue of the clitics' syntactic status, I will point out some problems of the bare-phrase-structure approach, which is tied to the head/phrase distinction between Czech and Polish clitics.<sup>161</sup> The core idea is that clitics are non-branching elements, as a result of which they are ambiguous between syntactic head and syntactic phrase possessing properties of both  $X^0$  and XP. For the approach to yield the desired distinction between both languages, the same ambiguity of Czech and Polish clitics needs to be resolved, such that we come to an articulated  $X^0$ /XP distinction. Franks (2010: 135) admits that under the assumption of bare phrase structure, it is not clear where to localise the switch from  $X^0$  to XP. The question arises of what the benefits of pursuing such an approach, which is grounded on unclear mechanisms, are. I recognise an issue similar to those addressed by Haspelmath (2015).

Several scholars argue that object clitics realise true arguments of a verbal predicate. Under this view, it is reasonable to treat them as XPs like argumental full pronouns or argumental NPs. (Re-)Assessing clitics in the Rigveda, Keydana (2011: 119) remarks that “clitics must be fully projected XPs in their base position.” That is, when being merged with their verbal predicate and receiving their theta-role, arguments should be of the same syntactic category, namely NP (or DP) (cf. Franks & King 2000: 312). A traditional point of view is related to paradigmaticity. A fundamental insight from structural linguistics is that linguistic entities, which are in paradigmatic relation, belong to the same category.<sup>162</sup> This insight persists in current syntactic research and theorising. It is also repeatedly found in introductions to syntax under the flag of the constituent test of *substitution*. Complex NPs can be substituted by orthotonic pronouns as well as by pronominal clitics. Given this paradigmatic relation, the phrasal status of pronominal clitics necessarily follows.

Clitics are indisputably notorious for having a restricted distribution in comparison to orthotonic pronouns or full NPs (cf. Dotlačil 2007; Junghanns 2002c: 83; Witkoś 1998: 159; inter alia). At the same time, NPs obey distributional restrictions, too. Only a subset of NPs seems to be able to undergo what has been called *heavy NP shift*.<sup>163</sup> Furthermore, recall from Section 3.8 that full

<sup>161</sup> Cf. also Keydana (2011: 199, also n. 25), who seems to be sceptical about the conception of bare phrase structure and its implications for clitic-hood.

<sup>162</sup> I am abstracting away from the subject and object clauses.

<sup>163</sup> I cite examples in (i)–(iii) for the sake of illustration.

(i) Jeeves introduced **him** to the guests.

(ii) \*Jeeves introduced to the guests **him**.

(iii) Jeeves introduced to the guests [<sub>NP</sub> **the famous detective from Belgium** ].

(Eng; Haegeman 1994: 418)

(Eng; Haegeman 1994: 419)

and clitic pronouns are not in strictly complementary distribution, as shown through the example of Czech. It is possible to account for the different distributional capacity of nominal elements without reliance on the head/phrase distinction in terms of set theory as presented in (3.66).<sup>164</sup> The notation in (3.66) reads as: the position of all NPs (potentially heavy NPs) is a superset of the position of plain (non-heavy) NPs, which is a superset of the positions that clitic pronouns can take.<sup>165</sup> Note that the labels are used for the sake of exposition only. They do not imply any theoretical content here.

$$(3.66) \text{ POS}(\text{NP}_{(\text{heavy})}) \supseteq \text{POS}(\text{NP}_{(\text{plain})}) \supseteq \text{POS}(\text{NP}_{(\text{clitic})})$$

If we accept this line of thinking, then it appears that the distinction between, e.g., Czech special clitics and Polish simple clitics is not well represented by the  $X^0/XP$  division. The formula in (3.66) presents a scale rather than mutually exclusive discrete categories.

Taking the perspective of dependency grammar, a head is a governor. That is, a head immediately dominates a set of syntactic elements or stemma nodes, i.e. its dependents. However, clitics do not dominate any element. Neither are they governing any other category, nor do they force any element to agree with them.<sup>166</sup> Consequently, clitics are non-heads in dependency grammar, but this does not render them phrases (i.e. subtrees) automatically. Instead, clitics are syntactic words/nodes in the first place, which are in a paradigmatic relation with noun phrases. The same analysis applies to orthotonic pronouns. Under this view, there is no structural difference between clitic and full pronouns, which could be utilised for a meaningful distinction or microtypology of West Slavic pronominal systems. It follows that the acknowledgement of underlying structural distinctions follows from the choice between theoretical constructions. It is not inherent in the elements under scrutiny themselves. Therefore, the world of clitics may look different depending on the chosen theory. In the dependency grammar scenario just sketched, the burden of description lies on the lexicon. This may be a welcome result, yet, for lexical specifications of clitic-hood are assumed in phrase structure grammars, too (e.g. Bošković 2001; Junghanns 2002a).<sup>167</sup>

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In fact, it is problematic to refer to the phenomenon of heavy NP shift in the context of the present thesis, because the notion does not have any significance in Slavic syntax due to the high degree of word order freedom. I am abstracting away from this issue for the sake of argumentation.

<sup>164</sup> The formula does not cover leftwards extraction of information structurally prominent constituent beyond the clause boundary (cf. Dotlačil 2007).

<sup>165</sup> This set-theoretic formula is compatible with Haspelmath's (2015: 294) Universal II, which is built upon the comparative concepts of plenimorph and minimorph: "In all languages, plenimorphs show greater ordering variability than minimorphs."

<sup>166</sup> This consideration seems to be in stark contrast to approaches which treat clitics as a kind of agreement markers (cf. Franks 2009b for a respective account of Macedonian clitics). However, it is a qualitative difference, if a clitic is a mere exponent of agreement or if it governs agreement between syntactic units.

<sup>167</sup> The fact that clitic properties are encoded on multiple levels reveals some degree of redundancy. If cliticisation is taken to be a purely syntactic species, why should one refer to the phonology/prosody or lexicon then and vice versa? On the other hand, if one can attribute relevant portions of clitic-hood and its observable consequences from the

I have argued that various classifications are vague and contradict each other. The present discussion focused on the head/phrase distinction, which does not provide unequivocal results with respect to the distinction between regular noun phrases, orthotonic pronouns, and pronominal clitics. If category distinctions are doubtful on conceptual and empirical grounds, then one might wish to avoid the substantial dichotomy of Czech special clitics vs. Polish simple clitics or more technical solutions like head  $X^0$  vs. phrase  $XP$ . Therefore, I shall continue to approach clitics in West Slavic without reliance on such uncertain notions. There are two studies on Czech clitics that explicitly refrain from the utilisation of the head/phrase distinction: On the one hand, Dotlačil (2007) accounts for the ban of clitic climbing across the clause boundary (= CP) in terms of information structure properties and explicitly refrains from hypothesising about the clitics' syntactic status. On the other hand, Adam (2019) proposes an account of Czech second position cliticisation in the framework of Optimality Theory utilising both syntactic and prosodic constraints without any reference to the clitics' underlying syntactic traits. The conceptual framework that arises from this procedure will be presented in Section 3.13.

## 3.12 Critical assessments of clitic-hood

### 3.12.1 Introductory remarks

While attempts have been made to delimit the concept of clitic-hood in some way, e.g. by distinguishing it from words and affixes or by establishing a notion of deficiency, different aspects of clitic-hood have been subject to criticism. The arguments of Klavans (1995), Bermúdez-Otero & Payne (2001), and Haspelmath (2007, 2011, 2015) are of particular interest here and shall be reviewed and discussed in the following sections.

Before turning to each critique individually, I would like to briefly point out a conceptual trait of “scalar” treatments of clitic-hood, according to which clitics are something intermediate between words and affixes. One can distinguish between two basic approaches, each with different implications. The “discrete approach” presupposes a scale of discrete categories with a more or less clear distinction of word, clitic, and affix. Basing its notions upon discreteness, this approach necessarily operates with a three-way distinction, at least, although technical implementations may differ significantly. This approach seems to prevail in linguistics irrespective of the theoretical

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grammar to the lexicon, why would one double information and implement clitic-specific generalisations into phonology and syntax, too? The application of Occam's razor favours a more frugal approach.

perspective (e.g. phonology, syntax, HPSG, Minimalism). On the other hand, the scale is a continuum from word to affix in the “continuous approach”. Accordingly, a distinct category of clitics does not exist, but only a range of clitic-hood spanning between two ends of the scale. An arbitrary element *x* can be closer to the affix end, closer to the word end, or anywhere in between. Suppose *x* does not exhibit a clear tendency towards any extreme point of the scale, then it may be located around the central range comparable to a standard deviation from the mean in statistics. Under the view of the continuous approach, the notions of *word* and *affix* become vague. It follows that the very distinction between these notions is not categorical but gradual. Then, least or even no elements may fully fit the definitional criteria of an abstract category of word or affix. Rather, they circulate somewhere in between. In this scenario, the term *clitic* becomes a mere auxiliary notion for elements far off the edges.<sup>168</sup>

Klavans (1995) in Section 3.12.2 and Bermúdez-Otero & Payne (2001) in Section 3.12.3 are representatives of the discrete approach. They assume three distinct classes of elements, namely words, clitics, and affixes. The scholars are sceptical about the distinctions usually made within the clitic category. Haspelmath (2007 and subsequent), whose proposal will be discussed in Section 3.12.4, is a proponent of the continuous approach and calls category distinctions into question. In particular, he is suspicious of the existence of an independent category of clitics.

### 3.12.2 Klavans: Abandoning the simple/special dichotomy

Klavans’s (1995) monograph is a slightly revised version of her 1980s PhD dissertation, which arose from a critical assessment of Zwicky’s (1977) clitic typology. Instead of referring to categorical concepts like simple and special clitics, Klavans (1995: 8–9) proposes an approach to cliticisation that is built upon three binary parameters.<sup>169</sup> I will first review Klavans’s critique of Zwicky’s simple/special dichotomy, before I turn to her parametric approach.

<sup>168</sup> Exemplar theory is a possible implementation of the continuous approach. A more traditional alternative would be a fine-grained feature matrix that distinguishes words from affixes by mutually exclusive feature values, figuratively [WORD] vs. [AFFIX]. Clitic-hood follows from sharing some values with either category. Individual elements could have more [WORD] features, whereas others could have more [AFFIX] features. Still, there would be a grey area of “clitic” elements, which comprise a potentially heterogeneous set, as they may significantly differ in their feature setup.

<sup>169</sup> Originally, Klavans proposed five parameters in her PhD thesis, which she later reduced to three (cf. Klavans 1995: 6–7, cf. also Klavans 1985). The abandoned parameters are CLITIC IDENTITY and DOMAIN OF CLITICIZATION. The former is responsible for the lexical specification of a linguistic element as a clitic by a feature [+clitic] (cf. Klavans 1995: 100–101). The latter specifies the syntactic domain a clitic is placed in by reference to immediate constituency of the host to a higher syntactic node: if a clitic attaches to a host *X* that is the immediate constituent of node *Y*, then *Y* is the domain of cliticisation (cf. Klavans 1995: 101–103). CLITIC IDENTITY is redundant because items are specified as clitic in their lexical entries. DOMAIN OF CLITICIZATION is dispensed with, since it follows from the parameter INITIAL/FINAL.

Klavans (1995: 51–59) identifies several problems with Zwicky’s (1977) original tripartite distinction of simple clitics, special clitics, and bound words. First, she argues that the diachronic cycle in (3.67), which Zwicky (1977: 6) proposes with reference to Givón’s (1971) analysis of Romance, is not testable and inadequate (cf. Klavans 1995: 51, 53–55). In order to test it, a paraphrase to a strong, falsifiable hypothesis was needed: “clitics are an obligatory intermediate step from word to affixe” (Klavans 1995: 53).<sup>170</sup> This hypothesis is false, as she argues, because Spanish future and conditional suffixes emerged from the verb *haber* ‘have.INF’ without an observable intermediate clitic stage (cf. Klavans 1995: 53–54). Similarly, the preverbal position of Romance clitics is argued to be unrelated to Latin SOV order contra Zwicky (1977: 6) and Givón (1971) (cf. Klavans 1995: 54–55).

(3.67) word → simple clitic → special clitic → affix (Klavans 1995: 51)

The second major block of Klavans’s critique concerns the distinction of the three clitic categories of simple clitic, special clitic, and bound word. The baseline of her argumentation is that the categories are vaguely determined and cannot be strictly told apart from one another. The distinction between special clitics and bound words misses the typological generalisation that enclitic pronouns in Walbiri (Pama–Nyungan) and enclitic particles in Tagalog (Austronesian) are second positions clitic (cf. Klavans 1995: 56). The relevance of the distinction is questionable, if it distinguishes two sets of elements, which pattern alike. Furthermore, the delimitation of simple clitics is weak as they exhibit essentially the same flexibility in host-selection like bound words (cf. Klavans 1995: 57–59). On the other hand, the distinction between simple and special clitics is argued to be unclear, such that an element may be assigned to either category (cf. Klavans 1995: 58–59). In this context, Klavans (1995: 57) criticises the criterion of “special syntax”, because it strictly refers to syntactic operations, while Romance clitics have been argued to be subject to morphological rather than syntactic procedures, as she notes. Towards the background of an indecisive status of syntactic and morphological notions, she concludes that the simple/special dichotomy is not useful. Finally, Klavans (1995: 59) takes issue with the assumption that clitics constitute a linguistic category on their own but are not localised in any particular level of grammar.

One can basically agree with Klavans (1995) in that Zwicky’s (1977) typology involves imprecise phrasing and misses restrictive criteria. However, it is unfortunate that she did not consult Zwicky & Pullum’s (1983) article.<sup>171</sup> Klavans (1995) appropriately raised the problems that appeared

<sup>170</sup> Klavans (1995: 53) contradicts herself in this respect. First, she judges that Zwicky’s (1977) claim about diachrony “is a weak one. If it is reformulated in a stronger form [...] then it at least is testable, but untrue.” Later, on the same page she states “the claim [= the strong one] is itself impossible to test and hence vacuous.”

<sup>171</sup> While Klavans’s original dissertation (1980) and the so-called Indiana Linguistics Club version (1982) were published prior to Zwicky & Pullum’s (1983) article, her 1995’s publication appeared 12 years after Zwicky & Pullum’s (1983) revision of the special clitic/bound word distinction.

with the notion of the bound word, but these were solved by Zwicky & Pullum (1983). Klavans's (1995) rejection of the grammaticalisation path in (3.67) is convincing, but she admits that there is no "evidence against the claim that a clitic stage intervened between word and affix." (Klavans 1995: 53). In fact, such grammaticalisation paths are observable. Contemporary Polish auxiliary clitics are argued to undergo grammaticalisation from clitic to affix (cf. Abramowicz 2008; Franks 2010: 133; Franks & Bański 1999; Franks & King 2000: 144; Rappaport 1988: 320; Ventayol Boada 2014). This development mirrors exactly the grammaticalisation path in (3.67), since Polish auxiliary clitics emerged diachronically from the copula verb *być* 'be.INF'. What is more, Nübling (1998: 274) regards the evolution from syntax over cliticisation to affixation as a prototypical grammaticalisation path.<sup>172</sup> Pertinent examples of this grammaticalisation path cited by Nübling (1998: 274) are the evolution of the North Germanic synthetic passive, the suffixation of the definite article in North Germanic, and the new person agreement markers in Walser German.<sup>173</sup> Therefore, as diachrony seems to be inconclusive, it may simply be irrelevant for the typology of clitics. On the other hand, this does not preclude treating Zwicky's (1977: 6) diachronic consideration as a purely observational statement: clitics may occur as a diachronic intermediate step from orthotonic word to affix, but it is not mandatory.

Klavans's (1995: 57) rejection of special syntax relies on the theory-dependent notion of syntactic rule, whose value for a global typology of clitic phenomena is debatable. Frameworks like HPSG, LFG, Construction Grammar, or Optimality Theory do not rely on the notion of rule or procedural operations. Remarkably, she does not comment on the phonological relation between clitic and strong forms when reviewing the simple/special distinction. Furthermore, Klavans (1995: 57) refers to clitics as verbal features, which renders them morpheme-like. Although such an analysis might have some plausibility in the case of Romance and especially French, there is no a priori reason that the same analysis is applicable to another data set, e.g. Slavic clitics. This is a matter of empirical testing. In a similar way, her implicit insistence on localising cliticisation in a component of grammar appears to be biased. A typology, especially a descriptive one, does not need to make theoretical confessions. It is also debatable whether clitic phenomena can or should be globally attributed to a particular component of grammar. There is no independent a priori reason for such a claim. Finally, Zwicky's (1977) concept of special syntax is surface-oriented, in the first place, and captures the observation that special clitic pronouns are in complementary distribution with their corresponding non-clitic expressions. This is demonstrated in Zwicky's (1977: 4–5) French example, but an empirical matter, again. In Section 3.8, I showed on Czech data that there is no justification for the

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<sup>172</sup> Orig: „Mit der Entwicklungsrichtung Syntax → Klise → Flexion betreten wir nun einen prototypischen Grammatisierungspfad.“ (Nübling 1998: 274)

<sup>173</sup> Furthermore, Nübling (1998: 274) cites the development of future tense and conditional mood in Romance, among which she also refers to Spanish, which appears to be at odds with Klavans's (1995) critical remarks on this matter.

idea of complementary distribution. As a result of her critical review, Klavans (1995) abandons Zwicky's (1977) typology and proposes an alternative model, which is meant to capture clitic distribution in genetically unrelated and typologically diverse languages in a unified way by the employment of the three following parameters:

1. **INITIAL/FINAL (DOMINANCE):** refers to syntactic dominance and selects the host. Clitics are positioned with respect to the initial or final constituent of a syntactic domain (e.g. sentence S, phrase XP).
2. **BEFORE/AFTER (PRECEDENCE):** a precedence parameter specifying a clitic's linear position in the surface string relative to the host (i.e. before or after it).
3. **PROCLITIC/ENCLITIC:** encodes the orientation of phonological dependence (i.e. which direction the clitic prosodically attaches to).<sup>174</sup>

Parameters 1 and 2 are syntactic, while parameter 3 is phonological. Klavans's (1995) proposal distinguishes prosodic from syntactic clisis by parameters 2 and 3, as they may be organised independently, though often being isomorphic (cf. Klavans 1995: 9). Klavans (1995: 9–10) exemplifies the point with an example from the First Nations language Kwakwala<sup>175</sup> (Wakashan) spoken in Western Canada, which I quote in (3.68) for the sake of illustration. Kwakwala is a VSO language with object and oblique NPs being marked by preposed particles coding their grammatical functions. Thus, the particles functionally resemble postpositions in Japanese or case suffixes in Slavic and are dependent upon the NP they mark. In other words, NPs in Kwakwala seem to be comparable to the determiner-noun sequences in English, e.g. [NP *the man* ], or German, e.g. [NP *den Männern* ] 'to the men'. However, the Kwakwala particles are prosodically enclitic leaning on the word preceding them (indicated by the equals sign). Mismatches between syntactic and phonological dependencies have been brought up for Slavic, too. Fried (1994: 167–168) informs that a clitic's prosodical orientation (marked by =) may deviate from the syntactic attachment (marked by ~), when following a parenthetical. The reflexive clitic *si* cannot prosodically cliticise to the subordinate clause in (3.69) because of a prosodic margin (marked by #). Consequently, the reflexive procliticises to the following verb. However, the clitic is regarded to be syntactically enclitic, as it occurs in clausal second position as expected.

<sup>174</sup> Klavans (1995: 8, 10, 11, 96) calls the phonological attachment of a clitic to its host *liaison*.

<sup>175</sup> Listed as *Kwak'wala* on Glottolog 4.3 (Hammarström et al. 2020).



- (3.68) nəp'id-i-da      gənənəm [NP =**x**a guk<sup>w</sup>] [NP =**s**a t'isəm]  
throw-DEIX-da child OBJ house OBL rock  
'The child hit the house with a rock by throwing.'  
(Kwakwala; Klavans 1995: 9)
- (3.69) [s Že živelně                  a narychlo připravené                  akce  
that spontaneously and hurriedly prepare.PTCP.NOM.PL.F event.NOM.PL  
ne-jsou k ničemu ]<sub>IP</sub> # ~**si**= ne-uvědomil                  nikdo.  
NEG-be.PRS.3PL to nothing.DAT REFL.DAT NEG-realise.PST.SG.M no\_one.NOM  
'Nobody realised that haphazardly and hurriedly prepared events are of no value.'  
(Cz; Fried 1994: 167)

(3.69) [s Že živelně a narychlo připravené akce  
that spontaneously and hurriedly prepare.PTCP.NOM.PL.F event.NOM.PL  
ne-jsou k ničemu ]<sub>IP</sub> # ~si= ne-uvědomil nikdo.  
NEG-be.PRS.3PL to nothing.DAT REFL.DAT NEG-realise.PST.SG.M no\_one.NOM  
'Nobody realised that haphazardly and hurriedly prepared events are of no value.'  
(Cz; Fried 1994: 167)

Klavans (1995: 12–15) suggests that the three parameters derive from clitics' subcategorisation requirements and the way phonological rules interact.<sup>176</sup> Clitics resemble affixes in that clitics are subcategorised for their host like affixes being subcategorised for stems. In contrast to affixes, clitics are not restricted to certain syntactic categories, but they are subcategorised for some phrasal constituent to appear in like NP, VP, or S (Klavans 1995: 13).<sup>177</sup> Since clitics are lexically specified for their cliticisation domain and the way they attach to their host, they are treated as phrasal affixes (cf. Klavans 1995: 13). Cliticisation resembles inflectional affixation rather than derivational affixation, because cliticisation does not change the grammatical category of the host (cf. Klavans 1995: 36). The main difference between cliticisation and inflectional affixation is that clitics attach to complete words, whereas affixes attach to roots and stems.

Klavans's (1995) parametric approach has linguistic appeal, as it can identify and model different parameter combinations. However, it is surprising that syntactic dependency, which essentially underlies the INITIAL/FINAL parameter, implies syntactic clisis. Taken seriously and pursued consequently, the proposal would also render English and German definite and indefinite articles as clitics. This analysis might be a welcome side-effect of the proposal at first sight. For one thing, the analysis accounts well for enclitic articles in the German PP. The German articles in (3.70) show a mismatch between syntactic and phonological clisis like Ngancara and Kwakwala examples.<sup>178</sup> The German definite and indefinite articles are syntactically dependent upon the head

<sup>176</sup> Alternatively, Klavans (1995: 12) considers the three parameters to be well-formedness conditions on clitic placement, but she disprefers this approach. Note that Bošković's (2001) account to cliticisation in terms of the syntax-prosody interface echoes Klavans's (1995) parameters by means of the well-formedness condition account she did not pursue.

<sup>177</sup> This statement holds for 2P clitics only, as VA clitics are categorically determined. Klavans's (1995) subcategorisation approach can deal with VA cliticisation (cf. Klavans 1995: 13).

178 An alternative analysis for the contracted sequences of preposition + article seems to be eligible in terms of inflected prepositions (cf. Nübling et al. 2017: 347–350 for a textbook overview). The German standard norm only allows a small number of preposition-article combinations to coalesce both in written and spoken form. So, a flexive analysis is rather untenable. In contrast, spoken substandard varieties of German exhibit a much higher degree of flexivity, as

123).<sup>179</sup>

- b. Ich bin inner Stadt. [inner [ʔinə] < in *der* Stadt]  
 I.NOM be.PRS.1SG in.DET.F.DAT.SG city.DAT.SG  
 ‘I am in the city [centre].’ (Ger, colloquial; constructed)

degree reduction appears in all remaining pretonic and posttonic syllables. In the so-called Akanye,

possessing inflecting prepositions:

flektierter Präpositionen beschreiben.“

'The paradigm of the coalesced forms in Ruhr German can be most aptly described as a system of inflecting

The morphosyntactic feature of “inflecting prepositions” has a wider distribution, in fact. Similar observations can be

<sup>179</sup> The incorporation of the article into the preposition is the reason for the emerging impression of an inflected

the phonemes /a/ and /o/ collapse qualitatively to [ɐ]<sup>180</sup> in the first degree and to [ə] in the second degree (cf. Lehmann 2013: 192–193; Timberlake 1993: 832; Yanushevskaya & Bunčić 2015: 225).<sup>181</sup> The prepositional head *po* ‘in, around, on’ undergoes reduction like any regular word-level syllable showing first degree reduction in (3.72b), but second degree reduction in (3.72c) due to stress shift in the inflectional paradigm of *gorod* ‘city’.

- (3.71) a. hory [ˈɦɔrɪ]<sub>ω</sub>  
           ‘mountain.NOM/ACC.PL’  
       b. na hory [ˈna ɦɔrɪ]<sub>ω</sub>  
           on mountain.ACC.PL  
           ‘to (the top of) the mountains’ (Cz; Čechová 1996: 32)

- (3.72) a. gorod [ˈgoˈrət]<sub>ω</sub>  
           ‘city.NOM/ACC.SG’  
       b. po gorodu [pɐ ˈgoˈrədɔ]<sub>ω</sub>  
           around city.DAT.SG  
           ‘in/around the city’  
       c. po gorodam [pɐ gərəˈdaˈm]<sub>ω</sub>  
           around city.DAT.PL  
           ‘in/around the cities’ (Rus)

Hence, Slavic prepositions cannot be described adequately in Klavans’s (1995) system, because they are clitics in phonology, but not in syntax. Parameter one (INITIAL/FINAL) is not applicable. Furthermore, Klavans’s parameters are not only binary (cf. Klavans 1995: 122), but also static. They cannot cope with variation in clitic positioning. Clitics are attested in a variety of positions delimited by the left periphery to the left and the verbal predicate to the right. For one thing, clitics are reported to occur in first and third clausal position in Czech (cf. Avgustinova & Oliva 1997; Lenertová 2004: 136–138, 148–149; Short 1993: 495; Toman 1996: 505–506). The Slovak example in (3.73) exhibits the reflexive clitic *sa* in third position in an independent clause and the pronominal clitic *mu* in fourth position in a subordinate clause. Such distributional properties cannot be accounted for by Klavans’s (1995) INITIAL/FINAL parameter.

<sup>180</sup> The allophonic realisation as [ɐ] follows the Moscow phonological school. In the tradition of the Petersburg phonological school the allophonic variant appears as [ʌ].

<sup>181</sup> Akanye occurs only after non-palatalised, i.e. velarised consonants. In the environment after palatalised consonants, so-called Ikanye occurs, whereby unstressed /i/, /e/, /a/, and /o/ collapse in [ɪ].



interrogative. No reference to a categorical parameter can be maintained with respect to the host, as Klavans (1995: 122) suggests for Romance. Finally, (3.74i–j) indicate that the placement of the clitic pronoun is not restricted by the verb it is argument of.<sup>183</sup> Both an orthotonic pronoun like *er* ‘he’ or a full NP like *Hannah* may perfectly intervene between the verb and the clitic.<sup>184</sup> Such occurrences have been reported to be ungrammatical in Slavic.<sup>185</sup>

- (3.74) a. 's= regnet.  
it.NOM rain.PRS.3SG  
'It's raining.'
- b. Regnet =s?  
rain.PRS.3SG it.NOM  
'Is it raining?'
- c. Da regnet =s immer.  
there rain.PRS.3SG it.NOM always  
'It's always raining there.'
- d. 's= hat ihn nicht gestört.  
it.NOM have.PRS.3SG him.ACC NEG bother.PTCP  
'It didn't bother him.'
- e. Hat =s ihn nicht gestört?  
have.PRS.3SG it.NOM him.ACC NEG bother.PTCP  
'Didn't it bother him?'
- f. Ihn hat =s nicht gestört.<sup>186</sup>  
him.ACC have.PRS.3SG it.NOM NEG bother.PTCP  
'It didn't bother him.'

<sup>183</sup> Cliticisation of a pronoun in the prefield works well only for *es*. It is hardly possible for other pronouns to cliticise in this position irrespective of their morphosyntax, e.g. *er* 'he.NOM', *ihn* 'him.ACC', *sie* 'she/they.NOM/ACC', *ihr* 'her.DAT' (cf. Haspelmath 2015: 274 ex. [2]). Interestingly, it seems that *ich* 'I.NOM' does not obey this restriction in contrast to *wir* 'we.NOM', cf. (i) and (ii). If this observation were confirmed independently, a hypothesis could be that cliticisation in the prefield is a function of morphophonological shape and/or phonetic/phonological features: only fricatives or sibilants can cliticise in the prefield, 'ch = [ç] < ich = [ʔɪç],

(i) 'ch= weiß nich'. (< Ich weiß nicht.)  
I.NOM know.PRS.1SG NEG  
'I don't know.'/'I'm unsure.'

(ii) \*Wa= wissen =s nich'. (< Wir wissen es nicht.)  
we.NOM know.PRS.1PL it.ACC NEG  
'We don't know it.'

(Ger, colloquial; constructed)

<sup>184</sup> The subject pronoun *er* 'he.NOM' in (3.74i) may optionally cliticise, too.

<sup>185</sup> See Section 4.8 for positional restrictions in Slavic.

<sup>186</sup> Note that, although the interpretation of *ihn* as contrastively focused is perhaps slightly more salient in (3.74f), it is perfectly possible to realise this utterance with the same plain intonation as (3.74h). Therefore, (3.74f) and (3.74h) are not necessarily distinct with respect to information structure.

- g. ..., weil =’s in London wirklich ständig in Strömen regnet.  
 because it.NOM in London.DAT really constantly in showers rain.PRS.  
 ‘..., because it’s always raining cats and dogs in London, in fact.’
- h. ..., obwohl =’s ihn nicht gestört hat.  
 although it.NOM him.ACC NEG bother.PTCP have.PRS.3SG  
 ‘..., although it didn’t bother him.’
- i. Weiß er =’s denn?  
 know.PRS.3SG he.NOM it.ACC actually  
 ‘Does he actually know it?’
- j. Weiß Hannah =’s denn?  
 know.PRS.3SG Hannah.NOM it.ACC actually  
 ‘Does Hannah actually know it?’ (Ger, colloquial; constructed)

Accepting Klavans’s (1995) proposal, we will not be able to account for cliticisation in Slavic and German. In contrast, the simple/special clitic dichotomy yields the desired distinction between German on the one hand and Slavic and Romance on the other. While Slavic and Romance clitics are characterised by opaque phonology as to their orthotonic counterparts and seem to exhibit something like “special syntax” in that they are not distributed like their orthotonic counterparts (if existent), German provides us with textbook examples of simple clitics. Despite the problems of Klavans’s argumentation, which I pointed out, there remain some severe problems for Zwicky’s typology.

### 3.12.3 Bermúdez-Otero and Payne: Against what is special about clitics

Bermúdez-Otero & Payne (2011) challenge the very concept of special-clitic-hood pointing their discussion particularly at Anderson’s (2005) approach. Their critique targets the assumption of what they call *Clitic Idiosyncrasy*:

“Certain clitics are neither words nor affixes, but constitute a separate type of object whose behaviour is partly governed by dedicated (i.e. clitic-specific) grammatical mechanisms.” (Bermúdez-Otero & Payne 2011: 57)

Clitic Idiosyncrasy assumes that clitics – more precisely: special clitics – need a special sort of theoretical device to account for their behaviour.<sup>187</sup> Bermúdez-Otero & Payne’s (2011) basic claim is that there is no category of special clitics and that no clitic-particular machinery is needed. Instead, putative special clitics can be reduced to either words obeying ordinary syntax or affixes concatenated by morphology. In other words, there is no “special syntax”. I will restrict myself to this particular argument, which makes the smallest portion of Bermúdez-Otero & Payne’s (2011) paper, however. Their main endeavour is to abandon the phrasal affixation approach to special cliticisation and to demonstrate that the alleged special clitic occurrences they scrutinise are plain affixes. It suffices here to note that Bermúdez-Otero & Payne (2011: 73–78) argue that Bulgarian postponed definite articles, among others, are inflectional affixes contra Anderson (2005: 111–114), who treats them as DP-internal second position clitics. Bermúdez-Otero & Payne (2011) independently arrive at the same conclusion as Franks & King (2000: 278–284, with references).

Referring to Anderson’s (2005: 75) exposition, Bermúdez-Otero & Payne’s (2011: 61) observe that “special syntax” comes in two variants. First, special clitics are subject to restrictions different from the general syntax, which corresponds to the clitics’ complementary distribution as to their orthotonic counterparts. Second, they exhibit some uncommon properties for their part of speech. This point refers to second position placement, clitic clustering, and clitic ordering, which contrast with the otherwise largely unrestricted distribution of syntactic constituents in “free word order” languages like Slavic, among others. Bermúdez-Otero & Payne (2011: 61) argue that it is common across languages to possess some “unusual” elements in syntax. They make their point through the example of the postpositions *notwithstanding*, *apart*, and *aside* in English, a language that has only prepositions otherwise. The argument goes that peculiarities in themselves do not justify the assumption of a distinct category. On the other hand, distribution patterns commonly associated with special clitics can be attributed to regular syntactic relations (cf. Bermúdez-Otero & Payne 2011: 61). Considering second position cliticisation in particular, Bermúdez-Otero & Payne (2011: 61–62) contend that clitic positioning can be handled in ordinary syntax provided that verbal placement in verb-second languages is analysed in purely syntactic terms. They continue that BCMS discontinuous constituents split by clitics likewise can be accounted for purely by syntax, as it has been observed that the respective phrases can be split either way. Clitics, thus, comply with a general syntactic pattern and there is no need to rely on the notions of a “special clitic” category or a “special syntax”, which operates exclusively over elements of this set. Thus, Bermúdez-Otero & Payne’s (2011) reasoning implicitly bears on Occam’s razor. What remains is the traditionally established distinction

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<sup>187</sup> Various instances of this special machinery can be found in the literature like, e.g., Klavans’s (1995) principles for clitic placement, placement rules targeting clitics (cf. Radanović-Kocić 1988), Bošković’s (2001) PF-filter, or morphological clitic cluster templates (cf. Simpson & Withgott 1986).

between words and affixes. While the notion of special-clitic-hood is entirely dropped, the concept of simple clitic is retained referring to *words* that lack prosodic structure. The authors conclude:

“The conclusion thus seems inescapable: there are words (including simple clitics) and there are affixes (including edge affixes), but there are no phrasal affixes and, in this sense, no special clitics.” (Bermúdez-Otero & Payne 2011: 90)

### 3.12.4 Haspelmath: Contesting clitics

In a series of papers, Haspelmath (2007, 2011, 2015) criticises what he calls the “restrictivist” assumption of universal linguistic categories. He contrasts it with the “non-aprioristic” approach, which does not presuppose linguistic categories, but attempts at defining comparative concepts that are intersubjectively valid across languages (cf. Haspelmath 2010, 2019 on comparative concepts). Haspelmath’s proposals have severe implications for the study of clitics for two reasons: First, it casts the validity of the universal maintenance of linguistic categories into doubt. In the non-aprioristic view, it is inappropriate to apply language-specific descriptive categories to the study and analysis of genetically and typologically disparate languages. Second, Haspelmath explicitly considers clitic-hood to be a pseudo-category, which is not eligible for cross-linguistic comparison. This position is particularly defended in Haspelmath (2015), to which I will limit myself here.

Haspelmath (2015) argues that clitic-hood is neither a natural category of human language nor a theoretical concept (cf. also Bermúdez-Otero & Payne 2011: 60, 89). A unique or uncontroversial set of properties, which defines clitic phenomena, does not exist. Entities, which are labelled clitics, show diverse properties and there is no agreement among scholars on how to classify clitic-hood. Haspelmath’s (2015) argumentation thus parallels the discussion in the previous sections. For one thing, Haspelmath (2015: 277–278) demonstrates that under reference to phonological phenomena like stress assignment (in English, Turkish) and vowel harmony (in Turkish), it is not possible to find a clear division line between cliticisation and affixation. This is why various terminologies exist with each notion implying some theoretical status of the entities under discussion (cf. Haspelmath 2015: 275). Although there seems to be agreement among linguists upon the very existence of clitic-hood and its intermediate position between independent words and affixes, this tripartite classification is not self-evident. The assignment of an element to one of these classes may differ from scholar to scholar. That is, the notion of clitic-hood is subjective. This is why Haspelmath (2015: 279–283) proposes to define criteria that allow to classify linguistic objects rigorously, instead of applying the “test-battery-method”, as he calls it (cf. Haspelmath 2015: 279).



Haspelmath (2015: 283–287) illustrates the problem of testing for the affix/clitic distinction through the example of Makassarese (Austronesian) spoken in South Sulawesi, Indonesia. Affixes attach to roots or inflected roots and are counted for assigning antepenultimate stress in Makassarese. On the other hand, clitics do not belong to the stress domain and have a freer distribution in the clause.<sup>188</sup> In addition to these two standardly assumed categories, an intermediate third category has been proposed for Makassarese sharing properties with both affixes and clitics. Haspelmath (2015: 283–284) coins the term *affclitics* for them. They are counted for determining the stressed syllable like affixes, while they do not attach to individual words but to phrases like clitics. The question arises of what prevents one from assuming affclitics to be a linguistic category *sui generis* (cf. Haspelmath 2015: 285)? Note that this is exactly the opposite situation as compared to Bermúdez-Otero & Payne’s (2011) conclusion in the previous Section 3.11.3. While they attempt to disprove the concept of special clitic as an intermediate category between word and affix, Haspelmath (2015) points out that the number of categories could increase in principle, if adopted by linguists. Accordingly, the affix/clitic distinction is uncertain, and linguists cannot take it for granted.

Haspelmath (2015: 287–289) adds an interesting discussion about the emergence and spread of the linguistic category of the clitic. He argues that most categories common in contemporary linguistics are artefacts of the historical course of events and, thus, accidental rather than the product of purposeful scientific research. Categories arise as labels for observations. If a label is found useful by a third party, the label may be adopted and eventually spreads. Haspelmath (2015: 288–289) portrays the career of the notion of *enclitic*, which was coined by Ancient Greek grammarians and later adopted by Latin grammarians. In modern times, the term *proclitic* appeared referring to elements like French stressless until the unprefix term *clitic* arose within American Structuralist linguistics. It was, however, not before the 1970s that the notion started its triumph in linguistics, which generative syntactic theory has been mainly responsible for. This is exactly when Zwicky’s (1977) seminal paper appeared, which “was found insightful by many linguists” (Haspelmath 2015: 288).

I would like to highlight an interesting argument of Haspelmath’s, according to which linguistic diagnosis of putative categories is subjective to a large degree (cf. Haspelmath 2015: 279). A phenomenon of particular interest is so-called *o*-raising in Polish, i.e. the alternation of [ɔ] <o> and [u] <ó> within a lexeme’s paradigm. The phenomenon has been used, among others, to assess affix vs. clitic status of Polish auxiliary clitics. The point is that *o*-raising is a word-level phonological process (cf. Franks 2010: 126–127; Franks & King 2000: 143; Witkoś 1998: 9–10; with references). It applies to the mid-open back rounded vowel in word-final syllables before an underlyingly voiced

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<sup>188</sup> Haspelmath’s (2015: 284) data show the clitic *ko* ‘PERFECTIVE’ in post-verbal or second position.

non-nasal consonant. A formal description is provided in (3.75) (cf. Kenstowicz 1994: 77 [cited after Buckley 2001: 2]) and a prototypical example in (3.76).

(3.75)  $\text{ɔ} \rightarrow \text{u} / \text{---} \text{C}_{\left[ \begin{smallmatrix} +\text{voice} \\ -\text{nasal} \end{smallmatrix} \right]} \#$

(3.76) wód ‘water.GEN.PL’                      woda ‘water.NOM.SG’                      (Pol; Gussmann 2007: 261)

Citing (3.77) from Booij & Rubach (1987), Franks & King (2000: 142–143) point out that the attachment of the 1SG auxiliary clitic *-m* to the masculine past tense form prevents *o*-raising. In contrast, the conditional clitic *by* in (3.78) does not affect vowel raising, which applies “as expected if *by* is a clitic” (Franks & King 2000: 148, n. 52). The same observation holds for the pronominal and reflexive clitics in (3.79). Consequently, the 1SG auxiliary clitic behaves as an inflectional affix rather than a clitic.

(3.77) a. mógł  
          can.PST.SG.M  
          ‘he could’  
      b. mógł-a  
          can.PST.SG-F  
          ‘she could’  
      c. mógł-e-m  
          can.PST.SG.M-M-1SG  
      d. \*mógł-e-m  
          ‘I could’                                              (Pol; Franks & King 2000: 143)

(3.78) O której                      mógł=by=ś                      dziś      odebrać      dziecko?  
      at which.LOC.SG.F can.PST.SG.M=COND=AUX.2SG today pick\_up.INF child.ACC.SG  
      ‘At what time could you pick up the kid today?’                      (Pol; Franks & King 2000: 148)

(3.79) a. [...] tylko nie      mógł                      mi                      sprzedać [...]  
          only NEG can.PST.SG.M me.DAT sell.INF  
          ‘... but he couldn’t sell [it] to me ...’

- b. a Rafał nie mógł go zaprosić do siebie[?]  
 and Rafał.NOM.SG NEG can.PST.SG.M him.ACC invite.INF to REFL.GEN  
 ‘But couldn’t Rafał (have) invite(d) him to his place?’
- c. [...] bo nie mógł się do mnie dodzwonić [...] [...]  
 because NEG can.PST.SG.M REFL to me.GEN reach\_by\_phone.INF  
 ‘... because he couldn’t reach me by phone ...’ (Pol, spoken; NKJP pelcra)

The analysis depends on the strict application of a synchronic phonological rule. The formal statement in (3.75) is imprecise and fails to capture (3.77a) with the word-final consonant cluster [gw] <gł>. This could still be accounted for by a more precise rephrasing of the statement. However, Baranowski & Buckley (2003), Buckley (2001), and Gussmann (2007: 261–269) note that there are many exceptions to the rule. I cite a few illustrative examples in (3.80)–(3.84). The examples in (3.80) comply with the description of *o*-raising (abstracting away from the need to account for the complex coda <źdź> /zdź/ in [3.80a]). In (3.81), *o*-raising does not apply despite showing the relevant environment. Example (3.82), in turn, exhibits *o*-raising, although it should be blocked. The diminutives in (3.83) differ with respect to *o*-raising.<sup>189</sup> While there is no raising in (3.83b), the raised vowel appears in non-final syllable in (3.83a). Furthermore, the lexical borrowings in (3.84), too, show divergent results as to rule (3.75).

- |        |                           |                         |                                 |
|--------|---------------------------|-------------------------|---------------------------------|
| (3.80) | a. wóźdź ‘nail.NOM.SG’    | gwoździe ‘nail.NOM.PL’  | (Pol; Gussmann 2007: 262)       |
|        | b. dom ‘house.NOM.SG’     | domy ‘house.NOM.PL’     | (Pol; Gussmann 2007: 265)       |
|        | c. kot ‘cat.NOM.SG’       | kota ‘cat.GEN.SG’       | (Pol; Buckley 2001: 1)          |
|        |                           |                         |                                 |
| (3.81) | a. węgorz ‘eel.NOM.SG’    | węgorza ‘eel.GEN.SG’    | (Pol; Gussmann 2007: 263)       |
|        | b. chodź ‘come.IMP.2SG’   | chodzić ‘go.INF’        | (Pol; after Gussmann 2007: 263) |
|        |                           |                         |                                 |
| (3.82) | a. stóp ‘foot.GEN.PL’     | stopa ‘foot.NOM.SG’     | (Pol; Gussmann 2007: 264)       |
|        | b. niósł ‘carry.PST.SG.M’ | niosła ‘carry.PST.SG.F’ | (Pol; after Gussmann 2007: 264) |
|        | c. sióstr ‘sister.GEN.PL’ | siostra ‘sister.NOM.SG’ | (Pol; Gussman 2007: 264)        |
|        |                           |                         |                                 |
| (3.83) | a. ogród ‘garden.NOM.SG’  | ogrodu ‘garden.GEN.SG’  | ogródek ‘garden.DIM.NOM.SG’     |
|        | b. miód ‘honey.NOM.SG’    | miodu ‘honey.GEN.SG’    | miodek ‘honey.DIM.NOM.SG’       |
|        |                           |                         | (Pol; Gussmann 2007: 267)       |

<sup>189</sup> An alternative diminutive form of *miód* ‘honey’ in (3.83b) is *miodzik*.

- (3.84) a. *mód* ‘fashion.GEN.PL’      *moda* ‘fashion.NOM.SG’  
 b. *konwój* ‘convoy.NOM.SG’ *konwoje* ‘convoy.NOM.PL’ (Pol; Buckley 2001: 5)  
 c. *gol* ‘goal.NOM.SG’      *gole* ‘goal.NOM/ACC.PL’ (Pol; after Buckley 2001: 5)  
 d. *kowboj* ‘cowboy.NOM.SG’ *kowboje* ‘cowboy.NOM.PL’ (Pol; after Gussmann 2007: 265)

Based on a quantitative examination of noun stems, Baranowski & Buckley (2003: 1) notice that there is a high degree of variation in the distribution of raised and unraised /o/ in the Polish lexicon. The picture emerges that we are dealing with an idiosyncrasy of individual lexemes rather than a productive rule of synchronic grammar (cf. Baranowski & Buckley 2003: 1; Buckley 2001: 4). Accordingly, one finds in Gussmann’s (2007) monograph that “we cannot dispense with lexical marking” (Gussmann 2007: 265) and “lexical marking is [...] inevitable” (Gussmann 2007: 266). If *o*-raising is not a phonological rule, then there is nothing the attachment of an auxiliary clitic could block in (3.77).<sup>190</sup> Consequently, the presence or absence of *o*-raising is inconclusive for the question of affix vs. clitic status and cannot serve as a diagnostic whatsoever.

In a recent paper, Haspelmath (2023) retains the term *clitic* attempting to define it on a typologically informed basis in terms of a comparative concept. While this procedure appears contradictory to Haspelmath’s earlier scepticism discussed above, it is nested in his current enterprise

<sup>190</sup> Beyond that, the argument grounded on example (3.77) hinges on certain assumptions about morphophonological structure. Note that the masculine past tense form *mógł* in (3.77a) is a monosyllable, whereas the feminine form *mogła* in (3.77b) is disyllabic and ends in an open syllable. Attachment of the nonsyllabic singular person markers *-m*, *-ś* provokes the realisation of the vowel *-e-* in (3.77c), which resembles an interfix in that it intervenes between two morphemes, whose concatenation would yield an illicit phonotactic string. As a result, the masculine verb form harmonises with the feminine one and becomes disyllabic. One could pursue, hence, the idea that it is the interfix vowel rather than the auxiliary clitic that affects the presence vs. absence of *o*-raising. It is difficult to test such hypotheses, because of the lack of relevant contrasts. For instance, plural person markers co-occur only with *l*-participles ending in open syllables, i.e. *o*-raising is necessarily absent in verb forms like *mogli* ‘can.PST.PL.VIR’, *mogły* ‘can.PST.PL.NVIR’. However, it is still possible to construct relevant examples with the caveat that they usually represent dialectal usage, such that they are not representative for the entire population of Polish speakers. Examples (i) and (ii) show plural auxiliary clitics following a lexeme, which surfaces with *o*-raising. Admittedly, there are alternatives to (i) and (ii), which are more likely, i.e. constructions with *że*-support like (iii) instead of plain auxiliary clitics. Such data points are indecisive as to the clitic-hood of *śmy* and *ście*, for the exact status of the concatenated structure with pleonastic *że* is unclear. Nevertheless, if we accept examples like (i)–(ii), then we have sufficient evidence for the clitic- rather than affix-status of the mobile person markers in Polish (or a closely-related Lechitic variety). Such arguments need to be empirically substantiated, i.e. by corpus data and/or experimental testing, which is not achieved here.

(i) Ogród      **śmy**      przygotowali      na zimę.  
 garden.ACC.SG AUX.1PL prepare.PST.PL.VIR on winter.ACC.SG  
 ‘We prepared the garden for winter.’

(ii) Grób      **ście**      odwiedziły?  
 grave.ACC.SC AUX.2PL visit.PST.PL.NVIR  
 ‘The grave, did you visit it?’

(iii) a. Ogród      **że=śmy**      przygotowali      na zimę.  
 garden.ACC.SG ŻE=AUX.1PL prepare.PST.PL.VIR on winter.ACC.SG  
 ‘We prepared the garden for winter.’

b. Grób      **że=ście**      odwiedziły?  
 grave.ACC.SC ŻE=AUX.2PL visit.PST.PL.NVIR  
 ‘The grave, did you visit it?’

(Pol, non-standard; constructed)

of “retro-definitions” aiming at rigorously defining well-established yet vaguely understood notions in linguistics like, e.g., “word” (cf. Haspelmath 2011 for discussion). According to his definition, “[a] clitic is a bound morph that is neither an affix nor a root.” (Haspelmath 2023: 2) He continues to define enclitics and proclitics as clitics which occupy a position at the end or beginning of a free form respectively but not vice versa (cf. Haspelmath 2023: 13). A free form may be larger than a lexeme (autosemantic) and correspond to phrase or clause. Correspondingly, “[a]n ambiclitic is a clitic that can occur at the end of a free form or at the beginning.” (Haspelmath 2023: 17) Several other criteria and phenomena are reviewed, like clitic clustering and word-class selectivity (= promiscuity). The main problem of Haspelmath’s approach is its broad granularity, which seems to be a consequence of the proposal’s global typological perspective. As a result, the conceptual framework does not neatly fit the Slavic facts discussed in the present thesis. First, the boundedness criterion at the definition’s core is undermined by clitics in Slovene (cf. Franks & King 2000: 43–44). Examples (3.85)–(3.87) show auxiliary, reflexive, and pronominal clitics being syntactically isolated. In fact, such constructions are problematic for any orthodox account of cliticisation, such that the Slovene data are peculiar in any framework currently available.

- (3.85) a. (Ali) **se**            dobro       počutiš?  
           Q    REFL.ACC good.ADV feel.PRS.2SG  
       b. Ja, **se**.  
           yes REFL.ACC (Slv; Franks & King 2000: 43)  
       c. **Se**.  
           REFL.ACC  
           ‘Yes, I do (feel well)’ (Slv; constructed after Franks & King 2000: 44)
- (3.86) a. Ali **si**            **si**            to            izmislil?  
           Q    AUX.2SG REFL.DAT this.ACC.SG.N think\_up.PST.SG.M  
       b. **Si**            **si**            izmislil?  
           AUX.2SG REFL.DAT think\_up.PST.SG.M  
       c. **Si**            **si**?  
           AUX.2SG REFL.DAT  
           ‘Did you think this up for yourself?’ (Slv; Franks & King 2000: 43)

- (3.87) a. Ali **mu**        **ga**        daješ?  
           Q   him.DAT him.ACC give.PRS.2SG  
       b. **Mu**        **ga**?  
           him.DAT him.ACC  
           ‘Are you giving it to him?’ (Slv; Franks & King 2000: 54)

Second, while Haspelmath (2023: 15) explicitly refers to Toman’s (1996) discussion of the prosodic neutrality of clitics in Czech, he does not consider the fact that Czech clitics are perfectly ambiclitic. The status of ambiclisism does not seem to be entirely clear. While Haspelmath (2023: 22, n. 9; 23, n. 10) repeatedly remarks that a clitic’s ability to encliticise at one time and to procliticise at another time renders it an ambiclitic, he asserts that “it is unclear how to treat ambiclitics in general, so it may be best to restrict the term *anchor* to enclitics and proclitics for simplicity.” (Haspelmath 2023: 18) Third, Haspelmath (2023: 18–21) presents second position cliticisation as a monolithic phenomenon when stating that clitics “must occur [...] directly after the first” eligible host be it a single word or a syntactic constituent (Haspelmath 2023: 19). The presupposed necessity (expressed by the modal verb *must*) misses facts of intra-linguistic and cross-linguistic variation, i.e. deviate positioning of clitics in so-called second position clitic languages.<sup>191</sup> Thus, we find a reproduction of the simplistic view of clitic phenomena which the present thesis explicitly attempts to argue against. Fourth, Haspelmath (2023: 29–30) contends that clitics are monomorphic, which he makes part of the definition of clitic-hood. Consider the Italian (Ita) clitics *le* ‘the.PL.F’ and *hanno* ‘AUX.PRF.3PL’ in (3.88). The clitic determiner and temporal auxiliary are decomposable into smaller units, which gives rise to Haspelmath’s (2023: 30) interpretation that we encounter actual clitic clusters.

- (3.88) **L-e**        donn-e        **ha-nno**        lavoro.  
           the.PL.F woman-PL AUX.PRF-3PL work.PTCP  
           ‘The women have worked.’ (Ita; Haspelmath 2023: 29)

A monomorphic treatment does not work for clitics in Slavic, which are portmanteau morphs straight from central casting, as argued in Section 3.6. Pronominal clitics like, e.g., Cz./Slk./Pol. *mu* and *ho/go* code person, number, and gender simultaneously but cannot be analysed into smaller morphological components. In sum, there is a number of inconsistencies in the comparative concept with respect to Slavic, which is particularly problematic provided that Haspelmath (2023) explicitly incorporates

<sup>191</sup> See Section 4.4 and Chapter 10 for relevant discussion, data, and references.

Slavic data points in this approach.<sup>192</sup> On the other hand, it provides us with the corroboration for a sociolinguistic or variationist approach that employs comprehensive empirical databases in researching individual languages.

### 3.13 Towards a workable notion of clitic-hood

The previous sections reviewed several conceptions of clitic-hood as well as multiple test conditions for determining an element's status as clitic rather than affix or word. As I have attempted to demonstrate, most criteria are inapplicable, e.g. Zwicky's (1985) Deletion test, or inconclusive, e.g. Hana's (2007) conditions [1P-Cl] and [Cl-Cl]. Similarly, major theoretical approaches to the notion of clitic turn out to face problems, once more data is taken into account. For instance, Klavans's (1995) parametric model of clitic distribution cannot cope with clitics in Slavic, while Cardinaletti & Starke's (1999a) typology is grounded on incorrect observations. Haspelmath's (2023) comparative concept of clitic-hood is grounded on a global typological perspective but lacks compatibility with the phenomenology in Slavic. These and similar arguments run like a common thread throughout scholarly debates on the topic of cliticisation. For one thing, the Prosodic Inversion account proposed by Halpern (1995) was meticulously criticised by Bošković (2001: 11–36), whose own proposal of distributing clitics in intonational phrases faces challenging arguments from Golden (2008), Lenertová (2004: 150–151), and Marušić (2008). Perhaps, it should not come as a surprise that some fundamental critique of clitic-hood has been pronounced. While Klavans's (1995) and Bermúdez-Otero & Payne (2011) basically agree in abandoning the concept of special clitics, the latter are united with Haspelmath (2015) in essentially questioning the very notion of clitic-hood in the first place.

I restricted the scope of the thesis to constant clitics in Section 2.8, but this was merely a delimitation of the object area. In the following, I will attempt to give a more principled account of the notion *clitic*. It may be regarded as an *ex post facto* definition of constant clitics. Since not only diverse classifications of clitics have been subject to criticism, but the very notion of clitic-hood itself, it should preferably be omitted as a basis for definitions. Therefore, I will draw upon the established notion of the *pronominal short form*, *short pronouns*, or *short pronominals* from traditional Slavic studies (cf. e.g. Janda & Townsend 2000: 26; Swan 2002: 155). I propose the definition in (3.89).

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<sup>192</sup> Alternatively, one could use Haspelmath's (2023) definition as a statistical (e.g. prototype-related) rather than an absolute concept or conclude that the linguistic elements examined in the present thesis are not clitics in the sense of the definition. But that is not what is originally intended (cf. Haspelmath 2023: 48). I will pursue neither alternative but continue to follow common opinion.

(3.89) *Definition: Constant clitic*

The set  $C_{CL}$  of constant clitics comprises those forms of a pronominal or reflexive dublet (pair) in a language  $L$ , which are morpho-phonologically smaller:

$$C_{CL} := \forall x \exists y (x < y \wedge x, y \in P \vee R), \text{ with } P = \{\text{'pronoun'}\} \text{ and } R = \{\text{'reflexive'}\}.$$

The formula in (3.89) reads as: The set  $C_{CL}$  is defined such that for every  $x$  there is a  $y$ , such that  $x$  is smaller than  $y$ , whereby  $x$  and  $y$  are elements of the pronominal set  $P$  or the reflexive set  $R$ . The definition is not perfect. It does not specify, how the relation between the constant clitic  $x$  and the corresponding orthotonic form  $y$  is established. Provided that the relation between  $x$  and  $y$  is formalisable under exclusion of the [n]-initial allomorph, I shall ignore this issue for now. The definition is sufficiently stated to operationalise the notion *clitic* for the purposes of the present thesis.

Note that the definition is language-specific. It is therefore fully compatible with Bermúdez-Otero & Payne's (2011) and Haspelmath's (2015) suggestions and conclusions. All notions included in the definition are retrievable from language-particular content, i.e. from the language's lexicon (perhaps, restated in set-theoretical terms). Thus, the sets  $P$  and  $R$  are perfectly determinable for a given Slavic language. The relation between elements of the doublets  $\{x, y\}$  is established on language-particular grounds, too. The definition is in principle applicable across the entire branch of Slavic, since most Slavic languages retained short and long pronouns. For any West or South Slavic language, we can find a set of elements which, first, can be assigned to a closed class of either 'pronouns' or 'reflexives', and, second, double with one form being morpho-phonologically smaller than the other one. Finally, the [n]-initial allomorph can be simply excluded by set-theoretical notation or by a generalisation with recourse to the context-sensitive alternation between the concerned long forms. The underlying notion of length is purely formal and can be measured in various ways, e.g. number of phonemic segments or, perhaps, average duration in speech production.<sup>193</sup>

One might wonder whether the utilisation of the traditionally employed notions of *short* vs. *long pronoun* is appropriate. The language-specific approach turns out to be an advantage here. Short and long elements belong to the pronominal inventory of the respective languages and are in systemic opposition. It is therefore necessary to contrast, e.g., short *mu* with long *jemu* (West Slavic) and *n(j)emu* (South Slavic). As any adequate description must acknowledge this fact, it is legitimate to employ it here. The form-based definition of constant clitics allows referring to a closed set of linguistic entities that bear various grammatical, referential, and pragmatic functions. It does not only

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<sup>193</sup> The number of syllables is not fully applicable, because the 1 SG.DAT pronominal forms both are monosyllabic in West Slavic. Compare the dative pronominal clitic *mi* with the corresponding long form, i.e. Czech *mně*, Polish *mnie*, Slovak *mne*.



cover pronouns with clear argument status, but also pragmatically motivated occurrences like the ethic dative. The reflexive covers formally identical elements with a wide range of functions, not all of which are manifest in short and long forms simultaneously. De-causative markers, impersonal markers, and lexical pleonasms of reflexiva tantum consistently realise the short form only and cannot take the long form (cf. Kupść 2000b: 88–90).

While the usage of the term *constant clitic* explicitly refers to (3.89), I will continue to use established vocabulary throughout the thesis. Therefore, the reader will find occurrences of *clitic*, *special clitic*, *short pronoun*, *orthotonic pronoun* etc. More generally, I do not abandon the notion of *clitic* for communicative reasons and continue to use it primarily for the corresponding set of auxiliary, conditional, pronominal, and reflexive forms (and *-li*).

## 4 Clitics in Slavic: Relevant phenomena and recurring problems

### 4.1 Introductory remarks

Despite the thesis's focus on West Slavic, clitic (or pronominal) phenomena in other branches have also been addressed already, e.g. East Slavic in Section 3.10. Chapter 4 is particularly broad in this respect and takes South Slavic languages into consideration as well. Consequently, the discussion of pertinent issues in Slavic clitic research will grow comparatively extensive. This procedure is, however, necessary to provide the reader with the context from which the thesis arose. Usually, generalisations about clitics in Slavic are drawn from a single or few individual languages. On the other hand, comparative studies both within and beyond the Slavic language family play an important role in Slavicist clitic research. However, certain data points have been neglected, marginalised, or simply missed. These discrepancies ultimately motivate the present study's empirical methodology.<sup>194</sup> In the following, I will attempt to provide a state-of-the-art report covering a range of clitic phenomena in Slavic, but I make no claim to completeness. The range of selected phenomena basically follows the presentation in Franks & King's (2000) seminal monograph, which is unsurpassed to date in breadth and coverage.<sup>195</sup> However, Bulgarian and Macedonian will be paid very little attention to, as I am focusing on second position cliticisation. Furthermore, the discussion will be restricted to a purely synchronic perspective.<sup>196</sup> Exemplary data is drawn primarily, but not exclusively, from research literature on Slavic clitics.<sup>197</sup> I will not review traditional grammaticography for practical reasons, i.e. due to time and space limitations as well as the lack of grammaticographic references for Goralish and Silesian. However, individual grammaticographic

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<sup>194</sup> The study's methodology and related issues will be presented in Part II of the thesis.

<sup>195</sup> This is why I do not discuss clitics in resumptive relative constructions, although such constructions have received considerable attention in recent years. Resumptive clitics are generally neglected in most research particularly dedicated to Slavic cliticisation. Rubadeau (1996: 130–131) is one of few notable exceptions I am aware of, and she provides examples for Common Czech plus a footnote example for BCMS (cf. Rubadeau 1996: 130, n. 15). For Slavic resumptive constructions, see Bondaruk (1995), Szczegielniak (2005) for Polish, Toman (1998) for Czech, Bošković (2007) for BCMS, and Krapova (2010) for Bulgarian. For comparative perspectives across multiple Slavic languages, see Chidambaram (2013) for BCMS, Slovene, Macedonian, and Slovak; Hladnik (2015) for Slovene, BCMS, and Polish (and to a minor extent Bulgarian and Russian). Klein (2017) takes a typologically broad data set into consideration involving 19 languages from 12 different language families, among which there are four Slavic languages, i.e. BCMS, Czech, Polish, and Ukrainian (the latter not possessing clitics in its standard variety).

<sup>196</sup> For diachronic perspectives on clitics in Slavic, cf. Pancheva (2005), Dimitrova (2021, 2023) for Old Bulgarian, Večerka (1985) for Old Church Slavonic, Zaliznjak (2008) for Old Novgorodian, Kosek (2009, 2011, 2015, 2017) for Old and Middle Czech. For a comparative view, cf. Gribble (1988) on Old Bulgarian and Old Russian. Migdalski (2006: 164–170, 2009a, 2009b, 2013, 2014, 2018) provides diachronic analyses within the framework of the Minimalist Program.

<sup>197</sup> Empirical attestations will be provided with their source, which usually is either an established electronic corpus or an actual utterance from a website. Corpora and sources are exhaustively listed in the references section.

references may occasionally occur, if they add interesting perspectives for the issue under discussion or relevant data points.

The majority of research output on Slavic clitics is nested within a formal linguistic theory. There is a wide range of linguistic theories and formalisms, sharing the common goal of describing and explaining linguistic structures and phenomena precisely and unambiguously. Most proponents of formal linguistic theory draw on Chomsky's (1957) early work on syntax (cf. Müller 2023: 6; Pollard & Sag 1994: 6–14; Przepiórkowski & Kupść 2006: 1; Rosen 2001: 17–22):<sup>198</sup>

“Precisely constructed models for linguistic structure can play an important role, both negative and positive, in the process of discovery itself. By pushing a precise but inadequate formulation to an unacceptable conclusion, we can often expose the exact source of this inadequacy and, consequently, gain a deeper understanding of the linguistic data. More positively, a formalized theory may automatically provide solutions for many problems other than those for which it was explicitly designed. Obscure and intuition-bound notions can neither lead to absurd conclusions nor provide new and correct ones, and hence they fail to be useful in two important respects.” (Chomsky 1957: 5)

However, Rosen (2001: 21) points out that “[a]lthough a formal language is usually preferable, plain natural language can also serve the purpose as well, provided that the statements can be interpreted only as intended.” (Cf. also Przepiórkowski & Kupść 2006: 1). As a legacy of Chomsky's work, formal linguistics is primarily associated with Generative Grammar. Indeed, Generative Grammar has been the most productive paradigm in Slavic clitic research beginning with Browne's (1968, 1974) analysis of BCMS clitics.<sup>199</sup> The reader will therefore notice the prevalence of generative publications, especially generative syntax, among the cited literature. Although I cannot do justice to all of them, I will also take work into account that is either nested in alternative theoretical frameworks or that belongs to purely empirical enterprises.

Chapter 4 is structured as follows: Section 4.2 provides notes on prosody. Section 4.3 discusses different second-position-placement patterns. The following section 4.4 deals with deviations from second-position placement, i.e. first-position placement and so-called Delayed Placement. Section 4.5 focuses on clitic cluster formation and Section 4.6 on the word order within the cluster. Section 4.7 touches upon clitic climbing. The following two sections are concerned with diverse restrictions clitics obey: Section 4.8 briefly looks into the constraints on rightwards clitic

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<sup>198</sup> Also consider Zwicky's (1977: 1) reference to Generative Grammar: “striving for precision and generalization in linguistic descriptions, has, in fact, led to the uncovering of a host of analytic problems”. Furthermore cf. Zwicky's foreword to Klavans's (1995) formal analyses of cliticisation: “Further advances in understanding the nature of clitics had to wait upon the development of formal grammatical frameworks, especially for syntax; a certain level of explicitness is necessary for problems even to be seen and articulated.” (Klavans 1995: xi)

<sup>199</sup> For the sake of completeness, note that at the same time, Koenitz & Walter (1968) independently propose a generative analysis of clitics with particular focus on BCMS, Bulgarian, Czech, and Macedonian.

placement. Section 4.9 discusses some co-occurrence restrictions that have been observed for clitics. Section 4.10 gives a quick discussion of clitic doubling. Section 4.11 is devoted to clitics beneath the sentence-level, particularly in the noun phrase. Section 4.12 offers an interim conclusion, which will segue into Part II of the thesis.

## 4.2 Notes on prosody

### 4.2.1 Prosodic orientation

Despite the thesis's essentially syntactic perspective, some comments on prosody are in order for the sake of completeness (and because we will encounter instances of a mismatch between theoretical claims and empirical observations).<sup>200</sup> Turning to prosodic orientation first, auxiliary and pronominal clitics in BCMS, Czech, Polish, Slovak, and Slovene have been traditionally regarded as enclitics. That is, they seek prosodic support to their left, such that they can never appear in initial position of the sentence or intonational phrase. Toman (1996) points out that the idea of inherent enclisis is misleading and does not reflect the actual phonological properties of clitics. He compellingly argues that Czech auxiliary and pronominal clitics are prosodically neutral being capable of enclisis as well as proclisis (cf. also Avgustinova & Oliva 1997: 32; Fried 1994: 167–168; Hana 2007: 68–71; Junghanns 2021: 178–179; Lenertová 2001: 295–296, 2004: 148; Rosen 2001: 207). I will adopt the term AMBICLITIC, which recently has appeared in the linguistic literature, for this property.<sup>201</sup> The actual phonological orientation of an ambiclitic is regarded as a function of the context. Enclisis is the default value, whereas proclisis occurs under certain conditions. Czech clitics can occupy a position after a prosodic break due to parenthetical material like relative clauses, such that they do not have prosodic support to their left. In this case, they phonologically attach to an eligible host to their right (4.1). The same holds for Polish (4.2), Slovak (4.3), and Slovene (4.4) (cf. Kraska-Szlenk 1995: 62–63; Kupść 2000b: 38; Rappaport 1988: 321 for Polish; Franks & King 2000: 133–134;

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<sup>200</sup> See Section 6.1 for a discussion of the tension between introspective data construction and empirical data elicitation.

<sup>201</sup> The origin of the term is unclear to me, but it seems to be common in recent typological literature (cf. Aikhenvald 2020: 165, 167–168; Aikhenvald et al. 2020: 13; Haspelmath 2023: 13, 17; Stump 2017: 26). The earliest mention I was able to trace is an abstract by Fabiola Henri & Jo Mackby (misspelled as *Machby*), both University of Kentucky on “Conditioned reordering in Louisiana Creole: the case of sentential negation” in the Meeting Handbook of the 90th Annual Meeting of the Linguistic Society of America in Washington, DC 7–10 January 2016 (p. 184). (<https://www.linguisticsociety.org/sites/default/files/LSA%202016%20Annual%20Meeting%20Handbook.pdf>, accessed 6 Nov 2023.) Perhaps, the conception of *ambifix* as an “an affix that can appear on either side” – cited by Hall (2000: 536) in reference to Malkiel (1978: 145) – could be regarded as a potential predecessor.

Ivanová et al. 2021: 355 for Slovak; Bennett 1986: 8–9, 1987: 272; Franks & King 2000: 41, 43; Golden 2003: 208, 214, 223 for Slovene).<sup>202</sup>

- (4.1) a. [NP Kniky, [s které tady vidíte ]], # **se=** dnes platí  
 book.NOM.PL which.ACC.PL.F here see.PRS.2PL REFL today pay.PRS.3SG  
 zlatem.  
 gold.INS.SG  
 ‘The books you can see here are paid for with gold today.’
- b. [NP Ta veverka, [s kterou krmíte ]], # **by=** vás  
 this.F.SG squirrel.NOM.SG which.ACC.SG.F feed.PRS.2PL COND you.ACC.PL  
 mohla kousnout.  
 can.PST.SG.F bite.INF  
 ‘The squirrel you are feeding might bite you.’ (Cz; Toman 1996: 506)
- (4.2) [NP Mój kolega, [s który tam był ]], #  
 my.NOM.SG.M colleague.NOM.SG who.NOM.SG.M there be.PST.SG.M  
**go=** odwiedził.  
 him.ACC visit.PST.SG.M  
 ‘My colleague, who was there, visited him.’ (Pol; Kupść 2000b: 36)
- (4.3) a. Samozrejme aj to, čo je na tanieri, **ma=** inšpiruje.  
 of\_course and this what be.PRS.3SG on plate me.ACC inspire.PRS.3SG  
 ‘Of course, everything on the plate inspires me, too.’ (Slk; Ivanová et al. 2021: 355)
- b. Knižky, ktoré tu vidíte, **sa=** dnes zaplatia  
 book.NOM.PL which.ACC.PL here see.PRS.2PL REFL.ACC today pay.PRS.PL  
 zlatom.  
 gold.INS.SG  
 ‘The books which you see here are now worth a fortune.’  
 (Slk; Franks & King 2000: 134)

<sup>202</sup> Similarly, Macedonian and Bulgarian clitics are ambiclitics (cf. Franks 2016: 94 explicitly on Macedonian, Franks & King 2000: Ch. 2.3 and Ch. 2.4 for various data points).

- (4.4) Kdo – za božjo voljo – ti= je= razbil  
 who.NOM for God.ADJ.ACC.SG.F will.ACC.SG you.DAT AUX.3SG ruin.PST.3SG.M  
 avto?  
 car.ACC.SG  
 ‘Who, for God’s sake, ruined your car?’ (Slv; Golden 2003: 215)

Interestingly, Avgustinova & Oliva (1997: 32–34) argue that enclisis cannot be regarded as the default value either, as proclisis can be observed in constructions without a prosodic break.<sup>203</sup> They say that Czech is undergoing a language change, according to which clitics tend to prosodically attach to the verb, if positioned contiguously (cf. Avgustinova & Oliva 1997: 33–34). As a result, example (4.5a) shows enclisis with the verb in clause-initial position, whereas proclisis comes with the clause-final verb following the clitic cluster in (4.5b).<sup>204</sup>

- (4.5) a. [v Koupil ] =jsem =mu =ho včera v Praze.  
 buy.PST.SG.M AUX.1SG him.DAT him.ACC yesterday in Prague.LOC  
 ‘I bought it for him in Prague yesterday.’  
 b. V Praze jsem= mu= ho= [v koupil ].  
 in Prague.LOC AUX.1SG him.DAT him.ACC buy.PST.SG.M  
 ‘In Prague, I bought it for him.’ (Cz; Avgustinova & Oliva 1997: 34)

In contrast, BCMS clitics have been repeatedly reported to be strictly enclitic, not capable of attaching to material following them (cf. Bošković 2001; Franks & King 2000; among others). If a prosodic boundary occurs after the first constituent, BCMS clitics must shift rightwards to encliticise to an

<sup>203</sup> Avgustinova & Oliva (1997) use the notion (*prosodically*) *autonomous clitics* for ambiclitics.

<sup>204</sup> Kraska-Szlenk (1995: 151) reports a similar situation for Polish, when discussing the diverse factors for a clitic’s prosodic orientation. For instance, she states that clitics preferably lean on what is “syntactically closer”, which is the verbal predicate for object clitics. On the other hand, clitics disprefer to attach to focused constituents. Thus, the object clitic *go* encliticises to the dative object *Marii* on its left in (i), because the verb receives default Focus in clause-final position, whereas it procliticises to the verb on its right in (ii), because the sentence-final adverb bears Focus.

(i) Janek Marii =go [ podarował ]<sub>FOC</sub>.  
 John.NOM Maria.DAT him.ACC give\_as\_present.PST.SG.M  
 ‘John gave it to Mary as a present.’  
 (ii) Janek Marii go= podarował [ wczoraj ]<sub>FOC</sub>.  
 John.NOM Maria.DAT him.ACC give\_as\_present.PST.SG.M yesterday  
 ‘John gave it to Mary as a present yesterday.’

(Pol; Kraska-Szlenk 1995: 151)

Admittedly, Avgustinova & Oliva (1997) do not claim that cliticisation to focused units is banned. In fact, one could hypothesise that it is precisely because of the verb’s information-structure-driven prosodic prominence, why the clitic in Czech (4.5) encliticises at one time but procliticises at another time. The verb receives default Focus in (4.5b), whereas example (4.5a) exhibits sentence-initial verb placement, which usually comes about with wide Focus reading. This seems to conflict with Kraska-Szlenk’s (1995) statement, but I contend that this statement is incorrect. Contrary to Kraska-Szlenk (1995: 151), I argue that the clitic *go* does lean on the following verb in (i) – or it is at least capable of doing so. Furthermore, her claim conflicts with the observation that clitics can be dislocated as a part of constituents that bear the discourse functions of either Topic or Focus.

eligible host.<sup>205</sup> However, strict enclisis applies only to the BCMS standard languages, which are built upon a Neo-Štokavian dialect basis. In contrast, Kajkavian (4.6) and Čakavian (4.7) possess ambiclitics (cf. Lončarić 2002 for Kajkavian; Nehring 2002 for Čakavian). Blažeka (2019: 129) asserts that Kajkavian clitics “may arbitrarily go before or after the tonic word.” Similarly, ambiclitics are attested in Burgenland Croatian (4.8), a minority language in present-day eastern Austria (cf. Neweklowsky 2012).

- (4.6) a. **poznaš** =**ga**  
 know.PRS.2SG him.ACC  
 b. **ga**= **poznaš**  
 him.ACC know.PRS.2SG  
 ‘you know him’  
 c. **poznao** =**je**  
 know.PST.SG.M AUX.3SG  
 d. **je**= **poznao**  
 AUX.3SG know.PST.SG.M  
 ‘he knew’

(Kaj; Lončarić 2002: 261)

- (4.7) a. **Si**= **bio** **u** **gradu**?  
 AUX.2SG be.PST.SG.M in city.LOC  
 ‘Were you in the city?’  
 b. **Ćeš**= **ići** **u** **Kastav**?  
 AUX.FUT.2SG go.PFV.INF in Kastav.ACC  
 ‘Will you go to Kastav?’

(Ča; Nehring 2002: 249)

- (4.8) a. **se**= **u** **dvoru** **pleše**  
 REFL on yard.LOC.SG dance.PRS.3SG  
 ‘there is dancing in the courtyard’ [lit.: ‘one is dancing in the courtyard’]  
 b. **me**= **koliko** **veseli**  
 me.ACC how delight.PRS.3SG  
 ‘I am very happy’ [lit.: ‘how [much it] pleases me’]

<sup>205</sup> See Section 4.2.3 for prosodic boundaries and Section 4.4.2 for delayed clitic placement.

- c. **ćemu=** **ju=** pitati  
 AUX.FUT.1PL her.ACC ask.INF  
 ‘we will ask her’ (BCr; Neweklowsky 2012: 269)
- d. [<sub>IP</sub> Ako imam svileni rubac, ] [<sub>IP</sub> **ga=** morem  
 if have.PRS.1SG silken scarf.ACC him.ACC can.PRS.1SG  
 motati okolo vrata ] [<sub>IP</sub> i= **ga=** sobomzeti ].  
 wind\_up.INF around neck.GEN and him.ACC take\_with\_me.INF  
 ‘If I have a silk scarf, I can wind it around my neck and take it with me.’  
 (BCr; Browne 2010: 27)

It is important not to marginalise these proclitic occurrences as “dialectalisms”. Kolaković et al. (2022) provide data for spoken Bosnian (Bos) with proclisis. In (4.9a), the reflexive clitic *se* follows the prosodic break after a relative clause, while (4.9.b) witnesses a procliticising cluster of pronominal *me* and auxiliary *je* after an appositive. By the same token, Milićević (2023: 108–110) informs that proclisis is available in the Serbian journalistic register and in colloquial speech. The clitic copula form *je* in (4.10) immediately follows the prosodic break after a “heavy” constituent and procliticises to the following lexeme *strach* ‘fear’. This is especially a property of the journalistic style according to Milićević (2023: 108), but also available in the standard language to some degree (cf. Milićević 2023: 70, 109).

- (4.9) a. [...] niko [s ko god dobije tu vojnu  
 nobody who ever get.PRS.3SG that.ACC.SG.F military.ACC.SG  
 obavezu ] **se=** ne treba javljat [...].  
 obligation.ACC.SG REFL.ACC NEG need.PRS.3SG apply.INF  
 ‘... nobody who gets invited to military service ever has to apply ...’
- b. Jedan drug, Musliman, **me= je=** zvao [...].  
 one.NOM.SG.M friend.NOM.SG muslim.NOM.SG me.ACC AUX.3SG call.PST.SG.M  
 ‘One friend, a Muslim, called me ...’ (Bos; Kolaković et al. 2022: 211)
- (4.10) [<sub>IP</sub> Jedna od najčešćih fobija ] [<sub>IP</sub> **je=** strach  
 one of most\_frequent.GEN.PL phobia.GEN.PL be.3SG.CL fear.NOM.SG  
 od mraka ].  
 from dark.GEN.SG  
 ‘One of the most frequent phobias is the fear of the dark.’  
 (Srp, non-standard; Milićević 2023: 109)



Finally, while the possibility of proclisis in BCMS has been noticed by formal linguists, it has received no further consideration. Bošković (2001: 164–165) acknowledges the existence of proclisis, but states that “[c]ircumstances under which this kind of construction are possible are still mysterious.” Among others, he cites examples (4.11) from Bennett (1987: 276). The hash mark (#) indicates a pause. Franks (2016: 99, n. 17), too, reports in a footnote that intonational-phrase-initial clitics are accepted by some native speakers of Croatian.

- (4.11) [NP Problemi [s o kojima ćemo razgovarati]] #  
 problem.NOM.PL about which.LOC.PL AUX.FUT.1PL converse.INF  
 su= kompleksni.  
 be.PRS.3PL.CL complex.NOM.PL  
 ‘The problems which we shall discuss are complex.’  
 (BCMS; Bennett 1987: 276 [cited after Bošković 2001: 164])

The general picture emerges that all Slavic 2P languages essentially allow enclisis as well as proclisis for the lexemes relevant here. Ambiclisism has repercussions for clitic placement, because it may be regarded as the prerequisite for sentence- or intonation-phrase-initial positioning as in Čakavian (4.7) and Burgenland Croatian (4.8).<sup>206</sup>

## 4.2.2 Clitics and stress

Clitics are considered prosodically deficient in lacking word level stress. However, it has been repeatedly noted that they can bear stress under certain circumstances (cf. Anderson 2005: 24; Cardinaletti & Starke 1999a: 162–163; Klavans 1995: 129–158; Rubadeau 1996: 3; Zwicky 1977: 14). Accordingly, stressed clitics have been reported for Slavic. Thorpe (1991: 4, n. 1) informs in her study of Czech clitics that “if a chain of enclitics occurs, the chain is considered a phonological unit whose first member carries stress.” Taking this remark at face value, it means that the clitic cluster in Czech bears a stress on the first clitic. Junghanns (2002a: 137) mentions clitics bearing secondary and emphatic stress among others. Palková (2017) informs that sequences of two or more monosyllabic words create their own *mluvní takt*, whereby the *takt*’s first element may bear stress.<sup>207</sup> There are clitics among these monosyllables, such that one encounters stressed clitics in (4.12).

<sup>206</sup> See Section 4.4.1 for sentence-initial clitics.

<sup>207</sup> The Czech term *mluvní takt* ‘phrase’ represents a more traditional terminology. It does not have a direct equivalent in contemporary Anglophone phonology like *foot*, *prosodic word*, or *intonation phrase*, but it is closest to the former two.

- (4.12) a. ZÍtra / SE tam / POdíváme.  
tomorrow REFL.ACC there look.PFV.PRS.1PL  
‘We will meet there tomorrow.’
- b. OBvykle / SE na / ZÁvody / TĚšili.  
usually REFL on competition.ACC.PL look\_forward.PST.PL.VIR  
‘Usually, they were looking forward to the contests.’
- c. PRAVděpodobně / SE na / KOUpališti / Uvidíme.  
probably REFL.ACC on swimming\_pool see.PFV.PRS.1PL  
‘Presumably, I’ll see you at the swimming pool.’
- d. Ale / CHTĚli / MU to / ODpustit.  
but want.PST.PL.VIR him.DAT this.ACC forgive.INF  
‘But they wanted to forgive him for that.’ (Cz; Palková 2017)

Kraska-Szlenk (1995: 81–96) argues that Polish clitics receive secondary and tertiary stress, when clustering together. The examples in (4.13b–c) have the foot structure in (4.13a), whereby each number represents a certain degree of prosodic prominence of a nucleus with 0 = no stress, 1 = primary stress, 2 = secondary stress, and 3 = tertiary stress. The finite verbs bear primary stress, whereas the second-to-last clitic is the locus of secondary stress. The actual stress pattern is not dependent upon the clitics themselves. On the one hand, *mu* ‘him.DAT’ bears secondary stress in (4.13b), but is unstressed in (4.13c). On the other hand, *-ście* ‘AUX.2PL’ is unaccented in (4.13b), but receives secondary stress in (4.13c). Any clitic may be stressed independently of its category, if it occupies the head-position of a prosodic foot (cf. Kraska-Szlenk 1995: 82).

- (4.13) a. 0 (1 0)<sub>F</sub> (3 0)<sub>F</sub> (2 0)<sub>F</sub> (Kraska-Szlenk 1995: 82)
- b. kuPIli=**BY**=**ście** =**MU** =**go**  
buy.PST.PL.VIR=COND=2PL him.DAT him.ACC  
‘you<sub>PL</sub> would buy it for him’
- c. kuPIli=**ŚCIE** =**mu** =**GO** =**tam**  
buy.PST.PL.VIR=2PL him.DAT him.ACC there  
‘you<sub>PL</sub> bought it for him there’ (Pol; Kraska-Szlenk 1995: 82)

It is generally assumed that patterns like those in (4.12)–(4.13) do not provide evidence that clitics possess lexical accent. Rather they are thought to receive stress on the surface, while being parsed into feet (cf. Anderson 2005: 24, 43–44; Franks 2016: 95; Kraska-Szlenk 1995: 82). Stressed clitics are thus an epiphenomenon of prosodic phonology. However, there are situations, when a clitic

receives stress, but an approach in terms of foot parsing is not available. Rosen (2001: 207) cites the example in (4.14) with a conditional clitic surrounded by intonational boundaries on either side, such that it is prosodically isolated. He argues that the criteria of being hosted by a prosodically strong element and the prohibition of prosodic isolation cannot be treated as mandatory classificatory criteria. Hana (2007: 71), in turn, assesses the example as marginal.<sup>208</sup> He points out that the conditional clitic *by* is idiosyncratic to some extent. Most strikingly, it can be contrastively stressed, cf. (4.15). On the other hand, Adam (2024: 47) cites examples from Junghanns (2003), which exemplify a pronominal, reflexive, auxiliary, and conditional clitic respectively in a prosodically isolated position between pauses.

<sup>208</sup> The conditional clitic seems to be peculiar in other Slavic languages, too. In Polish, it can appear in clause-initial position (I am thankful to Uwe Junghanns, p.c., for reminding me of this fact). Szczegielniak (1997: 314), Kupś (2000b: 74–75), and Kosta & Zimmerling (2014: 471) claim that *by* could not occupy the first clausal position, the latter of which contend that *by* were licensed in subordinate clauses only, where it “almost invariably” followed the complementizer, cf. (i). However, *by* can appear in root clauses, where it can take clause-initial position, cf. (ii). Note that example (ii.c) is from Borsley & Rivero (1994) and appeared in *Natural Language & Linguistic Theory*, which is among the highest received journals of generative linguistics. The examples in (iii) show *by* which is disjunct from the subordinator *gdy* ‘when; if’ and attached to the verb instead. Discussion in Kolaković et al. (2022: 96–97) indicates that cognate *bi* can occupy clausal-initial position in BCMS, too, and that it can receive stress according to some standard grammars.

- (i) a. \***By** on zabił go  
COND he.NOM.SG kill.PST.SG.M him.ACC  
‘He would kill him.’/‘He would have killed him.’/‘That he would kill him.’ (Pol; Szczegielniak 1997: 314)
- b. \***By=ś** Piotrowi pomagał.  
COND=AUX.2SG Peter.DAT help.PST.SG.M  
‘You would be helping Peter.’ (Pol; Kupś 2000b: 74)
- c. **Gdy=by=m** miał czas.  
when=COND=AUX.1SG have.PST.SG.M time.ACC.SG  
‘If I had time.’ (Pol; Kosta & Zimmerling 2014: 471, n. 29)
- d. \***Gdy miał=by=m** czas.  
‘If I had time.’ (Pol; Kosta & Zimmerling 2014: 471, n. 29)
- (ii) a. Ja **by=m** na pana uwagi nie zwrócił, **gdy=by** nie  
I.NOM.SG COND=AUX.1SG on mister.ACC.SG attention.GEN.SG NEG return.PST.SG.M when=COND NEG  
ten dach.  
this.NOM.SG.M roof.NOM.SG  
‘I wouldn’t have taken notice of you, [sir,] if there wasn’t the roof.’
- b. **By=m** cię nie straszył, **jak=by=m** cię nie  
COND=AUX.1SG you.ACC NEG frighten.PST.SG.M when=COND=AUX.1SG you.ACC NEG  
miał po co straszyć.  
have.PST.SG.M for what frighten.INF  
‘I would not frighten you, if I would not have a reason to frighten you.’ (Pol; NKJP poliarp)
- c. **By=ś** widział tę książkę.  
COND=AUX.2SG see.PST.SG.M this.ACC.SG.F book.ACC.SG  
‘You would see this book.’ (Pol; Borsley & Rivero 1994: 375)
- (iii) a. **Gdy miał=by=m** kupić wino na pewno stać  
when have.PST.SG.M=COND=AUX.1SG buy.INF wine.ACC.SG on certainly afford.INF  
było **by** mnie tylko na te do 5zł.  
be.PST.SG.N COND me.ACC.SG only on this.NOM.SG.N to 5\_zloties.GEN.PL  
‘If I would have to buy wine, I surely could afford only the one up to 5 zloties.’
- b. Możesz próbować nawet wtedy, **gdy miał=by=ś** zeznawać przeciwko  
can.PRS.2SG try.INF even then when have.PST.SG.M=COND=AUX.2SG testify.INF against  
wspólnikowi w interesach.  
partner.DAT.SG in business.LOC.PL  
‘You can try even if you would have to testify against a business partner.’ (Pol; NKJP poliarp)

- (4.14) a. My všichni, co spolu chodíme, **bychom**, jak říká  
 we.NOM all.NOM.VIR what together walk.PRS.1PL COND.1PL how say.PRS.3SG  
 Zilvar z chudobince, měli držet za  
 Zilvar.NOM from poorhouse.GEN.SG have.PST.PL.VIR hold.INF on  
 jeden provaz.  
 one.ACC.SG.M.INA rope.ACC.SG
- b. [<sub>IP</sub> My všichni ] [<sub>IP</sub> co spolu chodíme ] **bychom** [<sub>IP</sub> jak říká Zilvar z chudobince ] [<sub>IP</sub> měli  
 držet za jeden provaz ] .  
 ‘As Zilvar from the poorhouse says, all of us should be thick as thieves.’

(Cz; Rosen 2001: 207)

- (4.15) Speaker A: Takže Petr to u-dělá?  
 so\_that Petr.NOM this.ACC PFV-do.PRS.3SG  
 ‘So Petr will do it?’
- Speaker B: Říkal, že **BY** to udělal, kdyby ...  
 say.PST.3SG.M that COND this.ACC PFV-do.PST.SG.M if  
 ‘He said, he WOULD do it, if ...’

(Cz; Hana 2007: 85)

Yet another instance of “stressed clitics” can be observed in the Czech and Polish vernacular. It has been reported that the short pronominal form *ho* ‘him.ACC’ may occur under emphasis instead of its orthotonic long counterpart *jeho* in Czech (cf. Bubeník 1987; Veselovská 1995: §11.3.4). Exemplification of this matter is given in (4.16). Example (4.16a) provides an instance of the constant clitic *ho* ‘him.ACC’ that is modified by the restrictive particle *jenom* ‘only’ and stressed. In the colloquial Czech (4.16b), *ho* is sentence-initial and bears sentential stress. Example (4.16c) is from the ČNK, where *ho* is modified by *i* ‘even; also’ and thus focused with the implication that it is necessarily stressed. The Polish attestations in (4.17) provide equivalent cases. I will not defend any ad hoc analysis of such occurrences, but there are at least three possibilities to account for such instances. First, it could be clitics proper, which bear stress under certain circumstances (e.g. contrastive stress, cf. Cardinaletti & Starke 1999a). Second, we observe underlyingly orthotonic pronouns, in fact, which are somehow substituted by the short pronominal forms, such that the data represent a kind of allomorphy. Third, the short pronouns have ceased to be clitics and have become orthotonic forms (or inconstant clitics) themselves (cf. Jung & Migdalski 2022: 7–8 for comparable arguments for Macedonian and Slovene). Note that it is only Czech *ho* and Polish *mi*, which behave

this way.<sup>209</sup> Nevertheless, as a descriptive generalisation, it is not correct that “[t]he short, purely enclitic forms are never used for emphasis” as put by Thorpe (1991: 6).<sup>210</sup>

- (4.16) a. Včera **jsem** viděl jenom jeho / **HO**.  
 yesterday AUX.1SG see.PST.SG.M only him.ACC him.CL.ACC  
 ‘Yesterday I saw only HIM.’ (Cz; Veselovská 1995: §9.3.4, ex. [30b])
- b. **HO** =sem =tam ne-potkal.  
 him.CL.ACC AUX.1SG there NEG-meet.PST.SG.M  
 ‘I have not met HIM there.’ (Cz, colloquial; Bubenik 1987: 49)
- c. Zkusili **jste** kontaktovat i **HO**?  
 try.PST.PL.VIR AUX.2PL contact.INF and him.CL.ACC  
 ‘Did you try to contact HIM, too?’ (Cz; ČNK syn\_v8)

- (4.17) a. Jak masz wyrzucić to daj **MI**.  
 how have.PRS.2SG throw\_away.INF TO give.IMP.2SG me.DAT  
 ‘If you [will] have to throw it away then give it to ME.’  
 (Pol; auditory attestation, Iława, 2021)
- b. **MI** tego nie musisz mówić!  
 me.DAT this.GEN NEG must.PRS.2SG tell.INF  
 ‘Don’t tell ME!’ (Pol; auditory attestation, no date)

Turning to South Slavic, Franks (2016: 94–99) provides examples for stressed clitics in Macedonian, Bulgarian, and Slovene, cf. (4.18)–(4.20).

- (4.18) a. Kamo =**MI** =ti =go?  
 where me.DAT you.DAT him.ACC  
 ‘Where did that thing of yours get to on me?’

<sup>209</sup> Bańko (2010) informs in a post on the academic publisher’s PWN linguist consulting website that long *mnie* is in a process of being substituted by short *mi*: “Współcześnie jednak obserwujemy, że *mi* wypiera z celownika formę *mnie*, a ta ostatnia ogranicza swój zakres do dopełniacza, biernika i miejscownika.” ‘However, we are currently observing that *mi* is replacing the form *mnie* from the dative case and the latter [form becomes] restricted to the genitive, accusative, and locative case.’ (Bańko 2010)

<sup>210</sup> As a side note, clitics may occur both with emphasis and in isolation in meta-communicative contexts, i.e. when talking about particular clitics as objects in a discourse. Thus, examples (i) and (ii) are perfectly acceptable.

- (i) a. Řekl jsi MĚ nebo TĚ?  
 ‘Did you say *mě* or *tě*?’  
 b. TĚ. (Cz; constructed)
- (ii) a. Powiedziałaś SIĘ czy CIĘ?  
 ‘Did you say *się* or *cię*?’  
 b. CIĘ. (Pol; constructed)

b. **Ne= mu= GO= dal.**  
 NEG him.DAT it.ACC give.PRF.SG.M  
 ‘He has not given it to him.’

c. **Što= bi= MU= zel?**  
 What MOD him.DAT take.PRF.SG.M  
 ‘What should he take from him?’

(Mac; Franks 2016: 94)

(4.19) a. **Az mu= ja= DAdox.**  
 I.NOM him.DAT it.ACC give.AOR.1SG  
 ‘I gave it to him.’

b. **AZ ne= MU= ja= DAdox.**  
 I.NOM NEG him.DAT it.ACC give.AOR.1SG  
 ‘I did not give it to him.’

(Bg; Franks 2016: 95)

(4.20) a. **SI že končal delo?**  
 AUX.2SG already finish.PST.SG.M work.ACC.SG  
 ‘Have you finished the work?’

b. **Predvčerašnjim še NE, negvčera pa sem GA.**  
 day\_before\_yesterday still NEG yesterday but AUX.1SG him.ACC  
 ‘The day before yesterday I didn’t, but yesterday I did (finish it).’

c. (Ali) **se dobro počutiš?**  
 Q REFL well feel.PRS.2SG.  
 ‘Do you feel well?’

d. **Jā, SE.**  
 yes REFL  
 ‘Yes, I do (feel well).’

(Slv; Priestly 1993: 429 [cited after Franks 2016: 98])

As mentioned earlier, it is usually assumed that clitics lack lexical accent and receive stress in terms of regular stress assignment by the phonological grammar, if at all. Klavans (1995: 132–141) argues that one cannot always tell with certainty whether or not a clitic is unaccented underlyingly. Consequently, two hypotheses can be contrasted:

- Prosody Hypothesis 1 (PH1): Clitics are generally toneless but may sometimes receive stress.
- Prosody Hypothesis 2 (PH2): Clitics possess lexical stress but are usually deaccented and may show up stressed only under certain circumstances.

The question is whether one of these options apply and to what extent could be language- and item-specific. PH1 is a standard assumption in clitic research and applied to Slavic, too. Determining whether PH2 could apply somewhere will depend on the heuristics that are at one's disposal. Such heuristics do not always lead to unequivocal results, as we know from Chapter 3. Therefore, Talić's (2018) study deserves particular attention, for she found that auxiliary and pronominal clitics in (a variety of) Bosnian interact with their hosts' pitch accent under certain conditions.<sup>211</sup> Leaving technical details aside, Talić (2018) reports the patterns in (4.22)–(4.24) among others, where a grave accent (˘) marks a falling pitch and an acute (ˊ) a rising pitch. Based on previous work on BCMS phonology, she informs that stress assignment in BCMS follows the patterns in (4.21). Each pattern represents a combination of two syllables (two monosyllabic morphemes, i.e. a root and a suffix in fact) with or without an underlying, lexically specified high tone (<sup>H</sup>). The concatenation of a high-tone syllable and a toneless syllable yields a falling accent (4.21a) which is also the case, when combining two high-tone syllables (4.21b). The falling tone in (4.21b) results from the deletion of the second high tone. If two toneless syllables are concatenated (4.21c), then a rule of High Tone Insertion applies which places the high tone on the first syllable resulting in a falling accent again. Finally, if a toneless syllable is followed by a high-tone syllable, the prosodic prominence shifts to the pretonic syllable and we receive a rising accent (4.21d). This means that a rising accent only emerges, if a toneless syllable is combined with a following tonic syllable.

| (4.21) | Combination                    | Example                                          | Accent  |
|--------|--------------------------------|--------------------------------------------------|---------|
| a.     | $(\sigma^H + \sigma)_\omega$   | brà <sup>H</sup> -ta<br>brother.GEN.SG           | Falling |
| b.     | $(\sigma^H + \sigma^H)_\omega$ | lò <sup>H</sup> pt-a <sup>H</sup><br>ball.NOM.SG | Falling |
| c.     | $(\sigma + \sigma)_\omega$     | dà:n-a<br>day.GEN.SG                             | Falling |
| d.     | $(\sigma + \sigma^H)_\omega$   | strá:n-a <sup>H</sup><br>side.NOM.SG             | Rising  |

(Talić 2018: 312)

(Talić 2018: 313)

Now, observe that, in (4.22), the subordinator *da* is accentless, while the interrogative expressions are realised with a falling tone when followed by lexical verbs. Comparing the same elements in the environment of auxiliary and pronominal clitics in (4.23) and (4.24) respectively, we notice that they

<sup>211</sup> Talić (2018: 310) informs that her data is from “a dialect of BCMS spoken in central Bosnia and Herzegovina”. It is not clear how the term *dialect* is used in her paper. If one assumes a traditional diatopic variety of BCMS, then reference to either the *zapadni* or the *srednjobosanski* dialect is possible and – to a lesser extent – the *istočnohercegovački* dialect.

realise the rising tone instead. Based on these facts, Talić (2018: 365) concludes that auxiliary and pronominal clitics have an underlying accent (a lexical high tone, in fact) in the variety she scrutinises.

- (4.22) a. Volim        da    čitam.  
           like.PRS.1SG that read.PRS.1SG  
           ‘I like to read.’  
       b. Kò    dolazi?  
           who come.PRS.3SG  
           ‘Who is coming?’  
       c. Štà    kaže?  
           what say.PRS.3SG  
           ‘What did he say?’  
       d. Štò    plače?  
           why cry.PRS.3SG  
           ‘Why is he crying?’  
       e. Gdjè    borave?  
           where stay.PRS.3SG  
           ‘Where are they staying?’

(Bos; Talić 2018: 316)

- (4.23) a. Kó    **bi**        želio        doći?  
           who COND want.PST.3SG come.INF  
           ‘Who would like to come?’  
       b. Štá    **su**        rekli?  
           what AUX.3PL say.PST.PL  
           ‘What did they tell you?’  
       c. Štó    **si**        došao?  
           why AUX.2SG come.PST.SG.M  
           ‘Why did you come?’  
       d. Gdjé    **ćeš**        parkirati auto?  
           where AUX.FUT.2SG park.INF car.ACC.SG  
           ‘Where will you park the car?’

(Bos; Talić 2018: 318)

- (4.24) a. Dá    **mi**        je da provedem        ljeto        na planini.  
           that me.DAT is that spend.PRS.1SG summer.ACC.SG on mountain.LOC.SG  
           ‘I wish I could spend the summer on a mountain.’



- b. Kó ti je dao sladoled?  
 who you.DAT AUX.3SG give.PST.3SG.M ice\_cream.ACC.SG  
 ‘Who gave you ice cream?’
- c. Štá mu je rekao?  
 what him.DAT AUX.3SG say.PST.3SG.M  
 ‘What did he say to him?’
- d. Štó nas je zvao?  
 why us.ACC AUX.3SG call.PST.SG.M  
 ‘Why did he call us?’
- e. Gdjé vas je vidio?  
 where you.ACC.PL AUX.3SG see.PST.3SG.M  
 ‘Where did he see you?’
- f. Štó vas je bilo tako malo na predavanju?  
 why you.GEN AUX.3SG be.PST.SG.N that\_much little on lecture  
 ‘Why were there so few of you at the lecture?’
- g. Gdjé ih je bilo mnogo?  
 where them.GEN AUX.3SG be.PST.SG.N a\_lot  
 ‘Where was there a lot of them?’ (Bos; Talić 2018: 319)

Syntagmatically conditioned stress shift was already reported by Ivić (1958), who surveys the Serbo-Croatian dialects. He cites corresponding patterns for several Serbo-Croatian macro-dialects, i.e. Zeta-Lovćen (= Zetsko-južnosanžački/Zeta-South Sandžak in current terminology), Kosovo-Resava, Slavonian, and some BCMS exclaves in Romania and North Macedonia.<sup>212</sup> He observes that there are lexemes whose stress position – or sometimes stress quality – depends on the adjacency of auxiliary and pronominal clitics.<sup>213</sup> Unfortunately, Ivić (1958) provides only very incomplete examples, cf. (4.25).

- (4.25) a. sèstra ~ sestră je  
 sister.NOM.SG ~ sister.NOM.SG be.PRS.3SG
- b. rúka ~ rŭkă je  
 hand.NOM.SG ~ hand be.PRS.3SG (BCMS, Zeta-Lovćen; Ivić 1958: 205)

<sup>212</sup> The stress shift patterns do not occur throughout the entire macro-dialects but seem to occur in particular sub-dialects only.

<sup>213</sup> According to Ivić’s (1958: passim) discussion, these shift patterns are the result of the Neo-Štokavian stress shift. The syntagms containing clitics fossilise the diachronically older stress position.

|                      |   |                                  |                                                  |
|----------------------|---|----------------------------------|--------------------------------------------------|
| c. dóđi              | ~ | dōđi                             | <b>mi</b>                                        |
| come.IMP.2SG         | ~ | come.IMP.2SG                     | me.DAT                                           |
| ‘Come!’              | ~ | ‘Come to me!’                    | (BCMS, Kosova-Resava; Ivić 1958: 227)            |
| d. séstra            | ~ | sēstrā                           | <b>je</b>                                        |
| sister.NOM.SG        | ~ | ‘sister.NOM.SG                   | be.PRS.3SG’                                      |
| e. jézik             | ~ | jēzik                            | <b>je</b>                                        |
| tongue.NOM.SG        | ~ | tongue.NOM.SG                    | be.PRS.3SG                                       |
|                      |   |                                  | (BCMS, Kosova-Resava; Ivić 1958: 229)            |
| f. krāl              | ~ | krāl                             | <b>je</b>                                        |
| ‘king.NOM.SG’        | ~ | ‘king.NOM.SG                     | be.PRS.3SG                                       |
|                      |   |                                  | (BCMS, Slavonian; Ivić 1958: 287) <sup>214</sup> |
| g. žèna              | ~ | ženā                             | <b>je</b>                                        |
| ‘woman.NOM.SG’       | ~ | ‘woman.NOM.SG                    | be.PRS.3SG                                       |
| h. vèselit           | ~ | veselit                          | <b>se</b>                                        |
| be_happy.INF         | ~ | be_happy.INF                     | REFL.ACC                                         |
| i. ùčiteļevat        | ~ | učiteļevā=éu                     |                                                  |
| serve_as_teacher.INF | ~ | serve_as_teacher.INF=AUX.FUT.1SG |                                                  |
|                      |   |                                  | (BCMS, Slavonian; Ivić 1958: 288)                |

Ivić’s (1958) data points nicely supplement Talić’s (2018) ones, as they show accentual interactions of a clitic with autosemantica in contrast to Talić’s (2018) discussion of stress modulation in synsemantica. Both, Talić’s (2018) and Ivić’s (1958) work convincingly show that BCMS clitics interact with the prosodic structure of their hosts and suggest that clitics possess prosodic structure themselves in contrast to the common opinion cited by Dogil et al. (1999: 849), who state: “Since enclitics occur on the right edge of syllable and never have lexical tone themselves, they are never stressed and do not influence the position of stress.”

### 4.2.3 Prosodic boundaries

Several accounts have been proposed that give credit to the role the phonological domain plays for the distribution of clitics. Bošković (2001) proposes a syncretic account of syntactic structure building

<sup>214</sup> Ivić (1958: 100, n. 2) informs that the tilde (ã) marks a long rising tone reflecting the Proto-Slavic so-called *neoacute*. Synchronically, the accent differs from the long rising marked by the acute diacritic (á) in tonal contour. The acute (á) exhibits a continuously rising pitch, whereas the neoacute (ã) is characterised by a sudden tonal ascent from a low to a significantly higher tone (Ivić 1958: 285).

and prosodic filtering to capture 2P in BCMS and other Slavic 2P clitic languages like Slovene. The idea is that clitics are linearised with respect to intonation phrase (iP) boundaries adopting essentially Klavans's (1995) model. The conjunction of two prosodic requirements yields second position cliticisation as desired. First, clitics must be initial in the iP. Second, they must encliticise onto an eligible host. The mechanism is supposed to account for delayed clitic placement in BCMS, too.<sup>215</sup>

Marušič (2008: 275–276) argues that Slovene clitic climbing constructions provide multiple landing sites for clitics, which yields a high degree of distributional freedom. He tells that “[c]litics are actually not free to position themselves in the sentence. Each position demands a specific prosody. Clitics always follow a pause.” (Marušič 2008: 276) If one takes the quote at face value, it means that each clitic (cluster) in (4.26) immediately follows a pause as indicated by a hash mark (#).<sup>216</sup> Consequently, clitics would be always positioned first in the respective prosodic constituent, which would render Slovene auxiliary and pronominal clitics strictly proclitic.

- (4.26) a. Včeraj # **j**<sub>o2</sub>= **j**<sub>e1</sub>= sklenil<sub>1</sub> počasi kot polž odpeljati<sub>2</sub>  
 yesterday her.ACC AUX.3SG decide.PST.SG.M slowly like snail drive\_away.INF  
 proti domu.  
 towards house.DAT
- b. Včeraj # **j**<sub>e1</sub>= sklenil<sub>1</sub> # **j**<sub>o2</sub>= počasi kot polž odpeljati<sub>2</sub> proti domu.
- c. \*Včeraj # **j**<sub>e1</sub>= sklenil<sub>1</sub> počasi # **j**<sub>o2</sub>= kot polž odpeljati<sub>2</sub> proti domu.
- d. Včeraj # **j**<sub>e1</sub>= sklenil<sub>1</sub> počasi kot polž # **j**<sub>o2</sub>= odpeljati<sub>2</sub> proti domu.
- e. Včeraj # **j**<sub>e1</sub>= sklenil<sub>1</sub> počasi kot polž odpeljati<sub>2</sub> # **j**<sub>o2</sub>= proti domu.

‘Yesterday, he decided to take her home as slowly as a snail.’ (Slv; Marušič 2008: 276)

Provided Bošković's (2001) model and Marušič's (2008) judgement on pauses, it appears that the clitics are, at least partly, distributed by means of prosodic factors. However, information on prosodic boundaries is empirically weak for Slavic cliticisation (cf. Čamdžić & Hudson 2007: 21; Diesing

<sup>215</sup> See Section 4.4.2 for delayed clitic placement.

<sup>216</sup> I follow Hana (2007), Kolaković et al. (2022), and Rosen (2014) in using numerical subscriptions to visualise a clitic's affiliation with a particular predicate. For example, the pronominal clitic *ho* in (i) is subscribed with index 2, since it is licensed by the embedded infinitive *koupit*, which has the same index. The matrix verb *chtěl* ‘want.PST.SG.M’ has the index 1. The index 1 of the auxiliary clitic *jsem* indicates that it is affiliated with the *I*-participle. Note that this subscript convention approximates syntactic subcategorisation relations only but does not perfectly mirror syntactic dependencies in term of hierarchies, for temporal auxiliaries dominate lexical verbs of analytical tense forms in dependency grammar (cf. Mel'čuk 2008: 190, 206; Milićević 2023: 129–137). Nevertheless, I will continue to talk about dependency relations.

(i) Já **jsem**<sub>1</sub> **ho**<sub>2</sub> chtěl<sub>1</sub> koupit<sub>2</sub>.  
 I.NOM AUX.1SG him.ACC want.PST.SG.M buy.INF  
 ‘I wanted to buy it.’

(Cz; constructed)

2010: 121 for a similar assessment).<sup>217</sup> Most arguments on clitic placement and prosodic constituency are subjective and there have not been any serious attempts to investigate the prosody of clitic phenomena in Slavic empirically.<sup>218</sup> It remains yet to be shown, whether the prosodic boundaries in (4.26) are really distributed as indicated. For instance, Marušič's (2008) suggestion on the distribution of prosodic pauses conflicts with Golden (2003: 208), who explicitly states that *ga* is enclitic in (4.27) and “integrat[es] into the prosodic phrase of the phonological word to [its] left”.

- (4.27) Janez        =**ga**        srečuje        v parku.  
                  Janez.NOM him.ACC meet.PRS.3SG in park.LOC.SG  
                  ‘Janez meets him in the park.’ (Slv; Golden 2003: 208)

Especially the prosodic boundary following the first element *včeraj* seems dubious without additional information. Admittedly, there is no reason to deny the possibility of placing a pause at this position. On the other hand, we are allowed to cast into doubt that the occurrence of a prosodic break is a necessity here. Consider also that the number of prosodic boundaries is a function of structure and complexity. Parentheses usually parse their own intonation phrase and the longer an utterance becomes the more likely the number of boundaries grows. However, in rather simple and short utterances, there might be no intermediate prosodic boundary and there also might not be any clear relationship between clitic placement and prosodic boundary. In summary, the facts might turn out to be more complex than the simplified idealisations discussed so far. This is an empirical matter ultimately, which clearly needs further research.

## 4.3 Second position placement

### 4.3.1 Second constituent in root clauses

Second position cliticisation obeys Wackernagel's law, such that the clitics are placed after the first element of the clause. However, the exact definition of “the first element” is subject to language-specific conditions (cf. Klavans 1995: 29–33).<sup>219</sup> It may range from the first word to the first syntactic

<sup>217</sup> Diesing (2010: 121) notes that “while the notion of intonational phrase plays a prominent role in a number of theories of clitic placement [...], these claims have not been studied in prosodic terms or substantiated by acoustic evidence.”

<sup>218</sup> However, Tabachnik et al. (2024) submitted a preprint on LingBuzz that attempts to substantiate Section 4.2.1's topic of prosodic orientation for the case of Slovene empirically. They found in a phonetic perception experiment that proclisis is the default in Slovene.

<sup>219</sup> I shall ignore purely phonological notions like the syllable ( $\sigma$ ) or foot (F).

constituent, which can be arbitrarily complex as long as it is definable as the “leftmost node in the (surface) phrase marker” (Klavans 1995: 29, citing Pullum 1981: 444).<sup>220</sup> In traditional grammaticography, the second position has been regarded as ‘the least accented position in the sentence’ (cf. Gebauer & Ertl 1924: 52: *místo nejméně přízvuchné ve větě*). Accordingly, Vintr (2001: 26, 98) refers to the second position as *Akzentschatten* ‘accentual lee’. Clitic positioning has been determined in phonological terms by more traditional approaches, therefore. The clitics are described as following the first accentual unit or stressed expression of the clause irrespective of its size (cf. Čechová et al. 1996: 32–33; Gebauer & Ertl 1924: 52; Janda & Townsend 2000: 8–9; Vintr 2001: 98). However, the second position has been determined syntactically for Slavic in many recent approaches and clitics are positioned with respect to syntactic constituents. They exhibit what is called NON-SELECTIVITY (cf. e.g. Haspelmath 2023: 4) or figuratively PROMISCUITY (cf. e.g. Kupšć 2000b: 37, 100; Zwicky 1987: 136).<sup>221</sup> That is, second position clitics may lean on a clausal unit of any (morpho-)syntactic category as long as it is a prosodically eligible host (cf. Hana 2007: 95–97; Veselovská 1995: §4.6, §4.9 for Czech; Kupšć 2000b: 37 for Polish; Wilder & Čavar 1994: 42 for Croatian). Consider the non-exhaustive examples in (4.28)–(4.31) citing Czech, Polish, BCMS, and Slovene respectively for the sake of illustration, each of which provides a verb (a), a subject NP (b), an object NP (c), a *wh*-element (d), and an adverb (e) as firstly positioned element preceding the clitic or clitic cluster.

- (4.28) a. Dal                **mu**        **ho**        k vánocům.  
                  give.PST.SG.M him.DAT him.ACC to Christmas.DAT.PL  
                  ‘He gave it to him for Christmas.’ (Cz; Hana 2007: 68)
- b. Eva                **se**                dívá                na televizi.  
                  Eva.NOM REFL.ACC watch.PRS.3SG on television.ACC.SG  
                  ‘Eva is watching TV.’ (Cz; Rosen 2001: 214)
- c. Petra                **jsem**                pozval                na pondělí.  
                  Petr.ACC AUX.1SG invite.PST.SG.M on Monday.ACC.SG  
                  ‘I invited Peter to come on Monday.’ (Cz; Fried 1994: 163)
- d. Komu                **jsi**                **ho**                dal?  
                  whom.DAT AUX.2SG him.ACC give.PST.SG.M?  
                  ‘To whom did you give him?’ (Cz; Veselovská 1995: §9.2.2, ex. [11])

<sup>220</sup> Klavans’s (1995) misdates Pullum’s article to the year 1980.

<sup>221</sup> To my knowledge, Nevis (1986: 15, 24) was the first to use the notion of *promiscuity*. Zwicky (1987: 136) seems to have coined the collocation “promiscuous attachment”. The notion of promiscuity and precisely the phrase *promiscuous attachment* is also used by Spencer & Luís (2012a, 2012b). Samvelian & Tseng (2010: 218) cite the collocation *promiscuous attachment* without any reference, while Brookes (2014: 91) assigns authorship erroneously to Spencer & Luís (2012a). Miličević (2023: 61) refers to non-selectivity by the term *variable placement*.

e. Včera **jsem** prospal celé odpoledne.  
 yesterday AUX.1SG sleep\_through.PST.SG.M whole.ACC.SG.N afternoon.ACC.SG  
 ‘Yesterday I slept through the whole afternoon.’ (Cz; Fried 1994: 157)

(4.29) a. Widział=by=m **go** wczoraj.  
 see.PST.SG.M=COND=1SG him.ACC yesterday  
 ‘I would have seen him yesterday.’ (Pol; Witkoś 1998: 165)

b. Maria **go** wczoraj zaprosiła na obiad.  
 Mary.NOM him.ACC yesterday invite.PST.SG.F on lunch.ACC.SG  
 ‘Mary invited him for lunch yesterday.’ (Pol; Kupść 2000b: 37)

c. [NP Tę książkę ] **się** czuje, słyszy, wacha,  
 this.ACC.SG.F book.ACC.SG REFL feel.PRS.3SG hear.PRS.3SG smell.PRS.3SG  
 dotyka i widzi.  
 touch.PRS.3SG and see.PRS.3SG  
 ‘One feels, hears, smells, touches, and sees this book.’ (Pol; NKJP poliqarp full)

d. Kto **się** kiedy golił?  
 who.NOM REFL.ACC when shave.PST.SG.N  
 ‘Who is shaving when?’ (Pol; Witkoś 1998: 196)

e. Często **go** teraz spotykam.  
 often him.ACC now meet.IPFV.PRS.1SG  
 ‘I often meet him nowadays.’ (Pol; Kupść 2000b: 37)

(4.30) a. Predstavili **smo mu je** juče.  
 introduce.PST.PL.M AUX.1PL him.DAT her.ACC yesterday  
 ‘We introduced her to him yesterday.’ (BCMS; Bošković 2001: 9)

b. [NP Moj brat ] **je** došao.  
 my.NOM.SG brother.NOM.SG AUX.3SG come.PST.SG.M  
 ‘My brother came.’ (BCMS; Radanović-Kocić 1996: 435)

c. Knjigu **mi** sutra donesi.  
 book.ACC.SG me.DAT tomorrow bring.IMP.2SG  
 ‘Bring me the book tomorrow.’ (BCMS; Radanović-Kocić 1996: 438)

d. Koga **je** Stefan zbunio?  
 whom.ACC AUX.3SG Stefan.NOM confuse.PST.SG.M  
 ‘Whom did Stefan confuse?’ (BCMS; Progovac 1996: 412)

e. Juče        **sam**        **ti**                    **joj**        pomogla.  
 yesterday AUX.1SG you.DAT.ETH her.DAT help.PST.SG.F  
 ‘Yesterday, I helped her.’ (BCMS; Bošković 2001: 62)

- (4.31) a. Pokazal        **sem**        **ji**        rezultate        najnovešega        eksperimenta.  
 show.PST.SG.M AUX.1SG her.DAT result.ACC.PL newest.GEN.SG experiment.GEN.SG  
 ‘I have shown her the results of the newest experiment.’ (Slv; Marušič 2008: 268)
- b. Janez        **ga**        srečuje        v        parku.  
 Janez.NOM him.ACC meet.PRS.3SG in park.LOC.SG  
 ‘Janez meets him in the park.’ (Slv; Golden 2003: 208)
- c. Sestroj        **so**        predstavili        { Roku.        |        svojemuj        bratu. }  
 sister.ACC AUX.3PL introduce.PST.PL.M Rok.DAT self’s.DAT brother.DAT.SG  
 ‘They introduced the sister<sub>j</sub> to Rok / her<sub>j</sub> brother.’ (Slv; Stegovec 2020: 270)
- d. Kdo        **mi**        **je**        kje        kaj        posodil?  
 who.NOM me.DAT AUX.3SG where what.ACC lend.PST.SG.M  
 ‘Who lent what to me where?’ (Slv; Marušič 2008: 272)
- e. Veliko        **se**        **je**        govorilo,        storilo        malo.  
 a\_lot REFL.ACC AUX.3SG say.PST.SG.N do.PST.SG.N little  
 ‘A lot was said, little done’ (Slv; Golden 2003: 216)

The (a) type verb-initial clauses are considered a special sort of construction, since the verb seems to be dislocated from the presumably deep structural position to the topmost available clause position. Verb fronting, especially of the *I*-participle, has been assumed to reflect the structural requirement of clitics to be supported by an eligible host (cf. Ćavar & Wilder 1999: 434–435; Franks 2010: 16; Progovac 1996: 415–420). It has been sometimes considered under the umbrella of “last resort” operations like so-called Long Head Movement, whose purpose is to save the syntactic derivation (cf. Rivero 1991).<sup>222</sup> Such approaches attempt to simultaneously capture the fact that clitics are almost invariably postposed to subordinators in embedded clauses and that especially verbs do not intervene between subordinator and clitic cluster.<sup>223</sup>

Most examples in (4.28)–(4.31) present one-word expressions in clausal first position except (4.29c) and (4.30b). However, it is globally accepted that multi-word expressions can regularly precede Wackernagel clitics in Slavic, if the multi-word expression forms a single syntactic

<sup>222</sup> For critical discussions of Long Head Movement, cf. Franks & King (2000: 222–225, 300–302, 362–365), Wilder & Ćavar (1994), Veselovská (1995: §4.9), Bošković (1995), with references.

<sup>223</sup> See Section 4.3.3 for clitic second position placement in subordinate clauses and Section 4.4.2 for deviations from second position placement.

constituent. In other words, the syntactic constituent in the clausal first position can be of arbitrary complexity, in principle. The examples in (4.32)–(4.35) illustrate this point. We encounter a PP as well as two coordinate complement clauses in Czech (4.32), a PP and a complement clause in Slovak (4.33), a complex NP with a genitival dependent NP in BCMS (4.34), and a complex NP with an NP-internal relative clause in Slovene (4.35).

- (4.32) a. [PP<sub>Na</sub> [NP zahradě ]]<sup>1</sup> **jsem**<sup>2</sup> **si**<sup>2</sup> dělal poznámky.  
on garden.LOC.SG AUX.1SG REFL.DAT make.PST.SG.M note.ACC.PL  
‘In the garden I was taking notes.’ (Cz; Veselovská 1995: § 4.9, ex. [48a])
- b. [s [s Kam má jít ] a [s co má vyřídit ]]<sup>1</sup>,  
where have.PRS.3SG go.INF and what.ACC have.PRS.3SG get\_done.INF  
**jsem**<sub>1</sub><sup>2</sup> **mu**<sub>2</sub><sup>2</sup> zapomněl<sub>1</sub> připomenout<sub>2</sub>.  
AUX.1SG him.DAT forget.PST.SG.M remind.INF  
‘I forgot to remind him, where he must go and what he must get done.’  
(Cz; Rosen 2001: 214)

- (4.33) a. [PP<sub>Od</sub> [NP toho dňa ]]<sup>1</sup> **sme**<sup>2</sup> **sa**<sup>2</sup> vídali takmer  
from this.GEN.SG day.GEN.SG AUX.1PL REFL.ACC see.PST.PL almost  
denne.  
daily  
‘From this day on, we saw each other almost every day.’
- b. [s Kam položil knižku ]<sup>1</sup>, **si**<sup>2</sup> ne-mohol spomenúť.  
where put.PST.SG.M book.ACC.SG REFL.DAT NEG-can.PST.SG.M remember.INF  
‘He couldn’t remember, where he had put the book.’  
(Slk; Ivanová & Bashir Géčová 2021: 27)

- (4.34) [NP Kolutovi [NP plavičastog dima ]]<sup>1</sup> **su**<sup>2</sup> **se**<sup>2</sup> penjali.  
circle.NOM.PL bluish.GEN.SG.M smoke.GEN.SG AUX.3PL REFL.ACC rise.PST.PL.M  
‘Circles of bluish smoke were rising.’ (BCMS; Radanović-Kocić 1996: 453)

- (4.35) [NP Deček, [s katerega sem srečal včeraj, ]]<sup>1</sup> **me**<sup>2</sup> **je**<sup>2</sup>  
boy.NOM.SG whom.ACC.SG AUX.1SG meet.PST.SG.M yesterday me.ACC AUX.3SG  
prepoznal.  
recognise.PST.SG.M  
‘The boy that I met yesterday, recognised me.’ (Slv; Marušič 2008: 266)



### 4.3.2 Second word and discontinuous constituents in root clauses

(4.36) a. [<sub>S</sub> [<sub>NP</sub> Ovu zanimljivu knigu ]<sup>1</sup> sam<sup>2</sup> počitao. ]  
this.ACC.SG.F interesting.ACC.SG.F book.ACC.SG AUX.1SG read.PST.SG.M  
(BCMS; Franks & King 2000: 219)

b. [<sub>S</sub> [<sub>NP</sub> [ [ Ovu ]<sub>ω</sub><sup>1</sup> sam<sup>2</sup> ]<sub>ω</sub> zanimljivu knigu ] počitao. ]  
'This interesting book I read.'  
(BCMS; Franks & King 2000: 220)

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- (4.37) a. [S [NP Malý chlapec ]<sup>1</sup> ho<sup>2</sup> z-jedol. ]  
small.NOM.SG.M boy.NOM.SG him.ACC PFV-eat.PST.SG.M  
 b. \*[S [NP [ [ Malý ]<sub>ω</sub><sup>1</sup> ho<sup>2</sup> ]<sub>ω</sub> chlapec ] zjedol. ]  
 ‘A small boy ate him.’ (Slk; Franks & King 2000: 133)

- (4.38) a. [ [ X ]<sub>ω</sub> CL ]<sub>ω</sub> ( [ X ]<sub>ω</sub> ... )<sup>225</sup>  
 b. [S XP CL ( YP ... ) ]

A recurring example for the phonologically determined placement of BCMS clitics is the splitting of proper names. Browne (2004 [1975]: 262) provides the most famous example (4.39), which has been repeatedly cited ever since. The proper name *Lav Tolstoj* is split up by an intervening auxiliary clitic.<sup>226</sup> This option is absent in other Slavic languages but highlighted by several BCMS grammars, as Reinkowski (2001: 176) points out.<sup>227</sup> For instance, Kosta & Zimmerling (2014: 460) claim that Czech prohibits “splitting of a multi-word initial constituent”.

- (4.39) [NP Lav<sup>1</sup> je<sup>2</sup> Tolstoj ] veliki ruski  
Leo.NOM.SG be.CL.PRS.3SG Tolstoy.NOM.SG great.NOM.SG.M Russian.NOM.SG.M  
 pisac.  
writer.NOM.SG  
 ‘Leo Tolstoy is a great Russian writer.’ (BCMS; Browne 2004: 262)

However, clitics can occur in second position after elements of discontinuous constituents, which is called a “partial subtree” by Rosen (2001: 219–220) or “partial constituent” by Hana (2007: 98–103). It has been observed that clitics can occur in such constructions both within and beyond BCMS, if splitting a constituent is a general possibility non-contingent on cliticisation (cf. Kupść 2000b: 39, 106–107; Witkoś 1998: 160–161 for Polish; Franks & King 2000: 132–133 for Slovak; Dimitrova-Vulchanova 1999: 110; Franks & King 2000: 219–222; Progovac 1996: 414–415; Wilder & Čavar 1994: 37 for BCMS).<sup>228</sup> Some non-exhaustive data is provided in (4.40)–(4.44). The Czech example

<sup>225</sup> The parentheses and the three dots indicate that more material can be present in the utterance.

<sup>226</sup> Bošković (2009, with references) observes that phrase splitting is only possible, if either no part of the proper name is inflected or if both are inflected. Phrase splitting seems to be impossible, if only one element inflects.

<sup>227</sup> Kolaković et al. (2022: 120–121) cite examples for split proper names from BCMS grammatigraphy. They furthermore cite attestations from the dialectological literature (cf. Kolaković et al. 2022: 158). On the other hand, Reinkowski (2001: 176, 203–204) reports that she did not find any instance of split names in her newspaper corpus of 8000 sentences.

<sup>228</sup> Such split constituents are commonly accounted for as Left Branch Extraction from NP, which violates the Left Branch Condition originally formulated by Ross (1967: 207). The pattern is widespread across Slavic (cf. Borsley 1983: 340; Borsley & Rivero 1994: 407, also n. 9; Haider & Szucsich 2022: 18–20; Hana 2007: 50–53; Toman 1986: 136–137, Wilder & Čavar 1994: 34–39, Witkoś 1998: 160).

(4.40a) shows clitic placement after the nominal head of a complex NP, while the attributive adjective is left behind.<sup>229</sup> The reverse situation occurs in (4.40b–c) with cliticisation to an adjective extracted from NP.<sup>230</sup> Example (4.40d) presents what Rosen (2001: 224) calls a “non-tree” and Bošković (2005: 30) a “non-constituent”. The fronted portion does not form a coherent syntactic unit, which is indicated by the bracketing for the sake of clarity.<sup>231</sup> In (4.40e–f), the clitic leans on an extracted relativising interrogative expression (WH), while (4.40g) shows a corresponding continuous constituent in (4.40f). The Polish data show extracted adjectives in (4.41a–b) and a *wh*-word in (4.41c). We encounter more extracted adjectives in Slovak (4.42). The BCMS example in (4.43) resembles Czech (4.40d) in that it represents a non-tree. Slovene (4.44) shows an extracted interrogative. Structures of these kinds have been reported to be pragmatically loaded involving Topic and Focus marking (cf. Diesing 2010: 112–117; Fried 1994: 161–162; Hana 2007: 50–52; Wilder & Čavar 1994: 35–36).<sup>232</sup>

- (4.40) a. [NP  <sub>j</sub> Tulipány ]<sup>1</sup> **si**<sup>2</sup> objednal [AdjP červené ]<sub>j</sub>.  
tulip.ACC.PL REFL.DAT order.PST.SG.M red.ACC.PL  
‘As for tulips, he ordered red ones.’
- b. [AdjP Červené ]<sub>j</sub><sup>1</sup> **si**<sup>2</sup> objednal [NP  <sub>j</sub> tulipány ].  
red.ACC.PL REFL.DAT order.PST.SG.M tulip.ACC.PL  
‘As for red, he ordered tulips.’ (Cz; Fried 1994: 161)
- c. [AdjP Střízlivého ]<sub>j</sub><sup>1</sup> **jsem**<sup>2</sup> [NP  <sub>j</sub> Patrika ] ne-viděl, ani  
sober.ACC.SG AUX.1SG Patrik.ACC.SG NEG-see.PST.SG.M neither  
ne-pamatuju.  
NEG-remember.PRS.1SG  
‘I do not remember when I saw Patrik sober the last time.’ (Cz; Hana 2007: 102)
- d. [PP O<sub>i</sub> [NP [AdjP irským ]<sub>j</sub>  <sub>k</sub> ]]<sup>1</sup> **jsem**<sup>2</sup> slyšel jen  
about Irish.LOC.SG AUX.1SG hear.PST.SG.M only  
[PP  <sub>i</sub> [NP [AdjP  <sub>j</sub> ] pívuk ].  
beer.LOC.SG  
‘I have only heard about Irish beer.’ (Cz; Hana 2007: 53)

<sup>229</sup> Underscores and indices serve purely illustrative purposes and visualise the expected syntagmatic order of the elements with respect to one another. They are neither meant to describe syntactic structure nor to imply transformations.

<sup>230</sup> Kupść (2000b: 41) erroneously distinguishes Polish from BCMS and Czech claiming that only the former allows clitics as well as orthotonic words to appear in phrase-internal position and to split constituents. Conversely, Aguado & Dogil (1989: 108) mistakenly claim that clitics cannot occur within phrasal constituents in Polish.

<sup>231</sup> Bošković (2005: 30) calls this pattern “extraordinary Left Branch Extraction”.

<sup>232</sup> See Chapter 13 on information structure and clitics.

- e. [WH Co ]<sub>j</sub><sup>1</sup>                **ses**<sup>2</sup>                do-věděl                [AdjP \_j nového ]?  
what.ACC.SG REFL.ACC.AUX.2SG PFV-know.PST.SG.M                new.ACC.SG  
‘What did you learn about that is new?’                (Cz; Rosen 2001: 221)
- f. [...] kdoví,                [WH kolik ]<sub>j</sub><sup>1</sup>                **mu**<sup>2</sup>                zbejvá                [NP \_j dnů ].  
who\_knows,                how\_many him.DAT remain.PRS.3SG                day.GEN.PL  
‘... who knows, how many days he has left.’
- g. (Na prstech mně ukazoval,)                [NP kolik dnů ]<sup>1</sup>                **mu**<sup>2</sup>  
on finger.LOC.PL me.DAT show.PST.3SG.M                how\_many day.GEN.PL him.DAT  
zbývá.  
remain.PRS.3SG  
‘(He showed me on [his] fingers,) how many days he has left.’                (Cz; ČNK syn2020)

- (4.43) [PP U<sub>i</sub> [NP [AdjP velikuj ] ] \_<sub>k</sub> ] **je** Jovan ušao  
 in big.ACC.SG.F AUX.3SG Jovan.NOM walk.PST.SG.M  
 [PP \_<sub>i</sub> [NP [AdjP \_<sub>j</sub> ] sobu<sub>k</sub> ].  
 room.ACC.SG  
 ‘Jovan walked into a big room’ (BCMS; Franks & King 2000: 359)

- (4.44) [WH Česa ]<sub>j</sub> **so** politiki vedno [Adj lačni \_<sub>j</sub> ]?  
 what.GEN be.PRS.3PL.CL politician.NOM.PL always hungry.NOM.PL  
 ‘What are politicians always hungry for?’ (Slv; Golden 2003: 210)

Since clitic placement after complete subtrees, partial subtrees, and non-trees can be accounted for under reference to well-established properties of Slavic syntax, the major issue is to what extent discontinuous constituents are licensed in a particular language.<sup>233</sup> Such a perspective is compatible with Diesing et al.’s (2009) finding that the probability of 2W vs. 2D correlates with syntactic structure, indeed. While clause-initial arguments preferably show clitic second word positioning, clausal-initial predicates are more likely to exhibit clitic second phrase positioning. On the other hand, non-trees occupying clausal first position as in (4.40d) pose a problem to localise the clitic position by means of syntactic constituents from a traditional point of view.<sup>234</sup> BCMS remains special in allowing the splitting of proper names.<sup>235</sup>

### 4.3.3 Second position in embedded clauses

The discussion in the previous two sections has shown that establishing the second position is a complex matter. In contrast, subordinate clauses appear to represent trivial cases, because the second position is determined with respect to the subordinator.<sup>236</sup> The clitic or clitic cluster immediately follows the subordinator, which represents the embedded clause’s first position. However, the syntactic status of the subordinators differs. While complementizers proper are syntactic heads C<sup>0</sup>,

<sup>233</sup> There seems to be disagreement as to whether individual languages allow for phrase splitting, at all. For instance, Franks & King (2000: 360) state that neither Czech nor Slovene license split phrases in contrast to BCMS, which “allow[s] substantial splitting of constituents”. On the other hand, Kupšć (2000b: 39) tells that phrase-internal clitic placement “[i]n Polish, unlike in Serbian/Croatian or Czech, [...] is not a unique property of clitics. [...] Polish NPs and PPs can be split by full prosodic words [...]”

<sup>234</sup> Cf. Bošković (2005: 30–36) for a discussion of possible solutions.

<sup>235</sup> Franks & King (2000: 222) remark that there are cases of clitics splitting phrases in BCMS, which cannot be attributed to these kinds of extraction, but are still accepted by native speakers. I leave this issue open.

<sup>236</sup> I use the term *subordinator* (gloss: SUB) as a hypernym for elements introducing subordinate clauses of various kinds, i.e. the term covers complementizers proper introducing argument clauses, subordinate conjunctions introducing adjunct clauses, relative pronouns, interrogatives, and adverbs introducing relative clauses.

relative pronouns are syntactic phrases XP (cf. de Vries 2018: 8 for phrasal status). It is therefore possible to extend the 2W/2D distinction to subordinate clauses, too. Examples (4.45)–(4.49) present clitic second position placement in subordinate clauses for Czech, Polish, Slovak, BCMS, and Slovene respectively. Slovene (4.49b) has been obtained from the Slovene web corpus slWaC (Erjavec & Ljubešić 2014; Ljubešić & Erjavec 2011). Again, the examples serve only for illustrative purposes and are not exhaustive. The a-examples present clitics following a complementizer proper, while the b-examples provide clitic placement after a relative pronoun. I am cautious using the position superscripts in Polish (4.46a), for the status of *żeby* ‘that’ is less trivial. Suffice it to say that certain contexts allow for counting *że* as the sole complementizer in the first position, whereas others require the morphologically complex form *żeby* to be treated as a single element.

- (4.45) a. Helena říkala, [s že<sup>1</sup> se<sup>2</sup> Petr odstěhoval ].  
 Helen.NOM.SG say.PST.SG.F that REFL.ACC Petr.NOM.SG move.PST.SG.M  
 ‘Helen said that Peter had moved.’ (Cz; Fried 1994: 159)
- b. Chlapci, [s které<sup>1</sup> jsem<sup>2</sup> viděl na ulici ]  
 boy.NOM.PL which.ACC.PL.M AUX.1SG see.PST.SG.M on street.LOC.SG  
 ‘The boys, which I saw on the street’ (Cz; Veselovská 1995: §9.4.2, ex. [46a])
- (4.46) a. Nie chcę, [s że<sup>1</sup>=by<sup>(2)</sup>=ś<sup>(2)</sup> mi<sup>2</sup> ją<sup>2</sup> nim  
 NEG want.PRS.1SG that=COND=AUX.2SG me.DAT.ETH her.ACC him.INS  
 straszył ].  
 scare.PST.SG.M  
 ‘I don’t want you to threaten her (on me) with him.’ (Pol; Rothstein 1993: 725)
- b. [...] i trafia na wiersz, [s który<sup>1</sup> się<sup>2</sup>  
 and come\_across.PRS.3SG on poem.ACC.SG which.NOM.SG REFL.ACC  
 mu<sup>2</sup> spodobał ] [...]   
 him.DAT please.PST.SG.M  
 ‘... and comes across a poem that he liked ...’ (Pol; NKJP poliqarp full)
- (4.47) a. Zato som ti to povedal, [s lebo<sup>1</sup> sa<sup>2</sup> to  
 for\_that AUX.1SG you.DAT this.ACC.SG tell.PST.SG.M because REFL this.ACC.SG  
 aj tak dozvieš ].  
 even so find\_out.PRS.2SG  
 ‘The reason I told you is that you’re bound to find out anyway.’ (Slk; Short 1993: 574)

- b. “[...] slovenčina patrí medzi jazyky, [s [PP v ktorých ]]<sup>1</sup>  
 Slovak.NOM belong.PRS.3SG between language.ACC.PL in which.LOC.PL  
**sa**<sup>2</sup> uplatňuje tzv. Wackernagelovo pravidlo [...]”  
 REFL.ACC invoke.PRS.SG so\_called Wackernagel.ADJ.ACC.SG.N rule.ACC.SG  
 ‘... Slovak is among the languages that invoke Wackernagel’s law ...’

(Slk; Ivanová & Bashir Géčová 2021: 26)

- (4.48) a. Stefan tvrdi [s da<sup>1</sup> **mu**<sup>2</sup> **ga**<sup>2</sup> **je**<sup>2</sup> Petar  
 Stefan.NOM.SG claim.PRS.3SG that him.DAT it.ACC AUX.3SG Peter.NOM.SG  
 poklonio ].  
 give.PST.SG.M

‘Stefan claims that Peter has given it to him as a present.’ (BCMS; Progovac 1996: 412)

- b. Onaj brod, [s kojeg<sup>1</sup> **sam**<sup>2</sup> očekivao, ] je  
 that.NOM.SG.M ship.SG.M which.ACC.SG.M AUX.1SG expect.PST.SG.M AUX.3SG  
 uplovio u luku.  
 sail.PST.SG.M into harbour.ACC.SG

‘That ship that/which I have been expecting has sailed into the harbour.’

(BCMS; Arsenijević & Gračanin-Yuksek 2016: 4)

- (4.49) a. vsem navzočim je znano, [s da<sup>1</sup> **se**<sup>2</sup>  
 all.DAT.PL attendee.DAT.PL be.PRS.3SG know.PASS.PTCP.NOM.SG.N that REFL.ACC  
 učna ura začne čez pet minut]  
 teaching.ADJ.NOM.SG.F hour.NOM.SG begin.PRS.3SG in five.ACC minute.GEN.PL  
 ‘(the fact) that the lesson begins in five minutes is known to everyone present’

(Slv; Priestly 1993: 434)

- b. In v čisto Devico, [s katera<sup>1</sup> **ga**<sup>2</sup> **je**<sup>2</sup>  
 and in pure.ACC.SG.F Virgin.ACC.SG who.NOM.SG.F him.ACC AUX.3SG  
 porodila grešnemu svetu ]?  
 give\_birth.PST.SG.F sinful.DAT.SG world.DAT.SG

‘And [believe] in the pure Virgin, who brought him forth to the sinful world?’

(Slv; slWaC)

In contrast to subordinators, coordinating conjunctions (‘and’, ‘or’, ‘but’) have been reported not to count as first position (cf. Ivanová & Bashir Géčová 2021: 26; Rosen 2001: 214–215; Thorpe 1991: 22–23). This is not surprising at first sight provided that a coordinating conjunction may conjoin two

sentences and may therefore occur beyond the clausal domain (cf. Bennett 1986: 8). The expectation is that clitics are in second position within their clausal domain, while the clause is conjoined with some other syntactic unit. In this logic, placing a clitic immediately after a coordinating conjunction would leave the clitic in clausal first position, which is usually considered a severe and, hence, ungrammatical deviation. The next section is devoted to this deviation from second position placement.

## 4.4 Positional deviations

### 4.4.1 Sentence-initial clitics: Clitic first

In this and the following section I will draw attention to the fact that the distribution of 2P clitics exhibits intralinguistic variation in Slavic. Variation with respect to clitic placement is not news. For instance, already Reiter (1959: 378) notices that second position clitic languages such as Slovene and Czech allow for clitic placement in first and – in the case of Czech – third position. Franks & King (2000: 226–234) devote particular discussion to distributional variation based on BCMS, Czech, and – to a lesser extent – Slovene. However, variation has been mostly neglected or marginalised in the study of clitics, such that the issue has often been displaced into footnotes (e.g. in Franks 2016; Thorpe 1991). The discrepancy between common stereotypes of second position cliticisation and the evidenced possibilities of variable clitic placement has been noted, among others, by Dotlačil (2004). He states that the obedience to Wackernagel’s law is an “oversimplification” for Czech and “[t]he clitic cluster can occupy the first position in colloquial Czech [...] and in embedded clauses it can appear in the third position” (Dotlačil 2004: 73, n. 53). Hana (2007: 70), too, attests sentence-initial clitics in colloquial Czech, though he considers them infrequent. Sentence-initial clitic positioning was already cited by Toman (1996: 505–506), which I will return to below.

The clause-initial placement of clitics is commonly known under the notion CLITIC FIRST (abbr.: 1P) (cf. Franks & King 2000: 226; Lenertová 2004: 148). The opinion that sentence-initial clitics are ungrammatical in Slavic has prevailed until recently (cf. Fried 1994: 157; Veselovská 1995: §11.3.5 for Czech; Kupść 2000b: 34–35, 93, 95; Witkoś 1998: 150–151, 155–156 for Polish).<sup>237</sup> Although an explicit ban of Clitic First is not always explicated, it is implicit in every treatment that particularly emphasises the second position requirement (cf. Rezac 2005 for Czech, Golden 2003 for

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<sup>237</sup> Macedonian is an exception, as it allows clause-initial clitics (cf. Dimitrova-Vulchanova 1999: 102; Franks & King 2000: 82, 236–237; cf. also the examples in Franks & King 2000: 67–89 [Ch. 2.4]).



Toman (1996: 505–506) attributes Clitic First to some sort of ellipsis (cf. also Thorpe 1991: 4 [n. 2], 14 [n. 12]). Hence, the demonstrative *to* is thought to be elided in (4.50a). Similar arguments are summarised by Franks & King (2000: 40–42) and Franks (2010: 54) for Slovene. Lenertová (2001: 295–296, 2004: 149) contends that such an analysis is not tenable for Czech. She argues that it is not clear what kind of element should have been deleted in (4.50b). Sometimes several possibilities seem to exist (cf. Lenertová 2001: 295, n. 8). Franks (2010: 55) provides a similar reasoning for Slovene Clitic First, cf. (4.51a). He concludes that the clitic is in the top-most clausal position, which, indeed, yields clause-initial clitic placement due to the possibility of proclisis in Slovene. Note that the Clitic First occurrences in (4.51b–c) are reported to represent spoken Slovene (cf. Franks & King 2000: 45).

- As in Czech and Slovene, Clitic First is also attested in Polish (4.52) and Slovak (4.53).

- (4.52) a. **mu**<sup>1</sup> posunął że na Tamizie będzie kręcił  
 him.DAT propose.PST.SG.M that on Thames.LOC.SG be.FUT.3SG record.PTCP.SG  
 ‘He proposed to him that he record on the [River] Thames.’ (Pol; NKJP pelcra)
- b. (Wiesz co?) **Ci**<sup>1</sup> powiem tak: ...  
 know.PRS.2SG what you.DAT tell.PRS.1SG so  
 ‘(You know what?) Let me put it this way [to you]: ...’  
 (Pol; auditory attestation, Iława, no date)

- (4.53) **Som**<sup>1</sup> čítal doma.  
 AUX.1SG read.PST.SG.M home.GEN.SG  
 ‘I read at home.’ (Slk; Franks & King 2000: 123, n. 31)

Clause-initial clitic placement is linked with the occurrence of clitics after coordinating conjunctions, which is generally considered impossible or only marginally possible, cf. Czech (4.54).<sup>238</sup> Franks (2010: 142) notices that the coordinating conjunctions *i* and *a* may host pronominal clitics in Polish (4.55) and Bulgarian (4.56) respectively (cf. also Kupść 2000b: 37 for Polish). He finds this to be a “curious fact” for Bulgarian (Franks 2000: 37).<sup>239</sup> Furthermore, Franks & King (2000: 42) find it “surprising (from the perspective of, e.g., S[erbo-]C[roatian]) that [...] clitics can immediately follow the conjunction *in*” ‘and’ in Slovene (4.57). However, comparable data is attested in other Slavic 2P languages, too. The examples in (4.58)–(4.59) provide data for Czech and Slovak respectively.<sup>240</sup> Note that the Czech example (4.58) discussed by Thorpe (1991) is in conflict with Fried’s (1994) data point (4.54).

- (4.54) \*A **mu** nic ne-dávej.  
 and him.DAT nothing.ACC NEG-give.IMP.2SG  
 ‘And don’t give him anything.’ (Cz; Fried 1994: 162)

<sup>238</sup> In fact, grammatical descriptions of Czech differ here. According to Čechová et al. (1996: 310) and Vintř (2001: 98), clitics cannot follow coordinating conjunctions. In contrast Gebauer & Ertl (1924: 52) and Trávníček (1951: 866 [cited after Thorpe 1991: 24]) inform that such positioning may occasionally occur.

<sup>239</sup> The Bulgarian data is relevant insofar as Bulgarian clitics are regarded to be syntactically verb-adjacent, but phonologically enclitic (cf. Franks & King 2000: 63, but Kosta & Zimmerling 2014 for a different perspective).

<sup>240</sup> Thorpe (1991: 24) notes that, while Trávníček (1951: 866) provides the example in (4.58), her native speaker informants find this example unusual and prefer (i).

(i) Děvčě pilo a modlilo se.

‘The girl was drinking and praying.’

(Cz; Thorpe 1991: 24)

(4.55) I **ja** wczoraj widzieli=śmy.  
 and her.ACC yesterday see.PST.PL.VIR=AUX.1PL  
 ‘And we saw it yesterday.’ (Pol; Franks 2010: 142)

(4.56) (Tja ne mi dade da vidja pismo-to.)  
 she.NOM.SG NEG me.DAT give.AOR.3SG that see.PRS.1SG letter.SG-DEF  
 A **go** skri.  
 and him.ACC hide.AOR.3SG  
 ‘(She wouldn’t let me see the letter.) And [she] hid it.’ (Bg; Franks & King 2000: 63)

(4.57) in **som** prišli v pokrajino.  
 and AUX.1SG come.PST.PL.M in region.ACC.SG  
 ‘and we came (thus) to a region.’ (Slv; Bennett 1986: 11)

(4.58) Děvče pilo a se modlilo.  
 girl.NOM.SG drink.PST.SG.N and REFL.ACC pray.PST.SG.N  
 ‘The girl was drinking and praying.’ (Cz; Thorpe 1991: 24)

(4.59) A **som** sa ho spýtal.  
 and AUX.1SG REFL.ACC him.GEN ask.PST.SG.M  
 ‘and I (duly) asked him.’ (Slk; Short 1993b: 567 [cited after Franks & King 2000: 134])

Obviously, there is no general structural prohibition against clitics immediately following a coordinating conjunction in Slavic special clitic languages. This does not imply that the respective clitics lean on the conjunction. On the contrary, the coordinating conjunctions behave as proclitics themselves (cf. Thorpe 1991: 23 for Czech; Kraska-Szlenk 1995: 62 for Polish). The variation in clitic positioning after coordinating conjunctions is well illustrated by Ivanová & Bashir Gécová (2021) for Slovak. They inform that clitics can realise either 2P or 1P with the conjunction *ale* ‘but’, cf. (4.60). It occurs that first position cliticisation is mostly represented by single clitics and involves an auxiliary, reflexive, or pronominal with a sibilant onset, which is reminiscent of the German cliticisation in the prefield from Footnote 183 in Section 3.12.2. However, I refrain from assuming that a phonological constraint applies in Slavic, since the same configuration shows varying categories of clitics with a diverse segmental make-up as well as clitic clusters.

- (4.60) a. Cítila jeho dych za chrbtom, ale<sup>0</sup>  
 feel.PST.SG.F his.POSS breath.ACC.SG behind back.INS.SG but  
 ne-obrátila<sup>1</sup> sa<sup>2</sup>.  
 NEG-turn\_around.PST.SG.F REFL.ACC  
 ‘She felt his breath from behind, but she did not turn around.’
- b. Vyдалa sa za nejakého Bergera, ale<sup>0</sup> sa<sup>1</sup> s  
 give.PST.SG.F REFL.ACC for some.ACC.SG Berger.ACC.SG but REFL.ACC with  
 ním rozviedla.  
 him.INS divorce.PST.SG.F  
 ‘(She married a certain Berger) but she divorced him.’
- (Slk; Ivanová & Bashir Géčová 2021: 26)

On a different note, clitics can occur in coordinate structures as conjuncts themselves when used meta-linguistically.<sup>241</sup> Example (4.61a–b) exhibits the coordination of the two reflexive clitics. Example (4.61c) is quoted from Prejzová’s (2010) bachelor’s thesis and presents an asyndetic coordinate structure of three conjunct clitics *mi*, *ho*, *mu*. I am not aware that this observation has been made before.

- (4.61) a. Krátké tvary se a si jsou součástí zvratných  
 short.NOM.PL form.NOM.PL *se* and *si* be.PRS.3PL part.INS.SG reflexive.GEN.PL  
 sloves (malovat si).<sup>242</sup>  
 verb.GEN.PL (*malovat si*)  
 ‘The short forms *se* and *si* are parts of reflexive verbs (*malovat si* [= to put on make-up])’
- b. Setká-li se ve větě zvraté zájmeno  
 meet.PRS.3SG-if REFL.ACC in sentence.LOC.SG reflexive.ACC.SG.N pronoun.ACC.SG  
 se a si, ani jedno z nich vynechat ne-lze.  
*se* and *si* neither one.NOM.SG.N from them.GEN omit.INF NEG-possible  
 ‘If one encounters the reflexive pronoun[s] *se* and *si* in a sentence, none of them can be omitted.’
- (Cz; ČNK syn\_v8)

<sup>241</sup> See Footnote 158 in Section 3.10 and Footnote 210 in Section 4.2.2 for comparable observations with regard to occurrences inside PPs and isolation respectively.

<sup>242</sup> CQL query: [tag="P[H7] .\*" & word!="mě"] [word="a"] [tag="P[H7] .\*" & word!="mě"]

- c. Pozice                      klitik                      *mi, ho, mu* součástí                      infinitivní  
position.NOM.SG    clitic.GEN.PL *mi ho mu* part.INS.SG    infinitival.GEN.SG  
fráze  
phrase.GEN.SG  
‘The position(s) of *mi, ho, mu* [as] part [of] infinitival phrases’ (Cz; Prejzová 2010: 31)

Returning to Clitic First, it should be noted that this placement pattern is not limited to West Slavic and Slovene. Bosnian-Croatian-Montenegrin-Serbian exhibits it, too. This is remarkable, because BCMS clitics are commonly treated as a strictly enclitic. For instance, Franks & King (2000: 40) remark that Slovene “countenances a surprising degree of clitic first, a possibility completely unknown in S[erbo-]C[roatian]”. Clause-initial clitics being contingent on proclisis as discussed in Section 4.2.1 are attested for Kajkavian, Čakavian, and Burgenland Croatian. Furthermore, Kolaković et al. (2022: 160–162 [Sec. 7.6.4]) review BCMS dialectological literature and cite clitics in post-conjunctive and absolute clause-initial position from various Štokavian dialects, cf. (4.62). Equally strikingly, Milićević (2023: 109–110) informs that Clitic First occurs in colloquial Serbian (but is available only for a subset of clitics), cf. (4.63).

- (4.62) a. *se=                      mōžēš                      priládit*  
REFL.ACC can.PRS.2SG catch\_cold.INF  
‘you can catch a cold’  
b. *i=    je=                      rěkō*  
and AUX.3SG say.PST.SG.M  
‘and he told [them]’ (BCMS; Srednjobosanski; Kolaković et al. 2022: 162)  
c. (*Došo                      i    mi*  
come.PST.SG.M and we.NOM.SG  
*pusti ga,)                      i=    su=                      ga=                      strélali.*  
let.INF him.ACC and AUX.3PL him.ACC shoot.PST.PL.M  
‘(He came and we let him go,) and they shot him.’  
(BCMS; Šumadijsko-vojvođanski; Kolaković et al. 2022: 160)

- (4.63) a. *Su=    tu?*  
be.3PL here  
‘Are they here?’

- b. **Ste=** došli?  
 AUX.2PL come.PST.PL.M  
 ‘Did you arrive?’
- c. **Će=** da dođe?  
 AUX.FUT.3SG that come.PRS.3SG  
 ‘Will s\*he come?’
- d. **Sam= ti=** rek’o?!  
 AUX.1SG you.DAT say.PST.SG.M  
 ‘I told you (so), didn’t I?’
- e. **Ga=** znam ja?  
 him.ACC know.PRS.1SG I.NOM  
 ‘Do I know him?’
- f. **Se=** sećaš?  
 REFL remember.PRS.2SG  
 ‘Do you remember?’
- g. **Ti= se=** svidā ovo?  
 you.DAT REFL appeal.PRS.3SG this.NOM  
 ‘Do you like this?’ [lit.: ‘Does this please you?’] (Srp, colloquial; Milićević 2023: 110)

It appears that there is a sociolinguistic commonality across 2P languages with respect to the occurrence of 1P.<sup>243</sup> Note that 1P in BCMS is a phenomenon of spoken varieties like colloquial speech or traditional rural dialects. The situation resembles what has been reported for Czech, where 1P is reported as a phenomenon of spoken language (cf. Hana 2007: 70, explicitly referring to Common Czech; Lenertová 2001: 296, 2004: 149; Toman 1996: 505). Likewise, Franks & King (2000: 43) tell that clitics are licit in absolute initial position (i.e. without a preceding conjunction) in colloquial Slovene. They furthermore point out (based on personal communication with Marcela Molčanová) that the clause-initial auxiliary clitic in Slovak (4.53) is possible rather in the spoken language (cf. Franks & King 2000: 123, n. 31). Finally, Polish (4.52) provides attestations of oral speech.<sup>244,245</sup>

<sup>243</sup> See Section 4.2.1 on prosodic orientation.

<sup>244</sup> In this context, note that Dimitrova-Vulchanova (1999: 105) states that “structures with clause initial clitics obtained in Old Bulgarian, which is an attested clause 2nd position (Wackernagel) language.” Unfortunately, she neither provides a reference nor an example. The question arises, if putative 1P in Old Bulgarian is diachronically linked to 1P in contemporary Slavic 2P languages or if it is two independent phenomena.

<sup>245</sup> Steven Franks (5 Sep 2021, SLS-16 at Urbana Champaign) suggested the interesting idea that there is a general ban on clitics in initial position, which holds even for languages like Macedonian, where clause-initial clitics are grammatical, as there is no Tobler-Mussafia-effect like in Bulgarian. His point is that [pronominal] clitics are never initial in discourse, since pronominal clitics must always refer to an antecedent introduced earlier. Some open questions remain and require further consideration: Do we need to include cataphora into the picture? How should non-referential reflexives be treated? What is the significance for auxiliary and conditional clitics?

While Clitic First represents a placement pattern, which is primarily attributed to non-standard language, the next section will have a look into delayed clitic placement, which is recognised by language standards.

#### 4.4.2 Delayed Placement: Clitic Third and so Fourth

Another major complex of deviating clitic positioning, which received considerable attention in the literature, is subsumed under the label DELAYED (CLITIC) PLACEMENT (DP).<sup>246</sup> The term denominates the occurrence of a clitic or clitic cluster beyond second position, which poses a severe problem for the traditional view of Wackernagel cliticisation in Slavic. For instance, Bošković (2004: 39) explains (cf. also Bošković 2001: 65, with references):

“The traditional statement that SC clitics are second within their clause is actually incorrect. It is well known that certain elements, such as appositives, fronted heavy constituents, and parentheticals, can delay clitic placement, which results in clitics occurring further than the second position of their clause [...] clitics occur in the third and the fourth position of their clause.” (Bošković 2004: 39)

Even so, the interpretation of Delayed Placement is a contentious issue. Avgustinova & Oliva (1997) reinterpret Wackernagel’s law pragmatically for Czech and argue that the first position is a single communicative unit. Multiple syntactic constituents in front of the clitic cluster constitute a single constituent in terms of information structure. Consequently, clitics in putative Delayed Placement still occupy second position irrespective of the syntagmatic make-up of the preceding environment.<sup>247</sup> This generalisation basically applies to what Avgustinova & Oliva (1997) call sentential clitics, i.e. textbook instances of Wackernagel placement. Reviewing Czech grammar handbooks, they identify deviations of the Wackernagel rule in cases, where the clitics are contiguous with their verbal predicate. Based on this observation, Avgustinova & Oliva (1997: 26) treat clitic deviant positions as instances of verbal cliticisation similar (but not necessarily identical) to VA cliticisation in Bulgarian.<sup>248</sup>

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<sup>246</sup> Alternatively, one can find the terms *delayed (clitic) pronunciation* (cf. Franks 2009a: 729; Franks & King 2000: 342), *skipping* (cf. Milićević 2023: 69–72) or, more specifically, *embedded thirdness* (cf. Toman 1999: 213–214).

<sup>247</sup> There is resemblance with the occurrence of multiple constituents in the German prefield (cf. Müller 2003, 2007: 79–84)

<sup>248</sup> Avgustinova & Oliva (1997: 26) conjecture that Czech underwent a language change in the domain of cliticisation from “preferable” VA placement to “almost exclusive” 2P placement since the end of the 19th or beginning of the 20th century.

Kosta & Zimmerling (2014) adopt a strict notion of second position cliticisation and assume that clitics universally occur in CP. They basically follow Avgustinova & Oliva (1997) in that multiple pre-clitic constituents represent a single communicative unit. Among other things, they propose what they call “barrier rules” responsible for generating derived word orders, particularly verb-initial clauses and Delayed Placement (cf. Kosta & Zimmerling 2014: 483–484). Furthermore, Kosta & Zimmerling (2014: 461) argue that the syntactic configurations in (4.64) are generally prohibited in Slavic:

- (4.64) a. \*[V<sub>fin</sub> + O]=CL (ban of complex finite VPs [incl. *I*-participle] [V<sup>0</sup> + XP<sub>OBJ</sub>] before clitics)  
 b. \*[S + V<sub>fin</sub>]=CL (ban of subject NP and finite verb before clitics)  
 c. \*[V<sub>fin</sub> + Adv]=CL (ban of finite verb and an adverb before clitics)  
 d. \*[S + O]=CL (ban of subject and object NP before clitics)

The ban on syntagms consisting of verbal head + *x* (4.64a,c) trivially follows from a general constraint on rightwards clitic placement.<sup>249</sup> At the same time, the constraints \*[S + V<sub>fin</sub>]=CL and \*[S+O]=CL (4.64b, 4.64d) are incorrect, which will be discussed below.

Returning to the general issue of Delayed Placement, note that this construction is usually conceptualised as so-called CLITIC THIRD (3P), whereby the clitic cluster is not placed *in* the Wackernagel position but *after* it (cf. Franks and King 2000: 229–234 [Ch. 6.1.4.2]). Clitic Third has been generally thought to be restricted to subordinate clauses in Czech (cf. Fried 1994: 159–160; Toman 1999: 213–214; Short 1993a: 496; Veselovská 1995: §4.6, §11.3.5). As a result, a single clausal constituent appears between subordinator and clitics. It is well-known that Clitic Third correlates with distinctive information structure configurations (cf. Fried 1994: 159–160; Veselovská 1995: §4.6).<sup>250</sup> In (4.65a), the embedded subject *Petr* is topicalised, while the embedded object (*i revma* ‘(even) rheuma’ in (4.65b) is focused. This process of information structuring has been called *aktualizace* ‘actualisation’ in the Czech linguistic tradition (cf. e.g. Čechová et al. 1996: 310), such that the topicalised and focused constituents respectively in (5.65) are actualised under this view. The intervening constituent can be of arbitrary complexity, as exemplified in (4.65c). Note that the delayed reflexive clitic *se* is not adjacent to the verbal predicate but separated by the PP *s EU* ‘with the E.U.’ contra Avgustinova & Oliva (1997: 26).

<sup>249</sup> See Section 4.8 for restrictions on rightwards clitic placement.

<sup>250</sup> See Chapter 13 for the relation of information structure and clitic placement.



- (4.65) a. Helena říkala, že<sup>1</sup> [NP Petr ]<sub>TOP</sub><sup>2</sup> se<sup>3</sup> odstěhoval  
 Helen.NOM say.PST.SG.F that Petr.NOM.SG REFL.ACC move.PST.SG.M  
 ‘Peter, Helen said, had moved.’ (Cz; Fried 1994: 159)
- b. Věřil by=ste, že<sup>1</sup> [NP i revma ]<sub>FOC</sub><sup>2</sup> jsem<sup>3</sup> ztratil?  
 believe.PST.SG.M COND=AUX.2SG that and rheuma.ACC AUX.1SG lose.PST.SG.M  
 ‘Would you believe that I even got rid of rheume?’ (Cz; Lenertová 2004: 137)
- c. Grégr včera sdělil, že<sup>1</sup> [PP o [NP přechodném  
 Grégr.NOM yesterday say.PST.SG.M that about transitional.LOC.SG  
 období [PP při [NP liberalizaci [NP energetického  
 period.LOC.SG by liberalisation.LOC.SG energy.ADJ.GEN.SG  
 trhu ]]]]]<sup>2</sup> se<sup>3</sup> s EU stále jedná a ...  
 market.GEN.SG REFL.ACC with EU.INS still negotiate.PRS.3SG and  
 ‘Grégr said yesterday that the transitional period in energy markets liberalisation is still  
 being negotiated with E.U. and ...’ (Cz; Hana 2007: 108)

Beyond being constrained to subordinate clauses, some further restrictions have been put forward. It has been argued that Clitic Third is licensed only under the complementizer *že* ‘that’ like in (4.66). Veselovská (1995: §4.6, §11.3.5) is particularly affirmative about that. Toman (1999: 214) argues that it is impossible to separate *a(by)* and *-li* from other clitics. However, his example (4.67) solely presents the non-disintegrability of the subordinator *aby-* into its etymological components *a* ‘and’ + *by* ‘COND’. Abstracting away from this issue, Toman’s (1999: 214) conclusion is that some subordinators are lexically specified for allowing Clitic Third facultatively, while others prohibit it, e.g. *aby*. This makes perfect sense from the perspective that clitics form a compact, impenetrable syntagm and the subordinators *aby* and *kdyby* involve the conditional clitic *by* (cf. Veselovská 1995: §4.7).<sup>251</sup> This reasoning accounts for the ungrammaticality of (4.66) and (4.67c). Moreover, it has been repeatedly mentioned that only non-verbal elements may intervene between subordinator and the clitic (cf. Adam 2019: 14, 23; Lenertová 2004: 145; Veselovská 1995: §11.3.5). Verbs are assessed as ungrammatical in this position, cf. (4.68).

<sup>251</sup> See Section 4.5 for clitic clustering.

(4.66) \*Ptal se, jestli<sup>1</sup> [NP Petr]<sup>2</sup> **mu**<sup>3</sup> to  
 ask.PST.SG.M REFL.ACC whether Petr.NOM him.DAT this.ACC.SG.N  
 ne-dal.  
 NEG-give.PST.SG.M  
 ‘He asked whether Peter gave it to him.’ (Cz; Veselovská 1995: §9.3.5, ex. [39a])

(4.67) a. Řekli nám, a-**bychom** příště raději odešli.  
 tell.PST.PL.VIR us.DAT that-COND.1PL next\_time better leave.PST.PL.VIR  
 b. \*Řekli nám, a [Adv příště] **bychom** raději odešli.  
 ‘They told us we should leave next time.’ (Cz; Toman 1999: 214)  
 c. \*a-**by** [NP Petr] **mu** to ne-dal.  
 that-COND Petr.NOM him.DAT this.ACC.SG NEG-give.PST.SG.M  
 ‘... that Peter would not give it to him.’ (Cz; Veselovská 1995: §4.7, ex. [34a])

(4.68) \*Řekl, že<sup>1</sup> [V ne-přál]<sup>2</sup> **si**<sup>3</sup> to.  
 say.PST.3SG.M that NEG-wish.PST.SG.M REFL.DAT this.ACC.SG  
 ‘He said that he did not want it.’ (Cz; Veselovská 1995: §9.3.5, ex. [40c])

The structural restrictions that have been posited for 3P in the Czech embedded clause turn out to be incorrect on closer inspection. Hana (2007: 107–110) and Lenertová (2004: 137) provide counterexamples to the putative constraint on Clitic Third being confined to subordinate clauses introduced by the complementizer *že*. In (4.69a), the focused NP *třeba Petr* ‘perhaps Peter’ occurs between the subordinator *jestli* ‘if’ and the clitic cluster. In (4.69b), it is the subordinator *protože* ‘because’, which is separated from the pronominal clitic *mi* ‘me.DAT’ by the (contrastively) topicalised NP *Kanad’ané* ‘Canadians’ (cf. also Junghanns 2021: 177).

(4.69) a. Ptal se, jestli<sup>1</sup> [NP třeba Petr]<sub>FOC</sub><sup>2</sup> **by**<sup>3</sup> **mu**<sup>3</sup>  
 ask.PST.SG.M REFL.ACC whether perhaps Petr.NOM COND him.DAT  
 to ne-dal.  
 this.ACC.SG.N NEG-give.PST.SG.M  
 ‘He asked whether perhaps Peter would not give it to him.’

- b. Ne-jsem překvapen, že se na to ptáte,  
 NEG-be.PRS.1SG surprise.PASS.PTCP.SG.M that REFL.ACC on this.ACC.SG ask.PRS.2PL  
 protože<sup>1</sup> [<sub>NP</sub> Kanad'ané ]<sub>TOP</sub><sup>2</sup> **mi**<sup>3</sup> dávají tuhle otázku pořád  
 because Canadian.NOM.PL me.DAT give.PRS.3PL this question.ACC.SG always  
 dokola.  
 around  
 'I am not surprised you ask me about this, because Canadians ask me that question all the  
 time.'  
 (Cz; Hana 2007: 108)

Furthermore, Hana (2007: 107–110) demonstrates that it is possible to place the verb between subordinator and clitic. Adam (2019: 18–21) elaborates that infinitives + object can precede clitics in independent and subordinate clauses, while this is not possible with syntagms of *l*-participle + object and present tense verb + object (cf. also Hana 2007: 110; Lenertová 2004: 144–145; Rosen 2001: 215). Consider the contrast between (4.70) and (4.71). There is an important structural difference between the examples, however. The subordinate clauses in (4.71) are simple clauses, whereas (4.70) presents a complex clause. The simple clause examples are subject to general restrictions on rightwards clitic positioning.<sup>252</sup> The complex clause in (4.70) provides the environment for clitic climbing.<sup>253</sup> In short, the pronominal clitic *mu* is a dependent of the infinitive, but it is realised in the finite domain of the modal verb *chceme*. It is therefore possible to treat clitic placement and topicalisation of the VP *udělit cenu* '(to) award the price' as independent of one another. Indeed, the pronominal clitic *mu* occurs in front (to the left) of the matrix verb, which is usually taken as a clear signal for clitic climbing.

- (4.70) Říkají, že<sup>1</sup> [<sub>VP</sub> udělit<sub>2</sub> cenu ]<sup>2</sup> **mu**<sub>2</sub><sup>3</sup> chceme<sub>1</sub> tajně.  
 say.PRS.3PL that award.INF price.ACC.SG him.DAT want.PRS.1PL secretly  
 'They say that we want to award him the price secretly.'  
 (Cz; Adam 2019: 20)

- (4.71) a. \*Říkají, že<sup>1</sup> [<sub>VP</sub> udělíme<sub>1</sub> cenu ]<sup>2</sup> **mu**<sub>1</sub><sup>3</sup> tajně.  
 say.PRS.3PL that award.PRS.1PL price.ACC.SG him.DAT secretly  
 Intended: 'They say that we are awarding him the price secretly.'  
 b. \*Říkají, že<sup>1</sup> [<sub>VP</sub> udělili<sub>1</sub> cenu ]<sup>2</sup> **jsme**<sub>1</sub><sup>3</sup> **mu**<sub>1</sub><sup>3</sup> tajně.  
 say.PRS.3PL that award.PST.PL.VIR price.ACC.SG AUX.1PL him.DAT secretly  
 Intended: 'They say that we awarded him the price secretly.'  
 (Cz; Adam 2019: 19)

<sup>252</sup> See Section 4.8 for the rightwards positional restriction.

<sup>253</sup> See Section 4.7 for clitic climbing.

Accordingly, verbal placement is not restricted in pre-clitic placement per se. Examples (4.72) and (4.73) provide that finite verbs and *l*-participles are licit in the second position of the subordinate clause despite Clitic Third (cf. Hana 2007: 109–110 for more examples and discussion). This is irrespective of whether the verb is the clitic’s governor or not. The dative clitic *mu* in (4.72a) is a free dative dependent of the verb *sviští* ‘whiz.PRS.3PL’, while *ho* in (4.72b) is governed by the infinitive *objevit* but preceded by the finite matrix verb *stačí* ‘suffice.PRS.3SG’. The *l*-participle *vykonala* and the auxiliary clitic *jsem* are in a dependence relation in (4.73a). Example (4.73b) exhibits the dative clitic *mi* (which is a *dativus incommodi*) following the governing verb *prolétla*.<sup>254</sup> Finally, the clitic *ho* is preceded by the matrix verb *neváhali* in (4.73c), while being a dependent of the infinitive *vyměnit*. Data like (4.72)–(4.73) provide evidence against the last-resort account of pre-clitic verb placement. It has been proposed that positioning the verb to clause-initial position (so-called long head movement) occurs as a last-resort operation to provide the clitic with a prop, if no other material is available to host it (cf. Franks & King 2000: 300–302, 362–365 for discussion). This reasoning does not work, if verbs are positioned before the clitic cluster in subordinate clauses, because subordinators are eligible hosts for clitics, such that there is no need for repairing an otherwise ill-formed structure.

- (4.72) a. (V tom okamžiku zapomínal,) že<sup>1</sup> sviští<sup>2</sup> mu<sup>3</sup>  
in this.LOC.SG moment.LOC.SG forget.PST.3SG.M that whiz.PRS.3PL him.DAT  
nad hlavou granáty a šrapnely [...] <sup>255</sup>  
above head.INS.SG grenade.NOM.PL and shrapnel.NOM.PL  
‘(At that moment he forgot) that grenades and shrapnels were whizzing over his head ...’
- b. (Postupně jsme pochopili, že každý člověk  
successively AUX.1PL understand.PST.PL.VIR that every.NOM.SG.M human.NOM.SG  
má půvab v sobě a) že<sup>1</sup> stačí<sup>2</sup> ho<sup>3</sup> jen  
have.PRS.3SG grace.ACC.SG in self.LOC.SG and that suffice.PRS.3SG him.ACC only  
objevit<sub>2</sub>.  
discover.INF  
‘(Step by step we understood that every human has grace within and) that one only needs  
to unveil it.’ (Cz; ČNK syn\_v8)

<sup>254</sup> The traditional notions *dativus commodi* and *dativus incommodi* often are subsumed under the term *dative of interest* in contemporary Anglophone linguistics.

<sup>255</sup> CQL query: [word="že"] [tag="VB.\*"] & lemma!="být"]  
[word="mu|ho|mi|tě|jsem|jsi|jsme|jste"]

- (4.73) a. (Myslím,) že<sup>1</sup> **vykonala**<sup>2</sup> **jsem**<sup>2</sup> teď nejvyšší čin  
 think.PRS.1SG that achieve.PST.SG.F AUX.1SG now highest deed.ACC.SG  
 svého života!<sup>256</sup>  
 own's.GEN.SG life.GEN.SG  
 '(I think) that I've now accomplished the highest achievement of my life.'
- b. (Ihned sedl jsem si na kámen a  
 immediately sit\_down.PST.SG.M AUX.1SG REFL.DAT on stone.ACC and  
 s-hledal jsem,) že<sup>1</sup> **prolétla**<sup>2</sup> **mi**<sup>3</sup> kulka  
 PFV-search.PST.SG.M AUX.1SG that fly\_through.PST.SG.F me.DAT bullet.NOM.SG  
 lýtkem.  
 calf.INS.SG  
 '(At once I sat down on a stone and found) that a bullet shot through my calf.'
- c. (A to v takové míře,) že<sup>1</sup> **ne-váhali**<sup>2</sup> **ho**<sup>3</sup>  
 and this.NOM.SG.N in such.LOC.SG scale.LOC.SG that NEG-hesitate.PST.PL.VIR him.ACC  
 vyměnit<sub>2</sub> jen pár hodin před včerejší  
 replace only pair.ACC.SG hour.GEN.PL before yesterday.ADJ.INS.SG.F  
 relací!  
 broadcast.INS.SG  
 '(And to such an extent) that they didn't hesitate to replace him just a few hours before  
 yesterday's broadcast.' (Cz; ČNK syn\_v8)

Another contentious issue is the limitation of DP to 3P. Hana (2007: 110–111) demonstrates that clitics may be further delayed. His example (4.74a) shows Clitic Fourth (4P) and contradicts Kosta & Zimmerling's (2014: 461) restriction in (4.64d): \*[S + O]=CL. The NPs *Petr* and *Pavlovi* represent subject and indirect object. Junghanns (2021) cites evidence for Clitic Fifth (5P) echoed in (4.74b–c).

- (4.74) a. Helena říkala, že<sup>1</sup> [<sub>NP</sub> Petr ]<sup>2</sup> [<sub>NP</sub> Pavlovi ]<sup>3</sup> **by**<sup>4</sup> to  
 Helena.NOM tell.PST.SG.F that Petr.NOM Pavel.DAT COND this.ACC.SG.N  
 dal, ale Honza Marii ne.  
 give.PST.SG.M but Honza.NOM Marie.DAT NEG  
 'Helena said that Peter would give it to Pavel, but Honza would not to Marie.'  
 (Cz; Hana 2007: 111)

<sup>256</sup> CQL query: [word="že"] [tag="Vp.\*"] [word="mu|ho|mi|tě|jsem|jsi|jsme|jste"]

- b. [...] že<sup>1</sup> [NP pan B. ]<sup>2</sup> [PP na dotčené střeše ]<sup>3</sup>  
 that [ Mr. B.].NOM on respective.LOC.SG roof.LOC.SG  
 [AdvP vskutku ]<sup>4</sup> se<sup>5</sup> ocitl.  
 really REFL.ACC get.PST.SG.M  
 ‘... that Mr. B. really found himself on the respective roof.’
- c. ([AdvP Zatím ]<sup>1</sup> [NP polní kurát ]<sup>2</sup> si<sup>3</sup> v  
 meanwhile field.ADV.NOM.SG chaplain.NOM.SG REFL.DAT in  
 katechismu zopakoval,) co<sup>1</sup> [AdvP kdysi ]<sup>2</sup> [PP v  
 catechism.LOC.SG recapitulate.PST.SG.M what.NOM once in  
 semináři ]<sup>3</sup> [V ne-utkvělo ]<sup>4</sup> mu<sup>5</sup> v paměti ]].  
 seminary.LOC.SG NEG-get\_stuck.PST.SG.N him.DAT in memory.LOC.SG  
 ‘(Meanwhile, the military chaplain recapitulated,) what once didn’t linger in his memory  
 from the seminary.’ (Cz; Junghanns 2021: 178)

Finally, there seems to be predominant consensus that Delayed Placement is bound to subordinate clauses. However, Avgustinova & Oliva (1997: 41–46), Hana (2007: 103–106), and Junghanns (2021: 177–178) discuss the occurrence of multiple constituents before the clitic cluster in independent clauses.<sup>257</sup> Example (4.75a) provides 3P after subject NP + adverb in an independent root clause, while we encounter 5P in (4.75b).

- (4.75) a. [NP Plukovník ]<sup>1</sup> [AdvP vlastnoručně ]<sup>2</sup> ho<sup>3</sup> táhl do  
 colonel.NOM.SG by\_own\_hand him.ACC pull.PST.SG.M to  
 kasáren k potrestání [...]  
 barracks.GEN.SG towards punishing.DAT.SG  
 ‘The colonel pulled him with his own hands to the barracks to punish [him] ...’  
 (Cz; Junghanns 2021: 178)

<sup>257</sup> I abstract away from Avgustinova & Oliva’s (1997) approach to conflate all pre-clitic phrases in a single communicative constituent. From a purely syntactic perspective, the attested structures involve multiple, independently permutable constituents.

b. ([...] v tom druhém ne-odchází nikdo, protože  
in this.LOC.SG second.LOC.SG NEG-pass.PRS.3SG nobody.NOM.SG because  
v něm nikdo není,) [AdvP zde]<sup>1</sup> jen<sup>2</sup> [NP něco ]<sup>3</sup>  
in it.LOC.SG nobody.NOM.SG NEG.be.PRS.SG here only something.NOM  
[PP v něco ]<sup>4</sup> se<sup>5</sup> mění].  
in something.ACC REFL.ACC change.PRS.3SG  
‘... (in the second [world], nobody passes away, because there isn’t anyone inside,) here,  
only something changes into something.’ (Cz; Junghanns 2021: 177)

It appears then that Delayed Placement is not a rare peculiarity of subordinate clauses introduced by *že*. We encounter a range of Delayed Placement patterns in Czech both with subordinate and root clauses. In fact, “rareness” is a questionable concept here. Based on a simple corpus query, Hana (2007: 109) quantifies the absolute frequencies of 2P vs. 3P with an intervening noun for the reflexive clitic *se* in *že*-clauses. Table 4.1 summarises his results.<sup>258</sup>

Table 4.1: Frequency of reflexive *se* in 2P and 3P under subordinator *že* (Hana 2007: 109).

| Subordinator | Abs. freq. 2P<br>[SUB CL N] | Abs. freq. 3P<br>[SUB N CL] | Sum        | Rel. freq. 2P | Rel. freq. 3P |
|--------------|-----------------------------|-----------------------------|------------|---------------|---------------|
| <i>že</i>    | ca. 10,000                  | ca. 6,000                   | ca. 16,000 | 62.5%         | 37.5%         |

Based on the figures in Table 4.1, Hana (2007: 109) argues that “neither of them [= 2P, 3P] is rare”. I replicated Hana’s (2007) query in the ČNK’s SYN v8 database and included subordinators *jestli* and *protože*.<sup>259</sup> These queries were repeated for the most frequent pronominal constant clitic *ho*.<sup>260,261</sup> The results for the *se* and *ho* queries are summarised in Table 4.2 and Table 4.3 respectively.

<sup>258</sup> It is unclear, what set of data or which (sub)corpus the numbers exactly represent. Note also that the calculation is confined to the comparison of 2P and 3P and does not factor 4P and 5P in. Nevertheless, to my knowledge, Hana (2007) is the first to provide any quantification for variation in clitic placement patterns in contemporary Czech. Note that I. Kulik (2019: 140) found a statistical ratio of ca. 60% for 2P and ca. 28% for 3P with the Polish pronominal clitic *mu*. The measures suggest a convergent trend. I. Kulik’s (2019) ratios for 2P and 3P are necessarily smaller than Hana’s (2007), because the former examined a wider range of positions (1P to 8P). Note also that Hana (2007) examines written language, whereas I. Kulik (2019) surveys spoken language.

<sup>259</sup> CQL queries:

- (i) a. [word="že"] [word="se" & tag="P7.\*"] [tag="N.\*"]  
b. [word="že"] [tag="N.\*"] [word="se" & tag="P7.\*"]
- (ii) a. [word="jestli"] [word="se" & tag="P7.\*"] [tag="N.\*"]  
b. [word="jestli"] [tag="N.\*"] [word="se" & tag="P7.\*"]
- (iii) a. [word="protože"] [word="se" & tag="P7.\*"] [tag="N.\*"]  
b. [word="protože"] [tag="N.\*"] [word="se" & tag="P7.\*"]

<sup>260</sup> Compare the instances per millions (i.p.m.) from SYN v8: *ho* (779.33 i.p.m.), *mu* (676.62 i.p.m.), *mi* (669.17 i.p.m.), *ti* (54.05 i.p.m.), *tě* (31.78 i.p.m.).

<sup>261</sup> CQL queries:

- (i) a. [word="že"] [word="ho" & tag="PH.\*"] [tag="N.\*"]

Table 4.2: Frequency of reflexive *se* in 2P and 3P under subordinators *že*, *jestli*, *protože*.

| Subordinator   | Abs. freq. 2P<br>[SUB CL N] | Abs. freq. 3P<br>[SUB N CL] | Sum     | Rel. freq. 2P | Rel. freq. 3P |
|----------------|-----------------------------|-----------------------------|---------|---------------|---------------|
| <i>že</i>      | 499,123                     | 229,046                     | 728,169 | 68.5%         | 31.5%         |
| <i>jestli</i>  | 13,502                      | 41                          | 13,543  | 99.7%         | 0.3%          |
| <i>protože</i> | 14,762                      | 28,855                      | 43,617  | 33.8%         | 66.2%         |

Table 4.3: Frequency of pronominal *ho* in 2P and 3P under subordinators *že*, *jestli*, *protože*.<sup>262</sup>

| Subordinator   | Abs. freq. 2P<br>[SUB CL N] | Abs. freq. 3P<br>[SUB N CL] | Sum       | Rel. freq. 2P | Rel. freq. 3P |
|----------------|-----------------------------|-----------------------------|-----------|---------------|---------------|
| <i>že</i>      | 28,700                      | 3,675                       | 32,375    | 88.7%         | 11.3%         |
| <i>jestli</i>  | 879                         | (1)                         | 879 (880) | 100%          | 0%            |
| <i>protože</i> | 1,924                       | 919                         | 2,843     | 67.7%         | 32.3%         |

Patterns differ across clitics and subordinators. The reflexive preferably occurs in 2P under *že* (ca.  $\frac{2}{3}$  of occurrences), but in 3P under *protože* (ca.  $\frac{2}{3}$  of occurrences). The pronominal clitic, too, occurs predominantly in 2P under *že*, but with a different ratio than *se* (ca.  $\frac{9}{10}$  of occurrences). The 2P pattern is also favoured under *protože* ( $\frac{2}{3}$  of occurrences). Both clitics exhibit similar ratios only for *jestli* approximating 100% of 2P placement. Provided the ratios for 3P, Hana's (2007: 109) assessment that Clitic Third is not rare in contemporary Czech is justified. One can see that such a quantitative approach is promising. A more sophisticated query will be elaborated in Chapter 12.

As is commonly known, Delayed Placement is not a phenomenon specific for Czech. Particular attention has been also paid to BCMS, which perhaps has been the most discussed language in this respect in Slavic linguistics. Particular discussion and further references can be found in Franks & King (2000), Bošković (2001, 2004), Ćavar & Seiss (2011), Kolaković et al. (2022). I have little to say about Delayed Placement in BCMS, so I will restrict myself to citing a few observations from the literature. While DP has been attributed to processes of information structure in Czech, it seems to be widely accepted that delayed clitic placement is caused by prosodic factors in BCMS (but cf. Ćavar & Seiss 2011). BCMS clitics are assumed to be strictly enclitic, they cannot occur after

- 
- b. [word="že"] [tag="N.\*"] [word="ho" & tag="PH.\*"]  
(ii) a. [word="jestli"] [word="ho" & tag="PH.\*"] [tag="N.\*"]  
b. [word="jestli"] [tag="N.\*"] [word="ho" & tag="PH.\*"]  
(iii) a. [word="protože"] [word="ho" & tag="PH.\*"] [tag="N.\*"]  
b. [word="protože"] [tag="N.\*"] [word="ho" & tag="PH.\*"]

<sup>262</sup> There is exactly one hit for [*jestli* N *ho*], but it is a tagging error. The respective example does not present a noun in 2P. I give the occurrence frequency in parenthesis rather than to suppress it, because I cannot account for potential tagging errors in the other queries.



prosodic margins usually referred to as “pauses” (cf. Bennett 1986: 8, 1987: 272; Bošković 2001, esp. 80–96; Čamdžić & Hudson 2007: 7, 12–13; Franks & King 2000: 229–234; Rubadeau 1996: 30).

Example (4.76a) shows the future auxiliary clitic *će* in clausal third position, but prosodically, it is in second position after the pause indicated by the hash mark (#), which I have added for the sake of clarity. The same situation is found in (4.76b) for the reflexive clitic *se*. In both examples, the clitic follows the verb, but while the complex subject NP occupies clause-initial position in the former, it is a temporal adverbial NP in the latter. Franks & King (2000: 229) remark that “the initial phrase is often topicalised or otherwise receives special discourse status.” I indicated this point in (4.76b). The examples in (4.77) show the successive delay of the clitic cluster from second to fourth position (without marking of prosodic boundaries). Occurrences of a further delay are reported, too. The Croatian example (4.78a) is from Reinkowski’s (2001) newspaper corpus and provides Clitic Fourth, while (4.78b) with Clitic Fifth is provided by Milićević (2023) for Serbian (Srp).

- (4.76) a. [[NP Brat ] i [NP njegova žena ]]<sup>1</sup> # [v ići ]<sup>2</sup> **će**<sup>3</sup>  
 brother.NOM.SG and his.NOM.SG wife.NOM.SG go.INF AUX.FUT.3PL  
 u bioskop.  
 in cinema.ACC.SG  
 ‘My brother and his wife will go to the movies.’ (BCMS; Rubadeau 1996: 30)
- b. [NP Svaki dan ]<sub>TOP</sub><sup>1</sup> # [v dogodi ]<sup>2</sup> **se**<sup>3</sup> najmanje jedna  
 every day happen.PRS.3SG REFL.ACC at\_least one.NOM.SG.F  
 prometna nezgoda.  
 traffic.ADJ.NOM.SG.F accident.NOM.SG  
 ‘Every day at least one traffic accident took place.’ (BCMS; Franks & King 2000: 229)
- (4.77) a. [NP Moj student ]<sup>1</sup> **ga**<sup>2</sup> **je**<sup>2</sup> našao.  
 my student him.ACC AUX.3SG find.PST.SG.M  
 ‘My student found him.’ (BCMS; Browne 2014: 83)
- b. [NP Prošle ili pretprošle godine ]<sup>1</sup> [NP moj  
 last.GEN.SG or before\_last.GEN.SG year.GEN.SG my.NOM.SG.M  
 student ]<sup>2</sup> **ga**<sup>3</sup> **je**<sup>3</sup> našao.  
 student.NOM.SG him.ACC AUX.3SG find.PST.SG.M
- c. [NP Prošle ili pretprošle godine ]<sup>1</sup> [NP moj student ]<sup>2</sup> [v našao ]<sup>3</sup> **ga**<sup>4</sup> **je**<sup>4</sup>.  
 ‘Last year or the year before last, my student found him.’ (BCMS; Browne 2014: 84)

- (4.78) a. [PP U poslijepodnevnom radu ]<sup>1</sup> [NP sudionici ovoga  
in afternoon.ADJ.LOC.SG work.LOC.SG participant.NOM.PL this.GEN.SG.M  
skupa ]<sup>2</sup> [V razmotrili ]<sup>3</sup> **su**<sup>4</sup> modele  
gathering.GEN.SG consider.PST.PL.M AUX.3PL model.ACC.PL  
poslovne suradnje.  
business.ADJ.GEN.SG.F cooperation.GEN.SG  
'During work in the afternoon, participants of this gathering considered models of  
business cooperation.'  
(Hrv; Reinkowski 2001: 175)
- b. [PP U tim okolnostima ]<sup>1</sup> [NP ideje Čomskog ]<sup>2</sup>  
in this.LOC.PL.F circumstance.LOC.PL idea.NOM.PL Chomsky.GEN  
[NP dalju razradu ]<sup>3</sup> [V doživele ]<sup>4</sup> **su**<sup>5</sup> u  
further.ACC.SG.F elaboration.ACC.SG experience.PST.PL.F AUX.3PL in  
radovima ...  
in work.LOC.PL  
'Under those circumstances, Chomsky's ideas were further elaborated in the work ...'  
(Srp; Milićević 2023: 70)

Delayed Placement in subordinate clauses is presented in (4.79)–(4.80). We find Clitic Third with an intervening verb, noun, and adverb in (4.79a), (4.79b), and (4.79c) respectively (judgement for [4.79c] as in the original). While most examples in the literature show subordinate clauses introduced by the complementizer *da* 'that', (4.79a–b) provide the complementizer *jer* 'because'. Note that example (4.79a) contradicts Ćavar & Wilder's (1994: 30) statement that "V[erb] fronting before the clitic-cluster is a root-phenomenon – [...] not possible in embedded finite clauses, either for finite or non-finite verbs." This statement coincides with what has – incorrectly – been claimed for Czech above. Example (4.80) provides Clitic Fifth in an embedded environment. The prosodic margins follow the cited source and show that, while the clitic is in clausal fifth position, it is second in its prosodic domain.

- (4.79) a. Raduj se [s jer]<sup>1</sup> [V došao ]<sup>2</sup> **ti**<sup>3</sup> **je**<sup>3</sup>  
rejoice.IMP.2SG REFL.ACC because come.PST.SG.M you.DAT AUX.3SG  
brat ].  
brother.NOM.SG
- b. Raduj se [s jer]<sup>1</sup> [NP brat ]<sup>2</sup> **ti**<sup>3</sup> **je**<sup>3</sup> došao ].  
'Rejoice because your brother has come.'  
(BCMS; Radanović-Kocić 1988: 101)

c. <sup>?</sup> Marina misli [s da<sup>1</sup> [AdvP sutra ]<sup>2</sup> je<sup>3</sup> ključni  
 Marina.NOM think.PRS.3SG that tomorrow be.PRS.3SG.CL key.ADJ.NOM.SG  
 dan ].  
 day.NOM.SG

‘Marina believes that tomorrow is the decisive day.’ (BCMS; Progovac 1996: 414, n. 3)

(4.80) Jovan je objavio [s da<sup>1</sup> [Adv juče ]<sup>2</sup> # [pp na autobuskoj  
 Jovan.NOM AUX.3SG announce.PST.SG.M that yesterday on bus.ADJ.LOC.SG.F  
 stanici ]<sup>3</sup> # [NP neko]<sup>4</sup> ga<sup>5</sup> je<sup>5</sup> opljačkao ].  
 station.LOC.SG someone him.ACC AUX.3SG rob.PST.SG.M

‘Jovan announced that yesterday someone robbed him at the bus station.’

(BCMS; Franks & King 2000: 232)

It appears that the distribution of prosodic boundaries affects the distribution of clitics in BCMS, because they avoid post-pausal position. However, recall from Section 4.2.1 and Section 4.4.1 that BCMS clitics are attested in proclisis. The question thus arises of why BCMS clitics should avoid a position following a pause, at all. For the sake of completeness, two lines of thinking can be found in the literature for the emergence of proclisis in BCMS, namely language change and language contact (cf. Bošković 2001: 167 for the former; Kolaković et al. 2022: 162 for the latter). I leave this issue open here.<sup>263</sup>

The other South Slavic language of interest here, Slovene, has not been reported with Delayed Placement. Notably, as Slovene special clitics are ambiclitics, delayed positioning is not contingent on prosodic breaks (cf. Bennet 1986: 7; Franks & King 2000; Golden 2003: 214–215). However, Delayed Placement is attested in Slovene. Example (4.81) exhibits embedded Clitic Third with an intervening verb.<sup>264</sup>

(4.81) In prav zahrepenel, da<sup>1</sup> **daš**<sup>2</sup> **mu**<sup>3</sup> svoje  
 and really desire.PST.SG.M that give.PRS.2SG him.ACC one’s.POSS.ACC.SG.N  
 srce.  
 heart.ACC.SG

‘And he really desired that you give him your heart.’

(Slv; slWaC)

<sup>263</sup> Note that Kolaković et al. (2022: 106–108) discuss that the prosodic reasoning is consistently found in BCMS standard grammar books. Bennett (1986, 1987) grounds his analysis on literary language. See Section 6.3 for a discussion of the related issue of linguistic competence vs. prescriptive norm.

<sup>264</sup> CQL query: [tag="Vd.\*"] [tag="Gg.s.\*"] [word="mu|ga"]

Coming back to West Slavic finally, it is repeatedly mentioned that Polish clitics exhibit a high degree of distributional freedom. Abstracting away from the auxiliary clitics, consider (4.82) and (4.83), which provide various grammatical positions for the pronominal clitic *go* and the reflexive clitic *się* respectively. The round brackets in (4.83) indicate that any of the positions is licit for *się*, although clause-final position is unusual and sounds awkward for native speakers.<sup>265</sup> The position of the clitics seems to be arbitrary with second to fourth position in (4.82) and second to fifth position in (4.83). The latter exemplifies that placement flexibility is independent of the reflexive type, either a lexical pleonasm of an inherently reflexive verb (4.83a), impersonal marker (4.83b), anti-causative (middle) marker (4.83c), or argument clitic of a truly reflexive verb (4.83d). Example (4.84) is from the spoken component of the NKJP and presents Clitic Fourth of the dative clitic *mu* in a subordinate clause introduced by *że* ‘that’.

- (4.82) a. [AdvP Teraz ]<sup>1</sup> **go**<sup>2</sup> często spotykam.  
           now       him.ACC often meet.PRS.1SG  
           ‘I often meet him now.’  
       b. [AdvP Teraz ]<sup>1</sup> [AdvP często ]<sup>2</sup> **go**<sup>3</sup> spotykam.  
       c. [AdvP Teraz ]<sup>1</sup> [AdvP często ]<sup>2</sup> [V spotykam ]<sup>3</sup> **go**<sup>4</sup> w mieście  
           now               often               meet.PRS.1SG him.ACC in town.LOC.SG  
           ‘I often meet him now in the town.’ (Pol; Witkoś 1998: 159)

- (4.83) a. [NP Sytuacja ]<sup>1</sup> (**się**) [AdvP jutro ]<sup>2</sup> (**się**) [V poprawi ]<sup>3</sup> (**się**).  
           Situation.NOM.SG REFL.ACC tomorrow improve.PRS.3SG  
           ‘The situation will improve tomorrow.’  
       b. [NP Książkę ]<sup>1</sup> (**się**) [AdvP dobrze ]<sup>2</sup> (**się**) [V czytało ]<sup>3</sup> (**się**).  
           book.ACC.SG REFL.ACC well read.PST.SG.N  
           ‘One enjoyed reading this book.’  
       c. [NP Lody ]<sup>1</sup> (**się**) [AdvP prędko ]<sup>2</sup> (**się**) [V roztopiły ]<sup>3</sup> (**się**).  
           ice-cream.NOM.PL REFL.ACC quickly melt.PST.PL.NVIR  
           ‘The ice-cream melted quickly.’

<sup>265</sup> Standard and reference grammars agree in that clause-final position is dispreferred to impossible for clitics except in two-word utterances with clause-initial verb placement (cf. P. Bąk 1977: 434; Bartnicka et al. 2004: 293; Sadowska 2012: 279; Skibicki 2016: 452; Swan 2002: 155, 410).

d. [AdvP Wczoraj ]<sup>1</sup> (się) [NP Piotrek ]<sup>2</sup> (się) [AdvP późno ]<sup>3</sup> (się)  
 yesterday REFL.ACC Peter.NOM.SG late  
 [v obudził ]<sup>4</sup> (się).  
 wake\_up.PST.SG.M  
 ‘Yesterday Peter woke up late.’ (Pol; Kupść 2000b: 102)

(4.84) [...] mam wrażenie [s że<sup>1</sup> [NP firma apel ]<sup>2</sup>  
 have.PRS.1SG impression.ACC.SG that company.NOM.SG Apple.NOM  
 [v nie płaci ]<sup>3</sup> **mu**<sup>4</sup> w żaden sposób ]  
 NEG pay.PRS.3SG him.DAT in no.ACC.SG.M manner.ACC.SG  
 ‘... I have the impression that the company Apple does not pay him in any way.’  
 (Pol; NKJP pelcra)

Slovak examples of Delayed Placement are cited in (4.85). Clitic Third of a single reflexive clitic and a clitic cluster in a root clause are found in (4.85a) and (4.85b) respectively. The subordinate clause in (4.85c) exhibits embedded Clitic Third (again of the reflexive). Example (3.73) from Section 3.12.2 repeated in (4.85d) below attests embedded Clitic Fourth. It contradicts Franks & King’s (2000: 135) claim that “the verb is not a possible host for the clitics in subordinate clauses”. What is more, if one considers the examples cited in this section then there is evidence for verb-adjacent placement of clitics in post-verbal position in each language taken into account. Note that, by example (4.85b), Kosta & Zimmerling (2014: 461) cite a counterexample to their own generalisation in (4.64b), i.e. \*[S + V<sub>fin</sub>]=CL, which prohibits a syntagms of subject + finite verb before clitics.

(4.85) a. [pp Z iniciatívy Jozefa Kohúta ]<sup>1</sup> [v založil ]<sup>2</sup> **sa**<sup>3</sup>  
 of initiative.GEN.SG Jozef.GEN Kohút.GEN found.PST.SG.M REFL.ACC  
 v Martine ... hasičský zbor.  
 in Martin.LOC firefighter.ADJ.NOM.SG brigade.NOM.SG  
 ‘A fire brigade was founded in Martin on the initiative of Jozef Kohút.’  
 (Slk; Short 1993b: 567)

b. [NP Husté čierne vlasy ]<sup>1</sup> [v kaderili ]<sup>2</sup> **sa**<sup>3</sup> **mu**<sup>3</sup>  
 thick.NOM.PL black.NOM.PL hair.NOM.PL curl.PST.3PL REFL.ACC him.DAT  
 za ušami.  
 behind ear.INS.PL  
 ‘Thick black hair curled behind his ears.’ (Slk; Kosta & Zimmerling 2014: 465)

- c. S. Markovič                      spomína,                      že<sup>1</sup> [AdvP kedysi ]<sup>2</sup>                      **sa**<sup>3</sup>                      náklad  
[S. Markovič].NOM remember.PRS.3SG that                      once                      REFL.ACC load.NOM.SG  
nosil                      hore síce                      ťažšie,                      ale oveľa romantickejšie  
carry.PST.SG.M up                      albeit more\_difficult but far                      more\_romantic  
‘S. Markovič recalls that carrying the load up used to be more difficult, though much  
more romantic.’                      (Slk; Ivanová et al. 2021: 355, n. 1)
- d. [AdvP Tak ]<sup>1</sup> [V stalo ]<sup>2</sup>                      **sa**<sup>3</sup>,                      [S že<sup>1</sup> [NP cestu ]<sup>2</sup>  
so                      happen.PST.SG.N REFL.ACC                      that                      journey.ACC.SG  
[V skrížili ]<sup>3</sup>                      **mu**<sup>4</sup>                      i                      priatelia                      hraničiaru ]                      [...]  
cross.PST.3PL him.ACC and friend.NOM.PL border\_guard.NOM.PL  
So it happened that [our] “friends”, the border guards, too, crossed his trip ...’  
(Slk; Sketch Engine)

Clitic positional deviations have usually been subsumed under umbrella terms like *delayed placement*, which often have been equated with Clitic Third placement. Even quantitative statistical surveys of clitic placement do not distinguish different types of delay (e.g. Kolaković et al. 2022; Reinkowski 2001 for BCMS). However, the recognition of a clitic's exact position by a discrete numerical value allows picturing variation more precisely. I. Kulik's (2019) study suggests that the frequency distribution of the different clitic placement patterns exhibit a Pareto-like distribution, i.e. to some extent it resembles Zipf's law. To my knowledge, this observation has not been pronounced before.<sup>266</sup> The numerical examination of clitic positions requires further study in Slavic. Exactly such an approach will be presented in Chapter 10.

## 4.5 Clitic clustering

The notion of the CLITIC CLUSTER has appeared multiple times in preceding sections and chapters, as if it were presupposed, although it has not been discussed yet. This section will go into more detail. A clitic cluster can be defined as a syntagm of contiguous clitics. Its formation is regarded as a prototypical characteristic of special clitics. Usually, clitic clustering has been regarded as obligatory and the resulting syntagm as indivisible (cf. Veselovská 1995: §4.9 for Czech), cf. (4.86).<sup>267</sup> Kosta &

<sup>266</sup> I. Kulik (2019: 140) explicitly discusses the quantities of the different attested clitic clausal positions but does not take Zipf's law into account. In fact, 2P and 3P placement mirror Zipf's law with 59.8% vs. 27.9%. The remaining attested delayed positions do not fit exactly, because there is no underlying power law for the entirety of the attested data.

<sup>267</sup> In fact, one may hypothesise that, what is actually problematic about example (4.86b) is not the discontinuous clitics, but a more robust constraint on rightwards clitic dislocation. See Section 4.8 on this matter.

Zimmerling (2014: 446) summarise this basic property as follows: “A cluster is by definition a string of clitics that neither allows insertion of non-clitic elements nor permutation of clitics, when they are contiguous.”

- (4.86) a. (Ne-)dělal **jsem** **si** (dnes) poznámky (dnes).  
 (NEG-)make.PST.SG.M AUX.1SG REFL.DAT today note.ACC.PL  
 ‘Today I have (not) been taking notes.’ (Cz; Veselovská 1995: §4.9, ex [46a])  
 b. \*(Ne-)dělal **jsem** dnes **si** poznámky. (Cz; Veselovská 1995: §4.9, ex [46c])

Polish has been assessed to be strikingly different, since it exhibits hardly any restrictions on the adjacency or ordering of the clitics, cf. (4.87) (cf. Franks 2009, 2010: 133–136; Franks & King 2000: 338; Kupść 2000b: 41–42; Spencer 1991: 369).<sup>268</sup> Witkoś (1998: 157) informs that “Polish pronouns tend to cluster together with the auxiliary in preverbal position”, but also points out that “Polish clusters are not very firm and can be discontinuous” (Witkoś 1998: 158). In (4.87a–b), the auxiliary clitics *-ś* and *-śmy* and the pronouns *ja* and *go* appear separately.<sup>269</sup> In (4.87c), it is the pronominal clitic *mu* and reflexive *się*, which are disjoint. Finally, the two pronominal clitics *mu* and *go* are non-contiguous in (4.87d). These SPLIT-CLITIC constructions lead Franks & King (2000: 205) to the conclusion that “Pol[ish] [...] does not have a clitic cluster”. The discontinuous placement of multiple clitics has also been referred to as DIACLISIS (cf. Bubenik 2001; Janse 1998; Kolaković et al. 2022).<sup>270</sup> I shall use this term from now on.

- (4.87) a. Dlaczego **ja** kupiła=ś?  
 why her.ACC buy.PST.SG.F=AUX.2SG  
 ‘Why did you buy it?’  
 b. Kiedy=śmy zobaczyli **go**?  
 when=AUX.1PL see.PST.PL.VIR him.ACC  
 ‘When did I see him?’ (Pol; Franks & King 2000: 338)  
 c. Teraz **mu** takie książki **się** nie podobają.  
 now him.DAT such.ACC.PL.NVIR book.ACC.PL REFL NEG please.3PRS.SG  
 ‘He doesn’t like such books nowadays.’ (Pol; Spencer 1991: 369)

<sup>268</sup> See Section 4.6 for clitic ordering within the cluster.

<sup>269</sup> Example (4.87a) is unrevealing, for the feminine accusative pronoun *ja* is an inconstant clitic and the auxiliary clitic *-ś* occurs on the clause-final verb, such that an analysis as plain person-number affix is eligible.

<sup>270</sup> As far as I know, Janse (1998: 270–274) was the first to propose the notion of *diacsis*.

d. Piotr      **mu**<sub>2</sub>      chce<sub>1</sub>      **go**<sub>2</sub>      dać<sub>2</sub>.  
 Peter.NOM him.DAT want.PRS.3SG him.ACC give.INF  
 ‘Peter wants to give it to him.’

(Pol; Witkoś 1998: 158)

It is important to point out that diacsis is not a peculiarity of Polish, but it is encountered cross-linguistically. Janse (1998: 270–274) discusses diacsis for a variety of languages, i.e. Cappadocian Greek, Ancient and Medieval Greek, Spanish, early Modern French, Franco-Provençal, and East Flemish. Some of the respective constructions represent diacsis proper, i.e. the non-contiguous realisation of *co-dependents*, i.e. clitics that are in a dependency relation with the same verbal predicate. Others, in turn, represent what Kolaković et al. (2022: 34) call PSEUDO-DIACSIS, which is the disjoint occurrence of non-co-dependent clitics. I will not use the term *pseudo-diacsis*, but I acknowledge the concept underlying it.<sup>271</sup>

Diacsis has hardly received attention in Slavic clitic research, although the construction is attested in Slavic 2P languages. Čechová et al. (1996: 33) and Dotlačil (2017) provide examples for diacsis, cf. (4.88). The disjunct clitics are dependents of different governors in a complex clause. More intriguingly, Lenertová (2004) and Hana (2007) cite examples of diacsis with co-dependents, cf. (4.89a–d). The pronominal and reflexive clitics are not contiguous with the conditional clitic of the morphologically complex subordinator *a-by-* ‘(in order) to’.<sup>272</sup> Note that the pronominal and reflexive clitics necessarily occur in Delayed Placement. Example (4.89e) from Avgustinova & Oliva (1997) shows that *-li* ‘if’ and the reflexive can be detached, too.<sup>273</sup> Finally, I. Kulik (2023b) provides the Czech data in (4.89f–g) showing that reflexive and pronominal clitics do not need to cluster, irrespective of the argument status of the reflexive: argument in (4.89f), but lexicalised pleonasm in (4.89g).

(4.88) a. Ne-bál<sub>1</sub>      **bych**<sub>1</sub>      **se**<sub>1</sub>      říct<sub>2</sub>      **mu**<sub>2</sub>      to      do      očí.  
 NEG-fear.PST.SG.M COND.1SG REFL.ACC tell.INF him.ACC this.ACC to eye.GEN.PL  
 ‘I wouldn’t be afraid of telling him it right in the face.’ (Cz; Čechová et al. 1996: 33)

<sup>271</sup> Kolaković et al. (2022: 18, n. 13; 28; 34) do not provide a precise definition of diacsis and pseudo-diacsis but rather a vaguely worded description, which gives rise to confusion between the two types of diacsis.

<sup>272</sup> One could object that the dative pronoun *ji* ‘her’ in (4.89a) is an inconstant clitic. However, the argument is that constant clitics can occupy the same position as the reflexive clitics in (4.89b–d).

<sup>273</sup> Avgustinova & Oliva (1997: 30) argue that, unlike auxiliary, pronominal, and reflexive clitics, *-li* is not a sentential clitic and does not occur within the clitic cluster but next to it. They further tell that *-li* does not need to occur in the second position but within the first syntactic constituent of the clause. The participation in clitic cluster formation is thus a superficial impression.



b. Jana **se**<sub>1</sub> rozhodla<sub>1</sub> pomoci<sub>2</sub> **mu**<sub>2</sub>.  
 Jana.NOM REFL.ACC decide.PST.SG.F help.INF him.DAT

c. Jana *se*<sub>1</sub> rozhodla<sub>1</sub> *mu*<sub>2</sub> pomoci<sub>2</sub>.

‘Jana decided to help him.’

(Cz; Dotlačil 2017)

(4.89) a. [...] **a-by** [NP nějakým komplotem ] **jí** bylo  
 SUB-COND.3SG any.INS.SG conspiracy.INS.SG her.DAT be.PST.3SG.N  
 zebraněno ve svobodném rozhodování.  
 preclude.PASS.PTCP.SG.N in free.LOC.SG.N decision.LOC.SG

‘... that she would be precluded by some conspiracy from taking a free decision.’

b. [...] **a-bych** [NumP jeden ] **si** nechal a ostatní  
 SUB-COND.1SG one.M.ACC REFL.DAT leave.PST.SG.M and other.ACC.PL  
 podepsal.

subscribe.PST.SG.M

‘... that I kept one and subscribed the others.’

(Cz; Lenertová 2004: 138–139)

c. Chceme, **a-by** [NP stat ] **se** k těmto  
 want.PRS.1PL SUB-COND.3SG state.NOM.SG REFL.ACC to this.DAT.PL  
 závazkům přihlásil (a vyplatil nám ho  
 obligation.DAT.PL acknowledge.PST.SG.M and pay.PST.SG.M us.DAT him.ACC  
 například později v rámci státního rozpočtu.)  
 for\_example later in scope.LOC.SG state.ADJ.GEN.SG budget.GEN.SG

‘We want the state to acknowledge these obligations (and pay it to us later, for example, as a part of the budget.)’

d. Spíš chtějí **a-bych** [NP já ] **se** svěřoval  
 rather want.PRS.3PL SUB-COND.1SG I.NOM REFL.ACC confide.PST.3SG.M  
 jim.  
 them.DAT

‘They would prefer, if I confide to them.’

(Cz; Hana 2007: 112)

e. Lásce-**li** své **se** v žití budeš protiviti,  
 love.DAT.SG-if.CL one’s.POSS REFL.ACC in live.LOC.SG be.FUT.2SG oppose.INF  
 žebrákem půjdeš světem.  
 beggar.INS.SG walk.PFV.PRS.2SG world.INS.SG

‘If you oppose your love in your life, you will go through the world as beggar.’

(Cz; Avgustinova & Oliva 1997: 30)

- f. Lehce **si**<sub>2</sub>          uměla<sub>1</sub>          **ho**<sub>2</sub>          představ<sub>2</sub>, [...]   
easily REFL.DAT be\_able.PST.SG.F him.ACC imagine.INF   
‘She could easily imagine him, ...’
- g. [...] kdekdo                                  **se**<sub>2</sub>          začal<sub>1</sub>          **mi**<sub>2</sub>          smáti<sub>1</sub>.   
almost\_everybody.NOM REFL start.PST.SG.M me.DAT laugh.INF   
‘... almost everybody started to laugh at me.’ (Cz; I. Kulik 2023b: 224)

Diaclitic constructions are more widespread across Slavic. Example (4.90) cites a discontinuous sequence of clitics across the verbal predicates of a complex clause in Slovak resembling Czech (4.88).<sup>274</sup>

- (4.90) Báľ<sub>1</sub>                                  **by**<sub>1</sub>          **som**<sub>1</sub>          **sa**<sub>1</sub>          povedať<sub>2</sub> **mu**<sub>2</sub>          to                  dnes.   
be\_afraid.PST.SG.M COND AUX.1SG REFL.ACC tell.INF him.DAT this.ACC.SG.N today   
‘I would be afraid to tell him (it) today.’ (Slk; Short 1993b: 566)

Diaclisis is also no stranger to South Slavic 2P languages. We encounter different configurations of diaclisis proper in Slovene in paradigm (4.91) from Golden (2003).

- (4.91) a. Janez          **ji**          **ga**          **je**          prebral          včeraj.   
Janez.NOM her.DAT it.ACC AUX.3SG read.PST.SG.M yesterday   
b. Prebral **ji ga je** včeraj Janez.   
c. Prebral včeraj **ji ga je** Janez.   
d. Prebral **ga** včeraj **ji je** Janez.   
e. Prebral **ji ga** včeraj **je** Janez.   
f. Prebral **ji** včeraj **ga je** Janez.   
‘Janez read it to her yesterday.’ (Slv; Golden 2003: 219)

The issue of discontinuous clitics has been already recognised as early as Bidwell (1965: 258) [cited by Rubadeau (1996: 25)] and Popović (1979: 678–679).<sup>275</sup> The Serbian data in (4.92) present diaclisis of the auxiliary and reflexive clitic. Recently, Kolaković et al. (2022) cite comparable evidence from

<sup>274</sup> See Section 10.5.2 for diaclisis proper in Slovak.

<sup>275</sup> Although Popović’s (1979) examples are from the literary production of the 19th-century Voivodinian writers Bogoboj Atanacković and Jakov Ignjatović, he informs that diaclisis is observable in contemporary Serbo-Croatian, even though with low frequency (Popović 1979: 679). Example (i) is from Bogoboj Atanacković.

(i) Ja          **bih**          dragovoljno          **se**          kurtalisao          ovoga          života.   
I.NOM COND.1SG voluntarily REFL.ACC abandon.PST.SF.M this.GEN.SG.M life.GEN.SG   
‘I would willingly abandon this life.’ (BCMS; Popović 1979: 684)

corpora of contemporary Bosnian, Croatian, and Serbian, cf. (4.93)–(4.95).<sup>276</sup> In Bosnian (4.93), the embedded object clitic *joj* escapes the embedded infinitive and joins the matrix auxiliary clitic leaving behind the co-dependent object clitic *ih*. Croatian (4.94) also provides a complex clause with an embedded infinitive subcategorised for two clitic objects. While the dative clitic *joj* occupies sentential second position, the accusative clitic *ga* remains in situ (cf. also Ćavar & Seiss 2011: 147–148). The Serbian example (4.95a) is particularly interesting, because it shows diacsis across the subordinator *da*. The embedded object clitic *mi* joins the matrix auxiliary clitic *je*, whereas the embedded reflexive clitic *se* remains in situ. Diacsis across *da* has been already introduced by Stjepanović (1998, 1999, 2004), cf. (4.95b), but remained a contentious issue (cf. Aljović 2004: 192, n. 3; Franks & King 2000: 335). Kolaković et al.’s (2022: 307) empirical data thus corroborate Stjepanović’s claim.

(4.93) [...] kada **su**<sub>1</sub> **joj**<sub>2</sub> zbog opasne infekcije  
when AUX.3PL her.DAT because dangerous.GEN.SG.F infection.GEN.SG  
stafilokokom morali<sub>1</sub> izvaditi<sub>2</sub> **ih**<sub>2</sub> [...].  
staphylococcus.INS.SG must.PST.PL.M remove.INF them.ACC  
‘... when they had to remove them from her because of a dangerous staphylococcus infection  
...’ (Bos; Kolaković et al. 2022: 273)

<sup>276</sup> Kolaković et al. (2022: 273, 307) refer to these constructions as *pseudo-diaclisis*, although the non-contiguous clitics are co-dependents in each case as indicated by the indexes.

- (4.95) a. [...] i počelo<sub>1</sub>      **mi**<sub>2</sub>      **je**<sub>1</sub>      da **se**<sub>2</sub>      vrti<sub>2</sub>      u  
                  and start.PST.SG.N me.DAT AUX.3SG that REFL.ACC spin.PRS.3SG in  
                  glavi.  
                  head.LOC.SG  
                  ‘... and I started to feel dizzy.’ (Srp; Kolaković et al. 2022: 307)
- b. Marija      **mu**<sub>2</sub>      zeli<sub>1</sub>      da **ga**<sub>2</sub>      predstavi<sub>2</sub>.  
                  Marija.NOM him.DAT want.PRS.3SG that him.ACC introduce.PRS.3SG  
                  ‘Marija wants to introduce him to him.’ (BCMS; Stjepanović 2004: 182)

Patterns of diacsis are also attested in Štokavian dialects, as Kolaković et al. (2022: 166–167) inform, cf. (4.96).

- (4.96) a. [...] i      ònda ka      **se**      dòbro      **j**<sup>e</sup>      svârī      [...].  
                  and then when REFL.ACC good.ADV AUX.3SG cook.PST.3SG.M  
                  ‘... and then when it is cooked well ...’
- b. kad      **bi**      ònī      stârci      **se**      skùpili  
                  when COND.3PL that.NOM.PL.M old\_man.NOM.PL REFL.ACC gather.PST.PL.M  
                  ùveče      [...].  
                  in\_the\_evening  
                  ‘... when those old men would gather in the evening ...’  
                  (BCMS, Zapadni; Kolaković et al. 2022: 166)
- c. U dva sata      **smo**      noću      **se**      dizali.  
                  in two o’clock AUX.1PL at\_night REFL.ACC get\_up.PST.PL.M  
                  ‘At two o’clock at night we got up.’ (BCMS, Slavonski; Kolaković et al. 2022: 167)

Finally, Blažeka (2018/2019: 129) cites diacsis in Kajkavian, cf. (4.97). Here, we encounter discontinuous clitics across a complex clause, again.

- (4.97) T’i      **bi**<sub>1</sub>      št’el<sub>1</sub>      s’am      ‘igrat<sub>2</sub>      **se**<sub>2</sub>.  
                  you.NOM COND want.PST.SG.M alone play.INF REFL.ACC  
                  ‘You would like to play all by yourself.’ (Kaj; Blažeka 2019: 129)

To conclude, there are two environments for diacsis. The non-contiguous placement of clitics across multiple verbal domains of a complex clause is a rather trivial case, as the clitics are positioned with respect to their respective verbal governor. Diacsis is thus the result of embedded clitic in-situ

placement, i.e. the absence of clitic climbing.<sup>277</sup> The occurrence of diacsis within the domain of a simple clause is less trivial, because it is unexpected that clitics, which are co-dependents, do not cluster. However, such constructions are attested not only in Polish, but also in exemplary Wackernagel clitic languages like Czech and BCMS.

## 4.6 Clitic ordering

In connection with clitic cluster formation, it was mentioned in the previous Section 4.5 that the linear order of clitics is fixed in the clitic cluster (cf. Franks 2009: 732–734; Kosta & Zimmerling 2014: 446).<sup>278</sup> The ordering pattern is sometimes referred to as “templatic” (cf. Franks 2009: 732; Franks & King 2000: 205–206; Klavans 1995: 37; Kosta & Zimmerling 2014: 446; Perlmutter 1970: *passim*; esp. 208–210; Spencer 1991: 356, 374, 384). Leaving the diverse technical implementations aside, word order deviations are considered ungrammatical both in traditional grammaticography and formal syntactic literature. Franks & King (2000: 205) summarise the word order regularities for Slavic clitic clusters as in Table 4.4, which I have adapted to the glossing conventions used in this thesis.<sup>279</sup> Polish does not appear in the table, because it is considered neither to have a clitic cluster nor to exhibit strict word order among clitics. However, in the section on clitic placement and ordering in Polish, Franks & King (2000) provide the word order description in (4.98). Note that no cluster-internal word orders have been reported for Goralish, Silesian, or Kashubian in the literature, yet.

<sup>277</sup> See Section 4.7 for clitic climbing.

<sup>278</sup> Kosta & Zimmerling (2014: 446) state that clitics do not have to obey strict ordering if they do not constitute a single clitic cluster. This seems to be a contentious point, as it has been argued that diacsis by clitic climbing in BCMS obeys the cluster-internal word order, cf. the paradigm in (i) from Stjepanović (1998). The embedded pronominal clitics *mu* and *ga* can either remain in situ (i.a) or climb together (i.b). If only one clitic climbs, then it has to be the dative clitic *mu*. Otherwise, the utterance is ungrammatical. Compare (i.c) with (i.d). (Cf. Franks & King 2000: 247 for a similar pattern in Slovene.)

(i) a. Marija želi<sub>1</sub> da **mu**<sub>2</sub> **ga**<sub>2</sub> predstavi<sub>2</sub>.  
       Marija.NOM wish.PRS.3SG that him.DAT him.ACC introduce.PRS.3SG  
       b. ? Marija **mu**<sub>2</sub> **ga**<sub>2</sub> želi<sub>1</sub> da predstavi<sub>2</sub>.  
       c. ? Marija **mu**<sub>2</sub> želi<sub>1</sub> da **ga**<sub>2</sub> predstavi<sub>2</sub>.  
       d. \* Marija **ga**<sub>2</sub> želi<sub>1</sub> da **mu**<sub>2</sub> predstavi<sub>2</sub>.  
       ‘Marija wants to introduce him to him.’

(BCMS; Stjepanović 1998: 271–272)

I cite Stjepanović’s (1998b) original judgements. Despite the question mark (?), she unequivocally states that “climbing only one clitic, while leaving the other clitic in the embedded clause, is possible” (Stjepanović 1998: 271).

<sup>279</sup> I cite the full table, which is why Sorbian appears in it, although I have nothing to say about Sorbian, at all.

Table 4.4: Word order within Slavic special clitic clusters (Franks &amp; King 2000: 205).

| Language    | Linear order                                                                   |
|-------------|--------------------------------------------------------------------------------|
| BCMS:       | $li < AUX < DAT < ACC < GEN < se_{REFL} < je_{AUX}$ <sup>280</sup>             |
| Slovene:    | $naj < COND / AUX_{PST} < REFL < DAT < ACC < GEN < AUX_{FUT} / je_{AUX}$       |
| Bulgarian:  | $\dot{s}te_{AUX} < AUX < DAT < ACC < e_{AUX}$                                  |
| Macedonian: | $da < ne < \dot{k}e / COND < AUX < DAT_{ETH} < DAT < ACC < e_{COP} / se_{COP}$ |
| Czech:      | $li < COND / AUX < DAT_{ETH} / REFL < DAT < ACC / GEN$                         |
| Slovak:     | $COND < AUX < REFL < DAT_{ETH} < DAT < ACC < GEN$                              |
| Sorbian:    | $PART < AUX < REFL < DAT < ACC$                                                |

(4.98) Polish:  $PART < COND < AUX < DAT < REFL < ACC < GEN < INS$ <sup>281</sup>

(Franks & King 2000: 156)

Despite some language-particular characteristics, the order  $AUX < DAT < ACC < GEN$  is effective in all languages.<sup>282</sup> However, the ordering pattern is a mere tendency in Polish, such that deviate orderings occur (cf. also Kupść 2000b: 43). Recall the controversial examples (4.87a) from the previous Section 4.5, where a pronoun precedes a post-verbal putative auxiliary clitic. A more compelling example is (4.99) reported by Kupść (2000b: 43), who tells that the order of clitics with respect to one another is not fixed. Franks & King (2000: 339) cite (4.100a) with the order  $DAT < ACC$ , which, however, is assessed as highly infelicitous due to a presumed restriction on third person preceding first and second person  $*[3rd < 1st / 2nd]$ . The reversed order  $ACC < DAT$  repairs the utterance.<sup>283</sup>

<sup>280</sup> Cf. also Kolaković et al. (2022: 100). Some Croatian and Serbian authors regard the order among the clitics to be  $GEN > ACC$  (cf. Kolaković et al. 2000: 100–102; Reinkowski 2001: 73, 80; Reiter 1959: 381–382; Rubadeau 1996: 21).

<sup>281</sup> Rubadeau (1996: 139) provides a slightly different word order for Polish that does not possess an individual slot for REFL. Instead, reflexives occupy the position of the respective pronominal case in her scheme. Grochowski (2019) claims that REFL *się* occupies the cluster-final position: “Jeżeli jednak po czasowniku występuje szereg enklityk, to *się* zajmuje w tym szeregu pozycję ostatnią.” ‘However, if a string of clitics appears after the verb, then *się* occupies the last position in this string.’ (Grochowski 2019: 105) Witkoś (1998: 157) gives the word order in (i) without the conditional clitic.

(i)  $AUX > DAT > (REFL) > ACC > (REFL)$

<sup>282</sup> Bulgarian and Macedonian do not possess genitival clitics.

<sup>283</sup> The frequency of the ordered pair (*mu*, *go*) is attested 101× in the NKJP (full corpus, pelcra search engine), whereas the sequence (*go*, *mu*) occurs 45 times (incl. 6 false positives). The co-occurrence of *mu* and *cię* seems to be a rare pattern. However, the sequence (*mu*, *cię*) occurs 9×, while the reverse order (*cię*, *mu*) is attested only 3× in the NKJP (full corpus, pelcra search engine). Accordingly, the cogency of example (4.100b) is uncertain.

- (4.99) a. Wczoraj **mu go** przedstawiła.  
 yesterday him.DAT him.ACC introduce.PST.SG.F  
 b. Wczoraj **go mu** przedstawiła.  
 ‘She introduced him to him yesterday.’ (Pol; Kupść 2000b: 43)

- (4.100) a. ?? Pokazali **mu cię** wczoraj.  
 show.PST.PL.VIR him.DAT you.ACC yesterday  
 b. Pokazali **cię mu** wczoraj.  
 ‘They showed you to him yesterday.’ (Pol; Franks & King 2000: 339)

The mentioned restriction is reminiscent of Bonet’s (1994) Person Case Constraint (PCC), which was first observed for Romance and is considered a universal constraint on clitic pronoun combinations (cf. Bonet 1991; Perlmutter 1971).<sup>284</sup> The PCC requires accusative clitics to be 3rd person in the presence of an argumental dative clitic, or put differently: accusative clitics cannot be 1st or 2nd person in ditransitive clitic constructions: \*[DAT<sub>n</sub> ACC<sub>1+2</sub>]. Polish thus does not only allow the typologically unusual ACC < DAT order but also disobeys the PCC. Cetnarowska (2003: 17) explicitly states that “the PCC does not hold for Polish, since the sequences of 1st/2nd person Accusative pronouns and 1nd/2nd [sic] person Datives are allowed”, cf. (4.101). Note however that *cię* is the only constant clitic in example block (4.101). On the other hand, Rezac (2005: 124–126) claims that the PCC is active in Czech and provides example (4.102) (cf. also Franks & King 2000: 106 with examples).

- (4.101) a. On **mnie wam** polecił.  
 he.NOM me.ACC you.DAT recommend.PST.SG.M  
 ‘He recommended me to you.’  
 b. Ona **cię nam** przedstawiła.  
 she.NOM you.ACC us.DAT introduce.PST.SG.F  
 ‘She introduced you to us.’ (Pol; Cetnarowska 2003: 17)

<sup>284</sup> Perlmutter (1971) does not use the term *Personal Case Constraint* but generalises the observed restrictions on clitic sequences as “Surface Structure Constraint on Clitics” (Perlmutter 1971: 48, 53). Note that Bonet (1991: 176–221) initially coined the term *\*me lui/I-II Constraint* in her PhD thesis. She observes the constraint to work in various other languages, e.g. Greek, Arabic, and Georgian. To my knowledge, Bonet (1994) was the first to introduce the term *Person Case Constraint* to linguistics.

- (4.102) a. Ukážu                    **mu**            **ho**            zítra.  
                                          show.PFV.PRS.1SG him.DAT him.ACC tomorrow  
                                          ‘I will show him to him tomorrow.’
- b. ?? Ukážu                    **mu**            **tě**            zítra.  
                                          show.PFV.PRS.1SG him.DAT you.ACC tomorrow  
                                          ‘I will show you to him tomorrow.’ (Cz; Rezac 2005: 115)

In contrast, Lenertová (2004: 152–154) argues against the validity of the PCC in Czech and criticises it to be “a false Person Case Constraint” (Lenertová 2004: 152). Surveying empirical data, she observes that 1st and 2nd person accusative clitics appear with dative clitics, in fact, cf. (4.103). Likewise, Hana (2007: 132) observes violations of the PCC by both grammatical judgements and corpus data.

- (4.103) a. ... a potom **nám**    **tě**            dávají            za vzor.  
                                          and then us.DAT you.ACC give.PRS.3PL as example.ACC.SG  
                                          ‘... and then they present you to us as an example.’ (Cz; Lenertová 2004: 152)
- b. Chci                    **mu**            **tě**            ukázat.  
                                          want.PRS.1SG him.DAT you.ACC show.INF  
                                          ‘I want to show you to him.’
- c. Představil                    **mu**            **mě**            jako prodavačku                    z  
                                          introduce.PST.SG.M him.ACC me.ACC as saleswoman.ACC.SG from  
                                          bazaru.  
                                          market.GEN.SG  
                                          ‘He introduced me to him as a market saleswoman.’ (Cz; Lenertová 2004: 153)

In this context, Lenertová (2004: 153) observes the occurrence of the reversed ACC < DAT order in Czech, cf. (4.104). What is more, Lenertová (2004: 154, n. 20) tells that there are diverging preferences of case sequences for clitics depending on the person/number(/gender) combinations of paired clitics in Czech. Unfortunately, she does not provide any quantities. Thus, the order DAT < ACC is preferred for 3SG.M < 1SG/2SG/2PL and 3PL < 2SG, whereas the reverse order ACC < DAT is preferred for 3SG.F < 1SG. Both orders (DAT < ACC, ACC < DAT) are equally acceptable with 3SG.F < 2PL and 3PL < 1SG.



(4.104) a. Kdy **mě** **jí** konečně představíš?  
 when me.ACC her.DAT finally introduce.PFV.PRS.3SG  
 ‘When are you finally going to introduce me to her?’

b. ... a-by **mě** **jim** vydal.  
 SUB-COND me.ACC them.DAT hand.PST.SG.M

‘... that he should hand me over to them.’

(Cz; Lenertová 2004: 153)

On a general note, Zwicky (1977: 8–9) seems to take word order variation of special clitics as an unsurprising and rather natural fact. More concretely, the relative freedom in placing clitics with respect to one another is put forward as a property distinguishing special clitics from affixes (cf. Zwicky 1977: 9). Janse (1998: 267–270) refers to the inversion of clitic order as METACLISIS and discusses the phenomenon through the example of Cappadocian Greek, Romane, and Bernese German. Metacclisis has been repeatedly noted to occur in Romance and is known since Wanner’s (1977) discussion of Italian clitic sequences. More recently, Heap (2005) examines variable clitic orders in Spanish. He argues that

“many linguists have ‘idealised away’ from this pesky problem by sweeping it under the rug (or simply not noticed it), and dealt with only invariably fixed-order clitic sequences without taking into account variable sequences.” (Heap 2005: 86)

He continues that

“[t]his ‘inversion’ or ‘clitic metathesis’ has been a feature of nonstandard varieties, in both Spain and Latin America, for several centuries at least [...], and it is sufficiently widespread [...].” (Heap 2005: 87)

It appears then that one encounters some degree of idealisation in Slavic clitic research, too. Franks & King (2000: 111) provide the Czech examples in (4.105). The question mark (?) indicates that native speaker judgements may vary for a particular sentence token. Note that it is not only the ethical dative clitic *ti* that is freely positioned within the cluster, but also that reflexive *se* and dative *mi* may occur in the reversed order.<sup>285</sup> The variable placement of the ethical dative has been also observed by Hana (2007: 116–118), cf. (4.106). He furthermore discusses variable ordering of accusative and genitive clitics, cf. (4.107)–(4.108), which has been a contentious issue as indicated by the discussion

<sup>285</sup> I will not go into details on the ethical dative. Suffice it to note that it is attested in all West and South Slavic 2P languages and that it occurs at the front of the pronominal block of the clitic cluster.

in Franks & King (2000: 107–109). The genitival clitic in (4.107) is a partitive, whose case is governed by the noun *spousta* ‘plenty’, from which the clitic is dislocated as indicated by the indexed underscore (cf. Toman 1986). Example (4.108) shows accusative and genitive argument clitics in free word order variation.

- (4.105) a. On        **ti**                **se**                **mi**                ne-omluvil.  
                  he.NOM you.DAT.ETH REFL.ACC me.DAT NEG-apologise.PST.SG.M  
       b. ? On **ti mi se** neomluvil.  
       c. ? On **mi ti se** neomluvil.  
       d. ? On **mi se ti** neomluvil.  
       e. ? On **se mi ti** neomluvil.  
       f. On **se ti mi** neomluvil.  
                  ‘(I’m telling) you, he didn’t apologise to me.’                                (Cz; Franks & King 2000: 111)

- (4.106) a. On        **se**                **ti**                vůbec ne-bál.  
                  he.NOM REFL.ACC you.DAT.ETH at\_all NEG-fear.PST.SG.M  
       b. On **ti se** vůbec nebál.  
                  ‘You know, he wasn’t scared at all.’                                                (Cz; Hana 2007: 116)

- (4.107) a. Napadá        **mě**        **jich<sub>j</sub>**        tu        vždycky spousta <sub>j</sub>.  
                  occur.PRS.3SG me.ACC them.GEN here always plenty.NOM.SG  
       b. Napadá **jich<sub>j</sub> mě** tu vždycky spousta <sub>j</sub>.  
                  ‘I always come upon a lot of them [e.g. jokes] here.’                                (Cz; Hana 2007: 118)

- (4.108) a. Pak **jí**        **ho**                zbavili.  
                  then her.GEN him.ACC get\_riD.PST.SG.VIR  
       b. ? Pak **ho jí** zbavili.  
                  ‘Afterwards they took it/him away from her.’                                                (Cz; Petkevič 2009: 193)

A genitival clitic preceding an accusative one is also cited for Slovak by Franks & King (2000: 208), cf. (4.109). Like in Czech (4.107), we encounter a partitive genitive clitic dislocated from its governing head, a numeral above four.

- (4.109) Dnes **ich<sub>j</sub>** **ju** napísalo sedem **\_j**.  
 today them.GEN her.ACC write.PST.SG.N seven.NOM  
 ‘Seven of them wrote it today.’ (Slk; Franks & King 2000: 208)

The structural difference between the partitive genitive constructions and the other examples of metaclisis discussed is that the partitive genitive clitic is licensed by a different governing head than the other clitics it joins in the cluster. The respective syntagm is called a MIXED CLUSTER by Kolaković et al. (2022: 20). Such mixed clusters seem to provide a relevant environment for metaclisis, cf. Czech (4.110) with violated REFL < DAT < ACC order. Petkevič (2009: 191) observes that the clitic pronouns are licensed by the matrix verb, while the reflexive clitics are dependent of the embedded infinitive in (4.110). The same subcategorisation facts account for the ACC < DAT order in utterance (4.110d). Example (4.110e) also exhibits the linear precedence of accusative before dative, but this time it is a consequence of the topicalisation of the embedded infinitival VP. Petkevič (2009: 192) notes that the pronominal clitics constitute a single clitic cluster.<sup>286</sup>

- (4.110) a. Viděl<sub>1</sub> **ho<sub>1</sub>** **se<sub>2</sub>** brodit<sub>2</sub> po pás ve vodě.  
 see.PST.SG.M him.ACC REFL.ACC ford.INF up belt.ACC.SG in water.LOC.SG  
 ‘He saw him fording to the belt in the water.’  
 b. Přikázal<sub>1</sub> **mu<sub>1</sub>** **si<sub>2</sub>** obout<sub>2</sub> boty.  
 command.PST.SG.M him.DAT REFL.DAT put\_on\_a\_shoe.INF shoe.ACC.PL  
 ‘He commanded him to put on his shoes.’ (Cz; Petkevič 2009: 191)  
 c. Nařídil<sub>1</sub> **ti<sub>1</sub>** **si<sub>2</sub>** připravit<sub>2</sub> věci.  
 command.PST.SG.M you.DAT REFL.DAT prepare.INF thing.ACC.PL  
 ‘He commanded you to prepare your stuff.’  
 d. Pověřil<sub>1</sub> **ho<sub>1</sub>** **ti<sub>2</sub>** vynadat<sub>2</sub>.  
 authorise.PST.SG.M him.ACC you.DAT tell\_off.INF  
 ‘He authorised him to blast you.’ (Cz; Petkevič 2009: 195)  
 e. [<sub>VP</sub> Líbat<sub>2</sub> **tě<sub>2</sub>**]<sub>TOP</sub> **mi<sub>1</sub>** připadá<sub>1</sub> úplně skvělý.  
 kiss.INF you.ACC me.DAT strike.PRS.3SG completely wonderful.NOM.SG.M  
 ‘Kissing you seems [to be] fairly wonderful to me.’ (Cz; Petkevič 2009: 192)

<sup>286</sup> “[...] zájmena [...] rozvíjejí různá slovesa, přičemž ovšem patří do téhož příklonkového shluku.”  
 ‘[...] the pronouns [...] unfold distinct verbs, whereas they belong to the same clitic cluster though.’  
 (Petkevič 2009: 192)

Hitherto, the data suggests that clitic order is variable to some degree within both simple and complex clauses in West Slavic and beyond. Therefore, it should come as no surprise that word order variation has been reported for South Slavic, too. To begin with, Franks (2010: 137, n. 143) summarises his findings of an informal web search on Slovene suggesting that the Slovene clitic cluster exhibits a higher degree of word order variation than expected by then. Likewise, Rath (2019) finds corpus evidence for non-canonical clitic linearisations in Slovene. He enumerates several cluster-internal word order deviations, but neither gives examples nor specifies the source of his information (cf. Rath 2019: 109).<sup>287</sup> Stegovec (2020) offers the examples in (4.111) with ACC < DAT.

- (4.111) a. Odkar      **ga**      **mu**      **je**      žena      kupila.  
ever\_since him.ACC him.DAT AUX.3SG wife.NOM.SG buy.PST.SG.F  
‘Ever since his wife bought it for him.’
- b. Jana      **ga**      **mu**      **je**      dala.  
Jana.NOM him.ACC him.DAT AUX.3SG give.PST.SG.F  
‘Jana gave it to him.’ (Slv; Stegovec 2020: 276)
- c. **Te**      **mi**      **je**      pokazal?  
you.ACC me.DAT AUX.3SG show.PST.SG.M  
‘Has he shown you to me?’ (Slv; Stegovec 2020: 287)

By the same token, metaclisis has been reported for BCMS. Usually, auxiliary clitics occur at the front of the clitic cluster in BCMS like in West Slavic. However, the 3rd person singular auxiliary clitic *je* exhibits the particularity of occupying cluster-final position by default (cf. Table 4.4). Wilder & Čavar (1994: 28–29) cite the reverse order of reflexive and auxiliary in Croatian, compare (4.112a) exhibiting the order *se* < *je* with (4.112b) showing the opposite linearisation *je* < *se*.<sup>288</sup>

- (4.112) a. To      je      ono,      što **se**      **je**      moralo  
this.NOM.SG.N be.PRS.3SG that.NOM.SG.N what REFL.ACC AUX.3SG must.PST.SG.N  
dogoditi.  
happen.INF
- b. To je ono, što **je se** moralo dogoditi.  
‘This is what had to happen.’ (Hrv; Wilder & Čavar 1994: 28)

<sup>287</sup> Perhaps, Rath (2019) informally reports observations from corpus research he conducted in the Slovene reference corpus *FidaPLUS* (Arhar et al. 2007) and the reference corpus of standard written Slovene *Gigafida* (Krek et al. 2020).

<sup>288</sup> The occurrence of the syntagm *se*<sub>REFL</sub> > *je*<sub>AUX</sub> is less preferred, but not ungrammatical. See Section 4.9.2 on the deletion of unlikes.

This type of reverse ordering, whereby the auxiliary precedes the reflexive recurs across the BCMS-speaking area, but it is not the only attested rearrangement within the clitic cluster. Kokalović et al. (2022: 147–151) discuss metaclisis in BCMS dialects. Below, I cite one example per metaclitic type for the sake of exposition (I refer the reader to Kokalović et al. 2022: 147–151 for more examples). Utterance (4.113a) shows the auxiliary clitic *je* in cluster-medial position preceding reflexive *se*, while (4.113b) presents the order reflexive before a pronominal clitic. The conditional clitic follows a pronominal clitic in (4.113c) and a reflexive in (4.113d). We encounter metaclisis of auxiliary and pronominal clitics in (4.113e–f), with the third person singular auxiliary *je* preceding the pronominal clitic and third person plural auxiliary clitic *su* following it respectively.

- (4.113) a. *Majka        **mi**        **je**        **se**        zvala.*  
 mother.NOM me.DAT AUX.3SG REFL.ACC call.PST.SG.F  
 ‘My mother was called.’ (BCMS, Šumadijsko-vojvođanski; Kolaković et al. 2022: 148)
- b. *jâ        **se**        **ga**        sjjećām*  
 I.NOM REFL.ACC him.GEN remember.PRS.1SG  
 ‘I remember him.’ (BCMS, Istočnohercegovački; Kolaković et al. 2022: 149)
- c. *Önda   **ti**        **bi**        dōli        dōšlo [...].*  
 then you.DAT COND down come.PST.SG.N  
 ‘Then (they) would come down ...’ (BCMS, Zapadni; Kolaković et al. 2022: 150)
- d. *öna **se**        **bi**        prepāla*  
 she REFL.ACC COND get\_scared.PST.SG.F  
 ‘she would get scared’ (BCMS, Zetsko-južnosandžački; Kolaković et al. 2022: 150)
- e. *komèsija        **je**        **me**        pítala*  
 commission.NOM.SG AUX.3SG me.ACC ask.PST.SG.F  
 ‘the commission asked me’  
 (BCMS, Šumadijsko-vojvođanski; Kolaković et al. 2022: 151)
- f. *Dvá        **mu**        **su**        sína        žèneta.*  
 two.NOM.SG him.DAT be.PRS.3PL.CL son.GEN.SG married.GEN.SG  
 ‘Two of his sons are married.’  
 (BCMS, Prizrensko-južnomoravski; Kolaković et al. 2022: 151)

The metaclisis of reflexive and pronominal clitics is also attested in Čakavian. Example (4.114) is from the Buzet dialect in northern Istria.

(4.114) dok **sε** **γa** nε pupije  
 until REFL.ACC him.GEN NEG drink\_up.PRS.3SG  
 ‘until it is drunk up’ (Ča, Buzetski; Kolaković et al. 2022: 149)

Furthermore, Kolaković et al. (2022: 193–194) present metaclefts in a corpus of spoken Bosnian. Interestingly, they find that the metacleftic order *je se* (AUX.3SG < REFL.ACC) occurs 4× more ( $n = 25$ ) than order *se je* permitted by the standard language ( $n = 6$ ), cf. (4.115a).<sup>289</sup> Though lesser frequent ( $n = 5$ ), the reverse order of auxiliary *je* and a pronominal clitic is also attested, cf. (4.115b).

(4.115) a. [...] nego muž **mi** **je** **se** prep’o [...] [  
 but husband.NOM.SG me.DAT AUX.3SG REFL.ACC get\_scared.PST.SG.M  
 ‘... but my husband got scared ...’ (Bos, spoken; Kolaković et al. 2022: 193)

b. [...] koji **je** **me** dobro poznavao i  
 which.NOM.SG.M AUX.3SG me.ACC good.ADV know.PST.SG.M and  
 cijenio [...].  
 respect.PST.SG.M  
 ‘... who knew me very well and respected me ...’  
 (Bos, spoken; Kolaković et al. 2022: 194)

Summing up, there is word order variation in the Slavic clitic cluster, part of which has either not come to the linguists’ attention or has been ignored. Although certain linearisations are strongly preferred, they cannot be treated as if they were inviolable.<sup>290</sup> I do not agree with Heap (2005) in asserting that the variation concerns only a small set of tuples as long as there is no sufficiently large database, which allows for sound quantitative assessments. On the contrary, the various types of metaclefts in BCMS dialects cited by Kolaković et al. (2022: 147–151) suggest that in principle every clitic type can be subject to reordering.

<sup>289</sup> The standardly preferred syntagm of omitting the auxiliary clitic *je* under simultaneous occurrence of reflexive *se* is by far the most often realisation in Kolaković et al.’s (2022) corpus of spoken Bosnian (cf. Kolaković et al. 2022: 193–195).

<sup>290</sup> This point was acknowledged by Rosen (2001: 205), though less rigidly.

## 4.7 Clitic climbing

### 4.7.1 Climbing from verb to verb

CLITIC CLIMBING (CC) is the most common term for a construction, where a pronominal or reflexive clitic, which is licensed by an embedded verbal predicate, is realised in the syntactic domain of a hierarchically higher verb.<sup>291</sup> In the previous sections, occasional examples of clitic climbing have been cited without discussion. Clitic climbing is a cross-linguistically common construction that received particular attention in the study of Romance clitics (cf. Aissen & Perlmutter 1976 for Spanish; Kayne 1975, esp. Ch. 4, for French; Rizzi 1982 for Italian).<sup>292</sup> This fact is important, as research in Romance clitic climbing significantly inspired Slavic clitic research. Given the number of publications on this topic within Slavic and beyond, the phenomenon is very complex and deserves a publication on its own. Time and space considerations require a concise presentation, such that I will restrict myself to basic patterns, but occasionally I shall comment on differences between Slavic and Romance that have been mentioned in the literature.

Basically, CC can be schematised as in (4.116) adapted from Junghanns (2002c: 66).<sup>293</sup> The notation provides that a clitic CL is licensed in an embedded verbal domain (constituent)  $\beta$  but realised in a superordinate verbal domain (constituent)  $\alpha$ . The mismatch between linear position and licensing domain is captured by the gap  $e$ , which is co-indexed with CL. In principle, there can be  $n$  phrasal boundaries between  $\alpha$  and  $\beta$ , whereby  $n$  might be any natural number  $\mathbb{N}_0$  including zero (0) in theory.<sup>294</sup>

$$(4.116) [\alpha \text{ CL}_i [{}_n [\beta e_i ]]] \quad ; n = \mathbb{N}_0$$

<sup>291</sup> Alternatively, the phenomenon has been labelled *clitic promotion* (cf. Klavans 1995: 28; Luján 1980: 382), *clitic movement* (cf. Klavans 1995: 28), *clitic gliding* (cf. Rivas 1977: 133–134), and *clitic raising* (cf. Emonds 1978: 168–171; Mel'čuk 2008: 213; Rappaport 1998; Talić 2018). I use the term *clitic climbing* in a theory-neutral, purely descriptive way and do not presuppose a syntactic derivation that literally shifts the clitic from  $\beta$  to  $\alpha$ . Non-derivational theories like HPSG (cf. Crysmann 2003; Kupść 1999, 2000a, 2000b; Monachesi 1993) or LFG (cf. Caretero García 2017) use the term *clitic climbing*, too.

<sup>292</sup> Clitic climbing was also considered in Zwicky's (1977: 22) basic typology of cliticisation.

<sup>293</sup> The description in (4.116) is a theory-neutral reformulation of Junghanns's (2002c: 66) scheme in (i), which is grounded on a generative movement analysis.

(i)  $[\alpha \dots \text{CL} \dots [{}_n [\beta \dots t_{\text{CL}} \dots ] \dots ]]$

<sup>294</sup> The reference to the set of natural numbers (including zero)  $\mathbb{N}_0$  implies the possibility of an infinite or arbitrarily high number of  $n$ 's in theory. In reality, there are not infinite or arbitrarily many intervening predicates or nodes between a clitic's original domain and its final domain. The real upper limit is not of particular concern here, but might be set by constraints of diverse kinds, e.g. processing, biological (biological structures like the brain are finite), and communicative ones.

The earliest explicit remark on CC in Slavic I am aware of is Polish (4.117) from Nilsson (1904: 263), who describes the construction in the following way: “If the pronoun belongs to an infinitive, the pronoun even can be in front of the verb the infinitive depends on, [...]”<sup>295</sup>

(4.117) ... to **mi**<sub>2</sub>      proszę<sub>1</sub>      na-pisać<sub>2</sub>.

TO me.DAT request.PRS.1SG PFV-write.INF

‘... then I ask you to write to me.’/‘...please write it down for me.’

(Pol; Nilsson 1904: 263)

Franks & King (2000: 241) note that CC only appears in the Slavic 2P languages, but not in the VA languages Bulgarian and Macedonian. They conjecture that CC is absent there, “because these languages lack nonfinite complementation.” (Franks & King 2000: 241) As shown by Polish (4.117), CC occurs from infinitival complements.<sup>296</sup> It is generally accepted that infinitive-hood of the embedded verb is a necessary, but not a sufficient condition for CC in West Slavic and Slovene (cf. Junghanns 2002c: 69 for Czech; Kupść 2000b: 58; Witkoś 1998: 192 for Polish; Golden 2003: 221–222 for Slovene). Examples (2.118)–(4.120) show that, despite being infinitival, the clitics must not climb out of the complement clauses. The occurrence of a subordinator blocks CC.<sup>297</sup> Clitic climbing

<sup>295</sup> “Wenn das pronomen zu einem infinitiv gehört, kann das pronomen sogar vor dem verbum stehen, von welchem der infinitiv abhängt, [...]” (Nilsson 1904: 263). Nilsson’s observation does not particularly refer to the finite-hood of the governing verb. His remark entails (perhaps unintentionally) that a clitic may climb to another infinitive, if the latter governs the infinitive the clitic is licensed by. This nicely fits the discussion about deep embeddings and intermediate placement below, cf. especially examples (4.126) and (4.128). However, the verbal form *proszę* ‘ask.PRS.1SG’ is ambiguous between an inflected verb proper and a particle ‘please’ comparable to German *bitte* ‘ask.PRS.1SG; please’. The morphological shape of the co-occurring verb cannot serve as a diagnostic for verb-hood vs. particle-hood, since infinitives may freely function as imperatives, as in German again, cf. (i)–(ii). Consequently, *napisać* in (4.117) does not need to be analysed as an embedded infinitive synchronically.

(i) Ale szybko posprzątać pokój!

but quickly tidy\_up.INF room.ACC.SG

‘Tidy up the room quickly!’

(Pol; constructed)

(ii) Los, das Zimmer aufräumen!

go the.ACC.SG.N room.ACC.SG tidy\_up.INF

‘Go, tidy up the room!’

(Ger; constructed)

<sup>296</sup> There is an alternative construction in BCMS, whereby the clitic escapes a morphologically finite clause headed by the subordinator *da* (cf. Kolaković et al. 2022; Progovac 1993, 1996; Stjepanović 1998, 1999, 2004). It is generally argued that the type of *da*-clauses that allow CC in BCMS are subjunctive, as was also argued by an anonymous reviewer of I. Kulik (2023b). It will become clear from the data cited in this section that the notion of the subjunctive is not significant for CC in West Slavic (or Slovene). It is the subordinatorless infinitival complementation that provides the licensing domain for CC (cf. Kupść 2000b: 58), which is why I will disregard clitic climbing from *da*-clauses (I refer the reader to Kolaković et al. 2022: 297–309 [= Ch. 13] for data and discussion). An interesting data point on this matter is provided by Krapova & Cinque (2019: 181). They cite Torlak and dialectal Bulgarian examples, which exhibit CC and conclude that it is the absence of the subordinator/particle *da* ‘that’ which enables CC, cf. (i). This suits the observation for West Slavic. The mark  $\emptyset$  indicates the lack of *da* in (i). Note that the embedded verb *fanām* ‘catch.PRS.1PL’ is inflected.

(i) i **gu**<sub>2</sub>      nă      možaxmă<sub>1</sub>       $\emptyset$       fanam<sub>2</sub>

and him.ACC NEG could.AOR.1PL  $\emptyset$  catch.PRS.1PL

‘And we couldn’t catch him’.

(Bulgarian, non-standard; Krapova & Cinque 2019: 181)

<sup>297</sup> This is basic empirical evidence for the claim that clitics cannot escape the clausal domain of CP. Dotlačil (2004: 83–84) shows that phrases and orthotonic pronouns, in contrast, can be dislocated from a CP in Czech, cf. (i)–(ii).



may result in mixed clusters of matrix and embedded clitics as in Czech (4.118a) and Slovene (4.120a). The adjacency of the verbs is not necessary for CC in Slavic (cf. Golden 2008: 313; Junghanns 2002c: 66; Przepiórkowski & Kupść 1997 for the notion of *verb cluster*).<sup>298</sup>

- (4.118) a. Včera **jsem<sub>1</sub>** **se<sub>1</sub>** **mu<sub>2</sub>** **ji<sub>2</sub>** pokusil<sub>1</sub> představit<sub>2</sub>.  
 yesterday AUX.1SG REFL.ACC him.DAT her.ACC attempt.PST.SG.M introduce.INF  
 ‘Yesterday, I made an attempt to introduce her to him.’ (Cz; Rosen 2001: 101)
- b. \*Ne-vím<sub>1</sub> **ho<sub>2</sub>** jak / komu představit<sub>2</sub>.  
 NEG-know.PRS.1SG him.ACC how whom introduce.INF  
 ‘I don’t know how/to whom to introduce him.’ (Cz; Rezac 2005: 111)
- (4.119) a. Piotr **go<sub>2</sub>** chce<sub>1</sub> kupić<sub>2</sub> w Austrii.  
 Peter.NOM him.ACC want.PRS.3SG buy.INF in Austria.LOC  
 ‘Peter wants to buy it in Austria.’
- b. \*Piotr chce<sub>1</sub> **go<sub>2</sub>**, żeby zaprosić<sub>2</sub> na kolację.  
 Peter.NOM want.PRS.3SG him.ACC that invite.INF on dinner.ACC.SG  
 ‘Peter wants to invite him for dinner.’ (Pol; Kupść 2000b: 57–58)

- (i) [<sub>PP</sub> S takou krásnou ženou ]<sub>j</sub> **bych** **se** moc ne-rozmýšlel  
 with such.INS.SG.F beautiful.INS.SG.F woman.INS.SG COND.1SG REFL.ACC much NEG-bother.PST.SG.M  
 [<sub>CP</sub> co dělat t<sub>j</sub>].  
 what do.INF

‘I would not give much careful consideration what to do with such a beautiful woman.’

- (ii) Jeho<sub>j</sub> **bych** moc ne-věděl [<sub>CP</sub> jak potěšit t<sub>j</sub>].  
 him.ACC COND.1SG much NEG-know.PST.SG.M how please.INF

‘Him, I would not know how to please.’

(Cz; Dotlačil 2004: 84)

Note, however, that the full expressions can only cross *wh*-elements. Dotlačil does not provide any example of a crossed subordinator *že* ‘that’, because this type of dislocation is generally prohibited, cf. (iii).

- (iii) \*Řekla Karla<sub>j</sub>, že viděla <sub>j</sub> v parku.  
 tell.PST.SG.F Karl.ACC.SG that see.PST.SG.F in park.LOC

Intended: ‘She said that she saw Carl in the park.’

(Cz; constructed)

On the other hand, Rezac (2005: 134, n. 9) points out that there are apparent exceptions to the ban of CC from *wh*-infinitives like (iv).

- (iv) Ne-mám<sub>1</sub> **ti<sub>2</sub>** [<sub>CP</sub>(?) co dát<sub>2</sub>].  
 NEG-have.PRS.1SG you.DAT what.ACC give.INF

‘I don’t have anything to give to you.’

(Cz; Rezac 2005: 134, n. 9)

I will not propose any ad hoc explanation for these data. Instead, I suggest that a more careful examination of empirical data is still necessary. Anyway, the C<sup>0</sup>-classification of *wh*-elements is not trivial (in Slavic).

<sup>298</sup> The hierarchically ordered cascade of verbs has been also called *verb cluster* (cf. Kupść 2000b; Przepiórkowski & Kupść 1997) and *stacked infinitives* (Hansen et al. 2018). Sabel (1999: 419) uses the notion *coherent infinitives* (referring to Bech 1955) and compares what he calls Long Scrambling in German and Polish (i.e. the climbing of an embedded NP into the matrix clause) with CC in Spanish towards a generative background (cf. also Reis & Sternefeld 2004 for an overview of coherent infinitives in German).

- (4.120) a. Jaz **sem**<sub>1</sub> **ji**<sub>2</sub> **ga**<sub>2</sub> moral<sub>1</sub> pošiljati<sub>2</sub> po pošti.  
 I.NOM AUX.1SG her.DAT him.ACC must.PST.SG.M send.INF by mail  
 ‘I had to send it to her by mail.’ (Slv; Golden 2008: 311)
- b. \*Janez **jo**<sub>2</sub> sprašuje<sub>1</sub>, ali nagraditi<sub>2</sub> s knjigo.  
 Janez.NOM her.ACC ask.PRS.3SG whether reward.INF with book.INS.SG  
 ‘Janez asked whether to reward her with a book.’ (Slv; Golden 2003: 222)

Note however that clitic climbing is not restricted to complement infinitives, but may occur from adjunct infinitives, too, as has been noted by Junghanns (2002c: 67) and Golden (2003: 221, 2008: 313).<sup>299</sup> In (4.121)–(4.125), the clitics climbed from the facultative final clauses to the clausal domain of the finite verbs of motion.<sup>300</sup> Note that CC from adjuncts is available for both pronominal and reflexive clitics as attested by Polish (4.122) and contra Kupść (2000b: 106), who argues that reflexives cannot escape adjoined infinitives. The Croatian example (4.124) is from the hrWaC web corpus (Ljubešić & Erjavec 2011; Ljubešić & Klubička 2014).<sup>301</sup>

- (4.121) a. Chodějí<sub>1</sub> **se**<sub>2</sub> sem na tebe ptát<sub>2</sub>.  
 walk.PRS.3PL REFL.ACC here on you.ACC ask.INF  
 ‘They are coming here to ask for you.’ (Cz; Junghanns 2002c: 67)
- b. Děti **se**<sub>2</sub> šli<sub>1</sub> hrát<sub>2</sub>.  
 children.NOM REFL.ACC go.PST.PL.VIR play.INF  
 ‘The children have gone to play.’ (Cz; Rosen 2001: 106)

<sup>299</sup> In contrast, Rezac (2005: 111) argues that CC is impossible out of adjuncts and provides (i) as evidence.

(i) Dana (\***tě**<sub>2</sub>) radši uteče<sub>1</sub> než (**tě**<sub>2</sub>) políbí<sub>2</sub>.  
 Dana.NOM you.ACC rather run\_away.PRS.3SG than you.ACC kiss.INF  
 ‘Dana will run away rather than kiss you.’ (Cz; Rezac 2005: 111)

The ungrammaticality of CC out of *než* was noticed by Junghanns (2002c: 76–77). Similarly, CC out of a comparative phrase is ungrammatical in BCMS (cf. Kolaković et al. 2022: 239).

<sup>300</sup> CQL query:

(i) SNK: [word="mu|ho"] [lemma="ist"] [tag="VI.\*"]  
 (ii) hrWac: [word="mu|ga|jo"] [lemma="iti |doći|voziti"] [tag="Vmn.\*"]

<sup>301</sup> Kolaković et al. (2022: 35) accept CC from infinitives under verbs of motion citing Bosnian (i). However, they do not regard the embedded infinitive *očistiti* as an adjunct but consider it a “semantically obligatory complement[...] and not [...] a final clause”. This is in stark contrast to the general treatment of adverbial clauses as purely facultative elements, i.e. adjuncts. The authors do not explain their decision nor provide a reference. I do not follow such a treatment and stick to the orthodox view.

(i) [...] jer **mu**<sub>3</sub> **je**<sub>1</sub> u subotu trebala<sub>1</sub> doći<sub>2</sub> očistiti<sub>3</sub> peč  
 because him.DAT AUX.3SG in Saturday.ACC.SG have\_to.PST.SG.F come.INF clean.INF oven.ACC.SG  
 ‘... because she was supposed to come on Saturday to clean his oven.’ (Bos; Kolaković et al. 2022: 35)

- (4.122) a. [...] (to mu powiedziała-m,) że smok go<sub>2</sub>  
 TO him.DAT tell.PST.SG.F-AUX.1SG that dragon.NOM.SG him.ACC  
 przyjechał<sub>1</sub> odwiedzić<sub>2</sub>  
 come\_by\_vehicle.PST.SG.M visit.INF  
 ‘... (then I told him) that the dragon came to visit him’ (Pol; S1)
- b. On się<sub>2</sub> przyszedł<sub>1</sub> poskarżyć<sub>2</sub>, (że go zwodzicie.)  
 he.NOM REFL.ACC come.PST.SG.M complain.INF that him.ACC deceive.PRS.2PL  
 ‘He came to complain that you are deceiving him.’ (Pol; NKJP poliqarp balanced)
- (4.123) Musím<sub>1</sub> mu<sub>3</sub> íst’<sub>2</sub> zavolat’<sub>3</sub>.  
 must.PRS.1SG him.DAT walk.INF call.INF  
 ‘I have to go and call him.’ (Slk; SNK prim-10.0-public-all)
- (4.124) Prijatelji su<sub>1</sub> ga<sub>2</sub> došli<sub>1</sub> tražiti<sub>2</sub> i [...]  
 friend.NOM.PL AUX.3PL him.ACC come.PST.PL.M seek.INF and  
 ‘The friends came looking for him and ...’ (Hrv; hrWaC)
- (4.125) Janez jo<sub>2</sub> je<sub>1</sub> šel<sub>1</sub> obiskat<sub>2</sub> v Ljubljano.  
 Janez.NOM her.ACC AUX.3SG go.PST.SG.M visit.INF in Ljubljana  
 ‘Janez went to visit her in Ljubljana.’ (Slv; Golden 2003: 221/2008: 311)

The scheme in (4.116) also accounts for clitic climbing from what I will somewhat informally call DEEP EMBEDDING loosely following Hana (2007: 132, 190, 197).<sup>302</sup> The concept of *deep embedding* captures the fact that a clitic may climb across a cascade of multiple infinitives into the finite verb’s clausal domain in Slavic 2P languages like in examples (4.126)–(4.128) (cf. Hana 2007: 127; Rezac 2005: 110 for Czech).<sup>303</sup> Czech (4.126) exhibits several positions for the dative reflexive *si*.<sup>304</sup> The Polish example (4.127) shows that the embedded pronominal clitics *mu* and *go* may climb independently from one another. In Slovene (4.128), the feminine accusative pronoun *jo* may occupy a multitude of positions, which includes also variable positioning with respect to the 3rd person singular auxiliary clitic *je* in the clitic cluster. As can be seen, deeply embedded clitics do not need to climb up the entire syntactic tree. They may be situated in intermediate positions between the finite

<sup>302</sup> Hana (2007: 132, 190, 197) approaches syntactic depth in relational terms distinguishing between “more deeply embedded verb[s]/governor[s]” and “less deeply embedded verb[s]”.

<sup>303</sup> Hansen et al. (2018) refer to similar syntactic environment as *stacked infinitives*.

<sup>304</sup> Rosen (2001: 228, n. 24) remarks that the clitic’s lower realisations “sound better with a pause before the embedded domain with the clitic.” This is reminiscent of the discussion about prosodic boundaries in Section 4.2.3 through the example of Slovene. To avoid repetition, I merely suggest that “sound[ing] better” does not indicate grammaticality.

matrix verb and the deepest embedded infinitive (cf. Rosen 2001: 226, 228–229). Note that this intermediate placement also involves the position between finite matrix verb and its neighbouring infinitive, which has been reported to be ungrammatical in Romance (cf. Kayne 1975: 269–270 for French; Klavans 1995: 25 for Spanish; Rizzi 1982: 38 for Italian; more generally Barrie 2000: 92).<sup>305</sup>

- (4.126) Karel (si<sub>3</sub>) bude<sub>1</sub> ještě tento rok (si<sub>3</sub>) chtít<sub>2</sub>  
 Karel.NOM REFL.DAT AUX.FUT.3SG still this.ACC.SG.M.INA year.ACC.SG want.INF  
 (si<sub>3</sub>) koupit<sub>3</sub> (si<sub>3</sub>) nový počítač.  
 buy.INF new.ACC.SG.M.INA computer.ACC.SG  
 ‘Karel will want to buy a new computer even this year.’ (Cz; Rosen 2001: 228)

- (4.127) a. Jan może<sub>1</sub> mu<sub>3</sub> chceć<sub>2</sub> go<sub>3</sub> pokazać<sub>3</sub>.  
 Jan.NOM can.PRS.3SG him.DAT want.INF him.ACC show.INF  
 b. Jan może<sub>1</sub> go<sub>3</sub> chceć<sub>2</sub> mu<sub>3</sub> pokazać<sub>3</sub>.  
 c. Jan może<sub>1</sub> mu<sub>3</sub> go<sub>3</sub> chceć<sub>2</sub> pokazać<sub>3</sub>.  
 ‘John may want to show it to him.’ (Pol; Kupść 2000b: 61)

- (4.128) On (jo<sub>4</sub>) je<sub>1</sub> (jo<sub>4</sub>) hotel<sub>1</sub> (jo<sub>4</sub>) nehati<sub>2</sub> (jo<sub>4</sub>) hoteti<sub>3</sub>  
 he.NOM her.ACC AUX.3SG want.PST.SG.M stop.INF want.INF  
 (jo<sub>4</sub>) videvati<sub>4</sub> (jo<sub>4</sub>) vsak dan.  
 see.INF every.ACC.SG.M.INA day.ACC.SG  
 ‘He wanted to stop wanting to see her every day.’ (Slv; Marušič 2008: 275)

That deep embeddings are not purely hypothetical constructions but empirically attested is evidenced by Croatian (4.129), Czech (4.130), and Polish (4.131).

- (4.129) a. [...] i u svakome trenutku ga<sub>4</sub> možemo<sub>1</sub> odlučiti<sub>2</sub>  
 and in any.LOC.SG.M moment.LOC.SG him.ACC can.PRS.1PL decide.INF  
 prestat<sub>3</sub> držati<sub>4</sub> [...] stop.INF hold.INF  
 ‘... and in any moment, we can decide to stop holding him ...’

<sup>305</sup> As far as I am aware, this typological difference between Romance and Slavic clitic climbing has not been mentioned, yet. Clitic placement between finite verb and embedded infinitive is possible in European Portuguese (cf. Barrie 2000: 92, 100).

b. [...] samo **se**<sub>3</sub> ne smijem<sub>1</sub> zaboraviti<sub>2</sub> sjetiti<sub>3</sub> reći<sub>4</sub> **im**<sub>4</sub> [...] only REFL.ACC NEG may.PRS.1SG forget.INF remember.INF tell.INF them.DAT  
 ‘... I only must not forget to remember to tell them ...’ (Hrv; Hansen et al. 2018: 264)

c. Kako **si**<sub>1</sub> **ga**<sub>3</sub> mogla<sub>1</sub> dati<sub>2</sub> ubiti<sub>3</sub>? how AUX.2SG him.ACC can.PST.SG.F give.INF kill.INF  
 ‘How could you have him get killed?’ (Hrv; Kolaković et al. 2022: 21)

(4.130) Na gymplu najednou zjistil, že má in grammar\_school.LOC.SG suddenly realise.PST.SG.M that have.PRS.3SG čtyřky a asi **se**<sub>4</sub> bude<sub>1</sub> muset<sub>2</sub> začít<sub>3</sub> učit<sub>4</sub>. D-grade.ACC.PL and perhaps REFL.ACC AUX.FUT.3SG must.INF start.INF teach.INF  
 ‘In secondary school he suddenly realised that he had D grades and that he would have to start learning.’ (Cz; ČNK syn\_v8)

(4.131) (Dobrze, że są Dacie Logan, Reno Thalie, good.ADV that be.PRS.3PL Dacia\_Logan.NOM.PL Renealt\_Thalia.NOM.PL Suzuki Liana,) ktoś **się**<sub>4</sub> będzie<sub>1</sub> musiał<sub>2</sub> Suzuki\_Liana.NOM.SG someone.NOM.SG REFL.ACC AUX.FUT.3SG must.PTCP.SG.M zacząć<sub>3</sub> rozglądać<sub>4</sub> ... start.INF look\_around.INF  
 ‘([It’s] good that there are Dacia Logans, Renault Thalias, Suzuki Liana,) someone will have to start looking around ...’ (Pol; S9)

The examples in (4.126) and (4.128) imply that clitic climbing is syntactically optional in that the embedded clitic either may climb, which is also referred to as non-local realisation, or remain in situ, which is also called local realisation correspondingly. Consider the minimal pairs in (4.132)–(4.136), where the a-examples show clitic climbing, while the b-examples have in situ positioning of the embedded clitic.<sup>306</sup> Croatian (4.135) does not provide a minimal pair *sensu stricto*, because it is empirically attested data. However, important parameters are identical across (4.135a) and (4.135b), i.e. matrix verb incl. tense, person, and number, embedded infinitive, and reflexive clitic. If there is a matrix clitic (cluster), the local realisation of the embedded clitic (cluster) results in diacsis, which I will return to at the end of this section, cf. Slovak (4.134b) and Slovene (4.136b).

<sup>306</sup> There are different opinions on the question of whether CC is obligatory or optional.

- (4.132) a. Asi **ho**<sub>2</sub> chtěla<sub>1</sub> usušit<sub>2</sub> pomalu.  
 perhaps him.ACC want.PST.SG.F dry.INF slowly  
 b. Asi chtěla<sub>1</sub> usušit<sub>2</sub> **ho**<sub>2</sub> pomalu.  
 ‘Perhaps she wanted to dry it slowly.’ (Cz; Junghanns 2002c: 82)

- (4.133) a. Jan **go**<sub>2</sub> chciał<sub>1</sub> obudzić<sub>2</sub> o szóstej.  
 John.NOM him.ACC want.PST.SG.M wake\_up.INF at six  
 b. Jan chciał<sub>1</sub> obudzić<sub>2</sub> **go**<sub>2</sub> o szóstej.  
 ‘John wanted to wake him up at six o’clock.’ (Pol; Kupś 2000b: 60)

- (4.134) a. Dnes **by**<sub>1</sub> **som**<sub>1</sub> **sa**<sub>1</sub> **mu**<sub>2</sub> to<sub>2</sub> bál<sub>1</sub> povedat’<sub>2</sub>.  
 today COND AUX.1SG REFL him.DAT this.ACC.SG.N fear.PST.SG.M tell.INF  
 b. Dnes **by**<sub>1</sub> **som**<sub>1</sub> **sa**<sub>1</sub> bál<sub>1</sub> povedat’<sub>2</sub> **mu**<sub>2</sub> to<sub>2</sub>.  
 ‘I would be afraid to tell him (it) today.’ (Slk; Short 1993b: 566)

- (4.135) a. A da **se**<sub>2</sub> morate<sub>1</sub> odlučiti<sub>2</sub>?  
 and if REFL.ACC must.PRS.2PL decide.INF  
 ‘And if you have to decide?’ (Hrv; Kolaković et al. 2022: 312)  
 b. Pa ako već morate<sub>1</sub> odlučiti<sub>2</sub> **se**<sub>2</sub> za jaslice [...].  
 well if already must.PRS.2PL decide.INF REFL.ACC for nursery  
 ‘Well if you [already] have to opt for nursery ...’ (Hrv; Kolaković et al. 2022: 311)

- (4.136) a. Jaz **sem**<sub>1</sub> **ji**<sub>2</sub> **ga**<sub>2</sub> moral<sub>1</sub> pošiljati<sub>2</sub> po pošti.  
 I.NOM AUX.1SG her.DAT him.ACC must.PST.SG.M send.INF by mail  
 b. Jaz **sem**<sub>1</sub> moral<sub>1</sub> pošiljati<sub>2</sub> **ji**<sub>2</sub> **ga**<sub>2</sub> po pošti.  
 ‘I had to send it to her by mail.’ (Slv; Golden 2008: 310)

Hitherto, most examples represent trivial instances of clitic climbing, but there are linearisation patterns, where it is more complicated to determine whether one encounters true instances of climbing. This is the case for strings of the type [V<sub>1</sub> CL<sub>2</sub> V<sub>2</sub>], i.e. clitics placed in intermediate position between a matrix and an embedded verb. I concur with Junghanns (2002c: 67–68) in that it is impossible to determine on an ad hoc basis whether the clitic in question has climbed in fact. It depends on the heuristics applied, if this pattern is analysed as climbing or as clitic positioning within the governing predicate’s domain. On a purely syntagmatic basis, one cannot decide this matter. Referring to Fried (1994) and Toman (1996), Junghanns (2002c: 67) remarks that prosodic evidence

(i.e. the distribution of pauses) can be taken to determine which verb the clitic is attached to (recall the discussion in Section 4.2.1, especially Czech example [4.5]). He suggests that clear instances of CC can be identified in Czech by the diagnostics in (4.137) (cf. Junghanns 2002c: 68):

- (4.137) a. The clitic is positioned to the left of / higher than the matrix predicate.  
b. The clitic is positioned to the left of / higher than other syntactic material (complements) of the matrix clause.  
c. The clitic participates in cluster-internal morpho-phonological processes of the matrix clitic cluster (i.e. the fusion of AUX + REFL).

From (4.137a) it follows that example (4.138a) cannot be qualified as CC with absolute certainty (cf. also the Czech ex. [4.88c] from Section 4.5), whereas utterance (4.122a) repeated in (4.138b) is a safe instance of CC. Note that, within the formal syntactic literature nested, e.g., in the Minimalist Program or HPSG, linearisations as in (4.138a) are considered as CC by default. I will follow this idealised procedure for practical reasons and treat sequences of the type  $[V_1 \text{ CL}_2 \text{ V}_2]$  as instances of clitic climbing.

- (4.138) a. i przyjechał<sub>1</sub> **mu**<sub>2</sub> z-robić<sub>2</sub> awanturę  
and come\_by\_vehicle.PST.SG.M him.DAT PFV-make.INF quarrel.ACC.SG  
'and he came to make a fuss [in front of] him' (Pol; NKJP pelcra)
- b. [...] (to mu powiedziałam,) że smok **go**<sub>2</sub> przyjechał<sub>1</sub> odwiedzić<sub>2</sub>  
'... (then I told him) that the dragon came to visit him' (Pol)

Note that the criterion in (4.137a) is not sufficient by itself. Linear re-orderings due to information structure can interfere and increase uncertainty, e.g. when the embedded infinitival phrase is topicalised. A clitic in intermediate position is thus ambiguous between local and non-local placement. The issue is illustrated in the schemes in (4.139), where  $V_1$  represents the finite matrix verb and  $V_2$  the embedded infinitive, and the bracketing displays possible phrasal boundaries. The contrast between configuration (4.139a) and (4.139b) expresses the previously mentioned uncertainty with clitic intermediate placement. Structure (4.139c) shows the safe occurrence of clitic climbing. The contrast between structures (4.139d) and (4.139e) visualises the interference of information structure and the resulting uncertainty for determining CC again. Configuration (4.139d) displays clitic climbing with simultaneous fronting of the infinitival head, while (4.139e) provides the same linearisation, but the entire embedded VP is topicalised including the clitic.

- (4.139) a. [<sub>α</sub> V<sub>1</sub> CL<sub>2</sub> [<sub>β</sub> V<sub>2</sub> ]] non-local realisation = CC  
 b. [<sub>α</sub> V<sub>1</sub> [<sub>β</sub> CL<sub>2</sub> V<sub>2</sub> ]] local realisation = CL in governor's domain  
 c. [<sub>α</sub> CL<sub>2</sub> V<sub>1</sub> [<sub>β</sub> V<sub>2</sub> ]] non-local realisation = CC  
 d. [<sub>α</sub> [<sub>β</sub> V<sub>2</sub> ] CL<sub>2</sub> V<sub>1</sub> ] topicalisation of V<sub>INF</sub> + non-local realisation (CC)  
 e. [<sub>α</sub> [<sub>β</sub> V<sub>2</sub> CL<sub>2</sub> ] V<sub>1</sub> ] topicalisation of V<sub>INF</sub> + local realisation (CL in governor's domain)

The problem is not purely theoretical, which is evident from examples in (4.140)–(4.142). Furthermore, the reasoning applies also to criterion (4.137b).

- (4.140) a. [...] že soudit<sub>2</sub> **ho**<sub>2</sub> může<sub>1</sub> jen Bůh.  
 that judge.INF him.ACC can.PRS.3SG only God.NOM  
 ‘... that only God can judge him.’ (Cz; ČNK syn\_v8)  
 b. vodložit<sub>2</sub> **si**<sub>2</sub> **ho**<sub>2</sub> ne-může<sub>1</sub>  
 take\_off.INF REFL.DAT him.ACC NEG-can.PRS.3SG  
 ‘he cannot take it [= the coat] off’ (Cz; ČNK oral\_v1)
- (4.141) polać<sub>2</sub> **mu**<sub>2</sub> chciała-m<sub>1</sub>  
 pour.INF him.DAT want.PST.SG.F-1SG  
 ‘I wanted to pour him [a shot].’ (Pol; NKJP pelcra)
- (4.142) [...] splatit’<sub>2</sub> **ho**<sub>2</sub> chce<sub>1</sub> do 15 rokov.  
 pay\_off.INF him.ACC want.PRS.3SG to 15 year.GEN.PL  
 ‘... he wants to pay it [= loan] off within 15 years.’ (Slk; SNK prim-10.0-public-all)

A solution to the problem is to refine the criteria in (4.137a)–(4.137b) and to apply them simultaneously, which seems to be in line with Kolaković et al. (2022: 230), who define clitic climbing as the realisation of an embedded clitic “in a position contiguous to elements of the matrix.” Consequently, clitic climbing can be identified, if a clitic is surrounded by matrix elements. Consider examples (4.143) from Junghanns (2002c) and (4.144) from Ruda et al.’s (2023a) SIL corpus. Utterance (4.143) presents the embedded reflexive clitic *se* occurring in between the matrix verb *chodějí* ‘they are walking’ and the local/directive adverb *sem* ‘here’. The adverb is unequivocally associated with the matrix verb of motion. Thus, the clitic is surrounded by matrix elements on both sides. In the Silesian example (4.144), the embedded infinitive is pragmatically prominent and has



been fronted due to Contrastive Focus.<sup>307</sup> Simultaneously, the reflexive clitic occurs left-adjacent to the matrix verb. The clitic cluster *bych se* is placed between the fronted infinitive and the matrix verb. However, the conditional clitic *bych* is associated with the *l*-participle and, therefore, a matrix element. Accordingly, the reflexive clitic cannot be localised in the infinitive's domain. The utterances in (4.143) and (4.144) thus represent unequivocal instances of CC.

- (4.143) Choděj<sub>1</sub>      **se**<sub>2</sub>      sem na tebe      ptát<sub>2</sub>.  
 walk.PRS.3PL REFL.ACC here on you.ACC ask.INF  
 'They are coming here to ask for you.' (Cz; Junghanns 2002c: 67)

- (4.144) Bajtłowi      [zabić<sub>2</sub>]<sub>FOC</sub>      **bych**<sub>1</sub>      **sie**<sub>2</sub>      niy      doł<sub>1</sub>,  
 kid.DAT.SG kill.INF COND.1SG REFL.ACC NEG give.PST.SG.M  
 (nale dzieckom      bych      tysz niy      zezwolił na iść      śmierć.)      [sic]  
 but child.DAT.PL COND.1SG also NEG allow on go.INF death.ACC.SG  
 i. 'I would not let a kid kill me, (but I would also not allow children to walk into death.)  
 ii. 'I would not let a kid kill itself, ...' (Sil; SIL)

On the other hand, examples like (4.145) are safe instances of local placement. The embedded infinitive is fronted due to topicalisation. The embedded accusative clitic occurs medially between the topicalised and matrix verb and so does the matrix reflexive clitic. However, the putative clitic cluster shows metaclisis, i.e. *ho* < *se* instead of template-compliant *se* < *ho*. It is therefore likely that the entire VP including the pronominal clitic *ho* is fronted. However, in the absence of environmental evidence, I do not possess any further heuristics to disambiguate between structures of the type (4.139d) and (4.139e).

- (4.145) (Šéf      je      jen jeden)      a      [VP přerůst<sub>2</sub>      **ho**<sub>2</sub>]<sub>TOP</sub>      **se**<sub>1</sub>  
 boss.NOM.SG be.PRS.3SG only one.NOM.SG.M and grow\_taller.INF him.ACC REFL  
 ne-doporučuje<sub>1</sub>.  
 NEG-recommend.PRS.3SG  
 '(There is only one boss) and it is not advisable to outperform him.' (Cz; ČNK syn\_v8)

The remaining criterion (4.137c) is language-specific. While Czech and BCMS exhibit well-known morpho-phonological processes in the clitic cluster, Polish and Slovak do not show any such

<sup>307</sup> Example (4.144) is ambiguous, which is why it is provided with two translations.

phenomena.<sup>308</sup> Note that these processes do not apply consistently. Exceptions occur with increasing frequency in informal contexts and colloquial speech.

Finally, clitic climbing does not invariably target second position contrary to statements found in the literature (cf. e.g. Rezac 2005: 105; Tabachnik 2024: 2). Suffice it to consider the Czech example in (4.146), which shows embedded Clitic Third after climbing of the reflexive clitic. This is fully in accordance with the discussion of Delayed Placement in Section 4.4.2.

- (4.146) Jistě        namítnete,        ..., [S že<sup>1</sup> [NP něco        podobného ]<sup>2</sup> se<sup>3</sup>        může<sub>1</sub>  
                  certainly object.PRS.2PL        that        something similar        REFL can.PRS.3SG  
                  stát<sub>2</sub>        jen        v Americe ].  
                  happen.INF only        in America.LOC  
                  ‘You will certainly note that something similar can happen only in America.’

(Cz; Short 1993a: 495)

Now, let us have a look at the syntactic structures licensing clitic climbing. The relevant environments have been identified as raising, control, and the accusative with infinitive construction (ACI) (cf. Hana 2007: 128–130; Junghanns 2002c: 65–66 for Czech; Kupść 2000b: 58–59; Witkoś 1998: 191, for Polish; Kolaković et al. 2022: 34–40, for BCMS; Golden 2003: 221, 2008: 311–312 for Slovene [and Czech]).<sup>309</sup> Raising verbs lack external arguments, i.e. they do not have logical subjects.<sup>310</sup> Hence, they do not pose selectional restrictions on whatever element agrees with them. Raising verbs

<sup>308</sup> Czech exhibits the fusion of the reflexive clitics with the 2nd person singular auxiliary clitic (*jsi + se* → *ses*, *jsi + si* → *sis*), cf. (i). BCMS shows dissimilative allomorphy of the feminine accusative clitic when co-occurring with the 3rd person singular auxiliary clitic (*je je* → *ju je*), cf. (ii). Bear in mind that these processes are not obligatory in actual speech.

(i) Takže ty        ne-máš        nic,        co        by        sis        dneska        čtl        pro  
       so        you.SG        NEG-have.PRS.2SG        nothing.ACC        what        COND        AUX.REFL.DAT        today        read.PST.SG.M        for  
       zábavu?  
       amusement.ACC.SG

‘So, you don’t have anything you could read for amusement today?’ (Cz; ČNK syn2020)

(ii) Projektirao        ju        je        arhitekt        Aleksandar        Bašić        iz        Zagreba.  
       design.PST.SG.M        her.ACC        AUX.3SG        architect.NOM.SG        Aleksandar.NOM        Bašić.NOM        from        Zagreb.SG

‘Architect Aleksandar Bašić from Zagreb designed it [= the church].’ (Hrv; hrWaC)

<sup>309</sup> It has been a contentious issue, exactly which syntactic constructions license CC. Junghanns (2002c; cf. esp. p. 69, n. 9) and Thorpe (1991: 45–50, 60) preclude CC from object control in Czech. On the other hand, Thorpe (1991: 38) claims that Czech does not possess the ACI construction. Rezac (2005: 108) argues that Czech does not have true ACI verbs. Veselovská (1995: §9.5, §9.5.2) judges CC ungrammatical from ACI constructions in Czech, but her example blocks involve an ACI as well as an control construction (cf. Veselovská 1995: §9.5, ex. [57]–[58]). Furthermore, CC is obligatory in Czech according to Veselovská (1995: §9.5). Adam (2022: 63) claims that “[c]litic climbing is barred out of [...] control clauses with PRO subjects.”

<sup>310</sup> I shall only briefly comment on some terminological inconsistencies: the term *raising verb* might be considered ambiguous. Generally, it is used for *subject raising verb* or *raising to subject* respectively. The abbreviation ACI originates in classical philology and stands for *accusativus cum infinitivo* ‘accusative with infinitive’. In generative syntax the ACI is rather known as *Exceptional Case Marking* (ECM), which the terms *object raising* or *raising to object* are sometimes used for. *Control* is also be referred to as *equi*.

subcategorise for a proposition as their internal argument only.<sup>311</sup> Examples for CC in raising constructions are provided in (4.147)–(4.150).

- (4.147) [...] *že mu<sub>2</sub> major začal<sub>1</sub> tykat<sub>2</sub>.*  
 that him.DAT major.NOM start.PST.SG.M address\_by\_informal\_you.INF  
 ‘... that the major started to address him with his first name’ (Cz; Junghanns 2002c: 66)

- (4.148) *Maria się<sub>1</sub> go<sub>2</sub> wydaje<sub>1</sub> lubić<sub>2</sub>.*  
 Maria.NOM REFL him.ACC seem.PRS.2SG like.INF  
 ‘Maria seems to like him.’ (Pol; Kupść 2000b: 59)

- (4.149) [...] *ne smije<sub>1</sub> se<sub>2</sub> više baviti<sub>2</sub> čokoladom.*  
 NEG be\_allowed.PRS.3SG REFL.ACC more occupy.INF chocolate.INS.SG  
 ‘... he is not allowed to be in the chocolate business anymore.’  
 (Hrv; Kolaković et al. 2022: 317)

- (4.150) *Jaz sem<sub>1</sub> ji<sub>2</sub> ga<sub>2</sub> utegnil<sub>1</sub> opisati<sub>2</sub> slabo.*  
 I.NOM AUX.1SG her.DAT him.ACC may.PST.SG.M describe.INF badly  
 ‘I may have described him to her badly.’ (Slv; Golden 2003: 220)

In contrast, control verbs have an external argument as well as internal arguments, i.e. they subcategorise for a logical subject, objects, and propositions. There is a further distinction between subject and object control, whereby the logical (non-overt) subject of the embedded proposition is co-referential with the matrix subject under the former, but with one of the matrix objects under the latter.<sup>312</sup> Clitic climbing under subject control is found in (4.151)–(4.154).

<sup>311</sup> The basic concept stems from Generative Grammar, where it was observed that utterances like English (i) and (ii) are semantically equivalent. The subject pronoun *it* in (i) is an expletive without semantic content. It only satisfies the syntactic requirement that English clauses must have an overt syntactic subject. The raising verb *seem* subcategorises for the embedded proposition *like(Chris, heavy metal)*, but it does not have a logical subject expressed in terms of a theta-role in Government & Binding Theory (cf. Chomsky 1981: 56–58). In (ii), *Chris*, which is the logical subject of *like*, has become the syntactic subject of the matrix subject *seem* (via movement) agreeing with it.

(i) It seems [ that Chris likes heavy metal ].

(ii) Chris<sub>j</sub> seems [ t<sub>j</sub> to like heavy metal ]. (Eng; constructed)

On a different note, the requirement of having an overt subject does not hold in English in a strict sense. Subject pronouns can be dropped in colloquial English and certain written registers (cf. Weir 2012 and references therein). Schröter (2019) investigates the phenomenon from a variationist perspective. The existence of null subjects poses a problem for the assumption of the so-called Extended Projection Principle (EPP) in English.

<sup>312</sup> Subject control is exemplified in (i) and object control in (ii) for English. I adopt PRO from generative syntax to indicate the logical subject of the embedded proposition. The indexed letters display co-reference in either control construction.

(i) Mary<sub>j</sub> promised John<sub>k</sub> PRO<sub>j</sub> to invite Paul.

‘Mary promised John such that she [= Mary] will invite Paul.’

(4.151) Alena **ho**<sub>2</sub> slíbila<sub>1</sub> navštívit<sub>2</sub>, jakmile to bude  
 Alena.NOM him.ACC promise.PST.SG.F visit.INF as\_soon\_as this.NOM.SG.N be.FUT.3SG  
 možný.  
 possible.NOM.SG.M

‘Alena promised to visit him as soon as possible.’ (Cz; Hana 2007: 128)

(4.152) Marek **go**<sub>2</sub> obiecał<sub>1</sub> zaprosić<sub>2</sub>.  
 Mark.NOM him.ACC promise.PST.SG.M invite.INF

‘Mark promised to invite him.’ (Pol; Kupść 2000a: 95)

(4.153) No ne znaju<sub>1</sub> **se**<sub>2</sub> svi izraziti<sub>2</sub> lijepo [...].  
 but NEG know.PRS.3PL REFL.ACC all.NOM.PL.M express.INF nicely

‘But not everyone knows to express themselves nicely ...’  
 (Hrv; Kolaković et al. 2022: 263)

(4.154) Janez **se**<sub>1</sub> **ji**<sub>2</sub> **ga**<sub>2</sub> **je**<sub>1</sub> naveličal<sub>1</sub> hvaliti<sub>2</sub>.  
 Janez.NOM REFL her.DAT him.ACC AUX.3SG tire.PST.SG.M praise.INF

‘Janez grew tired of praising him to her.’ (Slv; Golden 2008: 311)

Examples for clitic climbing in object control constructions are cited in (4.155)–(4.157). The Czech examples in (4.155) show that object control comes in two flavours, either a direct object or an indirect object controls the embedded infinitive’s subject position. The former is shown in (4.155a) with the accusative controller *ji*, while (4.155b) provides the latter with the dative controller *nám*.<sup>313</sup> Note that two accusative argument clitics cluster in (4.155a).<sup>314</sup> Polish (4.156) gives an example of indirect object control with the dative NP *Piotrowi* as controller.<sup>315</sup> Utterance (4.157) shows indirect

(ii) Mary<sub>j</sub> forced John<sub>k</sub> PRO<sub>k</sub> to invite Paul.

‘Mary forced John such that he [= John] will invite Paul.’

(Eng; constructed)

<sup>313</sup> On a different note, Lenertová (2004: 154) tells that the reverse order *ho > ji* is not acceptable for (4.155a) and suggests that a phonological constraint is at work prohibiting *ho* to occur initially in the clitic cluster irrespective of its case (*ho* is ambiguous between accusative and genitive). However, the reverse order is unacceptable for the particular reading only, while the reading (i) is fine (cf. Rosen 2001: 233, who argues that linear order in a clitic cluster mirrors the hierarchic order of verbs in complex clauses).

(i) Stejně **ho**<sub>1</sub> **ji**<sub>2</sub> ne-nechali<sub>1</sub> dokončit<sub>2</sub>.  
 anyway him.ACC her.ACC NEG-let.PST.PL.VIR finish.INF

‘They didn’t let him finish it [= her] anyway.’

(Cz; constructed from Lenertová 2004: 154)

<sup>314</sup> Data with two clustering accusative clitics contradict Rezac (2005: 105, 118, 120), who argues that such double constructions are banned.

<sup>315</sup> Note that the matrix element *powinien* ‘should’ in (4.156) is an adjective etymologically, but synchronically analysed as a paradigmatically defective modal verb. It inflects for person, number, and gender, but is underspecified for tense. Past and future tense forms are built analytically with the copula *być* ‘be.INF’ serving as tense auxiliary, while present tense is the default value in the absence of an auxiliary. This is why it is not accompanied by a gloss for tense.

object control for Slovene, but it is ambiguous between two interpretations. In the first reading, the dative pronominal clitic *ji* is a matrix object and, hence, the controller. In the second interpretation, *ji* is an embedded object, while the phonetically unrealised controller gives rise to a generic reading.<sup>316</sup>

- (4.155) a. Stejně **ji**<sub>1</sub> **ho**<sub>2</sub> ne-nechali<sub>1</sub> dokončit<sub>2</sub>.  
 anyway her.ACC him.ACC NEG-let.PST.PL.VIR finish.INF  
 ‘They didn’t let her finish it anyway.’ (Cz; Lenertová 2004: 154)

- b. Pavel **nám**<sub>1</sub> **ho**<sub>2</sub> pomohl<sub>1</sub> najít<sub>2</sub>.  
 Paul.NOM us.DAT him.ACC help.PST.SG.M find.INF  
 ‘Paul helped us to find him/it.’ (Cz; Rosen 2001: 226)

- (4.156) Jan powinien<sub>1</sub> zabronić<sub>2</sub> **go**<sub>3</sub> Piotrowi zapraszać<sub>3</sub>.  
 John.NOM should.3SG.M forbid.INF him.ACC Peter.DAT invite.INF  
 ‘John ought to forbid Peter to invite him.’ (Pol; Kupść 2000a: 96)

- (4.157) Jaz **sem**<sub>1</sub> **ji**<sub>1/2</sub> **ga**<sub>2</sub> dovolil<sub>1</sub> poslati<sub>2</sub> po pošti.  
 I.NOM AUX.1SG her.DAT him.ACC allow.PST.SG.M send.INF by mail  
 i. ‘I allowed her to send it by mail.’  
 ii. ‘I allowed to send it to her by mail.’ (Slv; Golden 2008: 312)

Clitic climbing in object control provides another typological distinction between Slavic and Romance, because the latter has been reported to prohibit CC in object control constructions (cf. Golden 2008: 315 and references therein; yet Spanish allows CC under object control, cf. Kulick 1998: 90–91; Nishida 2013). Based on corpus and experimental data, Kolaković et al. (2022) conclude that clitic climbing is blocked in BCMS control constructions. However, their experiment only examines object control constructions with overtly realised matrix object clitics. Kupść (2000b: 58–59) discusses for Polish that the realisation of a matrix object may affect the possibility of clitic climbing, but this does not imply a general ban on CC under object control (cf. also Dyła 1983: 329–332 for Polish; Dotlačil 2004: 80–82; Hana 2007: 129–130; Rosen 2001: 227, for comparable Czech examples). Franks & King (2000: 335–336) discuss this blocking effect more generally with reference to the Head Movement Constraint, according to which a lower head cannot skip a higher one.

<sup>316</sup> Ambiguity is indicated by the subscript  $x_{1/2}$ . Cf. also a similar example from Golden (2008: 316, ex. [25]) in (i).

(i) Janez **ji**<sub>1/2</sub> **jih**<sub>2</sub> **je**<sub>1</sub> dovolil<sub>1</sub> kupiti<sub>2</sub>.  
 Janez.NOM her.DAT them.ACC AUX.3SG allow.PST.SG.M buy.INF  
 i. ‘Janez allowed her to buy them.’  
 ii. ‘Janez allowed (someone) to buy it/them for her/them.’ (Slv; Golden 2008: 316)

Translated to clitics it means that a lower (further right) clitic cannot leave a phrasal domain moving across a higher (further left) clitic (cf. Golden 2003: 223–226 for Slovene).<sup>317</sup> Returning to object control constructions in BCMS, consider the evidence from the Croatian web corpus in (4.158).<sup>318</sup> Example (4.158a) involves the past tense (*l*-participle) form of the prohibitive object control *zabraniti* and (4.158b) provides the passivised permissive object control verb *dozvoliti*. In both cases the preceding accusative pronominal clitic *ga* is a dependent of the embedded infinitive, while the matrix verb does not have an overtly realised object.

- (4.158) a. [...] (i dao nam mozak,) zašto **bi**<sub>1</sub> **nam**<sub>1</sub>  
 and give.PST.SG.M us.DAT brain.ACC.SG why COND us.DAT  
 onda **ga**<sub>2</sub> zabranio<sub>1</sub> koristiti<sub>2</sub>?  
 then him.ACC forbid.PST.SG.M use.INF  
 ‘... (and [He] gave us a brain,) [so] why should He then forbid us to use it?’
- b. [...] ali nije<sub>1</sub> **ga**<sub>3</sub> dozvoljeno<sub>2</sub> stavljati<sub>3</sub> u živežne  
 but NEG.be.PRS.3SG him.ACC allow.PASS.PTCP.SG.N put.INF in food.ADJ  
 namirnice.  
 groceries  
 ‘... but it isn’t allowed to put it into groceries.’ (Hrv; hrWaC)

In the ACI construction, the matrix verb subcategorises for a subject and a proposition (or alternatively for an object). Different from the control construction, the embedded subject is overtly realised, but its case is governed by the matrix predicate.<sup>319</sup> The ACI construction does not exist in (standard) Polish and is considered ungrammatical therefore (cf. Kupść 2000b: 59; Przepiórkowski & Rosen 2005: 33, n. 33; Urbańczyk 1976: 56). It is also not attested in contemporary Polish colloquial speech. Urbańczyk (1976: 56) notes that the ACI occurs in the Polish dialects of Pomerania, Greater Poland, Silesia, and south-western Lesser Poland.<sup>320</sup> It occurs in Kashubian, too (cf. Stone 1993: 785). The

<sup>317</sup> Correspondingly, Rosen (2001: 227) informs that embedded clitics cannot climb across matrix clitics.

<sup>318</sup> CQL query: [word="ga"] [lemma="zabraniti|dozvoliti"] [tag="Vmn.\*"]

<sup>319</sup> Constructions of this type have been named Exceptional Case Marking (ECM) in Generative Grammar.

<sup>320</sup> Urbańczyk (1976: 56) conjectures that the occurrence of the ACI construction is due to linguistic influence of German or Slovak (the latter for Podhale). Klemensiewicz (1999: 314–315) informs that the ACI construction existed in earlier stages of Polish. It is attested already in Old Polish, cf. (i). The frequency of occurrence reached its peak during the Middle Polish period in the 16th and 17th century but decreased afterwards during the 18th century. According to Klemensiewicz (1999: 627), the ACI was so infrequent in the 19th century that he qualifies it as an exceptionally occurring construction, which was not productive anymore.

(i) Panie, kiedy=**smy**<sub>1</sub> **cie**<sub>2+3</sub> widzieli<sub>1</sub> łaknąć<sub>2</sub> albo pragnąć<sub>3</sub> picia, [...]?  
 Lord.VOC when=AUX.1PL you.ACC see.PST.SG.VIR hunger.INF or be\_thirsty.INF drink.GEN.SG  
 ‘Lord, when did we see you hungry ... or thirsty, ...?’ (OPol; Rozmyślanie przemyskie)

ACI is rather rare in Croatian, but attested, cf. (4.163).<sup>321</sup> Data points for clitic climbing with the ACI are given in (4.159)–(4.164).<sup>322,323</sup> Example (4.160) provides dialectal data points for Polish for illustrative purposes. Note that there are two argumental accusative clitics in the clitic cluster in Slovene (4.164a), whereby one realises the embedded subject, while the other represents the embedded direct object.

- (4.159) a. V jaké poloze jste<sub>1</sub> ho<sub>2</sub> našel<sub>1</sub> ležet<sub>2</sub>?  
in what.LOC.SG.F position.LOC.SG AUX.2PL him.SBJ.ACC find.PST.SG.M lie.INF  
‘In what position did you find him lying?’ (Cz; Junghanns 2002c: 65)

- b. Když tě<sub>2</sub> slyším<sub>1</sub> popisovat<sub>2</sub> své problémy  
when you.SBJ.ACC hear.PRS.1SG describe.INF one’s.POSS.ACC.PL problem.ACC.PL  
s přítelem (mám o tebe strach a  
with boyfriend.INS.SG have.PRS.1SG about you.ACC fear.AGG.SG and  
obavy.)  
concern.ACC.PL  
‘When I hear you describing your problems with [your] boyfriend, (I am anxious and worried about you.)’ (Cz; ČNK syn2020)

- (4.160) a. a ja slyszal-e-m<sub>1</sub> go<sub>2</sub> tak mówić<sub>2</sub>  
and I.NOM hear.PST.SG.M-M-AUX.1SG him.SBJ.ACC so talk.INF  
‘and I heard him talking like this’  
b. znaleźli<sub>1</sub> go<sub>2</sub> pijanego w ryństoku na ulicy leżeć<sub>2</sub>  
found.PST.PL.VIR him.SBJ.ACC drunk.3SG.M in street\_gutter.LOC on street.LOC lie.INF  
‘They found him lying drunk on the street in the street gutter.’  
(Pol, non-standard; Urbańczyk 1976: 56)

<sup>321</sup> Cf. Horvat & Perić Gavrančić (2019) and Perić Gavrančić (2022) for the occurrence of the ACI in Croatian of the pre-standard era.

<sup>322</sup> The gloss SBJ.ACC is meant to mark the embedded infinitive’s accusative subject for ease of reference. It should not be confused with the matrix subject, which can be omitted.

<sup>323</sup> CQL query:

(i) ČNK: [word="ho|tě"] [lemma="slyšet"] [tag="Vf.\*"]  
(ii) SNK: [word="ma|ťa|ho"] [lemma="počut|vidiet"] [tag="VI.\*"]  
(iii) sIWaC: [word="ga"] [lemma="slišati"] [tag="Gg.n.\*"]

- (4.161) Widzioł<sub>1</sub>=že-ch<sub>1</sub>                go<sub>2</sub>                kurzić<sub>2</sub>                na placu.<sup>324</sup>  
see.PST.SG.M=ŽE-AUX.1SG him.SBJ.ACC smoke.INF on yard.LOC.SG  
‘I saw him smoking outside.’ (Sil; Tambor 2014: 156)
- (4.162) a. A ked’ som<sub>1</sub>                ho<sub>2</sub>                počula<sub>1</sub>                spievať<sub>2</sub>, [...]   
and when AUX.1SG him.SBJ.ACC hear.PST.SG.F sing.INF  
‘And when I heard him singing ...’  
b. (Ne-pláč,                prosím                ťa,)                ne-čhcem<sub>1</sub>                ťa<sub>3</sub>                vidiet’<sub>2</sub>  
NEG-cry.IMP.2SG request.PRS.1SG you.ACC NEG-want.PRS.1SG you.SBJ.ACC see.INF  
plakat’<sub>3</sub>.  
cry.INF  
‘(Don’t cry, I beg you,) I don’t want to see you cry.’ (Slk; SNK prim-10.0-public-all)
- (4.163) a. Mislim                da sam<sub>1</sub>                te<sub>2</sub>                čula<sub>1</sub>                govoriti<sub>2</sub> (kako je  
think.PRS.1SG that AUX.1SG you.ACC hear.PST.SG.F speak.INF how be.PRS.3SG.CL  
njohov tetak                otvjentik                u Merytonu.)  
their uncle.NOM.SG attorney.NOM.SG in Meryton.LOC  
‘I think I have heard you say (that their uncle is an attorney in Meryton).’  
(Hrv; Mihaljević 2009: 323)  
b. Vidim<sub>1</sub> /                čujem<sub>1</sub>                ga<sub>2</sub>                voziti<sub>2</sub> auto.  
see.PRS.1SG hear.PRS.1SG him.SBJ.ACC drive.INF car.ACC.SG  
‘I see/hear that him drive a car.’ (Hrv; Mihaljević 2009: 328)
- (4.164) a. Janez                te<sub>2</sub>                jo<sub>2</sub>                je<sub>1</sub>                vida<sub>1</sub>                dati<sub>2</sub> v  
Janez.NOM you.SBJ.ACC her.OBJ.ACC AUX.3SG see.PST.SG.M give.INF in  
žep.  
pocket.ACC.SG  
‘Janez saw you put it in your pocket.’ (Slv; Golden 2008: 311)

<sup>324</sup> The gloss ŽE is used here to mark the pleonastic element *že*, which is not a subordinator or intensifying particle/affix but a semantically empty element hosting the auxiliary clitic *-ch*. Franks & King (2000: 154–156) discuss this occurrence in Polish under the term “*že*-insertion”, while Franks (2010: 139–141) uses “*že*-support”, which I will adopt here for its theory-neutral flavour. Note that superficially similar sequences of non-subordinating *že* and auxiliary clitic occur in Czech, too, cf. (i)–(ii) from Veselovská (1995).

(i) Komu **že jsi** to dal?  
who.DAT ŽE AUX.2SG this.ACC.SG give.PST.SG.M  
'Whom did you give it?'

(ii) Marušce **že jsi** to dal?  
Mary.DIM.DAT ŽE AUX.2SG this.ACC.SG give.PST.SG.M  
'You gave it to Mary?'

(Cz; Veselovská 1995: §4.6, ex. [29])



b. Dostikrat sem<sub>1</sub>      **ga**<sub>2</sub>      slišal<sub>1</sub>      reči<sub>2</sub>:    [...]

often      AUX.1SG him.ACC hear.PST.SG.M say.INF

‘Often I heard him saying: ...’

(Slv; slWaC)

Witkoś (1998: 157, 193) observes that clitics may climb from negated infinitives in Polish, which contrasts with Romance, where negation is reported to block climbing. He provides the examples in (4.165). The same facts hold for Czech, Slovak, Croatian, and Slovene as shown by the examples in (4.166)–(4.169).<sup>325</sup> These data pose a problem for the hypothesis that CC occurs from restructuring bare infinitives (cf. Aljović 2004 for BCMS; Rezac 2005 for Czech).<sup>326</sup> The Croatian data point (4.168) directly contradicts Aljović’s (2004: 194) claim for BCMS that “[...] sentence negation [...] prevent[s] clitics from attaching to the matrix verb”. The same contradiction applies to Kupść’s (2000b: 62) argument “that negation stops syntactic CC.”<sup>327</sup>

<sup>325</sup> CQL queries:

- (i) ČNK: [word="ho"] [tag="V.\*"] [word="nemilovat"]
- (ii) ČNK: [word="mu"] [tag="V.\*"] [tag="Vf.\*" & word="ne.\*"]
- (iii) SNK: [word="ho"] [tag="G.\*"] [tag="VI.\*" & word="ne.\*"]
- (iv) hrWaC: [word="ga"] [tag="V.\*"] [word="ne"] [tag="Vmn.\*"]
- (v) slWaC: [word="ga"] [tag="G.\*"] [word="ne"] [tag="Gg.n.\*"]

<sup>326</sup> I thank Jacek Witkoś (p.c., 5 Sep 2021), who pointed this argument out to me. It is assumed in Generative Grammar that negation projects its own phrase NegP.

<sup>327</sup> Kupść (2000b: 62) basically follows Przepiórkowski’s (1999: 160–163) discussion of the interaction between negation and CC. He argues on the basis of examples like (i)–(ii) that “optional raising [= CC; I.K.] in Polish is blocked by intervening negation.” (Przepiórkowski 1999: 162) While fronting the infinitive without the clitic is possible and easily accounted for by CC in (i.b), preposing a negated infinitive without its clitic argument degrades the utterance in (ii.b). When the clitic is fronted with its governing infinitive, then the judgement is improved. The main problem with this reasoning is that Witkoś’s (1998) observation clearly holds in (4.165), which is a textbook example of CC across a negated verb. Additionally, Przepiórkowski (1999: 162) admits that “the judgements are not always clear”, which suggests that the data are somewhat controversial, and strong conclusions should not be drawn too eagerly. Consequently, one should consider alternative explanations for the contrasts in (ii). Neither can I do justice to such an endeavour, nor will I propose an ad hoc solution here. However, it seems that the problem is of a more basic nature and not tied to CC or any other aspect of cliticisation. Example (iii) suggests that the issue might be rather one of fronting a negated verb, which renders utterance (iii.b) awkward. This is why I contest Przepiórkowski’s (1999) and Kupść’s (2000b) conclusions here. Since there are neither convincing evidence nor sound arguments for a blocking effect of negation on CC, I continue to follow Witkoś’s (1998) observation.

- (i) a. Próbowałem<sub>1</sub> wystraszyć<sub>2</sub> **go**<sub>2</sub> wczoraj.  
try.PST.SG.M.1SG frighten.INF him.ACC yesterday
- b. Wystraszyć<sub>2</sub> próbowałem<sub>1</sub> **go**<sub>2</sub> wczoraj.  
    ‘I tried to frighten him yesterday.
- (ii) a. Próbowałem<sub>1</sub> **nie** wystraszyć<sub>2</sub> **go**<sub>2</sub> wczoraj.  
try.PST.SG.M.1SG NEG frighten.INF him.ACC yesterday
- b. ?\* **Nie** wystraszyć<sub>2</sub> próbowałem<sub>1</sub> **go**<sub>2</sub> wczoraj.
- c. ? **Nie** wystraszyć<sub>2</sub> **go**<sub>2</sub> próbowałem<sub>1</sub> wczoraj.  
    ‘I tried not to frighten him yesterday.

(Pol; Przepiórkowski 1999: 162)

- (iii) a. Wolę **nie** spać w hotelach.  
prefer.PRS.1SG NEG sleep.INF in hotel.LOC.PL
- b. ?/\* **Nie** spać wolę w hotelach.  
    ‘I prefer not to sleep in hotels.’

(Pol; constructed)

(4.165) a. Jan            **go**<sub>2</sub>            mógł<sub>1</sub>            **nie**    spotkać<sub>2</sub>    wczoraj.  
 John.NOM him.ACC can.PST.SG.M NEG meet.INF yesterday  
 ‘John may not have met him yesterday.’ (Pol; Witkoś 1998: 157)

b. Kapitan            **go**<sub>2</sub>            mógł<sub>1</sub>            **nie**    bić<sub>2</sub>.  
 captain.NOM.SG him.ACC can.PST.SG.M NEG beat.INF  
 ‘The captain might not beat him.’ (Pol; Witkoś 1998: 193)

(4.166) a. Člověk            **ho**<sub>2</sub>            **ne-může**<sub>1</sub>            **ne-milovat**<sub>2</sub>    [...]  
 human.NOM.SG him.ACC NEG-can.PRS.3SG NEG-love.INF  
 ‘One cannot not like him ...’

b. Případná            liknavost            by<sub>1</sub>            **se**<sub>2</sub>            **mu**<sub>2</sub>            mohla<sub>1</sub>  
 possible.NOM.SG.F tardiness.NOM.SG COND REFL.ACC/GEN him.DAT can.PST.SG.F  
**ne-vyplatit**<sub>2</sub>.  
 NEG-pay\_off.INF  
 ‘The possible tardiness could cost him dear.’ (Cz; ČNK syn\_v8)

(4.167) a. Ako by            **ho**<sub>2</sub>            mohla<sub>1</sub>            **ne-mat’**<sub>2</sub>            rada, (zvlášť    teraz, keď  
 how COND him can.PST.SG.F NEG-have.INF glad.F especially now when  
 ju            tak urputne háji            pred klebetami?)  
 her.ACC so fiercely defend.PRS.3SG from gossip.INS.PL  
 ‘How could she not like him, (especially now, when he is defending her from [all the] gossip?)’

b. (Tréner            mi            dal            dôveru)            a    ja            som<sub>1</sub>            sa<sub>1</sub>  
 coach.NOM.SG me.DAT give.PST.SG.M trust.ACC.SG and I.NOM AUX.1SG REFL  
**ho**<sub>2</sub>            snažil<sub>1</sub>            **ne-sklamat’**<sub>2</sub>            [...]  
 him.ACC try.PST.SG.M NEG-disappoint.INF  
 ‘(The coach put [his] trust in me) and I tried not to disappoint him ...’  
 (Slk; SNK prim-10.0-public-all)

(4.168) Također **ga**<sub>2</sub> pokušajte<sub>1</sub> **ne** upisati<sub>2</sub> u školu prije  
 likewise him.ACC try.IMP.2SG NEG enroll.INF in school.ACC.SG before  
 sedme [godine].  
 seventh.GEN.SG.F year.GEN.SG  
 ‘Try also not to enroll it [= child] in school before it is seven [years old].’<sup>328</sup> (Hrv; hrWaC)

(4.169) V tem primeru pa **ga**<sub>2</sub> skušaj<sub>1</sub> **ne** puščati<sub>2</sub> na  
 in this.LOC.SG.M case.LOC.SG but him.ACC try.PRS.3SG NEG leave.INF on  
 začetku predolgo časa tam.  
 beginning.LOC.SG overlong time there  
 ‘In this case, however, try not to leave it there at the beginning of an overlong time.’  
 (Slv; slWaC)

Besides a syntactic categorisation of complex clause constructions, the involved matrix verbs may be classified semantically. With reference to Romance, Golden (2008: 310–311) lists modals, phase verbs (= aspectual verbs), verbs of perception, and causatives, the latter of which is an umbrella term for various subtypes (cf. Spencer & Luís 2012a: 164). Climbing to (orthotonic) auxiliaries of periphrastic tense formations have been mentioned less frequently (but cf. Hana 2007: 126 for Czech; Kupść 2000b: 73–79 for Polish; I. Kulik 2020: 39 for Czech, Polish, and Slovene). I cannot go into details but simply put on record that we have already encountered many of these semantic categories. The reader should be aware that not all possible semantic classes are represented in the examples and that I cannot strive for completeness here. For instance, verbs of remembering, forgetting, and reminding have not been cited in this section.<sup>329</sup>

Before completing this section, let us briefly return to the matter of clitic clustering in the context of clitic climbing. Ideally, clitic climbing leads to mixed cluster formation, if there is a matrix clitic or matrix clitic cluster. As discussed in Section 4.5, clustering represents a deciding feature of special-clitic-hood and as such, it is expected to occur whenever possible. We have encountered two situations, when clustering with matrix clitics fails to appear. First, an embedded clitic (cluster) remains in situ, which is in accordance with the non-obligatoriness of clitic climbing. Second, a clitic (cluster) does not climb the entire syntactic tree from a deep embedding but targets an intermediate syntactic domain. The syntactic literature takes these structures for granted under the premise that co-dependent clitics behave uniformly. For instance, Rezac (2005: 111) tells that clitic climbing is an

<sup>328</sup> The original source is from: <https://dnevnik.hr/vijesti/zdravlje/rano-citanje-moze-djetetu-ostetiti-vid.html> (accessed 23 Nov 2023).

<sup>329</sup> A corresponding construction is cited in example (4.32b) in Section 4.3.1.

“all-or-nothing phenomenon”, such that either all clitics must climb, or all remain in situ.<sup>330</sup> Likewise, Dotlačil (2017) informs that a clitic cluster must act uniformly in this respect.<sup>331</sup> However, we know already from Section 4.5 that diacclisis may occur within the domain of the simple clause, which gives rise to split co-dependent clitics. This is generally expected for Polish clitics because of their peculiar status. Accordingly, we find diacclisis under clitic climbing in (4.87) from Section 4.5 and (4.127) from the present section repeated in (4.170) below. However, we have also encountered evidence for co-dependent clitics, which do not behave uniformly with respect to clitic climbing, in other languages.<sup>332</sup> I repeat Czech (4.89f–g) and BCMS (4.93)–(4.95) from Section 4.5 in (4.171)–(4.172) below. In sum, clitics subcategorised for by one and the same verbal predicate may split under clitic climbing in principle.

- (4.170) a. Piotr        **mu**<sub>2</sub>        chce<sub>1</sub>        **go**<sub>2</sub>        dać<sub>2</sub>.  
              Peter.NOM him.DAT want.PRS.3SG him.ACC give.INF  
              ‘Peter wants to give it to him.’ (Pol; Witkoś 1998: 158)
- b. Jan        może<sub>1</sub>        **mu**<sub>3</sub>        chceć<sub>2</sub>        **go**<sub>3</sub>        pokazać<sub>3</sub>.  
              Jan.NOM can.PRS.3SG him.DAT want.INF him.ACC show.INF
- c. Jan może<sub>1</sub> **go**<sub>3</sub> chceć<sub>2</sub> **mu**<sub>3</sub> pokazać<sub>3</sub>.  
              Jan może<sub>1</sub> **mu**<sub>3</sub> **go**<sub>3</sub> chceć<sub>2</sub> pokazać<sub>3</sub>.  
              ‘John may want to show it to him.’ (Pol; Kupść 2000b: 61)
- (4.171) a. Lehce **si**<sub>2</sub>        uměla<sub>1</sub>        **ho**<sub>2</sub>        představit<sub>2</sub>, [...]   
              easily REFL.DAT be\_able.PST.SG.F him.ACC imagine.INF  
              ‘She could easily imagine him, ...’
- b. [...] kdekdo        **se**<sub>2</sub>        začal<sub>1</sub>        **mi**<sub>2</sub>        smát<sub>1</sub>.  
              almost\_everybody.NOM REFL.ACC start.PST.SG.M me.DAT laugh.INF  
              ‘... almost everybody started to laugh at me.’ (Cz; I. Kulik 2023b: 224)

<sup>330</sup> A comparable point is made by Aljović (2004: 194) for BCMS, although she does not phrase it explicitly this way. However, her argument follows the same underlying logic as Rezac’s (2005) account, i.e. clitic climbing is not optional at all, but obligatory under so-called restructuring, while it is banned in case of non-restructuring. This implies that the entire embedded clitic cluster necessarily is treated as if it were a single unit.

<sup>331</sup> Original: “Ačkoliv je š[plhání] k[litik] převážně neobligatorní, platí pravidlo, že se klitický trs musí chovat jednotně: buďto všechny klitiky šplhají, n[ebo] žádná nešplhá.” (Dotlačil 2017) ‘Although clitic climbing is largely non-obligatory, the rule is in place that the clitic cluster must behave uniformly: either all clitics climb, or no one climbs.’

<sup>332</sup> As discussed by I. Kulik (2023b), this and other attested constructions challenge the hypothesis that clitic climbing follows from so-called restructuring.

- (4.172) a. [...] kada su<sub>1</sub> **joj**<sub>2</sub> zbog opasne infekcije  
 when AUX.3PL her.DAT because dangerous.GEN.SG.F infection.GEN.SG  
 stafilocokom morali<sub>1</sub> izvaditi<sub>2</sub> **ih**<sub>2</sub> [...].  
 staphylococcus.INS.SG must.PST.PL.M remove.INF them.ACC  
 ‘... when they had to remove them from her because of a dangerous staphylococcus  
 infection ...’ (Bos; Kolaković et al. 2022: 273)
- b. [...] a ja **joj**<sub>2</sub> nisam<sub>1</sub> imao<sub>1</sub> namjeru<sub>1</sub>  
 and I.NOM her.DAT NEG.AUX.1SG have.PST.SG.M intention.ACC.SG  
 mijenjati<sub>2</sub> **ga**<sub>2</sub> [...].  
 change.INF him.ACC  
 ‘... and I had no intention of changing it for her ...’ (Hrv; Kolaković et al. 2022: 273)
- c. [...] i počelo<sub>1</sub> **mi**<sub>2</sub> je<sub>1</sub> da se<sub>2</sub> vrti<sub>2</sub> u  
 and start.PST.SG.N me.DAT AUX.3SG that REFL.ACC spin.PRS.3SG in  
 glavi.  
 head.LOC.SG  
 ‘... and I started to feel dizzy.’ (Srp; Kolaković et al. 2022: 307)
- d. Marija **mu**<sub>2</sub> zeli<sub>1</sub> da **ga**<sub>2</sub> predstavi<sub>2</sub>.  
 Marija.NOM him.DAT want.PRS.3SG that him.ACC introduce.PRS.3SG  
 ‘Marija wants to introduce him to him.’ (BCMS; Stjepanović 2004: 182)

So far, this section has been dealing with the textbook example of clitic climbing from verb to verb, which has been discussed extensively in the literature. In the next section, we will turn to climbing configurations involving superordinate nouns, adjectives, and adverbs. Such structures have been paid less attention to in the study of clitic climbing yet.

## 4.7.2 Climbing across different categories

This section is about the dislocation of clitics across embedded structures of different categories. Two basic patterns can be distinguished as shown in the schemes in (4.173). On the one hand, clitics originating in an infinitival domain cross a non-verbal category when climbing, cf. (4.173a). On the other hand, clitics that are dependents of non-verbal categories climb the cascade of embeddings, cf. (4.173b). The key feature of the structures under examination is the difference between the  $\beta$ -domain and the  $n$ -domain.

- (4.173) a.  $[_\alpha V^0 \text{CL}_j [_n X^0 [_\beta V^0 e_j ]]]$  ;  $X^0 \neq V^0$  ( $X^0 = N^0, \text{Adj}^0, \text{Adv}^0$ )  
 b.  $[_\alpha V^0 \text{CL}_j [_n V^0 [_\beta X^0 e_j ]]]$  ;  $X^0 \neq V^0$  ( $X^0 = N^0, \text{Adj}^0, \text{Adv}^0, \text{Num}^0$ )

Let us consider the type in (4.173a) first. It has been observed that clitics may climb from infinitives embedded under nouns, predicative adjectives, and adverbs. These constructions have been somewhat controversial. For one thing, Junghanns (2002c: 72–78) considers CC to be generally impossible from these structures. However, he admits that counterexamples exist but assigns them a special status as these are light verb constructions (Ger. *Funktionsverbgefüge*), cf. (4.174) with nominal matrixes and (4.175) with an adjectival matrix domain. He proposes a minimalist analysis of incorporation of the nominal and adjectival head in  $V^0$ . Example (4.176) also cited by Junghanns (2002c) poses a problem, because CC is judged ungrammatical here irrespective of the same light verb construction with the adjectival head *schopný* ‘able’ as in (4.175).<sup>333</sup>

- (4.174) a. Já jsem<sub>1</sub> **mu**<sub>2</sub> to<sub>2</sub> ne-měla<sub>1</sub> [<sub>NP</sub> čas  
 I.NOM AUX.1SG him.DAT this.ACC.SG.N NEG-have.PST.SG.F time.ACC.SG  
 [<sub>VP</sub> vysvětlit<sub>2</sub> ]].  
 explain.INF  
 ‘I didn’t have time to explain it to him.’
- b. [...] zda **ho**<sub>2</sub> máte<sub>1</sub> [<sub>NP</sub> možnost [<sub>VP</sub> vidět<sub>2</sub> v  
 whether him.ACC have.PRS.2PL opportunity.ACC.SG see.INF in  
 televizi ]]  
 television.LOC.SG  
 ‘... whether you have the opportunity to watch it [= movie] on TV ...’  
 (Cz; Junghanns 2002c: 73)

<sup>333</sup> Comparable examples showing CC across an ADJ-head are empirically attested, cf. (i)–(ii). In (ii), the reflexive clitic precedes the finite copula verb. CQL query: [lemma!="být"] [word="jsme"] [word="se"] [word="schopni"]

(i) Na jejich hru jsme<sub>1</sub> **se**<sub>2</sub> [<sub>AdjP</sub> schopni [<sub>VP</sub> připravit<sub>2</sub> ]]  
 on their game.ACC.SG be.PRS.1PL REFL.ACC able.PRED.PL.VIR prepare.INF  
 ‘We are capable of getting ready for their game ...’ (Cz; ČNK syn\_v8)

(ii) [...] který dokladuje, že **se**<sub>2</sub> jsme<sub>1</sub> [<sub>AdjP</sub> schopni  
 which.NOM.SG.M document.PRS.3SG that REFL.ACC be.PRS.1PL able.PRED.NOM.PL.VIR  
 [<sub>VP</sub> adaptovat<sub>2</sub> téměř na cokoliv ]].  
 adapt.INF almost on anything.ACC  
 ‘... which documents that we are capable of adapting to almost anything.’ (Cz; S10)

Therefore, there is no obvious contrast between Czech and BCMS with respect to CC from infinitives under predicative adjectives, as supposed by Kolaković et al. (2022: 249).

(4.175) Já jsem<sub>1</sub> **mu**<sub>2</sub> ted' však ne-byla<sub>1</sub> [AdjP schopna  
 I.NOM AUX.1SG him.DAT now however NEG-be.PST.SG.F able.PRED.SG.F  
 [VP nic říct<sub>2</sub> ].  
 nothing.ACC tell.INF  
 'However, now I was't able not to tell him anything.' (Cz; Junghanns 2002c: 76)

(4.176) a. ... jsme [AdjP schopni [VP **se**<sub>2</sub> i k této  
 be.PRS.1PL able.PRED.PL.VIR REFL.ACC and towards this.DAT.SG.F  
 věci společně vyjádřit<sub>2</sub> ]]  
 thing.DAT.SG together express.INF  
 b. \*Jsme **se**<sub>2</sub> [AdjP schopni [VP i k této věci společně vyjádřit<sub>2</sub> ]]  
 'Are we able to speak with one voice on this matter, too?' (Cz; Junghanns 2002c: 75)

In contrast, Golden (2003: 221, 2008: 312–313) considers such structures to be generally eligible for CC in Slovene and Czech (cf. also Hana 2007: 125–126; Rosen 2001: 225–226 for Czech). She provides the data in (4.177)–(4.178), where the embedded clitics escape the infinitival phrase crossing a superordinate nominal head. She further argues that the modification of the matrix noun by attributive adjectives (bold) in (4.177b) and (4.178b) contradicts an incorporation analysis. Examples (4.179) and (4.180) show CC from infinitives under predicative adjectives.

(4.177) a. Ty jsi<sub>1</sub> **mu**<sub>2</sub> **ho**<sub>2</sub> měl<sub>1</sub> [NP příležitost  
 you.NOM AUX.2SG him.DAT him.ACC have.PST.SG.M opportunity.ACC.SG  
 [v doručit<sub>2</sub> osobně ]].  
 deliver.INF personally  
 'You had the opportunity to give it to him in person.' (Cz; Golden 2008: 312)  
 b. Ted' **ho**<sub>2</sub> máš<sub>1</sub> ještě [NP **poslední** možnost  
 now him.ACC have.PRS.2SG still last.ACC.SG opportunity.ACC.SG  
 [VP osobně pozdravit<sub>2</sub> ]] dříve, než odjedeš.  
 personally greet.INF earlier than depart.PRS.2SG  
 'Now you have the last opportunity to greet him personally, before you depart.'  
 (Cz; Golden 2008: 313)

- (4.178) a. Ti            si<sub>1</sub>            **mu**<sub>2</sub>            **ga**<sub>2</sub>            ime<sub>1</sub>            [NP priložnost  
 you.NOM AUX.2SG him.DAT him.ACC have.PST.SG.M            opportunity.ACC.SG  
 [VP izročiti<sub>2</sub>    osebno ]].  
 deliver.INF personally  
 ‘You had the opportunity to give it to him in person.’
- b. Sedaj **ga**<sub>2</sub>            imaš<sub>1</sub>            še    [NP **zadnjo**            priložnost  
 now him.ACC have.PRS.2SG still            last.ACC.SG opportunity.ACC.SG  
 [VP osebno            pozdraviti<sub>2</sub> ]], preden odpotuje.  
 personally greet.INF            before depart.PRS.3SG  
 ‘Now you have the last opportunity to greet him personally, before he departs.’  
 (Slv; Golden 2008: 312)
- (4.179) Nakonec **si**<sub>3</sub>            byl<sub>1</sub>            [AdjP ochoten<sub>2</sub>            [VP to            připustit<sub>3</sub> ]]  
 finally REFL.DAT be.PST.SG.M            willing.ADJ.PRED.M            this.ACC.SG.N admit.INF  
 i Tomáš.  
 and Tom.NOM  
 ‘In the end, even Tom was willing to admit it.’  
 (Cz; Rosen 2001: 226)
- (4.180) Jaz **se**<sub>1,2</sub>            **mu**<sub>2</sub>            ne    čutim<sub>1</sub>            [AdjP dolžnega  
 I.NOM REFL.ACC him.DAT NEG feel.PRS.1SG            obliged.ADJ.GEN.SG.M  
 [VP opravičevati<sub>2</sub> ]].  
 apologise.INF  
 ‘I do not feel obliged to apologise (for myself).’  
 (Slv; Golden 2008: 312)

Another contentious issue is climbing from what is called a right-extraposed subject clause in the generative literature.<sup>334</sup> According to Junghanns (2002c: 77–78), CC is impossible from such infinitival domains. The clitic cluster can only be realised locally as in (4.181a). Clitic climbing in (4.181b) renders the utterances ungrammatical.

- (4.181) a. [...] mám            pořád    pocit,            že    není<sub>1</sub>  
 have.PRS.1SG always feeling.ACC.SG that NEG.be.PRS.3SG  
 [AdjP vhodné            [SBJ ukazovat<sub>2</sub> **se**<sub>2</sub>            **mu**<sub>2</sub>            šťastní ]].  
 appropriate.NOM.SG.N            show.INF REFL.ACC him.DAT happy.NOM

<sup>334</sup> I will not discuss theoretical issues associated with these infinitival constituents. Suffice it to note that they function as syntactic subject clauses, while they simultaneously occur in complement position of an adjective or adverb.



- b. \*že **se**<sub>2</sub> **mu**<sub>2</sub> není<sub>1</sub> [AdjP vhodné [VP ukazovat<sub>2</sub> šťastní ]].

‘I always have the feeling that it is inappropriate to show one’s happiness to him.’

(Cz; Junghanns 2002c: 77)

However, Dotlačil (2004: 82–83) states that clitics may climb out of infinitival subject clauses with the limitation that the reflexive clitic is banned from climbing. Pronominal clitics, in turn, may escape the domain of the subject VP, cf. (4.182). Note that example (4.181) involves a reflexive clitic, such that this would explain the ungrammaticality of (4.181b) (cf. Dotlačil 2004: 83, n. 65).

- (4.182) a. Myslím že (**\*se**<sub>2</sub>) není<sub>1</sub> [AdjP možné [SBJ (**se**<sub>2</sub>)  
 think that REFL.ACC NEG.be.PRS.3SG possible.NOM.SG.N REFL.ACC  
 touhle zbraní zabít<sub>2</sub> ]].  
 this.INS.SG.F weapon.INS.SG kill.INF

‘I think that it is not possible to kill oneself with this weapon.’

- b. Myslím že (**mu**<sub>2</sub>) není<sub>1</sub> [AdjP možné [SBJ (**mu**<sub>2</sub>)  
 think that him.DAT NEG.be.PRS.3SG possible.NOM.SG.N him.DAT  
 pomoci<sub>2</sub> ]].  
 help.INF

‘I think that it is not possible to help him.’

- c. Myslím že (**tě**<sub>2</sub>) není<sub>1</sub> [AdjP možné [SBJ (**tě**<sub>2</sub>)  
 think that you.ACC NEG.be.PRS.3SG possible.NOM.SG.N you.ACC  
 touhle zbraní zabít<sub>2</sub> ]].  
 this.INS.SG.F weapon.INS.SG kill.INF

‘I think that it is not possible to kill you with this weapon.’

- d. Myslím že (**ho**<sub>2</sub>) není<sub>1</sub> [AdjP možné [SBJ (**ho**<sub>2</sub>)  
 think that him.ACC NEG.be.PRS.3SG possible.NOM.SG.N you.ACC  
 touhle zbraní zabít<sub>2</sub> ]].  
 this.INS.SG.F weapon.INS.SG kill.INF

‘I think that it is not possible to kill him with this weapon.’ (Cz; Dotlačil 2004: 82)

Golden (2008: 312) concurs with Dotlačil (2004), cf. the Slovene example (4.183).

(4.183) Takrat **ji**<sub>2</sub>        **ga**<sub>2</sub>        je<sub>1</sub>        bilo<sub>1</sub>        [AdjP mogoče  
then her.DAT him.ACC AUX.3SG be.PST.SG.N        possible.NOM.SG.N  
[<sub>SBJ</sub> poslati<sub>2</sub> po pošti ]].  
send.INF by mail  
‘Then it was possible to send it to her by mail.’ (Slv; Golden 2008: 312)

Based on Junghanns's (2002c) discussion, Kolaković et al. (2022: 244–251) examine the above-mentioned structures for BCMS and find that clitic climbing is possible from infinitives subordinate to nouns (cf. Kolaković et al. 2022: 245–246), from infinitives embedded under predicative adjectives (cf. Kolaković et al. 2022: 248–249), and from right-extraposed subject clauses (cf. Kolaković et al. 2022: 250–251). I cite one example per structure in (4.184)–(4.186) for illustrative purposes and refer the reader to the respective sections of Kolaković et al.'s (2022) monograph for more examples. Climbing crossing a noun is cited in (4.184), a crossed adjective occurs in (4.185), and extraction from a subject clause in (4.186).<sup>335</sup>

(4.184) Neki su<sub>1</sub> **ga**<sub>2</sub> imali<sub>1</sub> [NP potrebu  
some.NOM.PL.M AUX.3PL him.ACC have.PST.PL.M need.ACC.SG  
[VP braniti<sub>2</sub> od [...] ]].  
defend.INF from  
'Some had the need to defend him from ...' (Bos; Kolaković et al. 2022: 245)

(4.185) Jesam<sub>1</sub> li se<sub>2</sub> [AdjP spreman [VP preseliti<sub>2</sub> ]]?  
 be.PRS.1SG Q REFL.ACC ready.NOM.SG.M move.INF  
 ‘Am I ready to move?’ (Srp; Kolaković et al. 2022: 249)

(4.186) [...] [AdjP potrebno ih<sub>3</sub> je<sub>1</sub> [SBJ pokušati<sub>2</sub> osnovati<sub>3</sub> ]]  
 necessary.NOM.SG.N them.ACC be.PRS.3SG.CL try.INF establish.INF  
 osnovati<sub>3</sub> ]] [...].  
 establish.INF  
 ‘... it is necessary to try to establish them ...’ (Hrv; Kolaković et al. 2022: 251)

<sup>335</sup> Note that (4.186) is a safe case of placing the embedded pronominal clitic *ih* in the matrix phrase, since it is surrounded by material of the matrix clause, i.e. in the position between the predicative adjective *potrebno* and the (clitic) copula verb form *je*.

I am not aware of any study examining these patterns for Polish. I provide the data points in (4.187)–(4.189). Example (4.187) provides extraction of a clitic across a noun and (4.188) involves climbing across a predicative (short) adjective. In examples (4.189a–b), we encounter dislocation of a clitic from a right-extraposed subject clause. There is however a significant difference between (4.189a–b) and the previously mentioned Czech, Slovene, and BCMS examples. The non-verbal category crossed by the clitic is not an adjective but an adverb – at least from a synchronic point of view. Diachronically these forms are short, indefinite neuter adjectives. As the short adjective paradigms have gone completely lost in West Slavic except a small set of language-specific nominative forms used predicatively, such occurrences are not transparent anymore for native speakers, who perceive them as adverbs due to formal resemblance. This seems to be corroborated by example (4.189c), where the superordinate category is a genuine adverb (in the comparative). Note that this structure is ambiguous with the infinitive being analysable as complement proper and as a subject clause. The lack of an overt copula verb fosters a complement analysis.

- (4.187) nikt                nie   był<sub>1</sub>        **mi**<sub>2</sub>        w [NP stanie                [VP odpowiedzieć<sub>2</sub>  
nobody.NOM NEG be.PST.SG me.DAT in state.LOC.SG answer.INF  
na takie                podchwytliwe pytanie ]].  
on such.ACC.SG.N tricky.ACC.SG.N question.ACC.SG  
‘No one was able to provide me with an answer to such a trick question.’  
(Pol; NKJP poliqarp balanced)

- (4.188) (Kandydat                powiedział,    a-by-m                na-pisał  
candidate.NOM.SG tell.PST.SG.M SUB-COND-AUX.1SG PFV-write.PST.SG.M  
odpowiedni                tekst,)  
corresponding.ACC.SG.M text.ACC.SG  
a    on                **go**<sub>2</sub>                [Adj gotów                jest<sub>1</sub>                [VP podpisać<sub>2</sub> ]].  
and he.NOM.SG him.ACC ready.PRED.NOM.SG.M be.PRS.3SG sign.INF  
‘(The candidate told me to write a suitable text) and he will be willing to sign it.’  
(Pol; NKJP poliqarp balanced)

- (4.189) a. (Od razu przypomniał mi się ten post,)  
 from one.GEN remind.PST.SG.M me.DAT REFL.ACC this.ACC.SG.M posting.ACC.SG  
 choć nie było<sub>1</sub> go<sub>3</sub> [AdvP łatwo<sub>2</sub> [SBJ znaleźć<sub>3</sub> ]]  
 though NEG be.PST.SG.N him.ACC easily find.INF  
 (i z mrówkami nic wspólnego nie ma.)  
 and with ant.INS.PL nothing.ACC common.GEN.SG.M NEG have.PRS.3SG  
 ‘(I remembered this post immediately,) although it wasn’t easy to find it (and it doesn’t  
 have anything in common with ants.)’ (Pol; S11)
- b. (W zamian za to my pokazaliśmy Mu drogę  
 in exchange for this.ACC.SG.N we.NOM show.PST.PL.VIR.1PL him.DAT way.ACC.SG  
 do Bolesława,) gdyż w samym lesie wcale go<sub>2</sub>  
 to Bolesław.GEN because in lonely.LOC.SG forest.LOC.SG at\_all him.ACC  
 [AdvP niełatwo<sub>1</sub> [SBJ znaleźć<sub>2</sub> ]].  
 not\_easy find.INF  
 ‘(In exchange for that, we showed him the road to [the] Bolesław [oak],) as it isn’t easy,  
 at all, to find it in the deep woods.’ (Pol; S12)
- c. (Wyobraź sobie, ale... — przerwała i, spoglądając na  
 imagine.IMP.2SG REFL.DAT but interrupt.PST.SG.F and look.PTCP.ACT on  
 Franię, dodała: —) może ci<sub>2</sub> [AdvP lepiej<sub>1</sub> [VP nie mówić<sub>2</sub> ]],  
 Frania.ACC.SG add.PST.SG.F perhaps you.DAT better.ADV NEG say.INF  
 (ty zaraz przed ojcem wypaplasz.)  
 you.NOM at\_once before father.INS.SG blab\_out.PRS.2SG  
 ‘(“Imagine, but” ... she stopped and, looking at Frania, added:) “[it’s] perhaps better, not  
 to tell you, (you blab [it] out towards father immediately.)”’ (Pol; S13, p. 195)

Turning to the second type (4.173b) now, two things are important to note. First, pronominal clitics can be licensed by non-verbal predicates in Slavic. Second, these clitics can be extracted from their licensing domain. Rosen (2001: 225–227, 234) informs that clitics may climb from (quantified) nouns (4.190), (predicative) adjectives (4.191), adverbs (4.192), and quantificational expressions (4.193), i.e. numerals and quantifiers proper (cf. also Junghanns 2002c: 72; Hana 2007: 122, 125; cf. Toman 1986 for cliticisation from quantified NPs). Note that Czech *jich* ‘them.GEN’ in (4.190) and (4.193) is a partitive genitive and that the pronominal form is an inconstant clitic.

(4.190) Král<sub>mu1</sub> **jich**<sub>3</sub> přikázal<sub>1</sub> sníst<sub>2</sub> [<sub>NP</sub> dvě  
king.NOM.SG him.DAT them.GEN order.PST.SG.M PFV.eat.INF two.ACC.F  
plné mísy<sub>3</sub> ].  
full.ACC.PL.F bowl.ACC.PL  
'The king ordered him to eat up two bowls full of them.' (Cz; Rosen 2001: 234)

(4.191) a. Marie **si**<sub>4</sub> musí<sub>1</sub> začít<sub>2</sub> být<sub>3</sub> [<sub>AdjP</sub> vědoma<sub>4</sub>  
Marie REFL.DAT must.PRS.3SG start.INF be.INF aware.PRED.SG.F  
svých přednosti ].  
one's.POSS.GEN.PL advantage.GEN.PL  
'Marie must begin being aware of her qualities.' (Cz; Rosen 2001: 226)  
b. Už je<sub>1</sub> **vás**<sub>2</sub> ale [<sub>AdjP</sub> skoro plný<sub>2</sub> ] dům.  
already be.PRS.3SG you.GEN but almost full.NOM.SG.M house.NOM.SG  
'The house it almost [completely] filled with you [guys].' (Cz; Rosen 2001: 234)

(4.192) Je<sub>1</sub> **mi**<sub>2</sub> **tě**<sub>2</sub> [<sub>AdvP</sub> líto<sub>2</sub> ].  
be.PRS.3SG me.DAT you.GEN sorry  
'I feel sorry for you.' (Cz; Rosen 2001: 234)

(4.193) a. Petr **jich**<sub>2</sub> našel<sub>1</sub> tehdy [<sub>Num</sub> pět<sub>2</sub> ].  
Peter.NOM them.GEN find.PST.SG.M that\_time five.ACC  
'Peter found five of them that time.' (Cz; Toman 1986: 131)  
b. Ne-toulá<sub>1</sub> se<sub>1</sub> **jich**<sub>2</sub> tu zrovna [<sub>Num</sub> moc<sub>2</sub> ].  
NEG-wander.PRS.3SG REFL.ACC them.GEN here exactly much  
'Not many of them are walking around here.' (Cz; Rosen 2001: 234)

Golden (2008: 313) makes a similar observation for Slovene, cf. (4.194) with a pronominal clitic dislocated from its adjectival head.

(4.194) Jaz sem<sub>1</sub> **ji**<sub>2</sub> bil<sub>1</sub> vedno (neskončno) [<sub>AdjP</sub> hvaležen<sub>2</sub> ]  
I.NOM AUX.1SG her.DAT be.PST.SG.M always infinitely grateful.NOM.SG.M  
za vsak nasvet.  
for every advice.ACC.SG  
'I have always been (infinitely) grateful to her for every piece of advice.'  
(Slv; Golden 2008: 313)

Kolaković et al. (2022: 243–244) find similar structures in BCMS, cf. (4.195)–(4.197). They argue that such examples do not represent “CC *sensu stricto* because we are dealing with a clearly mono-clausal structure with a single predicative element.” (Kolaković et al. 2022: 243) This is certainly correct from the perspective of predication. There is however a difference between a mono-clausal verbal predication and structures like (4.195)–(4.197). In simple clauses with a single verbal predicate, the auxiliary, pronominal, and reflexive clitics are in a dependency relation with the very same verb. In the extractions discussed here, clitics are realised by different predicate heads, which gives rise to mixed clusters (cf. the indexes in the examples). However, I agree that most of the data points cited here do not show clitic climbing proper. So far, it is only the Czech examples (4.190) and (4.191a), which comply with the structure in scheme (4.173b).

- (4.195) I još ću<sub>1</sub> **mu**<sub>2</sub> biti<sub>1</sub> [AdjP zahvaln<sub>2</sub> ].  
and also AUX.FUT.1SG him.DAT be.INF grateful.NOM.SG.M  
'And I will also be grateful to him. (Bos; Kolaković et al. 2022: 243)
- (4.196) Sutra ćeš<sub>1</sub> **mi**<sub>2</sub> biti<sub>1</sub> [AdjP zahvaln<sub>2</sub> ].  
tomorrow AUX.FUT.2SG me.DAT be.INF grateful.NOM.SG.F  
'Tomorrow you will be grateful to me' (Hrv; Kolaković et al. 2022: 244)
- (4.197) Toliko sam<sub>1</sub> **im**<sub>2</sub> bila<sub>1</sub> [AdjP dužn<sub>2</sub> ].  
that\_much be.PRS.1SG.CL them.DAT be.PST.SG.F in\_debt.NOM.SG.F  
'I was so much in their debt.' (Srp; Kolaković et al. 2022: 244)

Finally, Polish (4.198) provides another instance of genuine clitic climbing with the pronominal clitic being dislocated from an adjective but realised in the finite matrix clause. Example (4.199) provides extraction of a pronominal clitic from an adverb, which yields a mixed cluster in second position.

- (4.198) [...] to ja mam<sub>1</sub> **mu**<sub>2</sub> za co być<sub>2</sub>  
 TO I.NOM have.PRS.1SG him.DAT for what.ACC.SG be.INF  
 [AdjP wdzięczny<sub>3</sub>].  
 thankful.NOM.SG.M  
 ‘... so I must be thankful to him for [all of this].’ (Pol; NKJP poliqarp full)

(4.199) Trochę **mu**<sub>2</sub> się<sub>1</sub> zrobiło<sub>1</sub> [AdvP przykro<sub>2</sub>] [...]   
a\_bit.ACC him.DAT REFL.ACC PFV.make.PST.SG.N upset.ADV   
‘He got a little upset ...’ (Pol; NKJP poliqarp)

After this overview, which is far from exhaustive, we may conclude that the clitic climbing works essentially similar in the Slavic 2P languages examined here – including Polish. Therefore, Dziwirek’s (1998: 89) assessment that “[c]litic climbing’, as a special phenomenon, does not exist in Polish” is unwarranted. It seems fair to say that clitic climbing in Slavic is still not entirely understood and requires further examination. This may be the reason for there being no consensus about an adequate analysis of this construction (cf. Aljović 2004; Bošković 2001; Dotlačil 2004; Golden 2008; Junghanns 2002c; Kolaković et al. 2022; I. Kulik 2023b; Lenertová 2004; Marušič 2008; Milićević 2023; Rezac 2005; Stjepanović 2004). For one thing, it has been proposed to analyse clitic climbing in terms of restructuring on the model of Romance (cf. Aljović 2004 for BCMS; Rezac 2005 for Czech). In contrast, Dotlačil (2004) and I. Kulik (2023b) show that clitic climbing is not contingent on restructuring in Czech and Polish.<sup>336</sup>

## 4.8 Rightwards positional restrictions

The discussion in Sections 4.2–4.4 showed that so-called Wackernagel clitics preferably occupy the second position of the clause. Yet, there is considerable variation with respect to how this second position could be specified on the one hand, and placement deviations showing a range of different positions from Clitic First to Clitic Fifth position placement on the other hand. This is not to suggest that everything is possible in clitic placement. So far, the discussion has been focused on the range of variation, i.e. possibilities. There are, however, some relevant restrictions for cliticisation in Slavic. As mentioned in Section 3.11, it has been noted that clitics do not enjoy the same distributional freedom as corresponding full expressions, be it orthotonic pronouns or (common/proper) noun phrases (cf. Dotlačil 2007; Junghanns 2002c: 83; Witkoś 1998: 159 *inter alia*). Although clitics can appear in Delayed Placement, i.e. shifted from 2P further to the right, there is a consistent syntactic restriction on the rightward positioning across languages: a clitic cannot be positioned further to the right than directly right-adjacent to its governor (cf. Eichler 1994: 78, n. 3; Popović 1979: 679;

<sup>336</sup> Note that both studies arrive at the same conclusion using disparate methodologies. As a native speaker of Czech, Dotlačil (2004) bases his analysis on introspective data using his linguistic intuition. On the other hand, I. Kulik (2023b) solely relies on corpus data. However, Lenertová (2004) comes to the contrary result (being a native speaker of Czech and simultaneously using corpus data). She concludes that restructuring environments are any multi-verbal constructions that do not involve a CP-layer.

Radanović-Kocić 1988: 102; Reinkowski 2001: 174 for BCMS; Franks & King 2000: 142, 158–159, 162, 269, 338; Kupść 2000b: 40; Witkoś 1998: 168–169 for Polish; Hana 2007: 122, less explicitly, for Czech).<sup>337</sup> The utterances in (4.200), (4.201), and (4.202) exemplify this structural restriction for Czech, Polish, and BCMS respectively.<sup>338</sup> The grammatical a-examples show what I will call *POSTPOSITION* following Kosek (2009, 2015), whereas the b-examples are ungrammatical, because the clitics are right-dislocated from the verbal head. Postposition overlaps with Kosek’s (2017) category of “final position”, but it is not identical with it, for Kosek refers to clitic placement in clause-final position. This is, however, only possible under references to the clitic’s verbal predicate. It needs to be emphasised that *postposition* as understood in the present thesis is not limited to clause-final position. The verb may occupy any position of a clause irrespective of the clause’s complexity. Thus, the possible right-most sites a clitic can take differ significantly from the right clause margin. Put differently, the set of permutations a clitic may be an element of is potentially smaller than the set of permutations for an entire clause.<sup>339</sup>

- (4.200) a. Udělíme **mu** cenu tajně.  
award.PRS.1PL him.DAT price.ACC.SG secretly  
'We are awarding him the price secretly.' (Cz; constructed from Adam 2019: 19)
- b. \*Udělíme [<sub>NP</sub> cenu ] **mu** tajně. (Cz; Adam 2019: 19)

<sup>337</sup> Franks & King (2000: 201–202) cite the same restriction for Hutsul.

<sup>338</sup> Note that this distributional restriction is easily overridden by extra-linguistic factors. For instance, in verse, clitic pronouns may appear right-dislocated from their governing head to fit the metre and/or rhyme scheme, cf. (i)–(iii) from different pieces of Polish music. Rhyming or assonant words are bold, the double slash ( // ) marks a line break.

(i) Nie znoszę, kiedy ktoś [VP dokucza [AdvP mocno] **mi**] // A kiedy pada  
 NEG bear.PRS.2 when someone.NOM tease.PRS.3SG strongly me.DAT and when fall.PRS.3SG  
 deszcz, zamykam w domu **drzwi**.  
 rain.NOM close.PRS.1SG in house.LOC.SG door.ACC

‘I cannot stand, when someone teases me a lot. // And I shut the door, when the rain is falling.’

(Pol; Zuziki: Miś, Album: *Wesołe historie*, 2018)

(ii) Więc uśmiechnij się do mamy // [VP przytul [AdvP mocno] [PP do niej]  
therefore smile.IMP.2SG REFL.ACC to mom.GEN.SG embrace.IMP.2SG strongly to her.GEN  
**się**] // I zaśpiewaj jej piosenkę // o tym jak kochana **jest**  
REFL.ACC and sing.IMP.2SG her.DAT song.ACC.SG about this.LOC.SG how beloved.F.SG be.PRS.3SG

'Therefore, smile at [your] mom // give her a heartfelt hug // and sing her a song // about how lovely she is.'

(Pol; Rodzina Treflików: Kochane Słońce, Album: *ABC dobrego wychowania w piosence*, 2018)

(iii) Nasza                      godka                      twardo                      **jest** // I    [VP niy    wstydza  
our.NOM.SG.F language.NOM.SG hard.NOM.SG.F be.PRS.3SG and    NEG feel\_ashamed.PRS.1SG  
[NP tego ]                      **sie** ]  
                                         this.GEN.SG REFL.ACC

'Our language sounds hard // and I do not feel ashamed because of this.'

(Sil; Oberschlesien: Szola, Album: *I*, 2013)

Note that examples (i)–(ii) are taken from children’s songs. It is an intriguing question whether exposition to such structures has an impact on a child’s emerging grammar, as it usually comes into touch with music already during first language acquisition. The issue is far beyond the present study and cannot be discussed here.

<sup>339</sup> See Section 10.9 for an analysis of what I will call the permutation space (or  $\pi$ -space), which accounts for the tension between a clitic's actual position and the sum of all potentially available placement sites.



- (4.201) a. Jan        oddał        **mu**        **go**        wczoraj    w parku.  
                  Jan.NOM   return.PST.SG.M   him.DAT   him.ACC   yesterday   in park.LOC.SG  
 b. \*Jan oddał [<sub>AdvP</sub> wczoraj ] **mu go** w parku.  
                  ‘Jan returned it to him yesterday in the park.’ (Pol; Witkoś 1998: 178)

- (4.202) a. Veruje        **se**        samo idiotskim    objašnjenjima.  
                  believe.PRS.1SG   REFL.ACC   only   idiotic.DAT.PL   explanation.DAT.PL  
 b. \*Veruje [<sub>AdvP</sub> samo ] **se** idiotskim objašnjenjima.  
                  ‘People believe only the idiotic explanations.’ (BCMS; Radanović-Kocić 1988: 103)

Despite the lack of explicit statements and minimal pair examples for other languages, it is plausible to assume that the same restrictions hold in other Slavic 2P clitic languages as well.<sup>340</sup> Kosta & Zimmerling (2014) are the only scholars I am aware of, who provide relevant generalisations for clitic systems in Slavic as a whole, cf. (4.64a) and (4.64c) from Section 4.4 repeated in (4.203) below. The ungrammatical b-examples in (4.200)–(4.202) match the structural descriptions in (4.203), which, however, are incomplete, accounting only for a subset of all prohibited constructions. For instance, they do not cover sequences of finite verb and subject or finite verb and a PP, the latter of which is exemplified in (4.204). On the other hand, they are meant to apply to fronted constituents only.

- (4.203) a. \*[V<sub>fin</sub> + O]=CL        ban of complex finite VPs (incl. *l*-participle) (V<sup>0</sup> + XP<sub>OBJ</sub>) before clitics  
 b. \*[V<sub>fin</sub> + Adv]=CL        ban of finite verb and an adverb before clitics
- (4.204) a. \*Maria przeczytała    [<sub>PP</sub> z        łatwością ]    **go**        wczoraj.  
                  Mary   read.PST.SG.F        with   ease.INS.SG   him.ACC   yesterday  
                  ‘Mary read it with ease yesterday.’ (Pol; Witkoś 1998: 192)  
 b. Jan        zobaczył        [<sub>PP</sub> w lustrze ]        siebie        / \***się**.  
                  John.NOM   see.PST.SG.M        in mirror.LOC.SG REFL.ACC   REFL.ACC.CL  
                  ‘John saw himself in the mirror.’ (Pol; Kupść 1999: 96)

The crucial generalisation mentioned above is captured in (4.205), which has two readings. First, non-clitic material intervening between a verb and a clitic in a simple clause is banned. Second, non-clitic

<sup>340</sup> The restriction trivially holds in VA languages.

elements must not intervene between a clitic and its governing verbal head (as indicated by indexing). The latter reading is underspecified for the domain it applies to, thus, potentially covering more data.

(4.205) \*[<sub>S</sub> V<sub>fin/j</sub> XP CL<sub>j</sub> ]

Note that both readings of (4.205) do not have a problem in accounting for the data in (4.206), which are discussed by Rosen (2001: 215). The reflexive clitics in (4.206a–b) are preceded by V+XP(+XP) complexes, but the utterances are grammatical. Both examples provide complex clauses, where the reflexive clitics are not dependents of the preceding verb forms. The situation is different in (4.206c), where we encounter a simple clause and the auxiliary clitic is in a dependency relation with the *l*-participle.

- (4.206) a. [<sub>VP</sub> Rozdávát<sub>2</sub> [<sub>NP</sub> ženám] [<sub>NP</sub> květy ]] **se**<sub>1</sub> před  
hand\_out.INF woman.DAT.PL flower.ACC.PL REFL before  
volbami musí<sub>1</sub>.  
election.INS.PL must.PRS.3SG  
‘One has to hand out flowers to women before the elections.’
- b. [<sub>VP</sub> Obdarovány<sub>3</sub> [<sub>NP</sub> květy ]] **si**<sub>1</sub>  
give\_present.PASS.PTCP.NOM.PL.NVIR flower.INS.PL REFL.DAT  
některé dámy být<sub>2</sub> ne-přály<sub>1</sub>.  
some.NOM.PL.F lady.NOM.PL be.INF NEG-wish.PST.PL.NVIR  
‘Some ladies did not wish to be presented with flowers.’
- c. \*[<sub>VP</sub> Rozdával<sub>1</sub> [<sub>NP</sub> ženám ] [<sub>NP</sub> květy ]] **jsem**<sub>1</sub> skoro  
hand\_out.PST.SG.M woman.DAT.PL flower.ACC.PL AUX.1SG almost  
denně.  
daily  
‘I handed out flowers to women almost daily.’ (Cz; Rosen 2001: 215)

The same applies to Adam’s (2019) examples (4.70)–(4.71) in Section 4.4 repeated in (4.207)–(4.208) below. The structures in (4.208) present subordinate complex clauses and the indexes reveal the corresponding dependency relations between the pronominal clitic and the verbal heads. In (4.207), the pronominal clitic is a dependent of the preceding verb again, but it is a complex clause. As we know from Section 4.7, this reasoning is based on the empirical observation that topicalisation of an embedded infinitival VP is possible independently of clitic climbing.

The plausibility of this reasoning is confirmed by the data in (4.209) involving the inherently reflexive verb *bać się* ‘(to) fear’ from Kupść (1999). Example (4.209a) is grammatical despite the inchoative verb *zaczniemy* intervening between the reflexive clitic and its verbal head. In contrast, (4.209b) is ungrammatical, although it exhibits essentially the same linearisation pattern. Kupść (1999: 97) concludes that this implies the relevance of syntactic structure instead of syntagmatic order, a view I adopt. In (4.209a), the inherently reflexive verb is fronted but subordinate to the inchoative verb. The reflexive cliticises on the matrix verb. This structure can be perfectly accounted for in terms of clitic climbing. Example (4.209b), in turn, shows the reverse syntactic hierarchy between the two verbs, such that the inchoative verb is now the embedded infinitive. Consequently, the reflexive clitic is realised in the embedded domain. The ungrammaticality of (4.209b) can be attributed to the “lowering” of the reflexive clitic, which represents a critical instance of non-contiguous postposition.

- (4.207) Říkají, že<sup>1</sup> [VP udělit<sub>2</sub> cenu ]<sup>2</sup> **mu**<sub>2</sub> chceme<sub>1</sub> tajně.  
 say.PRS.3PL that award.INF price.ACC.SG him.DAT want.PRS.1PL secretly  
 ‘They say that we want to award him the price secretly.’ (Cz; Adam 2019: 20)

- (4.208) a. \*Říkají, že<sup>1</sup> [VP udělíme<sub>1</sub> cenu ]<sup>2</sup> **mu**<sub>1</sub> tajně.  
 say.PRS.3PL that award.PRS.1PL price.ACC.SG him.DAT secretly  
 Intended: ‘They say that we are awarding him the price secretly.’  
 b. \*Říkají, že<sup>1</sup> [VP udělili<sub>1</sub> cenu ]<sup>2</sup> **jsme**<sub>1</sub> **mu**<sub>1</sub> tajně.  
 say.PRS.3PL that award.PST.PL.VIR price.ACC.SG AUX.1PL him.DAT secretly  
 Intended: ‘They say that we awarded him the price secretly.’ (Cz; Adam 2019: 19)

- (4.209) a. Bać<sub>2</sub> zaczniemy<sub>1</sub> się<sub>2</sub> po wyborach.  
 fear.INF begin.PRS.1PL REFL.ACC after election.LOC.SG  
 ‘We’ll begin to fear after the elections.’  
 b. \*Boimy<sub>1</sub> zacząć<sub>2</sub> się<sub>1</sub> wybory.  
 fear.PRS.1PL begin.INF REFL.ACC election.ACC.PL  
 ‘We fear to start the elections.’ (Pol; Kupść 1999: 97)

Finally, note that (4.205) is trivially fulfilled by all post-verbal examples in Section 4.4.2 on Delayed Placement. Examples matching postposition in Delayed Placement are attested for all languages mentioned there. Suffice it to repeat the Slovak Clitic Third example (4.85a) in (4.210).

- (4.210) [ppZ iniciativy Jozefa Kohúta]<sup>1</sup> [v založil]<sup>2</sup> sa<sup>3</sup> v Martine  
 of initiative.GEN.SG Jozef.GEN Kohút.GEN found.PST.SG.M REFL.ACC in Martin.LOC  
 ... hasičský zbor.  
 ... firefighter.ADJ.NOM.SG brigade.NOM.SG  
 ‘A fire brigade was founded in Martin ... on the initiative of Jozef Kohút.’  
 (Slk; Short 1993b: 567)

It is interesting to note that the rightwards positional restriction is the only restriction on clitic positioning in a strict sense. Recall that the previous sections provided that, in principle, clitics may occur in clause-initial position and that multiple clitics allow the realisation of diacsis even when they are co-dependent upon the same verbal head. The next section deals with restrictions of a different kind, namely referring to co-occurrence of clitics.

## 4.9 Co-occurrence phenomena

### 4.9.1 Haplology of the reflexive

In this and the following two sections, we shall look at restrictions concerning the co-occurrence of clitics. We will begin with the so-called HAPLOLOGY of the reflexive clitic. Haplology is an originally morphological notion denoting the omission of identical syllables or segmental sequences, which has been extended to the ban of multiple reflexive (and other) clitics within the same clausal domain.<sup>341</sup> Junghanns (2002c: 79) refers to this ban as *Beschränkung für Pseudo-Zwillinge* ‘restriction on pseudo-twins’. I will continue to use the term *haplology*, as it is already established in Slavic clitic research (cf. Franks & King 2000; Kolaković et al. 2022; Kupść 1999, 2000b; Rosen 2001, 2014). Some attention has been paid to haplology in complex clauses, while less so in simple clauses (cf. Avgustinova & Oliva 1997: 30–32; Rosen 2014 for Czech; Kiklewicz 2012; Kupść 1999, 2000: 133–155 for Polish; Kolaković et al. 2022 for BCMS; Franks & King 2000: 119–120, 248 more generally).<sup>342</sup> It has been observed that there can be only one reflexive clitic per clitic cluster or simple clause. This observation holds irrespective of function and case of the clitic. In the grammatical Czech example (4.211a) from Rosen (2014), there are two reflexives in a single clause, one of which is the

<sup>341</sup> Cf. Mayerthaler (1977: 53–78) on haplology and esp. Mayerthaler (1977: 73–78) on Spanish for an early analysis of haplology in syntax.

<sup>342</sup> Franks & King (2000: 120, 248) state that haplology is common throughout Slavic but cite examples only from Czech and Polish (cf. Franks & King 2000: 119–120, 248).

dative reflexive clitic (*si*), while the other is the orthotonic accusative reflexive. Grammaticality changes in (4.211b), however, if the full accusative reflexive is substituted by its clitic counterpart. Kupść's (1999) Polish example (4.212a) provides an impersonal construction with two reflexives, one of which is orthotonic again, while the clitic marks the impersonal construction. Substitution of the orthotonic reflexive by its clitic counterpart renders the utterance ungrammatical, cf. (4.212b). The utterance becomes grammatical under haplology, in turn, i.e. the elision of one clitic reflexive, cf. (4.212c). Kupść (1999: 103–104) notes that example (4.212c) is ambiguous. It has either what she calls an “object-drop” reading, whereby the construction is transitive with an unspecified object, or a reflexive reading, whereby the structure is analysed as truly haplogologised.

- (4.211) a. Představil jsem **si** **sebe**, (jak stojím na jeho  
 see.PST.SG.M AUX.1SG REFL.DAT.CL REFL.ACC how stand.PRS.1SG on his  
 místě).  
 place.LOC.SG  
 b. \*Představil jsem **si se**, (jak stojím na jeho místě).  
 ‘I imagined myself (in his position).’ (Cz; Rosen 2014: 101)

- (4.212) a. Janowi łatwo **się** myje **siebie** w ciepłej wodzie.  
 John.DAT easy.ADV REFL.CL wash.PRS.3SG REFL.ACC in warm.LOC.SG.F water.LOC.SG  
 b. \*Janowi łatwo **się** myje **się** w ciepłej wodzie.  
 ‘John finds it easy to wash himself in warm water.’  
 c. Janowi łatwo **się** myje w ciepłej wodzie.  
 John.DAT easy.ADV REFL.(ACC.)CL wash.PRS.3SG in warm.LOC.SG.F water.LOC.SG  
 i. ‘John finds it easy to wash in warm water.’  
 ii. ‘John finds it easy to wash himself in warm water.’ (Pol; Kupść 1999: 103)

Kiklewicz's (2012) examples in (4.213) show particularly well that haplology in simple clauses is motivated by syntactic operations applying to argument structure like impersonalisation. The verbal predicate *milczyć* ‘(to) be silent’ in (4.213a) is intransitive. Under impersonalisation, the subject becomes demoted and the reflexive is overtly realised instead, and the verb realises impersonal inflection morphology.<sup>343</sup> In (4.213b), the inherently reflexive verb *modlić się* ‘(to) pray’ occurs in an impersonalised construction but realises only one reflexive not two, compare (4.212b) with (4.213c). Rosen (2014: 110–111) furthermore states for Czech that the impersonal marker always

<sup>343</sup> Impersonal agreement realises the 3rd person singular neuter. As gender is marked only on the *l*-participle, other tenses than past show merely the 3rd singular.

supersedes the underlying reflexive. This is nicely shown in (4.214), where the verbal predicate is the inherently reflexive verb *stěžovat si* ‘(to) complain’ specified for the dative reflexive.

(4.213) a. Tu **się** milczy.

here REFL be\_silent.PRS.3SG

‘One is silent here.’

b. Tu **się** modli.

here REFL pray.PRS.3SG

‘One prays here.’

(Pol; Kiklewicz 2012: 145)

c. \*Tu **się** modli **się**.

(Pol; constructed from Kiklewicz 2012: 145)

(4.214) a. Vy **si** stále stěžujete.

you.NOM.PL REFL.DAT always complain.PRS.2PL

‘You keep complaining.’

b. Vám **se** to stěžuje!

you.DAT.PL REFL.ACC this.ACC.SG.N complain.PRS.3SG

c. \*Vám **si** to stěžuje!

you.DAT.PL REFL.DAT this.ACC.SG.N complain.PRS.3SG

‘For you, it’s easy to complain!’

(Cz; Rosen 2014: 110)

The main research focus has been put on haplology within complex environments that license clitic climbing. In fact, it has been suggested that haplology is the result of climbing of an embedded reflexive into the matrix clause (cf. Avgustinova & Oliva 1997: 30–31; Franks & King 2000: 119–120, 248; Kolaković et al. 2022: 267; Kupść 1999: 105; Rosen 2014: 104). Again, haplology is indifferent with respect to the morphological and syntactic properties, and function of the reflexive clitics involved, as Avgustinova & Oliva (1997: 31) emphasise. Both verbs realise a formally accusative reflexive clitic *se* in (4.215), whereby *snažit se* is inherently reflexive, such that *se* is a pleonasm, while *obléci se* has a true argument reflexive which is coreferential with the sentential subject. Each verb can realise its reflexive clitic individually as long as the clitics are realised in different domains or clusters, compare (4.215a) with (4.215c) (cf. also Junghanns 2002c: 79). The utterance becomes grammatical under haplology in (4.215b). The same pattern can be observed with the dative reflexive clitic in (4.216).

- (4.215) a. Jan **se**<sub>1</sub> snažil<sub>1</sub> **se**<sub>2</sub> elegantně obléci<sub>2</sub>.  
 John REFL.ACC try.PST.SG.M REFL.ACC elegantly dress.INF  
 b. Jan **se**<sub>1,2</sub> snažil<sub>1</sub> elegantně obléci<sub>2</sub>.  
 ‘John tried to dress up elegantly.’ (Cz; Avgustinova & Oliva 1997: 31)  
 c. \*Jan **se**<sub>1</sub> **se**<sub>2</sub> snažil<sub>1</sub> elegantně obléci<sub>2</sub>.  
 (Cz; constructed from Avgustinova & Oliva 1997: 31)
- (4.216) a. Jan **si**<sub>1</sub> ne-troufá<sub>1</sub> **si**<sub>2</sub> koupit<sub>2</sub> nové auto.  
 John REFL.DAT NEG-dare.PRS.3SG REFL.DAT buy.INF new.ACC.N car.ACC  
 b. Jan **si**<sub>1,2</sub> netroufá<sub>1</sub> koupit<sub>2</sub> nové auto.  
 ‘John does not dare to buy a new car.’ (Cz; Avgustinova & Oliva 1997: 31)  
 c. \*Jan **si**<sub>1</sub> **si**<sub>2</sub> netroufá<sub>1</sub> koupit<sub>2</sub> nové auto.  
 (Cz; constructed from Avgustinova & Oliva 1997: 31)

Example (4.217) shows that the relevant property is the status of being a reflexive clitic, not morphological form. The reflexive clitics *se* and *si* cannot occur in a cluster, irrespective of their morphological form and linear order (cf. also Junghanns 2002c: 80). Furthermore, Avgustinova & Oliva (1997: 32) point out that clustering is the crucial factor of the co-occurrence restriction, not syntagmatic adjacency. Example (4.218) is grammatical, because the respective clitics are not placed in the same clitic cluster. According to Rosen (2014: 105), clitics are separated prosodically in such cases, as indicated by the hash mark.

- (4.217) a. Stále **se**<sub>1</sub> snažím<sub>1</sub> získat<sub>2</sub> **si**<sub>2</sub> její přízeň.  
 constantly REFL.ACC try.PRS.1SG win.INF REFL.DAT her fondness.ACC.SG  
 b. Stále **se**<sub>1</sub> snažím<sub>1</sub> **si**<sub>2</sub> získat<sub>2</sub> její přízeň.  
 c. \*Stále **si**<sub>2</sub> **se**<sub>1</sub> snažím<sub>1</sub> získat<sub>2</sub> její přízeň.  
 ‘I am constantly trying to win her fondness.’ (Cz; Avgustinova & Oliva 1997: 31)  
 d. \*Stále **se**<sub>1</sub> **si**<sub>2</sub> snažím<sub>1</sub> získat<sub>2</sub> její přízeň. (Cz; Avgustinova & Oliva 1997: 32)
- (4.218) a. Snažil<sub>1</sub> =**se**<sub>1</sub> # **se**<sub>2</sub>= tam dostat<sub>2</sub>.  
 try.PST.SG.M REFL.ACC REFL.ACC there get.INF  
 ‘He tried to get there.’  
 b. Naučil<sub>1</sub> =**se**<sub>1</sub> # **si**<sub>2</sub>= vždy znovu získat<sub>2</sub> její přízeň.  
 teach.PST.SG.M REFL.ACC REFL.DAT always again win.INF her fondness.ACC.SG  
 ‘I learnt to win her fondness always anew.’ (Cz; Avgustinova & Oliva 1997: 32)

Moreover, haplology is not a blind phonological process. Reflexive clitics may be contiguous with phonologically identical elements as in (4.219). In (4.219a), the dative reflexive *si* clusters with the homophonous auxiliary clitic *jsi*. In (4.219b), the accusative reflexive *se* is adjacent to the formally identical comitative preposition *se*, which it procliticises to because of the prosodic boundary following the relative clause (cf. also Junghanns 2002c: 80).<sup>344</sup>

- (4.219) a. Ty                    (j)si<sub>1</sub>        si<sub>2</sub>            chtěl<sub>1</sub>            hrát<sub>2</sub>?  
                  you.NOM.SG AUX.2SG REFL.DAT want.PST.SG.M play.INF  
                  ‘Did you want to play?’
- b. Ti,                            [s co        mě        ne-znají ],                    se<sub>2</sub>=            [pp se        mnou ]  
                  these.NOM.PL.VIR        what me.ACC NEG-know.PRS.3PL REFL.ACC        with me.INS  
                  začnou<sub>1</sub>        hádat<sub>2</sub>,        ...  
                  start.PRS.3PL argue.INF  
                  ‘Those, who do not know me, start to argue with me, ...’                    (Cz; Hana 2007: 124)

Finally, haplology also occurs with deep embeddings. In (4.220), the matrix verb *rozhodli se* as well as the deepest embedded infinitive *projít [se]* are inherently reflexive, but there is only one reflexive clitic in the utterance.<sup>345</sup> Although I will turn to haplology in Polish further below, I cite resembling example (4.221), which shows an even more complex embedding with more instances of the reflexive haplogised. The matrix predicate *powinno* realises the grammatical reflexive marker of the de-causative construction, whereas the embedded infinitives *udawać [się]*, *obawiać [się]*, and *spotkać [się]* are inherently reflexive.<sup>346</sup>

- (4.220) Rodiče                    se<sub>1,3</sub>            rozhodli<sub>1</sub>            jít<sub>2</sub>        projít<sub>3</sub>.  
                  parent.NOM.PL REFL.ACC decide.PST.PL.VIR go.INF go\_for\_a\_walk.INF  
                  ‘The parents have decided to go for a walk.’                    (Cz; Rosen 2001: 106)

<sup>344</sup> Note that the 2SG form of the auxiliary clitic / copula *jsi* is pronounced [si] and phonologically identical with the dative reflexive clitic *si* ‘REFL.DAT’. Simplifying issues somewhat, the consonantal instrumental preposition *s* ‘with’ becomes syllabic *se* before a morpheme starting with a sibilant (irrespective of voicing: /s/, /z/, /ʃ/, or /ʒ/) or having a complex onset of flat sonority, e.g. *se mnou* ‘with me’, *se ptákem* ‘with (a/the) bird’ (cf. Dickins 1998 for a comprehensive survey of the syllabification of consonantal prepositions in Czech).

<sup>345</sup> The underlying construction [ jít [ projít se ] ] is a common collocation and has a parallel in contemporary Polish [ iść [ przejść się ] ], cf. (i).

(i) Koło 7-ej idziemy się przejść.  
                  around 7th go.PRS.1PL REFL.ACC go\_for\_a\_walk.INF  
                  ‘We (will) go for a walk around 7 [o’clock].’                    (Pol; NKJP poliqarp balanced)

<sup>346</sup> Originally, the lexeme *powinien/powinna/powinno* is an adjective, compare Cz. *povinný/povinná/povinné*, short predicative form *povinen/povinna/povinno*). Synchronically, *powinien* is treated as a verb with a defective inflectional paradigm in contemporary Polish grammaticography.



- (4.221) Po tych lekach powinno<sub>1</sub> mu się<sub>1,3,4,5</sub> zacząć<sub>2</sub> udawać<sub>3</sub>  
 after this.LOC.PL medicine.LOC.PL should.N him.DAT REFL.ACC begin.INF succeed.INF  
 mniej obawiać<sub>4</sub> spotkać<sub>5</sub> ze znajomymi sprzed wojny.  
 less fear.INF meet.INF with friend.INS.PL from\_before war.GEN.SG  
 ‘As a result of these pills he should begin to succeed to be less afraid of meeting friends  
 from before the war.’ (Pol; Kupść 1999: 105)

The question arises, which clitic is affected by haplology. Golden (2008: 312) considers the embedded reflexive to be deleted, while it is retained under the loss of the matrix reflexive in Hana’s (2007: 124–125) opinion. Franks & King (2000: 199) seem to be generally agnostic regarding this issue, though they refer once to the concept of “merger of two reflexive forms [...] into one” (cf. also Kupść 1999: 104). Considering the examples so far, it is impossible to determine whether the embedded or the matrix reflexive clitic is omitted under haplology, or if an analysis in terms of fusion is more suitable.<sup>347</sup> This is corroborated by Rosen’s (2014: 105–107) discussion of what he tentatively calls “haplology of unlikes”, i.e. haplology between the accusative reflexive *se* and the dative reflexive *si*.<sup>348</sup> The data in (4.222)–(4.223) imply that haplology is possible independently of which is the matrix reflexive and which is the embedded one, although “with a preference for the embedded reflexive overriding the matrix one” (Rosen 2014: 106).<sup>349</sup> However, things seem to differ, when purely grammatical markers come into play. Rosen (2014: 111) tells that the only possible realisation is (4.224), where the impersonal marker *se* being realised under haplology instead of the embedded *si*. We encountered this preference pattern in realisation of the impersonal marker rather than the dative reflexive already in (4.214).

- (4.222) a. Jan se<sub>1</sub> bál<sub>1</sub> vzít<sub>2</sub> si<sub>2</sub> kravatu.  
 Jan REFL.ACC fear.PST.SG take.PFV.INF REFL.DAT tie.ACC.SG  
 b. Jan si<sub>2</sub> bál<sub>1</sub> vzít<sub>2</sub> kravatu.  
 c. ? Jan se<sub>1</sub> bál<sub>1</sub> vzít<sub>2</sub> kravatu.  
 ‘Jan was afraid to put on a tie.’ (Cz; Rosen 2014: 106)

<sup>347</sup> This is why I use subscriptions of the type  $x_{1,2}$  as a neutral descriptive means of visualising the parallel dependency relations between the reflexive and the involved verbs. I do not use  $x_{1+2}$  as Kolaković et al. (2022: 265–269), for this appears to be suggestive of a “merger” or “fusion” of the reflexives. Note also the difference to the subscription type  $x_{1/2}$ , where the slash is meant to indicate alternative dependency relations resulting in ambiguity.

<sup>348</sup> The examples provide again complex clauses comprising a finite matrix verb with a single embedded infinitive. It would be interesting to examine haplology in deep embeddings with both accusative and dative reflexive clitics licensed. I must leave this for future research.

<sup>349</sup> In an earlier discussion, Rosen (2001: 232) presents deviating grammaticality judgements. Only embedded dative *si* is capable of haplogising the accusative *se* after CC, but not vice versa. If *si* were licensed in the matrix clause, haplology is not available, at all.

- (4.223) a. Troufla<sub>1</sub>      **si**<sub>1</sub>      usadit<sub>2</sub> **se**<sub>2</sub>      v první      řadě.  
              dare.PST.SG.F REFL.DAT sit.INF REFL.ACC in first.LOC.SG row.LOC.SG  
 b. ? Troufla<sub>1</sub> **si**<sub>1</sub> usadit<sub>2</sub> v první řadě.  
 c. Troufla<sub>1</sub> **se**<sub>2</sub> usadit<sub>2</sub> v první řadě.  
 ‘She dared to sit in the first row.’ (Cz; Rosen 2014: 106)

- (4.224) a. Na šéfa      **se**<sub>1</sub>      přestalo<sub>1</sub>      stěžovat<sub>2</sub>.  
              on boss.ACC.SG REFL.ACC stop.PST.SG.N complain.INF  
 b. \*Na šéfa      **si**<sub>2</sub>      přestalo<sub>1</sub>      stěžovat<sub>2</sub>.  
              on boss.ACC.SG REFL.DAT stop.PST.SG.N complain.INF  
 ‘People stopped complaining about the boss.’ (Cz; Rosen 2014: 111)

After this comparatively detailed discussion of Czech, I shall restrict myself to citing selected examples of haplology for other Slavic 2P languages (including Polish) in (4.225)–(4.231) below. The Polish data in (4.225a) show a complex clause with an inherently reflexive finite matrix verb and an embedded infinitive with a reflexive argument. Haplology occurs in (4.225b). Since both utterances are attestations from a corpus study by Kiklewicz (2012), they do not provide a minimal pair, but still involve the same matrix and embedded verbs.<sup>350</sup> Kiklewicz (2012: 138–142) finds that haplology, which he calls *konstrukcje zredukowane* ‘reduced constructions,’ is preferred over the *konstrukcje pełne* ‘full constructions’ with all semantically licensed reflexives realised overtly.

- (4.225) a. (Jeśli państwo      się      zgodzą,)      postaram<sub>1</sub>      **się**<sub>1</sub>  
              if ladies\_and\_gentlemen.NOM REFL.ACC agree.PRS.3PL try.PRS.1SG REFL.ACC  
              ustosunkować<sub>2</sub> **się**<sub>2</sub>      do tych      wątpliwości.  
              relate.INF REFL.ACC to this.GEN.PL doubt.GEN.PL  
              ‘(If you will agree,) I will try to take a stance on these doubts.’

<sup>350</sup> The relevance of this fact will be discussed, among other things, in the methodological discussion in Section 6.1.

- b. (Ze względu na skomplikowaną treść projektu  
 from reason.GEN.SG on complicated.ADJ.ACC.SG.F content.ACC.SG draft.GEN.SG  
 ustawy) postaram<sub>1</sub> się<sub>1,2</sub> ustosunkować<sub>2</sub> w kilku słowach do  
 law.GEN.SG try.PRS.1SG REFL.ACC relate.INF in few.LOC word.LOC.PL to  
 każdego punktu.  
 every.GEN.SG.M point.GEN.SG  
 ‘(Due to the complicated content of the law’s draft) I will try to take a position on every  
 point in a few words.’ (Pol; Kiklewicz 2012: 139)

Haplology involving the dative reflexive has not been taken into consideration so far. Rosen (2014: 107) is the only source I am aware of, who cites haplology involving a dative reflexive for Polish, cf. (4.226).<sup>351</sup> In (4.226a), the finite matrix verb is again an inherently reflexive verb, while the embedded infinitive has an indirect reflexive object *sobie*. The embedded reflexive is dropped in (4.226b). Note that *sobie* is the orthotonic form. It is however possible to find haplological constructions involving the dative reflexive clitic *se*. Example (4.227) is from the NKJP. Again, we encounter the same inherently reflexive matrix verb *starać* [*się*], which embeds the infinitive *przyrobić* ‘(to) earn something extra’. The infinitive is accompanied by the facultative datival reflexive clitic *se*, which is realised between both verbs, while the matrix reflexive *się* is absent.

- (4.226) a. Starła-m<sub>1</sub> się<sub>1</sub> sobie<sub>2</sub> wytłumaczyć<sub>2</sub> jego zachowanie.  
 try.PST.SG.F-AUX.1SG REFL.ACC.CL REFL.DAT explain.INF his behaviour.ACC.SG  
 ‘I tried to explain his behaviour to myself.’  
 b. ? Starła-m<sub>1</sub> sobie<sub>1,2</sub> wytłumaczyć<sub>2</sub> (jego zachowanie).  
 try.PST.SG.F-AUX.1SG REFL.DAT explain.INF his behaviour.ACC.SG  
 ‘I tried to explain (his behaviour) to myself.’ (Pol; Rosen 2014: 107)
- (4.227) Zawsze też starał-e-m<sub>1</sub> se<sub>1,2</sub> przyrobić<sub>2</sub>, (by  
 always also try.PST.SG.M-M-AUX.1SG REFL.DAT.CL earn\_a\_bit\_extra.INF[SUB.]COND  
 od czasu do czasu było na piwo lub wódkę.)  
 from time.GEN.SG to time.GEN.SG be.PST.SG.N on beer.ACC.SG or vodka.ACC.SG  
 ‘I always tried to earn something extra for myself (to allow a beer or vodka once in a while.)’  
 (Pol; NKJP poliqarp full)

<sup>351</sup> There are no inherently reflexive verbs involving the dative reflexive in Polish according to Rosen (2014: 107).

Kolaković et al. (2022: 264–270) provide corpus data for haplology in Bosnian and Croatian complex clauses, such that (4.228)–(4.229) do not provide minimal pairs, but relatable constructions. The Bosnian data in (4.228) involve the matrix verb *truditi se* ‘(to) try’ and the embedded infinitive *svidjeti se* ‘(to) like’ which are both inherently reflexive. Example (4.228a) exhibits the full construction, whereas (4.228b) shows haplology. Similar is shown in (4.229) for Croatian with the inherently reflexive matrix verb *bojati se* ‘(to) fear’. The embedded infinitive *odreći (se)* ‘(to) renounce’ realises a facultative reflexive clitic, whereby the direct object bears genitive case.<sup>352</sup> In (4.229a), the full construction with two non-contiguous reflexives is attested, whereas (4.229b) shows haplology.

- (4.228) a. [...] *toliko se<sub>1</sub> covjek trudi<sub>1</sub> svidjeti<sub>2</sub> se<sub>2</sub>*  
 that<sub>much</sub> REFL.ACC man.NOM.SG try.PRS.3SG like.INF REFL.ACC  
*drugima* [...].  
 other.INS.PL  
 ‘... one tries hard to be liked by others ...’
- b. [...] *da se<sub>1,2</sub> ne trudi<sub>1</sub> previse svidjeti<sub>2</sub> drugima* [...].  
 that REFL.ACC NEG try.PRS.3SG too<sub>much</sub> like.INF other.INS.PL  
 ‘... that s/he does not try too much to be liked by others ...’
- (Bos; Kolaković et al. 2022: 266)

- (4.229) a. [...] *koji se<sub>1</sub> boje<sub>1</sub> odreći<sub>2</sub> se<sub>2</sub> grijeha* [...].  
 which REFL.ACC fear.PRS.3PL give<sub>up</sub>.INF REFL.ACC sin.GEN.SG  
 ‘... who are afraid to renounce sin ...’
- b. [...] *jednostavno se<sub>1,2</sub> boji<sub>1</sub> odreći<sub>2</sub> studentskog*  
 simply REFL.ACC fear.PRS.3SG give<sub>up</sub>.INF student.ADJ.GEN.SG.M  
*načina života.*  
 way.GEN.SG life.GEN.SG  
 ‘... he is simply afraid to give up the student way of living.’
- (Hrv; Kolaković et al. 2022: 266)

Kolaković et al. (2022: 267) conclude that data in (4.230) suggests that haplology is contingent on clitic climbing. The example shows a deep embedding construction with two subsequently embedded infinitives. The higher infinitive *bojati se* is inherently reflexive, while the lower infinitive *približiti* is subcategorised for an argumental reflexive proper and a dative pronominal clitic, cf. (4.230a).

<sup>352</sup> The realisation of the reflexive has an impact on lexical semantics. The reflexive verb *odreći se* means ‘(to) renounce’, while plain *odreći* means ‘(to) deny’.

Haplology is possible under climbing of the lower cluster, cf. (4.230b). The utterance becomes ungrammatical, if only the reflexive clitic climbs and the dative pronominal clitic remains in situ, cf. (4.230c). However, the construction could be equally well analysed as violating the restriction on clitic climbing, which disallows lower clitics to cross higher ones and was discussed in Section 4.7.1 (e.g. under the label of the Head Movement Constraint). Example (4.230) is therefore inconclusive for illuminating the relation between clitic climbing and haplology.

- (4.230) a. Ne trebaju<sub>1</sub> se<sub>2</sub> bojati<sub>2</sub> približiti<sub>3</sub> mi<sub>3</sub> se<sub>3</sub>.  
 NEG need.PRS.3SG REFL.ACC fear.INF approach.INF me.DAT REFL.ACC  
 b. Ne trebaju<sub>1</sub> mi<sub>3</sub> se<sub>2,3</sub> bojati<sub>2</sub> približiti<sub>3</sub>.  
 c. \*Ne trebaju<sub>1</sub> se<sub>2,3</sub> bojati<sub>2</sub> približiti<sub>3</sub> mi<sub>3</sub>.  
 ‘They do not have to fear approaching me.’ (Hrv; Kolaković et al. 2022: 267)

An instance of haplology in Slovene is cited by Golden (2008: 312) in (4.180) from Section 4.7.2 repeated as (4.231) below. She informs that both the matrix verb *čutiti se* ‘(to) feel’ and the embedded verb *opravičevati se* ‘(to) apologize’ license the reflexive clitic *se* – the latter licenses also pronominal *mu*.<sup>353</sup> Similar to Croatian (4.230b), the embedded clitic cluster is realised non-locally with simultaneous haplology.

- (4.231) Jaz se<sub>1,2</sub> mu<sub>2</sub> ne čutim<sub>1</sub> [AdjP dolžnega  
 I.NOM REFL.ACC him.DAT NEG feel.PRS.1SG obliged.ADJ.GEN.SG.M  
 [VP opravičevati<sub>2</sub>]].  
 apologise.INF  
 ‘I do not feel obliged to apologise (for myself).’ (Slv; Golden 2008: 312)

## 4.9.2 Deletion of unlikes

The previous section reviewed haplology of the reflexive, whereby two configurations can be distinguished. Either the reflexives are formally identical, or they are morphologically different, the latter of which has been labelled “haplology of unlikes” by Rosen (2014). Kolaković et al. (2022: 22 *inter alia*) extend this notion to co-occurrence restrictions involving different clitic types in BCMS, particularly reflexive *se* and past tense auxiliary clitic *je* (cf. also Franks & King 2000: 30). Example

<sup>353</sup> Compare also the parallel German lexemes: transitive (*etwas*) *fühlen* vs. copulative *sich fühlen* ‘(to) feel’, (*jemanden*) *entschuldigen* ‘(to) excuse (someone else)’ vs. *sich entschuldigen* ‘(to) apologize’.

(4.232) provides a simple clause in the past tense and with a 3rd person singular subject. Different from West Slavic, past tense auxiliary clitics occur in the 3rd persons in South Slavic, too. In contrast, the auxiliary is omitted in reflexive constructions like in example (4.233).

(4.232) Vesna        **mi**        **ga**        **je**        kupila        juče(r).  
 Vesna.NOM me.DAT him.ACC AUX.3SG buy.PST.SG.F yesterday  
 ‘Vesna bought it for me yesterday.’ (BCMS; Franks & King 2000: 25)

(4.233) On        **se**        (**\*je**)        obradovao.  
 he.NOM REFL.ACC AUX.3SG gladden.PST.SG.M  
 ‘He was happy.’ (Srp; Kolaković et al. 2022: 104)

There are important differences between haplology of reflexives and the co-occurrence restrictions affecting *se + je*. Haplology applies to a (more or less) homogeneous class of expressions, namely reflexive clitics. The contrast between (4.232) and (4.233) reveals a restriction involving different categories, i.e. a reflexive and a temporal auxiliary clitic. The latter is restricted to a particular tense, whereas the former is consistently licensed by its verbal head or a particular grammatical construction irrespective of tense. Furthermore, Kolaković et al. (2022: 22, n. 21) inform that auxiliary deletion is a phenomenon primarily associated with the standard languages, but where some variation can be found, too (cf. Kolaković et al. 2022: 104–106). Franks & King (2000: 30) remark that “[s]equences of *se + je* [...] are often contracted to *se*”, which also implies an essentially variable nature of the phenomenon.<sup>354</sup> In contrast, haplology does not show optionality. The adjacency of two independently licensed reflexive clitics is always banned across languages. Therefore, I will not adopt the notion *haplology* for such occurrences but keep with Rosen’s (2014) narrower original usage and reserve it for the process involving reflexive clitics. I will refer to other types of co-occurrence restrictions provisionally by the term *deletion (of unlikes)*.

Deletion seems to be less trivial to capture than haplology. Rosen (2001: 227) states that “[t]wo phonologically identical clitics cannot co-occur in a single clitic cluster as a result of clitic climbing[.]” Two dative pronouns are licensed in the complex clause in (4.234a) the matrix one being clitic, while the embedded one being a focused orthotonic pronoun. The orthotonic pronoun may be substituted by its clitic counterpart when de-focused, but must remain in situ to be grammatical, cf. (4.234b). Co-occurrence of matrix and embedded dative clitic within a single cluster is not possible,

<sup>354</sup> For instance, Kolaković et al. (2022: 106) mention that “according to several Croatian authors [...], not only the verbal CL[itic] *je* but also *su* as an auxiliary can be omitted, even if there is no pronominal or reflexive CL[itic] in the sentence.” Furthermore, considerable variation in *je*-deletion (including reverse ordering) is encountered in dialects (cf. Kolaković et al. 2022: 147–149, 153) and spoken language (= Bosnian) (cf. Kolaković et al. 2022: 194–195)

cf. (4.234c). The grammatical climbing variant in (4.234d) involves deletion of one dative clitic.<sup>355</sup> Yet, Rosen (2001: 228) observes that multiple phonologically/morphologically identical clitics are grammatical, if they are licensed within the same clausal domain. Consider (4.235), where two argumental accusative clitics occur.<sup>356</sup>

- (4.234) a. Kamila **mi**<sub>1</sub> slíbila<sub>1</sub> to<sub>2</sub> vrátit<sub>2</sub> [ **MNĚ**<sub>2</sub> ]<sub>FOC</sub>.  
 Kamila.NOM me.DAT.CL promise.PST.SG.F this.ACC.SG.N return.INF me.DAT  
 b. Kamila **mi**<sub>1</sub> slíbila<sub>1</sub> **mi**<sub>2</sub>to<sub>2</sub> vrátit<sub>2</sub>.  
 Kamila.NOM me.DAT.CL promise.PST.SG.F me.DAT.CL this.ACC.SG.N return.INF  
 c. \*Kamila **mi**<sub>1</sub> **mi**<sub>2</sub>to<sub>2</sub> slíbila<sub>1</sub> vrátit<sub>2</sub>.  
 d. Kamila **mi**<sub>1,2</sub>to<sub>2</sub> slíbila<sub>1</sub> vrátit<sub>2</sub>.  
 ‘Kamila promised me to return it to me.’ (Cz; Rosen 2001: 227)

- (4.235) a. (A kdo pomůže [NP Heleně]<sub>j</sub> [PP s [NP dračí  
 and who.NOM help.PRS.3SG Helena.DAT with dragon.ADJ.INS.SG  
 smyčkou]<sub>k</sub> ]?)  
 loop.INS.SG  
 ‘(And who will help Helena with the bowline knot?)’  
 b. Lukáš **ji**<sub>k</sub> **ji**<sub>j</sub> už naučil.  
 Lucas.NOM her.ACC her.ACC already teach.PST.SG.M  
 ‘Lucas already taught it to her.’ (Cz; Rosen 2001: 228)

The data in (4.236) corroborate the impression that co-occurrence restrictions among pronominal clitics are vague. Deletion fails to appear, if two morphologically identical (by case) but phonologically distinct pronominal clitics cluster after climbing (cf. Rosen 2001: 232–233). In (4.236) two dative clitics licensed by different verbal heads are positioned in the same clitic cluster, while the utterance is acceptable. Note that their order is *mi* < *mu* in (4.236a) and *mu* < *mi* in (4.236b). Indexation indicates that the cluster-internal word order follows the hierarchy within verbal cascade. The leftmost dative clitic depends on the higher verb. The same is true for accusative pronominal clitics in (4.237).<sup>357</sup>

<sup>355</sup> The deletion pattern presented in example block (4.234) resembles the Croatian example (4.230) with haplogy from Section 4.9.1. As declared at the beginning of Section 4.9.2, I will refer to (4.234) as deletion, because I reserve the term *haplogy* exclusively for reflexive clitics.

<sup>356</sup> Data of this kind provide another contradiction of Rezac’s (2005: 105, 118, 120) claim that double accusatives are banned in Czech.

<sup>357</sup> Rosen (2001: 234) does not provide an explanation for why (4.237b) receives a double question mark. For instance, it could be due to the odd meaning.

- (4.236) a. Poslat<sub>2</sub> kurýrem se<sub>1</sub> mi<sub>1</sub> mu<sub>2</sub> ho<sub>2</sub> dnes  
 send.INF courier.INS.SG REFL.ACC me.DAT him.DAT him.ACC today  
 ne-podařilo<sub>1</sub>.  
 NEG-succeed.PST.SG.N  
 ‘Today, I did not manage to send it to him by courier.’
- b. Poslat<sub>2</sub> kurýrem se<sub>1</sub> mu<sub>1</sub> mi<sub>2</sub> ho<sub>2</sub> dnes  
 send.INF courier.INS.SG REFL.ACC him.DAT me.DAT him.ACC today  
 ne-podařilo<sub>1</sub>  
 NEG-succeed.PST.SG.N  
 ‘Today, he did not manage to send it to me by courier.’ (Cz; Rosen 2001: 233)
- (4.237) a. Třeba tě<sub>1</sub> ji<sub>2</sub> nechají<sub>1</sub> zahrát<sub>2</sub>.  
 maybe you.ACC.SG her.ACC let.PRS.3PL play.INF  
 ‘Maybe they let you play it [= her].’
- b. ?? Třeba ji<sub>1</sub> tě<sub>2</sub> nechají<sub>1</sub> zahrát<sub>2</sub>.  
 maybe her.ACC you.ACC.SG let.PRS.3PL play.INF (Cz; Rosen 2001: 233–234)

Let us briefly return to BCMS. Beside the co-occurrence restriction on the string *se* < *je*, Kolaković et al. (2022: 105) tell that “[i]n headlines or certain contexts of spoken language the auxiliary can be omitted for all persons irrespective of the presence of other CLs.” Franks & King (2000: 30) report that “*je* sometimes even drops after pronominal clitics *me* and *te*.” Kolaković et al. (2022: 105) provide the same information accompanied by example (4.238).

- (4.238) Gizela me čekala u posječenom parku.  
 Gizela.NOM me.ACC wait.PST.SG.F in trimmed.LOC.SG park.LOC.SG  
 ‘Gizela was waiting for me in the trimmed park.’ (Hrv; Kolaković et al. 2022: 105)

Strictly speaking, there does not seem to be any formal condition for deletion. Stanojević (2011) and Dickey (2017) examine the deletion of *je* for Croatian and BCMS respectively based on empirical data (corpus data and a novel respectively).<sup>358</sup> They propose a cognitive explanation by means of epistemic distance/immediacy, which is not relevant for the present discussion. However, it is worth noting that both authors declare that the deletion of the auxiliary has a comparatively low occurrence frequency in their data sets. Still, the phenomenon deserves careful examination. I cite some

<sup>358</sup> Stanojević (2011) uses the *Hrvatski nacionalni korpus* ‘Croatian National Corpus’ (HNK), while Dickey (2017) relies on Meša Selimović’s novel *Derviš i smrt* ‘Death and the Dervish’ from 1966.



representative examples from Dickey (2017) in (4.239). I follow Dickey’s notational representation of auxiliary deletion using the mark  $\emptyset$ , which I will apply within both the cited object linguistic example and the glosses. In (4.239a), *je* is missing after the pronominal accusative clitic *me* as well as after the accusative clitic *te* in (4.239b). Both examples display exactly the cases cited above from Franks & King (2000) and Kolaković et al. (2022). However, (4.239c) provides yet two other occurrences of *je*-deletion, namely one without any other contextual clitic (in the first orthographic sentence) and one in the presence of the pronominal dative clitic *mi* (in the third orthographic sentence).

A newspaper headline is cited in (4.240) from Stanojević (2011). Here, the plural auxiliary *su* is missing. Stanojević (2011: 223) informs that in his random sample of 978 occurrences “[a]ll of the auxiliary-less examples are limited to the third person.”

Sonnenhauser & Meermann (2016) analyse Serbian auxiliary deletion in a similar fashion. They investigate colloquial Serbian and Serbian dialect data from the transitional zone to Bulgarian.<sup>359</sup> An

example for singular and a plural auxiliary deletion in colloquial Serbian are cited in (4.241a) and (4.241b) respectively. Note that in (4.241b) a plural auxiliary clitic is realised in the preceding clause (surrounded by parentheses).

- (4.241) a. [...] (a glavna ličnost je jedna jevrejka  
 and main.NOM.SG.F person.NOM.SG be.PRS.3SG one.NOM.SG.F Jew.NOM.SG.F  
 Vera,) koja  $\emptyset$  preživala raznu strahotu  
 Vera.NOM who.NOM.SG.F  $\emptyset$  survive.PST.SG.F various.ACC.SG.F horror.ACC.SG  
 u logoru.  
 in camp.LOC.SG  
 ‘... (and the protagonist is the jewish [woman] Vera), who survived the many horrors of  
 the [concentration] camp.’
- b. (Ona ne zna kad su uhapsili Gavrila  
 she.NOM NEG know.PRS.3SG when AUX.3PL arrest.PST.PL.M Gavriilo.ACC  
 Principa,) i kaže oborili je  $\emptyset$  ispitu.  
 Princip.ACC and say.PRS.3SG fail.PST.PL.M her.ACC  $\emptyset$  exam.DAT.SG  
 ‘(She doesn’t know, when Gavriilo Princip was arrested) and says, they [deliberately] let  
 her fail the exam.’ (Srp, colloquial; Sonnenhauser & Meermann 2016: 97)

Like for missing *su* in (4.240), the omission of the auxiliary clitics *je* in (4.241a) cannot be explained by the syntactic environment or phonological constraints, because there are no clause-mate clitics. I reaffirm that the notion of *haplology* should not be extended to auxiliary deletion in BCMS. The deletion pattern in the third persons is reminiscent of the past tense formation in West Slavic. It has either gone unnoticed or been neglected in formal linguistic research.

To sum up, the co-occurrence restrictions discussed in this section cannot be consistently attributed to phonological or morphological identity. Pronominal clitics of identical morphological case can occupy the same clitic cluster. Phonological identity appears to be permitted only in case the clitics are co-dependents licensed by the same verbal head. On the other hand, it seems to be necessary to treat reflexive clitics differently. Multiple reflexives are generally banned from the clitic cluster irrespective of their actual case or function. Phonological form does not play a role, at all, as clustering with formally identical auxiliary clitics and prepositions is not excluded, as shown in the previous Section 4.9.1. Finally, the deletion of third person auxiliary clitics in BCMS seems to be a process that applies inconsistently. It cannot be regarded as motivated by any formal phonological, morphological, or syntactic feature. The next section will particularly focus on auxiliary deletions of a different kind.

### 4.9.3 Agreement drop (1st person auxiliary clitic omission)

The following discussion focuses on the non-canonical omission of auxiliary clitics in Slavic, which will be a relevant topic in the empirical Part III of the thesis.<sup>360</sup> Such phenomena have been referred to by different terms, i.e. *agreement drop* (cf. Ruda 2017, 2022), *auxiliary (clitic) drop* (Willer-Gold & Peti-Stantić 2022), and *truncated perfect* (Meermann 2015). I follow Ruda (2017, 2022) and shall prefer AGREEMENT DROP over its alternatives, but I generally regard these notions as synonyms.<sup>361</sup> Most Slavic standard languages relevant here do not allow auxiliary clitic omissions. BCMS is an exception as seen in the previous Section 4.9.2, because the standard languages allow (or require) the omission of the auxiliary clitic *je* under simultaneous occurrence of the reflexive clitic *se*. While this pattern is optional in the Bosnian and Croatian standard languages, it is considered mandatory in standard Serbian (cf. Kolaković et al. 2022: 104, with references). Relevant agreement drop facts from West Slavic shall be presented first, before I will turn to Slavic languages from other branches including BCMS.

To simplify somewhat, the West Slavic languages build the past tense analytically by the *l*-participle specifying tense, number, and gender, and an auxiliary clitic flagging person and number.<sup>362</sup> Czech, Polish, and Slovak possess auxiliary clitics only for the first and second persons in both numbers. There are no third person auxiliaries as opposed to the perfect tense of the South Slavic languages, all of which exhibit full paradigms of auxiliary clitics across all numbers and persons (except Macedonian).<sup>363</sup> The prescriptive standard norms of Czech and Polish consistently require the realisation of the auxiliary clitics in the past tense. However, a 1st person auxiliary clitic may be omitted in non-standard varieties like colloquial speech or dialects, if the corresponding subject pronoun is overtly realised.

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<sup>360</sup> See Section 10.4.4 and Section 10.6.4 for agreement drop in Goralish and Silesian respectively, and Section 10.8.4 for a comparative perspective.

<sup>361</sup> The usage of such notions might appear problematic in one case or another. In short, the term *auxiliary drop* has a rather wide scope, but the orthotonic auxiliary verb cannot be omitted in the analytical future tense. Furthermore, the status of the Polish so-called mobile inflection is controversial, and it is treated as an inflectional agreement suffix rather than a genuine auxiliary clitic. The *truncated perfect* is also not adequate for all Slavic languages, since the verb form built with the *l*-participle has become the standard and only past tense form in North Slavic (irrespective of the pluperfect, which is not a productive category of the native speakers' grammar). Furthermore, Meermann's (2015) examples exhaust in 3rd person past tense forms, while the phenomenon dealt with in this section, targets another morphosyntactic person category. Given that the corresponding elements in other Slavic languages code number and person, the notion *agreement drop* seems to be generally appropriate.

<sup>362</sup> I shall abstract from Kashubian, which is an outlier inasmuch as it allows for the omission of the whole auxiliary paradigm (cf. Ruda 2017, with further references). This pattern resembles the past tense formation in East Slavic.'

<sup>363</sup> The full paradigm with an overt auxiliary clitic for 3SG and 3PL is attested in older stages, e.g. in Old Czech and Old Polish (cf. the data discussed by Kosek 2015, 2017 for Czech; Witkoś 1998: 43–44 for Polish). Macedonian resembles West Slavic in that it lacks third person auxiliary clitics in the perfect tense of the so-called *sum* type (cf. Franks & King 2000: 69).

The occurrence of auxiliary drop in Czech has always been a mere side note in clitic research examining Czech. Example (4.242) cites Czech data, where I continue to apply Dickey’s (2017) notational convention for marking an absent auxiliary clitic by the mark  $\emptyset$ . Toman (1980: 307) provides only a first person singular example, but generalises that “*be* is optionally deleted in the first person” without mentioning further restrictions. In contrast, Veselovská (2008: 559) seems to regard agreement drop to be restricted to 1SG stating that

“there exists also a kind of partial complementary distribution between the  $AUX_{1sg}$  and an overt pronominal subject. [...] the verbal complex requires the presence of either overt  $AUX_{1sg}$  or a pronominal 1sg subject (or both).” (Veselovská 2008: 559)

However, agreement drop occurs in the 1st person plural, too, cf. (4.242c).

- (4.242) a. Já  $\emptyset$  už spál.  
 I.NOM  $\emptyset$  already sleep.PST.SG.M  
 ‘I was already sleeping.’ (Cz; Toman 1980: 307)
- b. Já  $\emptyset$  chvalil Petra.  
 I.NOM  $\emptyset$  praise.PST.SG.M Peter.ACC  
 ‘I praised Peter.’ (Cz; Veselovská 2008: 559)
- c. (O dvacet let později se ukázalo,) že  $\emptyset$  my  
 about twenty year.GEN.PL later REFL.ACC turn\_out.PST.SG.N that  $\emptyset$  we.NOM  
 měli pravdu.  
 have.PST.PL.VIR truth.ACC.SG  
 ‘(Twenty years later, it turned out) that we were right.’ (Cz; ČNK syn\_v8)

The reader should be aware that the languages relevant here allow “grammatically unconstrained null subjects of all person/number/gender combinations”, as Ruda (2022) cites with reference to Barbosa (2019) while showing that Polish, Kashubian, and Silesian exhibit undisputable pro-drop properties.<sup>364</sup> Ruda (2017) furthermore argues that the occurrence of the subject pronoun under agreement drop is motivated by grammatical factors, i.e. the overt realisation of  $\phi$ -features (i.e. morphosyntactic features). Thus, the omission of the auxiliary forces the overt realisation of the subject pronoun. It is difficult to determine the exact motivation of the agreement drop pattern, such

<sup>364</sup> I will omit Barbosa’s (2019) other criteria referred to by Ruda (2022), as it is sufficient for the purpose of the present thesis to recognise that pro-drop and agreement drop may occur in complementary distribution in the lects under scrutiny but are not mutually exclusive.

that the logical alternative, i.e. an overt subject pronoun licenses agreement drop, must also be treated as a possible scenario.<sup>365</sup> More research on the cause-effect chain is certainly needed.

Agreement drop in Polish is exemplified in (4.243), but it seems to be a comparatively rare construction. It occurs only in colloquial speech, but no information on diatopic or diastratic distribution of the construction is available at this moment. Note that lects, which traditionally have been considered dialects, exhibit agreement drop more often.<sup>366</sup> Ruda (2017, 2022) provides the Silesian examples in (4.244). Furthermore, I cite agreement drop in Goralish (4.245) anticipating the corpus study presented in Chapter 10.<sup>367</sup>

- (4.243) a. (i powtarzam) ja była  $\emptyset$  świadkiem  
 and repeat.PRS.1SG I.NOM be.PST.SG.F  $\emptyset$  witness.INS.SG  
 tych odejść [...]   
 this.GEN.PL quitting.GEN.PL  
 ‘(and I repeat) I witnessed these quittings ...’
- b. (ja szedł-e-m kiedyś do Marty) jeszcze my  
 I.NOM go.PST.1SG.M-M-AUX.1SG once to Marta.GEN.SG still we.NOM  
 nie byli  $\emptyset$  po ślubie [...]   
 NEG be.PST.PL.VIR  $\emptyset$  after wedding  
 ‘(once, I went to Marta) we weren’t a married couple, yet’ (Pol, spoken; NKJP pelcra)
- (4.244) a. Jo o nich słyszoł  $\emptyset$  – oni o mie niy.  
 I.NOM about them.LOC hear.PST.SG.M  $\emptyset$  they about me.LOC NEG  
 ‘I heard about them – they didn’t [hear] about me.’ (Sil; Ruda 2022: ex. [42])
- b. jag my dostali  $\emptyset$  tu pszidział  
 when we.NOM get.PST.PL.VIR  $\emptyset$  here allotment.ACC  
 ‘when we got allotment here’ (Sil; Ruda 2017: 338)

<sup>365</sup> As a third option, it is logically possible that agreement drop is a construction in its own right.

<sup>366</sup> It suffices to rely on very rough quantitative expressions from everyday language for the purpose of the present discussion. See Section 10.4.4 and Section 10.6.4 for quantitative measures of agreement drop in Goralish and Silesian respectively.

<sup>367</sup> It is an interesting research question, if the putative contrast in frequency between agreement drop in standard Polish and its vernacular on the one hand and agreement drop in Goralish and Silesian on the other hand could be attributed to the different status of the mobile inflection. While it is genuine inflectional affix for most native speakers of (standard) Polish, it retained clitic status in Goralish and Silesian. A related hypothesis would be that native speakers of (standard) Polish, who did not acquire a corresponding lectal grammar, but detach the mobile inflection from the *l*-participle, treat it as a clitic, too. I must leave it for future research.

- (4.245) a. ja        cię        szanowała         $\emptyset$   
 I.NOM you.ACC esteem.PST.SG.F  $\emptyset$   
 ‘I esteemed you.’
- b. my        mu<sub>2</sub>        też    nie    mieli<sub>1</sub>         $\emptyset$     z        czego        dawać<sub>2</sub>  
 we.NOM him.DAT also NEG have.PST.PL.VIR  $\emptyset$  from what.GEN give.INF  
 ‘We also did not have anything to share with him.’ (Gor; KoSpi)

I would like to draw the reader’s attention to the observation that agreement drop seems to come about with a word order particularity. Contrary to the comparatively free constituent order in Slavic, which is driven by information structure, the overt subject pronoun must precede both the *l*-participle and the clitic cluster, as exemplified by the Goralish example (4.245), where the subject is clause-initial and a pronominal clitic follows in second position. Accordingly, the permutations in (4.246) are supposed to be ungrammatical, irrespective of whether they comply with Wackernagel’s law.

- (4.246) a. \*szanowała         $\emptyset$     cię        ja  
 esteem.PST.SG.F  $\emptyset$  you.ACC I.NOM  
 Intended: ‘I esteemed you.’
- b. \*nie    mieli<sub>1</sub>         $\emptyset$     mu<sub>2</sub>        z        czego        dawać<sub>2</sub>    też    my  
 NEG have.PST.PL.VIR  $\emptyset$  him.DAT from what.GEN give.INF also we.NOM  
 Intended: ‘We also did not have anything to share with him.’  
 (Gor; constructed from [4.245])

Given this observation, delayed clitic placement is expected to be a default linearisation in subordinate clauses introduced by a subordinator. This is illustrated by the Goralish data in (4.247). As far as I am aware, the present thesis is the first to introduce this observation to scientific discourse and I do not know of any publication mentioning this pattern before.

- (4.247) a. [...] (zacniemy    od    tego)        że<sup>1</sup> właściwie<sup>2</sup> ja<sup>3</sup>    się<sup>4</sup>  
 begin.PRS.1PL from this.GEN.SG.M that actually I.NOM REFL.ACC  
 wychowała     $\emptyset$     przy babce        [...]  
 raise.PST.SG.F  $\emptyset$  at grandmother.LOC.SG  
 ‘... (to begin with) I grew up at my grandmother’s, in fact ...’

- b. (i staraj                      się              tak) żeby<sup>1</sup> my<sup>2</sup>      cię<sup>3</sup>              po prostu  
 and attempt.IMP.2SG REFL.ACC so      that      we.NOM you.ACC on plain.DAT.SG  
 z-rozumieli                      Ø  
 PFV-understand.PST.PL.VIR      Ø  
 ‘(and take care) that we will just understand you’ (Gor; KoSpi)

Facts are more complicated, indeed, and the proposed word order generalisation does not capture all clitic realisations under agreement drop in Slovak. In (4.248a), the conditional clitic *by* is placed in clausal second position after an interrogative expression outside of the agreement drop syntagm of subject pronoun and *l*-participle.<sup>368</sup> Examples (4.248b–c), in turn, show “well-behaved” singular and plural agreement drop with pronominal clitics in a root and subordinate clause respectively.<sup>369</sup> A potential source for positional variation across different clitic categories could be attributed to the emphatic particle *-že* following the interrogative *čo*, as it is formally identical and etymologically related to the contemporary homophonous subordinator *že* ‘that’. Again, things become more complicated once more data is included. Example (4.248d) provides a conditional clitic that clusters with the object clitic *ti* and precedes the agreement drop syntagm, but there is no emphatic particle or subordinator.

- (4.248) a. Ej, čože<sup>1</sup> by<sup>2</sup>      Ø      ja      bol.  
 ey what COND Ø I.NOM be.PST.SG.M  
 ‘Hey, what would I be?’ (Slk; SNK prim-10.0-public-sane)
- b. [...] ale ja<sup>1</sup>      Ø      mu<sup>2</sup>      chcela<sub>1</sub>      pomôcť<sub>2</sub>.  
 but I.NOM Ø him.DAT want.PST.SG.F help.INF  
 ‘... but I wanted to help him.’
- c. (Primator              ne-chce              po-besedovac, no mal              by,)  
 mayor.NOM.SG NEG-want.PRS.3SG PFV-talk.INF but have.PST.SG.M COND  
 bo<sup>1</sup>      Ø      [i      my]<sub>FOC</sub> ho<sup>3</sup>              volili.  
 because Ø and we.NOM him.ACC elect.PST.PL  
 ‘(The mayor doesn’t want to talk, but he should,) ’cause we elected him, too.’

<sup>368</sup> CQL queries:

(i) [word!="som"]{3} [word="ja"] [tag="VL.\*"] [word!="som"]  
 (ii) [word!="som"]{3} [word="ja"] [word="mi|mu|ho|ta|se|ma"] [tag="VL.\*"]  
 [word!="som"]  
 (iii) [word="my"] [word="mi|mu|ho|ta|se|ma"]{1,2} [tag="VL.\*"]

<sup>369</sup> Example (4.248c) exhibits features of the Eastern Slovak (southern) Šariš dialect around Prešov, e.g. the lexeme (*po*-)bešedovac ‘(to) talk’ including the infinitival ending *-(i)c* instead of *-(i)t’*, compare with standard Slovak *besedovať* ‘(to) chat’. (The lack of the caron *ň* in ex. [4.248c] could reflect either ordinary phonological variation or an interference from standard Slovak. The primary source of the utterance is the Slovak newspaper *SME*.)

- d. Ved<sup>1</sup> **by**<sup>2</sup> **ø** **ti**<sup>2</sup> ja po-radil, pomohol [...]  
 but COND **ø** you.DAT I.NOM PFV-advise.PST.SG.M help.PST.SG.M  
 ‘But I would advise you, help [you] ...’ (Slk; SNK prim-10.0-public-all)

It appears that Slovene permits a similar auxiliary drop pattern. In (4.249a), the conditional clitic *bi* and the pronominal clitic *te* occur, but they are realised in diacsis.<sup>370</sup> The former occupies the Wackernagel position, thus, preceding the actual agreement drop construction, while the latter is placed in fourth position between subject and verb. Example (4.249b–c) shows the pronominal clitic *ga* interfering in the agreement drop syntagm.

- (4.249) a. Potem<sup>1</sup> **bi**<sup>2</sup> **ø** jaz<sup>3</sup> **te**<sup>4</sup> vprašal, (zakaj rabimo  
 afterwards COND **ø** I.NOM you.ACC ask.PST.SG.M why need.PRS.1PL  
 ST forum) [...] ST forum.ACC.SG  
 ‘Afterwards I would ask you (why do we need the ST forum) ...’  
 b. Vsaj<sup>1</sup> **ø** jaz<sup>2</sup> **ga**<sup>3</sup> povezal v omrežje,  
 at\_least **ø** I.NOM him.ACC connect.PST.SG.M in network.ACC.SG  
 (tako dostopam še do slik in mp3 na računalniku.)  
 so access.PRS.1SG still to image.GEN.PL and MP3 on computer.LOC.SG  
 ‘At least I connected him with the network (so I still [can] access images and MP3s on the computer.)’  
 c. (Domen je pizdil), jaz<sup>1</sup> **ø** **ga**<sup>2</sup> miril [...]  
 Domen.NOM AUX.3SG complain.PST.SG.M I.NOM **ø** him.ACC calm.PST.SG.M  
 ‘Domen was complaining, I calmed him ...’ (Slv; slWaC)

The Slovak and Slovene data suggest that conditional and pronominal clitics behave differently under agreement drop. It remains an open question, whether conditional clitics pattern similarly in other West Slavic languages. On a different note, to my knowledge, nobody else has taken agreement drop in Slovak or Slovene into consideration before. Note that in contrast to Czech, Goralish, and Silesian, it is more difficult to obtain clear agreement drop examples for Slovak and Slovene from corpora.<sup>371</sup> More research is needed to clarify the actual range of linearisation patterns under agreement drop

<sup>370</sup> CQL query: [word!="sem"]{2} [word="jaz"] [word="mu|ga|jo|te|mi"] [tag="Gg.d.\*"]

<sup>371</sup> For one thing, I encountered several examples of agreement drop in the SNK that are Czech, in fact. As for Slovene, homography of the 1st person subject pronouns *mi* ‘we.M’, *me* ‘we.F/N’ and the pronominal clitics *mi* ‘me.DAT’, *me* ‘me.ACC’ renders the elicitation of agreement drop in plural environments even more difficult.



across West Slavic, and to evaluate the availability of agreement drop in West and South Slavic in more detail.

Boudovskaia (2017) examines auxiliary drop in (Carpatho-)Rusyn, but the pattern differs from West Slavic in that first as well as second persons participate in the construction. Examples (4.250)–(4.252) cite corresponding data for the first person singular (4.250) and plural (4.251). The latter deviates from the linearisation pattern  $PRN_{SBJ} < V$  showing reversed  $V < PRN_{SBJ}$ . The example block in (4.252) provides auxiliary drop for the second person plural. Note that each example block cites a coherent extract of a narrative or dialogue, such that auxiliary drop and pro-drop with an overt auxiliary clitic are realised subsequently by one and the same speaker.<sup>372</sup>

- (4.250) a. a **já** pišlá **ø** u čotýrynác'c'et' rókiw  
 and I.NOM go.PST.SG.F **ø** in fourteen.ACC years.GEN.PL  
 'and I went when I was fourteen'
- b. užé **m** koróvŷ pišlá w kolxóz dôjýty  
 already AUX.1SG cow.ACC.PL go.PST.SG.F in collective\_farm.ACC.SG milk.INF  
 'I already went to the collective farm to milk the cows'
- (Rue, Transcarpathian; Boudovskaia 2017: 14)

- (4.251) a. a pac'átko malóje býlo u nas ta  
 and pig.NOM.SG small.NOM.SG.N be.PST.SG.N by we.GEN and  
 'and we had a young pig'
- b. koxály **ø** mŷ totó pac'á  
 raise.PST.PL **ø** we.NOM that.ACC.SG.N pig.ACC.SG  
 'we fed this pig'
- c. ta ws'ó je... vŷopuščály sme  
 and always it.ACC let\_out.PST.PL AUX.1PL  
 'and we always let it out'
- (Rue, Transcarpathian; Boudovskaia 2017: 34)

- (4.252) Speaker 1: a. býla j gerendá  
 be.PST.SG.F and main\_beam.NOM.SG  
 'there was also the main beam;'

<sup>372</sup> Boudovskaia (2017) proposes an analysis of agreement drop in terms of discourse factors. Roughly speaking, the choice between agreement drop and pro-drop depends upon *newness* vs. *givenness*. Additionally, particular speech genres prefer one construction over the other.

b. ta vŷ užé čúly ø totó?  
 but you.NOM.PL already hear.PST.PL ø that.ACC.SG.N  
 ‘but you have already heard about it?’

(Rue, Transcarpathian; Boudovskaia 2017: 47)

Speaker 2: c. čúla m  
 hear.PST.SG.F AUX.1SG  
 ‘I have.’

Speaker 1: d. a dé s’té býly?  
 and where AUX.2PL be.PST.PL  
 ‘and where have you been?’ (Rue, Transcarpathian; Boudovskaia 2017: 48)

Finally, consider examples (4.253) from colloquial Serbian cited by Sonnenhauser & Meermann (2016). It presents an overt first-person subject pronoun and the simultaneous lack of the auxiliary clitic *sam*. The pattern reminds of agreement drop in West Slavic and Slovene. However, it is not clear to me at this moment if auxiliary deletion in BCMS is a different type of construction, if there are overlaps with agreement drop in West Slavic, or if different patterns need to be distinguished within BCMS.

(4.253) A ja ø kupila prvi put banane za nas [...]  
 and I.NOM ø buy.PST.SG.F first way banana.ACC.PL for us.ACC  
 ‘And for the first time I have bought bananas for us ...’

(Srp, colloquial; Sonnenhauser & Meermann 2016: 99)

To summarise, agreement drop is a phenomenon attested in but not limited to non-standard West Slavic speech, whereby a first-person auxiliary clitic (or mobile inflection marker) is absent, while, at the same time, a corresponding subject pronoun is present. The predominant linearisation puts the subject pronoun before the *I*-participle and, if occurring, the clitic (cluster). This pattern has repercussions for the positioning of pronominal and reflexive clitics, which are more likely to occur in Delayed Placement. This is especially true for subordinate clauses, where Wackernagel cliticisation is excluded due to the precedence of the subject pronoun. Finally, some refinements are pending, as it appears that clitics of different categories do not necessarily behave alike. We have seen that conditional clitics in Slovak and Slovene may precede the entire agreement drop construction. On the other hand, Slovak allows an entire clitic cluster to precede the agreement drop syntagm, while diacsis is attested in Slovene. I leave this for future research, which will have to broaden the database.

#### 4.9.4 Morpho-phonological alterations

Finally, this section briefly presents morpho-phonological alterations within the clitic cluster which I will label as FUSION and SUBSTITUTION respectively. It was mentioned in Section 3.6.4 that BCMS and Czech exhibit cluster-internal processes which formally alternate certain sequences of contiguous clitics. Beginning with Czech, recall that the 2nd person singular auxiliary clitic (*j*)*si* tends to be fused with the reflexive clitics *se* and *si*. It needs to be emphasised that this is a tendency rather than a strict grammatical rule. Example block (3.25) from Section 3.6.4 repeated in (4.254) illustrates that the concatenation of AUX + REFL may perfectly realise discrete clitics. Note that the separate pronunciation of AUX and REFL in a single clitic cluster is a feature especially of spoken language.

- (4.254) a. To **ses** ne-bála? Vážně?  
 TO REFL.ACC.AUX.2SG NEG-fear.PST.SG.F seriously  
 ‘So, you weren’t afraid? Seriously?’
- b. (Myslela jsem,) že už **jsi** se mě nabažil.  
 think.PST.SG.F AUX.1SG that already AUX.2SG REFL me.ACC have\_enough.PST.SG.M  
 ‘(I thought) that you’re already fed up with me.’
- c. Všechno **sis** vymyslel!  
 everything REFL.DAT.AUX.2SG make\_up.PST.SG.M  
 ‘You made up everything!’
- d. Chtěl **jsi** si hrát na hrdinu (a dobře  
 want.PST.SG.M AUX.2SG REFL.DAT play.INF on hero.ACC.SG and good.ADV  
 víš, že jsem tě varovala,) ale ty **jsi**  
 know.PRS.2SG that AUX.1SG you.SG.ACC warn.PST.SG.F but you.NOM.SG AUX.2SG  
**si** říct ne-dal.  
 REFL.DAT tell.INF NEG-give.PST.SG.M  
 ‘You wanted to act the hero (but you [are] well aware that I warned you,) but you didn’t  
 listen to reason.’ (Cz; ČNK syn2020)

Turning to BCMS, it is known that the standard languages display an alteration of the object clitic *je* ‘her.ACC’ in terms of a suppletion or dissimilation to *ju* when adjacent to the auxiliary clitic *je* ‘AUX.3SG’. The potential sequence *je je* overtly becomes *ju je*.<sup>373</sup> I prefer to describe this phenomenon more neutrally as the *substitution* of *je* by *ju*. Kolaković et al. (2022: 21, 94–95, 103–104) inform that

<sup>373</sup> I abstract away from the fact that the substitution may be conditioned by a broader environment including the negated copula *ni-je* ‘NEG-be.PRS.3SG’ and, more generally, verb forms ending in suffix *-(i)je* (Kolaković et al. 2022: 103).

the substitution is not consequently adhered to by native speakers in oral communication. Perusing BCMS dialectological literature, they furthermore find that the allegedly substituted string *ju je* manifests only in those dialects that solely employ the pronominal form *ju* irrespective of the phonological environment (Kolaković et al. 2022: 152). They cite example (4.255a) from the Zetsko-južnosandžački dialect, which is roofed by Serbian and Montenegrin but does not feature the feminine accusative singular form *je*. Therefore, they conclude that the occurrence of the accusative pronominal form *ju* cannot be caused by the adjacent auxiliary clitic *e*. Example (4.255b) from the Croatian web corpus (hrWaC) corroborates that the form *ju* surfaces independently of the phonological environment.<sup>374</sup>

- (4.255) a. [...] kə<sup>a</sup>(d) **ju** **e** vǎdio da idê k ñěmu [...].  
when her.ACC AUX.3SG see.PST.SG.M that go.PRS.3SG towards him.DAT  
‘... when he saw her coming towards him ...’  
(BCMS, Zetsko-južnosandžački; Kolaković et al. 2022: 152)
- b. Oko 6. mjeseca **ju** **je** prihvatio samo odjednom  
around sixth month.GEN.SG her.ACC AUX.3SG seize.PST.SG.M just suddenly  
navečer kad **sam** **mu** **ju** dala, i odonda  
in\_the\_evening when AUX.1SG him.DAT her.ACC give.PST.SG.F and since\_then  
dudaaaaa.  
pacifier.NOM.SG  
‘Around the sixth month, he suddenly took it [= pacifier] once in the evening, when I  
gave it to him, and since then[:] pacifier.’ (Hrv; hrWaC)

#### 4.10 Clitic doubling

This section deals with the so-called CLITIC DOUBLING construction. Usually, clitic doubling is regarded as an exclusive feature of VA-clitic languages like Bulgarian and Macedonian (cf. Franks & King 2000: 250–251). Hereby, an object clitic co-occurs with an orthotonic object pronoun or a referential object NP with the same thematic role and case respectively. The notion of *doubling* refers to the observation that a verbal predicate's syntactic argument is thus realised twice within a clause, which poses a challenge to any syntactic theory that recognise valency. This is the reason why clitics in VA clitic languages like Bulgarian and Macedonian are considered morphological markers rather

<sup>374</sup> CQL query: [word="sam"] [word="mu"] [word="ju"]

than syntactic entities (cf. Franks & Rudin 2005). This analysis seems to be corroborated by the mandatory contiguity of the clitics to the verbal predicate, which renders them more affix-like. It has been observed that clitics do not automatically double overt objects in every transitive or ditransitive construction. Instead, it has been suggested that clitic doubling is dependent on the specificity – plus topicality in case of Bulgarian – of the respective objects and/or their information structural properties (Franks & King 2000: 252–254). In contrast, Werkmann (2003: 222–238) argues that clitic doubling is licensed for non-focal (= [–FOCUS]) constituents, i.e. the doubled part of speech is either Background material itself or a (non-contrastive) Topic. Bulgarian clitic doubling is exemplified in (4.256) (with relevant notions of information structure indicated by subscript) and in (4.257) for Macedonian.<sup>375</sup>

- (4.256) a. [ Az<sub>TOPIC</sub> nego<sub>j</sub> go<sub>j</sub> vidjax ]<sub>BACKGROUND</sub> [ AdvP ošte včera ]<sub>FOCUS</sub>.  
 I.NOM him.ACC him.ACC.CL see.AOR.1SG already yesterday  
 b. [ Nego<sub>TOPIC/j</sub> az go<sub>j</sub> vidjax ]<sub>BACKGROUND</sub> [ AdvP ošte včera ]<sub>FOCUS</sub>.  
 him.ACC I.NOM him.ACC.CL see.AOR.1SG already yesterday  
 ‘I saw him already yesterday’ (Bg; Werkmann 2003: 224)

- (4.257) Marija go<sub>j</sub> poznavu učenik-ot<sub>j</sub> / Vlado<sub>j</sub> / nego<sub>j</sub>.  
 Mary him.ACC.CL know.PRS.3SG pupil-DEF Vlado him.ACC  
 ‘Mary knows the pupil / Vlado / him.’ (Mac; Franks & King 2000: 251)

Interestingly, clitic doubling is not restricted to VA languages, but has been reported for 2P languages, too. According to my knowledge, Marušič & Žaucer (2009, 2010) were the first to explicitly introduce data to scientific discourse that shows clitic doubling in a non-VA or article-less language. They exemplify clitic doubling in the Nova Gorica variety Slovene, cf. (4.258). Note that the doubled dative clitic *ti* in (4.258b) has climbed from the embedded infinitive *znižat* ‘(to) lower’ to the matrix clause, which is headed by the complex predicate *bo treba* ‘will have to’.

<sup>375</sup> The clitic and the orthotonic expression it doubles are co-indexed for the sake of clarity.

- (4.258) a. Jaz    **se**    **ga**<sub>j</sub>            njega<sub>j</sub>    spomnim            še    iz    srednje  
 I.NOM REFL him.ACC.CL him.ACC remember.PRS.1SG still from middle.GEN.SG.F  
 šole.  
 school.GEN.SG  
 ‘I remember him already from high school.’  
 (Slv, Nova Gorica; Marušič & Žaucer 2009: 287)
- b. Zdej **ti**<sub>j</sub>            bo            treba            tebi<sub>j</sub>            znižat            provizijo.  
 now you.DAT.CL be.FUT.3SG necessary you.DAT lower.INF commission  
 ‘Now we’ll have to cut your commission.’  
 (Slv, Nova Gorica; Marušič & Žaucer 2009: 290)

Similarly, Živojinović (2021) as well as Kolaković et al. (2022) bring up clitic doubling data from Serbian dialects. Example block (4.259) is from Živojinović (2021), who gathered her data in fieldwork in the Trgovište region close to the borders with North Macedonia and Bulgaria.<sup>376</sup> The data in (4.260) are cited by Kolaković et al. (2022) from dialectological studies.<sup>377</sup>

- (4.259) a. Mene<sub>j</sub>    **me**<sub>j</sub>            boli            stomak.  
 me.ACC me.ACC.CL hurt.PRS.3SG stomach.NOM.SG  
 ‘I have a stomachache.’
- b. Mene<sub>j</sub>    **me**<sub>j</sub>            je            jat.  
 me.ACC me.ACC.CL be.PRS.3SG anger.NOM.SG  
 ‘I am angry.’
- c. [ Na Marinu ]<sub>j</sub>    **gu**<sub>j</sub>            se            spije.  
 to Marina.DAT her.DAT.CL REFL.ACC sleep.PRS.3SG  
 ‘Marina is sleepy.’
- d. [ Na Marinu ]<sub>j</sub>    **gu**<sub>j</sub>            lkna            čim            ...  
 to Marina.DAT her.DAT.CL feel\_relief.PST.SG.M as\_soon\_as  
 ‘Marina felt relief as soon as ...’

<sup>376</sup> According to Živojinović (2021: 425), the genetic classification of the Torlak/Prizren-Timok dialect as either Serbian or Bulgarian is subject to controversy between “Serbian/Croatian and Bulgarian scholars”. This issue is irrelevant here, but it represents a common phenomenon in Slavic dialectology. Linguistic areas close to or on both sides of state borders are often disputed among philologists of different national affiliation due to nationalist agendas.

<sup>377</sup> Usages of the term *Torlak* vary. The respective variety is referred to as *Prizren-Timok* by Živojinović (2021), but *Zetsko-južnosandžački* by Kolaković et al. (2022). For the sake of transparency, Živojinović’s (2021) Prizren-Timok is synonymous to the Torlak macrodialect in the transitional zone from Serbian to Macedonian and Bulgarian. On the other hand, Kolaković et al. (2022) do not directly refer to Torlac (sic!) when citing examples but use dialect classifications one level below. According to their scheme (cf. Kolaković et al. 2022: 129 Figure 7.1), the Torlak macrodialect consists of the Prizrenski-Južnomoravski and the two smaller Svrliško-Zplanjski and Timočko-Lužnički dialect areas.

- e. Poznavam **ga**<sub>j</sub> [NP toga **čoveka** ]<sub>j</sub>.  
 know.PRS.1SG him.ACC.CL that.ACC.SG.M man.ACC.SG  
 ‘I know that man.’ (Srp, Torlak; Živojinović 2021: 431)

- (4.260) a. Dáj **mu**<sub>j</sub> njěmu<sub>j</sub>.  
 give.IMP him.DAT.CL him.DAT  
 ‘Give him.’ (Srp, Prizrensko-južnomoravski; Kolaković et al. 2022: 167)
- b. on **me**<sub>j</sub> mene<sub>j</sub> videja  
 he me.ACC.CL me.ACC see.PST.SG.M  
 ‘he saw me’ (Srp, Prizrensko-južnomoravski; Kolaković et al. 2022: 168)

Moreover, Kolaković et al. (2022: 167) provide an example of clitic doubling from the Zetsko-južnosandžački dialect, which is mainly distributed over south-eastern Montenegro and south-western Serbia along the borders with Albania and Kosovo, cf. (4.261).

- (4.261) Měne<sub>j</sub> **me**<sub>j</sub> zovê.  
 me.ACC me.ACC.CL call.PRS.3SG  
 ‘He calls me.’ (Srp, Zetsko-južnosandžački; Kolaković et al. 2022: 167)

Observe that the Slovene and Serbian non-standard varieties that exhibit clitic doubling are contact varieties. Nova Gorica Slovene is in contact with Italian. Torlak is a set of contact or transitional varieties from Serbian to Macedonian and Bulgarian. The Zetsko-južnosandžački dialect is in contact with Albanian. All these varieties’ linguistic neighbours are languages, which belong to the Balkan sprachbund and for which clitic doubling is attested (Tomić 2011: 314–315). Therefore, it is plausible to assume that the availability of clitic doubling in these varieties is the result of language contact. It is however important to note that Italian, Bulgarian, Macedonian, and Albanian are VA clitic languages, whereas the Slovene and Serbian non-standard varieties are 2P clitic languages like the standard varieties roofing them. Therefore, it has been pointed out by Marušić & Žaucer (2009: 291–292, 2010: 106–111) and Živojinović (2021: 432–433, 435–437) that the typological predictions proposed by Bošković (2016: 25, with references) are not borne out. For one thing, the proposal claims that only VA clitic languages possess clitic doubling, since they are determiner phrase (DetP) languages.<sup>378</sup> On the other hand, 2P clitic languages are regarded as noun phrase (NP) languages and

<sup>378</sup> Determiner phrase is commonly abbreviated *DP*, which I substitute by *DetP*, because *DP* is exclusively used for *delayed placement* in the present thesis.

deemed to lack clitic doubling accordingly.<sup>379</sup> As we have seen, evidence from different languages contradicts this prediction (but cf. Runić 2014 for an alternative view).

Importantly, clitic doubling may occur in languages that are not in contact with VA languages. Example block (4.260) provides Goralish data. Neither Goralish nor its contact languages Polish and Slovak possess determiners, with the latter being classified as a 2P language. One might object that the alleged clitic doubling could be a plain utterance with a parenthetical syntagma following, in fact. In this view, the putative parenthetical would be a corrective or clarifying expression external to the syntactic domain of the original utterance. The answer to this argument is based on phonological evidence. If an element is parenthetical, then it should be separated by a prosodic break from the syntactic environment. This is not the case in any of the utterances in (4.262). The Spisz dialect corpus provides users with audio recordings of the transcribed data and allows testing such hypotheses and there is no interruption of the speech flow in the respective utterances, which would qualify as a prosodic break. On the contrary, all the syntactic strings provided in (4.262) are parsed as a single prosodic unit (cf. Marušič & Žaucer 2009: 283–284 for Gorica Slovene).

- (4.262) a. <i   nieraz   **goj**       postrzelali       tegoj       bo   bo   ...>  
           and not\_once him.ACC shot.PST.PL.VIR this.ACC.SG.M because because  
           ‘and they shot at him not only once ’cause ’cause ...’
- b. <no   to   trza   **goj**       zaprzeć       tegoj       co   bije  
       well TO necessary him.ACC lock\_up.INF this.ACC.SG.M what beat.PRS.3SG  
       wszystkie   nie?>  
       all.ACC.NVIR NEG  
       ‘well, then one has to lock up that one who is beating all [others], right?’
- c. <no   potem   **goj**       trza       było       pozbierać [NP ten  
       well afterwards him.ACC necessary be.PST.SG.N gather.INF       this.ACC.SG.M.INA  
       len ]>  
       flax.ACC.SG  
       ‘well, afterwards one had to gather the flax’
- d. <[...] ino   **się goj**       wyowijało       dzieckoj   [...]>  
       only REFL him.ACC wrap\_up.PST.SG.N child.ACC.SG  
       ‘... the child was only wrapped up ...’

<sup>379</sup> Roughly speaking, the NP/DetP distinction assumes different internal syntactic structures for NPs and DetPs. DetPs are more complex than NPs, as they possess an additional layer for the realisation of the determiner. Importantly, the DetP-layer belongs to the extended projection of the nominal head N<sup>0</sup>. The NP/DetP distinction is meant to also account for the difference between grammatical determiners proper and superficially similar occurrences of genuine demonstratives in determiner-less languages. A true determiner occupies a position in the DetP-layer, while a demonstrative is an NP/DetP-external element essentially.



- e. <[...] i położył **mu** ją na ziemię [NP tą  
and put.PST.SG.M him.DAT her.ACC on earth.LOC.SG this.ACC.SG.F  
mapę ]>  
map.ACC.SG  
'... and he put the map on the ground for him'
- f. <zaś **mu**<sub>j</sub> gada kumotrze<sub>j</sub> wy tu sobie pod  
again him.DAT tell.PRS.3SG godfather.DAT.SG you.PL here REFL.DAT under  
krzak legnijcie [...]>  
straw.ACC.SG lay\_down.IMP.2PL  
'again, he tells the godfather: lay down here under the straw ...'
- g. <(weźmiesz go) to z nosa **mu**<sub>j</sub> się leje  
take.PFV.PRS.2SG him.ACC TO from nose.GEN.SG him.DAT REFL.ACC pour.PRS.3SG  
biedakowi<sub>j</sub>>  
poor\_thing.DAT.SG  
'([if] you will take him,) then the poor thing starts to have a runny nose' (Gor; KoSpi)

One might hypothesise whether non-standard varieties, which are predominantly oral varieties, are prone to the emergence of clitic doubling. Such doubled occurrences could be grammaticalised from originally parenthetical NPs or postponed NPs that are likely to be corrections or clarifications but then may be reanalysed and further grammaticalised by L1 acquirers.<sup>380</sup> Language contact with lects that already possess the relevant constructions may accelerate such a development.

## 4.11 Sub-sentence-level clitics

### 4.11.1 Introductory remarks

Until now, so-called clause-level or clausal clitics have been considered. Roughly speaking, it is clitics that are elements of the sentential domain governed by the finite verbal predicate. However,

<sup>380</sup> In fact, even German exhibits constructions that resemble instances of pronominal doubling. Importantly, utterances like (i) are encountered in oral communication rather than in written language. Note that the prosodic break between the full grammatical clause and the postponed NP *den Mann*, which, orthographically, is indicated by the comma, is not mandatory. In informal communication the entire string may be parsed into a single prosodic unit (which is why the comma is in parentheses).

(i) Ich habe ihn<sub>j</sub> nicht gesehen<sub>(,)</sub> [NP den Mann ]<sub>j</sub>.  
I.NOM have.PRS.1SG him.ACC NEG see.PTCP the.ACC.SG.M man.ACC.SG  
'The man, I didn't see him.'

(Ger; constructed)

clitics are not only licensed and positioned within clausal or verbal domains, but they can also occur in syntactic domains beneath the sentential or clausal level. Section 4.7.2 on clitic climbing across different categories already discussed clitics that are licensed by (quantified) nouns, (predicative) adjectives, adverbs, and quantificational expressions. The present Section 4.11 will focus on noun phrases for two reasons. First, this configuration has been particularly discussed in the literature.<sup>381</sup> Second, NPs may be regarded as the prototypical case of non-verbal domains below the clausal or sentence level.<sup>382</sup> NP-internal clitics are subcategorised for by a nominal head and, as an approximation, should be consistently placed within the phrasal projection of this head. It is established that the possibility of NP-internal cliticisation varies considerably within Slavic (cf. Franks & King 2000: 273, Kosta & Zimmerling 2014: 443). Polish allows NP-internal clitics without almost any restrictions, whereas languages like BCMS and Czech are considered to generally forbid NP-internal clitics (cf. Franks & King 2000: 160–161, 277; Kosta & Zimmerling 2014: 443–444). Bulgarian and (to a lesser extent) Macedonian are treated as the only notable exceptions to the general ban on NP-internal cliticisation in allowing NP-internal possessive dative clitics (cf. Franks & King 2000: 55–57, 74–75, 273–277; Kosta & Zimmerling 2014: 444–445).

Now, let us briefly look at some aspects of NP-internal clitics. There are basically three possible ways to approach the concept of *NP-internality*, two of which are structural, while the third one is semantic. From the structural point of view, which one may regard as broad, any clitic that is dominated by an NP-node is NP-internal, which is also true if it is only a mediate constituent of NP. Under this view, clitics licensed by and contained in complements of the nominal head are treated as NP-internal clitics, for the NP-node is part of the superstructure above them. We already encountered corresponding environments in Section 4.7.2, whereby the clitics have been extracted. Examples of locally positioned clitics are given in (4.264).<sup>383</sup> In other words, the clitics in (4.264) are NP-internal,

<sup>381</sup> Except for the authors cited in Section 4.7.2, clitics subcategorised for by adjectives are widely neglected in the research on Slavic clitics. Examples (i) and (ii) provide examples for adjective-internal clitics in Czech and Polish respectively. Note that it is attributive adjectives, which are islands for clitic extraction as observed by Junghanns (2002c: 71) on the basis of Czech (cf. Kolaković et al. 2022: 243 for BCMS).

- (i) V ten moment jsem uviděl [NP [AdvP **mi** neznámého ]  
in this.ACC.SG.M.INA moment.ACC.SG AUX.1SG see.PST.SG.M me.DAT foreign.ACC.SG.M.ANI  
muže ] [...] man.ACC.SG  
'At this moment, I saw a man unknown to me ...' (CZ; CNK syn\_v8)
- (ii) Niestety, kiedy po-szedł-e-m [PP po [NP [AdjP potrzebne **mi** ]  
unfortunately when PFV-go.PST.SG.M-M-AUX.1SG for necessary.ACC.SG.N me.DAT  
zaświadczenie płacowe ]], takowego nie było.  
confirmation.ACC.SG salary.ADJ.ACC.SG.N such.GEN.SG.N NEG be.PST.SG.N  
'Unfortunately, when I went for the certificate of employment that I need, such [a thing] did not exist.'  
(Pol; NKJP full, poliarp)

<sup>382</sup> This view incorporates the idealisation that clauses minimally consist of a verb and *n* nominal arguments, with *n* ∈ {0, 1, 2, 3}. I necessarily disregard one-word utterances, utterances heavily truncated by ellipsis, interjections etc.

<sup>383</sup> CQL queries:

- (i) [word="příležitost"] [tag="Vf.\*"] [word="ho"]  
(ii) [word="příležitost"] [word="mu"] [tag="Vf.\*"]

because they are literally within the NP, although only mediately. Imagine a matryoshka, whereby the smallest doll is contained in all bigger ones including the biggest one ultimately. NP-internality is thus expressed by a cascade of subset relations, cf. (4.263), which reads: if  $A$  is a subset of  $B$ , and  $B$  is a subset of  $C$ , then  $A$  is a subset of  $C$ .

(4.263)  $A \subseteq B \subseteq C \rightarrow A \subseteq C$

- (4.264) a. Bude to jedinečná [NP příležitost [VP vidět  
 be.FUT.3SG this.NOM.SG.N unique.NOM.SG.F opportunity.NOM.SG see.INF  
**ho** naživo ]].  
 him.ACC live.ADV  
 ‘This will be a unique opportunity to see him [= the pope] live.’
- b. Vyměnila bych několik let života za  
 exchange.PST.SG.F COND.1SG several.ACC year.GEN.PL life.GEN.SG for  
 [NP příležitost [VP **mu** říct ], (kolik pro mě  
 opportunity.ACC.SG him.DAT tell.INF how\_much.ACC for me.ACC.SG  
 znamenal.)  
 mean.PST.SG.M  
 ‘I would give several years of [my] life for the opportunity to tell him (how much he  
 meant to me.)’ (Cz; ČNK syn\_v8)

Under the narrow structural view, an NP-internal clitic is an immediate constituent of the nominal head, which means that it is positioned within the smaller domain headed by  $N^0$ . This configuration may be regarded as NP-internal cliticisation in a strict sense. It is particularly this configuration which played a prominent role in the analysis of Polish clitics as peculiar in Slavic. We will come back to these instances in Section 4.11.3.

Finally, from the semantic perspective, clitics are NP-internally licensed but may be realised NP-externally. In other words, a clitic is subcategorised for by a nominal head, hence, it is a dependent of the noun. Recall example (4.190) from Section 4.7.2 repeated in (4.265), where the genitive clitic *jich* is licensed by the quantified noun *mísy*. In contrast, (4.266) shows a clitic that escaped the plain NP *hanba* ‘shame’ and climbed into the finite matrix domain.<sup>384</sup> Example (4.267) shows that the dative clitic belongs to the domain of the noun, indeed, as no other element occurs in the respective minimal clause.<sup>385</sup>

<sup>384</sup> CQL query: [word="být"] [word="hanba"]

<sup>385</sup> CQL query: [word="hanba"] [word="mi"]

(4.265) Král                mu<sub>1</sub>        **jich**<sub>3</sub>        přikázal<sub>1</sub>        sníst<sub>2</sub>        [NP dvě  
king.NOM.SG him.DAT them.GEN order.PST.SG.M PFV.eat.INF two.ACC.F  
plné                mísy<sub>3</sub> ].  
full.ACC.PL.F bowl.ACC.PL  
'The king ordered him to eat up two bowls full of them.' (Cz; Rosen 2001: 234)

(4.266) (Ne-věděla                jsem,)        jestli **mi**<sub>3</sub>        má<sub>1</sub>                být<sub>2</sub>        hanba<sub>3</sub>,  
NEG-know.PST.SG.F AUX.1SG if me.DAT have.PRS.3SG be.INF shame.NOM.SG  
(nebo se<sub>2</sub>                mám<sub>1</sub>                smát<sub>2</sub>.)  
or REFL.ACC have.PRS.1SG laugh.INF  
'I didn't know if I should be ashamed, (or [if] I should laugh.)' (Cz; ČNK syn\_v8)

(4.267) (Ne-znal                jsem        ho,)        hanba                **mi**!  
NEG-know.PST.SG.M AUX.1SG him.ACC shame.NOM.DAT me.DAT  
'(I didn't know him,) shame on me!' (Cz; ČNK syn\_v8)

In a way, the structural and the semantic approaches to NP-internality are diametrical. Clitics are within NP, but not necessarily dependents of the nominal head under the structural view, while they may escape NP in the semantic approach, even though they are subcategorised for by the noun. Both points of view are complementary, but not entirely exclusive of one another, since clitics dependent on a noun may remain inside their licensing NP. The three configurations are summed up in (4.268).

- |            |                                                                                    |                                                                                        |
|------------|------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| (4.268) a. | [YP ... [NP ... N <sup>0</sup> ... [ZP ... CL ... ]]]                              | (NP-internal CL, local realisation,<br>mediate dominance by N <sup>0</sup> )           |
| b.         | [YP ... [NP ... N <sup>0</sup> CL ... ([ZP ... ])]                                 | (NP-internal CL, local realisation,<br>immediate dominance by N <sup>0</sup> )         |
| c.         | [YP ... CL <sub>j</sub> ... [NP ... N <sup>0</sup> e <sub>j</sub> ... ([ZP ... ])] | (NP-internally licensed CL, non-local<br>realisation, no dominance by N <sup>0</sup> ) |

One might object that the label *NP-internal* was descriptively inappropriate for the semantic perspective, if the clitic escapes the phrasal domain of its governing head, cf. (4.268c). However, the notion is not problematic in itself, if we accept that the dependence relation between a pronominal clitic and its nominal head is established internally. This is reflected by the common observation that,

in long distance dependencies, syntactic items are semantically interpreted with respect to the domain they are extracted from and not with respect to the material surrounding them.<sup>386</sup>

Nevertheless, it is common to refer to NP-internal clitic placement narrowly in structural terms of type (4.268b). This will be the perspective under which I will mainly review NP-internal clitics in the next two sections, although we shall keep in mind the two other configurations. Section 4.11.2 addresses the dative of possession. A discussion of clitics in deverbal nouns is offered in subsequent Section 4.11.3.

## 4.11.2 NP-internal clitics: Dative of possession

The so-called DATIVE OF POSSESSION is a possessive construction with the possessor bearing dative case. The availability of this construction across Slavic languages varies according to Šarić (2002: 5), but NP-internal possessive clitics are mentioned only for Bulgarian and Macedonian (cf. Franks & King 2000: 273–284 [Ch. 9], esp. 274–277 [Ch. 9.1.1]; Kosta & Zimmerling 2014: 443–445).<sup>387</sup> Bulgarian examples are cited in (4.269). The reflexive clitic *si* acts as possessor of *kuštata* and is co-referential with the subject *Ivan*.<sup>388</sup> Macedonian is exemplified in (4.270).

- (4.269) a. Ivan<sub>j</sub> ja<sub>1/k</sub> prodade<sub>1</sub> [<sub>NP</sub> k[ă]šta-ta<sub>2/k</sub> si<sub>2/j</sub>].  
 Ivan her.ACC sell.AOR.3SG house-DET REFL.DAT  
 ‘Ivan sold his house.’ (Bg; Dimitrova-Vulchanova 1999: 118, n. 39)

- b. [<sub>NP</sub> roditeli-te mu]  
 parent.PL-DET him.DAT  
 ‘his parents’ (Bg; Franks & King 2000: 274)

- (4.270) a. [<sub>NP</sub> žena mi]  
 woman.SG me.DAT  
 ‘my wife’

<sup>386</sup> This observation is implemented, e.g., in terms of traces in Government and Binding theory, copies in the Minimalist Program, non-saturated argument slots in HPSG, and by slashes in GPSG.

<sup>387</sup> In the tradition of Germanist linguistics, the term *pertinence dative* (Ger. *Pertinenzdativ*) is established for the dative of possession and has been recently introduced to the international discourse of theoretical syntax (cf. e.g. McIntyre 2006).

<sup>388</sup> Alternatively, the reflexive clitic can occur outside NP and cluster with the argument clitic *ja*, cf. (i).

(i) Ivan<sub>j</sub> si<sub>2/j</sub> ja<sub>1/k</sub> prodade<sub>1</sub> [<sub>NP</sub> kušta-ta<sub>2/k</sub>].  
 Ivan REFL.DAT her.ACC sell.AOR.3SG house-DET  
 ‘Ivan sold his house.’

(Bg; Dimitrova-Vulchanova 1999: 118, n. 39)

b. [<sub>NP</sub> *sakan-iot*     **si**             v[u]jko ]

beloved-DET REFL.DAT uncle.SG

‘his/her beloved uncle’

(Mac; Franks & King 2000: 275)

From a general viewpoint, the exact structure of this very construction is not as trivial as appears at first sight, as there are intersections with other dative constructions. Browne (1993: 371) notes that “[t]he transition to the dative of experience is gradual”. Short (1993a: 517–518) remarks that “[t]he possessive dative (often close to *dativus (in-)commodi*) is almost obligatory in co-occurrence with the names of body parts [...] but also with intimate possessions” (cf. also Short 1993b: 580).<sup>389</sup> This property is immanent in the Polish examples in (4.271). The dative clitics *mi* and *mu* respectively can be analysed both as datival possessors and *dativus incommodi* (which is also implicated by the two translations).<sup>390</sup>

(4.271) a. *Pies*             **mi**             *zdechł.*

dog.NOM.SG me.DAT die.PST.SG.M

i. ‘My dog died.’

ii. ‘To my misfortune, the/my dog died.’             (Pol; auditory attestation, Knurów, 2009)

b. <*nie wiem,*             *ale za kare*                             *ze-żre*             **mu**

*Nie wiem,*             *ale za karę*                             *ze-żrę*             **mu**

NEG know.PRS.1SG but for punishment.ACC.SG PFV-eat.PRS.1SG him.DAT

*cała*             *czekolade,*             *co*     *sobie*     *kupił*             (*swoja*

*całą*             *czekoladę,*             *co*     *sobie*     *kupił*             (*swoją*

entire.ACC.SG.F chocolate.ACC.SG what REFL.DAT buy.PST.SG.M one’s.POSS.ACC.SG.F

*już*             *zjadła-m).*>

*już*             *zjadła-m).*

already eat.PST.SG.F-1SG

i. ‘I don’t know, but as a punishment, I will eat all of his chocolate that he bought for himself (I’ve already eaten mine).’

ii. ‘... but, to his misfortune, I will eat all of the chocolate, as a punishment ...’

(Pol; S14)

<sup>389</sup> One may hypothesise about a grammaticalisation path from one of the dative constructions to the others, which supposedly could be perfectly accounted for by a listener-oriented approach along the lines of Ohala (1981, 1993, 2012 *inter alia*)

<sup>390</sup> The *dativus commodi* and *dativus incommodi* represent facultative benefactive and malefactive arguments respectively. They verbalise referents to whose fortune or misfortune the event coded by the verb happens.

Conceptualising the dative of possession as an NP-internal construction is a contentious issue. Discussing the features of Standard Average European, König & Haspelmath (1999) and Haspelmath (1999, 2001) argue that the dative of possession is an NP-external syntactic element and refer to it as *dative external possessor*.<sup>391</sup> König & Haspelmath (1999: 119) remark that external possessors are “a separate part of speech”. Haspelmath (1999: 109) states that the (pro-)nominal possessors “are not constituents of an NP, but contrary to the English translation, they occur NP-externally”. Accordingly, the external possessor is “a constituent of the clause.” (Haspelmath 1999: 109) Consider the German paradigm in (4.272). Example (4.272a) provides a dative possessor NP *dem Kind*, which is substituted by the dative pronoun *ihm* in (4.272b). A genuine possessive pronoun, *seine*, is found in (4.272c). Now observe that the possessive dative may perfectly co-occur with the possessive pronoun in (4.272d). Similar patterns are attested in, e.g. Czech and Polish, cf. (4.273) and (4.274) respectively. The possessive pronouns *jeho* and *jego* are facultative and could be omitted in principle.<sup>392</sup>

- (4.272) a. Die Mutter wäscht **dem** **Kind** die  
the.NOM.SG.F mother.NOM.SG wash.PRS.3SG the.DAT.SG.N child.DAT.SG the.ACC.PL  
Haare.  
hair.ACC.PL  
‘The mother is washing the child’s hair.’ [lit.: ‘The mother is washing the hair to the child.’]  
(Ger; Haspelmath 2001: 1498)
- b. Die Mutter wäscht **ihm** die Haare.  
the.NOM.SG.F mother.NOM.SG wash.PRS.3SG him.DAT the.ACC.PL hair.ACC.PL  
‘The mother is washing his hair.’
- c. Die Mutter wäscht **seine** Haare.  
the.NOM.SG.F mother.NOM.SG wash.PRS.3SG its.ACC.PL.N hair.ACC.PL  
‘The mother is washing his hair.’
- d. Die Mutter wäscht { **dem** **Kind** / **ihm** }  
the.NOM.SG.F mother.NOM.SG wash.PRS.3SG the.DAT.SG.N child.DAT.SG him.DAT  
**seine** Haare.  
its.ACC.PL.N hair.ACC.PL  
‘The mother is washing { the child’s / his } hair.’ [lit.: ‘The mother is washing { the child / him } its/his hair.’]  
(Ger; constructed from Haspelmath 2001: 1498)

<sup>391</sup> I am thankful to Stefan Heck for discussing Haspelmath (2001) and the issue of external possessors with me during our jointly taught class on *Language contact and heritage language* at the Friedrich Schiller University Jena in the summer term 2021.

<sup>392</sup> The statement needs to be weakened for the Polish example (4.272) due to the contrast by *zamiast twoich* ‘instead of yours’, but without such a contrastive constituent, the strong statement is valid again.

(4.273) [...] a zabrali **mu** (**jeho**) legendární kavárnu  
 and take\_away.PST.PL.VIR him.DAT his.POSS legendary.ACC.SG café.ACC.SG  
 U bezrukého Frantíka [...]   
 at handless.GEN.SG Frantík.GEN  
 ‘... and they took the/his legendary café “The Handless Frantík Inn” away from him ...’  
 (Cz; ČNK syn\_2020)

(4.274) ... i zabrali **mu** (**jego**) nuty, zamiast  
 and take\_away.PST.PL.VIR him.DAT his.POSS musical\_note.ACC.PL instead  
 twoich.  
 your.POSS.GEN.PL  
 ‘... and they took (the/)his [musical] notes away from him, instead of your ones.’  
 (Pol; NKJP polqiarp balanced)

Example (4.274) indicates that something more is involved than plain possession, however. The relation between the putative possessum *nuty* and the dative clitic *mu* refers to immediacy. Consider the modified utterances in (4.275). Substituting the pronominal clitic *mu* ‘him.DAT’ with *mi* ‘me.DAT’ yields a grammatical and completely senseful utterance (4.275a). Apparently, there are two referentially distinct possessors, which may be captured by the conceptual division common in the legal domain, i.e. between a physical notion of possession proper or control on the one hand and an abstract notion of ownership on the other hand (cf. Stern & Lewinsohn-Zamir 2020: 21). Thus, *mi* refers to the physical controller, while *jego* refers to the intellectual or legal owner of the musical notes.<sup>393</sup> This is corroborated by (4.275b), which can have only the reading (ii), where the notes are necessarily taken physically from the possessor. However, they are not necessarily the possessor’s legal property. Note that neither reading is capable of realising the contrastive reading with *zamiast twoich*. While the impossibility of contrast could be attributed to the clitic status of *mi*, this cannot be the only explanation, because substituting clitic *mi* with orthotonic *mnie* does not improve the utterance’s acceptability.

<sup>393</sup> Obviously, there are more levels of possession and ownership from a linguistic perspective. Kinship terms, e.g. *mother*, *father*, have a different quality than physical control, or intellectual and legal property. I am necessarily simplifying for the sake of presentation and argument.



(4.275) a. ... i zabrali **mi** **jego** nuty, zamiast  
 and take\_away.PST.PL.VIR me.DAT his.POSS musical\_note.ACC.PL instead  
 twoich.

your.POSS.GEN.PL

‘... and they took his [musical] notes away from me, instead of yours.’

b. ... i zabrali **mi** nuty (\*zamiast twoich).

i. \*‘... and they took away my notes (instead of yours).’ [intended]

ii. ‘... and they took away the [musical] notes from me (\*instead of yours).’

c. \*... i MNIE zabrali nuty, zamiast  
 and me.DAT take\_away.PST.PL.VIR musical\_note.ACC.PL instead  
 twoich.

your.POSS.GEN.PL

Intended: ‘... and they took away my notes instead of yours.’

(Pol; constructed from [4.274])

On the other hand, examples like Bulgarian (4.269) and Macedonian (4.270) instantiate physical possession, cf. (4.269a), and abstract ownership, cf. (4.269b) and (4.270). One may conjecture that the dative of possession is further grammaticalised in these languages. Aside from that, the dative clitic’s actual semantics may be subject to contextual resolution with the result of interpretational variation. I shall leave this issue here. Bear in mind that, provided the available interpretation of possession/ownership, we may pursue an NP-internal analysis of the syntagm *pies mi* in (4.269) in analogy to (4.269) and (4.270), cf. (4.276).

(4.276) [<sub>NP</sub> Pies [<sub>CL</sub> **mi** ]] zdechł.  
 dog.NOM.SG me.DAT.POSS die.PST.SG.M

‘My dog died.’

(Pol)

This essentially fits Browne’s (1993: 371) remark that “[s]uch clitics [= possessive dative CL] can also appear within the possessed phrase.” There should be no doubt that Browne refers to an NP-internal analysis for (4.277a). Analogically, the same applies to (4.277b) then. The Bosnian data in (4.278) exemplify the dative of possession in second phrase (4.278a–b) and second word (4.278c) placement. However, the placement of the dative clitic is essentially ambiguous in (4.277b–c) and (4.278a–b). The clitic can be either placed inside NP or it occupies clausal second position external to NP, cf. the alternative bracketing in (4.277c) and (4.278b). The 2W placement in (4.278c) can be regarded as a safe instance of NP-internal clitic placement as long as one does not pursue a split-NP

analysis that were syntactically independent of cliticisation. Example (4.279) provides the last data point for NP-internal clitic placement in BCMS from the Šumadijsko-vojvođanski dialect, which is roofed by standard Serbian.

- (4.277) a. Pozovi Marka i [NP sestru **mu** ].  
 invite.IMP.2SG Marko.ACC and sister.ACC.SG him.DAT  
 ‘Invite Marko and his sister.’ (BCMS; Browne 1993: 371)
- b. [NP Otac **mi** ] je umro prošle godine.  
 father.NOM.SG me.DAT AUX.3SG die.PST.SG.M last.GEN.SG.F year.GEN.SG
- c. [NP Otac ] **mi** je umro prošle godine.  
 ‘My father died last year.’ (BCMS; Browne 1993: 370)
- (4.278) a. [NP Starija sestra **mu** ] pjeva u horu.  
 older.NOM.SG.F sister.NOM.SG him.DAT sing.PRS.3SG in choir.LOC.SG
- b. [NP Starija sestra] **mu** pjeva u horu.
- c. [NP Starija **mu** sestra ] pjeva u horu.  
 ‘His older sister sings in a choir.’ (Bos; Kolaković et al. 2022: 23)
- (4.279) [NP pòkōjni **mi** mûž ]  
 deceased me.DAT husband.NOM.SG  
 ‘my deceased husband’ (Šumadijsko-vojvođanski; Kolaković et al. 2022:159)

Czech and Slovak examples of the dative of possession are presented in (4.280)–(4.282). The Czech example hardly provides clear instances of NP-internal cliticisation, unless one accepts NP-initial dative possessor positioning. Czech has NP-initial possessors in the form of possessive pronouns, e.g. (4.281a), and possessive adjectives, e.g. (4.281b). Post-nominal possessors regularly appear as adnominal genitives, e.g. (4.281c).<sup>394</sup> However, it is far from obvious, if the dative of possession occupies the same structural positions when linearised contiguously to the supposedly modified head noun. Dative possessor clitics enjoy a greater distributional freedom than the possessors in (4.281), which cannot occur outside NP. Still, from an external observer’s perspective, the structure is potentially ambiguous. The same considerations apply to Slovak (4.282). Perhaps we encounter NP-initial clitic dative possessors in (282a), although a clausal 2P structure is more likely. Example

<sup>394</sup> Cf. Křivan (2014) for an analysis of NP-internal possessor placement vs. adnominal genitival possession in Czech by means of information structure.

(4.282b) is ambiguous between NP-internal and -external cliticisation in post-nominal position, compare with (4.282c).

- (4.280) a. U-myla                      **mu**                      vlasy.  
                  PFV-wash.PST.SG.F him.DAT   hair.ACC.PL  
                  ‘She washed his hair.’
- b. Rozpáraly                      se                      **ti**                      kalhoty.  
                  split.PST.PL.NVIR REFL.ACC you.DAT   trousers.NOM.PL  
                  ‘Your trousers have split.’
- (Cz; Short 1993a: 518)

- (4.281) a. [NP [PRN můj ]                      otec ]  
my.NOM.SG.M father.NOM.SG  
‘my father’
- b. [NP [ADJ Karlův ]                      most ]  
Charles.ADJ.NOM.SG.M bridge.NOM.SG  
‘Charles Bridge’
- c. [NP Den                      [NP děti ]]  
day.NOM.SG child.GEN.PL  
‘Children's Day’
- (Cz)

- (4.282) a. Umyl { **jej** / **mu** / **si** } vlysy.  
 washed.PST.SG.M her.DAT him.DAT REFL.DAT hair.ACC.PL  
 ‘He<sub>j</sub> washed { her<sub>k</sub> / his<sub>k</sub> / his<sub>j</sub> } hair.’  
 b. [<sub>NP</sub> Syn **mu**] prerástol cez hlavu.  
 son.NOM him.DAT outgrow.PST.SG.M over head.ACC.SG  
 c. [<sub>NP</sub> Syn ] **mu** prerástol cez hlavu.  
 ‘His son has outgrown him.’ (Slk; Short 1993b: 581)

Finally, the data in (4.283) from Danylenko (2020a) relativises Šarić's (2002: 5) statement about the absence of the dative of possession in Ukrainian and attests possessive dative clitics in southwest Ukrainian dialects.<sup>395</sup>

<sup>395</sup> Bear in mind that I do not take a stance on questions of language status and cite glottonyms according to the source.

(4.283) a. zaslabla                    **my**        žinka  
               get\_sick.PST.F.SG   me.DAT wife.NOM.SG  
               ‘My wife got sick.’

b. ja        i        [NP syn                **my** ]  
               I.NOM and                son.NOM.SG me.DAT  
               ‘me and my son’

(Ukr, non-standard; Danylenko 2020a)

I will not go into further detail here but conclude this section.<sup>396</sup> It is sufficient for the purpose of the thesis, first, that the dative of possession is common across Slavic (though possibly not deployed in every language) and, second, that it can be realised by a pronominal clitic. The position of the clitic is, however, a matter of controversy. While it basically functions like an NP-internal possessive pronoun in Bulgarian, Macedonian, and – at least partly – in BCMS, this is less clear for other Slavic languages, especially Czech, Polish, and Slovak. Here, the clitic could be simply placed adjacent to the nominal head rather than within its maximal projection. However, there is still the possibility of structural ambiguity, which has a parallel in the semantic domain in that the dative-bearing expression can be interpreted either as a possessor *sensu stricto* or as a free dative (of the *commodi/incommodi* type). Note that, referring to Rutkowski (2002), Kosta & Zimmerling (2014: 445) conclude that

“the noun/pronoun asymmetries and the distinction between qualifying and classifying adjectives seem to confirm the hypothesis that both Bg/Mac and Pol allow for cliticization inside the DP domain as a consequence of the greater functional structure available in these particular languages or of the fact that they employ a Verb-adjacent (VA) rather than 2P clitic placement strategy”. (Kosta & Zimmerling 2014: 445)

These and comparable typological hypotheses (e.g. Bošković 2016: 25, with references) hinge on the question of whether possessive dative clitics are located within or outside NP. At least some of the BCMS data points suggest that NP-internal clitics are available in this 2P language or some of its dialects, such that these data contradict common opinion. The Czech and Slovak data are not unequivocal and may, therefore, turn out to be insignificant here. However, we shall see in the next section that there is still unmistakable evidence for NP-internal cliticisation in the Czech-Slovak branch of West Slavic. The deverbal noun is the domain relevant here.

<sup>396</sup> I refer the reader to Šarić (2002) for a comparative analysis of the dative of possession in Slavic within the framework of Cognitive Grammar.

### 4.11.3 NP-internal clitics: Deverbal nouns

There is a notable body of evidence that has been put forward to underline the distinguished typological position of Polish within Slavic. One of the phenomena setting Polish apart from its cognate languages is NP-internal cliticisation (cf. Franks 2009: 731–732; Franks & King 2000: 160–161, 277–278; Kosta & Zimmerling 2014: 443–444). The previous Section 4.11.2 provided data of – partially controversial – NP-internal clitics in connection with the dative of possession. Let us turn to NP-internal placement of argument clitics, now. Polish appears to be unequally liberal in this respect. The examples in (4.284) show that pronominal clitics freely occur within NP in Polish irrespective of the NP’s internal complexity.<sup>397</sup> An NP-internal reflexive clitic is presented in (4.285).

- (4.284)
- a. [NP zobaczenie **go** ]  
seeing him.GEN
- b. [NP twoje ciągle spotykanie **go** ]  
your constant meeting him.GEN (Pol; Franks & King 2000: 160)
- c. [NP moje pomaganie **mu** ]  
my helping him.DAT
- d. [NP nasz oskarżenie **go** o zdradę ]  
our accusing him.GEN of treason
- e. [NP ich zaproszenie **go** na obiad ]  
their inviting him.GEN to lunch
- f. [NP zrozumienie **cię** ]  
understanding you.GEN
- g. [NP przebaczenie **mi** ]  
forgiving me.DAT
- h. [NP nasze (**go**) szybkie (**go**) ratowanie (**go** ]  
our him.GEN speedy him.GEN saving him.GEN
- i. [NP twoje (??**go**) w nocy (?**go**) stale (**go**) oglądanie  
your him.GEN in night him.GEN constantly him.GEN looking.NOM.SG  
(**go**) ]  
him.GEN (Pol; Franks & King 2000: 161)

<sup>397</sup> It is unclear to me why example (4.282i) contains an adverb (*stale*) instead of an attributive adjective (*stale*), which would be expected instead. Perhaps, this is a typo.

j. [NP (\*go) nasze (go) szybkie (go) ratowanie  
 him.GEN our.NOM.SG.N him.GEN speedy.NOM.SG.N him.GEN saving.NOM.SG  
 (go) ]  
 him.GEN  
 ‘our speedy saving of him’ (Pol; Franks & King 2000: 277)

(4.285) Jednak mimo wszystko dzieciak dużo szybciej chłonnie  
 however despite everything.ADV child.NOM.SG much faster absorb.PRS.3SG  
 wszystko, to co jest związane ze  
 everything.ADV this.NOM.SG.N what.NOM be.PRS.3SG connect.PASS.PTCP.NOM.SG.N with  
 zmysłami i dlatego np. [NP uczenie się języka ]  
 sense.INS.PL and therefore e.g. learning.NOM.SG REFL.ACC language.GEN.SG  
 przychodzi im dużo łatwiej niż dorosłym, [...]  
 come.PRS.3SG them.DAT much easier than adult.DAT.PL  
 ‘Either way, a child absorbs everything, which is connected to the sense, far more rapidly  
 and this is why, for instance, learning languages is easier for it than for adults...’  
 (Pol; Janečka 2020: 132)

In complex NPs, several positions may be available for a clitic, which leads Franks & King (2000: 161) to draw a parallel with the distributional possibilities in the clause. They notice “that locating the clitic lower than immediately following the noun is impossible” (Franks & King 2000: 161), as evidenced by the examples in (4.286). This observation mirrors the rightwards positional restrictions in the clause discussed in Section 4.8. However, they cite the judgement by Piotr Bański in (4.287), according to whom the intervening NP *mojej siostrze* sounds “completely natural in the appropriate context” when focused (Franks & King 2000: 277). On the other hand, clitics cannot occur in NP-initial position, as shown by (4.284j) above and (4.286c) below. Franks & King (2000: 277) allude that this might be because of lacking prosodic support for the clitics in this position. Recalling the discussion in Section 4.2.1 on prosodic orientation, this argument is less likely.<sup>398</sup> Presumably, the

<sup>398</sup> It is unclear then, why the clitic cannot precede the possessive pronoun, when a demonstrative is realised additionally, cf. (i)–(ii). It is not clear to me, whether this restriction has implications for the NP/DetP distinction according to which there is a structural and typologically meaningful distinction between determinerless NP languages and determiner-possessing DetP languages (cf. Bošković 2016, with references). While under this premise one might be tempted to assume that the neuter demonstrative pronoun *to* is positioned NP-externally, such that the clitic cannot leave NP to join the demonstrative, it would be necessary to show that presumed determiner-languages allow such dislocation.

(i) Bez sensu to jest [NP to twoje gadanie ].  
 without sense.GEN.SG this.NOM.SG.N be.PRS.3SG this.NOM.SG.N your.NOM.SG.N talking.NOM.SG  
 ‘All your talking is pointless.’ (Pol; NKJP poliqarp balanced)

clitic cannot occur in this position, because it would be outside the NP-domain, which implicates that NP is an island for clitics.<sup>399</sup>

- (4.286) a. [NP moje (**mu**)      pomaganie (**mu**)      we wtorki    (**\*mu**) ]  
my him.DAT helping him.DAT on Tuesdays him.DAT  
b. [NP twoje (?go)      stale      (**go**)      oglądanie (**go**)      w nocy (**\*go**) ]  
your him.GEN constantly him.GEN looking him.GEN in night him.GEN  
(Pol; Franks & King 2000: 161)  
c. [NP (**\*mu**)      moje                  (**mu**)      pomaganie      (**mu**)      we wtorki  
him.DAT my.NOM.SG.N him.DAT helping.NOM.SG him.DAT on Tuesday.ACC.PL  
(**\*mu**) ]  
him.DAT  
'my helping him on Tuesdays' (Pol; Franks & King 2000: 277)
- (4.287) ?? podarowanie [NP mojej                  siostrze ]<sub>FOC</sub> go      na urodziny  
giving my.DAT.SG.F sister.DAT.SG him.GEN on birthday.ACC.PL  
'giving it to my sister for her birthday' (Pol; Franks & King 2000: 161)

In contrast to Polish, languages like Czech and BCMS do not tolerate clitics inside NP. Example (4.288a) is from Janečka 2020, while Kosta & Zimmerling (2014) cite (4.288b). Example (4.289) from Franks & King (2000) shows that BCMS allows NP-internal arguments in the form of orthotonic pronouns, whereas clitics are not tolerated. Thus, it appears that Polish is quite peculiar in comparison to other Slavic languages in the first place.

- (4.288) a. Ne-mám ráda [NP Karlovo čištění (\*ho)].  
 NEG-have.PRS.1SG glad.PRED.SG.F Karl.ADJ.ACC.SG.N cleaning.ACC.SG him.GEN  
 ‘I do not like Karel’s cleaning (of it).’ (Cz; Janečka 2020: 126)

[illegible]

<sup>399</sup> The term *island*, which is commonly said to be coined by Ross (1967), refers to syntactic domains which do not permit dislocation.

- b. [NP Potupa                    jehó / \*hó ]                    zahanbila                    celý  
                                          disgrace.NOM.SG he.GEN                    \*he.GEN.CL                    embarrass.PST.SG.F                    entire.ACC.SG.M  
                                          národ.  
                                          nation.ACC.SG  
                                          ‘His disgrace embarrassed the whole nation.’                    (Cz; Kosta & Zimmerling 2014: 444)

- (4.289) a. [NP Predstavljanje njega Mariji ]                    je                    iznenadilo svakoga.  
                                          introduction                    him.GEN Marija.DAT                    AUX.3SG surprised                    everybody.ACC  
                                          b. \* [NP Predstavljanje **ga** Mariji ] *je* iznenadilo svakoga.  
                                          c. \* [NP Predstavljanje Mariji **ga**] *je* iznenadilo svakoga.  
                                          ‘The introduction of him to Mary surprised everybody.’  
                                          (BCMS; Franks & King 2000: 273)

An interesting data set is discussed by Franks & King (2000: 55–57, 274), who notice that the possessive dative clitic can bear an argument reading in certain Bulgarian constructions: “Parasitic on the possessive function one also finds dative clitics that correspond to arguments of deverbal nouns [...].” (Franks & King 2000: 56) The dative clitics in (4.290) are ambiguous between the possessor and an argument of the nominal head, as indicated by the translation lines (cf. Franks & King 2000: 56).

- (4.290) a. [NP ran[n]o-to                    **mu**                    pre-pluvane                    na kanal-a ]  
                                          early.SG.N-DEF                    him.DAT                    through-swimming.SG                    of                    channel-OBJ  
                                          i. ‘his swimming of the channel early’  
                                          ii. ‘the channel’s early swimming’  
                                          b. [NP pisane-to                    **mu** ]  
                                          writing.SG-DEF                    him.DAT  
                                          i. ‘his writing’  
                                          ii. ‘the writing of it’  
                                          c. [NP peene-to                    **i** ]  
                                          singing.SG-DEF                    her.DAT  
                                          i. ‘her singing’  
                                          ii. ‘the singing of it’



d. [<sub>NP</sub> neprek[ǎ]snato-to **im** broene ]  
continuous.N-DEF them.DAT counting.SG

i. ‘their incessant counting’

ii. ‘the incessant counting of them’

(Bg; Franks & King 2000: 56)

There is an important structural parallel between the Bulgarian argument reading and NP-internal cliticisation in Polish. The Polish examples are deverbal nouns derived by the suffix *-nie* (or *-cie*) as noted by Franks & King (2000: 160, 277–278; cf. also Kupść 2000b: 40).<sup>400</sup> Similarly, Franks & King (2000: 56) note that NP-internal dative clitics with argument status occur in deverbal noun phrases in Bulgarian. In both languages, the deverbal nouns are derived by the cognate suffixes *-nie* and *-ne*. However, Franks & King (2000) do not relate these facts, while Kosta & Zimmerling (2014) missed them.

Heck (2022: 135–141, with references) discusses the deverbal noun in Polish. He notes that its derivation is very productive (cf. Heck 2022: 136–138) and states that the Polish verbal noun is “particularly verbal” (Heck 2022: 138; cf. also Franks & King 2000: 160 [incl. n. 66]). According to Bartnicka et al. (2004: 460–461), the verb is “transformed” into a noun while keeping its original semantics including aspect and reflexivity. Heck (2022: 155) describes the derivation as an extension of the original verb’s functional capabilities, because, when nominalised, it can do things that would be impossible otherwise. For instance, the verbal noun may occur after prepositions, which is impossible for a verb proper (cf. Heck 2022: 155–156). Thus, it appears that the verbal noun in Polish remains verbal in nature, by and large. It keeps the underlying verb’s argument structure, reflexivity, grammatical voice, and aspect (cf. Heck 2022: 139). On the other hand, it takes over some new features from the nominal category, e.g. certain distributional properties or genitive case assignment.

Accordingly, it is reasonable to assume some importance of the particular type of nominal derivation for the question of NP-internal clitic positioning in Polish as well as Bulgarian. This is also one reason why Kosta & Zimmerling’s (2014: 444) Czech example (4.288b) is unsuitable as a token for comparison with Polish, because *potupa* is not derived.<sup>401</sup> Note furthermore that Polish does not allow clitics within other nominal formations of the same lexical root irrespective of whether they

<sup>400</sup> Examples of deverbal nominalisations with *-cie* in Polish: *picie* ‘drinking’ < *pić* (to) ‘drink’, *mycie* ‘washing’ < *myć* ‘(to) wash’.

<sup>401</sup> Furthermore, the example confuses the personal pronoun *jeho* ‘him.ACC’/‘him/it.GEN’ with the homonymic possessive pronoun *jeho* ‘his’. Synchronically, the personal pronoun *jeho* is regarded as a suppletive inflectional form of *on* ‘he.NOM.SG’ and *ono* ‘it.NOM.SG’ respectively, whereas the possessive *jeho* is an invariant possessive particle. It is only the personal pronoun, which alternates with the clitic form *ho*.

also are deverbal derivations or simplex nouns.<sup>402</sup> Therefore, while clitics are grammatical in (4.284)–(4.286) above, they are impossible within nouns in (4.291) below.<sup>403</sup>

- (4.291) a. \*<sub>[NP moja **mu** pomoc (we wtorki) ]</sub>  
           b. \*<sub>[NP moja pomoc **mu** (we wtorki) ]</sub>  
               Intended: ‘my helping him’/‘my help for him’  
           c. \*<sub>[NP nasza **go** skarga o zdradę ]</sub>  
           d. \*<sub>[NP nasza skarga **go** o zdradę ]</sub>  
               Intended: ‘our accusation him of treason’  
           e. \*<sub>[NP rozum **cię** ]</sub>  
               Intended: ‘understanding you’  
           f. \*<sub>[NP ratunek **go** ]</sub>  
               Intended: ‘rescuing him’/‘his rescue’  
           g. \*<sub>[NP (stały) ogląd **go** ]</sub>  
           h. \*<sub>[NP (stały) **go** ogląd ]</sub>  
               Intended: ‘(constantly) looking at him’  
           i. \*<sub>[NP podarunek **go** mojej siostrze na urodziny ]</sub>  
           j. \*<sub>[NP podarunek **go** dla mojej siostry na urodziny ]</sub>  
               Intended: ‘giving it to my sister as a birthday present’

(Pol; constructed from Franks & King 2000: 161)

The data imply that the distinction between deverbal derivations by *-nie/-cie* and other nominal types is relevant. Franks & King (2000: 277–278) take this difference only very briefly into consideration and conclude that there are “many poorly understood factors contributing to the felicity of clitic pronouns inside NPs.” (Franks & King 2000: 278) However, Franks & King (2000: 277–278) cite data and native speaker judgements, which show some variability. According to their informants, the

<sup>402</sup> The noun *ogłqd* ‘overview’ is standardly described as a deverbal nominalisation, which is correct only from a diachronic perspective, because there is no corresponding productive nominalisation process in contemporary Polish.

<sup>403</sup> Note that (4.291a–b) is awkward even without the presence of pronominal clitic *go* ‘him’, cf. (i). However, utterances like in (ii) or are grammatical in principle (although they possibly hardly ever occur in speech). The latter could be realised with a pronominal clitic again, cf. (iii). Note that in this particular case *mu* is not subcategorised by the proper noun *pomocy* but by the deverbal noun *udzielanie*.

(i) <sup>?</sup> <sub>[NP moja pomoc we wtorki ]</sub>  
       ‘my help on tuesdays’  
 (ii) a. <sub>[S <sub>[NP pomoc we wtorki ] jest wskazana ]</sub></sub>  
       ‘help on Tuesdays is advisable’  
       b. <sub>[NP moje udzielanie pomocy we wtorki ]</sub>  
       ‘my providing of help on Tuesdays’  
 (iii) <sub>[NP moje (**mu**) udzielanie (**mu**) <sub>[NP pomocy ] we wtorki ]</sub>  
       ‘my providing him with help on Tuesdays’.</sub>

(Pol; constructed)

occurrence of NP-internal clitics with NPs derived by “less productive nominalization suffixes” is possible in spoken Polish, cf. (4.292). The data in (4.293) exhibit some fluctuations of native speaker judgements, such that the non-derived nouns are rejected by some informants, while still accepted by others (though assessed as being awkward). The small amount of data points is unrepresentative and obviously controversial.

(4.292) a. *modernizacja* **go**  
 modernisation it/him.GEN  
 ‘its modernisation’ (Pol; Franks & King 2000: 277)

b. *utrata* **was**  
 loss you.GEN  
 ‘your loss’ (Pol; Franks & King 2000: 278)

(4.293) a. (\*) *zwrot* **go**  
 return.NOM.SG him/it.GEN  
 ‘his/its return’

b. *Zwrot depozytu niczego nie zmienił.*  
 return.NOM.SG deposit.GEN.SG nothing.GEN NEG change.PST.SG.M  
 ‘The return of the deposit changed nothing.’

c. \**Zwrot go niczego nie zmienił.*  
 return.NOM.SG him.GEN nothing.GEN NEG change.PST.SG.M

d. *Zwrócenie go niczego nie zmieniło.*  
 returning.NOM.SG him.GEN.SG nothing.GEN NEG change.PST.SG.M  
 ‘Returning it changed nothing.’

e. *Zwrot go jest niezbędnym warunkiem*  
 return.NOM.SG him.GEN be.PRS.3SG necessary.INS.SG condition.INS.SG  
*dalszego uczestnictwa w konkursie.*  
 further.GEN.SG participation.GEN.SG in competition.LOC.SG  
 ‘His return is a necessary condition for further participation in the competition.’

(Pol; Franks & King 2000: 278)

Corroboration for the relevance of the derivational origin or path of the nominal head comes from additional data, which also show that Polish is less special in allowing NP-internal clitics than previously thought. A morpheme for deverbal nominalisation cognate with Polish *-nie* (*-cie*) and Bulgarian *-ne* is shared by all Slavic languages including Czech *-ní* (*-ti*) and Slovak *-nie* (*-tie*).

Although NP-internal pronominal clitics are considered impossible in Czech, cf. (4.288a) above, the data in (4.294) prove the existence of NP-internal reflexive clitics subcategorised for by a deverbal noun. Such evidence was put forth, for example, by Junghanns (2002c), Petkevič (2009), and Janečka (2020) for contemporary Czech, and Kosek (2009: 128) for baroque Czech, but it seems to have gone unnoticed by a wider academic audience. Example (4.295), which I encountered during my corpus research, confirms that NP-internal reflexive clitics occur in spoken Czech. Here, the reflexive clitic realises a true argument. The variety of deverbal nouns that occur with a reflexive clitic suggests that NP-internal cliticisation may be productive in Czech (and, perhaps, Slovak), even though it appears tied to the reflexive clitic *se* (*sa*).<sup>404</sup>

- (4.294) a. Kluky už dávno přestalo bavit [NP posmívání se  
 boy.ACC.PL already long\_ago stop.PST.SG.N enjoy.INF laughing.NOM.SG REFL.ACC  
 Fiškusovi ] [...]   
 Fiškus.DAT  
 ‘It has been a long time ago that laughing at Fiškus ceased to be amusing for the boys ...’  
 (Cz; Junghanns 2002c: 72).
- b. Na druhou stranu šance našich juniorů na  
 on second.ACC.SG.F side.ACC.SG chance.ACC.PL our.GEN.PL junior.GEN.PL on  
 [NP prosazení se ] jsme pokládali za velké[.]  
 winning\_out.ACC REFL AUX.1PL lay.PST.SG.VIR too high.ACC.PL.F  
 ‘On the other hand, we gave too high chance to our juniors to win out.’  
 (Cz; Petkevič 2009: 189)
- c. [NP Sdružování se ] je totiž logickým  
 grouping.NOM.SG REFL.ACC be.PRS.3SG actually logical.INS.SG.M  
 důsledkem...  
 consequence.INS.SG  
 ‘Uniting is a logical consequence, in fact ...’ (Cz; Petkevič 2009: 190)
- d. [...] postrádá některé zbytečnosti typu  
 lack.PRS.3SG some.ACC.PL.F useless\_thing.ACC.PL type.GEN.SG  
 [NP ptaní se ] na heslo  
 asking.GEN.SG REFL.ACC on password.ACC.SG  
 ‘[the version] lacks some useless things of the type [like] asking for a password’  
 (Cz; Janečka 2020: 126)

<sup>404</sup> It is unclear to me whether the dative reflexive clitic *si* can also appear inside deverbal nouns.

e. Uvědomte si, že všeobecně rozvíjející pohybová  
 realise.IMP.2PL REFL.DAT that generally progressive.NOM.SG movement.ADJ.NOM.SG.F  
 příprava se zaměřením na [NP učení se  
 training.NOM.SG with focus.INS.SG on learning.ACC.SG REFL.ACC  
 dovednostem ] má jiné prováděcí podmínky než  
 skill.INS.SG have.PRS.3SG other implementing.ACC.SG requirement.ACC.PL than  
 vrcholový sport [...] top.ADJ.NOM.SG.M sport.NOM.SG  
 ‘Realise that progressive movement training with a focus on learning skills generally has  
 different requirements for implementation than high-performance sport.’  
 (Cz; Janečka 2020: 128)

(4.295) se mně líbilo jak je dneska evropskej  
 REFL.ACC me.DAT like.PST.SG.N how be.PRS.2SG today European.NOM.SG  
 den [NP koupání se v řekách ]  
 day.NOM.SG bathing.GEN.SG REFL.ACC in river.LOC.PL  
 ‘Me, I liked that today is the European day of bathing in rivers.’ (Cz; ČNK oral\_v1)

Janečka (2020) surveys the (non-)occurrence of the accusative reflexive clitic *se/sa/się* with deverbal nouns in Czech, Slovak, and Polish searching the corresponding national corpora. Particularly, he investigates the deverbal nouns Cz. *učení se*, Slk. *učenie sa*, and Pol. *uczenie się* ‘learning’ and their non-reflexive counterparts Cz. *učení*, Slk. *učenie*, and Pol. *uczenie* ‘teaching’. In each language, deverbal nouns are capable of realising a reflexive clitic dependent. Czech examples are cited in (4.294b–d) above and a Polish one in (4.285) at the beginning of this section. A Slovak example is cited in (4.296) below.

(4.296) Pretože som [NP učenie sa jazykov ] považovala  
 because AUX.1SG learning.ACC.SG REFL language.GEN.PL consider.PST.SG.F  
 za veľmi nudné, učil sa so mnou.  
 as very boring.NOM.SG.N learn.PST.SG.M REFL.ACC with me.INS.SG  
 ‘He was learning with me, because I considered learning languages very boring.’  
 (Slk; Janečka 2020: 130)

In sum, NP-internal cliticisation has not been well understood, such that the claims concerning the occurrence of NP-internal clitics in Slavic turn out to be too simplistic. The limitation of the

phenomenon to Polish and Bulgarian/Macedonian cannot be held up, if one accepts clitic datives of possession to be instances of NP-internal cliticisation (provided the appropriate structural configuration). The dative of possession is inherent in most Slavic languages including BCMS, Czech, and Slovak. NP-internal possessive dative clitics are attested for BCMS – at least to some extent. Another important fact in disentangling the intricate matter of NP-internal clitics is the relevance of word formation. The hypothesis seems to be reasonable that the high degree of productivity of deverbal nominalisations by the suffix *-nie* rather than any property of the clitic system itself is the main factor for the free occurrence of short pronouns within the Polish NP. Various Slavic languages allow clitics inside NPs derived by the cognate suffix, i.e. Bulgarian (*-ne*), Czech (*-ní*), and Slovak (*-nie*). These deverbal nouns inherit the argument structure (which can be modified however) of the respective underlying verbs. On the other hand, clitics are not licensed with other noun types in Polish and examples of such NP-internal clitics are controversial, at least. Therefore, it is plausible to assume that differences in NP-internal clitic placement are fine-grained and gradual across Slavic, rather than categorical. Polish exhibits the greatest degree of freedom for clitics to appear in the domain of deverbal nouns, but it is not the only one of its kind. It is a quest for future research to examine the grammar of verbal nouns in more detail and in a comparative manner across Slavic.

## 4.12 Interim conclusion

Chapter 4 provided a review of pertinent issues in Slavic clitic research like positioning, clustering, and clitic ordering. Simultaneously, some lesser studied issues were given more space, e.g. agreement drop or NP-internal placement. The discussion has shown that linguistic variation with respect to various phenomena has been largely missed. Clitics are generally ambiclitic in Slavic and regarding them as entirely stressless is too simple, as they may be assigned stress or – as in case of the Bosnian stress shift patterns – seem to possess prosodic structure themselves (Section 4.2). Little news was told with respect to the preferred second position placement pattern. Wackernagel placement can be defined in different terms referring to a position after a syntactic head, a constituent, or in a split constituent/after a non-constituent (Section 4.3). On the other hand, evidence was given for deviations from clitic second position placement across Slavic showing a wide range of placement patterns from first to fifth position (Section 4.4). Importantly, there are no restrictions for the element(s) intervening between subordinator and clitic (cluster) in Delayed Placement. Categories as diverse as nouns, adjectives, adverbs, and verbs may precede the clitic (cluster) within root and subordinate clauses equally. The further discussion implies that clustering is a preferred linearisation option for clitics, but it is not mandatory or absolute (Section 4.5). Similarly, clitics allow for word order variation in

the cluster, which presumably is more likely to occur in complex clauses than in simple ones, but not limited to the former (Section 4.6). The construction of clitic climbing shares similar properties across Slavic being licensed in identical syntactic environments, which beside cascaded verbs involve the crossing of diverse categories (Section 4.7). Clitic climbing is an essentially optional realisation pattern from a syntactic perspective. We may take the rightwards positional restriction for clitic positioning for granted (at least for most languages), according to which the predicate poses a boundary for Delayed Placement (Section 4.8). Furthermore, clitics in Slavic are subject to diverse co-occurrence restrictions involving different categories of clitics (Section 4.9). Here, haplology of the reflexive clitic and agreement drop of the auxiliary clitic are particularly relevant. Clitic doubling has been argued not to be restricted to VA clitic (or determiner/DP) languages, although the construction's degree of grammaticalisation differs between languages (Section 4.10). Finally, clitic occurrences below the clause level are attested in several Slavic languages, but there is cross-linguistic variation as to what kind of clitic may occur within which domain. In this context, NP-internal cliticisation with the dative of possession and with deverbal nouns was looked at (Section 4.11).

After all, let us reconsider the typologically peculiar classification of Polish within Slavic. Basically, it is grounded on three criteria. First, Polish clitics are freely distributed within the clause, such that they are neither 2P nor VA (cf. Franks & King 2000: 158). Second, Polish clitics do not cluster and can be positioned independently of one another within the clause (cf. Franks & King 2000: 151). Third, Polish clitics do not obey a rigid order with respect to each other (cf. Franks & King 2000: 157–158). These findings have been taken as deciding evidence for assessing Polish as a language without special clitics. In addition, NP-internal cliticisation has been treated as a distinguishing property, too. The data discussed in the previous chapters do not support the view that cliticisation in Polish radically differs from other Slavic languages. On the contrary, we observe basically the same phenomena and constructions. This is not to deny variation across languages, but there is sufficient convergence to recalibrate the typology of clitics in Slavic. Notably, individual peculiarities can be found in Slavic special clitic languages, too. For instance, only BCMS allows name splitting, while Slovene allows isolated clitics (cf. Franks & King 2000: 43–44).

In the face of the present discussion, it should not come as a surprise that there is no overall agreement on certain aspects of the typology of Polish clitics. Admittedly, most authors consider Polish not to be a 2P language (e.g. Bošković 2015; Dziwirek 1998: 89; Franks 2010; Franks & King 2000; Grochowski 2019; Kupść 1999; Rappaport 1988; Witkoś 1998). In contrast, others regard Polish as a 2P language essentially (cf. Birnbaum & Molas 2012: 159; Dimitrova-Vulchanova 1999: 85; Rothstein 1993: 725; Veselovská 1995: §4.8, n. 23). Urbańczyk (1976: 51, 62) reports in his compendium of Polish dialects that auxiliary and pronominal clitics occupy the second position in

the clause in dialects, while they show the tendency to be verb-adjacent in standard Polish. Spencer (1991: 390) treats Polish as a special clitic language. Aguado & Dogil (1989: 108) state that Polish pronominal clitics build clitic clusters (which they call *clitic chains*). Recall from Section 4.6 that Lenertová (2004: 153) finds ACC < DAT orders in Czech. At the same time, Franks & King (2000: 339) cite comparable Polish data and deduce the phrasal status of Polish clitics from that. Furthermore, bear in mind that Polish short pronouns and reflexives have inflectional paradigms, which was put forward as a criterion for special-clitic-hood in Slavic by Franks & King (2000: 187, n. 2) and Franks (2010: 5, 2016: 114–115), as discussed in Section 3.9.

Now recall from Chapter 3 that, more generally, clitics are a controversial linguistic concept. Neither is there a commonly accepted definition of clitic-hood (except an intuitive and vague approach to the notion), nor are typologies consistent and unequivocal. Most tests proposed to identify clitics by distinguishing them from orthotonic words and affixes are not applicable (in Slavic) or turn out to be inconclusive. Consequently, there is neither compelling empirical nor sound theoretical-conceptual motivation for the typological distinction of Polish. From this it follows that research into Slavic clitics needs a more reliable database and must strive for empirical adequacy. In Part II, I shall reflect on some methodological shortcomings and argue that generalisations should be driven by a rich body of linguistic data, which explicitly include empirical data accumulated by corpus research and/or psycholinguistic experimentation.



## **PART II: METHODOLOGY**

## **5 Overview of Part II**

The second part is the shortest one of the thesis and methodological in nature. Two aspects will be addressed in this part of the dissertation. On the one hand, there are methodological considerations of the relation between theoretical and empirical linguistics. In this context, the thesis attempts to contribute to a lively methodological debate in linguistics. On the other hand, I shall present the study's actual methodology and explain and justify it. Chapter 6 presents major methodological considerations dealing with the tension between introspection and empirical data, prescriptivism and descriptivism, and the role of colloquial, non-standard, or spoken language data. Chapter 7 describes the individual sources for the data analysis in Part III. Chapter 8 illustrates the actual methodology used in the corpus analyses including data retrieval, data preparation, and annotation.

## 6 Methodological considerations

### 6.1 Theory and empirical evidence: A conflict?

Let us begin this section with a short prelude of two quotes from Broekhuis (2010) and Stefanowitsch (2020) representing a generative theorist's and a corpus linguist's stance respectively (emphases in italics are both mine):

“the arguments against the intuitionist approach to synchronic syntax I have seen so far strike me as being merely *dismissive* in nature” (Broekhuis 2010: 182)

“much of the formalist literature has been, and continues to be, at best *dismissive* of corpus data, at worst openly hostile.” (Stefanowitsch 2020: 2)

The methodological issues arising from the tensions between both factions, (formal) theory and (empirical) data, are the topic of this section. I cannot give a thorough discussion of this complex matter, which is worth a monograph in its own right. I will only touch upon selected points relevant for the purposes of the present study.<sup>405</sup>

The largest portions of literature on clitics in Slavic cited in the previous chapters rely on introspection. By *introspection* I refer to a widely held methodology in theoretical linguistics (Broekhuis's [2010] “intuitionist approach”), which involves the following components:

1. A set of examples *E* of an object language *L<sub>O</sub>* is constructed by individual native speaker linguists (and/or their peers).
2. The set *E* is assessed by the same individuals ( $\pm n$  peers) with respect to grammaticality (and/or acceptability) (= so-called *informal judgements*).
3. Steps 1 and 2 are executed with either explicit or implicit recourse to the native speaker's linguistic intuition.
4. The set *E* is utilised as evidence for scientific statements about *L<sub>O</sub>* (and possibly other natural languages), which usually involves the aim to corroborate, falsify, or defend a particular analysis.

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<sup>405</sup> I refer the reader to Kolaković et al. (2022: 51–67 [Ch. 3], with references) for some relevant discussion.

This procedure arose with the dawn of Generative Grammar from Noam Chomsky's mentalistic and rationalistic approach to the study of language and was subject to criticism quite early (cf. e.g. Labov 1975). The resulting methodological debate has continued to date but gained new dynamics with the spread of psycholinguistic experimental methodology in grammar research and the rise of large, annotated corpora. Advocates on both sides – rationalists and empiricists – argue for the advantages of one approach, while challenging the other. Müller (2007) summarises and evaluates the situation as follows:

“Bis in die 50er Jahre des letzten Jahrhunderts wurde in der Sprachwissenschaft mit Belegen gearbeitet, die zum großen Teil aus den Werken anerkannter Schriftsteller stammen. Ende der 50er Jahre änderte sich das in einem Teilbereich der Linguistik und zwar so radikal, dass ausschließlich introspektiv gewonnene Beispiele verwendet wurden. In den Anfangsjahren der generativen Linguistik war das unproblematisch, da Sätze wie *Das Buch schenkt Peter seinem Bruder zu Weihnachten* behandelt wurden, deren Akzeptabilität außer Frage steht. In den 70er und 80er Jahren des letzten Jahrhunderts erreichte die Theorie jedoch eine beachtliche Komplexität, und die jeweiligen Analysen wurden immer noch durch introspektiv konstruierte Beispiele gestützt bzw. auf Grund introspektiv konstruierter Beispiele verworfen. Dabei entstand eine unbefriedigende Situation, da mitunter der Einfluss eines einzelnen Autors über Kernbereiche einer syntaktischen Theorie entschied, ohne dass die Akzeptabilitätsurteile des Autors bzw. dessen Behauptungen in Bezug auf nicht vorkommende Muster je überprüft worden wären. Argumente gegen eine Theorie wurden mitunter mit der Begründung ‘man teile die Beurteilung nicht’ einfach beiseite gewischt [...]” (Müller 2007: 71)

‘Until the 50s of the last century [= until the 1950s; I. K.], linguistics worked with attestations, which come from the works of renowned writers. At the end of the 1950s, this changed in a subfield of linguistics so radically that introspectively gathered examples were used exclusively. In the early years of generative linguistics, this was unproblematic, because sentences as *Das Buch schenkt Peter seinem Bruder zu Weihnachten* [‘This book, Peter gives to his brother as a gift for Christmas.’] were dealt with, whose acceptability was beyond dispute. In the 70s and 80s of the last century, the theory acquired a remarkable [degree of] complexity and the respective analyses still were supported by introspectively constructed examples or rejected due to introspectively constructed examples. Thereby, an unsatisfactory situation arose, because, once in a while, the influence of a single author was deciding for core areas of a syntactic theory without ever reassessing the author's acceptability judgements or his claims about non-occurring patterns. Arguments against a theory were swept aside with the rationale that “one does not share the judgement” [...]’ [translation by I. K.]

Müller (2007) discusses three exemplary phenomena of German, which, based on introspectively constructed data, have been argued to be ungrammatical: the placement of phrasal verbs' particles in

the prefield, extraposition disobeying restrictions of subadjacency, and the occurrence of multiple constituents in the prefield (cf. also Müller 2003 for the latter). He shows that it is possible to obtain corpus attestations that contradict the theoretical claims and, thus, pose problems for analyses built upon constructed evidence. He concludes that “introspection does not suffice for supporting theories” (Müller 2007: 70) and “that, at times, we do not have access to our grammatical competence[s] and that we should be suspicious of ourselves therefore.” (Müller 2007: 72)<sup>406</sup> These and other points of critique have been put forward against introspective data by various scholars over the years. Some points of criticism are summarised in (6.1) (without any claim to completeness).

- (6.1)
- a. Introspective data are potentially biased towards a particular theoretical claim (cf. Jurkiewicz-Rohrbacher et al. 2017b: 49; Kolaković et al. 2022: 5; Polinsky & Kluender 2007: 275–276)
  - b. Introspective data can be incorrect (with theories being built upon false positives or negatives) (cf. Jurkiewicz-Rohrbacher et al. 2017b: 49).
  - c. Introspective judgements may be contradictory across scholars (cf. Kolaković et al. 2022: 53; Müller 2007: 71)
  - d. Introspective examples provide an insufficient database, as it is small and not representative in various ways (focusing on a limited number of constructions, being incomplete, subjective, and not replicable) (cf. Diesing 2010: 104–105; Jurkiewicz-Rohrbacher et al. 2017a: 180–181; Kolaković et al. 2022: *passim*; Polinsky & Kluender 2007: 275–276; Stefanowitsch 2020: 2).
  - e. Introspective data ignore intra-linguistic (sociolinguistic) variation and thus provides a static (either-or) picture of language (cf. Hentschel 2013: 93).

It appears that examples for the arguments in (6.1) can be found in Slavic clitic research, thus, supporting the critiques. Point (6.1a) manifests in Rezac’s (2005) account of clitic climbing in Czech, which is proposed to be dependent upon a binary restructuring environment with the embedded infinitival domain being a bare VP. Dotlačil (2007) and Skoumalová (2005) disagree with (some of) Rezac’s introspective judgements, which seem to be very controversial.<sup>407</sup> I. Kulik (2023b) shows that the predictions of the approach do not withstand a corpus-linguistic examination. An instantiation of case (6.1b) was presented in Section 4.4.2 citing Hana (2007), who shows that certain restrictions on Delayed Placement put forth by Veselovská (1995) are incorrect, e.g. the ban on verbs intervening

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<sup>406</sup> Orig.: “dass Introspektion als Stütze von Theorien nicht ausreicht” (Müller 2007: 70) and “dass uns unsere grammatischen Fähigkeiten mitunter nicht zugänglich sind und dass wir uns deshalb selbst misstrauen sollten.” (Müller 2007: 72)

<sup>407</sup> Additionally, Skoumalová (2005) is highly sceptical towards Rezac’s (2005) language proficiency.

between subordinator and clitics or the limitation of Delayed Placement to the subordinate clauses introduced by *že* ‘that’.<sup>408</sup> Case (6.1c), an instance of which is also reported in the quote from Müller (2007: 71) above, is exemplified by the different assessments of clitic climbing by Stjepanović (1998, 1999, 2004) and Aljović (2004).<sup>409</sup> Stjepanović (2001) accepts the possibility of split clitics and, in fact, (re-)introduces the data to scientific discourse, whereas Aljović (2004: 192, n. 3) disagrees on this issue.<sup>410</sup> Furthermore, Kolaković et al. (2022: 5) cite a comparable instance, whereby Stjepanović (2004) tells that clitic climbing from “subjunctive” *da*-clauses is possible, while Čavar & Wilder (1994) argue that climbing is strictly banned. Presumably, critiques of introspective data collections regard (6.1d) to be a general and recurring issue. For one thing, Veselovská (1995: §9.5) argues that CC is obligatory in Czech. Based on judgements by the same scholar, Franks & King (2000: 241–242) state that clitic climbing in Czech is limited to infinitival complements of modal verbs, where it is obligatory, while it is ungrammatical in object control constructions.<sup>411</sup> However, Section 4.7.1 provided that clitics may climb from any embedded infinitive in West Slavic as long as it is not headed by a subordinator. This includes the possibility of clitic climbing in subject and object control constructions as well as in the accusative with infinitive.<sup>412</sup> Adam (2022: 60, 63) adopts Veselovská’s (1995: §9.5, ex. [57a], [58c]) judgement that CC is ungrammatical in control environments for her optimality-theoretic analysis of clitic placement in Czech, while missing the counterevidence cited by Hana (2007: 128–130) and Golden (2008: 312, 315). With respect to point (6.1e), recall from Section 4.4.2 that clitic placement has been regarded as almost completely free of positional variation in Czech with Delayed Placement being restricted to Clitic Third in subordinate clauses (headed by *že*). However, data cited from Hana (2007) and Junghanns (2021) attest clitic placement in third, fourth, and fifth position in root and subordinate clauses. Furthermore, consider clitic first position placement from Section 4.4.1, which has been reported to be a feature of spoken Czech – with strikingly similar patterns occurring in other Slavic special clitic languages, too, as we saw. Accordingly, the degree of positional variation in Czech is significantly higher than usually acknowledged by introspective judgements or prescriptive-normative grammars.

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<sup>408</sup> The discussion of Cardinaletti & Starke’s (1999a) model of pronominal deficiency in Section 3.6 localises the tripartite approach in category (6.1b), too. Despite a wide range of languages, which have been cited, the proposal also falls into category (6.1d), as not all relevant data points have been covered.

<sup>409</sup> Kolaković et al.’s (2022) empirical study on BCMS clitics corroborates Stjepanović’s position, even though it must be acknowledged that clitic climbing out of *da*-clauses has low frequency.

<sup>410</sup> As to the reintroduction, recall the data provided by Bidwell (1968), which are cited in Section 4.5. With regard to disagreement on grammaticality, Aljović (2004: 192, n. 3) tells: “Stjepanović (2001) presents some SC examples contradicting the ban on splitting clitics. Since I do not share the same intuitions for her SC examples and judge them completely unacceptable, I leave this complication aside pending better understanding of this phenomenon.”

<sup>411</sup> Franks & King (2000: 241–242) acknowledge that the realisation of a matrix objects affects the felicity of CC under control and refer to the Head Movement Constraint.

<sup>412</sup> By this strong generalisation I am abstracting away from the constraints on CC introduced by overt matrix objects.

Another example that I will present in slightly more detail comes from Borsley & Rivero's (1994: 385–387) treatment of the Polish person and conditional markers, which also touches upon the issue of clitic pronouns. They compare the positional restrictions of Polish pronominal clitics with Bulgarian, in order to support their hypothesis that Polish participle + auxiliary (= person marker) sequences are the result of incorporation of the participle ( $V^0$ ) into the auxiliary ( $Aux^0$ ). Incorporation yields a complex head ( $X^0 = V^0 + Aux^0$ ), which cannot be interrupted by intervening elements. In contrast, other Slavic languages are thought to derive the respective sequence via Long Head Movement (LHM) with the consequence of participle and auxiliary being separable. Though being distributed quite freely, Polish pronominal clitics cannot intervene between participle and auxiliary (6.2), but this is possible in Bulgarian (6.3). This line of reasoning resembles other attempts to classify Polish as peculiar within Slavic. In the following, I will not attempt to refute Borsley & Rivero's (1994) analysis but simply show that the structural distinction made between Polish vs. Slavic (represented by Bulgarian) cannot be maintained by reference to the contrast between (6.2) and (6.3).

- (6.2) a. Widział-e=ś                      go.  
           see.PST.SG.M-M=2SG him.ACC  
       b. \*Widział go=ś.  
           saw        him.ACC=AUX.2SG  
           ‘You saw him.’ (Pol; Borsley & Rivero 1994: 386)

- (6.3) a. Toj go            e            viždal.  
           he him.ACC AUX.3SG see.PTCP.SG.M  
       b. Viždal            go            e.  
           see.PTCP.SG.M him.ACC AUX.3SG  
           ‘He saw/has seen him.’ (Bg; Borsley & Rivero 1994: 385)

First, remember that Bulgarian and Macedonian are the only Slavic languages possessing verb-adjacent clitics. Moreover, these languages exhibit typological properties typical for the Balkan sprachbund (cf. Fiedler 2012; Tomić 2011). Therefore, one could argue that structural differences between Bulgarian and Polish come as no surprise. Second, Bulgarian is famous for its strict word order in the clitic cluster such that  $AUX < DAT < ACC$ . However, the auxiliary's 3SG present form *e* exhibits the common South Slavic idiosyncrasy of being positioned at the right edge of the clitic cluster (cf. Franks & King 2000: 61).<sup>413</sup> This feature is completely absent in West Slavic, especially

<sup>413</sup> Macedonian is an exception within South Slavic, because its perfect tense is built with a reduced series of 1st and 2nd person auxiliary clitics like in West Slavic (cf. Franks & King 2000: 69; Pitsch 2023b: 320–321). The special

as 3SG/PL past tense auxiliaries were lost (except for standard Sorbian, cf. Franks & King 2000: 166–167; Scholze 2007: 188; and partly Kashubian, cf. Makùrôt 2016: 46–48; Pitsch 2023b: 318–319).<sup>414</sup> Third, Bulgarian is prone to the Tobler-Mussafia law, whereby the verb appears in sentence-initial position in lack of any other constituents to host the clitics.<sup>415</sup> Although Bulgarian clitics can procliticise and appear initially in their Intonation Phrase like in Polish or Macedonian, they cannot be positioned at the beginning of an utterance (cf. Franks 2016: 104). Macedonian does not exhibit the Tobler-Mussafia law (cf. Franks & King 2000: 82). Fourth, from the two previously mentioned facts it follows that linear contiguity of the participle and the 3SG auxiliary *e* is excluded in the presence of a pronominal clitic under the effect of the Tobler-Mussafia law. Fifth, it is thus a matter of data choice, whether one finds Bulgarian constructions corresponding to Polish (6.2). The Bulgarian example (6.4) is structurally equivalent to (6.2). Sixth, the comparison of Czech examples parallel to (6.2) exhibits the same kind of distributional restrictions recognised in Polish: The intervention of a pronominal clitic between a participle and the auxiliary is impossible and renders the utterance ungrammatical, cf. (6.5). Consequently, Borsley & Rivero's (1994) arguments of structural differences between Polish and Bulgarian being responsible for different distributions of pronominal clitics in both languages are vacuous.

(6.4) a. Dal **si** **mu** **gi**.  
gave AUX.2SG him.DAT them.ACC  
‘You gave them to him.’ (Bg; Franks & King 2000: 61)

b. \*Dal **mu** **gi** **si**.  
gave him.DAT them.ACC AUX.2SG (Bg; constructed from Franks & King 2000: 61)

(6.5) a. Dál **jsi** **mi** **ho**.  
give.PST.SG.M AUX.2SG me.DAT him.ACC

b. \*Dál **mi** **ho** **jsi**.  
give.PST.SG.M me.DAT him.ACC AUX.2SG  
‘You gave it to him.’ (Cz; constructed)

These and other results of the data review and literature survey in Section 4 provide that there is a lack of empirical coverage in Slavic clitic research giving rise to incorrect claims and predictions. It

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placement pattern is, however, visible with the plain copula form, whereby both 3SG and 3PL close off the clitic cluster (cf. Franks & King 2000: 81).

<sup>414</sup> According to Dotlačil (2004: 78, n. 62), the deaccented 3SG copula is positioned at the right margin of the clitic cluster in Czech, which is reminiscent of the idiosyncratic positioning of the auxiliary clitic *je/e* in South Slavic.

<sup>415</sup> The Tobler-Mussafia law is named after the two Romance linguists Adolf Tobler (1875) and Adolfo Mussafia (1886), who discovered the phenomenon for Old French and Old Italian respectively.



is a plausible assumption that the root cause of these discrepancies is the impressionistic databases, which are the result of the rationalistic approach to linguistic data and the postulates derived therefrom. Among other things, Chomsky (1957: 15–17) became famous for his rejection of the significance or importance of corpus study or statistical notions for grammar research:

“Despite the undeniable interest and importance of semantic and statistical studies of language, they appear to have no direct relevance to the problem of determining or characterizing the set of grammatical utterances.” (Chomsky 1957: 17)

Karlsson (2008) surveys early generative publications – predominantly (but not exclusively) authored by Chomsky – from 1953 to 1970. Based on quotations, he shows that Chomsky himself referred to notions of corpora and corpus data presupposing American Structuralist methodological standards in the early days of Generative Grammar. Simultaneously, Karlsson (2008) observes that the methodological reference to attested data occurred merely in script, while Chomsky and other early Generativists did not comply with Structuralist methodology in practice and almost exclusively utilised introspectively constructed example sentences. It is only in Chomsky’s (1965) monograph *Aspects of the Theory of Syntax*, where the primacy of introspection over attested data is explicitly articulated (cf. Karlsson 2008: 27–28) and the apparent contradiction between methodological references to corpora and the sole reliance on introspective constructions starts to cease.

Several arguments have been put forward in favour of introspective judgements and against corpus data. It is generally known that corpora do not contain negative evidence.<sup>416</sup> While a native speaker can tell apart well-formed utterances from unwell-formed ones, typically marked by an asterisk (\*), there are no asterisks in electronic corpora. Furthermore, even the biggest and best-balanced electronic corpus might miss linguistic structures that are well-formed, which is especially true for rare constructions (cf. Gilquin & Gries 2009: 9; Schütze 2016: 2; but cf. Juola 2018; Jurkiewicz-Rohrbacher et al. 2017b). Since Generative Grammar is occupied with linguistic competence, performance data like those encountered in a corpus, are not of primary interest. Chomsky (1965: 4) abstracts away from performance, which does not represent competence directly, because “[a] record of natural speech will show numerous false starts, deviations from rules, changes of plan in mid-course, and so on.” Admittedly, these hardly represent parts of the native speaker’s acquired mental grammar.

However, although speech undeniably shows different kinds of disfluencies, it does not solely consist of speech errors or stuttering. On the contrary, one should regard oral communication between

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<sup>416</sup> The argument of negative evidence is a subset of the so-called argument of the poverty of the stimulus introduced by Chomsky (1980). It is connected to the more general problem of learnability coined *Plato’s problem* by Chomsky (1986). Cf. Pullum & Scholz (2002) for discussion.

people in everyday situations as being rather well-behaved. Interestingly, in a given discourse situation, speakers appear to be able to predict an interlocutor's speech (cf. Huettig & Mani 2016, Mitterer et al. 2018, and references therein for discussion on prediction in language). For instance, if a speaker hesitates (e.g. because s/he cannot remember a word or collocation), then their interlocutor may continue the speaker's stream of speech like in the empirically attested (6.6). Speaker A is talking to Speaker B, who intervenes in Speaker A's flow of speech and continues Speaker A's utterance appropriately. That Speaker B correctly anticipates speaker A's utterance is evidenced by the continuation of Speaker A's flow of speech. The verbal predicate *vydělal* occurs in Speaker B's intervention as well as Speaker A's continuation.<sup>417</sup> One could hypothesise that people's ability to anticipate, what another person is going to say, reflects linguistic competence.

- (6.6) Speaker A: [...] takže bych úplně ty peníze  
 such\_that COND.1SG completely this.ACC.PL.NVIR money.ACC.PL  
 co mně dal bych teoreticky +  
 what me.ACC give.PST.SG.M COND.1SG theoretically  
 '... such that I completely would the money that he gave to me, I theoretically  
 would ...'  
 Speaker B: mu **vydělal**,  
 him.DAT earn.PST.SG.M  
 '... earn [for] him ...'  
 Speaker A: + **vydělal** a vdešel bych v klidu [...]  
 earn.PST.SG.M and go\_away.PST.SG.M COND.1SG in peace.LOC.SG  
 '... earn and I would go in peace ...' (Cz; ČNK oral\_v1)

Despite repeated criticism of introspective data construction, the approach has been defended, too. Phillips (2009) provides an interesting contribution to the methodological discussion. His point of departure is the question of whether linguistics faces a crisis in face of the enduring utilisation of introspective data for theorising.<sup>418</sup> From the perspective of the present section, one may translate the issue under discussion to the question of whether linguistic theory and empirical data are in conflict. Phillips (2009) remarks that first one would have to demonstrate that introspective data has led to erroneous generalisations and false conclusions about natural language structures.<sup>419</sup>

<sup>417</sup> The plus sign (+) is a conventionalised diacritic of ČNK's Oral subcorpus marking of related margins of an interrupted flow of speech.

<sup>418</sup> According to Herdan's (1967) perception, a crisis of general linguistics existed at the latest in the second half of the 1960s.

<sup>419</sup> The three criteria utilised by Phillips (2009: 51) are only one diagnostic tool of many. Its perspective is theoretical in essence. Another possibility to operationalise a diagnosis – and perhaps not the only alternative – may take a

“it would need to be the case that (i) Intuitive judgments have led to generalizations that are widely accepted yet bogus. (ii) Misleading judgments form the basis of important theoretical claims or debates. (iii) Carefully controlled judgment studies would solve these problems.” (Phillips 2009: 51)

In fact, he is sceptical about the correctness of the statements (i)–(iii). Phillips (2009) has an essentially positive attitude towards empirical (experimental) approaches, still he defends introspective assessments of linguistic constructions, because, as he argues, no false generalisations emerged from that procedure. In fact, according to Phillips (2009: 53), “carefully constructed tests of well-known grammatical generalisations overwhelmingly corroborate the results of ‘armchair linguistics’.” This statement is corroborated by Sprouse & Almeida’s (2012) and Sprouse et al.’s (2013) studies. Sprouse & Almeida (2012) conduct a judgement experiment, whereby all 469 example sentences found in Adger’s (2003) textbook on syntax were assessed by naïve respondents. A comparable experimental design was used by Sprouse et al. (2013), who tested a random sample of 296 data points collected from *Linguistic Inquiry* issued between 2001 and 2010. The formal acceptability judgements were further statistically tested with various inferential procedures. Most interestingly, the scholars found that the informal and formal judgements converge to 95–98% (ignoring the respective standard deviations here). Consequently, informal acceptability judgements seem to provide a considerably sound linguistic database, which is reliable to a degree comparable to formal data collection methods. However, good scientific conduct requires not to make any ad hoc conclusions. First, the experimental results are restricted to English. Sprouse & Almeida’s (2012) and Sprouse et al.’s (2013) findings cannot be simply transferred to other languages, say Slavic ones or German. The applicability to other languages needs to be shown by further testing. Provided that Adger’s (2003) textbook is an introductory coursebook for students, the question is whether the data presented therein are complex or subtle enough to suit a methodological controversy. However, it is difficult to extend these doubts to the data sample in Sprouse et al. (2013). An issue not addressed in the experimental studies is contradictory introspective judgements. Karlsson (2008: 28–29) cites introspective data points from early generative literature, whose (un-)grammaticality judgements are controversial. For one thing, he shows that structures that have been claimed to be ungrammatical, are empirically attested and retrievable from internet queries. Karlsson (2008: 30) concludes that “the method of introspective construction of questionable sentences often had detrimental consequences,

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sociological perspective and factor in those linguists who originally were occupied with Generative Grammar but abandoned this tradition. For instance, Webelhuth (2011) advocates an analysis of German relative clauses in terms of Construction Grammar, while arguing that a Minimalist analysis is inferior. Wolfgang Sternefeld informs on his personal website that he stopped doing syntax in favour of semantics, while referring to a pessimistic book review co-authored with Frank Richter, i.e. Sternefeld & Richter (2012) (orig.: “wobei im letzten Jahrzehnt die Syntax deutlich (und von ihr selbstverschuldet [...]) verschwand zugunsten von sehr abstrakten semantischen Überlegungen [...].”; <https://www.sternefeld.info/cv>, accessed 16 Feb 2024). Hubert Haider, once a generative syntactician, has become highly critical and polemic against Generative Grammar (cf. Haider 2018a, 2018b).

especially when full grammaticality was claimed for very strange made-up sentences [...] which then served as data for strong theoretical claims.”

Tajsner’s (2017) contribution to the debate is a “defense of linguists’ intuitions”, as the title of his article reveals, and carefully confronts a selection of arguments by critiques and advocates of introspection. Tajsner (2017) responds to four main points of criticism, summarised in (6.7).

- (6.7)
- a. Native speaker linguists’ judgements are unreliable.
  - b. Native speaker linguists construct biased data serving their agenda.
  - c. Introspectively constructed examples do not represent real linguistic data.
  - d. Introspective judgements operate on a binary scale distinguishing grammatical and ungrammatical utterances, but experimental responses exhibit gradience and even native speaker linguists provide intermediate judgements.

To cite only a few counterarguments, Tajsner (2017: 596) points out that there are not “many questionable examples in linguistic papers” contra (6.7a). He remarks that conference audiences and/or peer reviewers would counterbalance doubtful judgements.<sup>420</sup> As to (6.7b), a main thread of argument is that scientific discovery is made by a top-down process from theory to data (cf. Tajsner 2017: 598–599).<sup>421</sup> Tajsner (2017: 599–603) faces the critique in (6.7c) by reference to the natural sciences comparing introspective data construction with experimentation, more precisely thought experiments, which do not rely on already existing data, but invent phenomena for further study. Finally, Tajsner (2017: 603) qualifies the argument in (6.7d) by reminding that Generative Grammar’s objective is to reveal the categorical distinctions of the competence grammar. He furthermore argues that levelling out noise by calculating mean values for experimental responses may defeat the opportunity to unveil the sources for variation across respondents (cf. Tajsner 2017: 604).

It is helpful to take the natural sciences, to which reference is made, into consideration here. Tajsner draws on Chomsky’s repeatedly uttered argument to treat linguistics as a natural science and quotes the two following passages, in which Chomsky (2011) [cited after Tajsner 2017] comments on the recurring critique of introspection.

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<sup>420</sup> This argument is tenable in principle, but it is weak for the case of conference audiences. Only reviewers have a reasonable amount of time at their disposal to verify a contribution’s database (abstracting away from the fact that they cannot invest all their time into a review). The situation is completely different for conferences, where time slots for talks and the following questions and discussion are highly restricted.

<sup>421</sup> This argument is incorrect. Scientific progress is also achieved bottom-up from the data to the theory. For one thing, the so-called gold foil experiments conducted by the physicists Ernest Rutherford, Hans Geiger, and Ernest Marsden in 1909 gave rise to the so-called Rutherford model of the atom, which laid the ground for our contemporary understanding of the atom’s structure as consisting of a small but massive nucleus of positive charge and a sphere of negatively charged electrons constituting the majority of the atom’s actual size (cf. Demtröder 2016: 64–65). (Later elaboration by Niels Bohr, who introduced orbits for the electrons, did not change this basic structural description of the atom, which was concluded from the experimental results.)

“linguistic research resorts to idealization and abstraction, relying on invented examples [...] not keeping to unanalyzed data but rather creating evidence by design. In other words, linguistic research is like the sciences generally. The sciences typically rely on experiments, highly idealized and abstract, and theory-internal – and even on thought experiments, including classic discoveries.” (Chomsky 2011: 266 [cited after Tajsner 2017: 595])

“Perhaps physics should be revised the same way. No balls rolling down frictionless planes and other such abstractions and idealizations that have virtually defined the subject for centuries: rather, extensive statistical analysis of videotapes of leaves blowing in the wind and other natural events, which will surely give more successful predictions of what will happen outside the window than what the physics department can provide.” (Chomsky 2011: 266 [cited after Tajsner 2017: 595])

Although one cannot object Tajsner’s (2017: 638) assertion that “generative grammarians have a right to ask the type of research questions they wish”, the recurring reference to physics is misplaced. The way reference is made to the natural sciences is not how contemporary physics works.<sup>422</sup> Physics is not confined to a rationalistic endeavour by means of theory and calculus, but a fundamentally empirical discipline that utilises experiment, observation, measurement, and computer simulation to test hypotheses and refine theories and models.<sup>423</sup> What is more, Galileo Galilei, to whom Chomsky (2002, 2012) himself refers, is famous not only for his theoretical work, but also or especially for his experiments.<sup>424</sup> This is basic knowledge for any student of physics, as the quote from Demtröder’s (2021) textbook on experimental physics shows:

“Die Physik im heutigen Sinn begann [...] mit Galilei (1564–1642), [...] der zum ersten Male gezielte Experimente durchführte und damit an Stelle der vom Beobachter unbeeinflussbaren Naturerscheinungen einen kontrollierbaren und beliebig oft unter definierten Bedingungen wiederholbaren Vorgang untersuchte [...]” (Demtröder 2021: 3)

‘Physics in its current sense began [...] with Galilei (1564–1642), [...] who purposefully conducted experiments for the first time, thereby, examining a controllable and arbitrarily often repeatable process under definable conditions rather than natural phenomena impossible to be influenced by an observer.’ [translation by I. Kulik]

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<sup>422</sup> A comparable argument has been already put forward as early as Herdan (1967) and, most recently by Piantadosi (2024), who submits a comprehensive critique from the perspective of Large Language Models.

<sup>423</sup> Stefanowitsch (2020: 4) independently made a similar argument, saying that “physicists searching for gravitational waves do not reject the idea of observational data”. The fundamentally empirical agenda of physics is exactly why instruments like CERN’s Large Hadron Collider (LHC, a particle accelerator) or the James Webb Space Telescope (JWST) were developed. In the same spirit, Piantadosi (2024: 386) independently refers to the LHC and the JWST.

<sup>424</sup> Behme (2014) argues that the “Galilean style” of scientific practice, which is advocated for by Chomsky (2002), does not have anything in common with scientific methodology.

In a similar fashion, chemistry and biology, too, know and incorporate theory and experiment (or other sorts of empirical data) into their overall scientific framework.<sup>425</sup> Most importantly, experimentation in the natural sciences is not limited to thought experiments, which are hardly the default procedure in these disciplines. In addition, physicists are concerned with the mundane realisation of their abstract theories and notions counter Chomsky's negative assessment. For instance, they study how solid materials, gases, or electromagnetic waves behave under different conditions.<sup>426</sup> If an alignment of linguistics with the natural sciences were treated seriously, then one would first have to acknowledge the symbiosis of the theoretical and the empirical domain. Furthermore, if one took the proposal to adopt standards of physics as a science at face value, then replicability of experimental set-ups and results is a necessary condition for a research paradigm attempting to follow the natural sciences. The view of linguistics as a discipline of theory and calculus, which Chomsky seems to advocate, does not conceptualise the discipline in line with physics or any other natural science, but rather with mathematics or philosophy.<sup>427,428</sup> On a related note, Sternefeld & Richter (2012: 265–266) remark that abstraction and idealisation in linguistics cannot be paralleled with abstraction and idealisation in the natural sciences, because both are of a fundamentally different kind. The natural sciences operate on quantitative dimensions. This is not the case in linguistics, which is a qualitative research paradigm.

Furthermore, Tajsner's (2017: 602) claim that "[c]orpus analysis does not promote cross-linguistic studies" is incorrect. On the contrary, empirical methods like corpus linguistics are adequate for comparative studies. Corpus attestations allow retrieving data points for one or more languages that a researcher neither has native intuitions about nor has a native speaker at hand. This does not imply that the researcher can do without any practical knowledge of the language(s) in question. Otherwise, a corpus linguist needs to engage colleagues in a collaborative research environment. But this basically holds for rationalists, too, who consult one or more colleagues' linguistic intuition when constructing examples for multiple languages. Tajsner (2017: 602) points out that "[f]or a cross-linguistic analysis to be successful the confrontation must be very precise, so that it is evident that

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<sup>425</sup> A different question is why physics has been chosen as a model for linguistics and not biology. As a species, *Homo sapiens* belongs to the domain of biology. Kinsella (2009) argues that the strong(est) minimalist thesis underlying the Minimalist Program – that "[l]anguage is an optimal solution to legibility conditions" (Chomsky 2000: 96) as a result of human evolution – is not compatible with evolutionary biology. I owe this reference to Bodo Winter (p.c.).

<sup>426</sup> This is an area, where statistics come into play and where one encounters another misconception about the natural sciences, especially physics. Chomsky's marginalisation of statistical notions is not adequate towards the attempt to call on physics as a methodological reference model. He fails to recognise statistical physics or statistical mechanics, which deals with complex systems containing plenty of microscopic parts, e.g. "the atoms of a gas or liquid, the phonons of a solid state, or the photons in a plasma." (Fließbach 2018: 1; orig.: "Atome eines Gases oder einer Flüssigkeit, die Phononen eines Festkörpers oder die Photonen in einem Plasma.")

<sup>427</sup> I cannot do justice to the discussion of the formal, especially mathematical foundations of Generative Grammar, but one might wonder whether the assessment provided in the main text are a coincidence or have their grounding in the history of Generative Grammar. I refer the reader to Seuren (2009) and Pullum (2011) for relevant discussion.

<sup>428</sup> Note that in German-speaking countries, mathematics is (also) conceptualised as a *Geisteswissenschaft*, which essentially corresponds to the humanities.

equivalent syntactic phenomena from distinct languages are being juxtaposed”. However, it appears that the notion of precision is conflated with literal translation, when it is required that confronted data points were identical in “all relevant parameters; syntactic, lexical, and semantic” (Tajsner 2017: 603). Tajsner (2017) illustrates this argument with examples (6.8)–(6.10), whose similarity goes far beyond what is syntactically, semantically, and lexically relevant.

(6.8) His<sub>i</sub> latest movie really disappointed Kusturica<sub>i</sub>. (Eng; Tajsner 2017: 602)

(6.9) \*Njegov<sub>i</sub> najnovij film je zaista razočarao  
his.NOM.SG.M newest.NOM.SG.M movie.NOM.SG AUX.3SG really disappoint.PST.SG.M  
Kusturicu<sub>i</sub>.  
Kusturica.ACC.SG (BCMS; Tajsner 2017: 602)

(6.10) %Jego<sub>i</sub> najnowszy film rozczarował Kusturicę<sub>i</sub>.  
his newest.NOM.SG.M movie.NOM.SG disappoint.PST.SG.M Kusturica.ACC.SG  
(Pol; Tajsner 2017: 602)

Obviously, it is unlikely to obtain data with such a high degree of – especially lexical – similarity across languages from corpora. In actual fact, this is not mandatory, and it is not a problem contra what seems to be implied by Tajsner’s (2017: 602–603) argument.<sup>429</sup> For any linguistic – say syntactic – phenomenon it suffices to be only as precise as necessary to approach the issue under discussion. The syntactic problem addressed in (6.8)–(6.10) concerns the availability of binding between the subject-internal possessive pronouns (*his/njegov/jego*) and the direct object (*Kusturica/Kusturicu/Kusturice*). When scrutinising the syntactic conditions on binding, it is irrelevant, whether the examples involve translation equivalents or not. It is the structure that is relevant, and similar structures can be represented by arbitrarily many other utterances of diverse lexical make-up. It is irrelevant, whether the source of disappointment was a *movie*, a *book*, or a *wine* (including their respective grammatical gender), or whether the direct object is realised by a proper name or an appellative NP. Note also that the presence (6.8, 6.9) or absence of an intensifying adverb (6.10) does not matter, in fact. One may also call into question, whether the semantics of the verbs *disappoint/razočarati/rozczarować* is of any importance in this particular case, such that the lexical semantics of the verbal predicates do not seem to be meaningful in any way. As a means of illustration, consider Kolaković et al.’s (2022) examples (4.135) from Section 4.7.1 repeated in (6.11)

<sup>429</sup> If it were not Tajsner’s (2017) intention to make such a strong claim, then he did not present his point clearly enough to avoid ambiguities or misunderstandings.

showing clitic climbing vs. local placement clitic in Croatian. The only parameters relevant for direct comparison are matrix verb, embedded infinitive, reflexive clitic, and the linearisation of these three elements. Therefore, juxtaposition of (6.11a) and (6.11b) fully suffices for the scientific needs of precision in order to investigate the issue of clitic climbing in Croatian. There is no necessity to render the two confronted example sentences more similar and there would be no benefit, if they were. Recall also examples (4.166)–(4.169) from Section 4.7.1 showing clitic climbing across negation in Czech, Slovak, Croatian, and Slovene, which were structurally similar only in the relevant respects. From this it follows, it is perfectly possible to conduct cross-linguistic research on the basis of attested data. Sticking to the minimal requirements posed by the aforementioned parameters, the Czech data points in (6.12) can be obtained empirically and confronted with Croatian (6.11).<sup>430</sup> Note that the examples in (6.12) mirror Croatian (6.11) in the relevant portions only, which is perfectly good. Consequently, there are no principal obstacles for cross-linguistic comparison when applying empirical methods, especially corpus research.<sup>431</sup>

- (6.11) a. A da **se**<sub>2</sub> morate<sub>1</sub> odlučiti<sub>2</sub>?  
 and if REFL.ACC must.PRS.2PL decide.INF  
 ‘And if you have to decide?’ (Hrv; Kolaković et al. 2022: 312)
- b. Pa ako već morate<sub>1</sub> odlučiti<sub>2</sub> **se**<sub>2</sub> za jaslice [...].  
 well if already must.PRS.2PL decide.INF REFL.ACC for nursery  
 ‘Well if you [already] have to opt for nursery ...’ (Hrv; Kolaković et al. 2022: 311)
- (6.12) a. Samy **se**<sub>2</sub> musíte<sub>1</sub> rozhodnout<sub>2</sub>, (co mu řeknete.)  
 alone REFL.ACC must.PRS.2PL decide.PFV.INF what him.DAT say.PFV.PRS.2PL  
 ‘You will have to decide on your own, (what you will tell him.)’ (Cz; ČNK syn2020)
- b. (Hra vezme vás zpět do začátku  
 game.NOM.SG take.PFV.PRS.3SG you.ACC.PL back to beginning.GEN.SG  
 dvacátého století,) kdy musíte<sub>1</sub> rozhodnout<sub>2</sub> **se**<sub>2</sub>  
 twentieth.GEN.SG century.GEN.SG when must.PRS.2PL decide.PFV.INF REFL.ACC  
 kde budovat vaši první automobilku.  
 where build.INF your first.ACC.SG car.ACC.SG  
 ‘(The game will take you back to the beginning of the twentieth century,) when you will  
 have to decide where to build your first car.’ (Cz; S15)

<sup>430</sup> An extra-linguistic factor affecting the likelihood of data acquisition is corpus size. I was unable to obtain the clitic in-situ structure (6.12b–c) in the most recent SYN subcorpus and in the ORAL subcorpus. However, the World Wide Web is big enough to find corresponding examples, both of which represent non-standard language use.

<sup>431</sup> The present thesis is an actual example of corpus use in language comparison.



c. (zjistil jsem, že ne-zvládám moc angličtinu a realise.PST.SG.M AUX.1SG that NEG-master.PRS.1SG a\_lot English.ACC.SG and ne-mám pomoc od někoho ve španělštině,) NEG-have.PRS.1SG help.ACC.SG from someone.GEN.SG in Spanish.LOC.SG musím<sub>1</sub> rozhodnout<sub>2</sub> se<sub>2</sub> pro něco jednoduchého.... must.PRS.1SG decide.PFV.INF REFL.ACC for something.ACC.SG simple.M.ANIM.ACC.SG ‘(I realised that I don’t know a lot of English and I don’t get help from anybody with Spanish,) I will have to decide for something simpler...’ (Cz; S16)

Another defence of introspective data is offered by Broekhuis (2010), who advocates the viewpoint “that introspection research is a better method for collecting synchronic syntactic data than corpus research.” (Broekhuis 2010: 182) He opens his article in an interesting way:

“Not so long ago it seemed unnecessary to motivate the use of what is nowadays called the intuitionist approach to the study of syntax: the study of the internal structure of phrases/sentences with the help of data obtained by introspection. But times have changed and the reviews [...] leave no doubt that some of the reviewers consider this approach to linguistics inadequate (if not obsolete)[.]” (Broekhuis 2010: 181–182)

It seems as if Broekhuis were regretful about the need to provide the rationale for the choice of introspective data.<sup>432</sup> But it is not a bad thing that “times have changed”. It is a standard procedure in many other disciplines to provide a transparent description of the tools and methods in one’s study as well as a reasoning for the utilisation thereof. There is no reason why linguistics should not follow suit. In a field like linguistics, where diverse kinds of data like introspective judgements, experimental responses, and corpus attestations are at the researcher’s disposal, it is necessary to document what type of data has been used and how it was obtained. This is good scientific conduct.

Returning to Broekhuis’s (2010) original agenda, he concurs with Philips (2009) and Tajsner (2017) in that “[f]aulty introspection data simply do not have much chance to survive in the long run and are normally already weeded out before publication of the research.” (Broekhuis 2010: 190) It has been mentioned earlier that this is not always true. An interesting aspect of Broekhuis’s article is that he informs the (critical) reader with some basic aspects of intuitionists’ methodology. One important rule of thumb that he mentions is that generative theorising should be based only on

<sup>432</sup> Note that, again, similar threads of argumentation are presented by proponents of corpus linguistics, as Stefanowitsch (2020: 1) points out: “I cannot think of any other scientific discipline whose textbook authors would feel compelled to begin their exposition by defending the use of observational data, and yet corpus linguistics textbooks often do exactly that.”

indisputably grammatical and ungrammatical utterances while leaving aside controversial cases (Broekhuis 2010: 186–187). However, this rule does not seem to be adhered to consequently, which I provided ample evidence for throughout Part I. Broekhuis (2010: 187) adds that the aforementioned practice “let[s] the prevailing theory of the time decide whether the example should be considered grammatical or not.” But this procedure is unreliable, provided that “competence theory is continuously being updated, which entails that certain objects can be characterized as grammatical at one stage of the theory and as ungrammatical at another stage (and vice versa).” (Broekhuis 2010: 185) Furthermore, it promotes a somewhat reversed perspective of what a Generative Grammar is supposed to be, namely a formal model of the native speaker’s unconscious linguistic knowledge  $K_L$ , which successfully discriminates an utterance as grammatical, if it complies with  $K_L$ , but as ungrammatical, if it does not. The model is not supposed to justify or discard actual data in a scientific discourse. Instead, it is the result of an ongoing epistemic endeavour to represent the system underlying the data. It is the theory, which must comply with the data, not the other way round.

Finally, Broekhuis (2010: 190–192) emphasises in the last two sections of his article that the intuitionist approach is superior to corpus research and provides four concluding arguments summarised in the following:

1. Raw corpus data need to be filtered by a researcher’s intuition to remove questionable or distorted data points.
2. Corpus data potentially need to be expanded by introspective data due to incompleteness.
3. Negative evidence must be additionally provided by the researcher based on linguistic intuitions.
4. Parts-of-speech annotation in electronic corpora “is based on the results of centuries of intuitionist linguistic research” (Broekhuis 2010: 192) and further development of such systems will rely on introspection, too.

There is no doubt in argument no. 1, as procedures of data manipulation, data cleansing etc. are inevitable steps in preparing raw data for further analysis.<sup>433</sup> But this is independent of the discipline of linguistics or grammar research. Hence, it can hardly serve as an argument in the methodological debate on introspection and empirical data. Arguments no. 2 and 3 do not hold under the premise that recourse to linguistic intuitions is confined to introspection in the sense referred to here. First, it is a popular fallacy that introspective data were not finite or incomplete. As Stefanowitsch (2020: 12) argues, “just like a corpus, the linguistic experience of a [native] speaker is finite” and “limited to

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<sup>433</sup> See Section 8.1 for data preparation for the present thesis.

certain language varieties”, cf. also (6.1d).<sup>434</sup> We may further add that those introspective examples cited and discussed at length in articles and monographs comprise finite and limited sets, too. Second, it is particularly the data sampling account of introspection that has been a regular subject of critique, not the notion of linguistic intuition itself. There seems to be a fundamental misunderstanding regarding this point. Corpus linguists will readily accept results from psycholinguistic experimentation, which, e.g., elicits acceptability judgements from a population of experimental subjects. These judgements may provide negative evidence and expand the empirical database by constructions unattested in electronic corpora.<sup>435</sup> Argument no. 4 is not related to the methodological discussion dealt with here.

To conclude, data points from Slavic and German (and, perhaps, English) suggest that Phillips’s (2009: 51) first and second criterion are met. This is true, even if it applies only to a subset of the entirety of all introspective data that has been utilised. There is a number of constructions that have been shown to challenge introspectively motivated claims, e.g., the ban of clitic climbing across subjunctive *da* (BCMS) and the ban of extraposition across CP (Slavic; cf. Junghanns 2021 for the latter), extraposition disobeying subadjacency and the occurrence of multiple constituents in the prefield in German. Therefore, critique of informal acceptability judgements cannot be simply swept away. How should Phillip’s (2009) third criterion be approached? He is doubtful that “[c]arefully controlled judgment studies would solve these problems” (Phillips 2009: 51), which perhaps could be also representative for a similar stance towards corpus research. Even if one shares this scepticism, then one must acknowledge that empirical data collection methods are an integral part of modern scientific practice and objectivise the research process. In fact, Phillips (2009) himself seems to agree with this baseline:

“This does not mean that there is no value in careful testing of subtle contrasts. *It is good practice to do so*, and lends greater weight to any (justified) claims made based upon the subtle contrast.” (Phillips: 2009: 56; my emphasis)

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<sup>434</sup> This argument touches upon Chomsky’s (1965: 4) concept of the ideal speaker-hearer, which I cannot discuss here. Suffice it to mention that Chomsky’s idealisation is a – most likely monolingual – individual, who has complete mastery of their native language. In contrast, variationist linguistics in the tradition of Flydal (1952) and Coseriu (1986) acknowledge at least three axes of linguistic variation, i.e. diatopic (= geographic), diastratic (= social group), and diaphasic (= register). As a matter of fact, no individual has complete mastery over all linguistic varieties of their titular language along these axes.

<sup>435</sup> In exactly this spirit Kolaković et al. (2022) supplement their comprehensive corpus studies on clitic climbing in BCMS by a psycholinguistic experiment. See also Gilquin & Gries (2009) for a discussion of corpus linguistic vis-à-vis psycho-linguistic methods.

Replicability is an important criterion here. Empirical methods allow for systemic hypothesis testing and can provide new data to ongoing debates.<sup>436</sup> Empirical data may either corroborate or correct scientific views, feed arguments, and contribute to the development of the field.

What this all amounts to is that the claim is dubious “that introspection research is a better method for collecting synchronic syntactic data than corpus research” – or any other empirical data elicitation method – as argued by Broekhuis (2010: 182). Introspection is merely one of several means applied in contemporary linguistic practice. As a matter of fact, there is no such thing as *the* perfect method; each method suffers from some limitation. Empiricists, too, need to be and indeed are aware of the weaknesses of the methods and technologies they apply. It appears fruitful then, to synergise different methodological procedures to build a research design upon their advantages, while attempting to overcome their disadvantages in the best possible way.

## 6.2 Conflict resolution: Methodological pluralism and division of labour

The previous section discussed two conflicting approaches to linguistic data: rationalists draw upon individual native speaker linguists’ intuitions in constructing examples by introspection, whereas empiricists utilise bodies of recorded attestations or experimentally elicited data. In the following I will sketch a possible – hopefully constructive – way of bridging the gap between both methodologies: the native speaker’s intuitively constructed data may serve as a means of approximative observations about linguistic structure. On this basis, inductively inferred hypotheses may be stated. On the other hand, empirical data may be utilised to test falsifiable hypotheses as suggested by Karl Popper (cf. already Labov 1975 for a similar suggestion).<sup>437</sup> A falsifiable hypothesis  $H_f$  may either stand the test by observational (corpus) and/or experimental findings, which is usually taken as a corroboration of  $H_f$ , or it may be falsified by empirical results, which calls either for a restatement of  $H_f$  or its rejection in favour of an alternative statement  $H_g$ .<sup>438</sup> From this it follows that empirical data has a corrective function and allows scientists to abandon implausible hypotheses.

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<sup>436</sup> In this respect, Tajsner’s (2017: 601) defence exhibits another misconception about corpus linguistics (or empirical methodology, in general), when claiming that “[c]orpus analysis is incompatible with the deductive *modus operandi* of generative grammar”. He contends that a “‘corpus-only’ strategy would have to start from a search through a corpus without a faint idea what to look for [...], ready to grasp at any interesting syntactic phenomenon encountered in this search.” (Tajsner 2017: 601). Taken at face value, this argument suggests that corpus linguistic – and, in principle, any empirical – methodology could not be linked to a stated or statable scientific problem. Obviously, this is incorrect. Purposeful corpus analysis presupposes one or more research questions. Furthermore, the argument fully neglects the potential of serendipities (e.g. X-radiation and penicillin). See Section 10.4.4 on agreement drop in Goralish for such an unexpected discovery.

<sup>437</sup> According to Popper, only falsifiable hypotheses are empirical hypotheses.

<sup>438</sup> See Stefanowitsch (2020: 16–17), who independently comes to the same conclusion and proposes a similar approach.

In a similar fashion, Ohala & Jaeger (1986: 2) conclude that experimentation “destroy[s] knowledge” rather than to add more information to our understanding of the world.<sup>439</sup> The very same logic is applied by Toman (2010) and Progovac (2019), i.e. representatives of (Mainstream) Generative Grammar. Toman (2010: 435) concludes his brief treatment of the Old Czech clitic Focus particle *-t’* in the following manner:

“In a way, the result [= of the study] consists in showing that a certain type of theory ‘works’. That may, however, be a somewhat problematic result – progress is usually achieved in showing exactly the opposite.” (Toman 2010: 435)

Progovac (2019: 1) critically discusses postulates of Minimalism dispensable in the light of language evolution and the emergence of hierarchical syntax and argues for the need of “interdisciplinary testing”:

“The emphasis here is on specific, falsifiable hypotheses, rather than some vague assertions that cannot be subjected to falsification. It is true that many such specific hypotheses will be proven wrong, but after all, the nature of the scientific process is simply to *narrow down the range of possibilities*.” Progovac (2019: 1; my emphasis)

On a related note, Schröter (2019: 252) concludes that “the relevance of this approach [= a data-based analysis] lies in the systematization of structural descriptions as a test of anecdotal or impressionistic observations.”

It should be emphasised that empirical research is not restricted to a merely corrective function of theoretical reasoning, but may serve well for inductive hypothesising, i.e. in exploratory studies like the present thesis. It will be demonstrated in Part III that empirical methods are capable of introducing new data and phenomena, which have not been taken into consideration before. Based on such data, new hypotheses may be stated leading to further lines of research. Thus, empirical research has a positive impact on the developments within the field and may contribute to linguistic theory construction and the modelling of linguistic phenomena. Analyses taking into account empirical evidence will become more adequate, as they cover a broader data set than before. As a consequence, introspective data is still acknowledged even in an empiricism-centric approach. As Broekhuis’s (2010: 190–192) arguments imply, linguists will need to rely on their intuition either way. It is the scholar’s intuition, which is the driving force of scientific research. For one thing, linguists use their intuition when searching an electronic corpus for a particular linguistic structure Z,

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<sup>439</sup> Ohala & Jaeger (1986) are concerned with experimental phonology, but their argument applies to all areas of empirical research.

as they infer, whether Z is likely to occur, in which context, and how to state a query (plus an arbitrary sequence of trial and error). This might be subsumed under what Karlsson (2008: 29) calls “critically minded corpus work”.

Put differently, empiricists do not abandon linguistic intuitions as such but strive for more objective and transparent means of aggregating the data. There seems to be a general misunderstanding in this respect, as mentioned already in Section 6.1. The position defended here is that analyses, which are built upon a set of subjective intuitive data, need to adjust to empirical findings, which deviate from singular intuitions. It needs to be emphasised that it is, in fact, possible to classify constructed data as observational data, if the data are grounded on self-reflection about the observer’s (or his peers’) linguistic behaviour. In other words, I argue for a methodological pluralism that aims at synthesising different approaches for the overarching purpose of the gain of knowledge.<sup>440</sup> Kolaković et al. (2022: 8–9) independently come to a similar conclusion: “The corroborating results from studies using different kinds of data and methods provide more insightful and reliable linguistic evidence in comparison to studies using only one type of data.” They adopt the term *triangulation* from the social sciences for such a kind of approach (cf. Kolaković et al. 2022: 53–54).<sup>441</sup>

The need for such a methodological pluralism and its relevance – especially the utilisation of empirical data – becomes apparent, if one takes the goal of formal linguistics into consideration as it was defended by Noam Chomsky. Formal linguistics aspires to provide the formal description of a grammar that tells apart grammatically well-formed utterances of a language L from ungrammatical ones. This goal is jeopardised, if formal grammar relies on a biased or incomplete data set. For the sake of illustration, recall the restructuring analysis of clitic climbing in Czech advocated for by Rezac (2005). It proposes that clitics climb from functionally deficient, i.e. bare infinitival VPs to a matrix domain in order to license their morphosyntactic features. I. Kulik (2023b) is in accordance with Dotlačil (2004) in concluding that a restructuring analysis is not feasible for Czech (and Polish). If a formal grammar of Czech renders clitic climbing contingent on restructuring environments, then climbing of accusative clitics into passivised matrix domain need to be assessed as ungrammatical. According to *Burzio’s generalisation*, passives are unable to license accusative case (cf. Burzio 1986). But Lenertová (2004), Dotlačil (2004), and I. Kulik (2023b) provide data that show clitic

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<sup>440</sup> In fact, present-day linguistics is a highly diversified field such that there is already methodological pluralism. The problem I am addressing, is an unfruitful coexistence without synthesis.

<sup>441</sup> Kolaković et al. (2022: 8–9) refer to the trinity of “intuition/theory – observation – experiment” by this token. The term *experiment* is, however, meant as a cover term for empirical validation and, particularly, for the synthesis of corpus research and psycholinguistic experimentation. The combination of several empirical methods adds another dimension to the issue under survey. I agree with the scholars in principle, but I am not able to conduct psycholinguistic experiments due to limitations of time, financial resources, and manpower. A synthesis of corpus findings and intuitively evaluated data should still be an eligible approach and hopefully yield an increase in adequacy as compared to a solely introspective database.

climbing of accusative pronominal clitics into the domain of passivised matrix verbs. The introspectively justified grammar fragment does not comply with the linguistic facts. A formal grammar that classifies clitic climbing in Czech as emergent from restructuring must be deemed inadequate. Keep in mind that science strives for objectivity or intersubjectivity.<sup>442</sup> No intersubjectivity can be achieved between individuals disputing whose intuition and judgement is the correct one. Put differently, informal acceptability judgements remain subjective regardless of their hypothetical convergence with formally gathered data, because they provide generalisations from a single idiolect (or a small sample of idiolects). Intersubjectivity can be achieved only with the efforts that come about with an empirical grounding and a sound statistical basis.

To provide a short illustration of the problems of idiolectal databases, consider the fusion of preposition and article in German. According to the standard grammar of German, there is only a restricted number of possible combinations available for this type of cliticisation (Wöllstein & Dudenredaktion 2022: 820–821). As exemplified in (6.13), the fusion applies in case of definite articles of masculine and neuter gender only with a predominance of dative cases. The only fusion including feminine gender is (6.13h) as pointed out by Nübling (2005: 112).

|        |    |               |   |                     |       |
|--------|----|---------------|---|---------------------|-------|
| (6.13) | a. | am            | < | an + dem            |       |
|        |    | at.M/N.DAT    | < | at + the.M/N.DAT    |       |
|        | b. | ans           | < | an + das            |       |
|        |    | through.N.ACC | < | through + the.N.ACC |       |
|        | c. | beim          | < | bei + dem           |       |
|        |    | by.M/N.DAT    | < | by + the.M/N.DAT    |       |
|        | d. | im            | < | in + dem            |       |
|        |    | in.M/N.DAT    | < | in + the.M/N.DAT    |       |
|        | e. | ins           | < | in + das            |       |
|        |    | in.N.ACC      | < | in + the.N.ACC      |       |
|        | f. | vom           | < | auf + dem           |       |
|        |    | on.M/N.ACC    | < | on + the.M/N.ACC    |       |
|        | g. | zum           | < | zu + dem            |       |
|        |    | to.M/N.DAT    | < | to + the.M/N.DAT    |       |
|        | h. | zur           | < | zu + der            |       |
|        |    | to.F.DAT      | < | to + the.F.DAT      | (Ger) |

<sup>442</sup> As it is a controversy in the philosophy of science, whether something like true objectivity can be ever achieved, I give primacy to the notion of *intersubjectivity* here.

Nübling (2005) discusses the cliticisation of the article onto the preceding preposition from a diachronic perspective basically. However, she provides also synchronic empirical data from contemporary German. Importantly, her data are of southern German provenance and conform to a large extent with the standard grammar. That is, native speakers from southern Germany accept only a restricted set of fused preposition-article complexes as grammatical or natural. Most of the logically possible combinations are not accepted and do not occur. Now, as a native speaker of German, I have a different intuition about the availability of preposition + article fusion. In my idiolect, virtually every logical combination is acceptable and occurs in daily speech. Standard German and Nübling's south German data agree in that only the definite articles encliticise to the prepositional head, whereas indefinite articles freely combine with prepositions in my idiolect, cf. (6.14).

- (6.14) a. Wir sind **inner** Stadt. [inner [ʔɪnə] < in + *der*]  
 we.NOM be.PRS.1PL in.DEF.F.DAT.SG city.DAT.SG  
 'We're in the city centre.'
- b. Wir gehen **inne** Stadt. [inne [ʔɪnə] < in + *die*]  
 we.NOM walk.PRS.1PL in.DEF.F.ACC.SG city.ACC.SG  
 'We're going to the city centre.'
- c. Wir treffen uns **inner** Stunde. [inner [ʔɪnə] < in + *einer*]  
 we.NOM meet.PRS.1PL us.DAT in.INDF.F.DAT.SG hour.DAT.SG  
 'We'll meet in an hour.'
- d. Er ist **inne** Hecke gefahren  
 he.NOM be.PRS.3SG in.INDF.F.ACC.SG hedgerow.ACC.SG drive.PTCP  
 'He drove into a hedgerow.' [inne [ʔɪnə] < in + *eine*]
- e. Sie hat **annem** Bauzaun geparkt  
 she.NOM have.PRS.3SG at.INDF.M.DAT.SG hoarding.DAT.SG park.PTCP  
 'She parked next to a hoarding.' [annem [ʔanəm] < an + *einem*]  
 (Ger, colloquial; constructed)

One possible explanation for this discrepancy could be that my idiolect is simply idiosyncratic. However, I do not only have intuitions about highly liberal fusions of preposition + article as a native speaker, but I regularly observe evidence for such seemingly inflected prepositions in everyday speech on the streets. As a native language acquirer, I should have had exposure to corresponding data in order to acquire the respective structures. Corroboration comes from two studies by Hartmann (1980: 166–167) for Berlin German and Schiering (2005) for Ruhr German. They find increasing occurrences of such liberal preposition + article fusions. My idiolect thus converges with lects spoken



in the western and eastern parts of northern Germany. As additional metadata, note that my linguistic socialisation took place in the Hannover–Braunschweig–Göttingen–Wolfsburg Metropolitan Region (Ger. *Metropolregion Hannover-Braunschweig-Göttingen-Wolfsburg*), which is halfway between the Ruhr area and Berlin as a rule of thumb. Therefore, it appears that the Metropolitan Region shares linguistic properties with other linguistic areas of northern Germany. This observation complies with other diatopic distinctions commonly observed between the “linguistic north” and “the linguistic south” in Germany. In fact, the most recent Duden grammar (10th edition) acknowledges this richness in variation and informs that, in spoken language, much more patterns of fusion are observable than in the written standard. Fusions cover various prepositions, diverse morpho-syntactic determiner forms as well as the indefinite article (Wöllstein & Dudenredaktion 2022: 553). To conclude, native speakers of northern and southern varieties have different intuitions about the grammaticality of fused preposition + article complexes. The discussion illustrates that introspective data is not representative. Native speaker linguists need to be careful when drawing conclusions from their linguistic intuition. This does not imply that native speakers’ intuitions are generally false, but that caution is needed. The integration of empirical data helps in objectivising one’s generalisations and hypotheses.

In sum, the conflict between theory and data is resolved by the recognition of the co-existence of introspection, attestations, and experiments in the linguist’s methodological toolbox. Each data elicitation approach contributes to the gain of scientific knowledge in its own way. The second pillar of this bridging strategy is built upon a division of labour like in the natural sciences. For instances, both theoretical physics and experimental physics contribute to the knowledge acquisition about the physical world. A similar approach can be taken in linguistics, whereby introspective data can be localised on the theoretical side, while attested and psycholinguistically elicited data occupy the experimental or empirical sphere.<sup>443</sup> This is surely a somewhat simplified, but illustrative picture. Finally, it is not argued here that individual researchers would have to utilise all three data acquisition techniques (introspection, attestation, experiment) simultaneously. It is still possible to propose hypotheses and analyses on the basis of intuitively constructed examples, but researchers utilising introspection need to acknowledge the empirical data and recognise its impact on the theory. In a similar manner, corpus linguists are not obliged to conduct experiments, but they need to be aware of the gaps in their data and the very fact that these might be only filled by experimental results, if growing a corpus by aggregating more attestations does not involve the relevant constructions. Anyway, the situation might come up that one does neither have (enough) corpus data at one’s disposal nor is preparing and/or conducting an adequate experiment a possible endeavour. Then,

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<sup>443</sup> To continue making use of the comparison, corpus linguistics thus resembles rather astronomy than particle physics, as the former does not have laboratory set-ups and empirical discoveries are mainly made by observation. In a similar vein, Gilquin & Gries (2009: 8) emphasise that “corpus studies are observational studies”.

perhaps, recourse to introspection could be the only alternative available. All in all, the methods complement each other in one way or the other.

### 6.3 What are we talking about? Linguistic competence vs. norm

In the following I would like to consider the issue of written vs. spoken data as a supplemental justification for the decision to draw the present survey on spontaneous speech. Alternatively, the problem can be broadly addressed by labels like “standard vs. colloquial language” or “prescriptivism vs. descriptivism” etc. In the following, I will outline the theoretical and methodological baseline of my argumentation.

Recall from the previous Section 6.2 that the goal of (a certain branch of) linguistics is to model native speakers’ mental grammar, i.e. the unconscious knowledge of linguistic structure internalised during first or native language acquisition. Taking a closer look at Slavic, one must acknowledge a significant degree of divergence between the – predominantly written – codified standard languages and the colloquial code used in – especially private – everyday communication. For instance, since the studies of Zemskaja (1973) it is common knowledge in Russian and Slavic studies that there is “a gulf between [...] *razgovornaja reč*’ [= colloquial Russian, I. K.] and literary Russian”, as cited by Timberlake (1993: 828).<sup>444</sup> That is, on all linguistic levels, colloquial Russian bears features, which are absent from standard Russian. However, a great number of studies in theoretical linguistics or grammar research implicitly rests upon the standard norm.

Anticipating the reasoning in Section 6.4, spoken data with all its oddities must be considered the primary input source for infants’ first/native language acquisition. Support for this view comes from heritage linguistics. There is a high degree of variation among heritage speakers with respect to their linguistic skills, such that they are assigned to a continuum, which is roughly classified from least to most proficient into basilectal, mesolectal, and acrolectal (cf. Polinsky & Kagan 2007: 371–372, Polinsky 2018: 5–6). It is acknowledged in heritage linguistics that the comparison between a heritage linguistic system and the standard language as a reference system is inadequate. Usually, heritage speakers are not familiar with the standard language, which students become acquainted with during institutionalised education. The standard variety does not or only sparsely occur in the private

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<sup>444</sup> See also Hinrichs (1999: 592): “Nach heutigen Erkenntnisstand ist die russ. US [= Umgangssprache, I. K.] in ihrer Phänomenologie sehr weit vom schriftsprachlichen Standard [...] entfernt [...]”

‘According to the current state of knowledge, the Russ. colloquial speech is very divergent from the written standard with respect to phenomenology ... .’

Cf. Kaltseis & Stadler (2023: 62–72) for a recent and concise overview of colloquial Russian.

domain of language acquisition. Therefore, it is suggested to compare heritage speakers' proficiency with the so-called BASELINE (cf. Polinsky 2018: 6, 10–13; Polinsky & Kagan 2007: 372), which is the variety the primary linguistic data originate from.<sup>445</sup> Usually, it is colloquial speech or a dialect of the titular language. If one surveys, e.g., heritage speakers of Russian, it should come as no surprise then that their acquired Russian exhibits features of *razgovornaja reč'* or even child language (cf. Polinsky & Kagan 2007: 373; Polinsky 2018: 49 more generally for the latter). The heritage linguistic concept of baseline is useful in the present discussion and perfectly fits the argument I want to make in favour of a variationistically inspired approach to grammar research and language acquisition, which gives particular attention to non-standard varieties and colloquial speech.

Czech is a particularly insightful example of the issue under discussion. Written Czech (*spisovná čeština*), the prescriptive standard norm, is an archaised variety. Like in Russian, Common Czech (*obecná čeština*), which is the colloquial language of Bohemia and its cultural centre Prague, radically differs from the standard on all levels, i.e. “phonology, morphology, syntax, and lexicon” (Janda & Townsend 2000: 4).<sup>446,447,448</sup> The divergence between both varieties is the outcome of history. In 1618, the Defenestration of Prague resulted in the Thirty Years' War, which began as a religious war between the Protestant Bohemian estates and the Catholic Holy Roman Emperor. After the Battle of White Mountain in 1620, when the Bohemian estates were defeated, the Czech language ceased to play an important role in Bohemia, whose population was put under cultural pressure and enforced Germanisation (cf. E. Eckert 1993: 7; Janda & Townsend 2000: 1). However, several processes of language change beginning in earlier periods continued as well as new ones emerged in the extant Czech-speaking population. To name only two of the most prominent features (cf. Krčmová 2017): the long, closed front vowel [i:] (graphically mostly ⟨ý⟩, but sometimes also ⟨í⟩) underwent diphthongisation (*ubejvat* < *ubyvat* ‘decrease.INF’) since the 15th century. Diphthongisation spread across the spoken language in the following period of humanistic Czech (Kováříková 2015: 39–40). On the other hand, the gender distinction in the adjectival declension was lost in the plural (*dobr-ý*

<sup>445</sup> According to Polinsky & Kagan (2007: 372), the term was adopted from creole linguistics but possesses a different denotation in the discipline of heritage linguistics.

<sup>446</sup> The following discussion applies primarily to Bohemian Czech, since the linguistic situation in Moravia, where the traditional dialectal distinctions are still preserved, differs from that in Bohemia (cf. Daneš 2003: 12–13; Davidová & Jandová 1999: 146–147; Kováříková 2015: 40; Nekvapil 2005: 12). Therefore, it might appear more adequate to utilise the notion *Common Bohemian*. I shall refrain from introducing such a terminology and stick to the established *Common Czech*.

<sup>447</sup> I refer the reader to Townsend 1990 for an accessible descriptive overview of Common Czech (which he refers to as *Spoken Prague Czech* due to his focus on the speech of the Czech capital city). His survey covers phonology, morphology, syntax, and lexis.

<sup>448</sup> I do not take into consideration the notion of *hovorová čeština* ‘Colloquial Czech’, which is subject to controversy (cf. Hoffmannová 2017, Nebeská 2017, with references, for an overview). For instance, Townsend (1990: 11) takes the strong position that “*hovorová čeština* as the term most recently used to describe the so-called spoken literary Czech” is not “a realistic entity”.

*sportovci* ‘good sportsmen’, *rozbit-ý* *vokna* ‘broken windows’).<sup>449</sup> Correspondingly, E. Eckert (1993: 6–7) informs that the Kralice Bible itself was already subject to linguistic conservatism and did not reflect the contemporary central Bohemian Czech language but possesses Moravian features, which rendered the bible accessible for a Slovak audience, too. At the turn between the 18th and 19th century, Standard Czech was codified by agents of the National Revival, whose leading figures, Josef Dobrovský and Josef Jungmann, took the 16th century’s humanistic Czech language of the Kralice Bible as the basis for the standard norm (cf. E. Eckert 1993: 6–7; Janda & Townsend 2000: 4; Nekvapil 2005: 7). Consequently, “the Modern Standard Czech became the most archaic member of the family of numerous Slavic standard languages and got quite fatally estranged from colloquial Czech” (Mathesius 1933, cited after Daneš 2003: 11). Needless to say, the puristic agenda of the Czech National Revival also aimed at significantly reducing German borrowings in the Czech lexicon (e.g. by replacing *špitál* with *nemocnice* ‘hospital’ etc., cf. Krčmová 2017). Replacements of Germanisms were coined on the basis of either older Czech vocabulary, loans from other Slavic languages, or neologisms. But to date, a comparatively high amount of lexical Germanisms is significant for Common Czech, e.g. *fajn* ‘good, nice’, *fraj* ‘free’, *furt* ‘always’, which are absent from the normative lexicon of Written Czech (cf. Newerkla 2017). From a syntactic perspective, sentence-initial clitic placement is frequently linked with the spoken or colloquial language and is, thus, in stark contrast to the well-behaved Czech standard language (cf. Dotlačil 2004: 73, n. 53; Hana 2007: 70; Lenertová 2004: 148–149; Toman 1996: 505). The differences between Written Czech and Common Czech are located within both lexicon and grammar leading to a situation that has been described as diglossia (cf. Daneš 2003: 13–14; E. Eckert 1993: 6; Nekvapil 2005: 7, 13). Written Czech is the official written standard language used in literature, official communication, and national media. This variety is transmitted via instruction in the educational system and, as Nekvapil (2005: 7) puts it, “has no native speakers.” However, it is an important point of reference for grammar research.<sup>450</sup> On the other hand, Common Czech is “the mother tongue of practically all speakers” according to Nekvapil (2005: 7) and it is used among family and friends. Therefore, Common Czech

<sup>449</sup> Another morphological feature is the realisation of the 2SG present tense of the copula *být* in the form *seš* (cf. Lenertová 2004: 142, n. 9; Toman 1980: 307), which is by far the most frequent realisation in the ORAL subcorpus of the ČNK in comparison to the standard form *jsi* or the “official” Common Czech copula form *si*, although *si*, but not *seš*, frequently occurs in the literature. The token frequencies for ORAL v1 are: *seš*: 1,994; *si*: 21; *jsi*: 27. Note that the *si* cases were obtained by the query [word="si" & lemma="být"] to exclude the homonymic dative reflexive clitic. However, the query output contains a high number of erroneously tagged reflexive clitics (and to a minor amount auxiliary clitics, while a great many 2SG auxiliary clitics *si* are mistagged as reflexive clitics). Therefore, one should remain cautious with such type of statistics.

<sup>450</sup> For one thing, Toman’s (1996) article on Czech clitic prosody is drawn on Written Czech. Common Czech is mentioned only in passing as “informal colloquial Czech” (Toman 1996: 505).

may be considered the primary input for native language acquisition – at least in Bohemia (contra Written Czech, whose transmission begins only with schooling on a larger scale).<sup>451</sup>

The question arises what linguists talk about, when referring to native language and acquired grammar. What are we bearing upon when employing grammaticality judgements? Is it possible that these judgements are biased by the standard language? If linguists were to model a – presumably – fully codified standard language, then there would not be anything to survey and to model ultimately, since all rules/constraints of the language are fixed in prescriptive grammar books. In fact, it would not make sense to ask speakers for their judgements, if one follows Nekvapil's (2005: 7) assessment that there are actually no native speakers of the standard. It is easy to imagine how serious an issue it is when one recalls that Franks & King's (2000) monograph is a survey of cliticisation in the Slavic standard languages but also relies on native speaker judgements from fellow linguists. Of course, the standard language is a reality in a speech community's everyday life and phenomena and constructions found in the written language do also occur in the spoken domain. However, spoken data exhibit properties both lexically and structurally, which are absent from written or standard language texts (cf. Guz & Jędrzejowski 2020 on Polish; Weiss 1993 on Russian).<sup>452</sup> Recall, for instance, the prevailing absence of the Polish short dative reflexive *se* in both reference grammars of Polish and linguistic enquiries into Polish clitics.<sup>453</sup> It is plausible to assume that spoken data represent the baseline for language acquisition and, thus, approximate the native speakers' mental grammars more accurately than the codified standard. From a different angle, the standard language constitutes only a subset of the entire linguistic repertoire of a native speaker. Vogel (2019: 73–74) addresses this very issue in an investigation of what he calls *grammatical taboos*:

“But grammatical theory is interested in the grammar of *natural* languages. How ‘natural’, after all, are languages actually, especially standard languages? What do we mean by ‘natural’? And how does the linguist, just from the armchair, come to recognise and sort out the artificial properties of a standard language?” (Vogel 2019: 73–74)

The problem addressed by Vogel becomes immanent in the following quote from Hana (2007):

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<sup>451</sup> I cannot address the issue of interference between both codes (cf. E. Eckert 1993: 10–11 on code switching and mixing, and references therein, cf. also Nekvapil 2005: 13).

<sup>452</sup> Guz & Jędrzejowski's (2020) conference paper discusses the discourse element *že*, which has several pragmatic functions, e.g. quotative or discourse-linking. It is distinct from the homophonous subordinator (complementizer) *že* ‘that’ and the emphatic particle (or post-verbal clitic) *-že*. Czech has a polyseme or multifunctional morph *že*, too (cf. Gruet-Skrabalova 2012; Kaspar 2017). Again, the multiple, especially pragmatic functions frequently occur in informal speech, but are usually disregarded.

<sup>453</sup> Recall also the significant deviations between standard Sorbian and colloquial Sorbian speech from Footnote 19 in Section 2.1.

“It is also true that, apart from a very colloquial register [...], clitics cannot be sentence initial. However, it is sometimes hard to separate this and other registers when making grammaticality judgements in less common cases.” (Hana 2007: 76)

Certain registers may favour or disprefer the occurrence of certain linguistic structures, but these factors are stylistic and/or social in nature. The choice of a particular structure is a matter of appropriateness in a given social or communicative context and surely belongs to our social and world knowledge as a result of socialisation, but it is not a matter of the internalised mental lexicon or grammar.<sup>454</sup> Czech clitic-initial sentences might be felt inappropriate in a particular social environment or when writing an official document, but it would be scientifically inadequate to consider this very construction ungrammatical. The notion of (*un*)*grammaticality* is orthogonal to social acceptability or appropriateness. Linguistic constructions, whose occurrences are socially sanctioned, can be perfectly grammatical. Stylistic reasons or the choice of a particular register are not indicative of the existence of a construction in a language.<sup>455</sup> This line of thought is compatible with recent research in register variation in natural languages (cf. Pescuma et al. 2023, with references). For instance, Varaschin et al. (2024) model Topic Drop in German within the formal framework of HPSG with reference to what they call use-conditional constraints. Such a kind of constraint “imposes a contextual appropriateness condition” on linguistic structures (Varaschin et al. 2024: 10). Consequently, the realisation of a linguistic structure that is perfectly grammatical in a language may be appropriate in certain contexts but inappropriate in others.<sup>456</sup>

Junghanns (2021) touches upon the issue under discussion while reflecting on the role of explanation in linguistics and data samples that underly linguistic generalisations. Analysing the

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<sup>454</sup> This perspective is fully compatible with Pescuma et al.’s (2023: 2) definition of register as “as intra-individual variation, that is, the conventionalized and recurrent linguistic patterns of (individuals in) a speech community depending on the situational-functional context.” It also complies with Coseriu’s (1988: 50) notion of diaphasic variation, which involves “differences between the modalities of speech according to the situation [of speech] (including the participants of a conversation).” (Orig.: “Unterschiede zwischen den Modalitäten des Sprechens je nach Situation desselben (einschließlich der Teilnehmer am Gespräch).”, Coseriu 1988: 50.) Bear in mind that Coseriu uses the term *modality* differently from standard terminology in current research in sign languages, where signing is one modality of language vis-à-vis speech representing another modality.

<sup>455</sup> This might be thought of as an analogue of register-dependent lexical choices. Lexemes associated with a particular register are appropriate in a given communicative context or not. Some lexeme *LEX*<sub>1</sub> is likely to occur in context *CXT*<sub>1</sub>, whereas *LEX*<sub>2</sub> is restricted to *CXT*<sub>2</sub>. However, scrutinising *CXT*<sub>1</sub> does not preclude the existence of *LEX*<sub>2</sub>. For example, P. Bąk (1977: 435) tells that the “principle of stylistic clarity requires that expressions, which logically and grammatically belong to each other, are placed in *immediate neighbourhood*.” [emphasis in the original]. This stylistic restriction prohibits discontinuous constituents, although discontinuous constituents are a common syntactic feature of Slavic that relies on principles of both syntax and information structure.

<sup>456</sup> A relevant factor in this approach are social meanings, i.e. “any kind of non-at issue content that indexes some socially relevant property” (Varaschin et al 2024: 11). Social meanings are associated with diverse types of linguistic entities ranging from individual lexemes over phonetic-phonological features to grammatical structures of various complexity. According to Varaschin et al. (2024: 11–13), social meanings are consistently transmitted by senders and perceived by recipients but do not need to be conscious at all. One can think of diverse conceptual labels assigned to certain linguistic structures and/or their users that may have mild to severe social implications and, thus, significantly affecting the likelihood of occurrence of those structures in a variety of contexts or situations.

Delayed Placement of clitics in Czech, he convincingly argues that it is impossible to attribute delayed positions specifically to spoken language or substandard varieties (cf. Junghanns 2021: 178). Recall from Section 4.4.2 that 3P, 4P, and 5P cliticisation is attested in Czech for both independent and embedded clauses. Importantly, Junghanns's (2021) data points are taken from Czech high literature (authored by Jaroslav Hašek and Milan Kundera among others) and do not represent informal speech. Therefore, one cannot maintain the conception that delayed clitic placement equals Clitic Third in subordinate clauses of colloquial speech.

Another illustration of the issue under discussion is brought up by Kolaković et al. (2022: 311–326 [Ch. 14]), while examining clitic climbing in Croatian. They find that the occurrence or non-occurrence of clitic climbing does not only depend upon structural conditions (i.e. the distinction between raising and subject control verbs, the presence/absence of a matrix reflexive clitic), but also upon register.<sup>457</sup> On the one hand, clitic climbing occurs almost without exception in standard Croatian. In other words, it is close-to-obligatory. On the other hand, embedded clitics remained in situ significantly more often in non-standard Croatian, although climbing is still the predominant structure.<sup>458</sup> From this it follows that clitic climbing cannot be regarded as an obligatory syntactic construction, which appears once the relevant structural conditions are met. On the contrary, clitics may remain in situ in the same syntactic environments, which license clitic climbing. This finding suggests two things: first, clitic climbing is essentially optional in structural, i.e. syntactic terms. Second, the seemingly obligatory character of clitic climbing is an artifact of prescriptivism, which does not comply with the linguistic reality. The findings also imply that there are conditions on the presence and absence of clitic climbing in Croatian (or BCMS), which are still not well-understood.

From the previous discussion it follows that particular data sets or constructions must not be neglected on the basis of an – obviously unmotivated – impression of “substandardness” or “spokenness”. The reliance on standard language data may obscure the linguistic facts of mental grammar, because the standard norm favours rigid, well-behaved constructions over the dynamics of the linguistic system internalised by the native speaker. Keep in mind that grammar research aims to model native speakers' linguistic competence, rather than their success in adhering to the prescriptive norms of the standard language.<sup>459</sup>

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<sup>457</sup> See Jurkiewicz-Rohrbacher et al. (2017a: 188) for a similar note on Serbian.

<sup>458</sup> Kolaković et al. (2022: 324) provide the following figures: 96% and 94% likelihood for clitic climbing under raising and subject control verbs respectively in standard Croatian. Roundabout 85% likelihood for CC under both raising and subject control verbs in colloquial Croatian. The quantitative differences between standard and colloquial Croatian are reported to be statistically significant (Kolaković et al. 2019: 324).

<sup>459</sup> Pescuma et al. (2023: 25–26) cite an observation, which is interesting towards the present discussion. In an eye-tracking pilot study, respondents were asked to read sequences of sentences representing different registers in terms of more vs. less formal. Some of the experimental tokens were additionally manipulated in that they involved mismatches of formality (i.e. occurrence of a target sentence of a deviant register). Not only was the reading time of the mismatches systematically longer than in case of matches, but the reading time also rises with increasing formality. One could hypothesise that this latter pattern may reflect the nativeness of informal speech vs. the doubtful nativeness

In this respect, a point made by Boden (2008: 1955) about Noam Chomsky's attitude towards corpora and linguistic intuitions is of interest: She cites that Chomsky criticises Thorstein Veblen's usage of the string *performing leisure*, because the verb (*to perform*) were not able to take mass noun complements.<sup>460</sup> Chomsky's argument goes that he knew about the grammatical incompatibility of *perform* and *leisure*, because he was a native speaker of English. Boden (2008: 1955) ironically comments: "Veblen, seemingly, wasn't." As a matter of fact, Veblen was born in the United States to Norwegian immigrants and passed the entire US educational system.<sup>461</sup> Admittedly, the collocation *perform leisure* rarely appears in a simple Google search. Usually, the collocation *perform leisure activities* and comparable strings occur. But one can still find syntagms of the kind rejected by Chomsky, which are produced by native speakers of English, as exemplified in (6.15).

- (6.15) a. [...] young people **perform leisure** in those spaces available and accessible to them.  
(K. King 2009: 153)
- b. **Performing Leisure**, Making Place: Wilderness Identity and Representation in Online Trip Reports (Williams & Champ 2015)
- c. The Tourist State: **Performing Leisure**, Liberalism, and Race in New Zealand  
(Werry 2011)
- d. [...] and perceptions of their ability to **perform leisure** in retirement [...]  
(Lee et al. 2020: 152)
- e. **Performing leisure** as labour in the Queen of Spas: [...]  
(Freeman 2021)
- f. It argues that **performances of laborious leisure** [...] reveal the resort as a distinctively modern place [...]  
(Freeman 2021: 1)

The data points show that the VP *perform leisure* (or the related NP *performance of leisure*) does occur in the performance/parole of native speakers of English in different areas of the English-speaking world (Australia, Canada, UK, USA). One may conclude that the collocation *perform leisure* is rare, but it belongs to the linguistic repertoire of native speakers of English, and it is

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of formal language. Alternatively, differences in reading time might be caused by frequency, thus, hinting at the predominance of informal speech in the native speakers' linguistic experience.

<sup>460</sup> In fact, Veblen (1899) uses the collocation of verb *perform* + object *leisure* three times in his treatise *The Theory of the Leisure Class*: *perform leisure* (1x), *perform vicarious leisure* (1x), *perform a vicarious leisure* (1x). Furthermore, the corresponding nominal collocation of noun *performance* + PP *of leisure* occurs in the treatise: *performance of leisure* (4x), *performance of (a) vicarious leisure* (3x), *performance of (a) conspicuous leisure* (2x). Veblen's complete text is freely available, e.g., from Project Gutenberg: <https://www.gutenberg.org/files/833/833-h/833-h.htm>

<sup>461</sup> I am not aware, whether Veblen was naturally bilingual, but there cannot be any doubt that he was a native speaker of English.



appropriate to consider the string grammatical, then. Moreover, note that the data are obtained from academic writing and that Veblen's (1899) treatise belongs to the same genre. It could be the case that we encounter a register-dependent collocation, whose occurrence may be constrained by multiple, extra-grammatical factors. Bear also in mind that academic publications standardly undergo review and revision. It seems that neither authors nor reviewers nor editors took issue with *perform leisure* and related constructions in (6.15).<sup>462</sup> Boden's (2008: 1955) discussion hints at the issue of prescriptivism, which has two facets. On the one hand, linguists themselves might – perhaps unconsciously – obey a prescriptive authority. This may be an institution like a language regulator, a printed source like a grammar book or dictionary, or some sort of authoritative idea (or dogma) like a (pseudo-)scientific argument or generalisation, e.g. the logical affiliations between parts of speech or the distinction between countable nouns and mass nouns as objects of certain verbs.<sup>463</sup> On the other hand – and this seems to be what Boden (2008: 1955) has in mind, while claiming that Chomsky regarded his own intuition as “paramount” – linguists may act as authorities themselves. Native speakers might take their own linguistic intuitions as gold standard, while simultaneously overlooking their own idiosyncrasies and individual preferences. Such an attitude could also emerge through the institutionalised education in school, which imposes the standards of the literary standard language on students. As a consequence of such an internalised form of prescriptivism, the native speakers tend

<sup>462</sup> Perhaps, Chomsky's generalisation was wrong in several respects. What might be more important about *leisure* is that it is an abstract noun. In fact, the subcategorisation frame of the verb (*to*) *perform* is quite diverse with respect to its objects. It covers both count nouns and mass nouns. For instance, one can (a) *perform experiments*, (b) *perform remorse*, (c) *perform authority*, or (d) *perform music* (examples [a]–[c] are taken from <https://en.wiktionary.org/wiki/perform>, accessed 5 Jul 2023). Most mass nouns may become countable under different semantic conditions, which also seem to hold for *leisure*. It appears that *perform* preferably takes abstract noun objects irrespective of whether they are count nouns or mass nouns. The noun *leisure* is a good fit as a direct object. Perhaps, the issue with *leisure* could be a matter of frequency.

<sup>463</sup> An argument like the one by Chomsky was put forward by essayist Sebastian Sick with respect to the collocation *Sinn machen* ‘make sense’, which he considers an unwelcome Anglicism. Sick claims that the use of *Sinn machen* is corrupted German, for the verb *machen* ‘(to) make’ originally meant ‘(to) knead’ and necessarily refers to material objects that can be touched, but *Sinn* ‘sense’ denotes an abstract concept that cannot be touched. The linguist André Meinunger deconstructs Sick's argument convincingly in his popular scientific reply. Meinunger (2008) argues that the verb *machen* perfectly combines with other abstract nouns in German, e.g. *Spaß* ‘pleasure’ (i), *Freude* ‘joy’ (ii), *Sorgen* ‘worries’ (iii), or *Angst* ‘fear’ (iv). Furthermore, he points out that the etymological portion of the argument is weak, because English (*to*) *make* is a cognate of German *machen* and should not be able to combine with abstract nouns like *sense* or *love* by the same logic.

- (i) Lesen macht mir Spaß.  
reading.NOM make.PRS.3SG me.DAT pleasure.ACC.SG  
‘I enjoy reading.’
- (ii) Er wollte ihr eine kleine Freude machen.  
he.NOM want.PST.3SG her.DAT a.ACC.SG.F small.ACC.SG.F happiness.ACC.SG make.PRS.3SG  
‘He wanted to give her a small treat.’
- (iii) Das macht mir Sorgen.  
that.NOM make.PRS.3SG me.DAT concern.ACC.PL  
‘I'm worried about it.’
- (iv) Du machst mir Angst.  
you.NOM.SG make.PRS.2SG me.DAT fear.ACC.SG  
‘You are scaring me.’

(Ger; constructed)

to disprefer constructions departing from the language's standardised norm (cf. Vogel 2019: 73–75 for a comparable argument).

To conclude, the focus on standard language or some related abstract concept cannot but yield an incomplete or even incorrect picture of the grammatical structures available in a language. In order to decrease the bias which stems from implicit or explicit normative notions, I follow scholars like Guz & Jędrzejowski (2020) and utilise non-standard data in the form of spoken corpora. The arguments presented in this section are not new and I will close it with a quote from Townsend's (1990) survey of Common Czech:

“Part of the problem is what I consider a misunderstanding of what Literary Czech is. It is a written and not a spoken language. The notion purveyed by many Bohemists that there is a spoken Literary Czech (usually referred to as ‘hovorová čeština’) is not only a contradiction in terms, it is false. It induces the misconception that spoken Czech is a corrupted or colloquialized version of Literary Czech, which is the basic language of the Czech lands, whereas the opposite is true. The basic language of the Czech lands is *obecná čeština*. And the *obecná čeština* of Prague and its environs is the basic spoken Czech which is being willy-nilly regularized (if so far only colloquially) by centripetal forces, which include, of course, substantial influences from Literary Czech. There is nothing unusual in this; the literary language of a literate people always influences their speech, and since the Czechs are a very literate and well-educated people, it influences their speech a great deal. But it can neither replace it nor form the basis of it.” (Townsend 1990: 11)

## 6.4 Spoken language data: Arguments in its favour

It was suggested in the previous Section 6.3 that grammar research has preferred the prescriptive standard norm over colloquial speech giving rise to an imbalanced database. This is why the present thesis focuses on spoken language data, which was pointed out to the reader as early as in Chapter 2. In doing so, the study treads new paths, which is corroborated by Dobrushina & Sokur's (2022: 78) remark that “spoken corpora still have not occupied their niche in linguistic analysis.” Grammar research has not paid particular attention to spoken language or colloquial speech, yet. However, research that employs introspection as a data collection method tends to blur the distinctions between the standard and colloquial language by intermingling statements about standard grammar with intuitive judgements about colloquially acceptable forms. This is methodologically problematic, especially if colloquial data is not given appropriate space. Therefore, spoken language corpora are the preferred means, if one takes structures of colloquial speech into consideration. This is in

accordance with the objectives of grammar research, if one takes Broekhuis's (2010) remark into consideration that

“there seems to be a consensus that at least those utterances that are produced frequently in colloquial speech and are judged acceptable by native speakers should be contained in the set of grammatical sentences, and that utterances that occur infrequently (or never at all) in colloquial speech and are judged unacceptable by native speakers should be contained in the set of ungrammatical sentences.” (Broekhuis 2010: 185)

There are several arguments in favour of the use of spoken language data besides the more traditional approach of scrutinising written language data.

1. Pretheoretically, spoken language must be viewed as the quantitatively prevalent mode of language and interpersonal communication (cf. Pęzik 2012: 37).<sup>464</sup>
2. Spoken language is the primary language mode (cf. Koch & Oesterreicher 1985: 25–26, with references). That is, language is primarily transmitted and perceived through the acoustic-auditory channel (cf. Hockett 1963: 7).<sup>465</sup> Speech is acquired before script, which is a cultural technique (Ger. *Kulturtechnik*) secondary to language (cf. Lehmann 2013: 187).
3. The reliance on native speakers' intuition qua grammaticality judgement has been the dominant paradigm in linguistic research since the dawn of Generative Grammar in the 1950s and is still methodologically widely accepted. The motivation for this approach comes from the goal to model linguistic competence, i.e. the native speaker's mental grammar, which results from natural language acquisition in childhood. From the perspective of language acquisition, the primary – or perhaps only – data source available to children for grammar and lexicon construction (called *primary linguistic data*, PLD, in Generative Grammar or *baseline* in heritage linguistics) is spoken data. Hence, spoken data necessarily should be treated as a decent research object for grammar research.
4. If a native speaker's mental grammar is built/acquired by abstraction from spoken input data, then it is reasonable to assume that speech exhibits linguistic features, which are relevant for grammar modelling, which I label *competence through performance* here (cf. Pullum &

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<sup>464</sup> Pęzik (2012: 37) states '[...] that it is spoken language, which constitutes the largest variety of language in terms of the quantity of naturally occurring utterances.' (orig.: “[...], że to właśnie język mówiony stanowi największą pod względem ilości naturalnie występujących wypowiedzi odmianę języka.”)

<sup>465</sup> Admittedly, I abstract from sign languages, which have manual-visual modality and, thus, are transmitted and perceived through a completely different channel. I also omit the gestural and mimic activities accompanying spoken language.

Scholz 2002 for the relevance of empirical data for grammatical structure in their critical assessment of the so-called *poverty of the stimulus argument*).

5. If one assumes that linguistic performance in the form of speech is mapped from linguistic competence in some way (details put aside), then one may infer properties of linguistic competence from spoken data – hence, one approaches *competence through performance* (cf. Stefanowitsch 2020: 3–4).
6. In contrast to written language, spoken language is not or less influenced by standard grammar and normative/prescriptive rules. This is most likely in spontaneous informal speech among family members and close friends. Most importantly, spoken language is not subject to revision, while most written texts are edited before printing irrespective of genre and medium (fiction, newspapers etc. – even blog posts and forum posts may be revised, in principle).

The arguments put forward so far imply that shifting emphasis towards spoken data and colloquial speech increases data quality and leads to more realistic generalisations in grammar research. If linguists are truly interested in the psychological or cognitive capacities underlying natural human language, then there is no way around colloquial speech or other forms of non-standard language. This reasoning applies not only to grammar research but also to linguistic typology. As pointed out by Nomachi (2019: 453), areal typology has focused on the examination of “standard languages or well-established dialects/language varieties or generalized language types” but neglected “the micro-orientation on specific non-standardized languages”. This discrepancy is still present in current typological surveys, which are based on coarse-grained linguistic features. The problem is that this reproduces and reinforces clichés rather than providing substantiated claims about structures in natural languages. For example, Haspelmath (2023) misses variation in BCMS cliticisation discussed at length in Kolaković et al.’s (2022) comprehensive study. He thus uncritically reiterates the picture of BCMS as an inflexible second position clitic language. Variation appears to be a cross-linguistic matter under this perspective, which takes languages as monoliths. In contrast, the present thesis’ research agenda acknowledges that variation is also a language-particular matter that deserves study in its own right, while still being compatible with cross-linguistic comparison. A variationistically inspired typology is possible and desirable, therefore.

## 7 Utilised databases

### 7.1 Introductory remarks

The motivation for using corpus or attested data for the examination of clitics in West Slavic resembles the one by Russi (2008) for Italian. She notes “that this type of data displays a broad variation of parameters (lexical, syntactic, semantic, and contextual) with regard to the linguistic structure under investigation.” (Russi 2008: 18) With regard to the finiteness/incompleteness of corpora, Russi (2008: 18) concludes that “corpus data needs to be complemented by constructed data, evaluated through native speakers’ intuitions.” From the discussion in Chapter 6 it follows that, if “native speakers’ intuitions” means introspection in the sense of Section 6.1, the present thesis takes a diametrically opposed perspective.<sup>466</sup> There is a rich body of introspective, i.e. constructed data dominating the scientific discourse on clitic phenomena in Slavic. This data is, however, fragmentary and not replicable and therefore needs to be supplemented and expanded by empirically aggregated data.

A certain amount of corpus data has been cited until now for different languages that have been mentioned and discussed in the previous chapters. Chapter 7 presents the corpora that were exploited for the empirical core of the present study, on which Part III is grounded. From this it follows that I will describe only the databases for the six lects scrutinised here, i.e. Czech, Polish, Goralish, Slovak, Silesian, and Kashubian. The reader will not find any information about other corpora from which I cite data, e.g. Croatian or (Carpatho-)Rusyn. The utilised corpora will be presented in a concise manner with the aim of providing readers only with as much information as necessary, whereby I will also comment on problems that I encountered while preparing the thesis.

### 7.2 Czech: Český Národní Korpus (ORAL)

The Czech data were obtained from the ORAL subcorpus (oral\_v1) of the *Český Národní Korpus* ‘Czech National Corpus’ (ČNK). ORAL comprises informal spontaneous speech in conversational settings (Kopřivová et al. 2017a, 2017b; Lukeš et al. 2015). The data were elicited between 2002 and

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<sup>466</sup> Russi (2008) does not specify, whether she means introspection or psycholinguistic data elicitation by “native speakers’ intuitions”. However, since her approach is “maximally data driven” (Russi: 2008: 16), it is likely that the intended meaning of *intuition* refers to a psycholinguistic data acquisition mode.

2011 yielding 582 hours of speech and a corpus of 5,368,391 word tokens.<sup>467</sup> As with every subcorpus, data can be queried via ČNK's concordance tool *KonText* (Machálek 2020). An asset of the ČNK is the ability to search the corpus by aid of the Corpus Query Language (CQL), which allows processing queries of word forms, lexemes, and morphosyntactic classes simultaneously. Note that CQL allows only to query a specified linearisation. Searching for particular linguistic structures or patterns, it might be necessary to iterate queries with elaborated formulations in order to aggregate data. It is worth mentioning that scholars can purposefully search a specified corpus for patterns through CQL queries and do not have to rely on mere chance finds in a text.<sup>468</sup> As is standard for electronic corpora, KonText returns the query results in a concordance per default, which is known as key-word-in-context (KWIC) in contemporary Corpus Linguistics. Alternatively, it is possible to switch the view settings to a sentence-based format, according to which the search string appears in its clausal domain only without additional left or right context. The default view returns a non-formatted paragraph. Depending on the type of corpus – written or spoken – the default view may be changed to a view that displays the actual structure of a text or discourse. The “Formatted text” view mirrors the structure of the written source in line and paragraph breaking (e.g. a newspaper, fictional prose, or a poem). For spoken corpora, the “Speech view” is available instead, which displays individual utterances assigned to the respective (anonymized) speakers.

Query results might be downloaded as a tabular file in one of four different data types: CSV, XLSX, XML, and TXT. Data files are provided either in KWIC or in sentence style. There is a one-click download option, which gives a table limited to 10,000 result utterances. It is possible to customise the data exportation and exactly define the range of lines one wishes to obtain (i.e. more or less than 10,000 hits). I chose the one-click option for the sake of convenience, but the query results were shuffled by the search engine automatically.

Another interesting and very helpful feature of the ČNK is the possibility to have the query output supplemented by the morphosyntactic analysis of the tagger – either as a suffix to the token or as an aligned gloss in a separate line (the same option is available with lemmatisation). Thus, every segment (including orthographic words and punctuation) is assigned an annotation tag of variable detail. Grammatical tags may be returned in every subcorpus, which is morphosyntactically analysed. The annotation tags can be extracted and downloaded with the tabular data file. In this case the tags are always suffixed to the individual tokens. I use this possibility for the purpose of the thesis for

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<sup>467</sup> Cz: <https://wiki.korpus.cz/doku.php/cnk:oral>, Eng: <https://wiki.korpus.cz/doku.php/en:cnk:oral> (accessed 6 Apr 2024).

<sup>468</sup> Before large electronic corpora were available, the encounter of certain constructions was a matter of chance. For instance, it was not possible for a single scholar to search the entire Rigveda for all clitic occurrences. Therefore, a purposeful examination of delayed clitic placement was practically impossible. In contrast, a simple query in a contemporary electronic corpus may return all clitic occurrences within a reasonable amount of time. Scrutinising delayed placement can thus be handled in a study devoted to clitics. I am thankful to Tim Felix Aufderheide (p.c.), who discussed the issue of chance finds in some branches of Indo-European linguistics with me.

basic syntactic analysis based on the parts-of-speech analysis.<sup>469</sup> As the tags are based on the ČNK annotation scheme, some re-adjustments might be necessary for individual cases. This is also true in cases where the annotation might contain some inaccuracies due to ambiguities. However, I take the analysis provided by the tagger at face value as long as there is no obvious error.

Keep in mind that annotation-based corpus linguistics is still prone to error despite a high level of technical elaboration of contemporary electronic corpora. Kolaković et al. (2022: 287) discuss this issue for BCMS and illustrate it by means of the examples in (7.1) from the Croatian web corpus (hrWaC).<sup>470</sup> There are 2 erroneously tagged tokens out of 8 in (7.1a) (highlighted by underlining). Kolaković et al. (2022: 287) note that “the relative pronoun *koju* ‘which’ is misclassified as an indefinite pronoun” in this example. Furthermore, “the verbal CL *je* ‘is,’ which has the function of copula, is [...] misclassified as [...] the third person singular feminine personal pronoun in the accusative.” Example (7.1b) exhibits 3 erroneously tagged tokens out of 7. “The tagger assigned the wrong case and gender to the demonstrative [...] *onoga* ‘that’”, while “the relative pronoun *koga* ‘which’ is misclassified as indefinite.” Finally, “the verbal CL *si* [...] is falsely interpreted as the reflexive CL *si* in the dative.” (Kolaković et al. 2022: 287)

- (7.1) a. Prva stvar koju trabata zapamtiti je ne paničariti.  
 first thing which have\_to.2PRS remember.INF be.3SG NEG panic.INF  
 Mlofsn Ncfsn Pi-fsa\* Vmr2p Vmn Pp3fsa\* QZ Vmn  
 ‘The first thing you have to remember is not to panic.’
- b. Onoga, koga treba banirati si ipravo ti [...]   
that which have\_to.3PRS ban.INF be.3SG exactly you  
Pd-nsg\* Pi3m-a\* Vmr3s Vmn Px-sd\* Rgp Pp2-sn  
 ‘You are the one who should be banned ...’ (Hrv; Kolaković et al. 2022: 287)

Errors may be encountered in the ČNK, too. For instance, the CQL query in (7.2) gives among others the example in (7.3). The query searches for strings of punctuation < adjective in the accusative < short personal pronoun < noun in the accusative. In principle, such a simple corpus query may be seen as an instantiation of searching clause-initial discontinuous noun phrases.<sup>471</sup> However, the utterance in (7.3) is misanalysed, although it looks like an instance of the target structure at first

<sup>469</sup> Since KWIC and Sentence view have different advantages and disadvantages, I pulled both visualisation types for each data set obtained from the ČNK. I assigned each view an individual Excel map within a shared XLSX file as per query.

<sup>470</sup> I cite the examples from Kolaković et al (2022: 287) exhaustively and provide all four lines of the original for the sake of clarity: (i) object language examples from the corpus, (ii) linguistic interlinear gloss, (iii) annotation tags of the corpus’s POS tagger, and (iv) English translations line.

<sup>471</sup> Whether such constructions are possible in Czech, at all, is a different matter.

glance. The example does not contain a string of adjective < short pronoun < noun, but adjective < noun. What is analysed as an individual token *ho* ‘him.ACC’ and annotated as a personal pronoun, is the accusative case ending of the animate masculine declension, in fact. That is, there is no alleged discontinuous noun phrase *zbité ho chlapce*, but the plain NP *zbitého chlapce* ‘beaten boy’.

(7.2) [tag="Z.\*"] [tag="AA...4.\*"] [tag="PH.\*"] [tag="N...4.\*"]

(7.3) Zbité                      ho chlapce              převezla              sanitka                      k  
 beaten.ACC.SG.M.ANI              boy.ACC.SG    bring.PST.SG.F    ambulance.NOM.SG    towards  
 ošetření                      do nemocnice.  
 treatment.DAT.SG    to    hospital.GEN.SG  
 ‘An ambulance took the beaten boy to the hospital for treatment.’              (Cz; ČNK syn\_v8)

In sum, the query output needs further examination and cannot be taken at face value, even in case of an elaborate annotation scheme and a far developed tagger. Modern electronic corpora may provide linguists quite easily with large amounts of data due to automatisation, but researchers are still responsible for the qualitative evaluation of the queried results, the detection of tagging errors, and the elimination of the respective data points from their analysis by hand.

As the scrutinised corpus is the subcorpus of spoken language, it is noteworthy that the oral data are not only available as transcripts, but also as audio. The transcriptions are aligned with the corresponding part of the audio file, so that researchers may actually hear Czech speech. However, the audio track is not retrievable for the entire corpus, as a subset of the data is only available as sole transcript without an accompanying sound file. Obviously, such data points disqualify for a phonetic analysis. I provide two such examples in (7.4) for the sake of completeness. Finally, a sound file’s eligibility for phonetic analysis depends on its quality, which admittedly is very low sometimes.

(7.4) a. v tý italštině, to já ne-vim                      jestli sem              ti  
           in this Italian    TO I.NOM NEG-know.PRS.1SG if    AUX.1SG you.DAT  
           vyprávěl              když sme stopovali                      po Itálii [...]  
           tell.PST.SG.M when AUX.1PL hitch-hike.PST.PL.M over Italy  
           ‘With regard to Italian, I don’t know, if I told you, when we were hitch-hiking in Italy  
           ...’



b. ale co to je a jak - v podstatě  
 but what.NOM this.NOM.SG.N be.PRS.3SG and how in quintessence.LOC.SG  
 nikdo mi to ne-vysvětlil  
 nobody.NOM me.DAT this.ACC.SG.N NEG-clarify.PST.SG.M  
 ‘but what it is and how ... – essentially no one explained it to me’ (Cz; ČNK oral\_v1)

### 7.3 Polish: Narodowy Korpus Języka Polskiego (PELCRA)

The Polish data were retrieved from the spoken subcorpus of the *Narodowy Korpus Języka Polskiego* ‘National Corpus of Polish’ (NKJP) (Przepiórkowski et al. 2012). The NKJP is an exclusively synchronic reference corpus of contemporary Polish. It comprises around 1.8 billion annotated segments in total, whereby the balanced part of the corpus covers ca. 300 million segments. Speech data accounts for approximately 10% of the balanced NKJP, i.e. 30 million words (Pęzik 2012: 38–39). However, there are three different types of speech data, namely medial, conversational, and other. The latter one is based on stenographs from parliamentary sittings and represents the majority of spoken data in the NKJP (ca. 27.2 million segments = 9.1% of balanced NKJP). Conversational data amounts to roundabout 1.9 million segments (i.e. 0.6% of balanced NKJP), while another 900,000 segments (i.e. 0.3% of balanced NKJP) represent speech in media (Pęzik 2012: 39). The Polish database of Part III consists primarily of conversational data and involves medially transmitted speech only to a small extent.<sup>472</sup>

The NKJP arose out of independent corpus projects, which have been united to create the first reference corpus of the modern Polish language. Despite originally different plans, work on the corpus has not been continued after its release in 2011 (Kieraś et al. 2024). Therefore, it has not been updated since 2010, such that the database is older than 10 years at the time the present thesis is written. This problem may be neglected for my research question, as the data is still recent enough to investigate the grammar of clitic pronouns in contemporary Polish.<sup>473</sup> There are sufficiently large corpora of Polish with more recent data, in fact. On the one hand, *MoncoPL* (Pęzik 2020) provides up-to-date web-based language data. On the other hand, the *Korpus Współczesnego Języka Polskiego* (KWJP) ‘Corpus of the contemporary Polish language’ (Kieraś et al. 2024) was published in 2023 as a balanced reference corpus covering the second decade of the 21st century (thus continuing the

<sup>472</sup> I am not able to quantify the amount of medial speech in the analysed data sample of Polish.

<sup>473</sup> Recall that ČNK’s ORAL data have not been updated since 2011, too.

NKJP's agenda). Nevertheless, since both corpora are restricted to written media, they are not suitable for the present thesis' research agenda, which is grounded on spoken language data.<sup>474</sup>

The NKJP has two different search engines, Poliqarp and PELCRA, which (should) have access to the very same database.<sup>475</sup> Both search engines can search the entire NKJP. However, they differ with respect to several features and technical solutions. The Poliqarp search engine was primarily developed for the purposes of the IPI PAN corpus (Janus & Przepiórkowski 2007; Przepiórkowski 2004).<sup>476</sup> The query results are displayed as KWIC. Additionally, bigger portions of a particular result's environment can be inspected after ticking the respective output line similarly to ČNK's KonText engine. However, a small window pops up next to the mouse cursor in the ČNK, whereas Poliqarp prints the respective information at the bottom of the web page, which is an impractical solution, especially in case of extensive query outputs.

There is no user interface to delimit corpus queries to a subset of the corpus data (e.g. genre or date of publication). The Poliqarp search engine enables users only to choose from a small set of predefined subcorpora.<sup>477</sup> Further specifications need to be entered in the search box together with the actual query. Poliqarp supports wordform-, lexeme-, and tag-based queries, which may be combined within a query. The query language is based on CQP from the University of Stuttgart but was made "from scratch" (Przepiórkowski & Wilk 2011), with expressions different from ČNK's CQL syntax.<sup>478</sup> Basically, ČNK's expressions *word*, *lemma*, and *tag* correspond to the NKJP's *orth*, *base*, and *pos* respectively.

The Poliqarp search engine does not seem to be able to provide more than 1,000 results even if there is more data available in the corpus. This is especially problematic, if one wants to increase the empirical database on a particular kind of query, because it seems that reiterating corpus queries yields identical results.<sup>479</sup> That is, a particular corpus query gives exactly the same output each time it is conducted, although the search engine is supposed to randomise query results by default. The tick box in the settings section needs to be unchecked and checked again, in order to get actually

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<sup>474</sup> I also abstract away from the fact that the KWJP was released only after I finished data collection for my thesis.

<sup>475</sup> The parenthesised *should* implies some sort of restriction, which will be discussed in the following.

<sup>476</sup> The acronym IPI PAN stand for *Instytut Podstaw Informatyki Polskiej Akademii Nauk* 'Institute of Computer Science Polish Academy of Sciences' based in Warsaw.

<sup>477</sup> These are (i) the balanced NKJP subcorpus, (ii) the full NKJP corpus, (iii) a manually annotated corpus, (iv) the full IPI PAN corpus, (v) an IPI PAN corpus sample, and (vi) the Frequency Dictionary Corpus.

<sup>478</sup> See Przepiórkowski & Wilk (2011) statement on Poliqarp's help site, Section 3 "Query Language", Footnote 1 (<https://nkjp.pl/poliqarp/help/ense3.html#x4-50003>, accessed 27 Nov 2024): "Although the query syntax of Poliqarp is based on that of CQP, Poliqarp was implemented from scratch within the current project and it does not contain any CQP code." (<https://nkjp.pl/poliqarp/help/en5.html#fn1x0>, accessed 27 Nov 2024).

<sup>479</sup> Another problem, I would like to address, is that the number of results seems to be inaccurate in some instances. When searching the string *się mu* a total number of 40 results was displayed. Refreshing the page yielded an amount of 510 results, another refresh gave a total of 810 results. A repetition of the procedure with cleared browser cache yielded 60, 360, 461, and 1000 results after each iteration. In another cycle, I got quantities of 40, 600, 720, and 1000 results. These problems did not emerge, when first changing the settings before processing a query. I opened the NKJP website in the Firefox web browser, but did not check, whether this behaviour would be replicable in a different browser.

randomised results. Accordingly, the search engine returns the very same results each time the corpus website is opened anew without saving cookies. This is a troublesome situation for first-time corpus users. With respect to morphosyntactic tagging, I should mention that the Poliqarp engine returns a word-by-word analysis of the queried string by default with the grammatical tags being suffixed to the tokens. However, tags are only provided for the queried token or string in the KWIC box. The segments belonging to the left and right context are not provided with a visualisation of the morphosyntactic analysis. The morphosyntactic tags are also implemented in the downloadable tabular file.

The PELCRA tool has its own corpus query architecture and provides users with a concordancer, a collocator, and the speech tool, which was developed for spoken data only.<sup>480</sup> The output paragraphs provided by PELCRA's speech tool are more extensive than the ones from Poliqarp. This asset is of special interest for the conversational data of the spoken component, as several lines before and after the search string's paragraph – usually uttered by different speakers – are displayed in PELCRA's speech tool (but not in the concordance tool).<sup>481</sup> The PELCRA search engine is not based on CQP, but uses an individually developed system, which superficially shares some similarities with common web search engines (using regular expressions). The PELCRA search engine has a significantly higher upper limit in terms of graphical display: a page may contain up to 10,000 results of a given query, but it is possible to switch to subsequent pages in order to view the remaining results in case there are more than 10,000 output tokens. This is not possible with Poliqarp. A drawback of PELCRA in comparison to Poliqarp is the impossibility of randomising the query output. The output of a query is returned in such a manner that utterances belonging to the same conversation are arranged subsequently in chronological order. Randomisation must be done in an additional working step after obtaining (downloading) the data.<sup>482</sup>

It is generally possible to extract query results from the NKJP. The Poliqarp search enables the researcher to download the results as HTML or XLS file in a KWIC style.<sup>483</sup> Things are more complicated with the PELCRA's engines depending on the type of data. PECLRA's concordance tool, through which the balanced or the full NKJP is available in principle, allows for a direct XML

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<sup>480</sup> The collocator automatically extracts collocations. There is also a “word of the day”, whose function is not supported anymore. A help page describes the corpus and PELCRA's functions, but it is available in Polish only.

<sup>481</sup> This type of presentation resembles ČNK's “Speech view” for spoken data.

<sup>482</sup> In contemporary spreadsheet applications (e.g. Microsoft Excel), randomisation of data points may be achieved by a workaround procedure only. One has to add an additional column to the sheet and use a function returning random values between 0 and 1. Then, the tabular lines may be sorted increasingly or decreasingly with respect to the random column, which may be deleted afterwards.

<sup>483</sup> The respective functions are to be found at the bottom of the query page. After choosing the appropriate format from a dropdown list, one may obtain the files pressing *Extract*. Note that due to some sort of bug, the dropdown-menu and the extract-button are sometimes absent in the first place, as I found with the query strings *się mu* and *się go* with default settings, i.e. 10 printed results per page, but not with *się*. It turned out that refreshing the result page renders the menu and button visible again. Note that refreshing the web page does not conduct another corpus query, such that this circumstance is not a big issue.

file download. On the other hand, the speech tool does not provide users with an option to download data. This issue is significant insofar as I observed that PELCRA's concordancer and speech tool sometimes return different results for identical queries within the spoken language component. Although one may restrict the query in the concordance tool to spoken language only (selecting *kanal\_mowiony*) in the advanced settings area (Pol. *zaawansowane*), one does not get the same pool of (potential) results for the identical queries in the concordancer and the speech tool.<sup>484</sup> Table 7.1 below shows that the speech tool yields more results searching for the types *cię*, *go*, *mi*, *ci*, *mu*, *się*, and *se* than the concordancer restricted to spoken language. As a consequence, spoken data would have to be extracted manually from the graphical user interface of the speech tool, if one wished to access the full spoken database. I refrained from doing so for the sake of convenience and chose the time-saving option of extracting spoken data from the concordancer by downloading. I do not suppose this approach does any harm by influencing the research results in subsequent analyses, as the quantitative differences between the two data collections are scientifically justifiable. In other words: the output by the concordancer is quantitatively sufficient for addressing my research question, even though the relative difference between both collections runs up to roundabout 10%.<sup>485</sup>

Table 7.1: Quantitative output differences between PELCRA's concordance and speech tool.

| Query      | Absolute number of output tokens                     |                    | Difference in % |
|------------|------------------------------------------------------|--------------------|-----------------|
|            | PELCRA concordancer<br>(spoken channel, full corpus) | PELCRA speech tool |                 |
| <i>cię</i> | 1,079                                                | 1,192              | 9.48%           |
| <i>go</i>  | 2,839                                                | 3,184              | 10.84%          |
| <i>mi</i>  | 8,666                                                | 9,313              | 6.95%           |
| <i>mu</i>  | 2,310                                                | 2,505              | 7.78%           |
| <i>ci</i>  | 5,810                                                | 6,099              | 4.74%           |
| <i>się</i> | 30,004                                               | 32,655             | 8.12%           |
| <i>se</i>  | 1,022                                                | 1,050              | 2.67%           |

I observed similar inconsistencies when comparing query results of Poliqarp and PELCRA's concordancer. Although the IPI PAN subcorpus (= written corpus) can be searched through both search engines, query outputs can differ. For one thing, when I was searching for the word string *go*

<sup>484</sup> Note that this matter occurs independent of whether one chooses the balanced or the full corpus, which in each case yields a different number of results in the concordances – none of which corresponds to the amount of results in the speech tool. The situation persists even in case of the restriction to spoken data only.

<sup>485</sup> Keep in mind that the quantity of *ci*-tokens in Table 7.1. is disproportionately higher than for the other pronominal types queried due to the homonymy of the short pronoun *ci* 'you.SG.DAT' and the demonstrative *ci* 'this.PL.NOM.VIR'.

*mu* in the balanced corpus, Poliqarp displayed three positions (7.5), whereas PELCRA returned only two of them (7.6).<sup>486</sup> For reasons unknown to me, the PELCRA engine does not display example (1a). Moreover, when extending the query to the full NKJP corpus in the Poliqarp engine, then the output yields a total of 37 results, but example (7.5a) is missing as in case of PELCRA.

On the other hand, the PELCRA concordancer seems to have issues handling an amount of data over a certain – yet unknown – threshold. If the potential query output contains too many utterances, the query will not be conducted at all and the user will be asked to repeat using a more specific query. This circumstance makes it sometimes necessary to split queries into several sub-queries.

- (7.5) a. [...] żeby Tremaine nie mogła **go mu** odebrać.  
           that Termain.NOM NEG can.PST.SG.F him.ACC him.DAT take\_away.INF  
           ‘...that Tremain could not take it from him.’
- b. (Szpital Powiatowy otrzymał certyfikat  
     hospital.NOM.SG county.ADJ.NOM.SG.M receive.PST.SG.M certificate.ACC.SG  
     jakości ISO 9001.) Nadała **go mu** norweska  
     quality.GEN.SG ISO 9001 bestow.PST.SG.F him.ACC him.DAT. Norwegian.NOM.SG.F  
     firma Det Norske Verutas.  
     company.NOM.SG Det Norske Verutas  
     ‘(The county hospital received the quality certificate ISO 9001.) The Norwegian company  
     Det Norske Verutas awarded it [= the certificate] to it [= the hospital].’
- c. Kilkunastu kopie leżącego, pan  
     over\_ten.NOM.VIR kick.PRS.3SG lie.ADJ.ACC.SG.M.ANI mister.NOM.SG  
     wyciąga nóż i wbija **go mu** w  
     take\_out.PRS.3SG knife.ACC.SG and beat\_in.PRS.3SG him.ACC him.DAT in  
     udo.  
     thigh.ACC.SG  
     ‘More than a dozen people kick the lying [person], you pull a knife and stab it into his  
     (upper) leg.’ (Pol; NKJP balanced poliqarp)
- (7.6) a. Kilkunastu kopie leżącego, pan wyciąga nóż i wbija **go mu** w udo.  
       b. ([...]) Nadała **go mu** norweska firma Det Norske Verutas. (Pol; NKJP pelcra)

<sup>486</sup> Examples are given in the order they were displayed after the query was accomplished.

Other issues concern the provision of linguistic and extra-linguistic information. The PELCRA engine does not provide the query output with any morphosyntactic information. Morphological or syntactic analysis needs to be done for every relevant attestation by hand. Furthermore, the PELCRA concordancer does not contain metadata. Generally, there is no information on utterances' provenance contrary to what is claimed on the NKJP's website. Such information would allow check for potential diatopic variation. Since I do not investigate the impact of sociolinguistic variables on cliticisation, I shall neglect the lack of metadata. Another problem is that the downloadable XML tables do not contain all query results, as it seems that they provide at most 5,001 tokens even though more than 5,000 tokens are available in principle and the maximum of 10,000 visible results was chosen before.<sup>487</sup> Finally, there is no option in the entire NKJP to listen to the audio files underlying the transcribed speech of the speech component. This is a serious issue, for the transcription scheme of the speech data is not clear. Nor is the transcription consistent across the entire speech corpus. Therefore, it is sometimes impossible to disambiguate a string of text, which has multiple structural analyses.<sup>488</sup>

## 7.4 Goralish: Korpus Spiski

The *Korpus Spiski* 'Spsz Dialect Corpus' (KoSpi) by Grochola-Szczepanek et al. (2019) is an electronic corpus of the Goralish lect in Polish Spisz. The corpus contains roundabout 250 hours of speech – giving approximately 2 million segments – recorded between 2015 and 2018.<sup>489</sup> The KoSpi is not a conversational corpus but consists predominantly of narrative speech. Native speakers were recorded in interview-like settings (cf. Grochola-Szczepanek et al. 2019: 170).

Queries may utilise CQL for specifying word forms, lemmas, and annotation tags. The KoSpi provides two graphical layers. On the one hand, a normalised layer is based on the Polish standard language. All tokens are represented in standard orthography except for particularly Goralish lexemes absent in standard Polish. Correspondingly, any query output is returned in standard Polish. On the other hand, a non-standard layer exists, which is phonetically and morphologically more faithful to the recorded speech. However, this holds only for forms that (i) either contain morphologically deviating formatives, (ii) semantically differ from standard Polish, or (iii) exist only in Goralish

<sup>487</sup> This issue appeared with the queries for *mi*, *ci* and *się*.

<sup>488</sup> See Section 13.6 with a small selection of illustrative examples of this issue (towards the end of that section).

<sup>489</sup> Note that the 2 million segments do not correspond to corpus linguistic tokens in the sense of individual word forms between two empty spaces. For one thing, the token *chcialbyś* 'you would like to' is analysed into three segments *chcial-*, *-by-*, and *-ś* lemmatised as *chcieć* 'want.INF', *by* 'COND', and *być* 'be.INF' (cf. Grochola-Szczepanek et al. 2019: 168).

(Grochola-Szczepanek et al. 2019: 172–173). Phonetic-phonological isoglosses tracing back to regular sound correspondences are not represented, at all (cf. Grochola-Szczepanek et al. 2019: 171–173).<sup>490</sup> There is a special search box for querying non-standard forms. Query outputs can be downloaded as tsv-files but provide only the normalised forms. The data sheets do not contain the dialect forms. This is not a serious issue for an investigation of distributional properties, but it still decreases user-friendliness. The non-normalised data need to be collected manually from the corpus website.

This circumstance has one significant effect on the present study, as it affects the short dative reflexive *se*. The form *se* is rigidly coded as dialectal, although *se* is a common Polish item, yet not recognised by the prescriptive standard of the Polish literary language, which allows only the long form *sobie*. Therefore, the form *se* needs to be queried through the search box for non-standard or dialectal forms (Pol. *Forma gwarowa*). The normalised output returns consistently the long form *sobie*. Note that querying *se* in the standard search box does not provide any results and that entering the form *sobie* returns both long *sobie* and short *se*, which are graphically identical by default. When analysing the distribution of short *se* in the Spisz corpus, data elicitation needs to be carried out as follows:

1. search *se* in the dialectal search box
2. download the data sheet
3. analyse the distribution of *sobie*

As a matter of fact, the analysis was done on the normalised data set, but Goralish examples will be cited both in the non-standard form alongside the normalised form for the sake of transparency.<sup>491</sup> On a positive note, the downloaded data sheets involve all available metadata such as place of residence, nationality, sex, year of birth, or educational level.<sup>492</sup> The metadata allowed for an easy way to exclude particular data points automatically. I removed all tokens from informants, whose nationality was or contained the label *Slovak*. I identified three such labels in the data: *słowacka* ‘Slovak’, *polska/słowacka* ‘Polish/Slovak’, *polska/słowacka/spiska* ‘Polish/Slovak/Spisz’. The rationale for eliding these data is to minimise direct interference from Slovak in the data set. I took an informant’s self-identification as Slovak as indication for a hypothetically possible proficiency in the Slovak

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<sup>490</sup> This procedure is a disadvantage for any phonological or morphological investigation, and it is also a drawback for studies of linguistic variation, which aim at investigating variation across and within speakers.

<sup>491</sup> Normalised utterances will be cited in angle brackets <...>.

<sup>492</sup> In contrast, this kind of data needs to be extracted from the NKJP circumstantially.

language of a yet unknown degree.<sup>493</sup> All data points were randomised after removing the potentially Slovak(ized) tokens.<sup>494</sup>

Grochola-Szczepanek et al. (2019: 177) inform that the Spisz Corpus is suitable for investigating syntax and its size allows for quantitative analyses, whereby it is possible to distinguish typical linguistic phenomena from marginal ones. It therefore has to be acknowledged that the Spisz data are suitable for investigations into clitic phenomena, which in general are considered a part of syntax proper in Polish and other Slavic languages (cf. Franks & King 2000; Kupść 2000b). The accessibility of acoustic data in terms of audio recordings must be regarded as an asset of the Spisz corpus. These data do not only allow to explore prosody, but also to disambiguate equivocal word strings. The Spisz Corpus provides users with the option of adjusting the playback speed to 75% and 50%, which is a helpful feature.

The Spisz corpus website contains some helpful features, e.g. a concise overview of the dialect's main isoglosses, a map of Polish Spisz including the locales, where the data were gathered, and a dictionary. The latter is particularly useful for scholars working with the corpus but unfamiliar with the dialect, as the Spisz lexicon contains lexical idiosyncrasies absent from the Polish standard language or the Polish vernacular (but some of which are occurring in other south Polish variants or neighbouring languages).

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<sup>493</sup> Of course, the declaration of Slovak nationality neither implies that an informant was an actual native or heritage speaker of Slovak, nor does the declaration of Polish preclude the acquisition of the Slovak language. Still, the procedure is appropriate, if one wants to increase data quality. However, metadata of the kind used for exclusion testify the ethnically/nationally mixed character of the Spisz area. This may be taken as hint for some sort of convergence between Polish and Slovak in this region and feeds the yet to test hypothesis that cliticisation patterns in the Spisz dialect resemble the Slovak second position placement type.

<sup>494</sup> I encountered a technical obstacle in terms of the coding of Unicode characters. The data points can only be downloaded as tsv-files. However, when opening the data sheets directly with Microsoft Excel, a character coding error occurred, such that all special characters (of the Polish orthography) were illegible. Data files needed to be re-coded by saving them with a different coding scheme. After iterated trial and error, the only viable results were achieved after saving the data sheets anew with *UTF-16 LE* coding. Other coding schemes like *ANSI*, *UTF-16 BE*, *UTF-8* did not return useful outputs (*ANSI* coding would change all special characters into their plain correspondents, such that it would be the second-best solution in this scenario). This problem did not occur, when opening the data sheets directly in an ordinary text editor. It did still occur after loading the tsv-file through the respective import option in Excel's data table. However, the coding error occurred only on a particular personal computer but was not reproducible on another computer with the same Windows operating system and an identical Microsoft Office bundle. A test in Google Sheets confirmed the singular occurrence of the problem, as all special characters were displayed correctly.



## 7.5 Slovak: Slovenský Národný Korpus (slovenský hovorený korpus)

The *Slovenský Národný Korpus* ‘Slovak national corpus’ (SNK) is the overall corpus infrastructure for the Slovak language, comparable to the ČNK in structure and scope.<sup>495</sup> The SNK uses the NoSketchEngine, i.e. the non-commercial and technically limited version of the corpus management system Sketch Engine.<sup>496</sup> The SNK is fully annotated and searchable on the basis of word forms, lemmas, and annotation tags. Access to the corpus is free of charge, but there are two access modes. On the one hand, a limited version of the corpus may be publicly accessed via a web interface. On the other hand, the full corpus may be accessed after registration. The SNK has two spoken subcorpora, i.e. *Slovenský hovorený korpus* ‘Corpus of Spoken Slovak’ and *Korpus nárečí Slovenského národného korpusu* ‘Corpus of Dialects of the Slovak National Corpus’. For the present study, I used *Slovenský hovorený korpus* (SHK) which comprises *hovorená slovenčina* ‘colloquial Slovak’.<sup>497,498,499</sup> Its current version *s-hovor-7.0* is online since March 2022 and based upon roundabout 850 hours of transcribed speech. The corpus size is 7,852,469 tokens from 869 individual records.<sup>500</sup>

The colloquial Slovak corpus has two main sources or subcorpora. First, it contains original recordings made for the corpus project. Second, it comprises testimonies that were audio-visually recorded for the *Projekt Oral history – Svedkovia z obdobia neslobody* ‘Project Oral History – Witnesses of the Oppression Period’ under the supervision of the *Ústav pamäti národa* ‘Nation’s Memory Institute’.<sup>501</sup> All audio recordings underlying the speech corpus are available, but not every audio track is aligned with the respective piece of transcribed text, which may pose obstacles of various kinds (e.g. for researchers interested in the phonetics or phonology of a particular piece of utterance, or in case one wishes to countercheck the transcription). Sound quality may significantly

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<sup>495</sup> URL: <https://korpus.sk/>

<sup>496</sup> NoSketchEngine is also used for the Bosnian, Croatian, and Serbian webcorpora (bsWaC, hrWaC, srWaC), which Kolaković et al. (2022) utilised for a good portion of their study of BCMS clitics.

<sup>497</sup> Slovenský hovorený korpus – s-hovor-7.0. Bratislava: Jazykovedný ústav Ľ. Štúra SAV 2022. Online: <http://korpus.juls.savba.sk>.

<sup>498</sup> The idea of integrating Slovak data into the project was motivated by Ivan Levin (p.c., 15 Jun 2023), whom I owe special thanks.

<sup>499</sup> Though dialectal data would be an interesting object of study, I refrain from scrutinising it for limitations of time and work force. Furthermore, the dialect subcorpus is still in an early stage. For instance, it is not morphologically annotated, such that it allows only for very simple word-based queries.

<sup>500</sup> Information according to: (1, Slk) <https://korpus.sk/korpusy-a-databazy/korpusy-snk/verejne-pristupne-korpusy-snk/slovensky-hovoreny-korpus/>, (2, Eng) <https://korpus.sk/en/corpora-and-databases/snc-corpora/publicly-available-snc-corpora/corpus-of-spoken-slovak/> (accessed 4 Aug 2023).

<sup>501</sup> For more information on the *Project Oral History*, I refer the reader to the Nation’s Memory Institute’s corresponding website: (1, Slk) <https://www.upn.gov.sk/sk/projekt-oral-history-svedkovia-z-obdobia-neslobody/>, (2, Eng) <https://www.upn.gov.sk/en/project-oral-history-witnesses-of-the-oppression-period/> (accessed 6 Aug 2023).

differ between recordings. Another advantage of the SHK is the wide range of sociolinguistic metadata it provides (which, however, is not relevant for the purpose of the present study).

## 7.6 Silesian: Dialekty i Gwary polskie

There have been recent efforts to compile electronic corpora for Silesian. For instance, Ruda et al. (2023a) compiled the corpus *Teksty śląskie* ‘Silesian texts’ (SIL) of contemporary Silesian writing covering five different text types, i.e. fiction, news, feuilleton, blog entries, and facebook posts, summing up to 23,989 word tokens in total.<sup>502</sup> The SIL corpus is available in plain text files but not tagged. There is currently no electronic corpus of spoken Silesian available. Therefore, I use the dialectological online compendium *Dialekty i Gwary polskie* ‘Dialects and subdialects’ (DiG) edited by Karaś (2010) as main source for Silesian data.<sup>503</sup> The compendium provides information about the Polish dialects and folklore as of the beginning of the 21st century, and it gives relevant historical background information. Every dialect and its subdialects are illustrated by small collections of dialect texts, which are mostly narratives, but also jokes, poems, or conversations to a lesser extent. These texts are products of the traditional dialectological approach based upon audio recordings of (usually immobile) rural dialect speakers. As to the actuality of the data, there is no explicit information about the date when recordings were made. Provided the publication date (2010) and the conception of the compendium, it is plausible to assume that the DiG texts represent Silesian speech from the first decade of the 2000s, which is sufficiently recent. Most audio recordings are available online together with the transcribed text, which follows traditional Polish dialectological practice to code dialect features by orthographical means.

The Silesian section contains 32 individual texts (approx. 11,700 words in sum) from 14 informants in 11 villages distributed over the three main groups of Silesian (northern, central, southern). It should be noted that the Silesian data points are limited to the area of Poland, which is relevant with respect to the southern group, whose linguistic area spreads into present-day Czech Silesia.<sup>504</sup> The data had been extracted manually, for the dialect texts are represented only

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<sup>502</sup> The *Korpus Ślōnskij Mowy* ‘Corpus of the Silesian Language’ compiled by G. Kulik et al. (2018) is another electronic corpus project for Silesian and utilises ČNK’s KonText frontend. Its last freely available version involved 1,102,837 word tokens. (The corpus was deactivated since 6 December 2023, but it has been reactivated in the second half of 2024). (*Od 6 grudnia 2023 korpus je wyłączony*. <https://silling.org/informacje-o-korpusie-slonskij-mowy/>, accessed 6 Apr 2024).

<sup>503</sup> URL: <http://www.dialektologia.uw.edu.pl/> (accessed 4 Nov 2022).

<sup>504</sup> Bogoczová & Bortliczek’s (2014) monograph about Silesian in the Trans-Olza area in the contemporary Czech Republic provides transcripts of speech of children and adolescents but are not considered here. The omission of these data is justified, since Trans-Olza linguistically differs from the Cieszyn area in Poland. Trans-Olza is characterised by language contact between Czech and the local autochthonous dialect, which gave rise to a hybrid lect called *po naszymu*. (Standard) Polish represents another linguistic layer and enters the scene through schooling, but the Polish

orthographically on the compendium's website and cannot be downloaded automatically, nor is it possible to search the texts by a search engine, which would be able to return concordances. Each text was searched for occurrences of short pronominal forms in various spellings.

The DiG texts are not balanced in any way, e.g. with respect to size, genre or sociological parameters (e.g. the speakers' age, gender, education etc.), nor are the informants evenly distributed geographically.<sup>505</sup> As a matter of fact, the transcriptions do not consequently follow a common scheme and there is variation across different transcribers as well as within one transcriber. The transcriptions' degree of detail is not explicated and there is reason to suspect that certain information was not coded, for the texts exhibit a rather low degree of disfluency phenomena (cf. Kolaković et al. 2022: 133–135 for a brief discussion of data quality in dialectological sources through the example of BCMS). For one thing, listening to the audio of the northern Silesian text no. 3 from Kuniów, one will encounter several hesitation markers in the informant's speech flow, which are not represented in the transcript. Furthermore, in the very same text, I detected a transcription error, whereby the *že*-supported copulative 2PL auxiliary clitic *żeście* is misanalysed as a syntagm of the *že*-supported 2SG auxiliary clitic + REFL, i.e. <*żeś sie*> in (7.7a). I propose a corrected transcription line in (7.7b). Another example of the issue at hand is encountered in the central Silesian text no. 1 from the village of Halemba (now a district of the city of Ruda Śląska) in (7.8)–(7.9). In (7.8a), the PP *na coś* 'to something' occurs in repetition, in fact, but is not transcribed. That is, the disfluency in the informant's speech flow is not represented. I propose a more accurate transcription in (7.8b). Shortly afterwards, a transcription error occurs in the very same text, whereby (7.9a) lacks the 1SG auxiliary clitic *-ch*, which appears in clausal second position as indicated in corrected (7.9b).

- (7.7) a. <[...] o ze-ś znalaz taką Ewa, i dzisiaj  
o ŻE-AUX.2SG find.PST.2SG.M such.ACC.SG.F Eve.ACC and today  
**ze-ś sie** tu pojąć ją za żona, [...]>  
ŻE-be.PRS.2SG REFL.ACC here take.INF her.ACC as wife.ACC.SG
- b. [...] o ze-ś znalaz taką Ewa, i dzisiaj  
o ŻE-AUX.2SG find.PST.2SG.M such.ACC.SG.F Eve.ACC and today  
**ze-ście** tu pojąć ją za żona, [...]  
ŻE-be.PRS.2PL here take.INF her.ACC as wife.ACC.SG

Correct: 'o you found such an Eve, and today you are here, to take her to wife'

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minority's language proficiency is described as rather passive. Overall, the endemic ethnically Polish population is essentially bi- or multilingual (cf. Bogoczová & Bortliczek 2014).

<sup>505</sup> The villages of the northern and central group are all located in the southern half of their respective dialect areas. The villages of the southern group are maximally south close to the trijunction Poland-Czechia-Slovakia. Furthermore, they are also rather close to the neighbouring Goralish dialect area of Podhale. Therefore, the data points are only representative for a rather small subset of the respective dialects.

- (7.8) a. <[...] bo wtedy to się tak człowiek czuł  
because back\_then TO REFL.ACC so human.NOM.SG feel.PST.SG.M  
taki, że się **na coś** rychtuje.>  
such.NOM.SG.M that REFL.ACC on something.ACC prepare.PRS.3SG
- b. [...] bo wtedy to się tak człowiek czuł  
because back\_then TO REFL.ACC so human.NOM.SG feel.PST.SG.M  
taki, że się **na coś na coś** rychtuje.  
such.NOM.SG.M that REFL.ACC on something.ACC on something.ACC prepare.PRS.3SG  
‘... because, in those days, one felt that one is preparing for something.’
- (7.9) a. <[...] i iść do Komunie, **to** się chyba najwytniej  
and go.INF to Communion.GEN.SG TO REFL.ACC probably most  
w czasie tych Rorat nauczył, [...]>  
in time.LOC.SG this.GEN.PL Rorate.GEN.PL teach.PST.SG.M
- b. [...] i iść do Komunie, **to=ch** się chyba  
and go.INF to Communion.GEN.SG TO=AUX.1SG REFL.ACC probably  
najwytniej w czasie tych Rorat nauczył [...]>  
most in time.LOC.SG this.GEN.PL Rorate.GEN.PL teach.PST.SG.M  
‘... and to attend Communion, then I probably learnt most during the Rorate Mass ...’  
(Sil; DiG)

In sum, there is some level of inaccuracy in the transcription due to idealisation and mistakes. However, the DiG texts provide valuable data points of spoken Silesian. From a practical point of view, the texts are easily accessible and might be regarded as the only currently available stable corpus of Silesian speech. This is why I decided to take these data into consideration. Keep in mind that Silesian is not the key component of the data analysis part but serves as supplementary material providing additional aspects to the overall topic.

## 7.7 Kashubian: Dialekty i Gwary polskie

To date, no electronic corpus of Kashubian exists. Sources containing Kashubian data are generally available, but there is no central storage for them. Since Kashubian has a standardised linguistic norm, which is used as a written code and taught in school, and since numerous Kashubian sources are freely available online, it is possible to obtain written Kashubian with comparatively little effort. For one

thing, the educational website *Skarbnica Kaszubska* ‘Kashubian treasure’ stores free electronic language materials, incl. textbooks and primers.<sup>506</sup> A corpus of contemporary Kashubian writing (*Teksty kaszubskie* ‘Kashubian texts’) was compiled by Ruda et al. (2023b), which is comparable to their SIL corpus for Silesian in size and structure (24,195 word tokens across the same five text types).<sup>507</sup>

In contrast, it is not easy to access spoken Kashubian despite growing interest in the language by diverse linguistic disciplines and the growing need of language documentation due to the decreasing number of proficient speakers. Kashubian has always been subject to traditional dialectological study in Poland, but the only available spoken data sources are dated, e.g. Lorentz’s (1924) compilation of Pomeranian texts.<sup>508</sup> Data collections by individuals exist (e.g. gathered for the purposes of doctoral dissertations or master theses), but high efforts would be needed to find and access them. The only freely available source of contemporary spoken Kashubian, which provides transcripts, is Karaś’s (2010) dialectological compendium *Dialekty i Gwary polskie* already mentioned in the previous section on Silesian.

Admittedly, the database is very limited, as it comprises very short recordings of nine individuals amounting to approximately 2,400 segments (word tokens). This amount only represents the transcriptions on the *Dialekty i Gwary* website, which do not always coincide with the actual extent of speech available through the audio recordings. For instance, the Central Kashubian text sample from Kartuzy (Central Kashubian Text no. 3) is based on an uploaded audio file of 5:07 minutes length, but the transcript covers merely the first two minutes.

Furthermore, as pointed out to me by Lechosław Jocz (p.c.), some of the data exhibit high degrees of Polish interference.<sup>509</sup> Therefore, the Kashubian data do not allow for strong generalisations or sound conclusions. But this corpus may still provide a tentative impression of Kashubian short pronominals and reflexives. Anyway, it is an advantage of this data collection to cover all three macro-dialects of Kashubian, i.e. the northern, central, and southern one. Hopefully, the humble survey stimulates future research in this understudied Slavic language.

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<sup>506</sup> URL: <http://skarbnicakaszubska.pl/> (accessed 6 Apr 2024).

<sup>507</sup> URL: <https://osf.io/q49y6/> (accessed 6 Apr 2024).

<sup>508</sup> I am thankful to Andrzej Żak for providing me with this Kashubian data.

<sup>509</sup> I am thankful to Lechosław Jocz (p.c., 26 Apr 2024) for pointing these issues out to me. In fact, he argues that some of the data even represent rather Polish with Kashubian interferences.

## 8 Data elicitation and data preparation

### 8.1 Data samples and data cleansing

The present survey is built upon a rich empirical database, which covers six spoken West Slavic varieties of the Lechitic-Czechoslovak continuum described in Chapter 2: Polish, Goralish, Silesian, Kashubian, Czech, and Slovak. The data were drawn from contemporary electronic corpora except for Silesian and Kashubian as explained in the previous Chapter 7. Correspondingly, the amount of available data varies between the corpora. The basic methodological approach was to process approximately 500 sentence or clause tokens per clitic type and lect. However, this goal could not be reached in every case because of data limitations, especially in the domain of smaller lects. Table 8.1 summarises the quantities of processed data, which also occasionally gives the reader an idea of the differences in data volume. In a first working cycle, these raw data were cleansed (i.e. unusable data points were removed) and annotated.<sup>510</sup>

Table 8.1. Quantities of the raw corpus data processed in the first cycle.

| Clitic type    | Czech | Polish | Goralish | Slovak | Silesian | Kashubian |
|----------------|-------|--------|----------|--------|----------|-----------|
| ACC/GEN.1SG    | N/A   | N/A    | N/A      | 127    | N/A      | N/A       |
| ACC/GEN.2SG    | 500   | 505    | 320      | 126    | 4        | —         |
| ACC.3SG.M/N    | 501   | 500    | 506      | 120    | 15       | 2         |
| DAT.1SG        | 500   | 503    | 500      | 122    | 10       | —         |
| DAT.2SG        | 500   | 500    | 501      | 129    | 2        | —         |
| DAT.3SG.M/N    | 501   | 500    | 500      | 120    | 13       | —         |
| REFL.ACC(/GEN) | 500   | 501    | 500      | 118    | 285      | 53        |
| REFL.DAT       | 501   | 501    | 500      | 134    | 32       | —         |
| Sum:           | 3,503 | 3,510  | 3,327    | 996    | 361      | 55        |

Raw data had to be cleansed as a preparatory step for subsequent analysis. Speech contains diverse disfluencies like repetitions, self-corrections, mid-sentence breakups and resumptions, stuttering, slips of the tongue, and hesitation markers. Immediate repetitions of the same item were elided and the utterance was handled, as if it did not contain the repetition, because it does not have an impact on syntactic structure. The same approach was taken with respect to self-correction, stuttering, and hesitation markers. The core principle was to keep the original syntactic structure and not to change

<sup>510</sup> See Section 8.2 for details of the annotation.

anything that would affect structural analysis. Mid-sentence breakups usually lead to removal of the respective sentence token from the data set, except when combined with a following resumption by the same individual. This was done under the premise that the resulting string constituted a single syntactically and semantically coherent utterance. Slips of the tongue alter lexical items or parts of an utterance to a greater or lesser extent. As far as the intended lexical item or utterance was still recoverable (due to a low degree of deformation), utterances were kept in the data set in their corrected form. If the respective string was not recognisable at all, because of a high degree of deformation, the respective utterance was removed from the data set.

Attention needs to be paid to homonymies. The forms *ci* (Polish, Goralish) and *ti, se* (Czech) are ambiguous. The forms *ci* and *ti* represent pronominal dative clitics as well as demonstratives of the plural virile gender. The PELCRA search engine does not provide any technical solution for disambiguating queried items in such cases. Therefore, disambiguation needs to be processed manually by eliding output utterances containing unwanted tokens, i.e. demonstratives in this case.

In the case of Czech, the word form *se* is ambiguous between the accusative reflexive clitic and the syllabic variant of the instrumental preposition *s/se* ‘with’. Although prepositions were explicitly excluded from the query by restricting it exclusively to pronouns (by conjunction of [tag="P.\*"]), current taggers do not reach precision of 100%, such that in some instances erroneous annotations may occur. Such instances need to be removed by hand, too. However, I encountered only one erroneously tagged syllabic preposition *se* in my data set.<sup>511</sup>

Similarly, a tag-based approach was used to restrict *ci* tokens to constant clitics by exclusion of the homonymous demonstrative in the Spisz corpus. In theory, different strategies are possible, but they may yield different results in practice. If one searches *ci* as a plain word form through the CQL query [word='ci'], then the corpus search engine returns 2,264 non-disambiguated occurrences of *ci*. One can disambiguate by restricting the *ci* occurrences only to the set classified as personal pronoun by the tagger. The query [word='ci' & tag='ppron12.\*'] returns 1,426 tokens (62.99% of all *ci* occurrences).<sup>512</sup> This is the procedure I opted for. On the other hand, one can elicit constant clitic occurrences of *ci* by exclusive reliance on the parts-of-speech annotation. The query [tag='ppron12:sg:dat.\*.sec.\*'] searches for 2nd person singular dative personal pronouns, but yields with 1,450 tokens (64.05% of all *ci* occurrences) a slightly different amount of data. As with other electronic corpora, the part-of-speech tagging does not achieve 100% accuracy in the Spisz corpus, such that there are still demonstrative pronoun tokens in the query output, e.g. (8.1)

<sup>511</sup> I cite the respective data point in (i) for the sake of transparency.

(i) no se se zaměstnáním taky teda mezi náma  
well with with employment.INS.SG also then between us.INS

‘well, concerning employment, [it’s] the same between the two of us then’ (Cz; ČNK oral\_v1)

<sup>512</sup> The amount of 1,426 represents the output of the corpus query interface. The downloadable tsv-file contains 1,428 tokens, in fact (= 0.14% discrepancy).

|                |              |              |      |       |               |             |         |
|----------------|--------------|--------------|------|-------|---------------|-------------|---------|
| nas            | było         | śtyrów       | ino  | ja    | został        | <b>toci</b> | już     |
| <nas           | było         | czterech     | ino  | ja    | został        | <b>ci</b>   | już     |
| us.GEN         | be.PST.3SG.N | four.GEN     | only | I.NOM | stay.PST.SG.M | these.NOM   | already |
| pomarli        | nie          | żyją         |      |       |               |             |         |
| pomarli        | nie          | żyją>        |      |       |               |             |         |
| die.PST.PL.VIR | NEG          | live.PRS.3PL |      |       |               |             |         |

‘there were four of us, only I remained, the other ones already passed, are dead’

(Gor; KoSpi)

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## 8.2 Annotation

The syntactic analysis of the recorded cliticisation patterns in the scrutinised varieties is based on an annotation scheme that was developed for the purpose of the present study. The annotation used in this study codes distributional and syntagmatic information, which may be used for both quantitative and qualitative analyses.<sup>513</sup> In fact, there have already been approaches for analysing clitic distributional properties in Slavic based on sufficiently big empirical data samples, i.e. Reinkowski (2001), Kolaković et al. (2022), and Kosek (2011, *inter alia*). So why not take an approach that is already available and was applied in practice? Previous attempts to capture clitic distributional properties are too coarse to adopt them here.

Reinkowski (2001) analyses the distribution of clitics in Croatian and Serbian newspaper texts and distinguishes three distribution patterns, i.e. *Initialstellung* ‘initial position’, *Mittelstellung* ‘central position’, and *Endstellung* ‘final position’. However, the scheme is not precise enough. Not only is it restricted to three categories treated as primitives, but there are also imprecisions in the categories themselves. Both the initial and central position essentially cover what is considered Wackernagel placement but with a rather unclear distinction between second word and second phrase placement both including split phrases with intervening clitics. These shortcomings originate in the neglect of syntactic constituency. The final position is defined in terms of the verbal predicate and refers to the placement of the clitic immediately following it. Even though Reinkowski (2001) investigates only comparatively long or contentful sentences that display multiple potential sites for clitic placement, the approach is unable to assign a clitic placement pattern to verb-initial clauses, because it cannot differentiate between the three patterns in such cases. Kolaković et al.’s (2022) approach respects syntactic constituency, but they differentiate only between first position (1P), second position (2P) and Delayed Placement (DP). Phrase-splitting is treated as an individual dimension, which allows for a more elaborate distributional analysis. But what is missing is a fine-grained perspective on the variation within the category of DP. Registering the fine structure of clitic placement options in this DP-category is especially relevant towards the discussion in Section 4.4.2, which provides that there is a misconception about DP being essentially equated with 3P. Finally, Kosek’s (2011, 2015, 2017) scheme accounts not only for the distribution of clitics in the clause, but it also takes clitic placement relative to the verbal predicate into consideration. Importantly, Kosek considers which side of the verbal predicate a clitic is located on and whether the clitic is adjacent to

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<sup>513</sup> Categorical information, i.e. the syntactic category (parts of speech) of the host or left-adjacent orthotonic word form, will not be considered in the data analysis in Part III. Wackernagel clitic languages (as well as Polish) exhibit what is sometimes called “promiscuous attachment”, i.e. the non-selectivity for the syntactic category of the host. Readers will notice throughout Section 10 that this is fully borne out for all six lects under scrutiny.

its predicate or not. However, the scheme is still too coarse. It operates on two scales, which duplicate information to some extent, but does not recognise constituency. On the one hand, the distribution patterns are assigned to a five-point classification of clitic clausal positions: (i) initial, (ii) post-initial, (iii) medial, (iv) pre-final, and (v) final. On the other hand, it distinguishes between contact position (= verb-adjacency) and distance position. Note that clausal positions “pre-final” and “final” inherently rely on verb-adjacency as in Reinkowski’s (2001) scheme. Though relying on the interaction of diverse parameters, Kosek’s annotation treats the placement patterns as primitives, too. In contrast, my annotation scheme allows capturing properties on individual axes of analysis separately and to combine them later to derive placement patterns as needed.

I annotated each data point, i.e. clause or sentence token, manually with respect to the parameters summarised in Table 8.2. Importantly, I treat clitic clausal position, verb-adjacency, and predicate adjacency as parameters independent of one another. Three parameters are adopted from Kolaković et al.’s (2022) study, i.e. (i) Clustering, (ii) Cluster length, and (iii) Cluster structure. The annotation scheme is essentially compatible with Kosek’s (2011, 2015, 2017) approach, although it was developed independently. The advantage of the present thesis’s annotation is that Kosek’s categories can be derived from the parameters utilised here, as they represent atomic units of analysis, whereas Kosek applies certain combinations thereof as primitives for his analysis. Note that the annotation was completed by a single individual. There was no second annotator, such that no best practices for reducing subjective mistakes by inter-annotator agreement could have been applied. Some residual error needs to be taken into account, therefore.

Some remarks on the annotation scheme are in order, for such a type of annotation is far from trivial. The reader should consider that the annotation scheme is a compromise between surface-orientation and analytical depth. It cannot be used for a – perhaps automated – syntactic in-depth analysis for several reasons. On the one hand, the data points consist of linear strings. Therefore, a structural analysis of the kind contemporary treebanks offer is not available. On the other hand, the annotation scheme operates with general grammatical labels (e.g. *N* for nouns, *V* for verbs, *Prn* for pronouns).<sup>514</sup> A detailed analysis would need a more fine-grained labelling. Finally, the data quantity exceeds the possibilities of such an in-depth analysis by hand, which ideally would operate on a smaller data sample. However, quantity is one of the present thesis’s key aspects.

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<sup>514</sup> Admittedly, the host category does not play any role in the current analysis except in case of coordinating conjunctions and clause boundaries, which were considered for first position cliticisation, see Chapter 10. As might be expected for second position clitic languages, not meaningful restrictions apply to the left context, i.e. we observe so-called non-selectivity or promiscuity.

Table 8.2: Parameters of the corpus annotation.

| Parameter      | Detail                                                                                                                                                                                                                                                                                            |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Position       | Counted from the left sentence boundary, a numerical value, recorded for both the level of the syntactic/orthographic word and the level of the syntactic constituent) <sup>515</sup>                                                                                                             |
| Distance       | Distance to the syntactic governor, i.e. head or predicate (mostly verbs, but also nouns, adjectives, and adverbs, on the level of the syntactic/orthographic word and the level of the syntactic constituent)                                                                                    |
| Host category  | Category of the host, e.g. noun (N), verb (V), pronoun (Prn) etc., whereby the host is meant as the left context. The host is idealised as the nearest left non-clitic word token but also includes coordinating conjunctions and clause boundaries (the latter being annotated as empty context) |
| Verb-adjacency | The occurrence of a clitic or clitic cluster contiguous with one or two verbs irrespective of the underlying subcategorisation relation between the clitic(s) and the verb <sup>516</sup>                                                                                                         |
| Laterality     | The side relative to the clitic's predicate (left, right)                                                                                                                                                                                                                                         |
| Clustering     | The occurrence of a clitic cluster vs. diacsis                                                                                                                                                                                                                                                    |
| Cluster length | Number of clitics involved, incl. single clitic occurrences                                                                                                                                                                                                                                       |

<sup>515</sup> I am thankful to Ivan Levin (p.c., 15 Jun 2023) for confronting me with the question of why the clitic position is not counted from the right sentence boundary. There are several reasons for this: First, as a matter of empirical fact, clitics oscillate between 2P and VA placement both typologically and diachronically. Second, it has been generally observed that clitics (in Slavic) cannot occur further to the right in an utterance than immediately right-adjoined to their verbal governor, cf. Section 4.8. Third, I am not aware of any natural syntax, which places constituents clause-medially with respect to the right sentence boundary (cf. Bermúdez-Otero & Payne 2011: 61–65, esp. 63–64). The latter situation is different from phonology, where right margins are natural reference points for phonotactics (e.g. Havlík's law) or prosody (e.g. [ante-]penult stress). Note that this does not conflict with clause-final verb placement in German subordinate clauses, as the verb is put in the boundary itself and no further reference to a particular position counted from that boundary is necessary.

<sup>516</sup> Readers might wonder why information seems to be doubled here, because verb-adjacent clitic placement is retrievable from the category of the host and the distance to the predicate. In fact, it is not possible to derive all attestations of verb-adjacent placement with this information. Consider the two following situations. First, an object clitic occurs in second position in the topmost matrix phrase of a complex clause and intervenes between the subject and the finite verb after clitic climbing in (i). Second, an embedded clitic climbs to second position of the matrix phrase, which itself is a subordinate and headed by a subordinator, e.g. complementizer *že*, as in (ii). In either case, the clitic is propped by a non-verbal category and non-contiguous with its verbal predicate. However, the clitic is contiguous with the matrix verb, i.e. it occurs in a VA configuration (more precisely, 2P and VA simultaneously).

(i) A ja go<sub>2</sub> muszę<sub>1</sub> tam kiedyś zabrać<sub>2</sub>  
and I.NOM him.ACC must.PRS.1SG there some\_day take.INF  
'and I will have to take him there one day' (Pol; NKJP pelcra)

(ii) podtrhla, že se<sub>2</sub> chce<sub>1</sub> učít<sub>2</sub> jazyky  
underline.PST.SG.F that REFL.ACC want.PRS.3SG teach.INF tongue.ACC.PL  
'she stressed that she wants to learn languages' (Cz; ČNK syn\_v8)

Example (ii) was retrieved with the following CQL query: [word="že"] [word="ho|mu|tě|se"] [tag="V.\*" & word!="jsem|jsi|jsme|jste"] [tag="Vf.\*"]

|                   |                                                                                        |
|-------------------|----------------------------------------------------------------------------------------|
| Cluster structure | Structure of the clitic cluster, types of clitics involved, linear order among clitics |
| Clitic climbing   | Presence or absence of clitic climbing (CC vs. noCC)                                   |

The positional annotation is surface-oriented and primarily based on the orthographic form of the word tokens encountered in the data. However, not every occurrence of contiguous graphemes between two empty spaces is counted as a single word. With respect to the so-called mobile inflection in Polish, it is counted as an individual word, if occurring adjacent to other clitics, such that the resulting sequence of [mobile inflection + clitic(s)] virtually yields a clitic cluster. I follow this approach consistently, even in cases where the mobile inflection appears on an *l*-participle, which I treat as an ordinary inflected verb form in all other cases (i.e. when the mobile inflection is not part of a presumed clitic cluster). Furthermore, the Polish negative particle *nie* is treated as a prefix when having scope over a verb, which is the default case in sentential negation, e.g. *nie śpię* ‘I don’t sleep’ (glossed: ‘NEG’ + ‘sleep.PRS.1SG’). This treatment makes the Polish complex of [negation + verb] more similar to Czech, where the corresponding sequence is resolved as a single unit orthographically, e.g. Cz. *ne-spím* ‘I don’t sleep’ (glossed: ‘NEG-sleep.PRS.1SG’). Note that in both examples the stress shifts from the verbal stem to the negative element due to penult (Polish) and initial (Czech) stress respectively. On purely accentual grounds, there is nothing preventing a proclitic analysis of the negative particle. But I will keep the morphological analysis of NEG + verb, as there is no possibility for other elements (except prefixes) to intervene between negation and verbal stem.<sup>517</sup> Furthermore, T. King’s (1994: 288–299) observation that Russian negation cannot be elided by ellipsis in finite verb coordinations holds for Czech and Polish, too.<sup>518</sup>

<sup>517</sup> From a typological perspective, there are parallels for a treatment of sentential negation in terms of affixation. For instance, there are three negation types in Turkish: (i) negative interjections *hayır* or *yok* ‘no’, (ii) the nominal negativizer *değil*, and (iii) the negative morpheme *-me/-ma-* (exhibiting the two-fold vowel harmony), which is suffixed to the verb stem and then followed by other verbal morphemes coding, e.g., tense, mode, or grammatical person. On a related note, Zwicky & Pullum (1983) analyse the contracted negation *-n’t* in English as inflectional element, not as cliticised syntactic element. As far as Slavic is concerned, T. King (1994) argues that “in Russian, negation form unit with the tensed verb” (T. King 1994: 296) and argues that there were no separate NegP. Her theoretical considerations (nested within generative syntax of Government and Binding theory) may be translated into an approach compatible with the treatment of verbal negation as a prefix.

<sup>518</sup> One can infer from this a rather morphological than syntactic status of NEG. In contrast, German allows ellipsis of NEG in finite coordinate structures, cf. (i). (I provide simple examples for the sake of argumentation, since negation becomes more complicated in complex VPs in German.)

- (i) a. Peter raucht nicht und trinkt nicht.  
Peter.NOM smoke.PRS.3SG NEG and drink.PRS.3SG NEG  
b. Peter raucht und trinkt nicht.  
Peter.NOM smoke.PRS.3SG and drink.PRS.3SG NEG  
‘Peter does not smoke and [does not] drink [alcohol].’ (Ger; constructed)

The negation has only scope over both conjuncts, if it is realised in the so-called *right bracket* (Ger. *rechte Satzklammer*) of the entire sentence. Otherwise, it has only scope over the first conjunct, cf. (ii). Example (ii.a) is a little awkward for pragmatical reasons, which is why I additionally provide (ii.b).

At the phrase level, I consider every element a syntactic constituent XP, if it is independently movable across the clause. This approach is in full accordance with the often-referred-to “free word order” of Slavic, which motivates Haider & Szucsich (2022) to classify Slavic as *T3 language*, i.e. “type with *variable head positioning*” (Haider & Szucsich 2022: 25, italics in the original).<sup>519</sup> Thus, elements that comprise a syntactic constituent by dependency are treated as phrases on their own, which might be a welcome outcome for the analysis of discontinuous constituents. If the elements of a dependency tree or subtree are linearly contiguous, they are also syntactically treated as a compact unit and as such represent a coherent syntactic constituent. Each constituent is counted as one element on the syntactic surface level. Parentheticals are treated as syntactic constituents if they occur within a relevant syntactic string. This might be a surprising approach, as parenthetical material is usually analysed as not belonging to the syntactic domain of the clause, cf. (8.3). The decision is motivated by the surface-orientation of the study.

- (8.3) [AdvP potem ] [AdvP tam ] [NP jego kolega z Łapszanki ]  
 afterwards there his friend.NOM.SG from Łapszanka.GEN  
 [NP świętej pamięci Wojtek Holowa ]<sub>PARENTHETICAL</sub> **mu**  
 holy.GEN.SG.F remembrance.GEN.SG Wojtek.NOM Holowa.NOM him.ACC  
 pożyczył tego ula  
 lend.PST.SG.M this.GEN.SG.M beehive.GEN.SG  
 ‘Afterwards, his friend from Łapszanka, the late Wojtek Holowa, lent him the beehive for him.’  
 (Gor; KoSpi)

Note that an automatic analysis of clitic placement by a computer-linguistic model was impossible. A test analysis of Spisz dialect data with UDPipe for Polish (Wijffels et al. 2021) yielded no usable results.<sup>520</sup> Several reasons can be put forward for the ineligibility of the model for the task. First, the analysis of actual speech poses an intricate problem due to disfluency phenomena. For instance, it becomes difficult for a language model to determine the intended verbal predicate and the argument structure of an utterance that exhibits disfluencies like self-corrections or mid-sentence break-ups.

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- (ii) a. Peter raucht nicht und trinkt (stattdessen).  
 Peter.NOM smoke.PRS.3SG NEG and drink.PRS.3SG instead  
 ‘Peter does not smoke and drinks (instead).’  
 b. Peter fährt nicht und trinkt (stattdessen / deswegen).  
 Peter.NOM drive.PRS.3SG NEG and drink.PRS.3SG instead therefore  
 ‘Peter does not drive and drinks (instead/therefore).’ (Ger; constructed)

<sup>519</sup> The general idea is that Slavic and similar “free word order” languages represent a third type (T3 = type 3) of languages next to head-initial and head-final languages. The crucial feature of T3 languages is that they are *un(der)specified* for head-directionality (cf. Haider & Szucsich 2022: 25, 30, 35).

<sup>520</sup> I am thankful to George Moroz for guidance.

Second, speech corpora do not contain orthographic punctuation and individual sentences and clauses cannot be identified by means of robust orthographic cues like punctuation or capitalisation. The kind of punctuation usually employed in spoken language corpora rather reflects aspects of prosody (e.g. marking pauses), which do not necessarily coincide with syntactic boundaries. Furthermore, the quality of such a punctuation scheme depends on good documentation and a uniform application of transcription rules by the transcribers. Especially the latter cannot be taken for granted. It is also important to note that most NLP models are trained on written language. Models of written language cannot be successfully applied to speech data, obviously.

### **8.3      Distributional freedom: A problem for positional analyses**

Surveying clitic placement in spoken utterances comes with an analytical problem that is usually not taken into consideration in the literature on cliticisation: the degree of potential positional freedom. As mentioned earlier, the clitic system of Polish is deemed typologically peculiar in Slavic because of its great distributional freedom. However, the literary survey in Section 4.4 revealed that undisputed Wackernagel clitic languages exhibit some degree of freedom in clitic placement, too. But what is distributional freedom, anyway? This question has never been treated in detail, although free word order is a pertinent issue in Slavic. Witkoś (1998: 159) briefly addresses the issue by stating that “the more words in the clause the more available clitic positions.” The increasing range of clitic positions in successively growing utterances is an important matter. The question arises, how to connect the position that a clitic occupies in an utterance with the utterance’s actual size or complexity. The commonly acknowledged typological division between Slavic 2P clitic languages and Polish can be transposed into the way these languages use the multiple hypothetically available positions: while idealised Wackernagel cliticisation anchors a clitic or clitic cluster in clausal second position irrespective of an utterance’s length, Polish cliticisation seems to be a function of clause or sentence length. Such a hypothesis can be operationalised and tested, in fact. In the following I will propose a simple method for accounting for clitic placement against available clitic positions.

It is repeatedly mentioned in textbook examples and introductory remarks on Slavic word order that an utterance with three constituents like subject (S), verb (V), object (O) has six different permutations, all of which are grammatical. For instance, the scheme in (8.4) provides all word order possibilities of a minimal transitive independent declarative clause.

- (8.4)    a. S V O  
           b. S O V  
           c. V S O  
           d. V O S  
           e. O S V  
           f. O V S

That is, for an utterance with  $n$  elements, there are  $n!$  ( $n$  factorial) possible word orders (cf. e.g. Browne 1974: 40). In formal terms, Slavic word order may be characterised as a finite set  $A$  of  $n$  elements  $a$ . The set  $A$  has  $n!$  permutations  $\pi$ , whereby  $\pi$  is a bijection of  $A$  in  $A$  such that a permutation  $\pi: A \rightarrow A(n)$  is a unique  $n$ -tuple  $(x_1, \dots, x_n)$ , i.e. a unique linear order comprising exhaustively all elements  $a \in A$ . I provide a simple formalisation in (8.5).

- (8.5)         $a \in A$ ,                    [ $a$  is an element of set  $A$ ]  
            $\wedge |A| = n$ ,                [AND the cardinality of  $A$  is  $n$ ]  
            $\wedge \pi: A \rightarrow A$ ,            [AND a permutation  $\pi$  is a mapping of set  $A$  in itself]  
            $\wedge |\pi| = n!$                 [AND the cardinality of  $\pi$  is  $n!$ ]

An interesting property is that, according to common opinion, each  $\pi$  is fully grammatical in Slavic with information structure being the domain relevant for the realisation of a particular linearisation. This is typologically significant in that this property varies across languages. For instance, it is well-known that not every logically possible permutation of parts of speech is grammatical in English. Languages like German impose restrictions on permutations, although German exhibits more linear flexibility than English. This is exactly why Haider & Szucsich (2022) propose the word order type labelled T3 for Slavic and similar languages. The result is a tripartition of word order types, i.e. [S[VO]] (e.g. English, Romance), OV (e.g. German, Japanese), and T3 (e.g. Slavic).<sup>521</sup>

However, the case described in (8.4) is an idealisation insofar as it represents a special case of full isomorphism between syntactic words and syntactic constituents. Syntactic structure imposes further restrictions on the syntagmatic relations between elements of an utterance, which have repercussions for the underlying factorial function. For the sake of exposition, consider ideal Wackernagel placement, whereby a clitic in clausal second position represents a fixed point, i.e. the set of permutations is a so-called partial derangement  $D_{n,k}$ . Formally, a partial derangement is a set of

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<sup>521</sup> As a default, German declarative root clauses exhibit verb-second, while subordinate clauses are verb-final. However, I shall refrain from dealing with the subtleties of German syntax.

$n$  permutations with  $k$  fixed points, cf. (8.6a).<sup>522</sup> Let us assume that  $n = 4$  syntactic elements {S, V, O, CL}, then we receive (8.7). Even for low values of  $k$ , fixed points reduce the number of possible permutations dramatically due to the factorial growth.<sup>523</sup>

$$(8.6) \quad D_{n,k} = (n - k)!$$

$$(8.7) \quad (n - k)! = (4 - 1)! = 3! = 6$$

Likewise, let us assume an utterance with 4 elements, but one of which is a prepositional head with a simple, non-branching complement NP. It is generally known that a P-head cannot be split from its first complement in Slavic (and beyond), such that the PP in our example is impenetrable. Accordingly, the positions of  $P^0$  and NP are fixed with respect to one another, and we encounter a partial derangement as in (8.7). Finally, a similar situation occurs in embedded clauses introduced by a subordinator. If the subordinate clause comprises  $n = 4$  elements, then the subordinator constitutes a fixed point, and the possible word order rearrangements are represented by a partial derangement following (8.7). It is evident that merely counting words is methodologically poor and that syntactic constituency is an important factor in determining the degree of freedom in clitic positioning.<sup>524</sup>

Fortunately, we do not need to determine and calculate all possible permutations and partial derangements within the obtained corpus data. This is because we do not have to account for all possible – i.e.  $n!$  or  $(n - k)!$  – word order variants of a given utterance. It is only the distribution of clitics in a particular string of constituents, which is of interest here. This string is manipulated by a single parameter, which is the position of the clitic itself. Moreover, the left-most and right-most margins of this string are not isomorphic to the left and right clause- or sentence-boundary, but potentially smaller. This is because we know from Section 4.8 that the right-most position for a clitic is determined by its verbal predicate. I introduce the notion PERMUTATION SPACE or  $\pi$ -SPACE ( $\mathbb{S}^n$ ) for the available positions between clause- or sentence-initial position and verbal predicate. The  $\pi$ -space's size is its cardinality on the constituency level.

In the following, I shall illustrate the idea of the  $\pi$ -space. Consider hypothetical structures like those in (8.8) consisting of a syntactic string of 6 elements. The elements  $\alpha, \beta, \gamma, \delta$  denote potentially permutable elements, i.e. syntactic constituents of an arbitrary non-verbal category,  $V^0$  is the verbal predicate, and CL is a clitic. While the clitic position is licit in (8.8a–d) according to empirical

<sup>522</sup> For the idealised case of full isomorphism between words and constituents, this is also the formal description for permutations of a non-thetic declarative clause in German with the finite verb being a simplex verb, i.e. it is not an analytical tense form, modal or phase verb construction, or phrasal verb.

<sup>523</sup> Compare the difference between  $n = 4$  ( $n! = 4! = 24$ ) and (8.7).

<sup>524</sup> The discussion is necessarily simplified and does not take hierarchical structure into consideration.



evidence, the structures in (8.8e–f) are ruled out, because the clitic is positioned far too far to the right. The notation in (8.9) depicts the fact that not all configurations are available for CL. The white boxes indicate available clitic sites, whereas black boxes are ruled out as positions for clitics. The successive leftward shifting of  $V^0$  shows that the permutation space contracts (the blocked sites may be treated like fixed points and be disregarded).

- (8.8) a.  $\alpha$  CL  $\beta$   $V^0$   $\gamma$   $\delta$   
 b.  $\alpha$   $\beta$  CL  $V^0$   $\gamma$   $\delta$   
 c.  $\alpha$   $\beta$   $V^0$  CL  $\gamma$   $\delta$   
 d. CL  $\alpha$   $\beta$   $V^0$   $\gamma$   $\delta$   
 e. \*  $\alpha$   $\beta$   $V^0$   $\gamma$  CL  $\delta$   
 f. \*  $\alpha$   $\beta$   $V^0$   $\gamma$   $\delta$  CL

- (8.9) a.  $\square$   $\alpha$   $\square$   $\beta$   $\square$   $\gamma$   $\square$   $\delta$   $\square$   $V^0$   $\square$   
 b.  $\square$   $\alpha$   $\square$   $\beta$   $\square$   $\gamma$   $\square$   $V^0$   $\square$   $\delta$   $\blacksquare$   
 c.  $\square$   $\alpha$   $\square$   $\beta$   $\square$   $V^0$   $\square$   $\gamma$   $\blacksquare$   $\delta$   $\blacksquare$   
 d.  $\square$   $\alpha$   $\square$   $V^0$   $\square$   $\beta$   $\blacksquare$   $\gamma$   $\blacksquare$   $\delta$   $\blacksquare$   
 e.  $\square$   $V^0$   $\square$   $\alpha$   $\blacksquare$   $\beta$   $\blacksquare$   $\gamma$   $\blacksquare$   $\delta$   $\blacksquare$

In this way, one may calculate the  $\pi$ -space from the annotated parameters as the sum of a clitic's phrasal position  $p$ , its phrasal distance  $d$  to the governing head (predicate), and  $j$  (with  $j \in \{0, 1\}$ ) as in (8.10). The addition of  $j$  is contextually necessary to avoid an off-by-one error (also known as fencepost error). For an independent clause:  $j = 1$ . For an embedded clause introduced by a subordinator:  $j = 0$ .

$$(8.10) \quad \mathbb{S}^n = p + d + j, \quad j \in \{0, 1\}$$

In sum, this section has suggested to counterbalance the quantities of attested clitic placement with the cardinalities of the permutation space. The proposal is meant to evaluate a clitic's factual distribution in the clause towards the hypothetically available sites it can take. The approach presented here is exemplified in Chapter 10.9.

## **PART III: CORPUS STUDY**

## 9 Overview of Part III

Part III is the empirical core of the thesis and includes the corpus study referred to in the thesis's title. Although I essentially conducted a single corpus analysis (across six different data sets), part III is essentially organised into a series of case studies of different size and scope, each pursued in an individual chapter. However, the case studies are not completely independent of one another, since they operate on the very same data samples for each lect. However, Chapter 12 is an exception, as it is not based on the same portion of Czech corpus data as the other chapters. All corpus analyses presented here are quantitative in nature and nested within a basically descriptive statistical framework except Chapter 13. Statistical tests are not in the present study's focus and may occur only to a very small extent.

Chapter 10 is the most comprehensive chapter of Part III and examines the distribution of clitics in spoken Czech, Polish, Goralish, Slovak, Silesian, and Kashubian. Basically, this investigation focuses on the attested positions clitics take in the clause, clitic clustering, and clitic climbing. After a language-by-language analysis, there will be a cross-linguistic comparison among the lects under scrutiny followed by an application of the permutation space. Sections on individual lects may discuss further attested phenomena, i.e. the possessive resultative in Czech and agreement drop in Goralish and Silesian. Chapter 11 is devoted to occurrences of clitics which are not licensed by verbal predicates but by non-verbal categories. As such, it is rather documentary and catalogues various structures. Chapter 12 is a purely quantitative case study of register effects on cliticisation in Czech comparing deviations from Wackernagel positioning in written and spoken Czech. Chapter 13, in turn, presents a qualitative case study on the interrelation of delayed clitic placement and information structure in Polish. A summary of the thesis, discussion, and prospects for future research are to be found in the final Chapter 14.

## 10 Attested clitic distributions in West Slavic

### 10.1 Introductory remarks

Chapter 10 presents the results of the corpus study on clitic distribution in the clause across the six West Slavic lects introduced in Chapter 2. A major aspect of distribution is clausal positioning. It was argued in Section 8.2 that existing approaches to analysing clitic distribution on empirical grounds are too coarse, because they do not register the variance in clitic positioning transparently or do not distinguish between different properties of what is usually subsumed under the label *distribution*. I argue therefore that an approach is more promising that includes the phrase level in the syntactic analysis, even though it is surface-oriented, and assigns each clitic token a numerical value corresponding to the syntactic surface position. Such an approach was adopted in a case study on the positioning of the Polish pronominal clitic *mu* ‘him.DAT’ by I. Kulik (2019) and it has been further elaborated for the purpose of the present thesis to cover more distributional and syntagmatic features.<sup>525</sup>

In the following, I will provide the descriptive statistics for the attested distribution patterns. Each lect will be presented individually in a separate section before I will turn to a comparative perspective. Czech, Polish, and Goralish will be treated in Sections 10.2, 10.3, and 10.4 respectively. They represent the thesis’s core objects of study with broad empirical databases in terms of quantity. This is why they come first before turning to the supplementary material from Slovak, Silesian, and Kashubian in Sections 10.5, 10.6, and 10.7 respectively. The database for the latter three lects becomes successively scant. This is because of purely practical reasons (i.e. time restrictions) in the case of Slovak but due to limited availability of data in the case of Silesian and Kashubian. All sections will follow an identical structure. Subsection 1 deals with clausal position and placement relative to the verbal predicate. Subsection 2 reports on clustering. Subsection 3 addresses clitic climbing. The topics covered by these three recurring subsections represent “core” issues of Slavic cliticisation. Furthermore, there may be additional subsections subsuming miscellaneous phenomena beyond the core, if such phenomena were represented in the data. On the one hand, it is clitic positioning under agreement drop especially for Goralish and Silesian. On the other hand, clitic occurrences in the construction of the so-called possessive resultative were observed for Czech. After having presented each lect individually, selected issues will be compared across all six lects in Section 10.8. Finally, Section 10.9 will analyse attested clitic position against hypothetically possible

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<sup>525</sup> See Section 8.1 on the annotation.

positions utilising the notion of permutation space as introduced in Section 8.3. Chapter 10 will be concluded in Section 10.10.

While Section 10.2 on Czech will serve as a blueprint for all other sections, some clarifying notes are needed as to what the reader might expect. The sections on the individual West Slavic lects will provide numerical data organised in tables, whose purpose is a transparent presentation of occurrence frequencies.<sup>526</sup> The data will be furthermore described in prose. However, Section 10.2 will be particularly detailed in that readers will find the successive elaboration of the data presentation approach as well as the reasoning for why certain decisions were taken. A certain degree of redundancy cannot be avoided, therefore. However, having established the framework, it will be presupposed in subsequent sections.

The presentation of the empirical findings will be accompanied by only little rudimentary inferential statistics (chi-square test for homogeneity). However, bear in mind that an inferential statistical analysis is not the thesis's main objective, such that there are no sophisticated statistics (e.g. correlation analysis) employed in the corpus study. In order to keep tables as clear as possible, the lack of an attestation will not be quantified by zero but indicated by an em dash (—). In general, ratios will be provided with the first decimal place, except when this would yield a 0.0% figure, then the first decimal place showing a number different from zero will be cited. Adopting the style of Kolaković et al. (2022), the exploratory corpus study presented here is guided by the following questions:

Q1: How are clitics in spoken West Slavic lects distributed in the clause?

Q1a: Which positions do clitics occupy?

Q1b: What are the quantities of the respective positions?

Q1c: How are clitics positioned with respect to their licensing predicate?

Q1d: How are clitics positioned in relation to verbs?

Q2: What is the status of clitic clustering?

Q2a: Is clustering mandatory?

Q2b: Can clitic clusters be split and in which environments?

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<sup>526</sup> I focus primarily on figures in the following sections on the individual lects and postpone the presentation of diagrams to the comparative Section 10.8. There are two reasons for this procedure. First, as we are partly dealing with phenomena of low frequency, diagrams may turn out to be less informative when it comes to details. However, neither do I wish to affect readability negatively by putting too much information into a single diagram nor do I intend to overload the thesis with figures next to tables. Second, the comparison of the six lects in Section 10.8 demands a more concise presentation style that allows readers to capture facts with the naked eye. Tables are not practical in this case, at all, and intuitively comprehensible bar plots need to be employed.

- Q3: What is the status of the clitic cluster itself?
- Q3a: What is its internal structure?
- Q3b: How consistently does it comply with described word order templates?

Bearing in mind that the present study is exploratory, one needs to acknowledge that the data may reveal facts that are beyond the immediate scope of the research questions Q1–Q3. These questions may be regarded as driving the study, but the study does not exhaust itself in them.

## 10.2 Czech

### 10.2.1 Clitic placement

Out of a random sample of roundabout 3,500 raw data points from ČNK's corpus of spoken language ORAL v1, which were examined for the annotation cycle, a total of 2,933 clause tokens was viable for analysis.<sup>527</sup> Let us look at clitic positioning first. Table 10.1 provides the absolute and relative frequencies for attested clitic positions in spoken Czech. The column "Total" provides the accumulated quantities for the entire data sample, while the columns "Pronominal" and "Reflexive" allow us to further differentiate between those two categories of clitics. Approaching frequency distribution from the most coarse to the more fine-grained makes sense on two different grounds. First, pronominal and reflexive clitics share only some properties while still exhibiting their own special features. Therefore, one may suppose that the underlying distinctions between both categories could also be visible through different distributional properties. Second, from a methodological point of view, bear in mind that each clitic type was elicited in roughly the same amount. This gives rise to an arithmetic imbalance between pronominal and reflexive clitics with the consequence of the overall frequency distribution being biased towards pronominal clitics. This is in contrast with the actual occurrence frequencies of pronominal vs. reflexive clitics, as the latter are more frequent than the former in text and speech.<sup>528</sup> However, the data samples drawn from the corpus are not meant to

<sup>527</sup> The difference between elicited and analysed tokens traces back to the fact that not the entire pool of acquired data points met minimal requirements for analysis. Erroneously tagged words and distorted utterances are generally the main reason for dispensing with data points across corpora. For instance, consider the plural demonstrative *ti* being tagged as pronoun due to misinterpretation as homophonic pronominal clitic *ti* or think of an excessively disfluent speech flow, e.g. by disruption.

<sup>528</sup> ČNK's most recent balanced reference corpus of Written Czech, SYN2020, contains 1,589,053 tokens of non-subject pronouns vs. 3,099,618 reflexive tokens (CQL queries: (i) `[tag="P[PH].*" & word!="[Jj]á|[Tt]y|[Oo]n|[Oo]na|[Oo]no|[Mm]y|[Vv]y|[Oo]ni|[Oo]ny"`], (ii) `[tag="P[67].*"]`). Both figures involve all types irrespective of cliticity. A corresponding query for the spoken component ORAL v1 yields 114,067 pronouns vs. 142,116 reflexives (the query for pronouns was adapted to

statistically mirror the entire population of constant clitics on a smaller scale, but to provide a sufficiently large data basis for each constant clitic. Looking at pronominals and reflexives individually, allows us to level the bias. Table 10.1 highlights the 2P and 3P pattern, and – in the bottom-most row – the totalised Delayed Placement (DP) by grey shading for ease of reference.

Table. 10.1: Frequencies of attested clitic clausal positions in spoken Czech.

| Position | Total      |            | Pronominal |            | Reflexive  |            |
|----------|------------|------------|------------|------------|------------|------------|
|          | Abs. freq. | Rel. freq. | Abs. freq. | Rel. freq. | Abs. freq. | Rel. freq. |
| 1P       | 110        | 3.8%       | 79         | 3.8%       | 31         | 3.6%       |
| 2P       | 2,381      | 81.2%      | 1,717      | 82.7%      | 664        | 77.5%      |
| 3P       | 350        | 11.9%      | 227        | 10.9%      | 123        | 14.4%      |
| 4P       | 70         | 2.4%       | 42         | 2.0%       | 28         | 3.3%       |
| 5P       | 14         | 0.5%       | 6          | 0.3%       | 8          | 0.9%       |
| 6P       | 7          | 0.2%       | 4          | 0.2%       | 3          | 0.4%       |
| 7P       | 1          | 0.03%      | 1          | 0.05%      | —          | —          |
| Sum      | 2,933      | 100%       | 2,076      | 100%       | 857        | 100%       |
| DP       | 442        | 15.1%      | 280        | 13.5%      | 162        | 18.9%      |

To begin with, the data exhibit considerable variation with respect to position types with clitic in 1P–7P being attested. However, the frequencies for the individual positions show a statistical distribution with a clear central tendency. In total, more than 80% ( $\approx \frac{4}{5}$ ) of all elicited clitic occurrences exhibit Wackernagel positioning (2P).<sup>529</sup> Almost 12% ( $\approx \frac{1}{8}$ ) of all data points show Clitic Third (3P). Frequencies decrease rapidly with growing distance from the left periphery. The total of all Delayed Placement occurrences (DP) comes to 15% ( $\approx \frac{1}{7} - \frac{1}{6}$ , or  $\frac{2}{13}$ ). Notably, almost 4% ( $\approx \frac{1}{25}$ ) of clitics occur in first position (1P), which is more often than the totalised Delayed Placement  $\geq$  4P. The occurrences of 1P are unevenly distributed after clause boundaries and coordinating conjunctions. 88 tokens or 80% ( $= \frac{4}{5}$ ) of 1P occurrences are positioned following a clause boundary, while 22 tokens or 20% ( $= \frac{1}{5}$ ) are found in a syntagm after a conjunction, cf. Table 10.2.

3rd person nominative pronouns labialised onset v-). While the proportion of non-subject pronouns vs. reflexives in Written Czech is 1 : 2, spoken Czech exhibits a proportion of 4 : 5. The very different proportions notwithstanding, reflexives have a higher occurrence frequency than pronouns in any case.

<sup>529</sup> Fractional numbers are provided for the sake of transparency and are meant to give useful approximative numbers; read as “four out of five” etc.

Table 10.2: Frequencies for Clitic First in spoken Czech after clause boundary (#) and coordinating conjunction (in total).

| Type        | Abs. freq. | Rel. freq. w/i 1P |
|-------------|------------|-------------------|
| # 1P ...    | 88         | 80%               |
| CONJ 1P ... | 22         | 20%               |

I cite one attestation for each position in (10.1). Clitic First positioning is given with an example of absolute clause-initial and post-conjunction in (10.1a–b) respectively. The question of the phonological features of clitics in 1P was briefly mentioned in Section 4.4.1 (cf. also Section 3.12.2, Footnote 183), but discussion shall be postponed to the next Section 10.2.2 on clustering after the attested types of clitic clusters will be established. Example (10.1c) shows “ordinary” 2P placement. Delayed Placement is found in (10.1d–h). Note that Delayed Placement is not confined to subordinate clauses. The remaining examples show 3P (10.1d), 4P (10.1e), and 6P (10.1g) in a subordinate clause, whereas the utterances in (10.1f) with 5P and (10.1h) with 7P present independent main clauses. The sentence-initial elements *to* and *tak* respectively serve for the purpose of text coherence, but do not introduce subordinate clauses here. Note that examples (10.1g–h) involve diacsis, whereby one clitic (AUX, REFL) occupies third position, while the other (REFL, PRN) is further delayed to 6P and 7P respectively.

- (10.1) a. #**mi**<sup>1</sup> řekni (kdo mu dá prostěradlo za  
me.DAT tell.IMP.2SG who.NOM him.DAT give.PRS.3SG bedsheet.ACC.SG for  
míň jak patnáct korun)  
less how fifteen.ACC koruna.GEN.PL  
‘tell me (who [will] give him a bed sheet for less than fifteen korunas[?])’
- b. [...] **ale**<sup>0</sup> **bych**<sup>1</sup> **tě**<sup>1</sup> vzal třeba na lanko [...]   
but COND.1SG you.ACC take.PST.SG.M maybe on rope.ACC.SG  
‘... but maybe I could take you along with a rope ...’
- c. (to ne ne ne) to<sup>1</sup> **se**<sup>2</sup> mně ne-libi  
this.NOM.SG.N no no no this.NOM.SG.N REFL.ACC me.DAT NEG-like.PRS.3SG  
‘([not] this [one], no no no,) I don’t like it’
- d. [...] **enomže**<sup>1</sup> **já**<sup>2</sup> **sem**<sup>3</sup> **se**<sup>3</sup> k tomu eště  
just\_that I.NOM AUX.1SG REFL.ACC towards that.DAT.SG.N still  
ne-dostal teda [...]   
NEG-get.PST.SG.M then  
‘... just that I didn’t still get around to doing it by that time ...’



- e. (to                      úno                      sem                      si                      taky potom vzala)  
 this.ACC.SG.N Uno.ACC.SG AUX.1SG REFL.DAT also later take.PST.SG.F  
 protože<sup>1</sup> Maruna<sup>2</sup> mně<sup>3</sup> ho<sup>4</sup> tu nechala tenkrát  
 because Maruna.NOM me.DAT him.ACC here leave.PST.SG.F ten\_times  
 ‘(later, I also took that Uno [card game] for myself) because Maruna left it there for me  
 [about] ten times’
- f. to<sup>1</sup> prostě<sup>2</sup> [NP takovýdle                      věci ]<sup>3</sup>                      já<sup>4</sup>                      bych<sup>5</sup>                      mu<sup>5</sup>  
 TO simply this\_kind\_of.ACC.PL.F thing.ACC.PL I.NOM COND.1SG him.DAT  
 ne-řekla  
 NEG-tell.PST.SG.F  
 ‘so, I simply wouldn’t have told him such kinda things’
- g. [...] takže<sup>1</sup> já<sup>2</sup> sem<sup>3</sup> dycky<sup>4</sup> [PP V pět ]<sup>5</sup>                      si<sup>6</sup>                      vstal                      [...]  
 so\_that I.NOM AUX.1SG always in five.ACC REFL.DAT get\_up.PST.SG.M  
 ‘... so that I always got up at five o’clock ...’
- h. tak<sup>1</sup> třeba<sup>2</sup> se<sup>3</sup> to<sup>4</sup>                      někdy<sup>5</sup>                      povede<sup>6</sup>                      ti<sup>7</sup>                      zasadit  
 so maybe REFL this.NOM.SG.N some\_day succeed.PRS.3SG you.DAT plant.INF  
 ‘thus, you’ll maybe succeed in planting it some day’ (Cz; ČNK oral\_v1)

Table 10.3 reports the three measures of central tendency arithmetic mean ( $\bar{x}$ ), median ( $Md$ ), and mode ( $Mo$ ) for the observed frequency distribution of all clitic positionings in spoken Czech.<sup>530</sup> The arithmetic mean is provided with the standard deviation (SD) in parentheses; this gives information about the dispersion of the data. A smaller SD indicates a more condensed distribution, such that the observations rather cluster and extreme values are more likely to be outliers. In contrast, a higher SD implies that the values are scattered across a wider area with the consequence that more extreme values belong to the core data and are less likely outliers. The three measures agree in showing second position placement as the basic distribution pattern for clitics in Czech. The transitive relation based on frequential inequality (greater than: >) described by (10.2) is fully sufficient in a simple descriptive account of the observed frequencies. The 7P pattern is provided in parentheses, because it is only observed with pronominal clitics.

<sup>530</sup> Supposedly, the *arithmetic mean* ( $\bar{x}$ ) is the best-known measure of central tendency. It is the sum of all observed values divided by the sum of observations and has a high degree of *efficiency*, i.e. precision. On the other hand, it is prone to bias by few extreme values and thus exhibits a comparatively small degree of robustness. The median ( $Md$ ) is less precise but more robust. It divides an ordered data set into two halves (thus, one half is not higher than the median, while the other is not smaller). Finally, the *mode* ( $Mo$ ) represents the value or values occurring most often. All three values represent a kind of “average” that characterises a distribution (cf. Bortz & Schuster 2010: 25–29; Meindl 2011: 81–86 for an overview of the three measures of central tendency).

Table 10.3: Measures of central tendency for clitic positioning in spoken Czech.

| Measure:          | $\bar{x}$ (SD)      | <i>Md</i> | <i>Mo</i> |
|-------------------|---------------------|-----------|-----------|
| Value (position): | 2.15 ( $\pm 0.54$ ) | 2         | 2         |

(10.2) 2P > 3P > 1P > 4P > 5P > 6P (> 7P)

Distinguishing between the two major categories of clitics, the relative frequencies between both classes differ with the pronominal clitics exhibiting more 2P and less 3P occurrences than reflexive ones. The question arises whether these differences are an accidental outcome of the data sampling procedure or whether they are significant. A chi-square ( $\chi^2$ ) test of homogeneity is the preferred means for approaching this question. It tests whether the distributions compared with each other belong to the same population. The null hypothesis  $H_0$  presumes that both samples belong to the same population. If the test result is not significant, we may proceed under  $H_0$ . If the result is significant, then  $H_0$  is rejected and the alternative hypothesis  $H_1$  is supported, i.e. that both samples belong to different populations. Applying the chi-square test of homogeneity to the sample of pronominal and reflexive clitics allows us to ascertain with a given level of confidence whether both categories of clitics pattern the same way or not. I choose a significance level of 5%, which is a conventional value in many disciplines. For the chi-square test to work appropriately and give reliable results, one must meet three conditions: First, the variables under scrutiny must be categorical, i.e. either nominal or ordinal (cf. Meindl 2011: 162). Second, the data sample must be sufficiently large with at least 40 observations in total (cf. Meindl 2011: 166).<sup>531</sup> Third, the minimum frequency per cell is 5 (cf. Meindl 2011: 163, 166, 170). The first condition is met, as the relevant variable *clitic position* is ordinal, i.e. representing categories ordered in succession. Condition two is also fulfilled, given the size of the sampled database. Condition three is not met, however, because there are frequencies < 5 for 6P and 7P. In such cases it is appropriate to subsume categories with too little observations under a new label, if it is a meaningful one. Since a general division is made between Wackernagel and Delayed Placement in the literature, it is useful to subsume positions 3P–7P under the single category DP, such that the distribution is divided into 1P, 2P, DP for the purpose of the homogeneity test. With a calculated result for  $\chi^2 = 13.37$  ( $p = .05$ ,  $df = 2$ ) the  $H_0$  is rejected.<sup>532</sup> That is, the pronominal and reflexive clitics do not share an equal distribution.

<sup>531</sup> There are varying opinions on the necessary minimum size of a sample in the literature. I follow Meindl (2011) for the sake of convenience.

<sup>532</sup> The result reads as: the calculated or so-called empirical chi-square value ( $\chi^2_{\text{emp}}$ ) is 13.37. This result is compared with the critical chi-square value ( $\chi^2_{\text{crit}}$ ) for a significance level of 5% ( $p = .05$ ) with a degree of freedom of 2 ( $df = 2$ ). The corresponding critical value is  $\chi^2_{\text{crit}} = 5.991$ . If  $\chi^2_{\text{emp}} > \chi^2_{\text{crit}}$ , then the test result is significant and  $H_0$  is rejected. If  $\chi^2_{\text{emp}} \leq \chi^2_{\text{crit}}$ , then the test result is not significant and the  $H_0$  may be kept for now.

Table 10.4 further differentiates between the individual clitics as types. The general picture re-emerges, but there is considerable fluctuation in the quantities. Whereas the reflexive clitics appear quite similar, one might tentatively distinguish three groups of pronominal clitics on a subjective basis. The first group contains *tě*, which has almost 90% of occurrences in 2P. The second group covers *ho*, *mi*, and *mu* which share similarities for 2P and 3P and are also close with respect to the peripheral positions. Group three is made up by *ti*, whose pattern essentially resembles the reflexive clitics. In general, delayed clitic placement (DP) ranges from almost 7% to 20%.

Table 10.4: Frequencies of attested clitic clausal positions in spoken Czech by clitic.

| Pos | <i>tě</i> |       | <i>ho</i> |       | <i>mi</i> |       | <i>ti</i> |       | <i>mu</i> |       | <i>se</i> |       | <i>si</i> |       |
|-----|-----------|-------|-----------|-------|-----------|-------|-----------|-------|-----------|-------|-----------|-------|-----------|-------|
|     | Abs.      | Rel.  | Abs.      | Rel.  | Abs.      | Rel.  | Abs.      | Rel.  | Abs.      | Rel.  | Abs.      | Rel.  | Abs.      | Rel.  |
| 1P  | 16        | 3.9%  | 14        | 3.3%  | 17        | 4.3%  | 22        | 4.9%  | 10        | 2.5%  | 13        | 3.0%  | 18        | 4.3%  |
| 2P  | 368       | 89.3% | 353       | 83.6% | 320       | 81.2% | 348       | 77.3% | 328       | 82.4% | 338       | 77.3% | 326       | 77.6% |
| 3P  | 20        | 4.9%  | 45        | 10.7% | 47        | 11.9% | 67        | 14.9% | 48        | 12.1% | 63        | 14.4% | 60        | 14.3% |
| 4P  | 5         | 1.2%  | 8         | 1.9%  | 9         | 2.3%  | 11        | 2.4%  | 9         | 2.3%  | 19        | 4.3%  | 9         | 2.1%  |
| 5P  | 2         | 0.5%  | 1         | 0.2%  | 1         | 0.3%  | 1         | 0.2%  | 1         | 0.3%  | 4         | 0.9%  | 4         | 1.0%  |
| 6P  | 1         | 0.2%  | 1         | 0.2%  | —         | —     | —         | —     | 2         | 0.5%  | —         | —     | 3         | 0.7%  |
| 7P  | —         | —     | —         | —     | —         | —     | 1         | 0.2%  | —         | —     | —         | —     | —         | —     |
| Sum | 412       | 100%  | 422       | 100%  | 394       | 100%  | 450       | 100%  | 398       | 100%  | 437       | 100%  | 420       | 100%  |
| DP  | 28        | 6.8%  | 55        | 13.0% | 57        | 14.5% | 80        | 17.8% | 60        | 15.1% | 86        | 19.7% | 76        | 18.1% |

An overall test of homogeneity for all pronominal clitics  $\{tě, ho, mi, ti, mu\}$  yielded a significant result with  $\chi^2 = 28.22$  ( $p = .05$ ,  $df = 8$ ). Note that the test merely indicates that the group of pronominal clitics does not pattern alike as a whole. That is, at least one member of the class deviates, but it is uncertain how many members, which member(s), or how great the deviation is. This is why I test for homogeneity pairwise.<sup>533</sup> Table 10.5 provides the empirical  $\chi^2$ -value for each pair of clitics. The red-shaded cells give significant chi-square values. As can be seen very clearly, the pronominal clitic *tě* does not pattern with any other clitic. Another significant test result exists only for the pair  $\{ho, se\}$ . A homogeneity test for the set of pronominal clitics excluding *tě* – i.e. for  $\{ho, mi, ti, mu\}$  – yields a non-significant result in turn ( $\chi^2 = 8.00$ ,  $p = .05$ ,  $df = 6$ ). That is, the pronominal clitics *ho*, *mi*, *ti*, *mu* may be considered to share a common distribution pattern in lack of contradictory evidence. The same test applied under addition of *si* with  $\{ho, mi, ti, mu, si\}$  returns a significant result again ( $\chi^2 = 16.72$ ,  $p = .05$ ,  $df = 8$ ). In contrast, testing the triplet  $\{ti, se, si\}$  yields a non-significant result ( $\chi^2 = 2.00$ ,  $p = .05$ ,  $df = 4$ ) as expected, i.e. the three clitics share the same distribution pattern. There will be no more in-text tables presenting  $\chi^2$ -test results, in order not to negatively impact reader-friendliness (but see the Appendix for calculated  $\chi^2$ -values).

<sup>533</sup> The procedure of pairwise testing is inspired by Kolaković et al.'s (2022: 203) analysis of spoken Bosnian.

Table 10.5: Matrix of calculated  $\chi^2$ -values for the homogeneity test of clitic pairs ( $p = 0.5$ ,  $df = 2$ ).

|           | <i>tě</i> | <i>ho</i> | <i>mi</i> | <i>ti</i> | <i>mu</i> | <i>se</i> | <i>si</i> |
|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|
| <i>tě</i> |           | 9.11      | 12.88     | 24.92     | 15.08     | 29.45     | 24.74     |
| <i>ho</i> |           |           | 0.99      | 5.55      | 1.10      | 6.49      | 4.94      |
| <i>mi</i> |           |           |           | 1.97      | 1.97      | 4.09      | 1.97      |
| <i>ti</i> |           |           |           |           | 4.78      | 1.89      | 0.19      |
| <i>mu</i> |           |           |           |           |           | 3.32      | 3.60      |
| <i>se</i> |           |           |           |           |           |           | 0.88      |
| <i>si</i> |           |           |           |           |           |           |           |

Another aspect of clitic distribution is LATERALITY. By *laterality* I mean the linear precedence relation between a clitic and its predicate. Usually, this feature is not treated explicitly in the literature except by a few publications, e.g. Kosek (2015). The rationale for taking laterality into consideration is two-fold. On the one hand, Wackernagel clitics are expected to occur on their predicates' left side most often, provided that they strive towards the left periphery of the clause. The notion is thus linked to some extent with Wackernagel cliticisation in terms of likelihood of occurrence. However, a high ratio of "L" (for left lateral) alone is not indicative of 2P cliticisation per se. It is well-known that the VA-clitic languages Bulgarian and Macedonian also exhibit a significant degree of left lateral placement. An important difference between both types of languages is the availability of non-contingency between clitic (cluster) and predicate. Slavic 2P languages do have this option in contrast to their cognate VA languages. On the other hand, the question may be put, to what degree a language's Wackernagel clitic system may be subject to diachronic restructuring. Looking at East Slavic, one can hypothesise that a possible result of grammatical change in the clitic system could yield a similar pattern as in Russian, for instance. As pronominal clitics (or better: short pronouns) are treated as non-existent in East Slavic (often narrowed down to Russian), the postfixed reflexive marker *-sja* remains the only save candidate for comparison. Accounting for laterality thus provides us with information about the number of postfixed clitics, which could be indicative of a diachronic trend. Right lateral placement implies contiguity, for the predicate poses the rightwards boundary for clitic positioning.<sup>534</sup> The annotation tag "R" therefore reads as CLITIC POSTPOSITION (exceptions will be mentioned below), which is equivalent to what is labelled "final position" by Kosek (2009, 2011, 2015, 2017) and Reinkoswki (2001). Clitic postposition or right lateral placement is expected in verb-initial clauses trivially. Table 10.6 presents the frequencies for laterality. Over 80% of all clitic

<sup>534</sup> See Section 4.8 for the relation on right-lateral and verb-adjacent placement.

occurrences are placed to the left side of the predicate. Pronominal clitics are in 80% of cases left-lateral, while reflexive clitics approximate almost 90%.<sup>535</sup>

Table 10.6: Frequencies for laterality (left, right) of clitic occurrences in spoken Czech.

| Laterality | Total      |            | Pronominal |            | Reflexive  |            |
|------------|------------|------------|------------|------------|------------|------------|
|            | Abs. freq. | Rel. freq. | Abs. freq. | Rel. freq. | Abs. freq. | Rel. freq. |
| L          | 2,422      | 82.6%      | 1,672      | 80.5%      | 750        | 87.5%      |
| R          | 511        | 17.4%      | 404        | 19.5%      | 107        | 12.5%      |

A closer look at individual types in Table 10.7 reveals a more or less consistent pattern across individual clitics except for the pronominal type *tě*, which shows an almost equal distribution of left and right laterality (Lat). Recall that the short form *tě* also stood out regarding position. Beyond that, the pattern significantly differs within the class of reflexive clitics ( $\chi^2 = 5.71$ ,  $p = .05$ ,  $df = 1$ ), which exhibit extreme values within the otherwise homogeneous distribution patterns. While accusative *se* shows the highest frequency of left lateral placement with ca. 90%, its dative counterpart *si* presents the lower boundary with ca. 85%.

Table 10.7: Frequencies of laterality (left, right) of clitic occurrences in spoken Czech by clitic.

| Lat | <i>tě</i> |       | <i>ho</i> |       | <i>mi</i> |       | <i>ti</i> |       | <i>mu</i> |       | <i>se</i> |       | <i>si</i> |       |
|-----|-----------|-------|-----------|-------|-----------|-------|-----------|-------|-----------|-------|-----------|-------|-----------|-------|
|     | Abs.      | Rel.  | Abs.      | Rel.  | Abs.      | Rel.  | Abs.      | Rel.  | Abs.      | Rel.  | Abs.      | Rel.  | Abs.      | Rel.  |
| L   | 214       | 51.9% | 373       | 88.4% | 345       | 87.6% | 392       | 87.1% | 348       | 87.4% | 394       | 90.2% | 356       | 84.8% |
| R   | 198       | 48.1% | 49        | 11.6% | 49        | 12.4% | 58        | 12.9% | 50        | 12.6% | 43        | 9.8%  | 64        | 15.2% |

But is looking at portions of left lateral vs. right lateral placement really telling? It is to some extent, indeed, as we encounter statistically significant differences among clitics in Czech. However, this level of analysis misses the configurations of verb-adjacency vs. non-contiguity. Recall that the Slavic VA-clitic languages do not have a choice here. Apart from the typological dimension, the distinction between adjacent and non-adjacency is relevant towards Avgustinova & Oliva's (1997: 26) statement that Delayed Placement in Czech is confined to instances of adjacency with the clitic's predicate. More generally, the analysis cannot relate to the observation that clitics in Slavic oscillate between clausal second position and predicate-adjacency (erroneously labelled *verb-adjacency* in the literature) both synchronically and diachronically.<sup>536</sup> It is therefore advantageous to incorporate the dimension of DISTANCE (with  $d = 0$  vs.  $d > 0$ ) into the analysis. This is why Table 10.8 and Table 10.9 present a finer grid with a three-way classification of LEFT DISTANT (Ld = left lateral non-adjacent,

<sup>535</sup> It is thus safe to say that Czech does not show any traces of a diachronic change towards a system resembling Russian.

<sup>536</sup> I thank Uwe Junghanns (p.c.) for bringing the relatedness of 2P and VA to my attention (cf. also Franks & King 2000: 319).

$d > 0$ ), LEFT ADJACENT (La = left lateral adjacent,  $d = 0$ ), and RIGHT (R(a) = right lateral, trivially adjacent,  $d = 0$ ). Table 10.8 displays the quantities for the entire data set and for clitic categories. Table 10.9 differentiates between individual clitics and Figure 10.1 visualises the corresponding distribution patterns.

Table 10.8: Frequencies for laterality of clitic occurrences with left distant (Ld), left adjacent (La), right adjacent (R(a)) in spoken Czech.

| Laterality | Total      |            | Pronominal |            | Reflexive  |            |
|------------|------------|------------|------------|------------|------------|------------|
|            | Abs. freq. | Rel. freq. | Abs. freq. | Rel. freq. | Abs. freq. | Rel. freq. |
| Ld         | 1,099      | 37.6%      | 729        | 35.3%      | 370        | 43.2%      |
| La         | 1,315      | 45.0%      | 935        | 45.2%      | 380        | 44.3%      |
| R(a)       | 503        | 17.2%      | 397        | 19.2%      | 106        | 12.4%      |

Table 10.9: Frequencies for laterality with left distant (Ld), left adjacent (La), right adjacent (R(a)) in spoken Czech by clitic.

| Lat  | <i>tě</i> |       | <i>ho</i> |       | <i>mi</i> |       | <i>ti</i> |       | <i>mu</i> |       | <i>se</i> |       | <i>si</i> |       |
|------|-----------|-------|-----------|-------|-----------|-------|-----------|-------|-----------|-------|-----------|-------|-----------|-------|
|      | Abs.      | Rel.  | Abs.      | Rel.  | Abs.      | Rel.  | Abs.      | Rel.  | Abs.      | Rel.  | Abs.      | Rel.  | Abs.      | Rel.  |
| Ld   | 84        | 20.4% | 140       | 33.2% | 165       | 41.9% | 182       | 40.4% | 158       | 39.7% | 213       | 48.7% | 157       | 37.4% |
| La   | 129       | 31.3% | 230       | 54.5% | 179       | 45.4% | 209       | 46.4% | 188       | 47.2% | 181       | 41.4% | 199       | 47.4% |
| R(a) | 197       | 47.8% | 46        | 10.9% | 47        | 11.9% | 57        | 12.7% | 50        | 12.6% | 42        | 9.6%  | 64        | 15.2% |

The idiosyncratic behaviour of the pronominal clitic *tě* is clearly visible in Figure 10.1. The form *tě* has a disproportionate amount of right lateral placement when compared to other clitics. Time and space limitations prohibit diving deep into this peculiarity. Suffice it to disclose that the seemingly special status of accusative *tě* is caused by the high token frequency of the collocation *prosím tě* ‘please’ in informal Czech discourse. I will not account for this fact and take the observed frequencies at face value here.<sup>537</sup>

<sup>537</sup> From a methodological point, the issue arising from highly frequent collocations calls for some sort of counterbalancing in order to level the bias such units bring about. However, any such approach must distinguish at least two instances, i.e. frozen syntagms and mutable collocations. The string *prosím tě* belongs to the former. It is fixed in terms of morphological forms and sequential order and occurs without any alteration. In fact, nothing prevents us to analyse it as a lexicalised chunk. Mutable collocations do not pose any problem, for they show productive grammar, such that a respective collocation is freely mutable in terms of morphology and syntax irrespective of the high likelihood of co-occurrence of its elements. Presumably, these two cases represent two extreme poles on a scale with intermediate stages in between. Perhaps, so-called ‘phrase constructions’/‘phraseo-schemes’ (Ger. *Phrasemkonstruktionen/Phraseoschemata*; Pavlova 2022) or ‘phraseogenic matrices’ (Pol. *matryca frazeotwórcza*; Podlecki 2021) are located in this intermediate area.

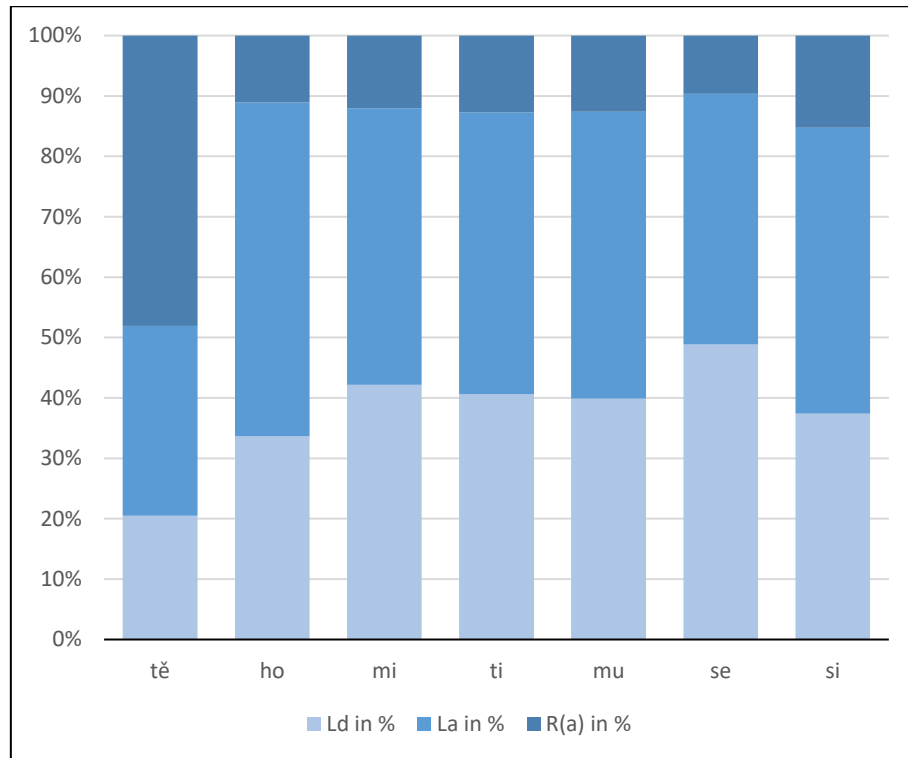


Figure 10.1: Proportion of laterality with left distant (Ld), left adjacent (La), right adjacent (R(a)) in spoken Czech by clitic.

Attestations for left-distant, left-adjacent, and right (adjacent) clitic placement are cited in example (10.3) for the sake of illustration. All examples show Delayed Placement, which is especially interesting with example (10.3a), which is not consistent with Avgustinova & Oliva's (1997: 26) hypothesis on the link between Delayed Placement and predicate-adjacent placement.

- (10.3) a. [...] že<sup>1</sup> já<sup>2</sup> sem<sup>3</sup> se<sup>3</sup> [Adv taky] ne-holil [...] [Ld]  
 that I.NOM AUX.1SG REFL.ACC also NEG-shave.PST.SG.M  
 '... that I didn't shave, too ...'
- b. (domníváme se) že<sup>1</sup> bud<sup>2</sup> doktor<sup>3</sup> ho<sup>4</sup> postrašil [...] [La]  
 reckon.PRS.1PL REFL that either doctor.NOM.SG him.ACC scare.PST.SG.M  
 '(we reckon) that either the doctor scared him ...'
- c. prostě<sup>1</sup> stačí<sup>2</sup> mu<sup>3</sup> [R(a)]  
 simply suffice.PRS.3SG him.DAT  
 'it simply suffices for him'
- (Cz; ČNK oral v1)

The revised grid may be further extended by yet another axis, i.e. POSITION. As a result, we receive a distributional analysis that distinguishes three individual parameters, i.e. *position*, *laterality*, and

*distance*.<sup>538</sup> Applying this enhanced grid to the entire Czech data set returns Table 10.10. The arithmetic differences in the frequencies between Table 10.10 and Table 10.1 total up to 0.6% and have two reasons. Either an instance of a truly right-disjoint clitic is missed, or the discrepancy arises from an annotation error. The error is small enough to be statistically negligible. From Table 10.10 we can read that Wackernagel placement is still the dominant pattern irrespective of (non-)adjacency to the predicate and laterality. What is more, we learn that clitics occur more frequently in deviant positions when they are in contact with the predicate. Over 10% of clitic occurrences represent positional deviations from Wackernagel placement (= 1P + DP) contiguous with the predicate. However, this does not preclude non-contiguity, such that non-contiguous deviations make over 7% of occurrences. Most importantly, Delayed Placement is only slightly less frequent with non-contiguous clitics than with verb-adjacent positioning (the difference amounts to 2.7%). Although we are dealing with phenomena of low frequency, it is justified to say that neither 1P nor DP are dependent upon syntagmatic contact with the predicate contra Avgustinova & Oliva's (1997: 26) supposition. From another angle, I would like to point out that the majority of attestations (= 62.2%) appears contiguously to the predicate. A good half of occurrences (= 51.1%) represents the intersection of Wackernagel placement and contact position [= 2P  $\cap$  (La  $\cup$  R(a))] with the predicate. The difference between the two left lateral placement patterns is not significant ( $\chi^2 = 2.64$ ,  $p = .05$ ,  $df = 2$ ), however. Right lateral placement may be regarded as rather marginal except for the trivial case of verb-initial clauses.

Table 10.10: Frequencies of clitic occurrences in spoken Czech by position, laterality, and distance.

| Pos | Left distant |       | Left adjacent |       | Right (adjacent) |       | Sum   |       |
|-----|--------------|-------|---------------|-------|------------------|-------|-------|-------|
| 1P  | 44           | 1.5%  | 66            | 2.3%  | N/A              | N/A   | 110   | 3.8%  |
| 2P  | 876          | 30.0% | 1,015         | 34.7% | 479              | 16.4% | 2,370 | 81.1% |
| 3P  | 146          | 5.0%  | 184           | 6.3%  | 15               | 0.5%  | 345   | 11.8% |
| 4P  | 28           | 1.0%  | 38            | 1.3%  | 4                | 0.1%  | 70    | 2.4%  |
| 5P  | 3            | 0.1%  | 7             | 0.2%  | 4                | 0.1%  | 14    | 0.5%  |
| 6P  | 2            | 0.1%  | 5             | 0.2%  | —                | —     | 7     | 0.2%  |
| 7P  | —            | —     | —             | —     | 1                | 0.03% | 1     | 0.03% |
| Sum | 1,099        | 37.6% | 1,315         | 45.0% | 503              | 17.2% | 2,917 | 99.8% |
| DP  | 179          | 6.1%  | 234           | 8.0%  | 24               | 0.8%  | 437   | 14.9% |

<sup>538</sup> I thank Edyta Jurkiewicz-Rohrbacher (p.c.) for discussion, which inspired the development of the grid.



The data in Table 10.11 differentiates between pronominal and reflexive clitics. The ratios for left-adjacent placement are comparable between both categories, but there are more obvious differences for left distant placement and clitic postposition. Reflexive clitics have a higher tendency to occur in left distant placement than pronominal clitics, which exhibit comparatively higher ratios of right placement. Bear in mind that the relative frequencies are calculated within the respective categories and not on the grounds of the overall Czech data sample. I will not zoom into individual clitic types here. Obviously, a table for each clitic would be necessary, which, multiplied with seven types, would provide an overload in data without benefit for the discussion. In other words, a reader-friendly presentation is not feasible here, such that the current level of detail must be regarded as sufficient.<sup>539</sup> From now on, I shall distinguish between *left-distant*, *left-adjacent*, and *right (adjacent)* placement for individual clitics without zooming into position, whereas position is accounted for as per clitic categories. Note that the data in Table 10.10 and Table 10.11 render Table 10.8 redundant, since the information presented in the latter is displayed in the respective rows “Sum”. The same applies to Table 10.1 at the section’s beginning thanks to the columns “Sum” irrespective of the deviations. I shall continue to use the enhanced grid as a default means of presenting the distribution frequencies.

Table 10.11: Frequencies of clitic occurrences in spoken Czech by position, laterality, and distance per clitic category.

| Pos | Pronominal |       |     |       |      |       |       |       | Reflexive |       |     |       |      |       |     |       |
|-----|------------|-------|-----|-------|------|-------|-------|-------|-----------|-------|-----|-------|------|-------|-----|-------|
|     | Ld         |       | La  |       | R(a) |       | Sum   |       | Ld        |       | La  |       | R(a) |       | Sum |       |
| 1P  | 31         | 1.5%  | 48  | 2.3%  | N/A  | N/A   | 79    | 3.8%  | 13        | 1.5%  | 18  | 2.1%  | N/A  | N/A   | 31  | 3.6%  |
| 2P  | 590        | 28.5% | 735 | 35.6% | 381  | 18.4% | 1,706 | 82.5% | 286       | 33.4% | 280 | 32.7% | 98   | 11.4% | 664 | 77.5% |
| 3P  | 92         | 4.5%  | 119 | 5.8%  | 12   | 0.6%  | 223   | 10.8% | 54        | 6.3%  | 65  | 7.6%  | 3    | 0.4%  | 122 | 14.2% |
| 4P  | 12         | 0.6%  | 27  | 1.3%  | 3    | 0.1%  | 42    | 2.0%  | 16        | 1.9%  | 11  | 1.3%  | 1    | 0.1%  | 28  | 3.3%  |
| 5P  | 2          | 0.1%  | 4   | 0.2%  | —    | —     | 6     | 0.3%  | 1         | 0.1%  | 3   | 0.4%  | 4    | 0.5%  | 8   | 0.9%  |
| 6P  | 2          | 0.1%  | 2   | 0.1%  | —    | —     | 4     | 0.2%  | —         | —     | 3   | 0.4%  | —    | —     | 3   | 0.4%  |
| 7P  | —          | —     | —   | —     | 1    | 0.05% | 1     | 0.05% | —         | —     | —   | —     | —    | —     | 0   | 0.0%  |
| Sum | 729        | 35.3% | 935 | 45.2% | 397  | 19.2% | 2,061 | 99.7% | 370       | 43.2% | 380 | 44.3% | 106  | 12.4% | 856 | 99.9% |
| DP  | 108        | 5.2%  | 152 | 7.4%  | 16   | 0.8%  | 276   | 13.4% | 71        | 8.3%  | 82  | 9.6%  | 8    | 0.9%  | 161 | 18.8% |

Tables 10.8–10.11 provide information about what might be called PREDICATE-ADJACENCY (PA). The notion is not conterminous with VERB-ADJACENCY (VA), even though both notions turn out to be usually conflated. This distinction is relevant on two grounds. First, clitics may be linearised contiguously to verbs, which are not their predicates. The recognition of clitic climbing relies on this very fact. Furthermore, the possibility of clitic climbing is not generally precluded by typological VA languages as evidenced by the Romance languages. Second, clitics that are licensed by a non-verbal

<sup>539</sup> The complexity of the data would render the data presentation part unclear, as it makes it impossible to compare the distribution with the naked eye. This is the consequence of a broad approach to linguistic analysis. I leave a more detailed analysis that zooms in for future research.

governor still may occur adjoined to a verb.<sup>540</sup> Hence, it makes sense to distinguish between *predicate-adjacency* and *verb-adjacency*, and to look at both configurations individually. Even though these notions are very close, they do not denote identical concepts.

Since the placement configurations of verb-adjacency (VA) vs. non-verb-adjacency (noVA) is relevant for clitic climbing, they will be considered in Section 10.2.3. Anticipating the discussion on clitic climbing, it is noteworthy that clitic verb-adjacent placement comes in two manifestations dependent on the number of verbs involved, i.e. mono-verbal (1VA) and bi-verbal (2VA), the latter of which I will also refer to as (VERBAL) CIRCUMPOSITION. But note that circumposition as a configuration is not restricted to clitics that undergo climbing. While it is true that the 2VA configuration is connected to complex clauses and analytical tense forms built with non-clitic auxiliaries, matrix clitics may find themselves in between two verbs, too. The most trivial case one can think of is a complex clause with sentence-initial verb positioning of the top-most matrix verb, which is associated with one or more clitics, cf. (10.4a–d).<sup>541</sup> However, clitics in verbal circumposition are attested in environments other than sentence-initial verb placement, too, cf. (10.4e). Recall also example (10.1h) with the sequence *povede<sub>1</sub> ti<sub>1</sub> zasadit<sub>2</sub>*, whereby the pronominal clitic *ti* occupies clausal 7P. I leave this general note and shall not return to the issue of verb-adjacency except in the sections on clitic climbing.

- (10.4) a. [...] **naučí<sub>1</sub>**        **tě<sub>1</sub>**        **programovat<sub>2</sub>** [...]   
           teach.PRS.3PL you.ACC program.INF   
           ‘... they teach you programming ...’
- b. (já    říkám)    Milen    **nech<sub>1</sub>**    **mi<sub>1</sub>**    **uříznout<sub>2</sub>**    čtyry    plátky   
           I.NOM say.PRS.1SG Milen.VOC let.IMP.2SG me.DAT cut.INF    four.ACC slice.ACC.PL   
           ‘(I’m saying:) Milen, let me cut four slices [thereof]’
- c. ([...] udála        se    docházka)        a    **šlo<sub>1</sub>**        **se<sub>1</sub>**    **hákovat<sub>2</sub>**   
           do.PRS.3SG REFL attendance.NOM.SG and go.PST.SG.N REFL hook.INF   
           ‘(... [one] gets the attendance [list] done) and [then one] went hooking ...’

<sup>540</sup> See Chapter 11 for clitics that are subcategorised for by non-verbal predicates.

<sup>541</sup> Example (10.4d) [Speaker B] can be analysed as an instance of haplology of the reflexive. So, this example is less trivial and might alternatively count as an occurrence of 2VA after CC depending on the exact analysis.

- d. Speaker A: *co deš taky do školy?*  
 what go.PRS.2SG also to school.GEN.SG  
 ‘What, are you going to school, too?’
- Speaker B: *ne du<sub>1</sub> si<sub>1,3</sub> nechat<sub>2</sub> vynadat<sub>3</sub> vod těch*  
 NEG go.PRS.1SG REFL.DAT let.INF tell\_off.INF from this.GEN.PL  
*holek (že sem se na to vysral)*  
 girl.GEN.PL that AUX.1SG REFL on this.ACC.SG.N shit.PST.SG.M  
 ‘No, I’m just going to let me telling off by those girls (that I didn’t give a shit about it.)’
- e. ([...] *maso my tam máme*) *i* [NP to  
 meat.ACC.SG we.NOM there have.PRS.1PL and this.ACC.SG.N  
*mletý]<sub>FOC</sub> ne-chce<sub>1</sub> se<sub>1</sub> mi<sub>1</sub> kupovat<sub>2</sub> [...]*  
 minced.ACC.SG.N NEG-want.PRS.3SG REFL.ACC me.DAT buy.INF  
 ‘(... we do have meat) and I don’t want to buy this minced one ...’ (Cz; ČNK oral\_v1)

Finally, out of 7 annotated tokens (0.24%) in the entire Czech data set, which were classified as right distant (Rd), I was able to verify only 1 attestation (0.03%) during the analysis part. Anyway, these occurrences are statistically neglectable. I tentatively label this configuration as CLITIC LOWERING, but I do not wish to insinuate any connection with the construction of *clitic climbing* here. I cite the only safe occurrence of clitic lowering in Czech I encountered, involving the reflexive clitic *se*, in (10.5).

- (10.5) *dělá [Adv takle] se celej život*  
 do.PRS.3SG so REFL whole.ACC.SG.M.INA life.ACC.SG  
 ‘One works like that the entire life.’ (Cz; ČNK oral\_v1)

## 10.2.2 Clitic clustering

Clitic clustering is a key feature of special clitic languages both within and beyond Slavic. This section will provide the aggregated quantities for the entire data sample covering all clitic types.<sup>542</sup>

<sup>542</sup> Zooming in by differentiation of clitic classes or individual clitics does not make sense for clustering, because of the high combinatorial capacities of clitics within a cluster. Furthermore, the occurrence of certain clitic categories is a function of grammar and discourse, e.g. valency of the lexical verb, conditionality, temporal reference points of a conversation, directionality of the utterance (i.e. the addressee in a broad sense, either as argument of a verb or as discourse anchor for ethical datives) etc. These features cannot be controlled for in a random sample.

Irrespective of the reported quantities, I will furthermore give a more qualitative discussion of the clustering data, for not all aspects can be quantified in a meaningful way in the present study. Table 10.12 reports the quantities for attested single clitic and clitic cluster occurrences. Single occurrences of clitics make almost 75% ( $\approx 3/4$ ) of the data points. Clitic clusters are significantly less frequent, whereby the majority consists of 2 clitic elements. Clusters of 3 and 4 sequential clitics are attested, but they are considerably rare. Example block (10.6) cites one instance per cluster length category ( $n = 2, 3, 4$ ). Note that *byste* ‘COND.AUX.2SG’ in (10.6c) is treated as compositional of two clitics due to its morphological transparency.

Table 10.12: Frequencies of single clitic and clitic cluster occurrences in spoken Czech by number of elements.

| Number of clitics | Absolute frequency | Relative frequency |
|-------------------|--------------------|--------------------|
| 1                 | 2,167              | 73.9%              |
| 2                 | 701                | 23.9%              |
| 3                 | 62                 | 2.1%               |
| 4                 | 3                  | 0.1%               |
| Sum               | 2,933              | 100%               |
| $n > 1$           | 766                | 26.1%              |

- (10.6) a. no ne my **ti** **ho** naschvál nechali na  
 well NEG we.NOM you.SG.DAT him.ACC purposely leave.PST.PL.VIR on  
 botníku než přišla mamka  
 shoe\_cabinet.LOC.SG before come.PST.SG.F mom.NOM  
 ‘Well, no. We left it for you on the shoe cabinet on purpose before mom came.’
- b. kdo **by** **si** **tě** vzal  
 who.NOM COND REFL.DAT you.SG.ACC take.PST.SG.M  
 ‘who would take you[?]’/‘who would have taken you[?]’
- c. to **by**<sub>1</sub>=**ste**<sub>1</sub> **se**<sub>2</sub> **ho**<sub>2</sub> mohli<sub>1</sub> zeptat<sub>2</sub>  
 TO COND=AUX.2PL REFL him.ACC can.PST.PL.VIR ask.INF  
 ‘then you could ask him’/‘then you could have asked him’ (Cz; ČNK oral\_v1)

Clusters can be classified with respect to the categories of clitics involved: auxiliary (AUX), conditional (COND), reflexive (REFL), and pronominal (PRN).<sup>543</sup> The richness of realisation patterns calls for a more lucid data presentation than a table, which is why I provide the diagram in Figure 10.2 following Kolaković et al. (2022: 192, Figure 8.2).

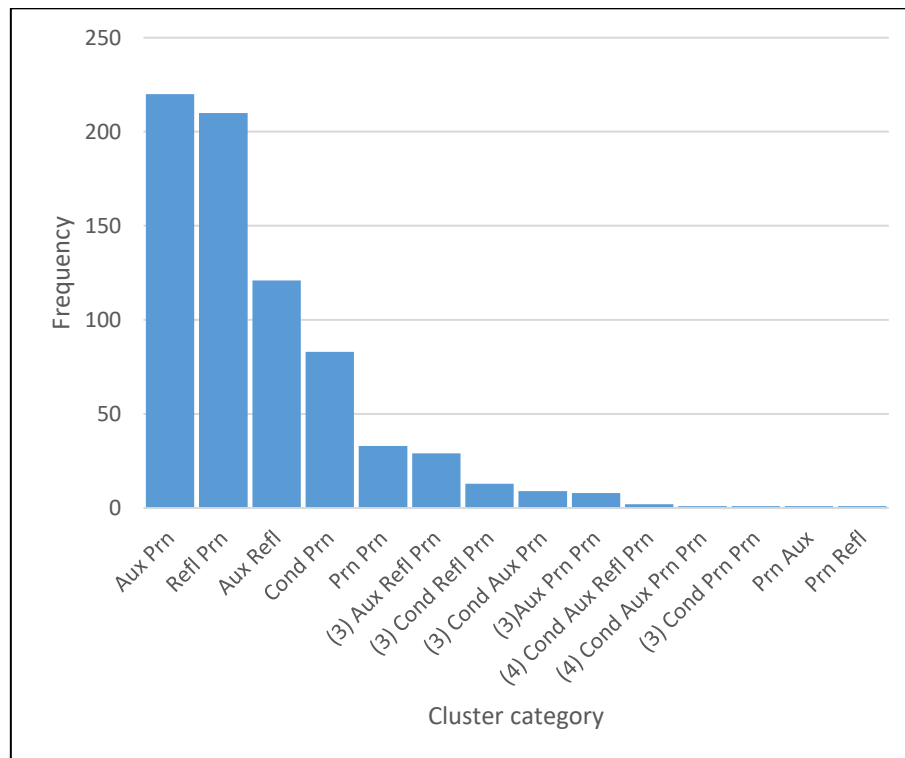


Figure 10.2: Frequency distribution of clitic cluster categories in spoken Czech.

The bar chart in Figure 10.2 shows the frequencies of all detected clitic cluster types. Types are defined in terms of the morphosyntactic (or morphosemantic) categories of auxiliary (Aux), conditional (Cond), reflexive (Refl), and pronominal (Prn) clitics. Clusters built from three and four clitics are additionally marked with the number of clitics given in parentheses for the sake of convenience.

<sup>543</sup> Note that a more fine-grained classification is possible but was not pursued for the purposes of the present thesis. Generally, reflexive clitics can be categorised into arguments, grammatical exponents, lexical elements, and pragmatic markers. Pronominal clitics can be divided into arguments and non-arguments, the latter comprising the set of free datives.

Table 10.13: Absolute frequencies of case-valued clitic clusters in spoken Czech.

| Case sequence           | Absolute frequency |
|-------------------------|--------------------|
| REFL < DAT              | 127                |
| REFL < ACC              | 118                |
| DAT < ACC               | 39                 |
| DAT < GEN <sup>?</sup>  | 4                  |
| ACC < REFL              | 1                  |
| REFL < GEN <sup>?</sup> | 1                  |

Figure 10.2 shows that clitic sequences obey the linearisation template within the cluster with COND < AUX < REFL < PRN. The same is true for pronominal clitics, when differentiating between individual case values in Table 10.13, which focuses on sequences of pronominal and reflexive clitics only. Although there has not been any attestation of the sequence REFL < PRN < PRN, we can infer from the table that clitics follow the order REFL < DAT < ACC/GEN. There is only one outlier with PRN = ACC preceding REFL, cf. (10.7).

- (10.7) [...] a tak **ho** se to dotklo že [...]  
 and so him.ACC REFL this.NOM.SG.N touch.PST.3SG.N that  
 ‘... and it affected him so much that ...’ (Cz; ČNK oral\_v1)

Admittedly, some structures have a controversial status. There are sequences of REFL < GEN and DAT < GEN, where the case value of the clitic (*ji*) is doubtful (this is the rationale behind the superscript question mark in Table 10.13). Perhaps, it could be a transcription error, because the pronominal form *ji* ‘ACC.SG.F’ is written without a diacritic according to the standard orthography.<sup>544</sup> The only unequivocal genitive in the cluster is the occurrence of *jich* in (10.8), whose form is governed by the quantifier *pár*, but admittedly it is an inconstant clitic.

- (10.8) jo tak **mi**<sub>1</sub> **jich**<sub>2</sub> pár<sub>2</sub> kup<sub>1</sub>  
 yeah so me.DAT them.GEN pair buy.IMP.2SG  
 ‘yes, buy a couple of them for me then’ (Cz; ČNK oral\_v1)

<sup>544</sup> I provide an example in (i). The tagger assigned the clitic *ji* dative case, which cannot be correct in face of the precedence of the unequivocal dative clitic *mu*. As a matter of fact, the subcategorisation frame of the verbal predicate *odstranit* ‘(to) remove’ requires a direct object in the accusative, which is why *ji* is glossed correspondingly despite its deviant morpho-phonological form (long vowel *i* instead of short vowel *i*).

(i) tak **mu** **ji** odstranili  
 so him.DAT her.ACC remove.PST.3PL  
 ‘so they cured him from it [= borreliosis]’ [lit.: ‘so they removed it to/for him’] (Cz; ČNK oral\_v1)

The attested 766 clitic clusters do not exhaust all occurrences of multiple clitics within a clause. A total amount of 21 clitics in diacclisis was detected in the data sample. Table 10.14 summarises the frequencies for multiple clitics. Obviously, diacclisis is a rare phenomenon both within the entire data sample and multiple clitic occurrences, which cluster in roundabout 97% of occurrences.

Table 10.14: Frequencies of multiple clitics with clustering and diacclisis in spoken Czech.

| Multiple clitics | Absolute frequency | Relative frequency<br>w/i multiple CL |
|------------------|--------------------|---------------------------------------|
| Clustering       | 766                | 97.3%                                 |
| Diacclisis       | 21                 | 2.7%                                  |
| Sum              | 787                | 100%                                  |

Recall from Section 4.5 that one may distinguish diacclisis of co-dependent clitics from diacclisis of non-co-dependent clitics. The term *pseudo-diacclisis*, coined by Kolaković et al. (2022) for the latter, is not used here. Since the very concept of diacclisis has implications for both theoretical linguistics and linguistic typology, it is worth taking closer inspection on the environments diacclisis occurs in. Table 10.15 gives the relevant quantities. Almost two-thirds ( $\approx 2/3$ ) of all tokens exhibit diacclisis within a simple clause, i.e. for co-dependent clitics. Approximately one-third ( $\approx 1/3$ ) of occurrences is attested in complex clauses, whereby the respective clitics are not co-dependents.

Table 10.15: Frequencies of diacclisis in spoken Czech by syntactic environment.

| Syntactic environment | Absolute frequency | Relative frequency w/i<br>diacclisis |
|-----------------------|--------------------|--------------------------------------|
| Simple clause         | 14                 | 66.7%                                |
| Complex clause        | 7                  | 33.3%                                |
| Sum                   | 21                 | 100%                                 |

Table 10.15 presents numbers for the distribution of environments among diacclitic tokens. However, this does not provide us with information about the frequencies of diacclisis *within* the respective environments. I calculate the ratios conventionally by  $\frac{\text{token frequency (diacclisis)}}{\text{cardinality (environment)}}$  for each environment individually. The next section on clitic climbing particularly relies on complex clauses and one can use the information therein for the calculation. A simple subtraction of the cardinality of complex clauses from the absolute frequency of all tokens from Table 10.1 gives the absolute frequencies of complex vs. simple clauses. As a result, diacclisis occurs with a ratio of 2.4% in complex clauses (7

within 294), whereas it occurs with a relative frequency of 0.5% in simple clauses (14 within 2,639). In other words, the relative frequency of diacsis is approximately five times higher in complex clauses than in simple clauses. This is a completely different result than the relative frequencies in Table 10.15 suggest. Therefore, tables on frequencies for diacsis shall display another column with the relative frequencies of diacsis within the respective environments, cf. Table 10.16. It needs to be pointed out that these figures are only approximations, because they operate on incomplete information. The ratio of diacsis within the respective syntactic environments is calculated as the quotient of the token frequency divided by the number of sentence tokens. However, a more accurate calculation would take into account only utterances with multiple clitics. This cannot be accomplished here, for my annotation scheme lacks the information needed for narrowing down the relevant cases. The present results are still sufficiently informative, so that I shall proceed with the approach described here.

Table 10.16: Frequencies of diacsis in spoken Czech by syntactic environment (revised).

| Syntactic environment | Absolute frequency | Relative frequency w/i diacsis | Relative frequency w/i syntactic environment |
|-----------------------|--------------------|--------------------------------|----------------------------------------------|
| Simple clause         | 14                 | 66.7%                          | 0.5%                                         |
| Complex clause        | 7                  | 33.3%                          | 2.4%                                         |
| Sum                   | 21                 | 100%                           | —                                            |

Diacsis has been mostly neglected in research on Slavic clitics (but cf. Kosta & Zimmerling 2014; Kolaković et al. 2022), except in case of Polish. Consequently, the question, in which environments the cluster is divided, has been rarely addressed. The Czech data show diacsis involving all clitic categories. A simple scheme of split patterns by environment is presented without frequencies in Table 10.17.

Table 10.17: Attested splits between clitic categories in spoken Czech by syntactic environment.

| Simple clause | Complex clause |
|---------------|----------------|
| COND    REFL  | COND    REFL   |
| REFL    COND  | AUX    PRN     |
| COND    PRN   | REFL    PRN    |
| AUX    REFL   | REFL    REFL   |
| AUX    PRN    | PRN    PRN     |
| REFL    PRN   |                |



We find splits (indicated by double vertical lines || ) between conditional and reflexive, conditional and pronoun, auxiliary and reflexive, auxiliary and pronoun, reflexive and auxiliary. There are no attestations of diaclisis between conditional and auxiliary, but it is unclear, whether this is for structural reasons or prompted by the research design, which focuses on pronominal and reflexive forms.<sup>545</sup> Two splitting types deserve particular attention, it is diaclisis among two reflexives on the one hand and between two pronouns on the other hand. The former competes with the construction of the *haplology of reflexives* (cf. Kupść 1999 for Polish; Avgustinova & Oliva 1997; Rosen 2014 for Czech; Kolaković 2022 et al. for BCMS), while the latter is rather unexpected.

- (10.9) a. a **estě a-by** do tři\* do třiceti na to [COND || REFL]  
and still that-COND to thir- to thirty.GEN on this.ACC.SG.N  
**se** čekalo  
REFL.ACC wait.PST.SG.N  
'even more that one would wait to the thir... to the thirtieth'
- b. to jo to **se** asi barvově **by** to [REFL || COND]  
TO yeah TO REFL.ACC perhaps colour.ADV COND this.NOM.SG.N  
vodpovídalo  
correspond.PST.SG.N  
'that's right, so it perhaps would correspond with respect to colour'
- c. to **bych** taky **mu** ne-řekla teda [COND || PRN]  
TO COND.1SG also him.DAT NEG-tell.PST.SG.F then  
'so, I also wouldn't tell him then'
- d. [...] takže já **sem** dycky v pět **si** [AUX || REFL]  
so\_that I.NOM AUX.1SG always in five.ACC REFL.DAT  
vstal [...] [COND || REFL]  
get\_up.PST.SG.M  
'... so that I always got up at five [o'clock] ...'

- e. to **sme** teda **tě** usvědčili z omylu [AUX || PRN]  
 TO AUX.1PL then you.SG.ACC convict.PST.PL.VIR of error.GEN.SG  
 ‘so, we’ve convicted you of error then’
- f. tak třeba **se**<sub>1</sub> to někdy povede<sub>1</sub> **ti**<sub>1</sub> [REFL || PRN]  
 so maybe REFL this.NOM.SG.N some\_day succeed.PRS.3SG you.DAT  
 zasadit<sub>2</sub>  
 plant.INF  
 ‘thus, you’ll maybe succeed in planting it some day’ (Cz; ČNK oral\_v1)

One needs to be aware of the low token frequencies, when comparing attested splits across syntactic environment. Almost all logically possible splits are attested in simple clauses. Complex clauses merely share the patterns (COND || REFL), (AUX || PRN) and (REFL || PRN) with them, but it appears reasonable to assume that the splitting types lacking in complex clauses could occur there anyway, if they are licit in simple environments. We find one example for a reflexive clitic preceding a conditional one, which is the reverse order of the clustering template and occurs in a simple clause, cf. (10.9b) above.

Only in complex clauses we find splits between identical categories of clitics. This is rather trivial for the reflexive clitics in (10.10a), as they are reported to avoid multiple instances in the same cluster and either haplologise or stay in situ therefore. Note that there are distinct reflexives with *se* being in the matrix and *si* in the embedding. As opposed to this, one cannot tell with certainty, whether the absence of (PRN || PRN) in simple clauses and its presence in complex ones implies a structural reason. The data is too scarce for sound conclusions but gives rise to interesting research questions that might be tackled more purposefully in future research.<sup>546</sup> It furthermore motivates the tentative hypothesis that pronominal argument clitics could be incapable of splitting within a single clause, i.e. when they are co-dependents of the very same predicate. Interestingly, we find an uncanonical case order in utterance (10.10b) with an accusative matrix clitic preceding an embedded dative clitic. Example (10.10c) shows diacsis between two ethical datives. Note that an ethical and an argument dative may freely co-occur in a clitic cluster in simple clauses.

<sup>546</sup> Some relevant questions might be: (i) What are the conditions for diacsis? (ii) Is there a link between diacsis and information packaging? (iii) How are diaclic constructions perceived by native speakers?

- (10.10) a. [...] tak **se**<sub>1</sub> vyplatí<sub>1</sub> koupit<sub>2</sub> **si**<sub>2</sub> ten  
 so REFL.ACC pay\_off.PRS.3SG buy.INF REFL.DAT this.ACC.SG.M  
 prostě nejlepší ten ten nejdražší  
 simply best this.ACC.SG.M this.ACC.SG.M most\_expensive  
 ‘... then it pays to simply buy the best one [= shaver], the most expensive one (for oneself)’
- b. a tak **ho**<sub>1</sub> napadlo<sub>1</sub> říkat<sub>2</sub> **mu**<sub>2</sub> názvem  
 and so him.ACC strike\_on.PST.SG.N say.INF him.DAT name.INS.SG  
 ‘and so he struck on the idea to call him by the name’
- c. a von **ti** začal něco **mi** tam jíst  
 and he.NOM you.DAT.ETH start.PST.SG.M something.ACC me.DAT.ETH there eat.INF  
 ‘and he started to eat just something, y’know’ (Cz; ČNK oral\_v1)

Most evidence of diacsis comprises exactly two clitics. Three instances of diacsis have been recorded, where a clitic is disjoint from a clitic cluster, i.e. (AUX REFL || PRN), (REFL PRN || PRN), and (PRN || PRN PRN) cf. (10.11). Example (10.11a) presents diacsis in a simple clause. The accusative clitic *ho* is separated from the clitic cluster by the parenthetical *blbec* ‘fool’. The two other examples show diacsis in complex clauses and exhibit the pattern (PRN || PRN), but they differ in an important respect. In example (10.11b), the embedded clitic *ho* climbed to the matrix verb, which it is right-adjoined to. However, it did not climb up the syntactic tree completely ending up in 4P, such that it does not join the matrix cluster, even though this were a licit sequence (REFL < DAT < ACC: *se mi ho*).<sup>547</sup> Example (10.11c) is different in that, first, it is the cluster involving pronominal *mu ho* (DAT < ACC), which originates in the embedded clause. Second, there is already a dative pronominal clitic in the matrix clause. It is reported in the literature that multiple dative arguments are banned from co-occurring in the same clitic cluster. I did not encounter evidence for multiple splits, i.e. three or more clitics, all of which were disjoint.

<sup>547</sup> Example (i) is the one and only utterance showing this very sequence in the spoken subcorpus ORAL v1, which is likely to be an artefact of the limited corpus size. The sequence is attested 89 times in the balanced SYN2020 corpus

(i) a je to asi že **se mi ho** ne-dostávalo v  
 and be.PRS.3SG this.NOM.SG.N perhaps that REFL me.DAT him.ACC NEG-receive.PST.SG.N in  
 dětství  
 childhood.LOC.SG  
 ‘perhaps this is, because I didn’t get it [= butter] in childhood’

(Cz; ČNK oral\_v1)

- (10.11) a. no já **sem** se blbec ne\* **ho** [AUX REFL || PRN]  
 well I.NOM AUX.1SG REFL idiot.NOM.SG [...] him.ACC  
 ne-zeptala  
 NEG-ask.PST.SG.F  
 ‘well, I stupidly didn’t ask him’
- b. to<sup>1</sup> **se**<sub>1</sub><sup>2</sup> **mi**<sub>1</sub><sup>2</sup> ne-chce<sub>1</sub><sup>3</sup> **ho**<sub>2</sub><sup>4</sup> tahat<sub>2</sub> přes [REFL PRN || PRN]  
 TO REFL me.DAT NEG-want.PRS.3SG him.ACC pull.INF across  
 rameno  
 shoulder.ACC.SG  
 ‘so I don’t want to drag him over the shoulder’
- c. to teda zase **mi**<sub>1</sub> bylo líto<sub>1</sub> **mu**<sub>2</sub> **ho**<sub>2</sub> [PRN || PRN PRN]  
 TO then again me.DAT be.PST.SG.N sorry him.DAT him.ACC  
 dát<sub>2</sub> za vodvoz  
 give.INF for disposal.ACC.SG  
 ‘so, I was sorry then to give him it for disposal’ (Cz; ČNK oral\_v1)

Finally, let us briefly mention the case of single clitics and clitic clusters in first position. Section 4.4.1 reported relevant data points from different Slavic languages. The question arises, what types of clitics and clusters may occur in this position. Is there any restriction observable, e.g., in terms of phonological features like having a sibilant onset or a syntagmatic constraint on clitic co-occurrences? The literary review in Section 4.4.1 provided that in principle any kind of clitic can occur in first clausal position across Slavic. It fully suffices here to summarise the kinds of clitic occurrences in absolute first clausal position, i.e. without a preceding coordinating conjunction. Such configurations are most informative about the syntactic variation within the first position. The following tuples of clitic and clitic cluster categories in 1P are attested in the Czech data set: (Prn), (Refl), (Cond, Prn), (Aux, Refl), (Aux, Prn), (Refl, Prn), (Prn, Prn), (Cond, Aux, Prn), (Aux, Refl, Prn), (Aux, Prn, Prn).

### 10.2.3 Clitic climbing

The construction known as clitic climbing (CC) received considerable attention in the literature. There are different assessments regarding its obligatoriness or optionality. CC is accounted for in terms of LOCALITY in the present thesis (cf. Kupść 2000b). The local or in situ placement is the

absence of clitic climbing (noCC), even though there were no structural restrictions prohibiting it.<sup>548</sup> Clitic climbing itself is the non-local realisation of a clitic, i.e. when it is placed in a superordinate clause it is not a dependent of. Table 10.18 provides quantities for CC vs. noCC in total and for clitic classes. CC is clearly the predominant pattern with 95% of complex clauses, although it makes merely 10% of the entire data set. Note that I follow scholars like Kupść (2000a) in treating orthotonic auxiliary verbs essentially as any other matrix verbs for the purpose of determining CC vs. noCC (cf. also Spencer & Luís 2012a: 163, who consider this configuration “the least interesting”). In total, CC to the domain of an orthotonic auxiliary amounts to 39 tokens or 13.9% of all CC instances.

Cases, in which clitic placement does not unequivocally allow for an interpretation as either CC or noCC, were recorded, too (CC?). There are three basic manifestations of this. First, a clitic occupies a position between two verbs without being contiguous with any of them. This holds only, if the surrounding environment does not provide clear evidence either. Second, the interplay of syntax and information structure yields a linearisation that does not permit any default analysis of clitic placement as either CC or noCC. Such instances occur with fronted infinitives. Third, clitics with ambiguous dependency relations are another source of case of uncertainty. This matter covers the possible licensing of a clitic by either matrix or embedded verb, which is possible in certain control constructions. The sum of CC and noCC (and possibly CC?) gives the entirety of complex clauses in the data set.

Table 10.18: Frequencies of clitic climbing and no climbing in spoken Czech.

| Locality        | Total      |                                | Pronominal |                                | Reflexive  |                                |
|-----------------|------------|--------------------------------|------------|--------------------------------|------------|--------------------------------|
|                 | Abs. freq. | Rel. freq. w/i complex clauses | Abs. freq. | Rel. freq. w/i complex clauses | Abs. freq. | Rel. freq. w/i complex clauses |
| CC              | 280        | 94.9%                          | 184        | 94.8%                          | 96         | 95.0%                          |
| noCC            | 10         | 3.4%                           | 6          | 3.1%                           | 4          | 4.0%                           |
| CC?             | 5          | 1.7%                           | 4          | 2.1%                           | 1          | 1.0%                           |
| Complex clauses | 294        | 100%                           | 193        | 100%                           | 101        | 100%                           |

I exemplify clitic climbing in (10.12a, 10.12c) vs. no-climbing in (10.12b, 10.12d) for pronominal and reflexive clitics respectively. An example for the insecure status of CC vs. noCC is cited in (10.12e).

<sup>548</sup> The label “noCC” is adopted from Kolaković et al. (2022).

- (10.12) a. ([...] a vona říká) dědo můžu<sub>1</sub> tě<sub>2</sub> vo  
 and she.NOM say.PRS.3SG grandpa.VOC can.PRS.1SG you.ACC about  
 něco poprosit<sub>2</sub>?  
 something.ACC request.INF  
 ‘(... and she says:) grandpa, can I ask you for something?’
- b. (na-trhám barborku) a pů-jdu<sub>1</sub> za Otíkem  
 PFV-pick.PRS.1SG winter\_cress.ACC.SG and PFV-go.PRS.1SG for Otakar.DIM.INS.SG  
 popřát<sub>2</sub> mu<sub>2</sub> k dvacátému výročí tévédé.  
 congratulate.INF him.DAT for twentieth.DAT.SG anniversary.DAT.SG TVD.GEN  
 ‘(I will pick [some] winter cress) and go to Otík [to] congratulate him on the twentieth anniversary of TVD [= a company].’
- c. to bych<sub>1</sub> se<sub>2</sub> měl<sub>1</sub> zeptat<sub>2</sub> táty (dyž teďka letí  
 to COND.1SG REFL.ACC have.PST.SG.M ask.INF dad.GEN.SG if now fly.PRS.3SG  
 pracovně do Chile)  
 work.ADV to Chile.GEN  
 ‘so, I would have to ask dad (if he’s flying to Chile now because of work)’
- d. no vona chce<sub>1</sub> ještě si<sub>2</sub> najít<sub>2</sub> nějakou práci [...]   
 well she.NOM want.PRS.3SG still REFL.DAT find.INF some.ACC.SG.F work.ACC.SG  
 ‘well, she wants still to find some [kind of] job (for herself) ...’
- e. (no pojd’) chce<sub>1</sub> tady babička ti<sub>2</sub> něco<sub>2</sub>  
 well come.IMP.2SG want.PRS.3SG here grandmother.NOM.SG you.DAT something  
 říct<sub>2</sub>  
 tell.INF  
 ‘(come on) grandma wants to tell you something [over] here’ (Cz; ČNK oral\_v1)

Table 10.19 presents the frequencies of CC vis-à-vis noCC per clitic. The occurrence frequency of CC oscillates between 93% and 100%. The absence of CC (i.e. noCC) is not attested for the pronominal clitics *tě* and *mi*. Note however the diverse counts of attestations of complex clauses and the resulting low frequencies as per individual clitic.

Table 10.19: Frequencies of clitic climbing and no climbing in spoken Czech by clitic.

| Local.           | <i>tě</i> |                      | <i>ho</i> |                      | <i>mi</i> |                      | <i>ti</i> |                      | <i>mu</i> |                      | <i>se</i> |                      | <i>si</i> |                      |
|------------------|-----------|----------------------|-----------|----------------------|-----------|----------------------|-----------|----------------------|-----------|----------------------|-----------|----------------------|-----------|----------------------|
|                  | Abs.      | Rel.<br>w/i<br>cmplx | Abs.      | Rel.<br>w/i<br>cmplx | Abs.      | Rel.<br>w/i<br>cmplx | Abs.      | Rel.<br>w/i<br>cmplx | Abs.      | Rel.<br>w/i<br>cmplx | Abs.      | Rel.<br>w/i<br>cmplx | Abs.      | Rel.<br>w/i<br>cmplx |
| CC               | 26        | 100%                 | 61        | 93.8%                | 20        | 100%                 | 51        | 94.4%                | 26        | 89.7%                | 43        | 97.7%                | 53        | 93.0%                |
| noCC             | —         | —                    | 3         | 4.6%                 | —         | —                    | 1         | 1.9%                 | 2         | 6.9%                 | 1         | 2.3%                 | 4         | 7.0%                 |
| CC?              | —         | —                    | 1         | 1.5%                 | —         | —                    | 2         | 3.7%                 | 1         | 3.4%                 | —         | —                    | —         | —                    |
| Cmplx<br>clauses | 26        | 100%                 | 65        | 100%                 | 20        | 100%                 | 53        | 100%                 | 29        | 100%                 | 44        | 100%                 | 57        | 100%                 |

A question that has not been tackled in the literature is the position clitic climbing targets. It is standardly assumed that climbing targets the second position (cf. e.g. Rezac 2005: 105; more recently Tabachnik 2024: 2). As far as I am aware, Marušič (2008) is the first to point out for a Slavic language that this is an oversimplification. Table 10.20 reports the frequencies for empirically attested target positions of CC in Czech, which predominantly complies with Wackernagel cliticisation. However, there is considerable variation, as the attested positions range from 1P to 6P. Reflexive clitics exhibit higher ratios for peripheral placement and, thus, show a greater flexibility in terms of likelihood of occurrence than pronominal ones.

Table 10.20: Frequencies of target positions under clitic climbing in spoken Czech.

| Position<br>after CC | Total      |                      | Pronominal |                      | Reflexive  |                      |
|----------------------|------------|----------------------|------------|----------------------|------------|----------------------|
|                      | Abs. freq. | Rel. freq.<br>w/i CC | Abs. freq. | Rel. freq.<br>w/i CC | Abs. freq. | Rel. freq.<br>w/i CC |
| 1P                   | 8          | 2.9%                 | 5          | 2.7%                 | 3          | 3.1%                 |
| 2P                   | 228        | 81.4%                | 156        | 84.8%                | 72         | 75.0%                |
| 3P                   | 32         | 11.4%                | 18         | 9.8%                 | 14         | 14.6%                |
| 4P                   | 7          | 2.5%                 | 2          | 1.1%                 | 5          | 5.2%                 |
| 5P                   | 3          | 1.1%                 | 1          | 0.5%                 | 2          | 2.1%                 |
| 6P                   | 2          | 0.7%                 | 2          | 1.1%                 | —          | —                    |
| DP                   | 44         | 15.7%                | 23         | 12.5%                | 21         | 21.9%                |

I cite attestations for clitic climbing targeting various positions from 1P to 6P in example block (10.13).

- (10.13) a. (já sem došel k baru[,] vzal sem  
 I.NOM AUX.1SG arrive.PST.SG.M towards bar.DAT.SG take.PST.SG.M AUX.1SG  
 si to pivo[.]) **mi**<sup>1</sup> **ho**<sup>1</sup> moh<sub>1</sub> přinýst<sub>2</sub>  
 REFL.DAT this.ACC.SG.N beer.ACC.SG me.DAT him.ACC can.PST.SG.M bring.INF  
 klidně (ale prostě přišel mě seřvat)  
 calmly but simply come.PST.SG.M me.ACC brawl\_out.INF  
 (I got to the bar, took the beer.) He could have brought it to me calmly (but he simply  
 came to yell at me.)
- b. mami já<sup>1</sup> **ti**<sup>2</sup> musím<sub>1</sub> něco říct<sub>2</sub>  
 mom.VOC I.NOM you.DAT must.PRS.1SG something say.INF  
 ‘Mom, I must tell you something.’
- c. [...] protože<sup>1</sup> oni<sup>2</sup> **ho**<sup>3</sup> chtěli<sub>1</sub> Němci zavřít<sub>2</sub> [...]   
 because they.NOM.VIR him.ACC want.PST.PL.VIR German.NOM.PL lock\_up.INF  
 ‘... because it were the Germans, who wanted to lock him up ...’
- d. (to kafe mám taky ráda horký  
 this.ACC.SG.N coffee.ACC.SG have.PRS.1SG also gladly.ADJ.PRED.SG.F hot.ACC.SG.N  
 ale většinou ho v tý práci piju studený)  
 but mostly him.ACC in this.LOC.SG.F work.LOC.SG drink.PRS.1SG cold.ACC.SG.N  
 protože<sup>1</sup> to<sup>2</sup> **ňák**<sup>3</sup> **se**<sup>4</sup> ne-stihne<sub>1</sub> vypít<sub>2</sub>.  
 because this.ACC.SG.N somehow REFL NEG-manage.PRS.3SG drink.INF  
 ‘(I also like the coffee hot, but mostly I drink it cold at work) because one somehow  
 doesn’t manage to drink it.’
- e. (no počkej) a<sup>0</sup> dyt<sup>1</sup> já<sup>2</sup> tam<sup>3</sup> musím<sub>1</sub><sup>4</sup> **si**<sup>5</sup>  
 PART.EMPH wait.IMP.2SG and after\_all I.NOM there must.PRS.1SG REFL.DAT  
 zbalit<sub>2</sub> věci taky  
 pack\_up.INF thing.ACC.PL also  
 ‘(now wait) after all, I have to pack up my stuff there, too’
- f. a<sup>0</sup> [pp ve čtvrtek]<sup>1</sup> už<sup>2</sup> tam<sup>3</sup> jenom<sup>4</sup> za-jdu<sub>1</sub><sup>5</sup> **mu**<sup>6</sup>  
 and on Thursday.ACC.SG already there only PFV-go.PRS.1SG him.DAT  
 to prostě dát<sub>2</sub>  
 this.ACC.SG.N simply give.INF  
 ‘and already on Thursday I will only go there [in order to] simply give it to him’  
 (Cz; ČNK oral\_v1)

A more fine-grained view is provided in Table 10.21 with target positions per clitic.



Table 10.21: Frequencies of target positions under clitic climbing in spoken Czech by clitic.

| Position<br>after<br>CC | <i>tě</i>     |                            | <i>ho</i>     |                            | <i>mi</i>     |                            | <i>ti</i>     |                            | <i>mu</i>     |                            | <i>se</i>     |                            | <i>si</i>     |                            |
|-------------------------|---------------|----------------------------|---------------|----------------------------|---------------|----------------------------|---------------|----------------------------|---------------|----------------------------|---------------|----------------------------|---------------|----------------------------|
|                         | Abs.<br>freq. | Rel.<br>freq.<br>w/i<br>CC | Abs.<br>freq. | Rel.<br>freq.<br>w/i<br>CC | Abs.<br>freq. | Rel.<br>freq.<br>w/i<br>CC | Abs.<br>freq. | Rel.<br>freq.<br>w/i<br>CC | Abs.<br>freq. | Rel.<br>freq.<br>w/i<br>CC | Abs.<br>freq. | Rel.<br>freq.<br>w/i<br>CC | Abs.<br>freq. | Rel.<br>freq.<br>w/i<br>CC |
| 1P                      | —             | —                          | 3             | 4.9%                       | 1             | 5.0%                       | —             | —                          | 1             | 3.8%                       | 1             | 2.3%                       | 2             | 3.8%                       |
| 2P                      | 20            | 76.9%                      | 56            | 91.8%                      | 18            | 90.0%                      | 41            | 80.4%                      | 21            | 80.8%                      | 31            | 72.1%                      | 41            | 77.4%                      |
| 3P                      | 4             | 15.4%                      | 1             | 1.6%                       | 1             | 5.0%                       | 10            | 19.6%                      | 2             | 7.7%                       | 7             | 16.3%                      | 7             | 13.2%                      |
| 4P                      | —             | —                          | 1             | 1.6%                       | —             | —                          | —             | —                          | 1             | 3.8%                       | 4             | 9.3%                       | 1             | 1.9%                       |
| 5P                      | 1             | 3.8%                       | —             | —                          | —             | —                          | —             | —                          | —             | —                          | —             | —                          | 2             | 3.8%                       |
| 6P                      | 1             | 3.8%                       | —             | —                          | —             | —                          | —             | —                          | 1             | 3.8%                       | —             | —                          | —             | —                          |
| DP                      | 6             | 23.1%                      | 2             | 3.3%                       | 1             | 5.0%                       | 10            | 19.6%                      | 4             | 15.4%                      | 11            | 25.6%                      | 10            | 18.9%                      |

We must acknowledge a considerable degree of variation with respect to target positions as well as statistical distribution even within the categories of pronominals and reflexives. While 2P is the predominant target position under CC for all clitics, they differ with respect to oscillation. The pronominal clitics *ho*, *mi* exhibit more than 90% of 2P targets. The same position reaches a total of roundabout 75% for pronominal *tě* and the reflexive clitics *se*, *si*, all of which show a considerable amount of 3P placement roughly at 15%. The pronominal clitic *ti* even reaches approximately 20% for 3P targets. Still, the disparity of the frequency distributions could be an artefact of the comparatively low volume of data, such that it is safer not to put too much emphasis on this level of detail for the remainder of the chapter.

Recall from Section 4.7.1 that clitics escape their licensing domain and enter a hierarchically higher one. Simultaneously, there is no a priori way to distinguish between local and non-local realisation, if a clitic occurs in verbal circumposition (V CL V), for the clitic is still contiguous with its licensing predicate (cf. Junghanns 2002c: 67–68). Consequently, two questions come up. First, does the clitic detach from its governor under clitic climbing? Second, does the clitic end up contiguously to a matrix verb? The interplay of the parameters underlying these questions give rise to a three-way pattern as in the case of lateral positioning from Section 10.2.1. I distinguish between non-verb-adjacency (noVA), mono-verbal contact (1VA), and bi-verbal contact or circumposition (2VA). Table 10.22 summarises the verb-adjacent and non-verb-adjacent occurrences of clitics after CC. Overall, clitics are predominantly verb-adjacent and preferably occur in mono-verbal configurations. The ratio of mono-verbal vs. bi-verbal placement is 3:2. In roundabout one of five ( $\approx 1/5$ ) cases, CC results in a noVA configuration which is comparable to the overall frequency of circumposition. Pronominal and reflexive clitics differ as to their frequency distributions. While both categories resemble each other with respect to 1VA, they appear to behave diametrically with respect to noVA vs. 2VA. Pronominal clitics prefer 2VA over noVA, whereas we find the opposite with

reflexive clitics. However, the disparity between both categories is not significant ( $\chi^2 = 5.54, p = .05, df = 2$ ).

Table 10.22: Frequencies of simultaneous occurrences of clitic climbing and (non-)verb-adjacency in spoken Czech.

| Placement<br>after CC | Total      |                      | Pronominal |                      | Reflexive  |                      |
|-----------------------|------------|----------------------|------------|----------------------|------------|----------------------|
|                       | Abs. freq. | Rel. freq.<br>w/i CC | Abs. freq. | Rel. freq.<br>w/i CC | Abs. freq. | Rel. freq.<br>w/i CC |
| noVA                  | 61         | 21.8%                | 31         | 16.8%                | 30         | 31.3%                |
| 1VA                   | 161        | 57.5%                | 107        | 58.2%                | 54         | 56.3%                |
| 2VA                   | 58         | 20.7%                | 46         | 25.0%                | 11         | 11.5%                |

The utterances in (10.14) provide examples for each adjacency pattern with distant placement in (10.14a), a mono-verbal configuration in (10.14b), and circumposition in (10.14c).

- (10.14) a. ([...] takže sem šla spát tak ve tři.) [noVA]  
so\_that AUX.1SG go.PST.SG.F sleep.INF so in three.ACC  
já sem<sub>1</sub> ho<sub>2</sub> [Adv pak] musela<sub>1</sub> vyhodit<sub>2</sub>.  
I.NOM AUX.1SG him.ACC then must.PST.SG.F throw\_out.INF  
‘(... so I went sleeping roundabout three o’clock.) Then, I had to throw him out.’
- b. já se<sub>2</sub> musím<sub>1</sub> učit<sub>2</sub> [1VA]  
I.NOM REFL.ACC must.PRS.1SG learn.INF  
‘I need to learn.’
- c. no tak ne-můžeme<sub>1</sub> tě<sub>2</sub> poslat<sub>2</sub> na nějakou [2VA]  
well so NEG-can.PRS.1PL you.ACC send.INF on some.ACC.SG.F  
vejšku.  
university.ACC.SG  
‘well, we cannot send you to some uni [just] like that’ (Cz; ČNK oral\_v1)

Accounting also for the target position, we receive Table 10.23 for the overall data set. The clausal second position is preferred across all three adjacency patterns. Targeting peripheral positions under climbing is not contingent on contact with a verb, although the likelihood for such positions is higher with verb-adjacency. Provided the comparatively small frequencies with increasing level of detail, I refrain from distinguishing between pronominal and reflexive clitics.

Table 10.23: Frequencies of target positions under clitic climbing by (non-)verb-adjacency in spoken Czech.

| Pos<br>after CC | noVA |       | 1VA |       | 2VA |       |
|-----------------|------|-------|-----|-------|-----|-------|
| 1P              | 3    | 1.1%  | 5   | 1.8%  | N/A | N/A   |
| 2P              | 53   | 18.9% | 129 | 46.1% | 46  | 16.4% |
| 3P              | 3    | 1.1%  | 21  | 7.5%  | 8   | 2.9%  |
| 4P              | 1    | 0.4%  | 5   | 1.8%  | 1   | 0.4%  |
| 5P              | —    | —     | —   | —     | 3   | 1.1%  |
| 6P              | —    | —     | 2   | 0.7%  | —   | —     |
| Sum             | 60   | 21.4% | 162 | 57.9% | 58  | 20.7% |
| DP              | 4    | 1.4%  | 28  | 10.0% | 12  | 4.3%  |

Table 10.24 distinguishes the three adjacency patterns by individual clitics but does not account for target position. By and large, the trends found with individual clitics are consistent with the respective clitic categories in Table 10.20.

Table 10.24: Frequencies of (non-)verb-adjacency under clitic climbing in spoken Czech by clitic.

| Placement<br>after CC | <i>tě</i> |                | <i>ho</i> |                | <i>mi</i> |                | <i>ti</i> |                | <i>mu</i> |                | <i>se</i> |                | <i>si</i> |                |
|-----------------------|-----------|----------------|-----------|----------------|-----------|----------------|-----------|----------------|-----------|----------------|-----------|----------------|-----------|----------------|
|                       | Abs.      | Rel.<br>w/i CC | Abs.      | Rel.<br>w/i CC | Abs.      | Rel.<br>w/i CC | Abs.      | Rel.<br>w/i CC | Abs.      | Rel. w/i<br>CC | Abs.      | Rel. w/i<br>CC | Abs.      | Rel.<br>w/i CC |
| noVA                  | 3         | 11.5%          | 8         | 13.1%          | 5         | 25.0%          | 11        | 21.6%          | 4         | 15.4%          | 12        | 27.9%          | 18        | 34.0%          |
| 1VA                   | 14        | 53.8%          | 44        | 72.1%          | 10        | 50.0%          | 23        | 45.1%          | 16        | 61.5%          | 26        | 60.5%          | 28        | 52.8%          |
| 2VA                   | 9         | 34.6%          | 9         | 14.8%          | 5         | 25.0%          | 17        | 33.3%          | 6         | 23.1%          | 5         | 11.6%          | 7         | 13.2%          |

Finally, the vast majority of CC occurs in bi-verbal environments, i.e. in utterances with a finite matrix verb and exactly one embedded infinitive the clitic originates from. This construction predominantly occurs in the literature. More complex constructions are rarely taken into account, although they present an intriguing interplay of phenomena like deep embeddings, intermediate positioning, non-canonical ordering, and diacsis (but cf. Hana 2007; Petkevič 2009; Rosen 2001 for Czech; Hansen et al. 2018; Kolaković et al. 2022 for BCMS; Marušič 2008 for Slovene). In fact, complex clauses with more than two hierarchically ordered verbs are rare in the scrutinised data sample. In sum, five such occurrences are attested in the data sample. I cite them in (10.15). Four of the CC constructions exhibit the orthotonic future auxiliary *být* embedding the modal verb *muset* ‘must’ (10.15a–d). Example (10.15e) shows the hierarchical order of modal verb < phase verb < lexical verb. In each case, the climbed clitic crosses the entire verbal sequence and occupies Wackernagel position except (10.15d) showing Clitic First (1P).

- (10.15) a. (to dyby střelíl medvěda?) tak **mu**<sub>3</sub> **ho**<sub>3</sub> budeš<sub>1</sub>  
 this if shoot.PST.3SG.M bear.ACC.SG so him.DAT him.ACC AUX.FUT.2SG  
 muset<sub>2</sub> nosit<sub>3</sub>  
 must.INF carry.INF  
 ‘(so, if he shoots a bear) you will have to carry it for him’
- b. ale pak **se**<sub>3</sub> sem budu<sub>1</sub> muset<sub>2</sub> vrátit<sub>3</sub>  
 but then REFL.ACC here AUX.FUT.1SG must.INF return.INF  
 ‘but afterwards I will have to come back here’
- c. to **si**<sub>3</sub> to budem<sub>1</sub> muset<sub>2</sub> asi **ňák** rozdělit<sub>3</sub>  
 this REFL.DAT this.ACC.SG.N AUX.FUT.1SG must.INF perhaps somehow split.INF  
 ‘then I will have to somehow split it for myself perhaps’
- d. **si**<sub>3</sub> prostě budu<sub>1</sub> muset<sub>2</sub> ukousnout<sub>3</sub> jazyk jako  
 REFL.DAT simply AUX.FUT.3SG must.INF bite.INF tongue.ACC.SG like  
 zašít pusu  
 sew\_up.INF mouth.ACC.SG  
 ‘I will simply have to bite my tongue like shutting up.’ [lit.: ‘...like sewing up my mouth’]
- e. [ tak ]<sub>TOP</sub> [ s to by<sub>1</sub> **si**<sub>3</sub> mamka taky měla<sub>1</sub> začít<sub>2</sub>  
 so TO COND REFL.DAT mom.NOM.SG also have.PST.3SG.F begin.INF  
 hledat<sub>3</sub> **ňáký** chlapy vid’ ]  
 search.INF some guy.ACC.PL right  
 ‘thus, mom would have also to look for some men, right?’ (Cz; ČNK oral\_v1)

## 10.2.4 Agreement drop

One occurrence of agreement drop was recorded in the entire data sample, cf. (10.16). The low occurrence frequency of agreement drop in spoken Czech is surprising, as the construction is particularly linked with informal speech.<sup>549</sup> Perhaps, the vanishingly small number of attestations

<sup>549</sup> It is easy to purposefully obtain agreement drop data from the corpus. Examples (i)–(ii) exemplify agreement drop of the 1st persons singular and plural respectively. CQL query: [word="že"] [word="já|my"] [word="ho|mu|tě|se|mi"] [tag="Vp.\*"] [word!="jsem|jsi|si|jsme|sme|jste"]

(i) (Mluvili jsme spolu, za sebe můžu říct,) že<sup>1</sup> já<sup>2</sup> **ø** **ho**  
 talk.PST.PL.VIR AUX.1PL together for REFL.ACC can.PRS.1SG tell.INF that I.NOM **ø** him.ACC  
 bral za přítele vždy.  
 take.IPFV.PST.SG.M for friend.ACC always  
 ‘(We talked, as far as I am concerned I can tell) that I always approached him as a friend.’

could be attributed to the unselective database, which covers the diatopically diverse area. Maybe, agreement drop is not evenly distributed across the linguistic varieties of the Czech Republic. Still, chance should have brought up more data, but this issue must be left for the future. The pronominal clitic *ho* ‘him.ACC’ in (10.16) occupies the fourth clausal position following the subject pronoun and the focused constituent *i přes ten telefon* ‘even through the telephone’. The pronominal clitic is placed left-adjacent to its verbal head.

- (10.16) tak<sup>1</sup> [ i přes ten telefon ]<sup>2</sup><sub>FOC</sub> ø já<sup>3</sup> ho<sup>4</sup> slyšela  
 so and through this telephone ø I.NOM him.ACC hear.PST.SG.F  
 ‘thus, I heard him even through the telephone’ (Cz; ČNK oral\_v1)

### 10.2.5 Possessive resultative

The possessive resultative has not been discussed earlier. A discussion of this construction shall be postponed to Section 10.8.5 when comparing the West Slavic lects. Suffice it to mention here that it is formally built upon the verb ‘(to) have’ and a phrasal complement comprising a direct object and passive participle both marked for accusative. In total, only 8 tokens (= 0.3%) of this construction were encountered in the data set. I cite them exhaustively in (10.17). Note that only the pronominal clitic *ho* ‘him.ACC’ occurs. Examples (10.17a–f) show single clitic occurrences in clausal second position. There is a mixed cluster as a result of clitic climbing in (10.17g). Utterance (10.17h) provides an attestation of Clitic Third. Bear in mind that I do not treat particles like *už* ‘already’ as clitics, which is the basic motivation for the Clitic Third analysis. In fact, the particle can be emphasised here.

- (10.17) a. koukám že<sup>1</sup> ho<sup>2</sup> máš<sub>1</sub> nastudovanýho<sub>2</sub>  
 see.PRS.1SG that him.ACC have.PRS.2SG study.PASS.PTCP.ACC.SG.M.ANI  
 ‘I see that you have studied it’

- 
- (ii) (Vtip je v tom,) že<sup>1</sup> my<sup>2</sup> ø se<sub>2</sub> začali<sub>1</sub> analýzou  
 point.NOM.SG be.PRS.3SG in this.LOC that we.NOM ø REFL.ACC start.PST.PL.VIR analysis.INS.SG  
 důchodové reformy zabývat<sub>2</sub> již před rokem.  
 pension.ADJ.GEN.SG reform.GEN.SG occupy.INF already before year.INS.SG  
 ‘(The point is) that we started to occupy ourselves with the analysis of the pension reform already a year ago.’  
 (Cz; ČNK syn\_v8)

- b. (a věnec teda mám) ale ne-mám<sub>1</sub><sup>1</sup> ho<sub>2</sub><sup>2</sup>  
 and wreath.ACC.SG then have.PRS.1SG but NEG-have.PRS.1SG him.ACC  
 nastrojený<sub>2</sub>  
 decorate.PASS.PTCP.ACC.SG.M.INA  
 ‘(and, so, the wreath, I have it) but I have not decorated it’
- c. taky<sup>1</sup> ho<sub>2</sub><sup>2</sup> mám<sub>1</sub> připravenej<sub>2</sub> že nasypu na  
 also him.ACC have.PRS.1SG prepare.PASS.PTCP.ACC.SG.M.INA that scatter.PRS.1SG on  
 zahrádku  
 garden.ACC.SG  
 ‘I have prepared it [= ash], too, [such] that I will scatter the garden’
- d. no mam<sub>1</sub><sup>1</sup> ho<sub>2</sub><sup>2</sup> tam master\* masterovanej<sub>2</sub>  
 yes have.PRS.1SG him.ACC there master.PASS.PTCP.ACC.SG.M.INA  
 ‘yes, I have him mastered [= equipped with a master hard drive]’
- e. (takže ho dycky umeju) a mám<sub>1</sub><sup>1</sup> ho<sub>2</sub><sup>2</sup> tam  
 so\_that him.ACC always wash.PRS.1SG and have.PRS.1SG him.ACC there  
 pověšenej<sub>2</sub> jako dekoraci  
 hang.PASS.PTCP.ACC.SG.M.INA as decoration.ACC.SG  
 ‘(so that I wash it [= iron-cast pan] each time) and I have it hanging as a decoration’
- f. a už<sup>1</sup> ho<sub>2</sub><sup>2</sup> mám<sub>1</sub> taky naloženej<sub>2</sub>  
 and already him.ACC have.PRS.1SG also pickle.PASS.PTCP.ACC.SG.M.INA  
 ‘I have pickled it, too’
- g. to jó ale<sup>0</sup> museli<sub>1</sub><sup>1</sup> by<sub>1</sub><sup>2</sup> ho<sub>3</sub><sup>2</sup> mít<sub>2</sub> někde  
 this yeah but must.PST.PL.VIR COND him.ACC have.INF somewhere  
 zavřenýho<sub>3</sub>  
 cage.PASS.PTCP.ACC.SG.M.ANI  
 ‘that’s right, but they would have to cage him up somewhere’
- h. (no ale musíš a-by se to obnovilo)  
 well but must.PRS.2SG that-COND REFL.ACC this.NOM.SG.N restore.PST.SG.N  
 tak<sup>1</sup> už<sup>2</sup> ho<sub>2</sub><sup>3</sup> tady máš<sub>1</sub> přijatej<sub>2</sub>  
 so already him.ACC here have.PRS.2SG receive.PASS.PTCP.ACC.SG.M.INA  
 ‘(well, but you must [do it] in order to restore it) thus you already have accepted it here’  
 (Cz; ČNK oral\_v1)

## 10.3 Polish

### 10.3.1 Clitic placement

From approximately 3,550 clausal tokens, which were screened for distributional properties of clitics in spoken Polish, there were 3,051 tokens eligible for analysis. Following the established framework from Section 10.2, Table 10.25 presents the overall distribution frequencies for clitic placement in Polish accounting for position, laterality, and distance. The overall sum of 3,022 tokens in Table 10.25 deviates from the quantity actually analysed and annotated by approximately 1%. Table 10.26 gives the respective counts for pronominal and reflexive clitics. Polish exhibits a varying range of clitic positions, 1P to 6P are attested for both pronominal and reflexive clitics. The occurrence of 10P for the dative reflexive clitic *se* is the only outlier. The frequency distribution is dominated by clausal second position, which makes approximately two-thirds ( $\approx \frac{2}{3}$ ) of all clitic occurrences. Pronominal clitics comply with this pattern, while reflexive clitics show less 2P, which is still the majority of cases with 60%. Clitic Third makes roughly one-fifth ( $\approx \frac{1}{5}$ ) of occurrences in each case. Generally, reflexive clitics show higher frequencies of peripheral positions than pronominal clitics. The differences in the overall distribution patterns between both categories are significant ( $\chi^2 = 20.54, p = .05, df = 2$ ). Still, they show identical tendencies with respect to positional preference, which is summarised in (10.19a). Note that the pronominal clitics' preference of 1P over 4P is insignificant, in fact, because the frequencies are identical.

Table 10.25: Frequencies of clitic occurrences in spoken Polish by position, laterality, and distance.

| Pos | Left distant |       | Left adjacent |       | Right (adjacent) |       | Sum   |       |
|-----|--------------|-------|---------------|-------|------------------|-------|-------|-------|
| 1P  | 55           | 1.8%  | 133           | 4.4%  | N/A              | N/A   | 188   | 6.2%  |
| 2P  | 463          | 15.3% | 979           | 32.3% | 548              | 18.1% | 1,990 | 65.7% |
| 3P  | 136          | 4.5%  | 369           | 12.2% | 131              | 4.3%  | 636   | 21.0% |
| 4P  | 30           | 1.0%  | 102           | 3.4%  | 36               | 1.2%  | 168   | 5.5%  |
| 5P  | 8            | 0.3%  | 19            | 0.6%  | 6                | 0.2%  | 33    | 1.1%  |
| 6P  | 2            | 0.1%  | 3             | 0.1%  | 1                | 0.03% | 6     | 0.2%  |
| 10P | —            | —     | —             | —     | 1                | 0.03% | 1     | 0.03% |
| Sum | 694          | 22.9% | 1,605         | 53.0% | 723              | 23.9% | 3,022 | 99.8% |
| DP  | 176          | 5.8%  | 493           | 16.3% | 175              | 5.8%  | 844   | 99.3% |

Overall, predicate-adjacency (ca. 77%) is the dominant placement option compared to Wackernagel cliticisation (ca. 66%). Interestingly, there is a strong tendency for left-adjacent placement in Polish, while left-distant placement and right placement resemble each other to some extent. Delayed pPlacement is most likely to occur in the position immediately preceding the verbal predicate. However, there is no causal relationship between Delayed Placement and predicate-adjacency, as deviant positioning is attested with distances  $d > 0$ . Still, we encounter a rare phenomenon. The intersection of Wackernagel cliticisation and predicate adjacency comes to exactly half of occurrences (= 50.4%).

Table 10.26: Frequencies of clitic occurrences in spoken Polish by position, laterality, and distance per clitic category.

| Pos | Pronominal |       |      |       |      |       |       |       | Reflexive |       |     |       |      |       |     |       |
|-----|------------|-------|------|-------|------|-------|-------|-------|-----------|-------|-----|-------|------|-------|-----|-------|
|     | Ld         |       | La   |       | R(a) |       | Sum   |       | Ld        |       | La  |       | R(a) |       | Sum |       |
| 1P  | 37         | 1.7%  | 76   | 3.6%  | N/A  | N/A   | 113   | 5.3%  | 18        | 2.0%  | 57  | 6.3%  | N/A  | N/A   | 75  | 8.2%  |
| 2P  | 335        | 15.8% | 660  | 31.2% | 447  | 21.1% | 1,442 | 68.1% | 128       | 14.1% | 319 | 35.0% | 101  | 11.1% | 548 | 60.2% |
| 3P  | 78         | 3.7%  | 244  | 11.5% | 105  | 5.0%  | 427   | 20.2% | 58        | 6.4%  | 125 | 13.7% | 26   | 2.9%  | 209 | 22.9% |
| 4P  | 18         | 0.9%  | 66   | 3.1%  | 28   | 1.3%  | 112   | 5.3%  | 12        | 1.3%  | 36  | 4.0%  | 8    | 0.9%  | 56  | 6.1%  |
| 5P  | 4          | 0.2%  | 10   | 0.5%  | 4    | 0.2%  | 18    | 0.9%  | 4         | 0.4%  | 9   | 1.0%  | 2    | 0.2%  | 15  | 1.6%  |
| 6P  | 1          | 0.05% | —    | —     | —    | —     | 1     | 0.05% | 1         | 0.1%  | 3   | 0.3%  | 1    | 0.1%  | 5   | 0.5%  |
| 10P | —          | —     | —    | —     | —    | —     | —     | —     | —         | —     | —   | —     | 1    | 0.1%  | 1   | 0.1%  |
| Sum | 473        | 22.3% | 1056 | 49.9% | 584  | 27.6% | 2,113 | 99.8% | 221       | 24.3% | 549 | 60.3% | 139  | 15.3% | 909 | 99.8% |
| DP  | 101        | 4.8%  | 320  | 15.1% | 137  | 6.5%  | 558   | 26.4% | 75        | 8.2%  | 173 | 19.0% | 38   | 4.2%  | 286 | 31.4% |

Looking closer at the two categories in Table 10.26, the frequencies for left-distant placement are comparable, but reflexives show a slightly increased frequency for Delayed Placement. There are more striking differences when it comes to the verb-adjacent placement patterns. Reflexive clitics show a higher tendency for left-adjacent placement than pronominal ones, which exhibit a higher ratio for right placement, in turn.<sup>550</sup>

For Clitic First, the distinction between adjacency to a clause boundary and a coordinating conjunction is quantified in Table 10.27. No pattern can be considered rare within this positional configuration, but there is an asymmetry, with 1P after clause boundaries representing roughly one-third ( $\approx 1/3$ ) of cases, while conjunctions preceding clause-initial clitics make almost two-thirds ( $\approx 2/3$ ) of occurrences.

<sup>550</sup> This is interesting against the background of East Slavic, since the Polish short pronouns and especially short reflexives do not resemble *-sja* with respect to their distributional properties.



Table 10.27: Frequencies for Clitic First in spoken Polish after clause boundary and coordinating conjunction (in total).

| Type        | Abs. freq. | Rel. freq. w/i 1P |
|-------------|------------|-------------------|
| # 1P ...    | 70         | 36.8%             |
| CONJ 1P ... | 120        | 63.2%             |

Examples for each clitic position are given in (10.18) with (10.18a–b) showing clause-initial Clitic First vs. post-conjunction Clitic First. Note that Clitic Tenth positioning (10P) in (10.18h) exhibits a large delay due to the occurrence of two interjections (INT), which are counted like regular constituents in the present thesis and even split constituents. The example shows, furthermore, diacsis of the conditional clitic *by* and the reflexive clitic *se*.

- (10.18) a. (podszedłem do pani) # **mi**<sup>1</sup> tam kazała podpisać  
 go.PST.SG.M to woman.GEN.SG me.DAT there tell.PST.SG.F sign.INF  
 ‘(I went to the woman.) She told me to sign [something].’
- b. [...] ale<sup>0</sup> **się**<sup>1</sup> do-wiedziała (że mu się życie nie  
 but REFL PFV-know.PST.SG.F that him.DAT REFL.ACC life.NOM.SG NEG  
 ułożyło)  
 arrange.PST.SG.N  
 ‘... but she found out (that he didn’t get his life together)’
- c. (przed bożym naro) [pp przed bożym narodzeniem]<sup>1</sup>  
 before God.ADJ.INS.SG Nati[vity] before God.ADJ.INS.SG Nativity.INS.SG  
**ci**<sup>2</sup> odpalę tą roletę  
 you.DAT fire\_up.PRS.1SG this.ACC.SG.F blind.ACC.SG  
 ‘(Before Christ...) before Christmas, I will set that blind in motion for you.’
- d. [...] że<sup>1</sup> ja<sup>2</sup> **by**<sup>3</sup>=**m**<sup>3</sup> **cię**<sup>3</sup> w tym widziała [...] <sup>1</sup>  
 that I.NOM COND=AUX.1SG you.ACC in this.LOC.SG.M see.PST.SG.F  
 ‘... that I can imagine you wearing that ...’ [lit.: ‘I would see you within that (clothing)’]
- e. ty<sup>1</sup> tam<sup>2</sup> [AdvP tym bardziej]<sup>3</sup> **by**<sup>4</sup>=**ś**<sup>4</sup> **go**<sup>4</sup> pewnie  
 you.NOM there the more COND=AUX.2SG him.ACC surely  
 znał  
 know.PST.SG.M  
 ‘All the more, you would know him for sure.’

- f. to<sup>1</sup> czemu<sup>2</sup> ja<sup>3</sup> mam<sup>4</sup> **mu**<sup>5</sup> nie dać na dziesięć minut  
 TO why I.NOM have.PRS.1SG him.DAT NEG give.INF for ten.ACC minute.GEN.PL  
 plejstejszyn  
 playstation.ACC.SG  
 ‘So why should I not give him the playstation for 10 minutes?’
- g. [...] bo<sup>1</sup> ja<sup>2</sup> [pp na przykład]<sup>3</sup> [v lubię]<sup>4</sup> [NP szczegóły]<sup>5</sup>  
 because I.NOM for example.ACC.SG like.PRS.1SG detail.ACC.PL  
 se<sup>6</sup> oglądać nie  
 REFL.DAT watch.INF NEG  
 ‘... because me, for instance, I just like to look at the details, you know’
- h. (i wiesz wybuchają by bomby) bo<sup>1</sup>  
 and know.PRS.2SG explode.PST.SG.NVIR COND bomb.NOM.PL because  
 [NP jakaś banda]<sup>2</sup> [INT rozumiesz]<sup>3</sup> [NP kretynów]<sup>4</sup> by<sup>5</sup>  
 some.NOM.SG.F mob.NOM.SG understand.PRS.2SG idiot.GEN.PL COND  
 [v chciała]<sup>6</sup> [INT wiesz]<sup>7</sup> [pp przez to]<sup>8</sup> [vp zrobić]<sup>9</sup>  
 want.PST.SG.F know.PRS.2SG though this.ACC.SG.N make.INF  
 se<sup>10</sup> wojnę ]  
 REFL.DAT war.ACC.SG  
 ‘(and – you know – bombs would explode) because a mob of idiots – you got it? – would  
 like to start a war just like that, you know’ (Pol; NKJP pelcra)

The measures of central tendency for clitic positioning in Polish are given in Table 10.28 (based on 3,051 analysed tokens). All three measures congruently show that the frequency distribution complies with second position placement, although Clitic Third lies within one standard deviation.

Table 10.28: Measures of central tendency for clitic positioning in spoken Polish.

| Measure:          | $\bar{x}$ (SD) | <i>Md</i> | <i>Mo</i> |
|-------------------|----------------|-----------|-----------|
| Value (position): | 2.31 (±0.75)   | 2         | 2         |

(10.19) a. 2P > 3P > 1P > 4P > 5P > 6P (> ... > 10P)

b. 2P > 3P > 4P > 1P > 5P > 6P (> ... > 10P)

The frequency distributions in Table 10.29 show considerable variance. Most clitics differ significantly. Only the pronominal clitic *mi* and the reflexive clitics adhere to the pattern in (10.19a), while all other pronominal clitics show the reversed preference of 4P over 1P, i.e. (10.19b).

Admittedly, 1P occurrences insignificantly outnumber 4P for the reflexive clitic *się* and are equinumerous (a difference of 0.2% may be neglected). The 2nd person clitics *cię* and *ci* exhibit the highest frequency of 2P with ca. 74% and 72% respectively. They also show the lowest occurrences of 1P. In contrast, the 1st person dative clitic *mi* presents the highest frequency of 1P with roundabout 12%. The ratio for 2P oscillates for *mi* and *go* around 65%. The lowest ratio for Wackernagel positioning is roundabout 60% for the reflexives and datival *mu*. Clitic Third is the highest for *go*, *mu*, and *się*, all of which approximate 25% ( $\approx \frac{1}{4}$ ). The dative clitics *ci* and *se* show roundabout 20% of 3P ( $\approx \frac{1}{5}$ ). Another 5% lower it is for *cię* and *mi* ( $\approx 15\%$  or  $\frac{1}{7}$ ). Farther delayed clitic placement becomes increasingly marginal. The overall frequency for DP ranges from ca. 20% to 34%. In short, only the pairs  $\{ci, cię\}$ ,  $\{go, mu\}$ ,  $\{go, się\}$ ,  $\{mu, się\}$ , and  $\{się, se\}$  have non-significant results, when tested for homogeneity. The same applies to the triplet  $\{go, mu, się\}$ , which share a common distribution as a whole ( $\chi^2 = 6.17, p = .05, df = 4$ ).

Table 10.29: Frequencies of attested clitic clausal positions in spoken Polish by clitic.

| Pos | <i>cię</i> |       | <i>go</i> |       | <i>mi</i> |       | <i>ci</i> |       | <i>mu</i> |       | <i>się</i> |       | <i>se</i> |       |
|-----|------------|-------|-----------|-------|-----------|-------|-----------|-------|-----------|-------|------------|-------|-----------|-------|
|     | Abs.       | Rel.  | Abs.      | Rel.  | Abs.      | Rel.  | Abs.      | Rel.  | Abs.      | Rel.  | Abs.       | Rel.  | Abs.      | Rel.  |
| 1P  | 14         | 3.1%  | 20        | 4.6%  | 50        | 12.3% | 13        | 3.0%  | 18        | 4.4%  | 32         | 6.6%  | 43        | 9.9%  |
| 2P  | 333        | 74.2% | 279       | 64.3% | 272       | 66.8% | 310       | 72.4% | 255       | 61.7% | 293        | 60.3% | 259       | 59.7% |
| 3P  | 68         | 15.1% | 103       | 23.7% | 67        | 16.5% | 86        | 20.1% | 109       | 26.4% | 121        | 24.9% | 92        | 21.2% |
| 4P  | 30         | 6.7%  | 27        | 6.2%  | 13        | 3.2%  | 18        | 4.2%  | 26        | 6.3%  | 31         | 6.4%  | 27        | 6.2%  |
| 5P  | 3          | 0.7%  | 5         | 1.2%  | 5         | 1.2%  | 1         | 0.2%  | 5         | 1.2%  | 7          | 1.4%  | 9         | 2.1%  |
| 6P  | 1          | 0.2%  | —         | —     | —         | —     | —         | —     | —         | —     | 2          | 0.4%  | 3         | 0.7%  |
| 10P | —          | —     | —         | —     | —         | —     | —         | —     | —         | —     | —          | —     | 1         | 0.2%  |
| Sum | 449        | 100%  | 434       | 100%  | 407       | 100%  | 428       | 100%  | 413       | 100%  | 486        | 100%  | 434       | 100%  |
| DP  | 102        | 22.7% | 135       | 31.1% | 85        | 20.9% | 105       | 24.5% | 140       | 33.9% | 161        | 33.1% | 132       | 30.4% |

In addition to positioning, we recognise considerable variation regarding the laterality patterns across individual clitics, too, cf. Table 10.30. The examples in (10.20) illustrate each of the basic pattern types. Among the pronominal clitics, the dative clitics *mi* and *ci* are the most deviant. The former has the highest frequency of left-distant placement and the lowest one for left-adjacent placement. This may be traced back to the incomparably high ratio of Clitic First placement. The clitic *ci* exhibits the highest ratio of right placement, in turn. Considering the reflexive clitics, they mirror their category's general behaviour noticed above, but differ especially as to left-distant placement, which is relatively prominent with dative *se* showing the second highest value for that configuration after *mi*.

Table 10.30: Frequencies for laterality with left distant (Ld), left adjacent (La), right adjacent (R(a)) in spoken Polish by clitic.

| Lat  | <i>cię</i> |       | <i>go</i> |       | <i>mi</i> |       | <i>ci</i> |       | <i>mu</i> |       | <i>się</i> |       | <i>se</i> |       |
|------|------------|-------|-----------|-------|-----------|-------|-----------|-------|-----------|-------|------------|-------|-----------|-------|
|      | Abs.       | Rel.  | Abs.      | Rel.  | Abs.      | Rel.  | Abs.      | Rel.  | Abs.      | Rel.  | Abs.       | Rel.  | Abs.      | Rel.  |
| Ld   | 82         | 18.3% | 99        | 22.8% | 119       | 29.2% | 75        | 17.5% | 98        | 23.7% | 99         | 20.4% | 122       | 28.2% |
| La   | 233        | 51.9% | 240       | 55.3% | 178       | 43.7% | 209       | 48.8% | 196       | 47.5% | 298        | 61.3% | 251       | 58.0% |
| R(a) | 133        | 29.6% | 91        | 21.0% | 103       | 25.3% | 144       | 33.6% | 113       | 27.4% | 83         | 17.1% | 56        | 12.9% |

- (10.20) a. [...] to<sup>1</sup> jak<sup>2</sup> tylko<sup>3</sup> się<sup>4</sup> [NP jedna pchła] [Ld]  
 TO if only REFL.ACC one.NOM.SG.F flea.NOM.SG  
 [PP w domu] **pojawiła** [...]   
 in house.LOC.SG appear.PST.SG.F  
 ‘... so, as soon as one flea appeared at home ...’
- b. i<sup>0</sup> co<sup>1</sup> [NP ta pani]<sup>2</sup> jeszcze<sup>3</sup> mi<sup>4</sup> **powiedziała** [La]  
 and what this.NOM.SG.F madam.NOM.SG still me.DAT tell.PST.SG.F  
 ‘and what else did the woman tell me[?]’
- c. o już<sup>1</sup> **wyjmę**<sup>2</sup> **cię**<sup>3</sup> [R(a)]  
 oh already take\_out.PRS.1SG you.DAT  
 ‘ah, I’m taking you out [of this] now’ (Pol; NKJP pelcra)

There have been 6 occurrences of clitic lowering (0.2%) counted in the entire annotated data set for Polish, two attestations of which are cited in (10.21). Example (10.21b) contains two instances of the right-disjoint reflexive *se*, which is an ethical dative. Both occurrences are uttered by the same speaker in a continuous string of speech.

- (10.21) a. (ujemny logarytm czyli tak jak=by=ś wzięła  
 negative.NOM.SG.M logarithm.NOM.SG that\_is so when=COND=AUX.2SG take.PST.SG.F  
 sobie wykładnik potęgi) i **zmieniła** [NP znak]  
 REFL.DAT exponent.ACC.SG power.GEN.SG and change.PST.SG.F sign.ACC.SG  
**mu**  
 him.DAT  
 ‘(the negative logarithm, in other words, [it is] as if you took the power’s exponent) and changed its [algebraic] sign’

b. **zostaw** [PRN to ] **se** do rana (rano  
keep.IMP.2SG this.ACC.SG.N REFL.DAT.ETH to morning.GEN.SG morning.ACC.SG  
**wrzucisz** [PRN to ] **se** na patelnię i podsmażysz)  
throw.PRS.2SG this.ACC.SG.N REFL.DAT.ETH on pan.ACC.SG and fry\_up.PRS.2SG  
‘Just keep it until [tomorrow] morning. (In the morning, you’ll just throw it onto the pan  
and fry it up.)’ (Pol; NKJP pelcra)

### 10.3.2 Clitic clustering

Single clitic occurrences have the highest density of attestations and represent over 80% of the database, cf. Table 10.31. Clitic clusters comprising two clitics are the most frequent, while those with increasing size become rarer. The various size categories of clitic clusters are illustrated in (10.22). I cite two examples of clusters with  $n = 3$  clitics in (10.22b–c) in order to demonstrate that the “inflected” conditional *bym* is treated like any other sequence of two clitics because of morphological transparency. Only a single occurrence of a cluster comprising four clitics has been detected, cf. (10.22d). Note that this is a mixed cluster that came about by clitic climbing of the embedded object clitic *go*, thus, joining the matrix clitic cluster, which itself involves the conditional form *bym* again.

Table 10.31: Frequencies of single clitic and clitic cluster occurrences in spoken Polish by number of elements.

| Number of clitics | Absolute frequency | Relative frequency |
|-------------------|--------------------|--------------------|
| 1                 | 2,486              | 81.5%              |
| 2                 | 503                | 16.5%              |
| 3                 | 60                 | 2.0%               |
| 4                 | 1                  | 0.03%              |
| Sum               | 3,051              | 100%               |
| $n > 1$           | 564                | 18.5%              |

(10.22) a. (no ty wiesz) że szerszenie to by cię  
well you.NOM know.PRS.2SG that hornet.NOM.PL TO COND you.ACC  
zagryzły  
bite\_to\_death.PRS.3PL.NVIR  
‘(well, you know) that hornets would sting(/have stung) you to death’

- b. [...] jutro może **by=m** się przeszedł [...]   
tomorrow maybe COND=AUX.1SG REFL.ACC walk\_around.PST.SG.M   
‘... I would go for walk tomorrow maybe ...’
- c. (widzisz) no to **by ci się** przydało takie   
see.PRS.2SG well TO COND you.DAT REFL.ACC be\_useful.PST.SG.N such.NOM.SG.N   
coś   
something.NOM.SG   
‘(you see) well, so something like this would be useful for you’
- d. (na przykład tam puszczają taki film)   
for instance there show.PRS.3PL such.ACC.SG.M.INA movie.ACC.SG   
to radziła<sub>1</sub>=**by<sub>1</sub>=m<sub>1</sub>** **ci<sub>1</sub>** **go<sub>2</sub>** nie oglądać<sub>2</sub>   
TO advise.PST.SG.F=COND=1SG you.DAT him.ACC NEG watch.INF   
‘(There, for instance, they show a movie like this.) So, I would advise you not to watch it.’   
(Pol; NKJP pelcra)

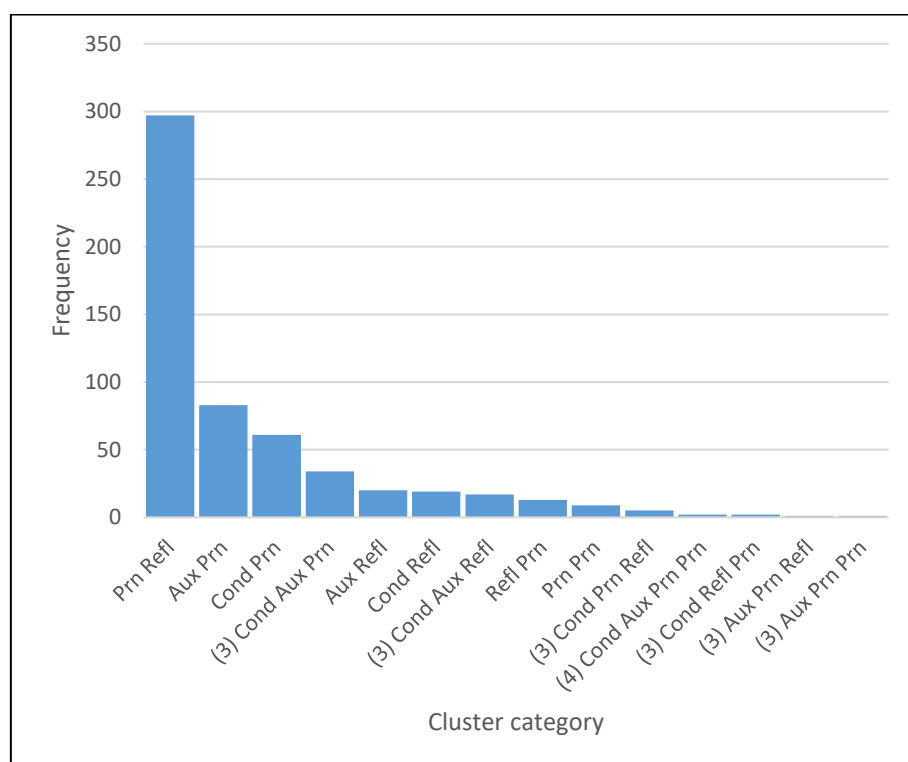


Figure 10.3: Frequency distribution of clitic cluster categories in spoken Polish.

Figure 10.3 visualises the occurrence frequency for different category combinations within a clitic cluster. The dominant linearisation pattern in the cluster is COND < AUX < PRN < REFL. The reverse order between pronominal and reflexive (REFL < PRN) clitics is rare.

Taking case into account in Table 10.32, dative is by far the most frequent case and dominates the PRN > REFL pattern. There is only one outlier showing a reflexive clitic preceding a dative pronoun (10.23a). This attestation involves the inconstant dative clitic *nam* ‘us.DAT’, however. More importantly, due to the strong asymmetry in attestation, cases other than dative occur in rather low frequencies. It is therefore impossible to say something meaningful about linearisation concerning other cases than dative. Generally, dative precedes accusative and genitive. The reflexive precedes the genitive, but the order between reflexive and accusative varies. Hence, the actual or preferred order in the cluster is inconclusive in the face of the scanty data. Examples (10.23b–c) confront the varying order between REFL and ACC.

Table 10.32: Absolute frequencies of case-valued clitic clusters in spoken Polish.

| Case sequence | Absolute frequency |
|---------------|--------------------|
| DAT < REFL    | 297                |
| REFL < ACC    | 8                  |
| REFL < GEN    | 5                  |
| DAT < ACC     | 4                  |
| DAT < GEN     | 3                  |
| ACC < REFL    | 3                  |
| REFL < DAT    | 1                  |

- (10.23) a. (to zajebiście           no)   bo.   w ogóle   też   **by**   **się**   **nam**  
 TO   fucking\_awesome EMPH   because by\_the\_way also   COND   REFL us.DAT  
 przydał                   ktoś           do pomocy  
 be\_useful.PST.SG.M someone.NOM to   help.GEN.SG  
 ‘([That’s] fucking awesome) ’cause we could need a helping hand, too, by the way.’
- b. jutro           **się**   **go**           będzie   trzeba spytać.  
 tomorrow REFL him.ACC be.FUT.3SG need   ask.INF  
 ‘one will have to ask him tomorrow’
- c. ([...] i   Piłsudski           przypadkowo doszedł)           o   coś  
 and Piłsudski.NOM by\_chance arrive.PST.SG.M about something.ACC.SG  
**go**           **się**   zapytał           (i poklepał           go           po ramieniu.)  
 him.ACC REFL ask.PST.SG.M and clap.PST.SG.M him.ACC on shoulder.LOC.SG  
 ‘(... and Piłsudski got there incidentally,) asked him about something (and gave him a  
 pat on the back.)’ (Pol; NKJP pelcra)

Table 10.33 provides that 94% of all occurrences of multiple clitics cluster, while diacclisis is attested with a low frequency of 6% only. Therefore, clitic clustering may be deemed a dominant pattern, while split clitic occurrences are a rare phenomenon.

Classifying split clitic occurrences by syntactic environment in Table 10.34, a majority of 75% is clause-bound. Diacclisis between clause boundaries or – more precisely – between verbs represents one-quarter of cases ( $\approx \frac{1}{4}$ ). Anticipating the figures from the next section on clitic climbing, the ratio for diacclisis in complex clauses is 2.8% (9 within 326), while the respective calculation for simple clauses yields 1.0% (28 within 2,725). Provided these numbers, the relative frequency for diacclisis is roughly three times higher in complex clauses than in simple ones.

Table 10.33: Frequencies of multiple clitics with clustering and diacclisis in spoken Polish.

| Multiple clitics | Absolute frequency | Relative frequency<br>w/i multiple CL |
|------------------|--------------------|---------------------------------------|
| Clustering       | 565                | 93.9%                                 |
| Diacclisis       | 37                 | 6.1%                                  |
| Sum              | 602                | 100%                                  |

Table 10.34: Frequencies of diacclisis in spoken Polish by syntactic environment.

| Syntactic environment | Absolute frequency | Relative frequency w/i<br>diacclisis | Relative frequency w/i<br>syntactic environment |
|-----------------------|--------------------|--------------------------------------|-------------------------------------------------|
| Simple clause         | 28                 | 75.7%                                | 1.0%                                            |
| Complex clause        | 9                  | 24.3%                                | 2.8%                                            |
| Sum                   | 37                 | 100%                                 | —                                               |

Diacclisis was registered between most clitic categories in simple clauses but recall that the majority of attested environments are simple ones. The diacclitic pair (PRN || REFL) involves only dative pronouns (10 tokens), while the split across the reverse order covers the same categories (2 tokens). Two splits showing uncanonical clitic order are attested in simple clauses and involve a conditional clitic each, i.e. (REFL || COND) and (PRN || COND). The respective data points are cited in (10.24). Note that the conditional morpheme is right-adjoined to the verb in (10.24a), where one might object that the present case could be affixal. This is, however, not the approach taken in the present study. Furthermore, the argument does not work with (10.24b), where the conditional clitic is left-adjoined.



Table 10.35: Attested splits between clitic categories in spoken Polish by syntactic environment.

| Simple clause | Complex clause |
|---------------|----------------|
| COND    REFL  | COND    REFL   |
| REFL    COND  | AUX    PRN     |
| COND    PRN   | REFL    PRN    |
| PRN    COND   | REFL    REFL   |
| AUX    REFL   | PRN    PRN     |
| AUX    PRN    |                |
| PRN    REFL   |                |
| REFL    PRN   |                |

- (10.24) a. (jeden                      drugiego                      nie sypnie)                      bo                      [COND || REFL]  
 one.NOM.SG.M    second.ACC.SG.M.ANI NEG    rat.PRS.3SG    because  
**by**    to                      **się**    tam. skończyło    niedobrze (i dla niego    i  
 COND    this.NOM.SG.N REFL    there end.PST.SG.N    badly                      and for him.GEN    and  
 dla tamtego)  
 for that\_one.GEN  
 ‘(one won’t rat on the other) ’cause it would just come to a bad end (for him as well as  
 for the other guy)’
- b. (wiesz                      co)    nawet **się**                      [v nie    opłacało ]                      [REFL || COND]  
 know.PRS.2SG what even    REFL.ACC    NEG    be\_worth.PST.SG.N  
**by**    że    tak powiem    bawić    w to.  
 COND    that so    say.PRS.1SG    play.INF in this.ACC  
 ‘(you know what) it wouldn’t even be worth spending one’s time on that, so to speak’
- c. ty                      jak **by**    teraz **cię**                      zwolnili                      [...]                      [COND || PRN]  
 you.SG.NOM when    COND    now    you.ACC    dismiss.PST.PL.VIR  
 ‘hey, if they would dismiss you now ...’
- d. co    **go**                      tutaj **by**    przywiodło ?                      [PRN || COND]  
 what him.ACC    here    COND    bring.PST.SG.N  
 ‘what would bring him here?’
- e. gdzie=ś                      ty                      **se**                      Mati                      tak                      [AUX || REFL]  
 where=AUX.2SG    you.SG.NOM    REFL.DAT Mateusz.DIM.VOC    so  
 nogi                      poodzierał ?  
 leg.ACC.PL    scrape.PST.SG.M  
 ‘Mati, where did you scrape your leg like that?’

- f. (albo może specjalnie) że=**by=m** ja **mu** [AUX || PRN]  
 or maybe purposefully that=COND=AUX.1SG I.NOM him.DAT  
 napisała (czy pójdziemy pograć)  
 write.PST.SG.F if go.PRS.1PL play.INF  
 ‘(Or maybe [he did it] on purpose) so that I would write to him, (if we went playing.)’
- g. (a jak włożysz do wody) to **mi** zaraz [PRN || REFL]  
 and when put.PRS.2SG to water.GEN.SG TO me.DAT immediately  
**się** zrobi twarde w środku  
 REFL.ACC make.PRS.3SG hard.NOM.SG.N in inside.LOC.SG  
 ‘(a when you put [it] into water) then it will immediately become hard inside [to my disadvantage]’
- h. oni **się** [PP w życiu] **ci** nie przyznają [...] [REFL || PRN]  
 they.NOM REFL.ACC in live.LOC.SG you.DAT NEG admit.PRS.3SG  
 ‘Under no circumstances they will admit ...’ (Pol; NKJP pelcra)

We find fewer splitting types within complex clauses, whereby we may attribute lower type frequency to lower token frequency. Three types mirror splits found in simple clauses, i.e. (COND || REFL), (AUX || PRN), (REFL || PRN). Two types are attested only for complex clauses and present instances of diacclisis among identical categories. There is one instance of diacclisis between two pronominal clitics with canonical order DAT < ACC, cf. (10.25a). The clause-final dative reflexive *sobie* is irrelevant for cliticisation, since it is the long orthotonic form. Two cases of diacclitic reflexive clitics have been encountered, whereby both times exhibiting the accusative/genitive form *się* next to the dative form *se*, cf. (10.25b–c). Example (10.25c) is particularly interesting, because the reflexive dative seems to have climbed into the matrix clause. However, none of the reflexive clitics has been subject to haplology. Note that the NP *łóżko* ‘bed’, which is intervening between the reflexive clitics, is part of a Focus domain that involves dative *se*. This is particularly interesting, since, from a phrase structure perspective, the dative clitic crossed the accusative reflexive by aid of the focused constituent. Note that both the dative reflexive *se* and the focused phrase *chociaż łóżko* are dependents of the embedded infinitive *pościelać* ‘(to) make bed’.

- (10.25) a. ([...] że im mniej mam czasu. tym więcej. [PRN || PRN]  
 that the less have.PRS.1SG time.GEN.SG the more  
 Znaczy) tym łatwiej<sub>1</sub> **mi**<sub>1</sub> jest zorganizować<sub>2</sub> **go**<sub>2</sub> sobie<sub>2</sub>  
 mean.PRS.3SG the easier me.DAT be.PRS.3SG organise.INF him.ACC REFL.DAT  
 ‘(... that the less time I have the more, I mean,) the easier it is for me to organise it for myself’
- b. a oni **se**<sub>1</sub> poszli<sub>1</sub> **się**<sub>2</sub> wysikać<sub>2</sub> [REFL || REFL]  
 and they.NOM.VIR REFL.DAT walk.PST.PL.VIR REFL.ACC pee.INF  
 za górkę wiesz?  
 behind hill.ACC know.PRS.2SG  
 ‘and they just left to pee behind the hill, you know?’
- c. niektórzy po takim wojsku to [ chociaż **se**<sub>2</sub> [REFL || REFL]  
 some after such.LOC.SG army.LOC.SG TO at\_least REFL.DAT  
 łóżko ]<sub>FOC</sub> **się**<sub>1</sub> nauczyli<sub>1</sub> pościelać<sub>2</sub>  
 bed.ACC.SG REFL.ACC learn.PST.PL.VIR make\_bed.INF  
 ‘it is in the army that some [people] learnt to make at least their bed’ (Pol; NKJP pelcra)

At the intersection of clitic clustering and clitic placement, there belongs the topic of clitic occurrences in clause-initial position, which covers also single clitic occurrences, of course. Continuing to follow the model of the Czech data presentation, I will give the attested categories of single clitics and clitic clusters in absolute 1P in the following: (Refl), (Prn), (Cond, Refl), (Cond, Prn), (Aux, Refl), (Prn, Refl), (Cond, Aux, Prn).

### 10.3.3 Clitic climbing

Clitic climbing represents circa 10% of the entire data set. It is the predominant pattern within complex clauses, where it makes 90% of occurrences, as can be seen in Table 10.36. There are few tokens with an unclear status as to locality (i.e. CC?).

Table 10.36: Frequencies of clitic climbing and no climbing in spoken Polish.

| Locality      | Total      |                                 | Pronominal |                                 | Reflexive  |                                 |
|---------------|------------|---------------------------------|------------|---------------------------------|------------|---------------------------------|
|               | Abs. freq. | Rel. freq.<br>w/i cmplx clauses | Abs. freq. | Rel. freq.<br>w/i cmplx clauses | Abs. freq. | Rel. freq.<br>w/i cmplx clauses |
| CC            | 298        | 91.4%                           | 215        | 92.3%                           | 39         | 89.2%                           |
| noCC          | 25         | 7.7%                            | 16         | 6.9%                            | 9          | 9.7%                            |
| CC?           | 3          | 0.9%                            | 2          | 0.9%                            | 1          | 1.1%                            |
| Cmplx clauses | 326        | 100%                            | 233        | 100%                            | 93         | 100%                            |

Example block (10.26) illustrates CC vs noCC for pronominal (10.26a–b) and reflexive clitics (10.26c–d). An unclear case is presented in (10.26e), where it is impossible to determine, whether the pronominal clitic *mu* has climbed into the matrix clause independently from the fronted infinitive, or whether it is realised locally within the infinitival VP.

- (10.26) a. to co ci<sub>2</sub> mam<sub>1</sub> powiedzieć<sub>2</sub>  
 TO what you.DAT have.PRS.1SG tell.INF  
 ‘So, what should I tell you?’
- b. i zaczynasz<sub>1</sub> później pokazywać<sub>2</sub> mu<sub>2</sub> (co mógł-by zrobić  
 and start.PRS.2SG later show.INF him.DAT what can.PST.SG.M-COND make.INF  
 lepiej)  
 better  
 ‘and, later, you start to show him (what he could improve)’
- c. to se<sub>2</sub> musisz<sub>1</sub> wynająć<sub>2</sub> kogoś  
 TO REFL.DAT must.PRS.2SG rent.INF someone.ACC  
 ‘Then you have to hire someone.’
- d. właśnie dlatego nie mogę<sub>1</sub> zorientować<sub>2</sub> się<sub>2</sub> (o co  
 exactly that\_is\_why NEG can.PRS.1SG orient.INF REFL.ACC about what  
 chodzi z tym planem nie)  
 go.PRS.3SG with this.INS.SG.M plan.INS.SG NEG  
 ‘This is exactly why I cannot figure out (what’s the matter with that plan, right?)’
- e. polać<sub>2</sub> mu<sub>2</sub> chciała-m<sub>1</sub> (ale tyle tylko)  
 pour.INF him.DAT want.PST.SG.F-1SG but so\_many only  
 ‘I wanted to pour him (but only this much)’ (Pol; NKJP pelcra)

The lowest frequencies of CC in Table 10.37 are found with pronominal *mu* and reflexive *se* with 84% and 88%, respectively. For the rest, CC occurs in at least 90% of complex clauses (cf. reflexive *się* with 91%), whereby figures approximate 95% for pronominal *cię*, *go*, and *ci*. The only clitic

without any attestation of local placement is *mi*, which, however, exhibits the lowest frequency of attestations.

Table 10.37: Frequencies of clitic climbing and no climbing in spoken Polish by clitic.

| Local.           | <i>cię</i> |                      | <i>go</i> |                      | <i>mi</i> |                      | <i>ci</i> |                      | <i>mu</i> |                      | <i>się</i> |                      | <i>se</i> |                      |
|------------------|------------|----------------------|-----------|----------------------|-----------|----------------------|-----------|----------------------|-----------|----------------------|------------|----------------------|-----------|----------------------|
|                  | Abs.       | Rel.<br>w/i<br>cmplx | Abs.      | Rel.<br>w/i<br>cmplx | Abs.      | Rel.<br>w/i<br>cmplx | Abs.      | Rel.<br>w/i<br>cmplx | Abs.      | Rel.<br>w/i<br>cmplx | Abs.       | Rel.<br>w/i<br>cmplx | Abs.      | Rel.<br>w/i<br>cmplx |
| CC               | 53         | 94.6%                | 68        | 94.4%                | 15        | 100%                 | 36        | 94.7%                | 43        | 84.3%                | 31         | 91.2%                | 52        | 88.1%                |
| noCC             | 3          | 5.4%                 | 4         | 5.6%                 | —         | —                    | 2         | 5.3%                 | 7         | 13.7%                | 3          | 8.8%                 | 6         | 10.2%                |
| CC?              | —          | —                    | 1         | 1.4%                 | —         | —                    | —         | —                    | 1         | 2.0%                 | —          | —                    | 1         | 1.7%                 |
| Cmplx<br>clauses | 56         | 100%                 | 72        | 100%                 | 15        | 100%                 | 38        | 100%                 | 51        | 100%                 | 34         | 100%                 | 59        | 100%                 |

When accounting for target positions under CC, there is a high degree of variation according to Table 10.38. Clausal second position (2P) is the most frequent target, but amounts to less than 50%, whereby pronominal and reflexive clitics differ considerably. Pronominal clitics show preference for 2P with 45% followed by 3P with almost 30% and 4P with 10%. Overall, DP and 2P placement are equally frequent, which is not exactly mirrored by the clitic categories. Pronominal clitics have a slight tendency to target 2P more often than DP. The situation is different with reflexive clitics, where 2P occurs in ca. 35% of cases, while DP outnumbers it with ca 45%. Within Delayed Placement, the frequency of 3P resembles the one of 2P, but 2P is still more frequent. Reflexive clitics, too, target 4P in approximately 10% of cases.

Table 10.38: Frequencies of target positions under clitic climbing in spoken Polish.

| Position<br>after CC | Total      |                      | Pronominal |                      | Reflexive  |                      |
|----------------------|------------|----------------------|------------|----------------------|------------|----------------------|
|                      | Abs. freq. | Rel. freq.<br>w/i CC | Abs. freq. | Rel. freq.<br>w/i CC | Abs. freq. | Rel. freq.<br>w/i CC |
| 1P                   | 6          | 2.0%                 | 3          | 1.4%                 | 3          | 3.6%                 |
| 2P                   | 131        | 44.0%                | 98         | 45.6%                | 29         | 34.9%                |
| 3P                   | 88         | 29.5%                | 61         | 28.4%                | 26         | 31.3%                |
| 4P                   | 32         | 10.7%                | 22         | 10.2%                | 9          | 10.8%                |
| 5P                   | 7          | 2.3%                 | 5          | 2.3%                 | 2          | 2.4%                 |
| 6P                   | 1          | 0.3%                 | 1          | 0.5%                 | —          | —                    |
| DP                   | 128        | 43.0%                | 89         | 41.4%                | 37         | 44.6%                |

Examples for clitic climbing targeting clausal first to sixth position are given in (10.27). The sixth position placement for *cię* in (10.27f) results from a disfluency, in fact, which is the repeated occurrence of the adverbial particle *już*.

- (10.27) a. (i idziesz do ZUS-u zgłaszasz że rozpoczynasz  
and go.PRS.2SG to ZUS.GEN.SG announce.PRS.2SG that start.PRS.2SG  
działalność) i<sup>0</sup> ci<sup>1</sup> dopiero zaczynają<sub>1</sub> naliczać<sub>2</sub>.  
activity.ACC.SG and you.DAT yet start.PRS.3PL count.INF  
'(and you go to the ZUS [= social insurance agency], announce that you start a  
[professional] activity) and they only start to count it [up]'
- b. ja<sup>1</sup> się<sup>2</sup> pójdę<sub>1</sub> myć<sub>2</sub>  
I.NOM REFL.ACC PFV-go.PRS.1SG wash.INF  
'I will go wash myself.'
- c. że<sup>1</sup> nigdzie<sup>2</sup> mu<sup>3</sup> nie chcą<sub>1</sub> z-operować<sub>2</sub> tego [...] [ ...]  
that nowhere him.DAT NEG want.PRS.3PL PFV-operate.INF this.GEN.SG.N  
'that they won't be willing to operate it (in his favour) anywhere ...'
- d. (czy wy wiecie że) jak<sup>1</sup> ja<sup>2</sup> poszła-m<sub>1</sub><sup>3</sup> go<sub>2</sub><sup>4</sup>  
if you.PL.NOM know.PRS.2PL that when I.NOM go.PST.SG.F-AUX.1SG him.ACC  
odwiedzić<sub>2</sub> (to przeszła-m wszystkie sale i nie  
visit.INF TO go\_through.PST.SG.F-1SG all.ACC.NVIR room.ACC.PL and NEG  
widziała-m go tam.)  
see.PST.SG.F-AUX.1SG him.ACC there  
'(do you know that) when I went visiting him (then I went through all rooms and did not  
see him there'
- e. (tak jak-by-m wygrała teraz milion baksów.) no  
so when-COND-AUX.1SG win.PST.SG.F now million.ACC.SG bucks.GEN.PL well  
to<sup>1</sup> [PP od razu ]<sup>2</sup> by<sub>1</sub><sup>3</sup>=m<sub>1</sub><sup>3</sup> poszła<sub>1</sub><sup>4</sup> se<sub>2</sub><sup>5</sup> za-jść<sub>2</sub> w  
TO from once COND=AUX.1SG go.PST.SG.F REFL.DAT PFV-go.INF in  
ciążę nie.  
pregnancy.ACC.SG NEG  
'(so when I would win a million bucks now) well, then I would get pregnant immediately,  
right'

f. *że<sup>1</sup> już<sup>2</sup> niekiedy<sup>3</sup> to<sup>4</sup> już<sup>5</sup> cię<sup>6</sup> zaczyna<sub>1</sub> łaskotać<sub>2</sub>*  
 that already sometimes this.NOM.SG.N already you.ACC start.PRS.3SG tickle.INF  
*po prostu.*  
 simply  
 ‘that it sometimes already starts simply tickling you’ (Pol; NKJP *pelcra*)

Table 10.39: Frequencies of clausal positions targeted by clitic climbing in spoken Polish by clitic.

| Position<br>after CC | <i>cię</i> |                   | <i>go</i> |                   | <i>mi</i> |                   | <i>ci</i> |                   | <i>mu</i> |                   | <i>się</i> |                   | <i>se</i> |                   |
|----------------------|------------|-------------------|-----------|-------------------|-----------|-------------------|-----------|-------------------|-----------|-------------------|------------|-------------------|-----------|-------------------|
|                      | Abs.       | Rel.<br>w/i<br>CC | Abs.      | Rel.<br>w/i<br>CC | Abs.      | Rel.<br>w/i<br>CC | Abs.      | Rel.<br>w/i<br>CC | Abs.      | Rel.<br>w/i<br>CC | Abs.       | Rel.<br>w/i<br>CC | Abs.      | Rel.<br>w/i<br>CC |
| 1P                   | 1          | 1.9%              | 1         | 1.5%              | —         | —                 | 1         | 2.8%              | —         | —                 | 1          | 3.2%              | 2         | 3.8%              |
| 2P                   | 29         | 54.7%             | 36        | 52.9%             | 7         | 46.7%             | 20        | 55.6%             | 23        | 53.5%             | 13         | 41.9%             | 26        | 50.0%             |
| 3P                   | 16         | 30.2%             | 21        | 30.9%             | 4         | 26.7%             | 10        | 27.8%             | 15        | 34.9%             | 12         | 38.7%             | 17        | 32.7%             |
| 4P                   | 5          | 9.4%              | 9         | 13.2%             | 2         | 13.3%             | 5         | 13.9%             | 4         | 9.3%              | 5          | 16.1%             | 5         | 9.6%              |
| 5P                   | 1          | 1.9%              | 1         | 1.5%              | 2         | 13.3%             | —         | —                 | 1         | 2.3%              | —          | —                 | 2         | 3.8%              |
| 6P                   | 1          | 1.9%              | —         | —                 | —         | —                 | —         | —                 | —         | —                 | —          | —                 | —         | —                 |
| DP                   | 23         | 43.4%             | 31        | 45.6%             | 8         | 2.0%              | 15        | 41.7%             | 20        | 46.5%             | 17         | 54.8%             | 24        | 46.2%             |

Individual clitics exhibit fluctuations as evidenced by the frequencies in Table 10.39. Most pronominal clitics target 2P in over 50% of occurrences. It is only the reflexive clitic *się* and the pronominal clitic *mi*, which are beneath this threshold. Generally, there is variation within DP, whereby it should be noted that the token frequencies are very low. Still, 3P is the most prominent position in DP under CC with roundabout 30% or more of CC instances per clitic. Only the dative clitics *mi* and *ci* show less than 30% of 3P with roughly 27% and 28% respectively.

Interestingly, we can read from Table 10.40 that more than 90% of all climbed clitics end up contiguous to a verb irrespective of clitic category. The overall distribution shows a clear preference for biverbal configurations over mono-verbal ones. The picture becomes more complex when zooming in. The 2VA pattern is prevalent especially for pronominal clitics, while reflexive clitics leave a different impression with 1VA outnumbering 2VA. CC without verb-adjacency occurs with a rather low frequency of less than 10% of occurrences independently of the clitic category. Correspondingly, one may hypothesise that 2P without VA occurs rather rarely, but an exact quantification is pending.

Table 10.40: Frequencies of simultaneous occurrences of clitic climbing and (non-)verb-adjacency in spoken Polish.

| Placement<br>after CC | Total      |                      | Pronominal |                      | Reflexive  |                      |
|-----------------------|------------|----------------------|------------|----------------------|------------|----------------------|
|                       | Abs. freq. | Rel. freq.<br>w/i CC | Abs. freq. | Rel. freq.<br>w/i CC | Abs. freq. | Rel. freq.<br>w/i CC |
| noVA                  | 23         | 7.7%                 | 16         | 7.4%                 | 7          | 8.4%                 |
| 1VA                   | 119        | 39.9%                | 78         | 36.3%                | 41         | 49.4%                |
| 2VA                   | 156        | 52.3%                | 121        | 56.3%                | 35         | 42.2%                |

- (10.28) a. [...] no ale trzeciego **mi**<sub>2</sub> [<sub>PART</sub> chyba ] **nie** **kazała**<sub>1</sub> [noVA]  
 well but third.GEN.SG me.DAT probably NEG tell.PST.SG.F  
**robić**<sub>2</sub> tego samego zęba  
 make.INF this.GEN.SG.M same.GEN.SG.M tooth.GEN.SG  
 ‘... but she didn’t tell to make me a third [x-ray] of the same tooth I guess’
- b. może w jakiś koc **się**<sub>2</sub> **chcesz**<sub>1</sub> **owinąć**<sub>2</sub> ? [1VA]  
 maybe in some.ACC.SG.M blanket.ACC.SG REFL.ACC want.PRS.2SG wrap.INF  
 ‘maybe you want to wrap yourself in some blanket?’
- c. ja **nie** **mogła**=**m**<sub>1</sub> **go**<sub>2</sub> **wyłączyć**<sub>2</sub> (i [2VA]  
 I.NOM NEG can.PST.SG.F=AUX.1SG him.ACC turn\_off.INF and  
 schowała-m go pod poduszkę.)  
 hide.PST.SG.F-AUX.1SG him.ACC beneath pillow.ACC.SG  
 ‘I couldn’t turn it [= alarm clock] off (and hid it beneath the pillow.)’ (Pol; NKJP pelcra)

The frequencies vary across clitics, but the overall pattern repeats in Table 10.41 by and large. The pronominal clitics *cię* and *mu* show the least frequencies of VA placement (87% and 88% respectively), while *go* and *ci* approximate exhaustive verb-adjacency (97% each). The prevalence of 2VA vs. 1VA is consistent among pronominal clitics, but the picture is more nuanced for reflexive clitics. The picture for *się* is inconclusive (14 vs. 15 tokens), but for *se*, we register a higher frequency of 1VA than for 2VA.



Table 10.41: Frequencies of simultaneous occurrences of clitic climbing and (non-)verb-adjacency in spoken Polish by clitic.

| Placement<br>after CC | <i>cię</i> |                | <i>go</i> |                | <i>mi</i> |                | <i>ci</i> |                | <i>mu</i> |                | <i>się</i> |                | <i>se</i> |                |
|-----------------------|------------|----------------|-----------|----------------|-----------|----------------|-----------|----------------|-----------|----------------|------------|----------------|-----------|----------------|
|                       | Abs.       | Rel. w/i<br>CC | Abs.      | Rel.<br>w/i CC | Abs.      | Rel.<br>w/i CC | Abs.      | Rel.<br>w/i CC | Abs.      | Rel.<br>w/i CC | Abs.       | Rel.<br>w/i CC | Abs.      | Rel.<br>w/i CC |
| noVA                  | 7          | 13.2%          | 2         | 2.9%           | 1         | 6.7%           | 1         | 2.8%           | 5         | 11.6%          | 2          | 6.5%           | 5         | 9.6%           |
| 1VA                   | 18         | 34.0%          | 26        | 38.2%          | 6         | 40.0%          | 14        | 38.9%          | 14        | 32.6%          | 14         | 45.2%          | 27        | 51.9%          |
| 2VA                   | 28         | 52.8%          | 40        | 58.8%          | 8         | 53.3%          | 21        | 58.3%          | 24        | 55.8%          | 15         | 48.4%          | 20        | 38.5%          |

Three instances of CC from deep embeddings were detected in the data sample, all of which concern reflexives and occur with a finite temporal auxiliary *być* ‘(to) be’ marking future tense, cf. (10.29). Note that the first embedded non-finite verb is not an infinitive, but an *l*-participle in each case. The reflexive clitic *se* in example (10.29c) is an ethical dative, in fact, which is why it is not clear which verb it is associated with. In principle, it can modify and thus be associated with any of the three verbs. Therefore, very different analyses are available.<sup>551</sup>

- (10.29) a. *nie będziemy się musieli stresować jakimiś*  
 NEG be.FUT.1PL REFL.ACC must.PST.PL.VIR stress.INF some.INS.PL  
*różnymi głupotami*  
 different.INS.PL nonsense.INS.PL  
 ‘we won’t have to bother about different kinds of nonsense’
- b. *a Marcin będzie se umiał z tym poradzić?*  
 and Marcin be.FUT.3SG REFL.DAT can.PST.SG.M with this.INS.SG cope.INF  
 ‘and will Marcin be able to cope with this?’
- c. *no to ja se<sub>1,2,3</sub> bede szła spać zara*  
 well TO I.NOM REFL.DAT be.FUT.1SG walk.PST.SG.F sleep.INF soon  
 ‘well, then I will just go to sleep soon’ (Pol; NKJP pelcra)

<sup>551</sup> Only if *se* were associated with the infinitive *spać*, we would face climbing from a deep embedding. If the reflexive were tied to the future auxiliary *bede*, then there would not be any climbing, at all. The fact that reflexive ethical datives can occur with the future form of the copula verb *być* is confirmed by (i) from an informal web search, where *będę* is the only verb present. Although I was not able to find an attestation with short *se*, it is perfectly possible to replace the orthotonic reflexive *sobie* by its clitic counterpart *se*.

(i) *(zawsze myślał-e-m,) spoko będę sobie lekarzem, (bede siedzieć*  
 always think.PST.SG.M-M-1SG no\_problem be.FUT.1SG REFL.DAT physician.INS.SG be.FUT.1SG sit.INF  
*gdzies w szpitalu i leczyc ludzi, [...])*  
 somewhere in hospital.LOC.SG and cure.INF people.ACC.PL  
 ‘(I always thought), no problem, I will be a physician, (I will sit somewhere in a hospital and treat people ...)’  
 (Pol; S17)

## 10.4 Goralish

### 10.4.1 Clitic placement

Roughly 3,900 elicited data points were examined for the Spisz variety of Goralish, of which 2676 tokens were eligible for analysis. The Goralish data in Table 10.42 show varying positional realisations for clitics ranging from 1P to 7P. Example block (10.30) cites evidence for every attested position in the normalised orthographic form, which is the default output of the KoSpi. In quantitative terms, clitic positioning is dominated by the Wackernagel pattern (2P), which makes two-thirds ( $\approx \frac{2}{3}$ ) of occurrences. One-fifth ( $\approx \frac{1}{5}$ ) of all tokens is placed in third position (3P). The frequency distribution then shows the successive ordering of 4P, 1P and increasing distance from the left periphery becoming marginal. For pronominal clitics, the relative frequency of second position placement is slightly higher than described and approximates 70% (it is close to 69%). The difference between 1P and 4P is less pronounced, but 1P is effectively more frequent. However, both positions do not exceed 5%. In sum, DP comes to one-quarter ( $\approx \frac{1}{4}$ ) of pronominal tokens.

Table 10.42: Frequencies of attested clitic clausal positions in Spisz Goralish.

| Pos | Left distant |       | Left adjacent |       | Right (adjacent) |       | Sum   |       |
|-----|--------------|-------|---------------|-------|------------------|-------|-------|-------|
| 1P  | 29           | 1.1%  | 101           | 3.8%  | N/A              | N/A   | 130   | 4.9%  |
| 2P  | 490          | 18.3% | 899           | 33.6% | 376              | 14.1% | 1,765 | 66.1% |
| 3P  | 161          | 6.0%  | 304           | 11.4% | 116              | 4.3%  | 581   | 21.7% |
| 4P  | 25           | 0.9%  | 81            | 3.0%  | 43               | 1.6%  | 149   | 5.6%  |
| 5P  | 6            | 0.2%  | 19            | 0.7%  | 9                | 0.3%  | 34    | 1.3%  |
| 6P  | —            | —     | 3             | 0.1%  | 3                | 0.11% | 6     | 0.2%  |
| 7P  | —            | —     | 1             | 0.04% | —                | —     | —     | —     |
| Sum | 711          | 26.6% | 1408          | 52.7% | 547              | 20.5% | 2,666 | 99.8% |
| DP  | 192          | 7.2%  | 408           | 15.3% | 171              | 6.4%  | 771   | 28.9% |

Turning to reflexive clitics, we observe a significantly different pattern ( $\chi^2 = 20.12$ ,  $p = .05$ ,  $df = 2$ ). In comparison to clitic pronouns, the ratio for 2P is lower showing 60%, while 3P is slightly higher with 24%. 1P is close to 5%, but there are more 4P tokens with a ratio of over 8%. DP is more strongly represented and adds up to over one-third ( $\approx \frac{1}{3}$ ) of all attestations. The frequency distribution is described by (10.31), where the proportion of 4P and 1P is treated as inconclusive.

With respect to laterality and predicate-adjacency, contact with the predicate makes almost three-quarters of occurrences (= 73.2%), whereas second position placement comes to two-thirds (= 66.1%). Altogether, left lateral placement sums up to four-fifths of occurrences (= 79.5%). Delayed Placement preferably occurs in pre-predicate position. Left-side placement without contact and right-side placement exhibit similar distributions, but the former is slightly more frequent. Though being a low frequency phenomenon, Clitic First positioning and Delayed Placement are available without contiguity to the predicate.

Table 10.43: Frequencies of clitic occurrences in Spisz Goralish by position, laterality, and distance per clitic category.

| Pos | Pronominal |       |     |       |      |       |       |       | Reflexive |       |     |       |      |       |     |       |
|-----|------------|-------|-----|-------|------|-------|-------|-------|-----------|-------|-----|-------|------|-------|-----|-------|
|     | Ld         |       | La  |       | R(a) |       | Sum   |       | Ld        |       | La  |       | R(a) |       | Sum |       |
| 1P  | 23         | 1.3%  | 66  | 3.6%  | N/A  | N/A   | 89    | 4.9%  | 6         | 0.7%  | 35  | 4.1%  | N/A  | N/A   | 41  | 4.8%  |
| 2P  | 364        | 20.0% | 622 | 34.3% | 263  | 14.5% | 1,249 | 68.8% | 126       | 14.7% | 277 | 32.4% | 113  | 13.2% | 516 | 60.3% |
| 3P  | 97         | 5.3%  | 202 | 11.1% | 79   | 4.4%  | 378   | 20.8% | 64        | 7.5%  | 102 | 11.9% | 37   | 4.3%  | 203 | 23.7% |
| 4P  | 14         | 0.8%  | 41  | 2.3%  | 25   | 1.4%  | 80    | 4.4%  | 11        | 1.3%  | 40  | 4.7%  | 18   | 2.1%  | 69  | 8.1%  |
| 5P  | 2          | 0.1%  | 11  | 0.6%  | 2    | 0.1%  | 15    | 0.8%  | 4         | 0.5%  | 8   | 0.9%  | 7    | 0.8%  | 19  | 2.2%  |
| 6P  | —          | —     | 1   | 0.1%  | 2    | 0.11% | 3     | 0.2%  | —         | —     | 2   | 0.2%  | 1    | 0.12% | 3   | 0.4%  |
| 7P  | —          | —     | —   | —     | —    | —     | —     | —     | —         | —     | 1   | 0.12% | —    | —     | 1   | 0.1%  |
| Sum | 500        | 27.5% | 943 | 51.9% | 371  | 20.4% | 1,814 | 99.9% | 211       | 24.6% | 465 | 54.3% | 176  | 20.6% | 852 | 99.5% |
| DP  | 113        | 6.2%  | 255 | 14.0% | 108  | 5.9%  | 476   | 26.2% | 79        | 9.2%  | 153 | 17.9% | 63   | 7.4%  | 295 | 34.5% |

Table 10.44 shows that clause-initial clitics (1P) are recorded less frequently after clause-boundaries than after coordinating conjunctions.

Table 10.44: Frequencies for Clitic First in Spisz Goralish after clause boundary and coordinating conjunction (in total).

| Type        | Abs. freq. | Rel. freq. w/i 1P |
|-------------|------------|-------------------|
| # 1P ...    | 56         | 43.1%             |
| CONJ 1P ... | 74         | 56.9%             |

- (10.30) a. <[...] to szukali dziewczęta na miejscu a dzisiaj a  
 TO search.PST.PL.VIR girl.ACC.PL on place.LOC.SG and today and  
 gdzie) # **cię**<sup>1</sup> sram (siadam w auto [...])>  
 where you.ACC shit.PRS.1SG sit\_down.PRS.1SG in car.ACC.SG  
 ‘(... so they were looking for girls nearby. And today? No way!) I don’t give a shit, (I get in the car ...)’

- b. <(dwadzieścia pięć metrów odmierzył em i  
twenty.ACC five.ACC metre.GEN.PL measure.PST.SG.M AUX.1SG and  
dał em córce) i<sup>0</sup> **sobie/se**<sup>1</sup> wybudowali  
give.PST.SG.M AUX.1SG daughter.DAT.SG and REFL.DAT build.PST.PL.VIR  
domek nie?>  
house.DIM.ACC.SG NEG  
'(I measured twenty-five metres and gave [them] to [my] daughter) and they built a house,  
right?'
- c. <no to<sup>1</sup> **m**<sup>2</sup> **go**<sup>2</sup> wzięła do szkoły>  
well TO AUX.1SG him.ACC take.PST.SG.F to school.GEN.SG  
'well, so I took him to school'
- d. <że<sup>1</sup> dziadek<sup>2</sup> **mu**<sup>3</sup> potem gada (wiesz co  
that grandfather.NOM.SG him.DAT afterwards say.PRS.3SG know.PRS.2SG what  
idź ty do księdza)>  
go.IMP.2SG you.SG.NOM to priest.ACC.SG  
'that grandpa tells him later (you know what? You go to the priest)'
- e. <a potem<sup>1</sup> zaś<sup>2</sup> [PP za dziesięć roków ]<sup>3</sup> **się**<sup>4</sup> **mi**<sup>4</sup>  
and afterwards again after ten.ACC year.GEN.PL REFL me.DAT  
urodziła siostra>  
be\_born.PST.SG.F sister.ACC.SG  
'and then, another ten years later, my sister was born'
- f. <(namyśleli) że<sup>1</sup> **by**<sup>2</sup> trza<sup>3</sup> jakosi<sup>4</sup> **go**<sup>5</sup> **ściągnąć**>  
think.PST.PL.VIR that COND need somehow him.ACC take\_down.INF  
'(they came up [with the thought]) that one would have to take it down somehow'
- g. <[...] bo<sup>1</sup> to<sup>2</sup> tak<sup>3</sup> tu<sup>4</sup> [PP na tym ]<sup>5</sup> **ci**<sup>6</sup>  
because this.ACC.SG.N so here on this.LOC.SG.N you.DAT  
poprzyprawiali [...]>  
attach.PST.PL.VIR  
'... because they attached it (for you) like this here on that [thing] ...'

h. <[...] bo<sup>1</sup> naprawdę<sup>2</sup> ciężko<sup>3</sup> było<sup>4</sup> y czasami<sup>5</sup> [pp na  
because really difficult.ADV be.PST.SG.N uh at\_times on  
te próby]<sup>6</sup> się<sup>7</sup> spotykać (na tych  
this.ACC.PL.NVIR rehearse!.ACC.PL REFL.ACC meet.INF on this.LOC.PL  
próbach [...]>  
rehearse!.LOC.PL  
‘... because it was really difficult at times to meet for the rehearsals (at the rehearsals ...)’  
(Gor; KoSpi)

The measures of central tendency in Table 10.45 identify Goralish as a second-position clitic language in spite of Clitic Third being within one standard deviation.

Table 10.45: Measures of central tendency for clitic positioning in Spisz Goralish.

| Measure:          | $\bar{x}$ (SD) | <i>Md</i> | <i>Mo</i> |
|-------------------|----------------|-----------|-----------|
| Value (position): | 2.34 (±0.74)   | 2         | 2         |

(10.31)2P > 3P > 4P / 1P > 5P > 6P (> 7P)

The comparison of absolute and relative frequencies between individual clitics in Table 10.46 yields a complex picture without a consistent pattern.

Table 10.46: Frequencies of attested clitic clausal positions in Spisz Goralish by clitic.

| Pos | <i>cię</i> |       | <i>go</i> |       | <i>mi</i> |       | <i>ci</i> |       | <i>mu</i> |       | <i>się</i> |       | <i>se</i> |       |
|-----|------------|-------|-----------|-------|-----------|-------|-----------|-------|-----------|-------|------------|-------|-----------|-------|
|     | Abs.       | Rel.  | Abs.      | Rel.  | Abs.      | Rel.  | Abs.      | Rel.  | Abs.      | Rel.  | Abs.       | Rel.  | Abs.      | Rel.  |
| 1P  | 19         | 7.5%  | 19        | 4.4%  | 21        | 5.3%  | 22        | 6.5%  | 8         | 2.0%  | 10         | 2.3%  | 31        | 7.2%  |
| 2P  | 180        | 70.6% | 283       | 65.4% | 269       | 67.8% | 242       | 71.6% | 276       | 69.5% | 282        | 65.9% | 234       | 54.7% |
| 3P  | 48         | 18.8% | 91        | 21.0% | 86        | 21.7% | 63        | 18.6% | 93        | 23.4% | 94         | 22.0% | 110       | 25.7% |
| 4P  | 5          | 2.0%  | 33        | 7.6%  | 18        | 4.5%  | 9         | 2.7%  | 17        | 4.3%  | 31         | 7.2%  | 40        | 9.3%  |
| 5P  | 3          | 1.2%  | 7         | 1.6%  | 2         | 0.5%  | 1         | 0.3%  | 2         | 0.5%  | 9          | 2.1%  | 10        | 2.3%  |
| 6P  | —          | —     | —         | —     | 1         | 0.3%  | 1         | 0.3%  | 1         | 0.3%  | 1          | 0.2%  | 2         | 0.5%  |
| 7P  | —          | —     | —         | —     | —         | —     | —         | —     | —         | —     | 1          | 0.2%  | 1         | 0.2%  |
| Sum | 255        | 100%  | 433       | 100%  | 397       | 100%  | 338       | 100%  | 397       | 100%  | 428        | 100%  | 428       | 100%  |
| DP  | 56         | 22.0% | 131       | 30.3% | 107       | 27.0% | 74        | 21.9% | 113       | 28.5% | 136        | 31.8% | 163       | 38.1% |

Wackernagel placement oscillates in a comparatively small range of values from 65% observed for *go* and *się* to 72% attested for *ci*. The short reflexive dative *se* presents an outlier in this respect with less than 55% of occurrences in second position. Correspondingly, it exhibits the highest frequency of Delayed Placement including Clitic Third. Remarkably, *se* also shows the highest frequency of

Clitic First occurrences (7.2%) except when compared with pronominal *cię*, which shows a slightly higher ratio in this position (7.5%). Delayed Placement is quite common for *go* and *się* with 30% and more. The smallest number of occurrences in Delayed Placement is observed with *cię* and *ci* (22%). The DP frequencies of *mi* and *mu* are located in between (27% and 28.5% respectively), but close to *go*. When tested for homogeneity, non-significant results come with the pairings {*cię*, *mi*}, {*cię*, *ci*}, {*go*, *mi*}, {*go*, *mu*}, {*go*, *się*}, {*mi*, *ci*}, and {*mu*, *się*}.

Table 10.47 shows a high degree of variation in lateral placement. Generally, left lateral placement prevails for each clitic type, while the same holds for predicate-adjacent placement. The intersecting configuration of left-adjacent placement is the strongest pattern, which represents over 50% in most cases (except for the dative pronominal clitic *mi*, *ci*, *mu*). The highest ratio for left-distant placement is found for the datival clitics *mi* and *se* being close to 30%. On the other hand, the lowest ratio is also attested for a dative clitic, i.e. second person *ci* with approximately 18% as is for its accusative counterpart *cię*. With respect to right lateral placement, the lowest frequency occurs with the reflexive clitics *się* and *se* with 17% and 13% respectively.

Table 10.47: Frequencies for laterality with left distant (Ld), left adjacent (La), right adjacent (R(a)) in spoken Goralish by clitic.

| Lat  | <i>cię</i> |       | <i>go</i> |       | <i>mi</i> |       | <i>ci</i> |       | <i>mu</i> |       | <i>się</i> |       | <i>se</i> |       |
|------|------------|-------|-----------|-------|-----------|-------|-----------|-------|-----------|-------|------------|-------|-----------|-------|
|      | Abs.       | Rel.  | Abs.      | Rel.  | Abs.      | Rel.  | Abs.      | Rel.  | Abs.      | Rel.  | Abs.       | Rel.  | Abs.      | Rel.  |
| Ld   | 82         | 18.3% | 99        | 22.8% | 119       | 29.2% | 75        | 17.5% | 98        | 23.7% | 99         | 20.4% | 122       | 28.2% |
| La   | 233        | 51.9% | 240       | 55.3% | 178       | 43.7% | 209       | 48.8% | 196       | 47.5% | 298        | 61.3% | 251       | 58.0% |
| R(a) | 133        | 29.6% | 91        | 21.0% | 103       | 25.3% | 144       | 33.6% | 113       | 27.4% | 83         | 17.1% | 56        | 12.9% |

(10.32) a. <i tak **cię** [PP do kościoła] [NP kumotrowie] [Ld]

and so you.ACC to church.GEN.SG godparent.NOM.PL

**zanieśli>**

carry.PST.PL.VIR

‘and in this way the godparents brought you to church’

b. <[...] i potem **em** **sobie/se** **wzięła** lekcje [La]

and afterwards AUX.1SG REFL.DAT take.PST.SG.F lesson.ACC.PL

**muzyczne>**

music.ADJ.ACC.PL.NVIR

‘... and later I attended music lessons.’

- c. <(no psa że-m zamykał) bo to on [R(a)]  
 well dog.ACC.SG ŻE-AUX.1SG lock\_up.PST.SG.M because TO he.NOM  
**zabił mi już parę zajęcy**>  
 kill.PST.SG.M me.DAT already pair.ACC.SG rabbit.ACC.PL  
 ‘(well, I locked the dog up,) because it was him who killed already a couple of my rabbits’  
 (Gor; KoSpi)

Considering clitic lowering, which is attested in the Spisz Goralish data sample, I came across 6 tokens, two of which involve pronominal clitics (i.e. *go*, *mi*), while the four remaining ones are confined to the dative reflexive clitic *se*, cf. (10.33).

- (10.33) a. <na i pyta się [NP liszka ] go [...]>  
 well\_and and ask.PRS.3SG REFL vixen.NOM.SG him.ACC  
 ‘well, and the vixen is asking him ...’
- b. <no a kto nie dał [NP rady] sobie/se w  
 well and who.NOM NEG give.PST.SG.M advice.GEN.SG REFL.DAT in  
 szkole [...]>  
 school.LOC.SG  
 ‘and whoever did not get along in school ...’  
 (Gor; KoSpi)

## 10.4.2 Clitic clustering

As with Czech and Polish, the vast majority of pronominal and reflexive clitic attestations occurs alone in Goralish, cf. Table 10.48. Clitic clusters make less than 20% of the examined database. The more clitics are involved in a cluster the lower their frequency. Most clusters comprise two clitics; three clitics are rare. There is only a single attestation of a quadruplet.

Table 10.48: Frequencies of single clitic and clitic cluster occurrences in Spisz Goralish by number of elements.

| Number of clitics | Absolute frequency | Relative frequency |
|-------------------|--------------------|--------------------|
| 1                 | 2,206              | 82.4%              |
| 2                 | 433                | 16.2%              |
| 3                 | 36                 | 1.3%               |
| 4                 | 1                  | 0.04%              |
| Sum               | 2,676              | 100%               |
| $n > 1$           | 470                | 17.6%              |

- (10.34) a. <za chwilę pytam się go (już mi  
after moment.ACC.SG ask.PRS.1SG REFL.ACC him.GEN already me.DAT  
gada siódma)>  
tell.PRS.3SG seventh.NOM.SG.F  
‘after a while I ask him (he tells me [that it’s] already seven [o’clock])’
- b. <(by ci chwycił palec) to by ci go  
COND you.DAT grab.PST.SG.M finger.ACC.SG TO COND you.DAT him.ACC  
nawet i mógł uciąć>  
even and can.PST.SG.M cut\_off.INF  
‘(if it [= crab] would grab your finger) then it [= crab] even could cut it [= finger] off’
- c. <(pamiętaj) że=byś mi się nie  
remember.IMP.2SG that=COND AUX.2SG me.DAT REFL.ACC NEG  
o-pił>  
PFV-drink.PST.SG.M  
‘(bear in mind) that you just shan’t become drunk’ (Gor; KoSpi)

Figure 10.4 provides a graphical representation of the frequency distribution of clitic categories co-occurring in a cluster. The diagram does not only visualise the frequential dominance of clitic clusters as pairs but shows also that there is great variability with respect to the linear order of pronominal and reflexive clitics. The order REFL < PRN is the most-frequent, but the reverse linearisation PRN < REFL is very frequent, too.



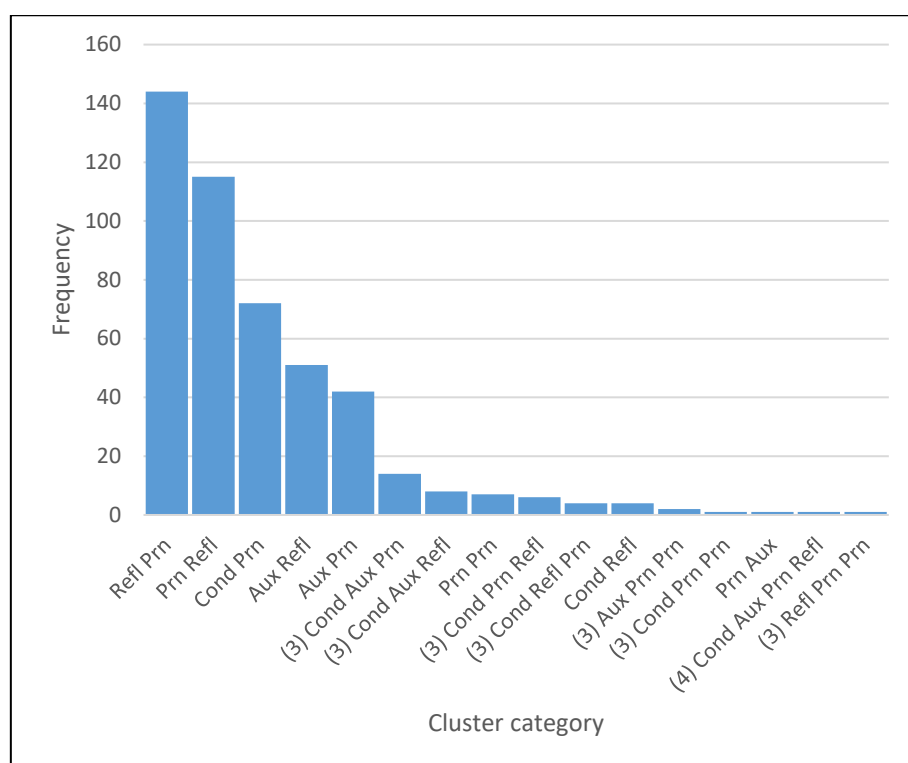


Figure 10.4: Frequency distribution of clitic cluster categories in Spisz Goralish.

When taking case into consideration in Table 10.49, dative occurs most often. The numbers reveal that it is primarily the dative case, which is affected, when it comes to variable ordering among pronominal and reflexive clitics. While varying linearisations are observed for accusative, too, the ratios are strikingly different.

Table 10.49: Absolute frequencies of case-valued clitic clusters in Spisz Goralish.

| Case sequence    | Absolute frequency |
|------------------|--------------------|
| DAT < REFL       | 118                |
| REFL < DAT       | 87                 |
| REFL < ACC       | 61                 |
| DAT < ACC        | 9                  |
| ACC < REFL       | 3                  |
| REFL < DAT < ACC | 1                  |
| DAT < DAT        | 1                  |

The relation (DAT, REFL) : (REFL, DAT) = 112 : 89 may be arithmetically broken down to 6 : 5. In contrast, the relation (REFL, ACC) : (ACC, REFL) = 60 : 3 corresponds to 20 : 1 or 6 : 0.3. While the most frequent order is DAT < REFL, the frequency of REFL < DAT is only slightly lower and not statistically significant ( $\chi^2 = 2.63$ ,  $p = .05$ ,  $df = 1$ ).

- (10.35) a. <noji to **mu** **się** rzeczywiście udało [...]>  
 well\_and this.NOM.SG.N him.DAT REFL.ACC indeed succeed.PST.SG.N  
 ‘well and it worked out for him indeed ...’
- b. <(mamy mu coś zrobić) co **się** **mu** nie  
 have.PRS.1PL him.DAT something.ACC make.INF what REFL.ACC him.DAT NEG  
 będzie podobało nigdy>  
 be.FUT.3SG appeal.PST.SG.N never  
 ‘(we will have to do something towards him) what he won’t be ever okay with’
- c. <pyta **się** **go** (ile chcesz za te  
 ask.PRS.3SG REFL him.ACC how\_many want.PRS.2SG for this.ACC.PL.NVIR  
 konie)>  
 horse.ACC.PL  
 ‘[he] asks him (how much do you want for those horses?)’
- d. <[...] i potem **go** **się** polewało przez dzień tak  
 and afterwards him.ACC REFL water.PST.SG.N through day.ACC.SG so  
 parę razy [...]>  
 pair.ACC.SG one.GEN.PL  
 ‘... and then it was watered several times a day ...’
- e. <ale może **się**<sub>1</sub> **mi**<sub>1</sub> **ją**<sub>2</sub> uda<sub>1</sub> namówić<sub>2</sub> teraz  
 but maybe REFL me.DAT her.ACC succeed.PRS.3SG convince.INF now  
 latem>  
 summer.INS.SG  
 ‘but maybe I will be able to convince her this summer’
- f. <a a on **ci** **mi** gada że [...]  
 and and he.NOM you.DAT.ETH me.DAT say.PRS.3SG that  
 ‘and he just tells me that ...’ (Gor; KoSpi)

Based upon the most frequent patterns, and including case distinctions, the overall linearisation pattern for a maximal clitic cluster in Goralish can be summarised as in (10.36a). This pattern complies with the linearisation template cited by Franks & King (2000: 156) for Polish. However, this pattern competes with (10.36b), where the order among dative and reflexive is reversed, thus, resembling Czech and Slovak in turn (cf. Franks & King 2000: 105, 129).

- (10.36) a. COND < AUX < DAT < REFL < ACC  
 b. COND < AUX < REFL < DAT < ACC

In total, there are 519 instances of multiple clitics. We know from Table 10.48 that 470 occurrences are clitic clusters, which represent over 90%, while another 49 cases of diacclisis are attested, which amount to roughly 10%. Table 10.50 summarises the figures.

Table 10.50: Frequencies of multiple clitics with clustering and diacclisis in Spisz Goralish.

| Multiple clitics | Absolute frequency | Relative frequency<br>w/i multiple CL |
|------------------|--------------------|---------------------------------------|
| Clustering       | 470                | 90.6%                                 |
| Diacclisis       | 49                 | 9.4%                                  |
| Sum              | 519                | 100%                                  |

According to Table 10.51, 46 out of 49 diacclitic configurations ( $\approx 94\%$ ) are recorded in simple clauses, i.e. between clitics with a shared verbal predicate. Only three instances of diacclisis occur across the multiple verbs of a complex clause ( $\approx 6\%$ ). The resulting ratio is 1.9% for simple clauses (46 within 2,451) and 1.3% for complex clauses (3 within 225). These figures are very close but provided the small absolute frequency of diacclisis in complex clauses, some caution is recommended when interpreting the numbers.

Table 10.51: Frequencies of diacclisis in Spisz Goralish by syntactic environment.

| Syntactic environment | Absolute frequency | Relative frequency w/i<br>diacclisis | Relative frequency w/i<br>syntactic environment |
|-----------------------|--------------------|--------------------------------------|-------------------------------------------------|
| Simple clause         | 46                 | 93.9%                                | 1.9%                                            |
| Complex clause        | 3                  | 6.1%                                 | 1.3%                                            |
| Sum                   | 49                 | 100%                                 | —                                               |

There is a great variety of splits across clitic categories attested for simple clauses, but only two splitting types in complex environments, as summarised in Table 10.52. The most frequent case value with pronominal clitics is dative. The complementary orders (PRN || REFL), (REFL || PRN) are limited to this morphological case. There were no reverse orderings found with accusative or genitive pronominal clitics. Accusative clitics are always preceded by a reflexive clitic in diacclisis. So far, diacclisis mirrors the facts from clustering. The attested reverse orderings (REFL || COND), (REFL || AUX), and (COND || PRN) contradict the order observed in clitic clusters, but share the common feature that the auxiliary and conditional clitic are in postposition, i.e. right-adjoined to verb. Critiques might therefore object that these are instances of inflection and not diacclisis. However, the mobile person markers are principally treated as clitics in the approach taken here. Bear also in mind that the

Goralish auxiliary clitic system differs from standard Polish in several respects, even though the original system is strongly influenced by (Standard) Polish. Therefore, one can still encounter the retained aorist ending in 1SG (*-k/-f < -ch*), syllabic auxiliary clitics in the singular (*jek/jef, jeś*), 1PL homonymous with corresponding nominative pronoun (*my*), free occurrence of *że*-support, and, most importantly, retained (or higher degree of) mobility of the person markers. The examples in (10.37) exhibit the relevant features. However, morphological isoglosses typical of Goralish are not encountered in all attestations. On the contrary, there is a great deal of common Polish forms.

Table 10.52: Attested splits between clitic categories in Spisz Goralish by syntactic environment.

| Simple clause | Complex clause |
|---------------|----------------|
| COND    REFL  | AUX    REFL    |
| REFL    COND  | REFL    AUX    |
| COND    PRN   |                |
| AUX    REFL   |                |
| REFL    AUX   |                |
| AUX    PRN    |                |
| PRN    AUX    |                |
| PRN    REFL   |                |
| REFL    PRN   |                |
| PRN    PRN    |                |

- (10.37) a. <[...] no to tam **by** [Adv już] [V zrobili] [COND || REFL]  
 well TO there COND already make.PST.PL.NVIR  
**się** działki [...]>  
 REFL plot\_of\_land.ACC.PL  
 ‘... well, so there would have been made plots (of land) already ...’
- b. <(choć kto ma we krwi) to [REFL || COND]  
 although who.NOM.SG have.PRS.3SG in blood.LOC.SG TO  
**sobie/se** [V zostawił]=**by** i pojechał>  
 REFL.DAT leave.PST.SG.M=COND and go.PST.SG.M  
 ‘(but in whoever’s DNA [it] is), [s\*he] would just leave [everything] and go’

- c. <(tata pojechał tak) jak=**by** [PRN ona ] **go** [COND || PRN]  
 dad.NOM.SG go.PST.SG.M so how=COND she.NOM him.ACC  
 zaś tam wzięła nie?  
 again there take.PST.SG.F NEG  
 ‘(dad went [there]) as if she [= mother] would take him there, in turn, right?’
- d. <a my **śmy** [NP to wszystko ] [AUX || REFL]  
 and we.NOM AUX.1PL this.ACC.SG.N everything.ACC.SG  
**sobie/se** suszyli>  
 REFL.DAT dry.PST.PL.VIR  
 ‘and we just dried all of that [= gras]’
- e. <[...] bo **sobie/se** [V uchwiał ] **eś** swoje [REFL || AUX]  
 because REFL.DAT hide.PST.SG.M AUX.2SG one’s.ACC  
 i miał eś>  
 and have.PST.SG.M AUX.2SG  
 ‘... because you stashed your [money away] and you had [it (for emergency)]’
- f. <[...] bo **m** [NP coś ] **ci** przywiózł> [AUX || PRN]  
 because AUX.1SG something.ACC you.DAT bring.PST.SG.M  
 ‘... because I (have) brought something for you’
- g. <a że osobiście **go** [V znał ] **em**> [PRN || AUX]  
 and that personally him.ACC know.PST.SG.M AUX.1SG  
 ‘and that I knew him personally’
- h. <[...] jak **ci** [NP nogę ] **się** złamie (to ci [PRN || REFL]  
 how you.DAT leg.ACC.SG REFL break.PRS.3SG TO you.DAT  
 wys... wyśrubują wytego... i już jest dobrze)>  
 ... screw.PRS.3PL ... and already be.PRS.3SG good.ADV  
 ‘... when you break your leg (then they scr... screw it [together], ’n’ stuff... and then [it]’s okay again)’
- i. <[...] że **się** [PP na blachach ] **go** grzało> [REFL || PRN]  
 that REFL on baking\_plate.LOC.PL him.ACC heat.PST.SG.N  
 ‘... that one heated it on baking plates.’
- j. <[...] i wtedy **go** [V dawaj ] **mu** kulą [PRN || PRN]  
 and that\_time him.ACC give.PRS.3SG him.DAT staff.INS.SG  
 tu przez głowę [...]>  
 here through head.ACC.SG  
 ‘... and then you give him [a stroke] at his head with the staff ...’ (Gor; KoSpi)

Before closing this section and going to clitic climbing in Goralish, let us sum up which categories of single clitic and clitic cluster occurrences were detected in non-coordinated clitic-first configurations. We find the following clitic categories and abstract tuples: (Prn), (Refl), (Cond, Refl), (Cond, Prn), (Aux, Refl), (Aux, Prn), (Prn, Refl), (Refl, Prn).

### 10.4.3 Clitic climbing

A subset of 225 clause tokens comprises complex clauses, i.e. 8.4% of the entire Goralish data set. Clitic climbing occurs in virtually 95% of occurrences, cf. Table 10.53. There is one instance with an uncertain status. Pronominal clitics follow this pattern by and large with a slightly higher ratio for CC. Here, we encounter the intermediate clitic. Reflexive clitics differ in that they show a reduced ratio for CC (91%) and more occurrences of local placement (9%).

Table 10.53: Frequencies of clitic climbing and no climbing in Spisz Goralish.

| Locality        | Total      |                                | Pronominal |                                | Reflexive  |                                |
|-----------------|------------|--------------------------------|------------|--------------------------------|------------|--------------------------------|
|                 | Abs. freq. | Rel. freq. w/i complex clauses | Abs. freq. | Rel. freq. w/i complex clauses | Abs. freq. | Rel. freq. w/i complex clauses |
| CC              | 213        | 94.7%                          | 142        | 96.6%                          | 71         | 91.0%                          |
| noCC            | 11         | 4.9%                           | 4          | 2.7%                           | 7          | 9.0%                           |
| CC?             | 1          | 0.4%                           | 1          | 0.7%                           | —          | —                              |
| Complex clauses | 225        | 100%                           | 147        | 100%                           | 78         | 100%                           |

Utterance (10.38a) shows clitic climbing of the embedded dative clitic *mu* co-occurring with agreement drop of the 1SG auxiliary clitic.<sup>552</sup> Another instance of CC is exemplified by (10.38c) with the reflexive clitic *się*. In contrast, the pronominal clitic *go* is placed locally in (10.38b) as is the reflexive dative clitic *se* in (10.38d). The only example of an unclear construction encountered in the data is cited in (10.38e). The embedded object clitic *go* occupies clausal third position following the subject. The embedded infinitive *prowadzić* precedes the orthotonic future auxiliary. However, it is unclear, if *go* has climbed independently of its governing verb or if it remained in situ and is dislocated as part of the infinitival VP. Note that neither adjacency between clitic and infinitive nor their linear order provide strong clues for or against a clitic-climbing-analysis, for the matrix verb is focused. Therefore, information structure obscures the syntactic structure on this matter.

<sup>552</sup> See Section 10.4.4 for agreement drop in Goralish.

- (10.38) a. <(on do mnie klockiem rzucił) a ja **mu**<sub>2</sub>  
 he.NOM to me.GEN brick.INS.SG throw.PST.SG.M and I.NOM him.DAT  
 chciała<sub>1</sub> oddać<sub>2</sub> [...]>  
 want.PST.SG.F return.INF  
 ‘(he threw a brick at me) and I wanted to get back at him ...’
- b. <[...] i musiał<sub>1</sub> nie przyjmować<sub>2</sub> **go**<sub>2</sub> [...]>  
 and must.PST.SG.M NEG hire.INF him.ACC  
 ‘... and he was forced to not hire him ...’
- c. <ja **się**<sub>2</sub> z tym nie mogę<sub>1</sub> zgodzić<sub>2</sub> że [...]>  
 I.NOM REFL.ACC with this.INS.SG.N NEG can.PRS.1SG agree.INF that  
 ‘I cannot agree on that ...’
- d. <[...] że chcą<sub>1</sub> coś zarobić<sub>2</sub> **sobie/se**<sub>2</sub> [...]>  
 that want.PRS.3PL something.ACC earn.INF REFL.DAT  
 ‘... that they want to earn [money] for themselves ...’
- e. <dokąd ja **go**<sub>2</sub> prowadzić<sub>2</sub> będę<sub>1</sub> (to ja go nigdy  
 as\_long\_as I.NOM him.ACC lead.INF be.FUT.1SG TO I.NOM him.ACC never  
 nie zmienię)>  
 NEG change.PRS.1SG  
 ‘As long as I will lead it [= dancing group] (I won’t ever change it[s character])’
- (Gor; KoSpi)

Zooming in yields a complex picture, cf. Table 10.54. Pronominal clitics pattern mostly homogeneously; all but *go* climb in 100% of cases. It is only *go*, which shows various locality patterns, but it is also the clitic with the highest absolute frequency of complex clause occurrences ( $n = 49$ ). The reflexive clitics exhibit some variation, too. Both *się* and *se* are attested with CC and noCC, but the ratio for noCC is higher for *se* ( $\approx 10\%$ ). Notably, the dative reflexive clitic exhibits the second-most common absolute frequency ( $n = 48$ ).

Table 10.54: Frequencies of clitic climbing and no climbing in Spisz Goralish by clitic.

| Locality          | <i>cię</i> |                   | <i>go</i> |                   | <i>mi</i> |                   | <i>ci</i> |                   | <i>mu</i> |                   | <i>się</i> |                   | <i>se</i> |                      |
|-------------------|------------|-------------------|-----------|-------------------|-----------|-------------------|-----------|-------------------|-----------|-------------------|------------|-------------------|-----------|----------------------|
|                   | Abs.       | Rel. w/i<br>cmplx | Abs.      | Rel. w/i<br>cmplx | Abs.      | Rel. w/i<br>cmplx | Abs.      | Rel. w/i<br>cmplx | Abs.      | Rel. w/i<br>cmplx | Abs.       | Rel. w/i<br>cmplx | Abs.      | Rel.<br>w/i<br>cmplx |
| CC                | 21         | 100%              | 44        | 89.8%             | 18        | 100%              | 31        | 100%              | 28        | 100%              | 28         | 93.3%             | 43        | 89.6%                |
| noCC              | —          | —                 | 4         | 8.2%              | —         | —                 | —         | —                 | —         | —                 | 2          | 6.7%              | 5         | 10.4%                |
| CC?               | —          | —                 | 1         | 2.0%              | —         | —                 | —         | —                 | —         | —                 | —          | —                 | —         | —                    |
| Cmplx.<br>clauses | 21         | 100%              | 49        | 100%              | 18        | 100%              | 31        | 100%              | 28        | 100%              | 30         | 100%              | 48        | 100%                 |

The observed frequencies for target positions under CC are reported in Table 10.55. Attested target positions range from 1P to 6P, but there are significant differences in frequency. Most common is the Wackernagel position (60%) followed by Clitic Third (27%). There is a considerable number of clitics climbing to fourth position (8%). Clitic occurrences in the fifth, sixth, and first position become increasingly rare under climbing (in sum, less than 5%). Pronominal and reflexive clitics pattern quite differently. Two-thirds of climbed pronominal clitics occur in 2P, while one-third occurs in DP. Clitics in 3P constitute the biggest portion making a good quarter of occurrences. Reflexive clitics exhibit greater variation in quantitative terms. Roughly speaking, half of the climbed clitics take 2P, while the other half is delayed. More precisely, 2P amounts to 48%, while DP exceeds 50%. Within DP, several positions are attested with noteworthy ratios (3P: ca. 30%, 4P: ca. 11%, 5P: ca. 9%).

Table 10.55: Frequencies of target positions under clitic climbing in Spisz Goralish.

| Position<br>after CC | Total      |                      | Pronominal |                      | Reflexive  |                      |
|----------------------|------------|----------------------|------------|----------------------|------------|----------------------|
|                      | Abs. freq. | Rel. freq.<br>w/i CC | Abs. freq. | Rel. freq.<br>w/i CC | Abs. freq. | Rel. freq.<br>w/i CC |
| 1P                   | 1          | 0.5%                 | —          | —                    | 1          | 1.4%                 |
| 2P                   | 128        | 60.1%                | 94         | 66.2%                | 34         | 47.9%                |
| 3P                   | 58         | 27.2%                | 37         | 26.1%                | 21         | 29.6%                |
| 4P                   | 17         | 8.0%                 | 9          | 6.3%                 | 8          | 11.3%                |
| 5P                   | 7          | 3.3%                 | 1          | 0.7%                 | 6          | 8.5%                 |
| 6P                   | 2          | 0.9%                 | 1          | 0.7%                 | 1          | 1.4%                 |
| DP                   | 84         | 39.4%                | 48         | 33.8%                | 36         | 50.7%                |

(10.39) a. <[...] ale<sup>0</sup> się<sup>1</sup> mogą<sub>1</sub> kłócić<sub>2</sub> [...]>

but REFL.ACC can.PRS.3PL argue.INF

‘... but they can argue ...’

b. <([...] że ja nie wiem) co ci<sup>2</sup> mam<sub>1</sub> powiedzieć<sub>1</sub>

that I.NOM NEG know.PRS.1SG what you.DAT have.PRS.1SG tell.INF

[...]>

‘(... that I don’t know) what I should tell you ...’

c. <no potem<sup>1</sup> zaczęło<sup>2</sup> mi<sup>3</sup> się<sup>3</sup> to nie podobać<sub>2</sub>

well afterwards start.PST.SG.N me.DAT REFL.ACC this.NOM.SG.N NEG appeal.INF

[...]>

‘well, later I started to not like it ...’



- d. <[...] bo<sup>1</sup> trza<sup>2</sup> było<sup>3</sup> go<sup>4</sup> czyścić<sub>2</sub> no nie?>  
because need be.PST.SG.N him.ACC clean.INF well NEG  
‘... because it was necessary to clean it [= house], right?’
- e. <no to<sup>1</sup> [NP wszystkie słowa]<sup>2</sup> właśnie<sup>3</sup> musiała<sup>4</sup> m<sup>5</sup>  
well TO all.ACC.NVIR word.ACC.PL actually must.PST.SG.F AUX.1SG  
sobie/se<sup>5</sup> przeinterpretować<sub>2</sub>>  
REFL.DAT translate.INF  
‘well, so I actually needed to translate all words for myself’
- f. <bo<sup>1</sup> przy czemu<sup>2</sup> tam<sup>3</sup> [PP przy tej ławie]<sup>4</sup> będziemy<sup>5</sup>  
because [...] why there by this.LOC.SG.F coffee\_table.LOC.SG be.FUT.1PL  
się<sup>6</sup> gnieździć<sub>2</sub>>  
REFL.ACC gather.INF  
‘because why shall we gather there around the coffee table?’ (Gor; KoSpi)

Scrutiny of CC with distinguished clitics yields no consistent pattern, cf. Table 10.56. Only 2P and 3P are attested across all clitics. The initial position (1P) is absent from all clitics except reflexive *się*, which provides the only attestation of a clause-initial clitic under CC. The distribution frequencies in DP are quite diverse across clitics.<sup>553</sup> The reflexive clitic *se* shows the lowest ratio for 2P (≈ 40%) followed by the pronominal clitic *mi* (≈ 56%). The highest ratios are found with *mu* (75%) and *ci* (71%). The clitics *się*, *go*, and *cię* are in an intermediate range from 60% to 67%. Clitics vary as to how many different positions, and which ones are attested within DP.

Table 10.56: Frequencies of clausal positions targeted by clitic climbing in Spisz Goralish by clitic.

| Position<br>after CC | <i>cię</i>    |                            | <i>go</i>     |                            | <i>mi</i>     |                            | <i>ci</i>     |                            | <i>mu</i>     |                            | <i>się</i>    |                            | <i>se</i>     |                            |
|----------------------|---------------|----------------------------|---------------|----------------------------|---------------|----------------------------|---------------|----------------------------|---------------|----------------------------|---------------|----------------------------|---------------|----------------------------|
|                      | Abs.<br>freq. | Rel.<br>freq.<br>w/i<br>CC | Abs.<br>freq. | Rel.<br>freq.<br>w/i<br>CC | Abs.<br>freq. | Rel.<br>freq.<br>w/i<br>CC | Abs.<br>freq. | Rel.<br>freq.<br>w/i<br>CC | Abs.<br>freq. | Rel.<br>freq.<br>w/i<br>CC | Abs.<br>freq. | Rel.<br>freq.<br>w/i<br>CC | Abs.<br>freq. | Rel.<br>freq.<br>w/i<br>CC |
| 1P                   | —             | —                          | —             | —                          | —             | —                          | —             | —                          | —             | —                          | 1             | 3.6%                       | —             | —                          |
| 2P                   | 14            | 66.7%                      | 27            | 61.4%                      | 10            | 55.6%                      | 22            | 71.0%                      | 21            | 75.0%                      | 17            | 60.7%                      | 17            | 39.5%                      |
| 3P                   | 4             | 19.0%                      | 12            | 27.3%                      | 7             | 38.9%                      | 8             | 25.8%                      | 6             | 21.4%                      | 5             | 17.9%                      | 16            | 37.2%                      |
| 4P                   | 2             | 9.5%                       | 5             | 11.4%                      | —             | —                          | 1             | 3.2%                       | 1             | 3.6%                       | 1             | 3.6%                       | 7             | 16.3%                      |
| 5P                   | 1             | 4.8%                       | —             | —                          | —             | —                          | —             | —                          | —             | —                          | 3             | 10.7%                      | 3             | 7.0%                       |
| 6P                   | —             | —                          | —             | —                          | 1             | 5.6%                       | —             | —                          | —             | —                          | 1             | 3.6%                       | —             | —                          |
| DP                   | 7             | 33.3%                      | 17            | 38.6%                      | 8             | 44.4%                      | 9             | 29.0%                      | 7             | 25.0%                      | 10            | 35.7%                      | 26            | 60.5%                      |

<sup>553</sup> Remarkably, there is no correspondence between absolute frequency and degree of variation on a purely phenomenological level. The clitics with the highest token frequencies, *go* ( $n = 49$ ) and *se* ( $n = 48$ ), occupy less positions than *się* ( $n = 30$ ). On the other hand, the dative clitic *mi* has the smallest token frequency ( $n = 18$ ) but is recorded with varying placement over 3 distinct positions like *go*. The accusative clitic *cię* has the second lowest tokens frequency ( $n = 21$ ), but occurs in four different clausal positions in combination with CC.

Table 10.57 reports on clitic climbing with the simultaneous contiguity to verbs. The vast majority of 85% exhibits verb-adjacency when climbed. Half of all climbed clitics are placed next to one verb, while one-third of occurrences is surrounded by two verbs. The low amount of non-verb-adjacent clitics climbs to 2P in most cases, which comes to circa 11% of all CC tokens. Despite some minor variation, pronominal and reflexive clitics obey the same pattern with respect to verb-adjacency ( $\chi^2 = 0.48$ ,  $p = .05$ ,  $df = 1$ ). The same holds true for the frequencies of mono-verbal and bi-verbal configurations across clitic categories ( $\chi^2 = 0.59$ ,  $p = .05$ ,  $df = 1$ ).

Table 10.57: Frequencies of simultaneous occurrences of clitic climbing and (non-)verb-adjacency in Spisz Goralish.

| Placement<br>after CC | Total      |                      | Pronominal |                      | Reflexive  |                      |
|-----------------------|------------|----------------------|------------|----------------------|------------|----------------------|
|                       | Abs. freq. | Rel. freq.<br>w/i CC | Abs. freq. | Rel. freq.<br>w/i CC | Abs. freq. | Rel. freq.<br>w/i CC |
| noVA                  | 31         | 14.6%                | 19         | 13.4%                | 12         | 16.9%                |
| 1VA                   | 112        | 52.6%                | 78         | 54.9%                | 34         | 47.9%                |
| 2VA                   | 70         | 32.9%                | 45         | 31.7%                | 25         | 35.2%                |

- (10.40) a. <jak    **go**<sub>2</sub>        [PART ino ] [NP baki ]                    **zaczęły**<sub>1</sub>                    [noVA]  
               when him.ACC            only        horsefly.NOM.PL start.PST.PL.NVIR  
               **gryźć**<sub>2</sub>    [...]>  
               bite.INF  
               ‘When horseflies just started to sting him [= horse] ...’
- b. <Boże        **nie**    **mógł**<sub>1</sub>                **mu**<sub>2</sub>        [NP jakiś        zabaweczkę ]                    [1VA]  
               God.VOC NEG can.PST.SG.M him.DAT            some.ACC toy.DIM.ACC.SG  
               **kupić**<sub>2</sub>?>  
               buy.INF  
               ‘[dear] God, couldn’t have he bought a toy for him?’
- c. <(dziś    naganiamy    do    nauki)                    a    **nie**    **chcą**<sub>1</sub>                **się**<sub>2</sub>                    [2VA]  
               today chase.PRS.1PL to    learning.GEN.SG and NEG want.PRS.3PL REFL.ACC  
               **uczyć**<sub>2</sub>    [...]>  
               teach.INF  
               ‘(Today we chase [the children] to learn), but they don’t want to learn ...’ (Gor; KoSpi)

Individual clitics differ more radically. The lowest frequencies for VA placement come with *go* (77%) and *się* (79%), whereas the highest frequencies are found with *cię* (100%). The second-most common

and third-most common occurrences of VA are attested for *mi* (94%) and *mu* (93%) respectively. We find *ci* (80%) and *se* (86%) in the middle. The ratios between mono-verbal and bi-verbal placement are diverse, too. The overall tendency is that there is more 1VA than 2VA. However, the distinction between both placement types is the smallest for *se*, where no clear difference is visible. The same applies to *mi*. On the other hand, there is a pronounced distinction especially for *mu*, but also for the other clitics (*cię*, *go*, *ci*, *się*). As far as observations can be turned into meaningful generalisations, 2P placement is prevalent in noVA configurations. However, the picture is not as clear as the claim suggests, because, strictly speaking, it applies only to *go*, *się*, and *se*. The question is indecisive for the remainder of the clitics.

Table 10.58: Frequencies of simultaneous occurrences of clitic climbing and (non-)verb-adjacency in Spisz Goralish by clitic.

| Placement after CC | <i>cię</i> |                   | <i>go</i>  |                   | <i>mi</i>  |                   | <i>ci</i>  |                   | <i>mu</i>  |                   | <i>się</i> |                   | <i>se</i>  |                   |
|--------------------|------------|-------------------|------------|-------------------|------------|-------------------|------------|-------------------|------------|-------------------|------------|-------------------|------------|-------------------|
|                    | Abs. freq. | Rel. freq. w/i CC | Abs. freq. | Rel. freq. w/i CC | Abs. freq. | Rel. freq. w/i CC | Abs. freq. | Rel. freq. w/i CC | Abs. freq. | Rel. freq. w/i CC | Abs. freq. | Rel. freq. w/i CC | Abs. freq. | Rel. freq. w/i CC |
| noVA               | —          | —                 | 10         | 22.7%             | 1          | 5.6%              | 6          | 19.4%             | 2          | 7.1%              | 6          | 21.4%             | 6          | 14.0%             |
| 1VA                | 12         | 57.1%             | 21         | 47.7%             | 9          | 50.0%             | 17         | 54.8%             | 19         | 67.9%             | 15         | 53.6%             | 19         | 44.2%             |
| 2VA                | 9          | 42.9%             | 13         | 29.5%             | 8          | 44.4%             | 8          | 25.8%             | 7          | 25.0%             | 7          | 25.0%             | 18         | 41.9%             |

#### 10.4.4 Agreement drop

I came across 78 tokens of agreement drop in the Goralish data. It is only 3% of the entire Goralish data sample but amounts to 39.6% of the subset of 1st and 2nd person past tense and conditional constructions ( $n = 197$ ), which constitute the environment for licensing *l*-participles and auxiliary clitics (or mobile person markers).<sup>554</sup> Clitics occur in 2P to 5P under agreement drop. Notably, there is not a single occurrence of 1P. As can be seen in Table 10.59, Delayed Placement is the most frequent with a clear predominance of Clitic Third (3P), which occurs almost twice as often as 2P. This is in stark contrast to the frequency distributions reported in Section 10.4.1. The order among frequencies is summarised in (10.41).

<sup>554</sup> Since the annotation scheme does not cover grammatical features like tense, mood, person, or number, the frequencies can only be calculated by other means. Here, the total amount of 1st and 2nd person past tenses and conditionals is calculated upon the occurrences of AUX within a clitic cluster ( $n = 119$ ) + the amount of agreement drop tokens ( $n = 78$ ). Note that an expansion of the annotation would have needed another processing cycle for the relevant tokens but could not be maintained due to time restrictions. Admittedly, I did not anticipate the occurrence of agreement drop and its repercussions for cliticisation, such that the findings presented here came up rather unexpectedly. On the other hand, it corroborates the suitability of the exploratory approach taken in this study and shows that corpus research can provide new insights and relevant data contra Tajsner (2017: 601).

Table 10.59: Frequencies for attested clitic clausal positions under agreement drop in Spisz Goralish.

| Position | Absolute frequency | Relative frequency |
|----------|--------------------|--------------------|
| 1P       | —                  | —                  |
| 2P       | 24                 | 30.4%              |
| 3P       | 45                 | 57.0%              |
| 4P       | 9                  | 11.4%              |
| 5P       | 1                  | 1.3%               |
| Sum      | 79                 | 100%               |
| DP       | 55                 | 69.6%              |

(10.41) 3P > 2P > 4P > 5P

Diverse clitics co-occur with agreement drop: the short pronouns and reflexives *cię*, *go*, *ci*, *mu*, *się*, and *se* are attested. I provide examples for different clitics in agreement drop with both numbers covering diverse positions in (10.42).

- (10.42) a. <ja<sup>1</sup>    **cię**<sup>2</sup>        szanowała        **ø**>  
           I.NOM you.ACC esteem.PST.SG.F **ø**  
           ‘I esteemed you.’
- b. <ale<sup>0</sup> rok    [ w zeszłym        roku ]<sup>1</sup>        my<sup>2</sup>        **go**<sup>3</sup>        sprzedali        **ø**>  
           but year    in last.LOC.SG year.LOC.SG we.NOM him.ACC sell.PST.PL.VIR **ø**  
           ‘but last year we sold it [= horse]’
- c. <ja<sup>1</sup>        **go**<sup>2</sup>        potem        poprawiała        **ø**>  
           I.NOM him.ACC afterwards correct.PST.SG.F **ø**  
           ‘I corrected it later’
- d. <ale<sup>0</sup> rok                            [pp w zeszłym        roku ]<sup>1</sup>        my<sup>2</sup>        **go**<sup>3</sup>  
           but year.NOM/ACC            in last.LOC.SG.M year.LOC.SG we.NOM him.ACC  
           sprzedali        **ø**>  
           sell.PST.PL.VIR **ø**  
           ‘but ... last year, we sold it’
- e. <mamo        ale ja<sup>1</sup>        **ci**<sup>2</sup>        już        to                    gadała        **ø** kiedy?>  
           mom.VOC but I.NOM you.DAT already this.ACC.SG.N say.PST.SG.F **ø** once  
           ‘Mom, but did I tell you that once?’

- f. <to<sup>1</sup> ja<sup>2</sup> **mu**<sup>3</sup> musiała<sub>1</sub> **ø** z tym mlekiem robić<sub>2</sub>>  
 TO I.NOM him.DAT must.PST.SG.F **ø** with this.INS.SG.N milk.INS.SG do.INF  
 ‘so I had to prepare it for him with that milk’
- g. <(jak my mogły **ø**) to<sup>1</sup> my<sup>2</sup> **mu**<sup>3</sup> pomagały **ø**>  
 when we.NOM can.PST.PL.NVIR **ø** TO we.NOM him.DAT help.PST.PL.NVIR **ø**  
 ‘(when[ever] we could,) we helped him’
- h. <[...] i<sup>0</sup> my<sup>1</sup> **sobie/se**<sup>2</sup> kupiły **ø**>  
 and we.NOM REFL.DAT buy.PST.PL.NVIR **ø**  
 ‘... and we bought [them (= cigarettes)]’
- i. <potem<sup>1</sup> my<sup>2</sup> **sobie/se**<sup>3</sup> i szły **ø** i zarobić tam  
 afterwards we.NOM REFL.DAT and go.PST.PL.NVIR **ø** and earn.INF there  
 po parę groszy>  
 about pair.ACC.SG grosz.GEN.PL  
 ‘later we also went there to even make some (small) extra money’
- j. <[...] że<sup>1</sup> właściwie<sup>2</sup> ja<sup>3</sup> **się**<sup>4</sup> wychowała **ø** przy babce>  
 that actually I.NOM REFL.ACC raise.PST.SG.F **ø** at grandmother.LOC.SG  
 ‘... that I was raised by [my] grandmother actually.’
- k. <no to<sup>1</sup> [pp do jednego autobusu]<sup>2</sup> my<sup>3</sup> **się**<sup>4</sup> wpakowali **ø**>  
 well TO to one.GEN.SG.M bus.GEN.SG we.NOM REFL.ACC load.PST.PL.VIR **ø**  
 ‘Well, then we squeezed into one bus.’ (Gor; KoSpi)

The lack of *mi* with agreement drop rather trivially follows in the singular number and can be attributed to the feature composition of the respective elements. In case of a 1SG subject the co-referentiality of subject and objects requires a reflexive object in Slavic, which is generally known to be indifferent for person, number, and gender, i.e. Czech *sebe/se*, *sobě/si*, Polish *siebie/się*, *sobie/se*, Slovak *seba/sa*, *sebe/si* etc.<sup>555</sup> Therefore, the environment for singular agreement drop and the occurrence of short dative pronoun *mi* are mutually exclusive. But this reasoning does not apply to 1PL, because subject and object are not strictly co-referential. There are no obvious grammatical features or semantic conditions, which would prohibit the co-occurrence of SBJ.PRN.1SG (*my*) + OBJ.PRN.DAT.CL (*mi*). However, a cursory corpus query for the co-occurrence of a 1PL subject *my* and dative *mi* covering both written and spoken language yielded no results for Czech and Polish. Maybe

<sup>555</sup> In contrast, e.g., German reflexive constructions distinguish person and number in subject-oriented anaphoric pronouns. Furthermore, the 1st and 2nd person forms are homophones of true object pronouns (*mich*<sub>ACC</sub>/*mir*<sub>DAT</sub> ‘me’, *dich*<sub>ACC</sub>/*dir*<sub>DAT</sub> ‘you.SG’, *uns*<sub>ACC/DAT</sub> ‘us’ *euch*<sub>ACC/DAT</sub> ‘you.PL’, which does not apply to 3rd persons, which occur with the uninflected form *sich*<sub>ACC/DAT</sub> (vs. *ihn*<sub>ACC</sub>/*ihm*<sub>DAT</sub> ‘him’, *sie*<sub>ACC</sub>/*ihr*<sub>DAT</sub> ‘her’, *es*<sub>ACC</sub>/*ihm*<sub>DAT</sub> ‘it’).

partial coreference could be a sufficient co-occurrence restriction.<sup>556</sup> Alternatively, one might suppose pragmatic conditions or mere chance. Another possibility could be the phonological resemblance between both pronominal forms (Pol. *my* [mi] vs. *mi* [mi]), which are even homophones in Czech (*my* [mi] vs. *mi* [mi]).<sup>557</sup> This issue requires further study.

It is predominantly single clitic occurrences that occur under agreement drop, but there are four attestations of clitic clusters, cf. (10.43). Bear in mind that the slot for auxiliary clitics is not filled, such that single clitics as well as clusters are evaluated against hypothetically larger clusters.

- (10.43) a. <(ja jej gadała ø) ja ci ją poszczypkiwała ø>  
 I.NOM her.DAT say.PST.SG.F ø I.NOM you.DAT.ETH her.ACC pinch.IPFV.PST.SG.F ø  
 ‘(I told her) I was just pinching her.’
- b. <(jakby mi ktoś tak zrobił) to ja by go  
 if me.DAT someone.NOM so make.PST.SG.M TO I.NOM COND him.ACC  
 tam znaszedł ø w nocy nawet>  
 there find.PST.SG.M ø in night.LOC.SG even  
 ‘(if someone would have done that to me) then I would just haunt him even at night’
- c. <ja tak [s jak=by ø się urodził dziś ] (nic  
 I.NOM so how=COND ø REFL.ACC be\_born.PST.SG.M today nothing.NOM  
 mnie nie boli)>  
 me.DAT NEG hurt.PST.SG.M  
 ‘I [feel] so as if [I] would have been born [only] today (I don’t have any aches)’

<sup>556</sup> The only occurrence of *my* > *mi* I could detect in the ČNK (SYN\_v11) is a present tense syntagm from the novel *Drahý příteli*, cf. (i). (*Drahý příteli* is a translation from French: *Et je danse, aussi*, by Anne-Laure Bondoux and Jean-Claude Mourlevat.) The utterance is pragmatically marked. The subject pronoun *my* ‘we’ is topicalised, and the clitic occurs in third position (3P). Furthermore, the pronoun’s semantics are peculiar, because it does not just refer to a plural subject, but particularly to a loving couple.

(i) Protože [ *my* ]<sub>TOP</sub> *mi* chybíme taky, Pierre-Marie!  
 because we.NOM me.DAT lack.PRS.1PL also, Pierre-Marie.VOC  
 ‘Because I miss us, too, Pierre-Marie!’

(Cz; ČNK syn\_v11)

<sup>557</sup> The probability of phonological conditions to play a role seems to be rather low, however. No such phenomena have been reported for Polish. On the other hand, sequences of identical syllables are generally licit in (non-standard) Czech, e.g. AUX.2SG + REFL.DAT *jsi si* [si si] or REFL.ACC + syllabic comitative preposition *se se* [se se].

- d. <(on chciał mnie zwalić) ja [s kiedy się  
 he.NOM want.PST.SG.M me.ACC overthrow.INF I.NOM when REFL.ACC  
 zwrótnęła  $\emptyset$ ] [s kiedy go złapała  $\emptyset$ ] [s pod ... tutaj pod  
 turn.PST.SG.F  $\emptyset$  when him.ACC catch.PST.SG.F  $\emptyset$  under ... here under  
 boki ścisnęła  $\emptyset$  se go] [...]>  
 flank.ACC.PL squeeze.PST.SG.F  $\emptyset$  REFL.DAT him.ACC  
 ‘(he wanted to overthrow me) when I turned [around], when I caught him, [then] under  
 ... I put him here in the headlock ...’ (Gor; KoSpi)

Note that examples (10.43c–d) show the left-dislocation of the subject from a clausal domain. The extracted subject has scope over three clauses in (10.43.d). This is closely linked to information structure, as the extracted subject is doubtlessly discourse-prominent and topicalised, which is also corroborated by the data in (10.44) below.

- (10.44) a. <Marynko [ my ]<sub>TOP</sub> [s jak cię chrzcili  $\emptyset$ ] (to my  
 Maryna.DIM.LOC we.NOM how you.ACC baptise.PST.PL.VIR  $\emptyset$  TO we.NOM  
 ci jacy na pierzynkę zarzucili  $\emptyset$  moją  
 you.DAT only on duvet.DIM.ACC.SG throw.PST.PL.VIR  $\emptyset$  my.ACC.SG.F  
 świętą białą sztofkę)>  
 festive.ACC.SG.F white.ACC.SG.F kerchief.ACC.SG  
 ‘Maryna, when we were baptising you, (we only put my festive white kerchief on you  
 duvet)’  
 b. <[ ja ]<sub>TOP</sub> [s jak chciała<sub>1</sub>  $\emptyset$  go<sub>2</sub> pocieszyć<sub>2</sub> coś ]>  
 I.NOM how want.PST.SG.F  $\emptyset$  him.ACC cheer\_up.INF something  
 ‘when I wanted to cheer him up a little’ (Gor; KoSpi)

It is remarkable that the data almost consistently shows the order  $\text{SBJ}_{\text{PRN}} < \text{CL} < \text{V}$ . In fact, there is no reverse ordering of overt subject pronoun and clitic, or subject pronoun and finite verb attested. This could be indicative of quite a fixed linearisation pattern, which is rather unexpected for constituent order in Slavic. While it is not clear to me at this moment, what the exact conditioning factors for agreement drop are, the examples in (10.43c–d) and (10.44) show that information structure is involved and needs to be considered for a full picture. As the data show, the notorious precedence of the subject pronoun has a severe impact on clitic positioning, for it raises the likelihood of DP and eliminates the possibility of 1P.

## 10.5 Slovak

### 10.5.1 Clitic placement

Slovak provides the first of three supplemental data sets, which are characterised by a comparatively small size. This is why one needs to be cautious when drawing conclusions for the Slovak data. My goal was to annotate at least 100 tokens per clitic type to reach a reasonable amount of data. An interesting feature of Slovak is the set of constant clitics, which is unique in West Slavic insofar as clitics are evenly distributed across case (accusative/genitive vs. dative) and person (1SG vs. 2SG vs. 3SG). That is, for each singular person, there is one morphologically distinct clitic per case. Consequently, there are six pronominal clitics and two reflexive ones yielding a lower boundary of  $8 \times 100 = 800$  annotated data points. Effectively, 810 tokens were analysed with three-quarters of pronominal ones and one-quarter of reflexive ones.

The aggregated frequency distributions in Table 10.60 provide that Wackernagel positioning occurs in over three-quarters of cases. Clitic Third is the second-most common distribution pattern for clitics in Slovak with circa 14% and makes the majority of Delayed Placement, which amounts to circa 18%. Clitic First represents a good 5% of occurrences, Clitic Fourth is already rare with 3%. Despite decreasing frequency with successive distance from the left periphery, clitics in fifth and sixth position are attested in Slovak. There is also an instance of a clitic in seventh position. The preference pattern among clitic positions is given in (10.46). The overall distribution is equally represented among pronominal and reflexive clitics ( $\chi^2 = 0.16, p = .05, df = 2$ ).

The Slovak data exhibits a clear difference between Wackernagel placement, which amounts to three-quarters of occurrences (= 76.6%), and contact with the predicate, which occurs in approximately two-thirds of the cases (= 68.0%). The intersecting set of 2P and PA represents half of the Slovak data set (= 51.9%). Left lateral placement represents more than four-fifths of cases (= 82.2%). As a tendency, Delayed Placement occurs preferably before the predicate, it is most probable when the clitic is adjacent to the predicate. Delayed Placement may occur independently of contact, even though it is rather rare, but post-predicate Delayed Placement is even rarer.



Table 10.60: Frequencies of clitic occurrences in spoken Slovak by position, laterality, and distance.

| Pos | Left distant |       | Left adjacent |       | Right (adjacent) |       | Sum |       |
|-----|--------------|-------|---------------|-------|------------------|-------|-----|-------|
| 1P  | 15           | 1.9%  | 29            | 3.6%  | N/A              | N/A   | 44  | 5.5%  |
| 2P  | 199          | 24.7% | 305           | 37.8% | 114              | 14.1% | 618 | 76.6% |
| 3P  | 39           | 4.8%  | 54            | 6.7%  | 19               | 2.4%  | 112 | 13.9% |
| 4P  | 5            | 0.6%  | 12            | 1.5%  | 8                | 1.0%  | 25  | 3.1%  |
| 5P  | —            | —     | 2             | 0.2%  | 2                | 0.2%  | 4   | 0.5%  |
| 6P  | —            | —     | 2             | 0.2%  | 1                | 0.1%  | 3   | 0.4%  |
| 7P  | —            | —     | 1             | 0.1%  | —                | —     | 1   | 0.1%  |
| Sum | 258          | 32.0% | 405           | 50.2% | 144              | 17.8% | 807 | 100%  |
| DP  | 44           | 5.5%  | 71            | 8.8%  | 30               | 3.7%  | 145 | 18.0% |

The figures for lateral placement in Table 10.61 demonstrate the dominance of left-side positioning with more than 80%. Right lateral clitic placement is predominantly represented by Wackernagel cliticisation and may be traced back to verb-initial positioning. Clitic postposition with Delayed Placement is statistically marginal. The frequencies of pronominal and reflexive clitics differ significantly ( $\chi^2 = 6.49$ ,  $p = .05$ ,  $df = 1$ ). While pronominal clitics still precede their predicate in 80% of occurrences, the frequency almost reaches 90% (exactly 88%) in case of reflexive clitics.

Table 10.61: Frequencies of clitic occurrences in spoken Slovak by position, laterality, and distance per clitic category.

| Pos | Pronominal |       |     |       |      |       |     |       | Reflexive |       |     |       |      |       |     |       |
|-----|------------|-------|-----|-------|------|-------|-----|-------|-----------|-------|-----|-------|------|-------|-----|-------|
|     | Ld         |       | La  |       | R(a) |       | Sum |       | Ld        |       | La  |       | R(a) |       | Sum |       |
| 1P  | 12         | 2,0%  | 20  | 3,3%  | N/A  | N/A   | 32  | 5,3%  | 3         | 1,5%  | 9   | 4,5%  | N/A  | N/A   | 12  | 5,9%  |
| 2P  | 140        | 23,1% | 230 | 38,0% | 95   | 15,7% | 465 | 76,9% | 59        | 29,2% | 75  | 37,1% | 19   | 9,4%  | 153 | 75,7% |
| 3P  | 27         | 4,5%  | 43  | 7,1%  | 15   | 2,5%  | 85  | 14,0% | 12        | 5,9%  | 11  | 5,4%  | 4    | 2,0%  | 27  | 13,4% |
| 4P  | 3          | 0,5%  | 6   | 1,0%  | 8    | 1,3%  | 17  | 2,8%  | 2         | 1,0%  | 6   | 3,0%  | —    | —     | 8   | 4,0%  |
| 5P  | —          | —     | 2   | 0,3%  | 1    | 0,2%  | 3   | 0,5%  | —         | —     | —   | —     | 1    | 0,5%  | 1   | 0,5%  |
| 6P  | —          | —     | 1   | 0,2%  | 1    | 0,2%  | 2   | 0,3%  | —         | —     | 1   | 0,5%  | —    | —     | 1   | 0,5%  |
| 7P  | —          | —     | 1   | 0,2%  | —    | —     | 1   | 0,2%  | —         | —     | —   | —     | —    | —     | —   | —     |
| Sum | 182        | 30,1% | 303 | 50,1% | 120  | 19,8% | 605 | 100%  | 76        | 37,6% | 102 | 50,5% | 24   | 11,9% | 202 | 100%  |
| DP  | 30         | 5,0%  | 53  | 8,8%  | 25   | 4,1%  | 108 | 17,9% | 14        | 6,9%  | 18  | 8,9%  | 5    | 2,5%  | 37  | 18,3% |

The frequencies of Clitic First after clause boundaries and coordinating conjunctions are summarised in Table 10.62. Clitics occur most frequently after clause boundaries in 1P. Simplifying quantitative relations, post-conjunctive and post-boundary Clitic First are more or less evenly distributed (ratio of 5 : 6).

Table 10.62: Frequencies for Clitic First in spoken Slovak after clause boundary and coordinating conjunction (in total).

| Type        | Abs. freq. | Rel. freq. w/i 1P |
|-------------|------------|-------------------|
| # 1P ...    | 24         | 54.5%             |
| CONJ 1P ... | 20         | 45.5%             |

- (10.45) a. #**Mi**<sup>1</sup>      niečo              povedz      trochu o      tom              [...]  
                  me.DAT something.ACC tell.IMP.2SG a\_bit about this.LOC.SG.N  
                  ‘Tell me a bit about it ...’
- b. ([...] čis    či              to                      zjest’              hneď po      obede)  
                  ... whether this.ACC.SG.N eat\_up.INF right after lunch.LOC.SG  
                  alebo<sup>0</sup> **si**<sup>1</sup>              to                      nejak              nechať      [...]  
                  or              REFL.DAT this.ACC.SG.N somehow leave.INF  
                  ‘(... wheth... whether to eat it up right after lunch) or to save it somehow [for later] ...’
- c. tak<sup>1</sup> **by**<sub>1</sub><sup>2</sup>    **som**<sub>1</sub><sup>2</sup>      **mu**<sub>2</sub><sup>2</sup>              chcel<sub>1</sub>              za t      za to  
                  so      COND AUX.1SG him.DAT want.PST.SG.M for ... for this.ACC.SG.N  
                  gesto                      poďakovať<sub>2</sub>  
                  gesture.ACC.SG thank.INF  
                  ‘I would like to thank him for th... for this gesture so [much]’
- d. Hneď na začiatok                      **by**<sub>1</sub><sup>3</sup>    **som**<sub>1</sub><sup>3</sup>      **sa**<sub>2</sub><sup>3</sup>    **ť**<sub>a</sub><sup>3</sup>              chcela<sub>1</sub>              teda  
                  right on beginning.ACC.SG COND AUX.1SG REFL you.ACC want.PST.SG.F thus  
                  spýtať<sub>2</sub>.  
                  ask.INF  
                  ‘Right at the outset I would thus like to ask you.’
- e. (keď    keď    si              ich              tak spíšem)                      a<sup>0</sup>    budem<sup>1</sup>      vedieť<sup>2</sup>  
                  when when REFL.DAT them.ACC so write\_down.PRS.1SG and be.FUT.1SG know.INF  
                  dopredu<sup>3</sup> **ti**<sup>4</sup>              poslať      [...]  
                  forward you.DAT send.INF  
                  ‘(when I will write them down for myself) and be able to send it to you in advance ...’
- f. Tak<sup>1</sup> **som**<sup>2</sup>      potom<sup>3</sup>      odvážil<sup>4</sup>      **sa**<sup>5</sup>      (a hľadal              som  
                  so      AUX.1SG afterwards dare.PST.SG.M REFL.ACC and search.PST.SG.M AUX.1SG  
                  pomoc      u      spolužiakov,      známym.)  
                  help.ACC.SG by classmate.GEN.PL acquaintance.GEN.PL  
                  ‘So I later gathered [my] courage and sought help from classmates, acquaintances.’

- g. (povedali,) že<sup>1</sup> oni<sup>2</sup> mali<sup>3</sup> príkaz<sup>4</sup> [AdvP už od  
tell.PST.PL.VIR that they.VIR have.PST.PL order.ACC.SG already from  
prvého januára ]<sup>5</sup> ma<sup>6</sup> sledovať<sup>2</sup>.  
first.GEN.SG January.GEN.SG me.ACC track.INF  
'(They said) that they were ordered to track me already from January first on.'
- h. [...] ktorý<sup>1</sup> potom<sup>2</sup> sme<sup>3</sup> mali<sup>4</sup> teda<sup>5</sup> záujem<sup>6</sup> ho<sup>2</sup><sup>7</sup>  
who.ACC.SG afterwards AUX.1PL have.PST.PL thus interest.ACC.SG him.ACC  
zobrať<sup>2</sup> [...] <sup>558</sup>  
take.INF  
'... whom we were later interested in taking ...' (Slk; SNK s-hovor.7.0)

Table 10.63: Measures of central tendency for clitic positioning in spoken Slovak.

| Measure:          | $\bar{x}$ (SD) | <i>Md</i> | <i>Mo</i> |
|-------------------|----------------|-----------|-----------|
| Value (position): | 2.18 (±0.65)   | 2         | 2         |

(10.46) 2P > 3P > 1P > 4P > 5P > 6P (> 7P)

The frequency distributions for individual clitics in Table 10.64 show variation. The pronominal clitics *ma*, *ho* and *mu* as well as the reflexive clitics *sa* and *si* follow the general pattern. Reflexive *sa* shows a slight increase of 4P, while *si* exhibits increased 1P. Pronominal *ho* exhibits less 3P, but more 4P. Different patterns may be observed for *t'a* and *ti* with over 80% of 2P (81% and 85% respectively) and for *mi* with 2P decreased to 68% and higher ratio for 3P. Overall, Delayed Placement is attested in a wide range of ratios from 11% (*ti*) to 23% (*mi*). Individual clitics have not been tested for homogeneity. The low absolute frequencies of 1P ( $n < 5$ ) for one half of the types render the chi-square test inapplicable. The alternative Fisher exact test has not been employed for the practical reason that Slovak data is used as a supplement, such that the level of descriptivism presented here is sufficient for the purposes of the present thesis.

<sup>558</sup> The positional interpretation of the object clitic *ho* as occupying either 7P of the entire clause or 2P of the NP headed by *záujem* depends on the ability of the clitic to escape the NP. This is perfectly possible. The reflexive clitic *sa* in (i) is a dependent of the embedded infinitive *chrániť* '(to) protect' but appears beyond the NP's boundaries. The clitic is a true reflexive argument here. Note that the accusative clitic *ho* in (10.45h) doubles the argument role of the infinitive's (*zobrať*) direct object, which is already realised by the relative pronoun (*ktorý*).

(i) [...] že ako majiteľ sa<sup>2</sup> mám<sup>1</sup> [NP záujem [S chrániť<sup>2</sup> takým  
that as owner.NOM.SG REFL.ACC have.PRS.1SG interest.ACC.SG protect.INF such.INS.SG.M  
spôsobom]], (aby som v konečnom dôsledku ne-bol  
manner.INS.SG such\_that AUX.1SG in final.LOC.SG.M consequence.LOC.SG NEG-be.PRS.1SG  
poškodený.)  
damaged.NOM.SG

'... that I have an interest in protecting myself in such a way (in order not to be the victim in the final analysis.)'  
(Slk; SNK s-hovor.7.0)

Table 10.64: Frequencies of attested clitic clausal positions in spoken Slovak by clitic.

| Pos | <i>ma</i> |       | <i>t'a</i> |       | <i>ho</i> |       | <i>mi</i> |       | <i>ti</i> |       | <i>mu</i> |       | <i>sa</i> |       | <i>si</i> |       |
|-----|-----------|-------|------------|-------|-----------|-------|-----------|-------|-----------|-------|-----------|-------|-----------|-------|-----------|-------|
|     | Abs.      | Rel.  | Abs.       | Rel.  | Abs.      | Rel.  | Abs.      | Rel.  | Abs.      | Rel.  | Abs.      | Rel.  | Abs.      | Rel.  | Abs.      | Rel.  |
| 1P  | 4         | 4.0%  | 4          | 3.9%  | 4         | 4.0%  | 10        | 9.8%  | 4         | 4.0%  | 6         | 5.9%  | 5         | 5.0%  | 7         | 6.9%  |
| 2P  | 76        | 75.2% | 83         | 81.4% | 78        | 77.2% | 69        | 67.6% | 86        | 85.1% | 76        | 75.2% | 75        | 74.3% | 78        | 77.2% |
| 3P  | 17        | 16.8% | 14         | 13.7% | 9         | 8.9%  | 21        | 20.6% | 10        | 9.9%  | 14        | 13.9% | 12        | 11.9% | 15        | 14.9% |
| 4P  | 3         | 3.0%  | 1          | 1.0%  | 7         | 6.9%  | 2         | 2.0%  | 1         | 1.0%  | 3         | 3.0%  | 7         | 6.9%  | 1         | 1.0%  |
| 5P  | —         | —     | —          | —     | 2         | 2.0%  | —         | —     | —         | —     | 1         | 1.0%  | 1         | 1.0%  | —         | —     |
| 6P  | 1         | 1.0%  | —          | —     | —         | —     | —         | —     | —         | —     | 1         | 1.0%  | 1         | 1.0%  | —         | —     |
| 7P  | —         | —     | —          | —     | 1         | 1.0%  | —         | —     | —         | —     | —         | —     | —         | —     | —         | —     |
| Sum | 101       | 100%  | 102        | 100%  | 101       | 100%  | 102       | 100%  | 101       | 100%  | 101       | 100%  | 101       | 100%  | 101       | 100%  |
| DP  | 21        | 20.8% | 15         | 14.7% | 18        | 17.8% | 23        | 22.5% | 11        | 10.9% | 19        | 18.8% | 21        | 20.8% | 16        | 15.8% |

Taking laterality into consideration in Table 10.65, one notes that left placement amounts to roughly 80% for *ho*, *mi*, *ti*, *mu* and reflexive *si*. The frequencies increase for pronominal *ma* and reflexive *sa* with 89% and 94% respectively. On the other side, pronominal *t'a* shows less left lateral placement with a ratio of merely 70%. Even without a statistical test it is obvious that a gap of 20% and more in distribution is huge, such that a common pattern for all one cannot be assumed.

Table 10.65: Frequencies for laterality with left distant (Ld), left adjacent (La), right adjacent (R(a)) in spoken Slovak by clitic.

| Lat  | <i>ma</i> |       | <i>t'a</i> |       | <i>ho</i> |       | <i>mi</i> |       | <i>ti</i> |       | <i>mu</i> |       | <i>sa</i> |       | <i>si</i> |       |
|------|-----------|-------|------------|-------|-----------|-------|-----------|-------|-----------|-------|-----------|-------|-----------|-------|-----------|-------|
|      | Abs       | Rel   | Abs        | Rel   | Abs       | Rel   | Abs       | Rel   | Abs       | Rel   | Abs       | Rel   | Abs       | Rel   | Abs       | Rel   |
| Ld   | 34        | 33.7% | 28         | 27.5% | 26        | 25.7% | 29        | 28.4% | 35        | 34.7% | 30        | 29.7% | 40        | 39.6% | 36        | 35.6% |
| La   | 56        | 55.4% | 44         | 43.1% | 54        | 53.5% | 50        | 49.0% | 49        | 48.5% | 50        | 49.5% | 55        | 54.5% | 47        | 46.5% |
| R(a) | 11        | 10.9% | 29         | 28.4% | 20        | 19.8% | 23        | 22.5% | 17        | 16.8% | 20        | 19.8% | 6         | 5.9%  | 18        | 17.8% |

(10.47) a. [...] tak **t'a** [ADV už] [NP žiadne kúzlo] [Ld]  
 so you.ACC already no.NOM.SG.N spell.NOM.SG

**ne-ohúri** [...]

NEG-stun.PRS.3SG

‘... then no spell will stun you ...’

b. [...] a že potom odtiaľ **sa** **dostal** až domov [...] [La]

and that afterwards thence REFL.ACC get.PST.SG.M to home.ACC.SG

‘... and that from there [= the woods] he got back home ...’

c. [PART Ešte ] [AdvP ako dieťa ] [PP v ľudovej škole ] [R(a)]  
 still as child.ACC.SG in folk.ADJ.LOC.SG.F school.LOC.SG  
**pripravoval ma na prvé sväté prijímanie.**  
 prepare.PST.SG.M me.ACC on first.ACC.SG.N holy.ACC.SG.N communion.ACC.SG  
 ‘[While I was] still a child in Volksschule [= primary school], he prepared me for the  
 First Communion.’ (Slk; SNK s-hovor.7.0)

In contrast to Czech, Polish, and Goralish, the spoken Slovak sample represents the first data collection, in which not a single occurrence of clitic lowering has been detected. Provided the low token frequencies of clitic lowering in the previous chapters, it is likely that this might be due to the small sample size of the Slovak portion. Recall that, on average, the Slovak data sample represents roughly 28% of the amount of data scrutinised in the three previous chapters.

## 10.5.2 Clitic clustering

We learn from Table 10.66 that approximately three-quarters of the elicited tokens present single clitic occurrences. Clitic clusters constitute one-quarter of occurrences, which are mostly clitic pairs. Sequences of three clitics are quite rare (3.5%), while more complex clusters have a vanishingly low frequency.

Table 10.66: Frequencies of single clitic and clitic cluster occurrences in spoken Slovak by number of elements.

| Number of clitics | Absolute frequency | Relative frequency |
|-------------------|--------------------|--------------------|
| 1                 | 601                | 74.2%              |
| 2                 | 179                | 22.1%              |
| 3                 | 28                 | 3.5%               |
| 4                 | 2                  | 0.2%               |
| Sum               | 810                | 100%               |
| $n > 1$           | 209                | 25.8%              |

(10.48) a. Nič **sa**<sub>2</sub> **mu**<sub>2</sub> ne-mohlo<sub>1</sub> stať<sub>2</sub>.  
 nothing REFL.ACC him.DAT NEG-can.PST.SG.N happen.INF  
 ‘He couldn’t have been harmed by anything.’

b. (Ne-viem,) čo by som ti k tomu ešte  
 NEG-know.PRS.1SG what COND AUX.1SG you.DAT towards this.DAT.SG.N still  
 povedala [...] say.PST.SG.F

‘(I don’t know), what else I could tell you about it ...’

c. Hned’ na začiatok by<sub>1</sub> som<sub>1</sub> sa<sub>2</sub> ťa<sub>2</sub> chcela<sub>1</sub> teda  
 right on beginning.ACC.SG COND AUX.1SG REFL you.ACC want.PST.SG.F thus  
 spýtať<sub>2</sub>. ask.INF

‘Right at the outset I would thus like to ask you.’

(Slk; SNK s-hovor.7.0)

Figure 10.5 shows the bar plot of the clusters’ frequency distributions. One can infer from the graph that clitic clusters almost strictly follow the linearisation pattern  $COND < AUX < REFL < PRN$ , which is attested twice in a quadruple, in fact, and is cited in (10.48c).

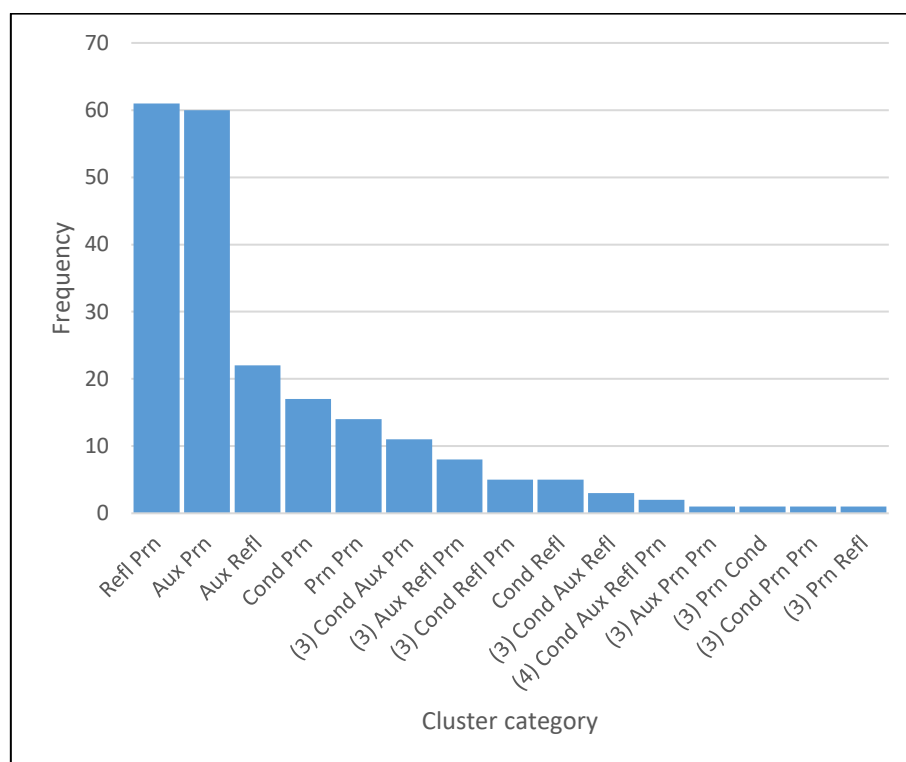


Figure 10.5: Frequency distribution of clitic cluster categories in spoken Slovak.

Linearisation among reflexive and pronominal clitics under consideration of the latter’s case value is presented in Table 10.67. There is nothing special about case sequences. Two occurrences of double datives are attested. There are two attested deviations. On the one hand, there is one occurrence of  $PRN < COND$ . On the other hand, we find one instance of  $PRN < REFL$ . Given the low frequencies, no

conclusions about variation in ordering can be drawn. The recorded case of a pronominal clitic preceding a reflexive one involves a dative pronoun, cf. (10.50a). For the conditional clitic preceded by a pronominal one, the respective case value is accusative, cf. (10.50b).

Table 10.67: Absolute frequencies of case-valued clitic clusters in spoken Slovak.

| Case sequence | Absolute frequency |
|---------------|--------------------|
| REFL < DAT    | 40                 |
| REFL < ACC    | 35                 |
| DAT < ACC     | 14                 |
| DAT < DAT     | 2                  |
| DAT < REFL    | 1                  |

Two instances of uncanonical clitic order were detected. The pronominal clitic *mu* precedes the reflexive clitic *sa* in (10.50a). Example (10.50b) shows pronominal *ho* ahead of conditional *by*. Example (10.50c) does not concern uncanonical order, but the co-occurrence of two dative clitics within a single cluster. The second person dative *ti* is an ethical one, whereas the first person dative *mi* is a genuine object clitic realising the verbal predicate's argument.

- (10.50) a. [...] v televízii **mu** **sa** zatvárali [...]   
in television him.DAT REFL.ACC grimace.PST.PL   
‘... at the television they screwed up their faces (towards him) ...’
- b. [...] tak čo **ho** **by** asi napadlo [...]   
so what him.ACC COND perhaps spring\_to\_mind.PST.SG.N   
‘... so what would have come to his mind perhaps ...’
- c. [...] A on **ti** **mi** zahlási (, Paľo a   
and he.NOM you.DAT.ETH me.DAT say.PRS.3SG Pavol.DIM.NOM and   
borovičku ne-máš?)   
borovička.ACC.SG NEG-have.PRS.2SG   
‘... And he just says to me[:] (Paľo, but don’t you have any borovičku [= a spirit from juniper berries]?)’ (Slk; SNK s-hovor-7.0)

Diaclisis was detected 11 times vis-à-vis the 209 clitic cluster tokens. Table 10.68 shows that multiple clitic occurrences add up to 220 attestations. Clustering amounts to 95% and diaclisis to 5%, which is rather rare.

Table 10.68: Frequencies of multiple clitics with clustering and diacsis in spoken Slovak.

| Multiple clitics | Absolute frequency | Relative frequency<br>w/i multiple CL |
|------------------|--------------------|---------------------------------------|
| Clustering       | 209                | 95.0%                                 |
| Diaclisis        | 11                 | 5.0%                                  |
| Sum              | 220                | 100%                                  |

At first sight, the diaclitic occurrences in Table 10.69 appear evenly distributed among simple and complex clause environments, but bear in mind that the absolute frequencies are extremely low. The picture changes when calculating the relative frequencies for the respective environments. Diaclisis has a ratio of 0.7% in simple clauses (5 out of 719), but a ratio of 6.6% in complex clauses (6 out of 91). That is, diaclisis has an over 9 times higher relative frequency in complex than in simple clauses.

Table 10.69: Frequencies of diaclisis in spoken Slovak by syntactic environment.

| Syntactic environment | Absolute frequency | Relative frequency w/i<br>diaclisis | Relative frequency w/i<br>syntactic environment |
|-----------------------|--------------------|-------------------------------------|-------------------------------------------------|
| Simple clause         | 5                  | 45.5%                               | 0.7%                                            |
| Complex clause        | 6                  | 54.5%                               | 6.6%                                            |
| Sum                   | 11                 | 100%                                | —                                               |

According to Table 10.70, most split types are complementarily distributed across environments, which possibly is an epiphenomenon of small sample size. Only the pattern (REFL || PRN) is recorded in both environments. In simple clauses, diaclisis occurs between clitics with different functional properties and licensing backgrounds. The appearance of the auxiliary clitics in (10.51) is induced by discourse properties, i.e. the temporal relation between a verbalised event and a reference point (prototypically, speaking time). There are, of course, formal dependencies between auxiliary clitic and *l*-participle, but the auxiliary is not semantically dependent on the verb in a strict sense. The reflexive dative *si* in (10.51a) is a facultative element. In contrast, the dative clitic *mu* is strictly dependent on the verb *dali* in (10.51b), as it is a true argument. Example (10.51c) provides a genuine example of diaclisis between co-dependent clitics in the strictest sense, whereby the lexical element *sa* of the inherently reflexive verb *bát'* '(to) fear' and the genitival object *t'a* are disjunct.



Table 10.70: Attested splits between clitic categories in spoken Slovak by syntactic environment.

| Simple clause | Complex clause |
|---------------|----------------|
| AUX    REFL   | REFL    PRN    |
| AUX    PRN    | PRN    REFL    |
| REFL    PRN   | REFL    REFL   |
|               | PRN    PRN     |

- (10.51) a. [...] ale **som** tiež **si** myslel, že [...] [AUX || REFL]  
 but AUX.1SG also REFL.DAT think.PST.SG.M that  
 ‘... but I also was thinking that ...’
- b. [...], čo **by** **sme** dali **mu**, [...] [AUX || PRN]  
 what COND AUX.1PL give.PST.PL him.DAT  
 ‘... what would we give to him ...’
- c. [...] lebo **sa** boja **ťa**. [REFL || PRN]  
 because REFL.ACC fear.PRS.3PL you.GEN  
 ‘... because they are afraid of you.’ (Slk; SNK s-hovor.7.0)

The complex environment exhibits some interesting types of disjunct clitics. The type (PRN || REFL) occurs twice and involves both times a matrix dative clitic, while the embedded reflexive clitic is accusative *sa* in (10.52a), but dative *si* in (10.52b). The realisation of the diacritic configuration can be attributed to the occurrence of the reflexive clitic *sa* in the matrix clauses, as we know that two reflexive clitics cannot occupy positions within a single clitic cluster. Therefore, we can associate the examples in (10.52) with a more general pattern, which is apparent in example (10.53)

- (10.52) a. Tam **sa** **mi** podarilo stretnúť **sa** so [...] [AUX || REFL]  
 there REFL.ACC me.DAT succeed.PST.SG.N meet.INF REFL.ACC with  
 ‘There I managed to meet with ...’
- b. Mne **sa** **ti** raz podarilo na jednej [...] [AUX || PRN]  
 me.DAT REFL.ACC you.DAT.ETH once succeed.PST.SG.N on one.LOC.SG.F  
 dovolenke v Grécku **si** zlomiť jednotku.  
 holiday.LOC.SG in Greece.LOC REFL.DAT break.INF central\_incisor.ACC.SG.  
 ‘I once succeeded in breaking my central incisor on one holiday in Greece, you know.’  
 (Slk; SNK s-hovor-7.0)

The two splitting types in (10.53) have been recorded in the same environments in Czech and Polish. Diaclisis between reflexives (REFL || REFL) in (10.53a) complies with the same well-known co-occurrence restrictions applying in (10.52), which otherwise result in the haplology of reflexives. In (10.53b–c) we encounter attestation for diaclisis between pronominals (PRN || PRN). Example (10.53b) shows the accusative clitic *ho* disjunct from the matrix clitic cluster *sa mi*, although the sequence *sa mi ho* is grammatical. The embedded clitic *mu* in (10.53c) does not occupy a position in the matrix clitic cluster, where another dative clitic (*mi*) is already present.

- (10.53) a. [...] a ona tak **sa** snažila k nám **sa** dostat'.  
 and she.NOM so REFL.ACC try.PST.SG.F towards us.DAT REFL.DAT get.INF  
 '... and she tried so [much] to get to us.'
- b. Stále **sa** **mi** podarilo **ho** niekde  
 always REFL.ACC me.DAT succeed.PST.SG.N him.ACC somewhere  
 vpašovať do nejakého zášvu.  
 smuggle\_in.INF to some.GEN.SG.M pleat.GEN.SG  
 'I always managed to smuggle it into some pleat.'
- c. Podarilo **sa** **mi** ten aparat **mu**  
 succeed.PST.SG.N REFL.ACC me.DAT this.ACC.SG.M camera.ACC.SG him.DAT  
 vytrhnúť [...] wrench.INF  
 'I managed to wrench the camera from him ...' (Slk; SNK s-hovor-7.0)

Occurrences of clitic clusters next to single clitic occurrences in first position have been found in Slovak, which, however, show comparatively less variation in terms of attested patterns because of decreased sample size: (Prn), (Aux, Refl), (Aux, Prn), (Refl, Prn), (Aux, Refl, Prn).

### 10.5.3 Clitic climbing

There are 91 complex clause tokens, which equals 11% of the entire Slovak data set. Clitic climbing is presented in Table 10.71. The construction occurs in the majority of cases with nearly 87%. The local realisation of clitics makes only a small subset. There are few attested cases without a clear classification as either CC or noCC.

Table 10.71: Frequencies of clitic climbing and no climbing in spoken Slovak.

| Locality           | Total      |                                         | Pronominal |                                         | Reflexive  |                                         |
|--------------------|------------|-----------------------------------------|------------|-----------------------------------------|------------|-----------------------------------------|
|                    | Abs. freq. | Rel. freq.<br>w/i<br>complex<br>clauses | Abs. freq. | Rel. freq.<br>w/i<br>complex<br>clauses | Abs. freq. | Rel. freq.<br>w/i<br>complex<br>clauses |
| CC                 | 79         | 86.8%                                   | 53         | 85.5%                                   | 26         | 89.7%                                   |
| noCC               | 7          | 7.7%                                    | 6          | 9.7%                                    | 1          | 3.4%                                    |
| CC?                | 5          | 5.5%                                    | 3          | 4.8%                                    | 2          | 6.9%                                    |
| Complex<br>clauses | 91         | 100%                                    | 62         | 100%                                    | 29         | 100%                                    |

Abstracting away from the low absolute frequencies, pronominal clitics exhibit a slightly lower occurrence rate for CC with 85.5% and an increased ratio for noCC with almost 10%. In contrast, reflexive clitics show circa 90% CC and a very low frequency for noCC with roughly 3%.

- (10.54) a. (To nie,) to **t'a**<sub>2</sub> ne-chcem<sub>1</sub> stresovať<sub>2</sub> s tým.  
 TO NEG TO you.ACC NEG-want.PRS.1SG stress.INF with this.INS.SG.N  
 ‘(Then don’t) then I don’t want to stress you with that.’
- b. [...] či **sa**<sub>1</sub> niekomu z bratov podarilo<sub>1</sub>  
 whether REFL.ACC someone.DAT from brother.GEN.PL succeed.PST.SG.N  
 naliat<sub>2</sub> **mu**<sub>2</sub> do kávy trochu brandy.  
 pour.INF him.DAT to coffee.GEN.SG a\_bit brandy.GEN.SG  
 ‘... whether someone of the brothers managed to pour a bit of brandy into his coffee.’
- c. Toto **som**<sub>1</sub> **si**<sub>2</sub> dovtedy ne-vedel<sub>1</sub> predstaviť<sub>2</sub> [...]  
 this.ACC.SG.N AUX.1SG REFL.DAT till\_then NEG-know.PST.SG.M imagine.INF  
 ‘I wasn’t able to imagine that until then ...’
- d. [...] a ona tak **sa**<sub>1</sub> snažila<sub>1</sub> k nám **sa**<sub>2</sub> dostať<sub>2</sub>.  
 and she.NOM so REFL.ACC try.PST.SG.F to us.DAT REFL.DAT get.INF  
 ‘... and she tried so [much] to get to us.’
- e. [...] že určite **ho**<sub>2</sub> nosiť<sub>2</sub> môžeš<sub>1</sub> [...]  
 that surely him.ACC wear.INF can.PRS.2SG  
 ‘... that you certainly can wear it ...’ (Slk; SNK s-hovor-7.0)

The CC frequencies for individual clitics are provided in Table 10.72 for the sake of completeness only.

Table 10.72: Frequencies of clitic climbing and no climbing in spoken Slovak by clitic.

| Locality         | <i>ma</i> |                        | <i>ʔa</i> |                        | <i>ho</i> |                        | <i>mi</i> |                        | <i>ti</i> |                        | <i>mu</i> |                        | <i>sa</i> |                        | <i>si</i> |                        |
|------------------|-----------|------------------------|-----------|------------------------|-----------|------------------------|-----------|------------------------|-----------|------------------------|-----------|------------------------|-----------|------------------------|-----------|------------------------|
|                  | Abs.      | Rel.<br>w/i<br>complex | Abs.      | Rel.<br>w/i<br>complex | Abs.      | Rel.<br>w/i<br>complex | Abs.      | Rel.<br>w/i<br>complex | Abs.      | Rel.<br>w/i<br>complex | Abs.      | Rel.<br>w/i<br>complex | Abs.      | Rel.<br>w/i<br>complex | Abs.      | Rel.<br>w/i<br>complex |
| CC               | 5         | 100%                   | 13        | 92.9%                  | 15        | 75.0%                  | 6         | 100%                   | 11        | 91.7%                  | 9         | 81.8%                  | 12        | 85.7%                  | 14        | 93.3%                  |
| noCC             | —         | —                      | —         | —                      | 3         | 15.0%                  | —         | —                      | 1         | 8.3%                   | 2         | 18.2%                  | 1         | 7.1%                   | —         | —                      |
| cc?              | —         | —                      | 1         | 7.1%                   | 2         | 10.0%                  | —         | —                      | —         | —                      | —         | —                      | 1         | 7.1%                   | 1         | 6.7%                   |
| Cmplx<br>clauses | 5         | 100%                   | 14        | 100%                   | 20        | 100%                   | 6         | 100%                   | 12        | 100%                   | 11        | 100%                   | 14        | 100%                   | 15        | 100%                   |

The scanty data gives rise to a slightly chaotic picture. For *ma* and *mi*, there are only CC construction attested, but note that these clitics come with the lowest token frequency. We encounter all three annotated locality patterns only with the clitics *ho* and *sa*, whereby *ho* has the highest token frequency. The four remaining types exhibit two of three locality patterns: *ti* and *mu* provide attestations for CC and noCC, while *ʔa* and *si* provide us with examples for CC and unclear locality (CC?). Obviously, the more configuration types are attested, the lower the ratio for CC is.

Table 10.73 shows target positions of clitic climbing with the predominance of second position placement amounting to over 80%. Clitic Third is the second-most common position with roughly 15%. Clitics in initial position are rare with an overall ratio of 2.5%. Accounting for clitic categories, pronominal clitics show a ratio of ca. 77% for Wackernagel positioning and an increased amount of Clitic Third with over 20%. On the other hand, reflexive clitics show over 90% of 2P, while both 3P and 1P are rare occurrences. We may again infer from what we know from Czech, Polish, and Goralish that the distribution patterns of pronominal and reflexive clitics differ significantly. Remarkably, no Delayed Placement beyond 3P is attested, but this could perhaps be due to the quantitative limitations of the data set.

Table 10.73: Frequencies of target positions under clitic climbing in spoken Slovak.

| Position<br>after CC | Total      |                      | Pronominal |                      | Reflexive  |                      |
|----------------------|------------|----------------------|------------|----------------------|------------|----------------------|
|                      | Abs. freq. | Rel. freq.<br>w/i CC | Abs. freq. | Rel. freq.<br>w/i CC | Abs. freq. | Rel. freq.<br>w/i CC |
| 1P                   | 2          | 2.5%                 | 1          | 1.9%                 | 1          | 3.8%                 |
| 2P                   | 65         | 82.3%                | 41         | 77.4%                | 24         | 92.3%                |
| 3P / DP              | 12         | 15.2%                | 11         | 20.8%                | 1          | 3.8%                 |

- (10.55) a.  $Aj^0$   $t'a_3^1$  mám<sub>1</sub> príst'<sub>2</sub> pozrieť<sub>3</sub> potom?  
 and you.ACC have.PRS.1SG come.INF watch.INF afterwards  
 ‘And shall I later come to look [in on] you?’
- b. [...]  $a^0$  že<sup>1</sup>  $sa_2^2$  s nami môžu<sub>1</sub> porozprávať<sub>2</sub> [...]   
 and that REFL.ACC with us.INS can.PRS.3PL talk.INF  
 ‘... and that we are good to talk with ...’
- c. [...] ved'<sup>1</sup> ja<sup>2</sup>  $ti_2^3$   $ich_2^3$  môžem<sub>1</sub> odniesť<sub>2</sub>.  
 indeed I.NOM you.DAT them.ACC can.PRS.1SG carry\_away.INF  
 ‘... indeed, I can carry them [= shoes] away for you,’ (Slk; SNK s-hovor-7.0)

A detailed look at individual clitics can only provide some rough impressions due to data limitations, cf. Table 10.74. Clitic climbing to second position varies across clitics. It is the only attested pattern for reflexive *sa*, while *si* is positioned more flexible irrespective of the low occurrence frequency for 1P and 3P with one attestation each. Turning to pronominal clitics, only *ta* exhibits 1P under CC providing one attestation, but it appears to be quite flexible with respect to positioning provided the comparatively high amount of 3P (30%). The other pronominal clitics most likely climb to 2P with at least 77% of occurrences (*mu*) up to 86% (*ho*).

Table 10.74: Frequencies of clausal positions targeted by clitic climbing in spoken Slovak by clitic.

| Pos<br>after<br>CC | <i>ma</i>   |                         | <i>ta</i>   |                       | <i>ho</i>   |                       | <i>mi</i>   |                       | <i>ti</i>   |                       | <i>mu</i>   |                       | <i>sa</i>   |                       | <i>si</i>   |                       |
|--------------------|-------------|-------------------------|-------------|-----------------------|-------------|-----------------------|-------------|-----------------------|-------------|-----------------------|-------------|-----------------------|-------------|-----------------------|-------------|-----------------------|
|                    | Abs<br>freq | Rel.<br>freq.<br>w/i CC | Abs<br>freq | Rel<br>freq<br>w/i CC | Abs<br>freq | Rel<br>freq<br>w/i CC | Abs<br>freq | Rel<br>freq<br>w/i CC | Abs<br>freq | Rel<br>freq<br>w/i CC | Abs<br>freq | Rel<br>freq<br>w/i CC | Abs<br>freq | Rel<br>freq<br>w/i CC | Abs<br>freq | Rel<br>freq<br>w/i CC |
| 1P                 | —           | —                       | 1           | 7,7%                  | —           | —                     | —           | —                     | —           | —                     | —           | —                     | —           | —                     | 1           | 7,1%                  |
| 2P                 | 4           | 80,0%                   | 8           | 61,5%                 | 13          | 86,7%                 | 5           | 83,3%                 | 9           | 81,8%                 | 7           | 77,8%                 | 12          | 100%                  | 12          | 85,7%                 |
| 3P/DP              | 1           | 20,0%                   | 4           | 30,8%                 | 2           | 13,3%                 | 1           | 16,7%                 | 2           | 18,2%                 | 2           | 22,2%                 | —           | —                     | 1           | 7,1%                  |

The examination of verb-adjacency under CC gives Table 10.75. Clitics occur most frequently in a verb-adjacent configuration when they climb. More than 80% of all climbed clitics end up

contiguously to one or two verbs. Mono-verbal configurations prevail and represent two-thirds of verb-adjacent CC tokens ( $\approx 2/3$ ), but bi-verbal ones are quite common given that they make one-third of occurrences ( $\approx 1/3$ ). A minority of almost 18% climbs to a position disjoint from a verb, whereby 2P without VA makes the largest portion. The VA vs. noVA pattern replicates for pronominal and reflexive clitics ( $\chi^2 = 0.76, p = .05, df = 1$ ).

Table 10.75: Frequencies of simultaneous occurrences of clitic climbing and (non-)verb-adjacency in spoken Slovak.

| Placement<br>after CC | Total      |                      | Pronominal |                      | Reflexive  |                      |
|-----------------------|------------|----------------------|------------|----------------------|------------|----------------------|
|                       | Abs. freq. | Rel. freq.<br>w/i CC | Abs. freq. | Rel. freq.<br>w/i CC | Abs. freq. | Rel. freq.<br>w/i CC |
| noVA                  | 14         | 17.7%                | 8          | 15.1%                | 6          | 23.1%                |
| 1VA                   | 43         | 54.4%                | 30         | 56.6%                | 13         | 50.0%                |
| 2VA                   | 22         | 27.8%                | 15         | 28.3%                | 7          | 26.9%                |

- (10.56) a. Nie každý **sa**<sub>2</sub> [ s tým ] [ potom ] [noVA]  
 NEG every.NOM REFL.ACC with this.INS.SG.N afterwards  
**vedel**<sub>1</sub> **vyrovnať**<sub>2</sub> po novembri.  
 know.PST.SG.M cope.INF after November.LOC.SG  
 ‘Not everyone knew then [how] to cope with it after November.’
- b. [...] keď **sa**<sub>2</sub> **ho**<sub>2</sub> **prišli**<sub>1</sub> ľudia **pýtať**<sub>2</sub> (, [1VA]  
 when REFL him.ACC come.PST.PL people.NOM.PL ask.INF  
 čo máme robiť.)  
 what have.PRS.1PL do.INF  
 ‘... when people came to ask him (, what we should do.)’
- c. [...] **môžem**<sub>1</sub> **ti**<sub>2</sub> **poslať**<sub>2</sub> link na tie veci. [2VA]  
 can.PRS.1SG you.DAT send.INF link.ACC.SG on this.ACC.PL thing.ACC.PL  
 ‘... I can send you the link to these things.’ (Slk; SNK s-hovor-7.0)

The overall homogeneous distribution pattern is not effective for each clitic type, as shown by Table 10.76, which, however, might be an artefact of the low token frequencies. Comparatively low VA ratios – based on the totalised frequencies of 1VA + 2VA – are encountered for *ma* (60%), *mi* (67%), and *ta* (69%). The pronominal clitic *ti* exhibits the highest frequency (100%) followed by *ho* (93%). The remaining clitics *mu*, *sa*, *si* are widely distributed in the intermediate range between 70% and 90%

Table 10.76: Frequencies of simultaneous occurrences of clitic climbing and (non-)verb-adjacency in spoken Slovak by clitic.

| Placement<br>after CC | <i>ma</i>     |                            | <i>i'a</i>    |                            | <i>ho</i>     |                            | <i>mi</i>     |                            | <i>ti</i>     |                            | <i>mu</i>     |                            | <i>sa</i>     |                            | <i>si</i>     |                            |
|-----------------------|---------------|----------------------------|---------------|----------------------------|---------------|----------------------------|---------------|----------------------------|---------------|----------------------------|---------------|----------------------------|---------------|----------------------------|---------------|----------------------------|
|                       | Abs.<br>freq. | Rel.<br>freq.<br>w/i<br>CC | Abs.<br>freq. | Rel.<br>freq.<br>w/i<br>CC | Abs.<br>freq. | Rel.<br>freq.<br>w/i<br>CC | Abs.<br>freq. | Rel.<br>freq.<br>w/i<br>CC | Abs.<br>freq. | Rel.<br>freq.<br>w/i<br>CC | Abs.<br>freq. | Rel.<br>freq.<br>w/i<br>CC | Abs.<br>freq. | Rel.<br>freq.<br>w/i<br>CC | Abs.<br>freq. | Rel.<br>freq.<br>w/i<br>CC |
| noVA                  | 2             | 40.0<br>%                  | 4             | 30.8<br>%                  | 1             | 6.7<br>%                   | 2             | 33.3<br>%                  | —             | —                          | 1             | 11.1<br>%                  | 2             | 16.7<br>%                  | 4             | 28.6<br>%                  |
| 1VA                   | 2             | 40.0<br>%                  | 4             | 30.8<br>%                  | 9             | 60.0<br>%                  | 4             | 66.7<br>%                  | 7             | 63.6<br>%                  | 8             | 88.9<br>%                  | 7             | 58.3<br>%                  | 6             | 42.9<br>%                  |
| 2VA                   | 1             | 20.0<br>%                  | 5             | 38.5<br>%                  | 5             | 33.3<br>%                  | —             | —                          | 4             | 36.4<br>%                  | —             | —                          | 3             | 25.0<br>%                  | 4             | 28.6<br>%                  |

## 10.6 Silesian

### 10.6.1 Clitic placement

The data sample for spoken Silesian is relatively small in comparison to the data sets in the previous sections. It comprises 339 tokens in its entirety, whereby the polyfunctional reflexive *się* is the most-frequent type ( $n = 270$ ). Other short pronominal forms and the dative reflexive clitic have few attestations (*cię*: 2, *go*: 15, *mi*: 9, *ci*: 0, *mu*: 12, *se*: 33), such that the discussion will be restricted to the total number of occurrences and – where useful – the two major clitic categories. There are varying graphical representations for some clitics due to phonological variation and different transcription conventions. Data will be cited as presented in the original source. However, in-text references to particular clitic types will use standard Polish orthography for the sake of convenience (if reference to actual tokens is necessary, angle brackets will indicate graphic form).

We learn from Table 10.77 that the most common position is represented by Wackernagel placement (61% in total). Clitic Third represents circa one fourth of occurrences. Placement in 4P and 1P is attested with 6% and 4% respectively. Other positions become increasingly marginal. The transitive frequency relation among positions is given in (10.58). In sum, one-third of the data displays Delayed Placement. Pronominal clitics make circa 11 % of the Silesian data. Abstracting away from the low token frequencies, Wackernagel cliticisation is more pronounced (68%), whereas Delayed Placement occurs in roundabout one-fifth of cases. The comparatively high ratio for 1P (10.5%) is likely to be an artefact of the small sample size. Possibly, the same could apply to the general lack of DP occurrences beyond 3P, although there is one attestation of 4P. Reflexive clitics add up to almost 89% of the Silesian data. The attested positions range from 1P to 7P with a clear

central tendency for 2P (ca. 61%). Clitic Third amounts to over one-quarter (ca. 27%). The third-most common placement is 4P (ca. 6%) followed by 1P (ca. 4%). The conflicting ratios for 1P and 4P between reflexive and pronominal clitic make it impossible to claim a transitive distribution relation for these positions. Increasing positional delay becomes marginal. A good third of reflexive occurrences are attested in Delayed Placement. The frequency distributions of the two clitic categories cannot be compared in a meaningful way, because the pronominal distribution pattern could be biased by reason of the low token frequencies.

Clitics predominantly occur with left lateral placement. Right lateral placement is attested most often for 2P with 11.5%. The figures for Delayed Placement are close with 9%, whereby the general pattern of decreasing ratios with growing distance repeats. The differences between pronominal and reflexive clitics are less pronounced. Reflexive clitics show four-fifths of left-side placement, while the amount slightly decreases for pronominal clitics towards three-quarters.

As far as one can tell from the Silesian data, contact position with the predicate is the dominant pattern representing almost three-quarters of occurrences (= 72.6%) vis-à-vis second position placement with three-fifths (= 61.4%). Their intersection yields two-fifths of occurrences (= 41.6%). Although left lateral placement is the clearly dominant pattern, one must note that Delayed Placement is most likely to occur with predicate adjacency. The highest frequency for Delayed Placement occurs with left lateral contact followed by right lateral contact. Delayed Placement without contact is rarer. Interestingly, left lateral distant placement and right lateral contact placement behave contrarily. While the former exhibits more 2P than the latter, it works the other way round with DP. Admittedly, it is unclear whether this is an artefact of the small database.

Table 10.77: Frequencies of clitic occurrences in spoken Silesian by position, laterality, and distance.

| Pos | Left distant |       | Left adjacent |       | Right (adjacent) |       | Sum |       |
|-----|--------------|-------|---------------|-------|------------------|-------|-----|-------|
| 1P  | 3            | 0.9%  | 12            | 3.5%  | 0                | 0.0%  | 15  | 4.4%  |
| 2P  | 67           | 19.8% | 102           | 30.1% | 39               | 11.5% | 208 | 61.4% |
| 3P  | 20           | 5.9%  | 46            | 13.6% | 23               | 6.8%  | 89  | 26.3% |
| 4P  | 2            | 0.6%  | 12            | 3.5%  | 6                | 1.8%  | 20  | 5.9%  |
| 5P  | 1            | 0.3%  | 2             | 0.6%  | 2                | 0.6%  | 5   | 1.5%  |
| 6P  | —            | —     | 1             | 0.3%  | —                | —     | 1   | 0.3%  |
| 7P  | —            | —     | 1             | 0.3%  | —                | —     | 1   | 0.3%  |
| Sum | 93           | 27.4% | 176           | 51.9% | 70               | 20.6% | 339 | 100%  |
| DP  | 23           | 6.8%  | 62            | 18.3% | 31               | 9.1%  | 116 | 34.2% |



Clitic First is attested for both environments relevant here, i.e. after clause boundaries as well as after coordinating conjunctions. The proportion is  $\frac{2}{3}$  boundary-adjacent vs.  $\frac{1}{3}$  post-conjunctive placement, cf. Table 10.78.

Table 10.78: Frequencies of Clitic First in spoken Silesian after clause boundary and coordinating conjunction (in total).

| Type        | Abs. freq. | Rel. freq. w/i 1P |
|-------------|------------|-------------------|
| # 1P ...    | 5          | 33.3%             |
| CONJ 1P ... | 10         | 66.7%             |

The diverse clausal positions of clitics attested in the data are illustrated in example block (10.57).<sup>559</sup>

- (10.57) a. (a potym sie biere masło,) # **sie**<sup>1</sup> rozpuści  
 and afterwards REFL.ACC take.PRS.3SG butter.ACC.SG.N REFL melt.PRS.3SG  
 i cukier [...] and sugar.ACC.SG  
 ‘(and then one takes butter) sugar is melting, too ...’
- b. [...] i<sup>0</sup> **mu**<sup>1</sup> **sie**<sup>1</sup> chajcnął (, jag za dugo sie  
 and him.DAT REFL.ACC burn.PST.SG.M when too long REFL.ACC  
 świeciła.) shine.PST.SG.F  
 ‘... and his [lantern] burnt (when it [= candle] shined too long.)’
- c. [...] bo jak<sup>1</sup> **ctie**<sup>2</sup> už’ mama wziyni <sup>l</sup>od\_  
 because, when you.ACC already mom.NOM.SG take.PST.PL.VIR from  
 malutka, (to<sup>u</sup>=żie-sz’ už’ widzioł wsziecko, co  
 small.DIM.GEN.SG TO=ŻE-AUX.2SG already know.PST.SG.M everything what  
 robili, ni?) do.PST.PL.VIR NEG  
 ‘... because, when [your] mom took you from early on, (then you already knew everything what they did, right?)’

<sup>559</sup> Examples are cited as encountered in the original, i.e. sticking to the transcription conventions applied in the individual dialect texts. For instance, the adverbial PP <sup>l</sup>od\_*malutka* ‘from early on’ in (10.57c) uses the superscript letter *l* < <sup>l</sup> > to transcribe labialisation, the stroke < | > marks an accent position deviating from Polish antepenult stress, and the underscore < \_ > is used to indicate linking, i.e. absence of a pause. In the present example, the underscore further represents the fact that the PP as a whole is the domain of stress assignment.

- d. [...] ze<sup>1</sup> [NP Pan Bóg ]<sup>2</sup> **mu**<sup>3</sup> stworzył Ewa [...]   
that Lord.NOM God.NOM him.DAT create.PST.SG.M Eve.ACC   
‘... that God, the Lord, created Eve for him [= Adam] ...’
- e. [...] i<sup>0</sup> [ na przykład ]<sup>1</sup> [NP tako wielko kraje   
and for example.ACC.SG such.ACC.SG.F big.ACC.SG.F slice.ACC.SG   
chleba ]<sup>2</sup> smarowała<sup>3</sup> **se**<sup>4</sup> takim masłem świeżo   
bread.GEN.SG spread.PST.SG.F REFL.DAT such.INS.SG.N butter.INS.SG freshly   
stłuczonym [...]   
beat.PASS.PTCP.INS.SG.N   
‘... and, for example, she [= mother] spread freshly beaten butter on such a big slice of   
bread (for herself) ...’
- f. a piyrwej<sup>1</sup> baby<sup>2</sup> [PP u nos ]<sup>3</sup> tyż<sup>4</sup> **sie**<sup>5</sup> tak oblykały   
and once woman.NOM.PL by us.GEN also REFL.ACC so dress.PST.PL.NVIR   
‘but women used to dress like this, too, over here’
- g. jak<sup>1</sup> to<sup>2</sup> sam<sup>3</sup> [PP u nas ]<sup>4</sup> to<sup>5</sup> **sie**<sup>6</sup> godo   
how this.ACC.SG.N here by us.GEN this.ACC.SG.N REFL say.PRS.3SG   
‘like they say over here’
- h. to<sup>1</sup> już<sup>2</sup> wo<sup>u</sup>m<sup>3</sup> [PP w życiu ]<sup>4</sup> nigdy<sup>5</sup> [NP nic złego ]<sup>6</sup> **sie**<sup>7</sup>   
TO already you.PL.DAT in life.LOC.SG never nothing.ACC bad.GEN.SG.N REFL   
nie przytrefi   
NEG happen.PRS.3SG   
‘then nothing bad will ever happen to you [as long as you live]’ (Sil; DiG)

Table 10.79 gives the measures of central tendency, all of which concur in that second position clitic placement is the dominant pattern in Silesian. The preference hierarchy for clitic positioning is summarised in (10.58).

Table 10.79: Measures of central tendency for clitic positioning in spoken Silesian.

| Measure:          | $\bar{x}$ (SD) | <i>Md</i> | <i>Mo</i> |
|-------------------|----------------|-----------|-----------|
| Value (position): | 2.41 (±0.80)   | 2         | 2         |

(10.58) 2P > 3P > 4P / 1P > ( 5P > 6P / 7P )

The examples in (10.59) illustrate left-distant, left-adjacent, and right (adjacent) placement. There are no occurrences of right-distant clitic placement, which could be due to the small sample size, which reduces the likelihood to encounter rare phenomena.

- (10.59) a. [...] to<sup>1</sup> ona<sup>2</sup> go<sup>3</sup> [ADV tam] **zaprowadziła** [...] [Ld]  
 TO she.NOM him.ACC there lead.PST.SG.F  
 ‘... so she led him there ...’
- b. ale to<sup>1</sup> już<sup>2</sup> [PP w piekarni]<sup>3</sup> se<sup>4</sup> **piekło** [La]  
 but this.NOM.SG.N already in bakery.LOC.SG REFL.DAT bake.PST.SG.N  
 ‘but this was just baked in the bakery already’
- c. no to<sup>1</sup> **przyprowadzają**<sup>2</sup> mu<sup>3</sup> tedy ta młodo<sup>†</sup> [R(a)]  
 well TO bring.PRS.3PL him.DAT then this.ACC.SG.F young.ACC.SG  
 pani  
 lady.ACC.SG  
 ‘well, so they bring the bride to him then’ (Sil; DiG)

## 10.6.2 Clitic clustering

The frequencies of single clitic vis-à-vis cluster occurrences are summarised in Table 10.80. Given the small data set, clitic clusters are exclusively attested in the form of pairs, cf. (10.60), while single clitic occurrences prevail with 90%, both of which could be an artefact of the restricted database.

Table 10.80: Frequencies of single clitic and clitic cluster occurrences in spoken Silesian by number of elements.

| Number of clitics | Absolute frequency | Relative frequency |
|-------------------|--------------------|--------------------|
| 1                 | 306                | 90.3%              |
| 2                 | 33                 | 9.7%               |
| Sum               | 339                | 100%               |
| $n > 1$           | 33                 | 9.7%               |

- (10.60) a. nie udało mu się  
 NEG succeed.PST.SG.N him.DAT REFL.ACC  
 ‘it did not work out for him’

- b. przede wszystkim      rzytowali      **my**      **się**      właśnie na  
before everything.LOC.PL prepare.PST.PL.VIR AUX.1PL REFL.ACC actually on  
to      przyjyncie      Pon      Bóczka  
this.ACC.SG.N reception.ACC.SG Lord God.GEN.SG  
‘above all, we actually prepared ourselves for the reception of God, the Lord’
- c. ni<sup>o</sup>kiedy      **by**      **się**      taka      jazda,      takom  
sometimes COND REFL.ACC such.NOM.SG.F ride.NOM.SG such.INS.SG  
maszafto<sup>u</sup>m      przydała      i      kibicom      Legii  
mine\_railway.INS.SG be\_useful.PST.SG.F and fan.DAT.PL Legia.GEN.SG  
‘sometimes, even the fans of Legia [Warszawa] could use such a ride with the mine  
railway’ (Sil; DiG)

The low token frequencies render diagrams for attested clitic cluster categories futile. Therefore, frequency distributions are provided in Table 10.81 with absolute quantities only. Due to the scarce data, it is difficult to provide generalisations for the internal ordering. The linearisation  $COND < AUX < PRN < REFL$  may be regarded as an informed guess. It complies with the basic pattern found in other West Slavic languages. However, the order among pronominal and reflexive clitics is inconclusive provided the low amount of data.

Table 10.81: Frequency distribution of clitic cluster categories in spoken Silesian.

| Clitic category | Absolute frequency |
|-----------------|--------------------|
| PRN < REFL      | 15                 |
| AUX < REFL      | 6                  |
| REFL < PRN      | 5                  |
| COND < REFL     | 4                  |
| COND < PRN      | 2                  |
| AUX < PRN       | 1                  |

Table 10.82 reports that the most occurring case value preceding a reflexive clitic is dative. The order among reflexive and accusative pronominal clitics is subject to fluctuations, cf. (10.61).

Table 10.82: Absolute frequencies of case-valued clitic clusters in spoken Silesian.

| Case sequence | Absolute frequency |
|---------------|--------------------|
| DAT < REFL    | 13                 |
| REFL < ACC    | 5                  |
| ACC < REFL    | 2                  |

- (10.61) a. (on co[ś<sup>?</sup>]                      ta[m<sup>?</sup>] ni      za    bardzo   rozumioł)  
                   he something.ACC there NEG too much understand.PST.SG.M  
                   i    **go**                    **sie**    pytoł<sup>†</sup>  
                   and him.ACC REFL ask.PST.SG.M  
                   ‘(he didn’t understand something or other too much) and asked him’
- b. ([...] bo            lnu                    szie            nie    koszi,)            ale **szie go**  
                   because flax.GEN.SG REFL.ACC NEG cut.PRS.3SG but REFL him.ACC  
                   wrywo                    z            k|or<sup>z</sup>yniami  
                   wrench.PRS.3SG with root.INS.PL  
                   ‘(... because one doesn’t cut flax) but harvests it [together] with the root’      (Sil; DiG)

There is only one attestation of diacclisis in the Silesian data as summarised by Table 10.83. Diacclisis appears in a complex clause (representing 5.9% of all complex clause tokens) between the auxiliary clitic *my* ‘AUX.1PL’ and the reflexive argument clitic *sie* ‘REFL.ACC’ and is caused by the reflexive’s in-situ realisation, cf. (10.62). Bear in mind that 1PL person marker *my* (corresponding to standard Polish *-śmy*) is homophonous to the 1PL subject pronoun *my* (corresponding to standard Polish *my*) but is distinguished from it. Provided the single occurrence of diacclisis in Silesian and the overall small database, a calculation of ratios is not very informative, indeed.

Table 10.83: Frequencies of multiple clitics with clustering and diacclisis in spoken Silesian.

| Multiple clitics | Absolute frequency | Relative frequency<br>w/i multiple CL |
|------------------|--------------------|---------------------------------------|
| Clustering       | 33                 | 97.1%                                 |
| Diacclisis       | 1                  | 2.9%                                  |
| Sum              | 34                 | 100%                                  |

(10.62) poradzili<sub>1</sub>                    **my**<sub>1</sub>            na przykład tyż    <sup>1</sup>oblykać<sub>2</sub> **sie**<sub>2</sub>            w stroje  
 achieve.PST.PL.VIR   AUX.1PL   on   example   also   dress.INF   REFL.ACC   in   costume.ACC.PL  
 ‘we were also capable of, for example, putting on [traditional] costumes’                    (Sil; DiG)

Due to the low amount of Silesian data, only two types were found in absolute Clitic First: (Refl), (Prn, Refl).

### 10.6.3 Clitic climbing

The Silesian data involves a total of 17 complex clauses, cf. Table 10.84. Clitic climbing is the prevalent construction, but the overall token frequency of pronominal clitics in the relevant environment is extremely small with merely two occurrences. Unsurprisingly, the only instance of local placement is attested for the reflexive clitic type *sie*, cf. (10.62) above.

Table 10.84: Frequencies of clitic climbing and no climbing in spoken Silesian.

| Locality        | Total      |                                | Pronominal |                                | Reflexive  |                                |
|-----------------|------------|--------------------------------|------------|--------------------------------|------------|--------------------------------|
|                 | Abs. freq. | Rel. freq. w/i complex clauses | Abs. freq. | Rel. freq. w/i complex clauses | Abs. freq. | Rel. freq. w/i complex clauses |
| CC              | 16         | 94.1%                          | 2          | 100%                           | 14         | 93.3%                          |
| noCC            | 1          | 5.9%                           | —          | —                              | 1          | 6.7%                           |
| CC?             | —          | —                              | —          | —                              | —          | —                              |
| Complex clauses | 17         | 100%                           | 2          | 100%                           | 15         | 100%                           |

Table 10.85 provides multiple target positions for clitic climbing with attested 2P, 3P, and 4P, cf. (10.63). Delayed clitic placement occurs most often with 3P being predominant, but these numbers are little informative in face of the low token frequencies. For the same reason, a differentiation between pronominal and reflexive clitics is not very useful, but it is kept for the sake of completeness.

Table 10.85: Frequencies of target positions under clitic climbing in spoken Silesian.

| Position after CC | Total      |                   | Pronominal |                   | Reflexive  |                   |
|-------------------|------------|-------------------|------------|-------------------|------------|-------------------|
|                   | Abs. freq. | Rel. freq. w/i CC | Abs. freq. | Rel. freq. w/i CC | Abs. freq. | Rel. freq. w/i CC |
| 2P                | 5          | 31.3%             | 1          | 50.0%             | 4          | 28.6%             |
| 3P                | 9          | 56.3%             | 1          | 50.0%             | 8          | 57.1%             |
| 4P                | 2          | 12.5%             | —          | —                 | 2          | 14.3%             |
| DP                | 11         | 68.8%             | 1          | 50.0%             | 10         | 71.4%             |

- (10.63) a. Ja<sup>1</sup>    **cie**<sub>2</sub><sup>2</sup>    nie    chce<sub>1</sub>    martwić<sub>2</sub>  
 I.NOM you.ACC NEG want.PRS.1SG worry.INF  
 ‘I don’t want to worry you.’
- b. ale    družbowie<sup>1</sup>    [v nie    pozwalajom]<sub>1</sub><sup>2</sup>    **go**<sub>2</sub><sup>3</sup>  
 but    bridesman\_and\_groomsman.NOM.PL.VIR    NEG allow.PRS.3PL him.ACC  
 zdjąć<sub>2</sub>  
 take\_off.INF  
 ‘but bridesmen and groomsman do not allow to take it [= wreath] off’
- c. Jak<sup>1</sup>    ju<sup>2</sup>    zacymy<sub>1</sub><sup>3</sup>    **się**<sub>2</sub><sup>4</sup>    robić<sub>2</sub>    [...]  
 when    already    start.PRS.1PL REFL.ACC work.INF  
 ‘When we are about to start working ...’ (Sil; DiG)

Climbed clitics are attested exclusively in verb-adjacency evenly distributed across mono-verbal and bi-verbal contact configurations, cf. Table 10.86. Examples for mono-verbal contact and verbal circumposition have been already cited earlier and I additionally provide (10.64a) for the former and (10.64b) for the latter configuration. Again, the limited amount of data can be considered responsible for the restricted picture in terms of gaps.

Table 10.86: Frequencies of simultaneous occurrences of clitic climbing and (non-)verb-adjacency in spoken Silesian.

| Placement<br>after CC | Total      |                      | Pronominal |                      | Reflexive  |                      |
|-----------------------|------------|----------------------|------------|----------------------|------------|----------------------|
|                       | Abs. freq. | Rel. freq.<br>w/i CC | Abs. freq. | Rel. freq.<br>w/i CC | Abs. freq. | Rel. freq.<br>w/i CC |
| noVA                  | —          | —                    | —          | —                    | —          | —                    |
| 1VA                   | 8          | 50.0%                | 1          | 50.0%                | 7          | 50.0%                |
| 2VA                   | 8          | 50.0%                | 1          | 50.0%                | 7          | 50.0%                |

- (10.64) a. i    Edyta    **se**<sub>2</sub>    **dała**<sub>1</sub>    **uszyć**<sub>2</sub>    czorno,    [1VA]  
 and Edyta.NOM REFL.DAT give.PST.SG.F sew.INF black.ACC.SG.F  
 jedwobno    kiecka  
 silken.ACC.SG.F skirt.ACC.SG  
 ‘and Edyta let sew a black, silken skirt (for herself)’
- b. w    badyhauzie    **muszymy**<sub>1</sub>    **się**<sub>2</sub>    **przewlyc**<sub>2</sub>    [2VA]  
 in pithead\_bath.LOC.SG must.PRS.1PL REFL.ACC change\_clothes.INF  
 ‘in the pithead bath we must change our clothes’ (Sil; DiG)

## 10.6.4 Agreement drop

There were 10 occurrences of agreement drop in the data representing ca. 3% of the total amount or 63% within 1st and 2nd person past tense tokens ( $n = 18$ , conditionals are not attested). Like in other West Slavic languages, agreement drop is licensed in Silesian for 1SG/PL only. Clitic positioning ranges from 2P to 4P. There is little to say about frequencies, such that Table 10.87 reports absolute quantities only. Obviously, Delayed Placement is the dominant pattern under agreement drop.

Table 10.87: Frequencies for attested clitic clausal positions under agreement drop in Silesian.

| Position | Absolute frequency |
|----------|--------------------|
| 2P       | 1                  |
| 3P       | 8                  |
| 4P       | 1                  |
| Sum      | 10                 |
| DP       | 9                  |

Given the unequal overall frequencies among the individual clitics, the reflexive type *się* ‘REFL.ACC/GEN’ prevails with 7 tokens. There is one occurrence of *go* ‘him.ACC’ and two of *se* ‘REFL.DAT’ in addition. The examples in (10.65) present agreement drop for both numbers and with varying clitic positioning. The only 2P attestation occurs in an independent clause. In embedded clauses headed by a subordinator, the clitic is found in DP following the subject pronoun (10.65b–e). The 3P placement of *się* in (10.65f) is not motivated by the presence of a subordinator, for it is an independent clause, but by the topicalised PP *na Śląsku* ‘in Silesia’. Utterance (10.65g) presents a subordinate clause again and shows the only instance of 4P under agreement drop in the Silesian data. The anaphoric adverb *wtedy* ‘(back) then’ intervening between the subordinator *bo* ‘because’ and the subject pronoun *my* ‘we’ is pragmatically prominent and topicalised. Generally, all attestations exhibit the precedence pattern  $\text{SBJ}_{\text{PRN}} < \text{CL} < \text{V}$ . Subject pronoun and clitic are always adjacent, whereas the verb may occur contiguously to the clitic or disjoint.

- (10.65) a. *my*<sup>1</sup>    *mu*<sup>2</sup>    *pedzieli*     $\emptyset$ , (że *przyjechali*    *my*    *po*  
           we.NOM him.ACC tell.PST.PL.VIR  $\emptyset$  that come.PST.PL.VIR AUX.1PL for  
           *pani*            *młodo*)  
           lady.ACC.SG young.ACC.SG.F  
           ‘we told him (that we came for the bride)’



- b.  $bo^1$   $jo^2$   $sie^3$   $u\dot{z}$   $\dot{t}od$   $taty$   $nie$   $dopyta\acute{la}$   $\emptyset$   
 because I.NOM REFL already from dad.GEN.SG NEG enquire.PST.SG.F  $\emptyset$   
 ‘because I didn’t enquire of [my] father anymore’
- c. Ciotko,  $bo^1$   $my^2$   $se^3$   $tak$   $my\acute{s}la\acute{y}$   $\emptyset$   $z$   $Jadzio^m$  [...]  $[\dots]$   
 aunt.VOC because we.NOM REFL.DAT so think.PST.NVIR  $\emptyset$  with Jadwiga.DIM.INS.SG  
 ‘Aunt, because we’ve been wondering, Jadzia and me, ...’
- d. (tak jak na przykład)  $jag^1$   $jo^2$   $sie^3$   $\acute{z}eni\acute{l}$   $\emptyset$   $w$   
 so how on example how I.NOM REFL marry\_a\_woman.PST.SG.M  $\emptyset$  in  
 Mizerowie [...]  $[\dots]$   
 Mizerów.LOC.SG  
 ‘(like for example) when I got married [to my wife] in Mizerów ...’
- e. [...],  $zanim^1$   $jo^2$   $sie^3$   $urodzi\acute{l}$ ,  $\emptyset$  [...]  $[\dots]$   
 before I.NOM REFL give\_birth.PST.SG.M  $\emptyset$   
 ‘..., before I was born, ...’
- f. Interviewer: ( $\acute{s}lub$   $przez$   $ksi\acute{e}dza$   $te\acute{z}$   $by\acute{l}$   $gwar\acute{a}$   
 wedding.NOM.SG by priest.ACC.SG also be.PST.SG.M dialect.INS.SG  
 udzielany?)  
 grant.PASS.PTCP.NOM.SG.M  
 ‘(did the priest marry couples using dialect, too?)’
- Informant: (Nie, nie.) [ $Na$   $\acute{S}l\acute{a}sku$ ]  $^1_{TOP}$   $my^2$   $si\acute{e}^3$   $nie$   
 NEG NEG on Silesia.LOC.SG we.NOM REFL NEG  
 dorobili  $\emptyset$   $tego$   
 make\_some\_more.PST.PL.VIR  $\emptyset$  this.GEN.SG  
 ‘(No, no.) In Silesia, we didn’t come so far’
- g. (i  $nojbardziej$   $to$   $pomiyntom$   $takie$   $Roraty$   
 and most TO remember.PRS.1SG such.ACC.PL.NVIR Rorate\_Mass.ACC.PL  
 w\acute{la}śnie  $w$   $drugiej$   $klasie$ ,  $jak$   $\acute{z}e$ - $ch$   $by\acute{l}$ ,)  $bo^1$   
 actually in second.LOC.SG.F class.LOC.SG when \acute{Z}E-AUX.1SG be.PST.SG.M because  
 [ $AdvP$   $wtedy$ ]  $^2_{TOP}$   $my^3$   $si\acute{e}^4$   $przygotowywali$   $\emptyset$   $do$   $Ko^m$ munii  
 back\_then we.NOM REFL.ACC prepare.PST.PL.VIR  $\emptyset$  to communion.GEN.SG  
 \acute{S}wiyntej  
 holy.GEN.SG  
 ‘(and I remember such Rorate Masses best, when I was attending second class, actually,) because, at that time, we were preparing for the Holy Communion’ (Sil; DiG)

## 10.7 Kashubian

### 10.7.1 Clitic placement

The Kashubian database is even scantier than the Silesian one and comprises 49 data points in total. The reflexive type *się* is attested in the forms <so>, <sa>, <śe>, <sã>, and occurs 47 times. The pronominal type *go* ‘him.ACC’ represents the two only instances of pronominal clitics. Frequencies will be provided in total, therefore. I apply the same writing convention as in Silesian, i.e. due to varying graphic representations of the reflexive, I will cite the forms as encountered in the source in case of tokens but use the standard Polish form *się* on type-level for the sake of convenience.

Table 10.88 reports that 2P is the most common position in Kashubian but amounts to roundabout a half of the data set (exactly 49%). Delayed Placement makes another 47%. The cardinality of empirically attested delayed positions ranges from 3 to 5. There are two attestations of Clitic First both of which occur after a clause boundary, cf. Table 10.89. See (10.66) for examples. The frequency distribution for clitic positioning resembles a linear function with lower *n* by growing distance from the left periphery. The pattern is summarised by (10.68).

Table 10.88 shows that most clitics occur in left lateral placement (ca. 78%). Examples are given in (10.67). The second position prevails in right lateral placement; the only attested delayed position is 3P.

The sparse database calls for caution when analysing the data. Therefore, the following inspection can only be tentative. Predicate adjacency is the most frequent pattern exceeding three-quarters of occurrences (= 77.9%) compared to half of occurrences representing Wackernagel cliticisation. The latter is evenly distributed across the three categories distinguished here. In fact, left lateral distant placement and right lateral placement behave quite identically. The dominant placement pattern is left lateral contact position, which also covers the majority of Delayed Placement cases. Therefore, Delayed Placement preferably occurs in this configuration, although not exclusively. It is noteworthy that Delayed Placement outnumbers Wackernagel placement within the left lateral contact pattern; especially the frequency for 3P is higher than for 2P. In this respect, this placement category strikingly differs from the two other ones. The intersection of second position and predicate adjacent placement amounts to approximately one-third of cases (= 32.7%).

Contiguity to the verb is the most common placement type (ca. 88%) dominated by mono-verbal configurations (ca. 84%). Two out of three clitics in non-verb-adjacent placement occur in second position.

Table 10.88: Frequencies of clitic occurrences in spoken Kashubian by position, laterality, and distance.

| Pos | Left distant |       | Left adjacent |       | Right<br>(adjacent) |       | Sum |       |
|-----|--------------|-------|---------------|-------|---------------------|-------|-----|-------|
| 1P  | —            | —     | 2             | 4.1%  | N/A                 | N/A   | 2   | 4.1%  |
| 2P  | 8            | 16.3% | 8             | 16.3% | 8                   | 16.3% | 24  | 49.0% |
| 3P  | 1            | 2.0%  | 11            | 22.4% | 3                   | 6.1%  | 15  | 30.6% |
| 4P  | 2            | 4.1%  | 4             | 8.2%  | —                   | —     | 6   | 12.2% |
| 5P  | —            | —     | 2             | 4.1%  | —                   | —     | 2   | 4.1%  |
| Sum | 11           | 22.4% | 27            | 55.1% | 11                  | 22.4% | 49  | 100%  |
| DP  | 3            | 6.1%  | 17            | 34.7% | 3                   | 6.1%  | 23  | 46.9% |

Table 10.89: Frequencies of Clitic First in spoken Kashubian after clause boundary and coordinating conjunction (in total).

| Type        | Absolute frequency | Relative frequency w/i 1P |
|-------------|--------------------|---------------------------|
| # 1P ...    | —                  | —                         |
| CONJ 1P ... | 2                  | 100%                      |

- (10.66) a. [...] i<sup>0</sup> so<sup>1</sup> wëprostowac ni młęgła.  
and REFL.ACC straighten NEG can.PST.SG.F  
‘... and she couldn’t straighten up.’
- b. [...] łen<sup>1</sup> go<sup>2</sup> miłuje [...]   
he.NOM him.ACC love.PRS.3SG  
‘... he loves it ...’
- c. [NP Kaszëbska lëteratura]<sup>1</sup> [NP cali czas]<sup>2</sup>  
Kashubian.NOM.SG.F literature.NOM.SG complete.ACC.SG.M time.ACC.SG  
so<sup>3</sup> rozwijy, [...]   
REFL.ACC develop.PRS.3SG  
‘The Kashubian literature is developing all the time ...’
- d. ([...] i jim mówił) co<sup>1</sup> tam<sup>2</sup> jemu<sup>3</sup> bę<sup>4</sup> sã<sup>4</sup>  
and them.ACC say.PST.SG.M what there him.DAT COND REFL.ACC  
podołało [...]   
please.PST.SG.N  
‘(... and he told them) what he would just prefer ...’

- e. ([...] i tam trzeba to pilnowac,) ż'e=**bě**<sup>1</sup> jeszcze<sup>2</sup> [ przez  
and there need this.ACC.SG.N take\_care.INF that=COND still through  
wiele, wiele lat ]<sup>3</sup> młęgło<sup>4</sup> **so**<sup>5</sup> rozwijac [...]   
many many year.GEN.PL can.PST.SG.N REFL.ACC develop.INF  
'(... and one just needs to take care) that it [= Kashubian literature] could develop for  
many many years to come ...' (Csb; DiG)

- (10.67) a. ([...], ale mje jidze łe to,) że mē **sā** [Ld]  
but me.DAT go.PST.3SG about this.ACC.SG.N that we.NOM REFL.ACC  
[<sub>ADV</sub> lepij ] [<sub>ADV</sub> terysku ] **dogadujemē** ze sobóm, [...]   
better now get\_along.PRS.1PL with self.INS  
'(... but my point is) that we are now finding a common language between us more  
easily...'  
b. [...] do teg<sup>1</sup>e **sa brėk<sup>1</sup>uje** olej, terpentyna, [La]  
to this.GEN.SG.N REFL.ACC use.PRS.3SG oil.NOM.SG turpentine.NOM.SG  
sykatywa [...]   
siccative.NOM.SG  
'...oil paint, turpentine, siccative are used for that [= painting] ...'  
c. [...] tedy kiedy **je so** kawalerem [...] [R(a)]  
then when be.PRS.3SG REFL.ACC young\_man.INS.SG  
'... at a time, when one is a young man ...' (Csb; DiG)

Table 10.90: Measures of central tendency for clitic positioning in spoken Kashubian.

| Measure:          | $\bar{x}$ (SD) | <i>Md</i> | <i>Mo</i> |
|-------------------|----------------|-----------|-----------|
| Value (position): | 2.63 (±0.91)   | 2         | 2         |

(10.68) 2P > 3P > 4P > 1P / 5P

## 10.7.2 Clitic clustering

Kashubian exhibits the well-known observation that single clitic occurrences make the vast majority of the empirical databases. Table 10.91 shows that roughly 92% of occurrences are represented by single clitics. Clitic clusters are observed only in four cases and are limited to clitic pairs, cf. (10.69).

Table 10.91: Frequencies of single clitic and clitic cluster occurrences in spoken Kashubian by number of elements.

| Number of clitics | Absolute frequency | Relative frequency |
|-------------------|--------------------|--------------------|
| 1                 | 45                 | 91.8%              |
| 2                 | 4                  | 8.2%               |
| Sum               | 49                 | 100%               |
| $n > 1$           | 4                  | 8.2%               |

- (10.69) a. To **bë** **so** pszëda jisz wicy [...]  
 TO COND REFL.ACC be\_useful.PST[.3SG.F<sup>2</sup>/DU<sup>2</sup>] already more  
 ‘So, some more of that [= writers workshops] would be advantageous ...’

- b. [...] stała **so** **ji** niesz’cz’escy.  
 happen.PST.SG.F REFL.ACC her.DAT misfortune.NOM.SG  
 ‘... she had an accident.’

(Csb; DiG)

Based on the scarce data, one can merely register three instances of COND preceding REFL and one instance of REFL preceding PRN in Table 10.92, whereby the pronoun is a feminine dative form that might be an orthotonic pronoun.

Table 10.92: Frequency distribution of clitic cluster categories in spoken Kashubian.

| Clitic category | Absolute Frequency |
|-----------------|--------------------|
| COND < REFL     | 3                  |
| REFL < PRN      | 1                  |

Besides the four clitic clusters, there are two occurrences of diacclisis, cf. Table 10.93. These extremely small token frequencies give a proportion of  $\frac{2}{3}$  clustering vs.  $\frac{1}{3}$  diacclisis and should be treated with caution.

Table 10.93: Frequencies of multiple clitics with clustering and diacclisis in spoken Kashubian.

| Multiple clitics | Absolute frequency | Rel. freq. w/i multiple CL |
|------------------|--------------------|----------------------------|
| Clustering       | 4                  | 66.7%                      |
| Diacclisis       | 2                  | 33.3%                      |
| Sum              | 6                  | 100%                       |

The two split clitic occurrences are attested with both a simple and a complex clause environment, cf. Table 10.94. As in the case of Silesian, it does not really make sense to calculate the relative frequency of diacsis for the respective environments.

Table 10.94: Frequencies of diacsis in spoken Kashubian by syntactic environment.

| Syntactic environment | Absolute frequency | Rel. freq. w/i diacsis | Rel. freq. w/i syntactic environment |
|-----------------------|--------------------|------------------------|--------------------------------------|
| Simple clause         | 1                  | 50%                    | 2.0%                                 |
| Complex clause        | 1                  | 50%                    | 12.5%                                |
| Sum                   | 2                  | 100%                   | —                                    |

In the simple clause (10.70a), the short reflexive *so* does not adjoin to the complex subordinator that involves the conditional element *by*. Note that the reflexive *so* is a lexicalised element and does not represent a true argument. The complex clause (10.70b) exhibits a split between the conditional clitic *bě* related to the modal verb *młegło* ‘can.PST.SG.N’ and the reflexive clitic *sã*, which is a lexical element of the reflexive tantum *stac sã* ‘(to) happen’.

- (10.70) a. No ale gdy=**by** to jesz’cze raz cz’as **so** copnął [...]   
 well but when=COND TO still once time.NOM.SG REFL go\_back.PST.SG.M   
 ‘Well, but if time would turn back once again ...’
- b. ([...] to nie wiadomo) [<sub>S</sub> co **bě**<sub>1</sub> młegło<sub>1</sub> **sã**<sub>2</sub> stac<sub>2</sub>] [...]   
 TO NEG known.ADV what COND can.PST.SG.N REFL happen.INF   
 ‘(... so, no one knows,) what could happen ...’ (Csb; DiG)

No clitic clusters were attested in clausal first position in Kashubian.

### 10.7.3 Clitic climbing

Clitic climbing is recorded for Kashubian six times vs. one locally placed clitic and an unclear case, cf. Table 10.95. See example (10.69b) above for clitic in-situ placement. Utterance (10.65a) repeated in (10.71) below represents the inconclusive case of an embedded clitic adjacent to the topicalised infinitive it is dependent on.

Table 10.95: Frequencies of clitic climbing and no climbing in spoken Kashubian.

| Locality        | Absolute frequency | Relative frequency<br>w/i complex clauses |
|-----------------|--------------------|-------------------------------------------|
| CC              | 6                  | 75.0%                                     |
| noCC            | 1                  | 12.5%                                     |
| CC?             | 1                  | 12.5%                                     |
| Complex clauses | 8                  | 100%                                      |

(10.71) [...] i so<sub>2</sub> wëprostowac<sub>2</sub> ni młegła<sub>1</sub>.

and REFL.ACC straighten NEG can.PST.SG.F

‘... and she couldn’t straighten up.’

(Csb; DiG)

Several target positions have been encountered for CC, but the small data sample leaves us with gaps in the distribution continuum, cf. Table 10.96.

Table 10.96: Frequencies of target positions under clitic climbing in spoken Kashubian.

| Position after CC | Absolute frequency | Relative frequency w/i CC |
|-------------------|--------------------|---------------------------|
| 1P                | —                  | —                         |
| 2P                | 5                  | 71.4%                     |
| 3P                | —                  | —                         |
| 4P                | 1                  | 14.3%                     |
| 5P                | 1                  | 14.3%                     |
| DP                | 2                  | 28.6%                     |

Most tokens ( $n = 5$ ) appear in 2P, which, at best, could be indicative of an underlying distributional pattern. Example (10.71a) shows that the reflexive clitic climbed from the embedded infinitive, whereby the collocation *sztrydowac so* ‘(to) argue’ is borrowed from and modelled after German *sich streiten* ‘(to) argue’. Note that this example seemingly shows a clitic occurrence under agreement drop but recall that Kashubian utilises an East-Slavic-type past tense formation without an overt auxiliary element across all numbers and persons as one of several competing means. This is why agreement drop is not taken into consideration for Kashubian. Examples (10.72b) and (10.72c) are abridged repetitions of (10.70b) and (10.66e) respectively.

- (10.72) a. Jy<sup>1</sup> so<sup>2</sup> [PP z niom] zaczął<sup>1</sup> sztrydowac<sup>2</sup> [...] [noVA]  
 I.NOM REFL.ACC with her.INS start.PST.SG.M argue.INF  
 ‘I started to argue with her ...’
- b. [...] co<sup>1</sup> bē<sup>2</sup><sub>1</sub> mległo<sup>3</sup> sã<sup>4</sup> stac<sup>2</sup> [...] [2VA]  
 what COND can.PSG.SG.N REFL.ACC happen.INF  
 ‘... what could happen ...’
- c. [...] ż’e=bē<sup>1</sup> jeszcze<sup>2</sup> [przez wiele, wiele lat]<sup>3</sup> mległo<sup>4</sup> so<sup>5</sup>  
 that=COND still through many many year.GEN.PL can.PST.SG.N REFL.ACC  
 rozwijac<sup>2</sup> [...] develop.INF  
 ‘... that it [= Kashubian literature] could develop for many many years to come ...’  
 (Csb; DiG)

Table 10.97: Frequencies of simultaneous occurrences of clitic climbing and (non-)verb-adjacency in spoken Kashubian.

| Placement after CC | Abs. freq. | Rel. freq. w/i CC |
|--------------------|------------|-------------------|
| noVA               | 1          | 14.3%             |
| 1VA                | 4          | 57.1%             |
| 2VA                | 2          | 28.6%             |

The only example for a non-verb-adjacent after clitic climbing is already cited in (10.72a), while the two instances of circumposition are presented in (10.72b–c). I provide (10.73) as an example for a mono-verbal contact configuration of clitic climbing for the sake of completeness.

- (10.73) ([...] bēlebē ten efekt <sup>1</sup>ysygnónc,) no [1VA]  
 as\_long\_as this.ACC.SG.M.INA effect.ACC.SG reach.INF well  
 jaki sa<sup>2</sup> chce<sup>1</sup> <sup>1</sup>esygnónc<sup>2</sup>  
 which.ACC.SG.M.INA REFL want.PRS.3SG reach.INF  
 ‘(... as long as the effect is achieved), well, which one wants to achieve’  
 (Csb; DiG)



## 10.8 A comparative view on West Slavic

### 10.8.1 Clitic placement

Describing each lect successively on an individual basis, the reader might have already taken notice of some resemblances and differences among the languages. In this and the following sections, I will provide a systematic comparison (supported by diagrams) for the patterns across languages. I try to cover all six scrutinised varieties, whenever possible. However, especially Silesian and Kashubian data are scarce, such that a healthy portion of caution is needed, when integrating them into the big picture. Remember that Czech, Polish, and Goralish are the three “core” languages of the present study, while Slovak, Silesian, and Kashubian are treated as supplemental data only. Finally, as the comparison of three up to six lects renders a typological and, thus, comparatively broad account, the degree of detail necessarily must be reduced and the level of abstraction raised (cf. Klavans 1995: 17–18). Therefore, direct comparison will be based on the overall data sets. Neither will I provide a detailed discussion for individual clitics across languages, nor for clitic categories. Furthermore, I will focus on phenomena, which may be regarded as most important for the typology of cliticisation in Slavic.

As previously mentioned, clausal positioning is one of the main properties of cliticisation in Slavic (and beyond) and a key feature for typological classification. First and foremost, a wide range of different positions is attested for all languages, as summarised in Table 10.98.

Table 10.98: Range of attested clitic positions (from 1 to *n*) per lect.

| Slk | Cz | Pol         | Gor | Sil | Csb |
|-----|----|-------------|-----|-----|-----|
| 1P  | 1P | 1P          | 1P  | 1P  | 1P  |
| ⋮   | ⋮  | ⋮           | ⋮   | ⋮   | ⋮   |
| 7P  | 7P | 6P<br>(10P) | 7P  | 7P  | 5P  |

As we already saw, the positions are not equally distributed with respect to frequency. Let us take measures of central tendency into account, which are given in Table 10.99 for the six lects. There are differences in arithmetic means ( $\bar{x}$ ), which result from differences in the individual frequency distributions.<sup>560</sup> But the numbers themselves are not telling, as the arithmetic mean is known for being

<sup>560</sup> The standard deviation (SD) is provided for the sake of completeness. It indicates whether there is more or less variation in the data sample in terms of deviations from the arithmetic mean ( $\bar{x}$ ). Observe that Czech has the lowest SD, while the Lechitic lects show higher ones. The Silesian and especially Kashubian data exhibit very high standard

prone to bias from extreme values. In contrast, the median (*Md*) and the mode (*Mo*) are more robust against outliers. The median is the value that splits the ordered values of a data sample in two halves. Accordingly, one half contains values that are not higher than the median, while the other half contains values that are not smaller than the median. The median's value is 2 across all lects, i.e. at least half of the data sample represents second or first position placement. Finally, the mode simply shows which value occurs most often in a data set. Again, it is 2 in all scrutinised data samples. Thus, we can say that the three measures of central tendency converge, as the arithmetic mean approximates the value of 2 in most cases (Kashubian seems to be an outlier). From this it follows that the overall tendency is to position short pronouns and reflexives in accordance with Wackernagel's law in West Slavic spoken lects.

Table 10.99: Measures of central tendency (mean  $\bar{x}$ , median *Md*, mode *Mo*) for clitic positioning across lects.

|     | $\bar{x}$ (SD)      | <i>Md</i> | <i>Mo</i> |
|-----|---------------------|-----------|-----------|
| Slk | 2.18 ( $\pm 0.65$ ) | 2         | 2         |
| Cz  | 2.15 ( $\pm 0.54$ ) | 2         | 2         |
| Pol | 2.31 ( $\pm 0.75$ ) | 2         | 2         |
| Gor | 2.34 ( $\pm 0.74$ ) | 2         | 2         |
| Sil | 2.41 ( $\pm 0.80$ ) | 2         | 2         |
| Csb | 2.63 ( $\pm 0.91$ ) | 2         | 2         |

The diagram in Figure 10.6 shows the proportions of 1P, 2P, and DP in the examined languages. It nicely illustrates that Wackernagel positioning is the predominant pattern across all languages, although the frequencies are less pronounced for Kashubian. The diagram furthermore shows that second position placement is not exclusive in any of the languages. The crucial differences are in the proportions, especially between 2P and DP, which brings us back to the varying mean values. One may hypothesise on the basis of the bar chart that Czech and Slovak pattern alike, but not Czech and Polish. The latter should cluster with Goralish and Silesian, in turn. The Kashubian data appear to be rather idiosyncratic. A pairwise test of homogeneity for the distribution patterns from Figure 10.6 shows that only the language pairs Polish-Goralish ( $\chi^2 = 4.73$ ,  $p = .05$ ,  $df = 2$ ), Polish-Silesian ( $\chi^2 = 5.98$ ,  $p = .05$ ,  $df = 2$ ), and Goralish-Silesian ( $\chi^2 = 3.44$ ,  $p = .05$ ,  $df = 2$ ) can be deemed to exhibit considerable resemblance, as the chi-square test is not significant here. Remarkably, the distribution frequencies of Czech and Slovak are distinct despite appearances; the test for homogeneity yields a

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deviations here, which is less surprising towards the background of the strongly restricted data samples. Slovak is in an intermediate area between Czech and Polish/Goralish as to the amount of variation that we encounter. However, bear in mind that Slovak, too, is represented by a smaller data set than our core languages.

significant result ( $\chi^2 = 8.19$ ,  $p = .05$ ,  $df = 2$ ). As could already be expected from the similarities between Goralish and Silesian with Polish, neither of these lects pattern with either Czech or Slovak.<sup>561</sup> However, the variation among the triplet {Polish, Goralish, Silesian} is significantly high ( $\chi^2 = 9.65$ ,  $p = .05$ ,  $df = 4$ ), so that no common distribution pattern for all three lects can be assumed.

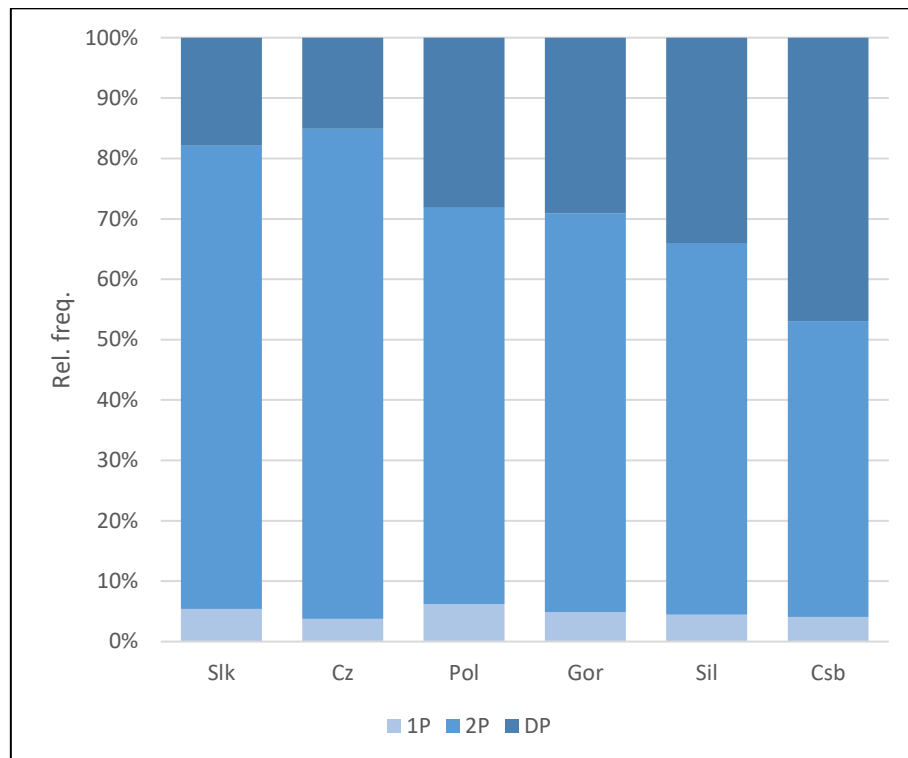


Figure 10.6: Proportions of position (1P, 2P, DP) across lects

As can also be read from Figure 10.6, clitic first position placement (1P) represents a marginal pattern in the examined languages and ranges roughly between 4% (Czech) and 6% (Polish). Two environments for a clitic in first position have been detected. It either follows a coordinating conjunction or a clause boundary. Both environments are attested in all languages except Kashubian, for which only two 1P tokens have been encountered due to scanty data. The diagram in Figure 10.7 shows, however, that languages differ as to how often either configuration occurs. While post-conjunctive clitics occur in 20% of cases in Czech, it is almost 70% in Polish. Slovak (45.5%) comes closest to an even distribution. As a tendency, Goralish and Silesian show more post-conjunctive than boundary-adjacent placement, but the tendency is less pronounced for Goralish, while for Silesian this is based on a small data set. The evidence shows that clitics can be placed after coordinating conjunctions, which contradicts prescriptive norms and has been regarded – erroneously – as a

<sup>561</sup> In short, all chi-square tests give significant results: Czech-Goralish ( $\chi^2 = 174.95$ ,  $p = .05$ ,  $df = 2$ ), Czech-Silesian ( $\chi^2 = 78.97$ ,  $p = .05$ ,  $df = 2$ ), Slovak-Goralish ( $\chi^2 = 38.84$ ,  $p = .05$ ,  $df = 2$ ), Slovak-Silesian ( $\chi^2 = 34.33$ ,  $p = .05$ ,  $df = 2$ ).

peculiarity of Polish in the formal literature (cf. Franks & King 2000: 39, 63, 134, Franks 2000: 37, Franks 2010: 142 for contradicting data points).

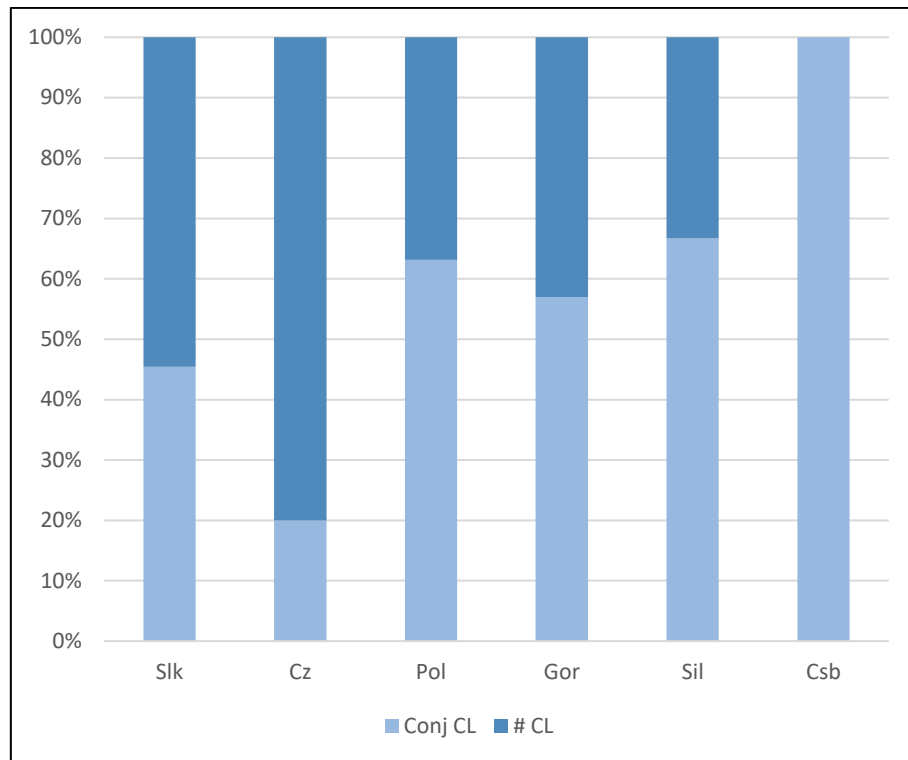


Figure 10.7: Proportion of CL following coordinating conjunction (CONJ) vs. clause boundary (#) in 1P across lects

The distribution patterns plotted in Figure 10.6 share the disadvantage of abstracting away from the different placement options that are subsumed under the label of Delayed Placement (DP). We saw in the respective sections on clitic placement that the putative DP category is internally nuanced. The individual clitic positions are graded regarding their likelihood of occurrence. Simply put, the greater the distance from 2P the lower the occurrence frequency. This pattern recurs across lects as illustrated in Table 10.100, which summarises and confronts the frequency distributions encountered in the studied lects in the form of a ranking.

Table 10.100: Ranking of positional placement patterns per lect.

|      |    |   |    |   |    |   |    |     |    |   |    |     |      |   |      |
|------|----|---|----|---|----|---|----|-----|----|---|----|-----|------|---|------|
| Slk: | 2P | > | 3P | > | 1P | > | 4P | >   | 5P | > | 6P | ( > | 7P ) |   |      |
| Cz:  | 2P | > | 3P | > | 1P | > | 4P | >   | 5P | > | 6P | ( > | 7P ) |   |      |
| Pol: | 2P | > | 3P | > | 1P | / | 4P | >   | 5P | > | 6P | ( > | ...  | > | 10P) |
| Gor: | 2P | > | 3P | > | 4P | / | 1P | >   | 5P | > | 6P | ( > | 7P ) |   |      |
| Sil: | 2P | > | 3P | > | 4P | / | 1P | ( > | 5P | > | 6P | /   | 7P ) |   |      |
| Csb: | 2P | > | 3P | > | 4P | > | 1P | /   | 5P |   |    |     |      |   |      |

Although there are minor differences in the details (especially considering Clitic First positioning) and some uncertainties, some of which might be artefacts of low frequencies of occurrence, the overall pattern is shared by all lects. It is interesting to note that the pattern repeats quite robustly even with small sample sizes like in Kashubian.

Laterality is a feature, which has not been discussed extensively in the literature is linked to the issue of verb-adjacency and restrictions on rightwards clitic placement. Three basic configurations were identified, i.e. left laterality with two manifestations depending on whether there is contiguity with the predicate and right laterality: left-distant [Ld], left-adjacent [La], and right (adjacent) placement [R(a)]. The latter corresponds to clitic post-position on the verb. Figure 10.8 shows the proportions of the three configurations without considering position for the sake of clarity.

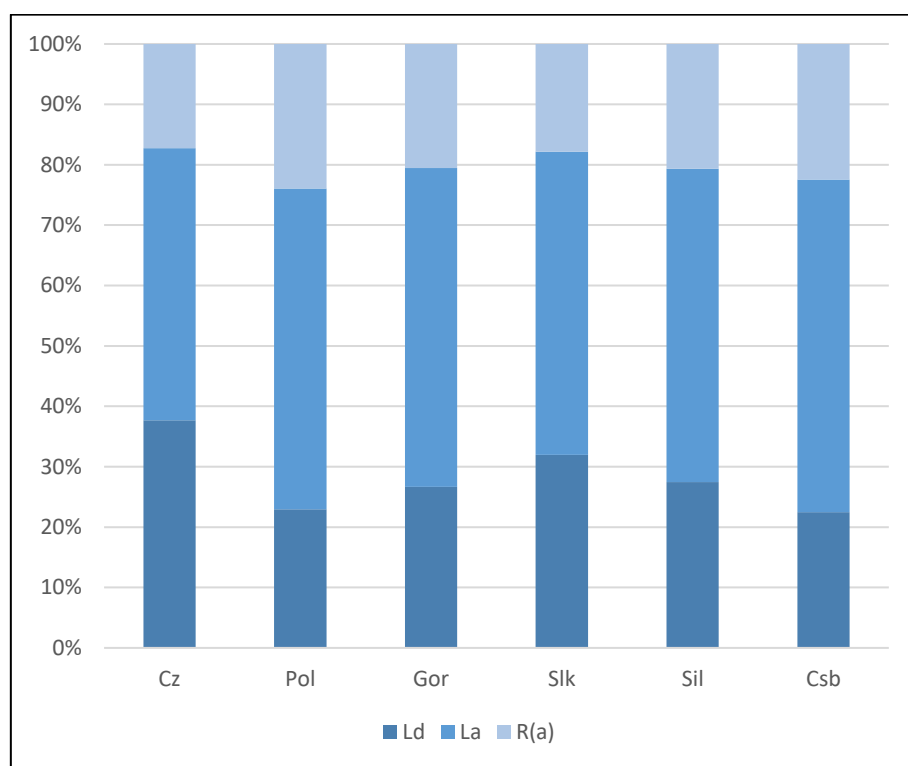


Figure 10.8: Proportion of laterality configurations [Ld, La, R(a)] per lect

Across all examined West Slavic languages, left lateral placement is the most frequent placement pattern and oscillates around 80%. Again, Polish (R: 75.9%) and Czech (R: 82.6%) appear to present extreme points of the continuum. There are differences in the proportions of left-distant and left-adjacent placement, but left-adjacent placement always exceeds left-distant placement. Czech exhibits the lowest discrepancy between both configurations, while we find the highest in Polish (and Kashubian). The other lects are in an intermediate area, but the trend for Goralish and Silesian tends towards Polish. The ratios of Left distant to Left adjacent are given in Table 10.101 for the sake of convenience. The smaller the value the more dominant is verb-adjacency within left laterality.

Table 10.101: Ratio of left-distant placement to left-adjacent placement per lect.

| Cz   | Pol  | Gor  | Slk  | Sil  | Csb  |
|------|------|------|------|------|------|
| 0.84 | 0.43 | 0.50 | 0.64 | 0.53 | 0.41 |

The bar chart in Figure 10.8 suggests that there is a considerable amount of right lateral placement in all lects. Czech and Slovak exhibit the lowest frequencies with 17.2% and 17.8% respectively. Polish and Kashubian represent the upper bound with 23.9% and 22.4% respectively. Goralish and Silesian are located in between with 20.5% and 20.6% respectively. Recall however that, while right placement corresponds almost entirely to clause-initial verb placement in Czech (the intersection of DP and R(a) makes merely 0.8%), there are higher frequencies of post-posed clitics in Delayed Placement attested in the other lects (Pol: 5.8%, Gor: 6.4%, Slk: 3.7%, Sil: 9.1%, Csb: 6.1%).

### 10.8.2 Clitic clustering

Clitic clusters can be of considerable length according to the literature. Franks & King (2000: 102, 105, 129) cite Czech and Slovak examples with up to five clitics in a cluster, while the Czech example (10.74) from Rezac (2005) provides a maximal cluster of seven clitics.<sup>562</sup> However, it has been observed that such large clusters are rather rare in language usage (cf. Kolaković et al. 2022: 100, 191–192; Ćavar & Seiss 2011: 147 for BCMS; Ivanová et al. 2021: 357 for Slovak). Ćavar & Seiss (2011: 147, n. 6) remark “that clitic clusters with more than two [...] clitics are very rare in language use data and corpora, e.g. in the [Riznica] Croatian Language Corpus.” Kolaković et al. (2022: 191–192) found clusters of up to 3 clitics in a corpus of spoken Bosnian. Similarly, Ivanová et al. (2021: 357) report clusters of at most 3 clitics for written Slovak as a foreign language. By and large, this observation holds for the data samples analysed for the present thesis, too. The maximal cluster either does not occur in actual speech or it is so rare that the likelihood of observing one in a random sample is rather low. The longest clusters that were encountered in the present study’s data samples comprise 4 clitics attested in Czech (0.1%), Polish (0.03%), Goralish (0.04%), and Slovak (0.2%). Such cases must be considered rare. They always involve a conditional and an auxiliary clitic followed either by a reflexive and pronominal clitic or two pronominal clitics. Silesian and Kashubian only exhibit clusters of at most 2 clitics, which is likely to be an artefact of small sample size. Finally, I would like to point out that the length of clitic clusters depends also on which elements are treated as clitics within a study’s framework. Recall that the present thesis only recognises conditional, auxiliary,

<sup>562</sup> The sequence *by (j)sem* (or *bysem*) is colloquial and alternates with *bych*, the latter being the only form accepted by the written standard norm of Czech.

reflexive, and pronominal clitics. Thus, other elements that are often considered clitics like demonstrative *to* or adverbs like *prý* and *však* in Czech are out of scope.

- (10.74) Já by jsem si ti mu je však  
 I.NOM COND AUX.1SG REFL.DAT you.DAT<sub>ETH</sub> him.DAT them.ACC however  
 přála vysvětlit.  
 wish.PST.SG.F explain.INF  
 ‘You know though, I would wish to explain them to him.’ (Cz; Rezac 2005: 105)

More importantly, clitic cluster formation is treated as one of the defining properties of special clitic languages in Slavic and beyond and thus plays a decisive role in the typological evaluation of a language’s clitic system. Diaclisis has been a contentious issue in the literature. Figure 10.9 compares the proportions of clustering and diaclisis among occurrences of multiple clitics across languages.

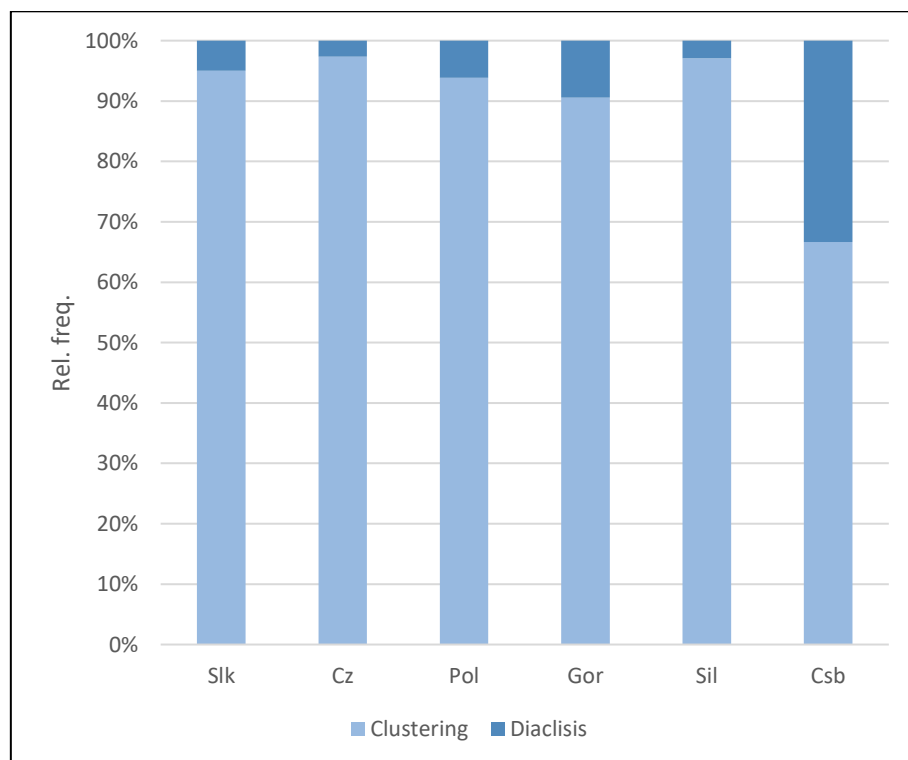


Figure 10.9: Proportion of clustering vs. diaclisis in environments with multiple clitics

It needs to be highlighted that clustering is highly preferred in all studied lects including Polish, where clitics cluster in circa 94% of cases. In contrast, diaclisis is clearly dispreferred in both Czech-Slovak and Lechitic lects. However, the occurrence of non-contiguous clitics is attested for all languages. We find the highest ratio of 9.4% in Goralish, while the lowest is found in Czech with 2.7%. Polish

(6.1%) and Slovak (5.0%) are in between. The numbers for Silesian (2.9%) and Kashubian (33.3%) are rather uninformative due to low token frequencies.

We saw that diacclisis is licensed in two different environments. Either it occurs in a simple clause as split of co-dependent clitics, or it occurs in a complex clause with split clitics licensed by different verbal heads. As mentioned in Section 10.2.2, the mere classification of diacclisis by either environment is vacuous. Instead, the relative frequency of diacclisis within each environment must be calculated. Table 10.102 recapitulates the relative frequencies of diacclisis in simple and complex clauses for the six lects studied here. Silesian and Kashubian are provided for the sake of completeness only, for they do not provide us with sufficient information. Briefly, diacclisis is attested in both environments for all languages (except Silesian), but the ratio is generally higher for complex clauses than for simple ones. The table also provides the factor as an approximative measure for the different probability of diacclisis in the respective environments. It indicates how many times higher the likelihood is for diacclisis to occur in complex environments compared to simple ones. Only Goralish aberrates from the other languages in showing a slightly higher likelihood of diacclisis in simple clauses, but the difference between both environments is marginal. To conclude, diacclisis is extremely rare – even in Polish, which exhibits frequencies comparable to Czech. The likelihood of splitting clusters is higher for clitics that are not dependents of the same predicate, i.e. pseudo-diacclisis in Kolaković et al.’s (2022) words, than for co-dependent clitics. Goralish appears as an outlier due to a more or less even distribution across environments. Slovak deviates, too, but because of an incomparably high ratio of diacclisis in complex clauses. A similar observation can be made for Kashubian. The question is then whether such biases are a result of the small sample sizes.

Table 10.102: Relative frequencies of diacclisis in simple and complex clauses across lects.

|                              | Slovak  | Czech   | Polish  | Goralish | Silesian | Kashubian |
|------------------------------|---------|---------|---------|----------|----------|-----------|
| Simple clause<br>Rel. freq.  | 0.7%    | 0.5%    | 1.0%    | 1.9%     | —        | 2.0%      |
| Complex clause<br>Rel. freq. | 6.6%    | 2.4%    | 2.8%    | 1.3%     | 5.9%     | 12.5%     |
| Factor                       | 9 (9.4) | 5 (4.8) | 3 (2.8) | 1 (0.7)  | —        | 6 (6.3)   |

Let us have a look at the categories between which splits have been discovered. We will focus on the environment of simple clauses, because these are of special interest for the issue of clitic clustering. In contrast, diacclisis among non-co-dependent clitics is a trivial case and narrows down to the question of whether clitic climbing has occurred or not. Table 10.103 shows split types per lect; attested splits are marked by a plus sign (+), while empty cells represent a gap in the data. Due to the



richer database, there is more to say about the three core lects, Czech, Polish, and Goralish than for the supplemental lects Slovak, Silesian, and Kashubian. Focusing on commonalities, one observes that Czech, Polish, and Goralish share six split types: (COND || REFL), (REFL || COND), (COND || PRN), (AUX || REFL), (AUX || PRN), and (REFL || PRN), the latter three of which are also found in Slovak. Note also that the split type (COND || REFL) is attested in Kashubian, while the split type (AUX || REFL) occurs in Silesian. The table's rows are arranged such that more commonly shared patterns cluster. We may infer from the data that, while clitics highly prefer to cluster, splits may occur between any two categories of clitics, i.e. conditional, auxiliary, reflexive, and pronominal. The only exception so far is the sequence of conditional and auxiliary clitic, for which no split has been attested in any of the data sets. Perhaps, this could reflect some sort of special cohesiveness of the two categories.<sup>563</sup>

Table 10.103: Attested splits between clitic categories in simple clauses across lects.

| Split type   | Slovak | Czech | Polish | Goralish | Silesian | Kashubian |
|--------------|--------|-------|--------|----------|----------|-----------|
| AUX    REFL  | +      | +     | +      | +        | +        |           |
| AUX    PRN   | +      | +     | +      | +        |          |           |
| REFL    PRN  | +      | +     | +      | +        |          |           |
| COND    REFL |        | +     | +      | +        |          | +         |
| REFL    COND |        | +     | +      | +        |          |           |
| COND    PRN  |        | +     | +      | +        |          |           |
| PRN    REFL  |        |       | +      | +        |          |           |
| PRN    COND  |        |       | +      |          |          |           |
| REFL    AUX  |        |       |        | +        |          |           |
| PRN    AUX   |        |       |        | +        |          |           |
| PRN    PRN   |        |       |        | +        |          |           |

Finally, taking cluster-internal word order into consideration, we should acknowledge that the linearisation templates usually described for the languages under scrutiny are confirmed by the empirical record. Instances of reverse ordering are rare and negligible, which is especially interesting in the case of Polish, which appears to be quite well-behaved, thus, resembling special clitic languages in this respect. On the other hand, two conflicting linearisation patterns were recognised for Goralish with either “Polish type” DAT < REFL or “Czech-Slovak type” REFL < DAT. While the former is quantitatively predominant, the latter is almost equally frequent (ratio of 6 : 5). It is fair to hypothesise

<sup>563</sup> Accordingly, one might put forward the hypothesis that this special cohesiveness mirrors the fact that the sequence of COND + AUX does not represent a syntactic linearisation but rather a morphological concatenation. Put differently, the conditional clitic by could be inflected for number and person, in fact.

that the competing linearisations may reflect the areal-linguistically intermediate position of Goralish between Polish and Slovak.<sup>564</sup> In contrast, Silesian does not seem to show here a mixed or double nature with respect to its areal-linguistic proximity to Czech. Table 10.104 summarises the ascertained linearisation patterns for each lect. Note that the cluster-internal word orders reported in Table 10.104 is based only on the empirical attestations, i.e. on the data elicited from the corpora and dialectological compendium. Neither are they intended to illustrate maximal clitic clusters nor are they intended to make theoretical statements about the internal organisation of the clitic cluster in the respective lects. Accordingly, there is no information on *-li* for Czech, simply because *-li* is absent from colloquial speech. I do also not provide details for the linearisation of ethical dative and argument dative, because there have been hardly any co-occurrences of ethical and argument datives in the data. I encountered only three instances, whereby two tokens occurred in the Slovak and one in the Goralish data sample, cf. (10.75). Finally, the Kashubian data are scarce, such that the table gives only a small segment of a hypothetically bigger cluster. Although the scanty Kashubian data do not allow for solid generalisations, the observed linearisations perfectly fit to the bigger picture of West Slavic or Slavic in general.

- (10.75) a. (Sa        stalo        toto ,        že    vieš ,)  
REFL.ACC happen.PST.SG.N this.NOM.SG.N that know.PRS.2SG  
čo    **ti**        **mu**        povie ?  
what you.DAT.ETH him.DAT tell.PRS.3SG  
‘(It happened that you know) what he is telling him, right?’
- b. (výber        som        mal ,        som        si        myslel .)  
selection.ACC.SG AUX.1SG have.PST.SG.M AUX.1SG REFL.DAT think.PST.SG.M  
A    on        **ti**        **mi**        zahlási ,        Pa’o        a  
and he.NOM you.DAT.ETH me.DAT say.PRS.3SG Pavol.DIM.NOM and  
borovičku        ne-máš ?  
borovička.ACC.SG NEG-have.PRS.2SG  
‘(I had a [good] selection [of beverages], I thought.) And he is saying to me – Pavol, but don’t you have borovička [= a spirit from juniper berries]?’ (Slk; SNK s-hovor.7.0)
- c. a    a    on        **ci**        **mi**        gada        že [...] (Gor; KoSpi)  
and and he.NOM you.DAT.ETH me.DAT say.PRS.3SG that  
‘and you know, he tells me that ...’

<sup>564</sup> One may furthermore speculate that the higher frequency of DAT > REFL might be indicative of a change in progress in the Spisz dialect due to the roofing by Polish. Such a hypothesis requires empirical testing, but it should be possible to approach this question with the Spisz dialect corpus by differentiating the respective linearisations patterns by age. I will not address this question here for it is beyond the present thesis’s scope.

Table 10.104: Attested (dominant) word orders within West Slavic clitic clusters.

| Language   | Linear order                                                          |
|------------|-----------------------------------------------------------------------|
| Czech:     | COND < AUX < REFL < DAT < ACC                                         |
| Slovak:    | COND < AUX < REFL < DAT < ACC                                         |
| Polish:    | COND < AUX < DAT < REFL < ACC / GEN                                   |
| Goralish:  | i. COND < AUX < DAT < REFL < ACC<br>ii. COND < AUX < REFL < DAT < ACC |
| Silesian:  | COND / AUX < DAT < REFL < ACC                                         |
| Kashubian: | COND < REFL < DAT                                                     |

Returning to the issue of single clitics and clitic clusters in first position, we saw that in principle any category of clitic or clitic cluster can occupy clausal first position in West Slavic. As before, the size of the data sample has not only a direct influence on the token quantities, but also the cardinality of different types. The matrix in Table 10.105 confronts the attested categories of clitics and tuples of clitics in absolute first clausal position (# 1P ...) across lects, i.e. without a coordinating conjunction in the left context.

Table 10.105: Single clitic categories and tuples of clitic categories occurring in absolute first clausal position (# 1P ...) across lects.

| Category/Tuple   | Czech | Polish | Goralish | Slovak | Silesian |
|------------------|-------|--------|----------|--------|----------|
| Prn              | +     | +      | +        | +      |          |
| Refl             | +     | +      | +        |        | +        |
| (Cond, Prn)      | +     | +      | +        |        |          |
| (Cond, Refl)     |       | +      | +        |        |          |
| (Aux, Refl)      | +     | +      | +        | +      |          |
| (Aux, Prn)       | +     |        | +        | +      |          |
| (Refl, Prn)      | +     |        | +        | +      |          |
| (Prn, Refl)      |       | +      | +        |        | +        |
| (Prn, Prn)       | +     |        |          |        |          |
| (Cond, Aux, Prn) | +     | +      |          |        |          |
| (Aux, Refl, Prn) | +     |        |          | +      |          |
| (Aux, Prn, Prn)  | +     |        |          |        |          |

### 10.8.3 Clitic climbing

Although it has been particularly discussed for Slavic, clitic climbing has sometimes been treated in purely dichotomic terms of occurring or not (cf. the discussion in Franks & King 2000: 241–248). A question, which barely has been addressed, is the target position a climbed clitic can take (but cf. Marušič 2008 on Slovene). Usually, the target of clitic climbing is idealised as a higher or matrix clause (cf. Junghanns 2002c: 57) or, more specifically, as a higher or matrix second position (cf. Rezac 2005: 105; Tabachnik 2024: 2). However, we observed various positions that a climbed clitic may occupy. Table 10.106 summarises the attested positions of climbed clitics in the individual lects. We encounter a good deal of variation on purely phenomenological grounds. Clitic climbing is attested with target positions from 1P to 6P in Czech, Polish, and Goralish. The range is smaller for the three remaining languages. It is 1P to 3P for Slovak and 2P to 4P in Silesian. There is no coherent range for Kashubian, but 2P, 4P, and 5P are attested. It is reasonable to assume that the reduction of target positions in Slovak, Silesian, and Kashubian could be an artefact of small sample size. Therefore, the supplemental databases need to be treated more cautiously, such that I will especially focus on the core data.

Table 10.106: Range of attested target positions (from ... to ...) under clitic climbing per lect.

| Slk | Cz | Pol | Gor | Sil | Csb |
|-----|----|-----|-----|-----|-----|
| 1P  | 1P | 1P  | 1P  | 2P  | 2P  |
| ⋮   | ⋮  | ⋮   | ⋮   | ⋮   | 4P  |
| 3P  | 6P | 6P  | 6P  | 4P  | 5P  |

We already know that certain constructions and languages differ with respect to occurrence rate, which is also the case for CC target positions. Figure 10.10 provides the bar charts for target positions per language distinguishing between 1P, 2P, and DP. Generally, 2P is the most frequent target position, but we encounter striking differences among the examined lects. The climbing to 2P occurs in over 80% of cases in Czech, while it makes 44% in Polish and 60% in Goralish. DP occurs more frequently in Goralish and especially Polish under clitic climbing. Slovak patterns with Czech, but provided the smaller data set and the limited range of recorded target positions, this observation is rather tentative. Silesian and Kashubian do not provide conclusive information here and occur in the chart bar for the sake of completeness only.

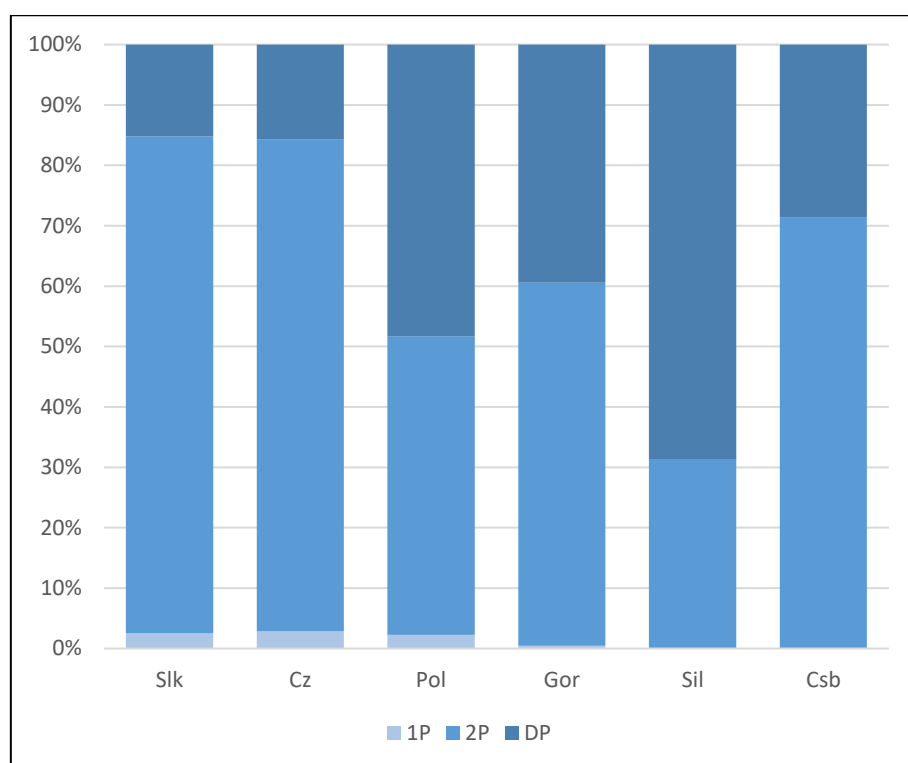


Figure 10.10: Proportion of target positions under clitic climbing per lect

As mentioned earlier, e.g. in Section 10.8.1, the tripartite distinction between 1P, 2P, and DP is rather coarse, as the subsumption of late clitic positions under the DP label comes about with a loss of information. Taking the distinctions within DP into consideration, we arrive at a preference distribution or placement hierarchy of target positions under CC across languages in Table 10.107. By and large, the overall preference pattern known from general clausal placement repeats. Clausal second position is the dominant or preferred option to place a clitic after it has climbed. Delayed positions become less likely with increasing distance. Some minor vagueness occurs with respect to the position of 1P in the hierarchy. Furthermore, some caution is needed regarding the preference of 3P over 2P in Silesian (due to gaps in the data).

Table 10.107: Ranking of target positions under clitic climbing per lect.

|      |    |   |    |   |    |   |    |   |    |   |    |
|------|----|---|----|---|----|---|----|---|----|---|----|
| Slk: | 2P | > | 3P | > | 1P |   |    |   |    |   |    |
| Cz:  | 2P | > | 3P | > | 4P | / | 1P | > | 5P | / | 6P |
| Pol: | 2P | > | 3P | > | 4P | > | 1P | / | 5P | > | 6P |
| Gor: | 2P | > | 3P | > | 4P | > | 5P | > | 1P | / | 6P |
| Sil: | 3P | > | 2P | > | 4P |   |    |   |    |   |    |
| Csb: | 2P | > | 4P | / | 5P |   |    |   |    |   |    |

Besides target position, the co-occurrence of clitic climbing and verb-adjacency is of particular interest for a micro-typological survey. The relevant data are visualised in Figure 10.11. Generally, we find that clitic climbing may result in contiguity as well as distant placement to a verb. The three verb-adjacency configurations [noVA, 1VA, 2VA] are attested in all lects except Silesian, which does not provide evidence of noVA. We observe a great portion of variation and distinct distribution patterns again. Verb-adjacency is the dominant pattern under clitic climbing in West Slavic. The smallest ratio is 78.2% found with Czech (followed by Slovak with 82.3%). Polish provides the upper frequency bound for VA with 92.3% (and disregarding Silesian without any noVA placement). Goralish is closer to Slovak and exhibits a ratio of 85.4% for VA. Considering the proportion of mono-verbal (1VA) vs. bi-verbal configurations (2VA), the former generally outnumber the latter ones. Polish is exceptional, because circumposition (2VA) is the most-frequent pattern here. In fact, we can trace this back to the larger sample size of pronominal clitics, which exhibit a pattern quite distinct from the reflexive ones. Abstracting away from reflexives, the Polish facts are insofar remarkable as we observe a significantly different situation from the other data sets (except the scanty Silesian data showing a 50-50 distribution).

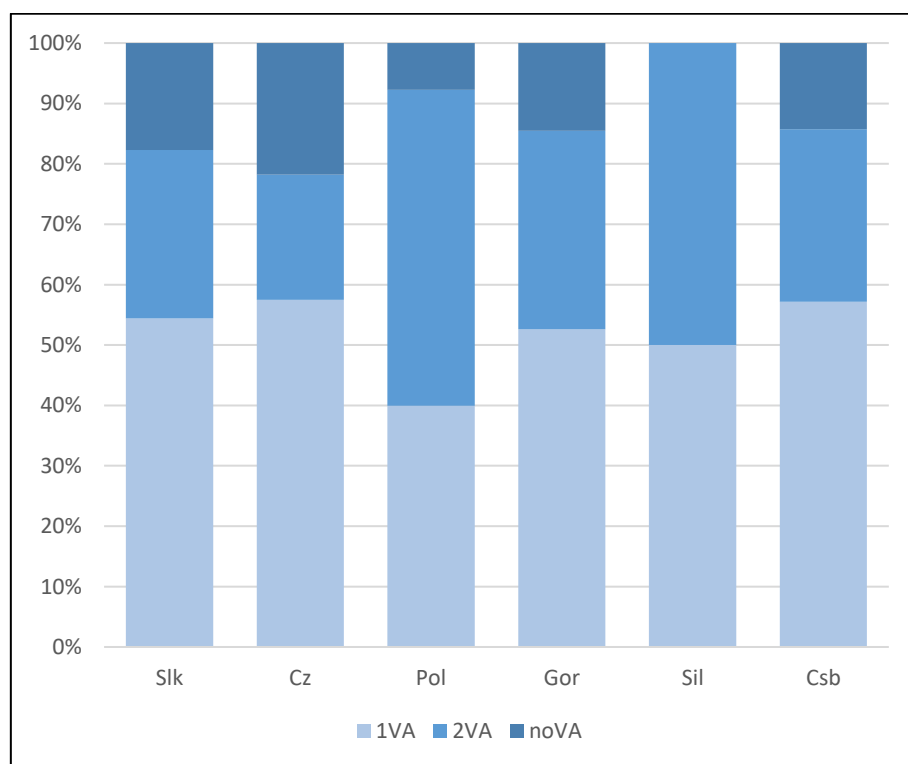


Figure 10.11: Proportion of verb-adjacency configurations under clitic climbing across lects

Finally, the attested cases of clitic placement in an intermediate position within complex clauses without verb-adjacency (labelled CC?) are of particular interest, because they elude an unequivocal structural interpretation as either CC or in situ placement. These syntagms are in stark contrast to the

well-behaved constructions usually found in the syntactic literature on the phenomenon in Slavic, which abounds in introspectively constructed examples. Actual language use displays far more complexity and variation than those occurrences the human mind purposefully invents to make the case for certain arguments or generalisations.

#### 10.8.4 Agreement drop

The construction of agreement drop occurs in several West Slavic lects, i.e. Czech, Polish, Goralish, Silesian.<sup>565</sup> However, the construction was only attested in the Silesian and especially Goralish data sample. I encountered exactly one occurrence for Czech and none for Polish and Slovak. However, it is not difficult to obtain more data for Czech, Slovak, and Polish by purposeful corpus queries. The present section is rather brief, for the majority of agreement drop data is Goralish and was already discussed in Section 10.4.4. In short, the data from the different spoken lects converge so far. Agreement drop is attested only for 1st person singular and plural; second person agreement drop does not occur. The most frequent linearisation is  $SBJ_{PRN} < CL < V$ , which is attested in Czech, Goralish, and Silesian. There is only one instance of a reversed order of clitic and verb yielding  $SBJ_{PRN} < V < CL$  from Goralish, which can be attributed to information structure, for the verb is focused, in fact. So, the linearisation may be captured by the two ordered pairs  $(SBJ_{PRN}, CL)$  and  $(SBJ_{PRN}, V)$ . Most occurrences of agreement drop present a compact syntagm of subject pronoun, clitic, and *l*-participle being linearised contiguously (attested in Czech, Goralish, and Silesian). If the syntagm is not contiguous, then subject and clitic remain adjacent in all these cases, while the verb is disjoint. The precedence and contiguity patterns seem to be very robust. The linearisation pattern  $SBJ_{PRN} < CL$  has repercussions for clitic positioning. The left-most position for a clitic is 2P, it cannot occur in 1P, at all. Furthermore, the clitic occupies third position compulsorily in embedded clauses headed by a subordinator. As far as I know, this observation has not been made before. Bear in mind that a similar effect (although less pronounced) is observable in independent main clauses, if pragmatically prominent elements occupy the left periphery. For instance, left-peripheral Focus phrases as in (10.16) repeated as (10.76) below precede the entire agreement drop construction and shift the  $SBJ_{PRN} < CL$  syntagm to the right with the consequence of delayed clitic placement.

<sup>565</sup> Kashubian also has an agreement drop construction, which, however, patterns differently resembling the East Slavic past tense formation (cf. Ruda 2017, 2022; Stone 1993: 776–777 with references). Therefore, agreement drop in Kashubian should be treated separately from the 1st person agreement drop in other West Slavic lects.

(10.76) tak<sup>1</sup> [ i přes ten telefon ]<sup>2</sup><sub>FOC</sub> ø já<sup>3</sup> ho<sup>4</sup> slyšela  
 so and through this telephone ø I.NOM him.ACC hear.PST.SG.F  
 ‘thus, I heard him even through the telephone’ (Cz; ČNK oral\_v1)

As in standard Czech, agreement drop is regarded as a violation of the normative rules of Polish, but it can be encountered in non-standard speech. Ruda (2017, 2022) does not consider agreement drop in Polish, but occurrences of agreement drop can be elicited from the spoken subcorpus of the NKJP. However, agreement drop seems to have a low frequency. Recall example (4.243) from Section 4.9.3 repeated in (10.77) below.

(10.77) a. (i powtarzam) ja była ø świadkiem tych  
 and repeat.PRS.1SG I.NOM be.PST.SG.F ø witness.INS.SG this.GEN.PL  
 odejść [...]   
 quitting.GEN.PL  
 ‘(and I repeat) I witnessed these quittings ...’  
 b. (ja szedł-e-m kiedyś do Marty)  
 I.NOM go.PST.1SG.M-M-AUX.1SG once to Marta.GEN.SG  
 jeszcze my nie byli ø po ślubie [...]   
 still we.NOM NEG be.PST.PL.VIR ø after wedding  
 ‘(once, I went to Marta) we weren’t married, yet’ (Pol)

There were no data points exhibiting agreement drop in the Slovak data sample, although agreement drop is attested in Slovak as exemplified in Section 4.9.3. Coming back to East Slavic once again, particularly short pronouns in colloquial Russian, there is an interesting parallel with the observed agreement drop patterns in West Slavic. Consider the syntactic strings in (10.78) extracted from (3.53b, 3.53d), (3.54), (3.55a–b), and (3.56b–c) in Section 3.10. The basic pattern seems to be  $SBJ_{PRN} < PRN_{SHORT} < V$  in the form of a compact syntagm, cf. (10.78a–e). If the syntagm’s elements are realised discontinuously, then subject pronoun and short object pronoun remain adjacent while the verb is left behind, cf. (10.78f–h). Example (10.78h) furthermore shows the syntagm in an embedded environment headed by a subordinator. The short pronoun occurs in third position and does not precede the subject. The same linearisation patterns are found in West Slavic agreement drop. While this observation does not automatically imply that West Slavic agreement drop and colloquial Russian short pronominal distribution necessarily require a common analysis, it highlights commonalities, which have gone unnoticed so far and deserve careful study in future research. It also confirms the relevance of contemporary East Slavic for the research in Slavic clitics.



- (10.78) a. [...] ja **te** napominaju [...]  
           ‘... I’m reminding you ...’  
       b. [...] ja **te** grila [...]  
           ‘... I told you ...’  
       c. Ja **te** pokažu [...]  
           ‘I’ll show you ...’  
       d. My **tja** ne boimsja [...]  
           ‘We’re not afraid of you ...’  
       e. [...] ja **te** govorju [...]  
           ‘... I’m telling you ...’  
       f. [...] ja **te** [ večerkom ] pozvonju.  
           ‘... I’ll call you in the evening.’  
       g. [...] on **tja** [ sam ] povedët.  
           ‘... it’ll lead you itself.’  
       h. [...] i čtob<sup>1</sup> ja<sup>2</sup> **tja**<sup>3</sup> [ tut ] [ bol’she ] ne videl.  
           ‘... and don’t let me see you [around] here again.’ (Rus)

Before closing this section, I would like to highlight that the delayed-placement-under-agreement-drop pattern has been discovered thanks to the exploratory corpus study. I was unaware of this pattern, while preparing the study and analysing the data, and had no expectations or hypothesis to come across such configurations.

### 10.8.5 Possessive resultative

First and foremost, the possessive resultative shall be described in short, as it has been treated only cursorily before. According to Giger (2009: 271–272), it is a weakly grammaticalised resultative construction that is attested across the whole of Slavic.<sup>566</sup> Formally, it consists of the present tense form of the verb ‘(to) have’ and a phrasal complement bearing accusative case consisting of an NP and a passive participle, the latter of which is language-specifically overtly realised as either etymologically short or long adjectival form (Pitsch 2023a: 142–144). Pitsch (2023a) analyses the accusative complement of ‘(to) have’ as a small clause (cf. Pitsch 2023a: 154), which shall be discussed briefly at this section’s end. Out of the six data sets that have been examined, it is only the

<sup>566</sup> Giger (2009: 273) informs that the possessive resultative was grammaticalised to a genuine tense form in Macedonian, Kashubian, north-western Russian dialects, and, perhaps, some Bulgarian dialects.

Czech one, where possessive resultatives involving a pronominal clitic were encountered. Therefore, the examples from (10.17) in Section 10.2.5 are repeated in (10.79) below for the sake of convenience. Provided that the possessive resultative's verbal head '(to) have' is subcategorised for a direct object, pronominal clitics necessarily bear accusative case when occurring in this construction.

- (10.79) a. *koukám že<sup>1</sup> ho<sup>2</sup> máš<sub>1</sub> nastudovanýho<sub>2</sub>*  
 see.PRS.1SG that him.ACC have.PRS.2SG study.PASS.PTCP.ACC.SG.M.ANI  
 'I see that you have studied it'
- b. (a *věnec teda mám) ale ne-mám<sub>1</sub> ho<sup>2</sup>*  
 and wreath.ACC.SG then have.PRS.1SG but NEG-have.PRS.1SG him.ACC  
*nastrojený<sub>2</sub>*  
 decorate.PASS.PTCP.ACC.SG.M.INA  
 '(and, so, the wreath, I have it) but I have not decorated it'
- c. *taky<sup>1</sup> ho<sup>2</sup> mám<sub>1</sub> připravenej<sub>2</sub> že nasypu na*  
 also him.ACC have.PRS.1SG prepare.PASS.PTCP.ACC.SG.M.INA that scatter.PRS.1SG on  
*zahrádku*  
 garden.ACC.SG  
 'I have prepared it [= ash], too, [such] that I will scatter the garden'
- d. *no mam<sub>1</sub><sup>1</sup> ho<sup>2</sup> tam master\* masterovanej<sub>2</sub>*  
 yes have.PRS.1SG him.ACC there master.PASS.PTCP.ACC.SG.M.INA  
 'yes, I have him mastered [= equipped with a master hard drive]'
- e. (takže *ho dycky umeju) a mám<sub>1</sub><sup>1</sup> ho<sup>2</sup> tam*  
 so\_that him.ACC always wash.PRS.1SG and have.PRS.1SG him.ACC there  
*pověšenej<sub>2</sub> jako dekoraci*  
 hang.PASS.PTCP.ACC.SG.M.INA as decoration.ACC.SG  
 '(so that I wash it [= iron-cast pan] each time) and I have it hanging as a decoration'
- f. a *už<sup>1</sup> ho<sup>2</sup> mám<sub>1</sub> taky naloženej<sub>2</sub>*  
 and already him.ACC have.PRS.1SG also pickle.PASS.PTCP.ACC.SG.M.INA  
 'I have pickled it, too'
- g. *to jó ale<sup>0</sup> museli<sub>1</sub><sup>1</sup> by<sub>1</sub><sup>2</sup> ho<sub>3</sub><sup>2</sup> mít<sub>2</sub> někde*  
 this yeah but must.PST.PL.VIR COND him.ACC have.INF somewhere  
*zavřenýho<sub>3</sub>*  
 cage.PASS.PTCP.ACC.SG.M.ANI  
 'that's right, but they would have to cage him up somewhere'

h. (no ale musíš a-by se to obnovilo)  
 well but must.PRS.2SG that-COND REFL.ACC this.NOM.SG.N restore.PST.SG.N  
 tak<sup>1</sup> už<sup>2</sup> ho<sup>3</sup> tady máš<sub>1</sub> přijatej<sub>2</sub>  
 so already him.ACC here have.PRS.2SG receive.PASS.PTCP.ACC.SG.M.INA  
 ‘(well, but you must [do it] in order to restore it) thus you already have accepted it here’  
 (Cz)

There were no data points exhibiting possessive resultatives in the Slovak data sample. However, possessive resultatives are common across Slavic including Slovak (cf. Giger 2009: 271–273, 2016; Wiemer & Giger 2005, esp. pp. 97–100). I cite (10.80) for illustrative purposes only as an example of a possessive resultative construction from spoken Slovak, which, however, does not involve clitics.

(10.80) ([...], že najprv otec biskup sa ma spýtal,)   
 that first father bishop REFL.ACC me.ACC ask.PST.SG.M  
 ktoré mám ukončené predmety.  
 which.ACC.PL.NVIR have.PRS.1SG complete.PTCP.ACC.PL.NVIR subject.ACC.PL  
 ‘(... that the bishop asked me first,) which subjects I completed.’ (Slk; SNK s-hovor.7.0)

Pitsch (2023a) analyses the possessive resultative as a construction consisting of a fully-fledged lexical verb ‘(to) have’ that takes a small clause (SC) complement and assigns accusative case to it. The accusative-marked small clause comprises a participle and an agreeing object NP, the latter of which may be phonetically empty. Using a generative syntactic framework, Pitsch (2023a: 154) assumes the structure in (10.81).

(10.81) [FP F<sup>0</sup> [<sub>VP</sub> v<sup>0</sup> [<sub>VP</sub> V<sup>0</sup> [<sub>SC</sub> ACC + PTCP ]]] (Pitsch 2023a: 154)

Leaving technical details aside, the small clause projects a clausal, yet minimal structure on its own. This minimal clause is embedded under the matrix verb ‘(to) have’. Now, if an accusative clitic originates in the small clause and leaves it to enter the matrix verb’s domain, we find a configuration, which resembles clitic climbing.<sup>567</sup> Although I cannot elaborate on this issue here, there are some points to consider in future research. Provided that the small clause is the domain of accusative case licensing, the clitic’s dislocation cannot be motivated by the need to receive case in the matrix phrase. This fits observations by Lenertová (2004), Dotlačil (2004), and I. Kulik (2023), who argue that clitic

<sup>567</sup> See Section 4.7 for clitic climbing.

climbing in Czech (and Polish) cannot be explained by considerations of feature-checking. The resemblance between clitics escaping small clauses and clitic climbing *sensu stricto* furthermore could widen the perspective on clitics in long-distance dependencies. Particularly, the constructions of clitic climbing, dislocation from small clauses, and the so-called possessor raising construction shares the core property of licensing a clitic in an embedded domain, which the clitics escape.<sup>568</sup> However, I am not aware of any thread of research that attempts to investigate these three constructions under a common research question or to propose a unified analysis. While such an endeavour may turn out to be unsuccessful, it is a worthwhile enterprise to deepen our understanding of syntactic structures in Slavic and the principles underlying clitic climbing.

## 10.9 Clitic positioning vs. available permutations

The present section is a follow-up to the distributional analyses presented in the subsections on clitic placement in the individual lects. We will deal with the relation between the clitics' actual position and the hypothetically available positions as a function of an utterance's complexity. This topic has not received much attention in the literature. Admittedly, it has been noticed that clitics can occupy various sites in a clause. For instance, Marušič (2008) informs that clitics in Slovene do not take any specific syntactic position and that several placement options are available. Recall also the quotation from Witkoś (1998: 159) in Section 8.3, who notes that increasing complexity raises the number of potential clitic positions. However, no one has ever attempted to put a clitic's actual position and the hypothetically available positions into relation (at least, I am not aware of such an approach). I will attempt to take a closer look at this question with recourse to the PERMUTATION SPACE or  $\pi$ -SPACE that was introduced in Section 8.3. The  $\pi$ -space is the range of positions available for a clitic between the left clause boundary and the verbal predicate. It is common knowledge in Slavic clitic research that 2P clitics oscillate between the so-called left periphery of the clause and predicate post-position. These two positions mark two terminal points of a one-dimensional space, in which clitics may more or less freely permute with other constituents. The more complex, i.e. the more materially contentful a clause is and the lower or further to the right the verbal predicate occurs in the structure, the wider the permutation space becomes.

Although relying on linearisation rather than hierarchic structure, approaching the  $\pi$ -space needs careful preparation. In fact, one must know first which elements in an utterance are permutable. Furthermore, information about clitic position, distance to the verbal predicate, and laterality is

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<sup>568</sup> The notion of possessor raising refers to the idea that possessive dative clitics originate in the possessum NP's domain but are subsequently moved out of NP. See Section 4.11.2 for the dative of possession.

needed. Finally, clausal structure plays a role, for independent clauses may in principle allow Clitic First, which is unavailable in embedded clauses headed by a subordinator. Fortunately, most of these parameters are already accounted for in the present study's annotation scheme. The only parameter missing is clause type (independent vs. subordinate). Therefore, the following discussion cannot but be approximative, for some imprecision remains due to the lack of information. Nevertheless, this does no harm to the approach for two reasons. First, information about clause type is missing for every data sample, such that languages may still be compared on the very same grounds (i.e. with the same uncertainty). Second, both prescriptive grammarians and formalists generally do not take Clitic First into consideration. Therefore, independent and subordinate clauses are treated in the very same way in the literature, for 2P is considered the left-most position in both syntactic environments. Third, 1P placement is rare, representing not more than 5% of the respective data samples (except for Polish with 6.2% and Slovak with 5.5%). We may therefore increase the level of abstraction and tentatively use a simplified approach, which does not integrate 1P, yielding a more conservative picture. I calculated the  $\pi$ -space by the simple arithmetic procedure in (10.81). The permutation space is the sum of phrasal position ( $Pos_{(XP)}$ ) and distance from the predicate ( $d_{(Pred)}$ ) and laterality ( $L$ ). Laterality  $L$  takes the value of 1 for left lateral placement and 0 for right lateral placement, for no further position to the right is available for the clitic. The general unavailability of the first position was taken into consideration, too, which implicates a reduction by 1. In fact, it is more precise to say that the calculation in (10.82) gives the *cardinality* – or, if one wishes, the *valency*, *arity*, or *adicity* – of the permutation space, but not the  $\pi$ -space itself, which is an abstraction over a syntactic string.

$$(10.82) \quad \pi = Pos_{(XP)} + d_{(Pred)} + L - 1$$

Having calculated the cardinality of the  $\pi$ -space for every token, the problem arises that a simple comparison of the accumulated figures for position and permutation space like in Table 10.108 for Czech and Polish does not provide us with a lot of meaningful information.

Table 10.108 reproduces already known facts: Clitic occurrences become rarer with increasing cardinality of Position. Likewise, increasing size of the permutation space yields decreasing frequencies. But what does this mean at all? Although the figures seem to imply a tendency for short utterances in spoken language, it must be kept in mind that the permutation space is not isomorphic to the overall utterance. Imagine a contentful utterance with 6 freely permutable constituents, one of which was a clitic (e.g. a ditransitive verb with three overtly realised arguments and two adverbs). According to the scheme, the cardinality of the  $\pi$ -space is 5 for a clitic, if the verbal predicate occurred clause-finally. Shifting a verb towards the left periphery successively minimises the permutation

space for the clitic, but it does not change the overall complexity of the clause in terms of its constituents.

Table 10.108: Absolute frequencies of permutation space values and clitic position in spoken Czech and spoken Polish.

| $\pi$ -space value | Czech | Polish | Position | Czech | Polish |
|--------------------|-------|--------|----------|-------|--------|
| 1                  | 479   | 549    | 2P       | 2,381 | 2,001  |
| 2                  | 1,100 | 1,243  | 3P       | 350   | 646    |
| 3                  | 853   | 809    | 4P       | 70    | 172    |
| 4                  | 334   | 316    | 5P       | 14    | 35     |
| 6                  | 115   | 87     | 6P       | 7     | 6      |
| 5                  | 28    | 18     | 7P       | 1     | —      |
| 7                  | 10    | 4      | 10P      | —     | 1      |
| 8                  | 3     | —      |          |       |        |
| 9                  | 1     | 1      |          |       |        |
| 10                 | 1     | —      |          |       |        |
| 12                 | —     | 2      |          |       |        |

So, what we need is the intersection of individual positions and particular  $\pi$ -space values to approach the relation of clitic position and permutation space size. This is why I will use cross tables in the following. Each cell provides the absolute occurrence frequency of a clitic position in utterances with a calculated  $\pi$ -value  $n$ . The tables' colour code follows the model of a heat map as a convenient means of illustration. The two bottom-most rows summarise all occurrences within a column, i.e. of a certain  $\pi$ -space, as well as the relative frequencies per cardinality. Consider Table 10.109, which presents the accumulated values of clitic position vis-à-vis permutation space for Czech. The first column represents  $\pi$ -spaces with cardinality one ( $\pi = 1$ ), i.e. utterances where exactly one position is available for clitics. It should be therefore no surprise that we find only one position in this row and that this position is 2P. The second column covers an increased permutation space with  $\pi = 2$ . There are two positions available for clitics and we find attestations for both positions. However, there is a high concentration of occurrences in 2P, but a low frequency for the next position 3P. The heat map shows two very distinct shades in the respective cells. High frequencies are red, while low frequencies are pale yellow. Column 3 specifies a  $\pi$ -space with 3 clitic sites and so on. Below, Table 10.110 presents the Polish and Table 10.111 the Goralish data. The supplemental object languages Slovak and Silesian are provided in a shared Table 10.112. Kashubian is not represented here due to the small sample size.

Table 10.109–10.112 show that clitic occurrences spread over delayed positions with increasing permutation space size. The expectation for an ideal(ized) Wackernagel clitic language is that clitic positioning would not be affected by the complexity of an utterance. However, we observe that each language makes use of a wide range of positions, if available. This tendency is less pronounced for Czech, where clitics gravitate towards the left periphery of the clause more often even in case of a high cardinality of the permutation space. However, this does not prevent clitics from occurring in the latest available position in principle. In column 6 of Table 10.109, we see that there are attestations for all six available positions of the hexavalent  $\pi$ -space. On the other hand, it does not imply that occupying a late position is mandatory within  $\pi$ -spaces of high cardinality. Such a pattern, whereby clitics would be regularly in the latest (or second-to-last) position, would be expected for verb-adjacent clitics, in fact, but not for second position clitic languages. Therefore, the approach presented here makes testable predictions. Notably, Polish and to some extent Goralish show a pattern with a shift in the frequency distribution. Nevertheless, they do not show a strict VA-clitic pattern. As can be read from the heat map, there are high concentrations of clitic occurrences in the left periphery. Silesian basically follows the Polish pattern, but one must bear in mind that the database is small. Slovak appears to be somewhat of a hybrid between the Czech and Polish type. Perhaps, this case could also be made for Goralish.

Table 10.109: Frequencies of clitic position against permutation space size in spoken Czech.

| Pos \ $\pi$ | 1     | 2     | 3     | 4     | 5    | 6    | 7    | 8    | 9     | 10    |
|-------------|-------|-------|-------|-------|------|------|------|------|-------|-------|
| 2P          | 479   | 1,019 | 633   | 188   | 42   | 10   | 2    | 1    | —     | 1     |
| 3P          | N/A   | 15    | 186   | 94    | 40   | 8    | 3    | —    | 1     | —     |
| 4P          |       | N/A   | 4     | 38    | 24   | 2    | 1    | 1    | —     | —     |
| 5P          |       |       | N/A   | 4     | 7    | 2    | 1    | —    | —     | —     |
| 6P          |       |       |       | N/A   | —    | 5    | 1    | 1    | —     | —     |
| 7P          |       |       |       |       | N/A  | 1    | —    | —    | —     | —     |
| Sum         | 479   | 1,034 | 823   | 324   | 113  | 28   | 8    | 3    | 1     | 1     |
| Relative    | 17.0% | 36.7% | 29.2% | 11.5% | 4.0% | 1.0% | 0.3% | 0.1% | 0.04% | 0.04% |

Table 10.110: Frequencies of clitic position against permutation space size in spoken Polish.

| Pos \ $\pi$ | 1     | 2     | 3     | 4     | 5    | 6    | 7    | 9     | 12   |
|-------------|-------|-------|-------|-------|------|------|------|-------|------|
| 2P          | 549   | 979   | 356   | 89    | 13   | 5    | —    | —     | —    |
| 3P          | N/A   | 131   | 373   | 109   | 23   | 2    | —    | —     | 2    |
| 4P          |       | N/A   | 36    | 103   | 28   | 1    | 1    | —     | —    |
| 5P          |       |       | N/A   | 6     | 20   | 7    | 1    | —     | —    |
| 6P          |       |       |       | N/A   | 1    | 3    | 2    | —     | —    |
| 10P         |       |       |       |       | N/A  | N/A  | N/A  | 1     | —    |
| Sum         | 549   | 1,110 | 765   | 307   | 85   | 18   | 4    | 1     | 2    |
| Relative    | 19.3% | 39.1% | 26.9% | 10.8% | 3.0% | 0.6% | 0.1% | 0.04% | 0.1% |

Table 10.111: Frequencies of clitic position against permutation space size in Spisz Goralish.

| Pos \ $\pi$ | 1     | 2     | 3     | 4     | 5    | 6    | 7    |
|-------------|-------|-------|-------|-------|------|------|------|
| 2P          | 376   | 899   | 351   | 113   | 21   | 5    | —    |
| 3P          | N/A   | 116   | 306   | 127   | 30   | 4    | —    |
| 4P          |       | N/A   | 43    | 83    | 24   | —    | 2    |
| 5P          |       |       | N/A   | 9     | 19   | 6    | —    |
| 6P          |       |       |       | N/A   | 3    | 3    | —    |
| 7P          |       |       |       |       | N/A  | —    | 2    |
| Sum         | 376   | 1,015 | 700   | 332   | 97   | 18   | 4    |
| Relative    | 14.8% | 39.9% | 27.5% | 13.1% | 3.8% | 0.7% | 0.2% |

Table 10.112: Frequencies of clitic position against permutation space size in spoken Slovak and Silesian.

| Pos \ $\pi$ | Slovak |       |       |       |      |      |      | Silesian |       |       |       |      |      |      |
|-------------|--------|-------|-------|-------|------|------|------|----------|-------|-------|-------|------|------|------|
|             | 1      | 2     | 3     | 4     | 5    | 6    | 7    | 1        | 2     | 3     | 4     | 5    | 6    | 7    |
| 2P          | 115    | 305   | 140   | 41    | 14   | 3    | 1    | 39       | 102   | 53    | 12    | 2    | —    | —    |
| 3P          | N/A    | 19    | 54    | 22    | 11   | 4    | 2    | N/A      | 23    | 46    | 19    | 1    | —    | —    |
| 4P          |        | N/A   | 8     | 12    | 4    | 1    | —    |          | N/A   | 6     | 12    | 1    | 1    | —    |
| 5P          |        |       | N/A   | 2     | 2    | —    | —    |          |       | N/A   | 2     | 2    | 1    | —    |
| 6P          |        |       |       | N/A   | 1    | 2    | —    |          |       |       | N/A   | —    | 1    | —    |
| 7P          |        |       |       |       | N/A  | —    | 1    |          |       |       |       | N/A  | —    | 1    |
| Sum         | 115    | 324   | 202   | 77    | 32   | 10   | 4    | 39       | 125   | 105   | 45    | 6    | 3    | 1    |
| Rel.        | 15.1%  | 42.4% | 26.4% | 10.1% | 4.2% | 1.3% | 0.5% | 12.0%    | 38.6% | 32.4% | 13.9% | 1.9% | 0.9% | 0.3% |



The heat maps offer us yet another interesting perspective. Imagine the hypothesis that Wackernagel placement might be an artefact of low complexity. Thus, the frequent recurrence of second position cliticisation would be a consequence of a high frequency of small  $\pi$ -spaces in a sample. If this were true, then we would encounter a bias in the data. The argument could continue such that the Czech data, which show the highest ratio of second position cliticisation, contains a disproportionately high amount of short or simple utterances whereby 2P is the only option available in most cases. In contrast, the Polish data sample, which exhibits greater flexibility in statistical terms, involves a more diversified set of sentence tokens in terms of complexity. More sentences of higher complexity tend to result in a higher frequency of Delayed Placement. We can falsify such a claim based on the data from Tables 10.109–10.112. All lects exhibit quite complex utterances with a maximal cardinality  $\pi_{max} = 7$  or higher like in Czech ( $\pi_{max} = 10$ ) and Polish ( $\pi_{max} = 9, 12$ ). Furthermore, the respective  $\pi$ -categories (= columns) have a comparable share across languages – given some tolerance range for deviations within the limits of approximately 5%. Therefore, we can pursue the assumption that the data samples are comparably complex. Differences in frequency distributions of particular clitic positions are thus not concomitant with complexity in the sense used here, but an incarnation of variation among closely-related languages.

To conclude, this section utilised the concept of the permutation space, which proposes an operationalisation of an utterance's complexity, to analyse attested clitic positions against clitics' hypothetically available positions. The data were presented in cross-tables as heat maps, which allow for two conclusions: First, unfolding the permutation space distributes clitic occurrences over a variety of available positions in all languages including Czech and Slovak. Obviously, the languages under scrutiny do not pattern identically, but we can observe a considerable degree of variation in all of them. Most importantly, we find an increase of delayed clitic placement, when clitics are given the necessary space for late placement in either language. Second, the languages examined here do not exploit all positions available. Even in an increasingly growing permutation space, clitics gravitate towards the left periphery. Second position placement is the preferred configuration and the likelihood of occupying the latest available position is rather low. This also holds for Polish, where we observe considerable variation, but no preference for late clitic positioning.

## 10.10 Interim conclusion

As the most comprehensive portion of Part III, Chapter 10 gave an empirical analysis of the distributional properties of clitics in six West Slavic spoken lects. Individual sections presented the study's core languages Czech, Polish, and Goralish as well as supplemental material from Slovak,

Silesian, and Kashubian. Clitic placement, clustering, and clitic climbing were systematically analysed for each lect. Furthermore, miscellaneous topics attested only in a subset of the data samples were addressed, i.e. agreement drop (Cz, Gor, Sil) and the possessive resultative construction (Cz). After discussing each lect individually, section 10.8 offered a comparative perspective on the most important issues followed by an application of the concept of permutation space in section 10.9. While I must admit minor drawbacks due to some gaps in the annotation scheme, the approach taken in the present thesis proved adequate, as it provides us with new insights and data both on a language-particular as well as comparative-typological level.<sup>569</sup> The following major conclusions can be drawn for *all* lects (exceptions are specified):

- Clitics predominantly occupy second position in simple clauses and under clitic climbing.
- Positional deviations involving 1P to 5P (and beyond) do occur, even though their occurrence frequencies may differ across lects. In general, an increase in delay comes with a decrease in likelihood.
- Clitics are preferably predicate-adjacent, which, however, is not mandatory.
- Multiple clitics are predominantly organised in clusters of contiguous clitics.
- The clusters are internally organised and sequentially regular to a very high degree.
- Diaclisis is possible between any morphosyntactic categories of clitics in principle (except between COND and AUX).
- Diaclisis may occur both within a single verbal or clausal domain (i.e. diaclisis of co-dependent clitics) and between different verbal or clausal domains (i.e. diaclisis of non-co-dependent clitics), but lects differ with respect to the probabilities.
- In principle, any category of clitic (COND, AUX, REFL, PRN) or clitic cluster can occupy absolute clausal first position.
- Clitics may climb to a deviate position including diverse cardinalities of Delayed Placement, whereby the likelihood becomes smaller with growing cardinality.
- Climbed clitics preferably remain adjacent to a verb (two verbs in Polish), while verb-adjacency as such is not obligatory in any lect.
- Agreement drop (attested in Czech, Goralish, Silesian) comes with a fixed linear precedence pattern ( $\text{PRN}_{\text{SBJ}} < \text{V}$ ,  $\text{PRN}_{\text{SBJ}} < \text{CL}$ ).

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<sup>569</sup> For instance, two shortcomings of the annotation scheme are (i) the missing distinction between independent and dependent clauses, and (ii) the non-consideration of number and person. The problems stemming from (i) were addressed in Section 10.9 on clitic positioning vs. available permutations. Issues arising from (ii) were discussed in Section 10.4.4 on agreement drop in Goralish.

- Agreement drop affects the occurrence frequencies of clitic positions, such that it significantly increases the likelihood for 3P/DP while eliminating the possibility of 1P (= probability of 0.0%).
- Possessive resultative constructions resemble clitic climbing environments in that an accusative clitic is licensed in a small clause complement but can leave it and enter the matrix domain.
- An utterance's increasing complexity in terms of a growing permutation space licenses delayed clitic placement, and languages might take advantage of this space. However, this does not override second position placement.
- Czech and Polish seem to represent two extreme poles on a scale with the remaining lects being located in between.
- Second position cliticisation or Wackernagel's law is best understood as a statistical distribution pattern – at least, in non-standard spoken West Slavic lects.
- More generally, the defining properties of special clitics are not categorical in nature, but bundles of statistical preferences – at least, within the limits of the lects under scrutiny.
- Polish is a second position clitic or special clitic language under this probabilistic view.
- Correspondingly, second position clitic languages show some degree of flexibility and allow a certain amount of variation, which can be captured by statistical frequencies. Language use may differ quantitatively across the individual varieties, but they intersect qualitatively in that they exhibit essentially the same phenomena.

# 11 Clitics governed by non-verbal heads

## 11.1 Introductory remarks

Although cliticisation in Slavic has become a well-established research area in its own right since the mid-1990s, little progress has been made with respect to variation (but cf. Kolaković et al. 2022 for an explicitly variationist approach to BCMS clitics). One incarnation of this monolithic point of view is the sole consideration of verbal predication under the almost full neglect of clitics licensed by non-verbal heads (but cf. Junghanns 2002c; Golden 2008 for broader perspectives). Eliciting empirical data, one encounters various instances of clitics governed by categories other than the verb: Clitics licensed by nouns, adjectives, and adverbs are attested across West Slavic. Their relative frequencies are comparatively low and vary across data samples and clitics. In the following, non-verbally governed clitics will be presented in a purely documentary character. Each lect will be treated individually, but there will not be a section on Kashubian, as I did not find respective attestations. The discussion will be of a mostly qualitative nature, because frequencies are too low for a genuinely quantitative account.

## 11.2 Czech

Non-verbal governors were encountered for the three pronominal clitics *ho*, *mi*, *mu* ( $n = 47$ ). There are only 3 attestations for *ho*, but 20 and 24 attestations for *mi* and *mu* respectively. The ratio for non-verbally licensed clitics in the complete data sample, i.e. covering verbal and non-verbal predicates, is 1.6%. The categories of predicates are adjective [ADJ], adverb [ADV], noun [NN], numeral [NUM], and interrogative expression [WH] with adverbs being the most frequent ones ( $n = 20$ , 42.6%). Examples will be accompanied by a tag for the category of the governor in square brackets and provided with further structural information, where relevant. While it might seem surprising to regard interrogatives as heads of pronominals, this step is motivated by the surface-orientation approach. Critiques might object that we are dealing with elliptic constructions, e.g., in (11.3f). The interrogative PP *k čemu* ‘what for’ co-occurs with the future copula form *bude*, and we would be prone to assume that there is an underlying predicate, which is phonetically null, e.g. the adjective *nutné* ‘necessary’. It may well be that this is the diachronic source of this type of construction, but the reasoning is problematic. Similar to Lenertová’s (2001: 295–296; esp. 295, n. 8; 2004: 149) counterargument to Clitic First being the result of ellipsis, it is hardly possible to tell what element

was deleted. Often, there are multiple possibilities but no heuristics to determine which is the correct one. Now, consider a child encountering such occurrences during first language acquisition. Under no circumstances is the child able to reconstruct the underlying, supposedly elided element. The least costly assumption is that the child will take the structure it encounters at face value in the sense of Ohala (1989: 186).<sup>570</sup> What is more, we are dealing with highly conventionalised constructions. So, there is a high likelihood of occurrence for “ellipses” against a comparatively low probability of fully saturated constructions. Consider further that such seemingly elliptical constructions are passed over to subsequent generations. There is justification for the assumption that true ellipses might get lost after three generations (similar to a language shift scenario): Generation *n* possesses the full construction and produces it alongside ellipsis. Generation *n*+1 acquires the full and elliptic construction but produces only the latter. Generation *n*+2 acquires the reduced construction without any evidence of elliptical material. Provided these arguments – and irrespective of the idealisation underlying them – it is plausible to treat interrogatives as predicative heads of pronominal clitics for the purposes of the present thesis.

Returning to the corpus attestations, I encountered four occurrences of *ho*, three of which are accusative arguments of the adverb *lito* as in (11.1a), whereas one instance shows a partitive genitive governed by a quantifying expression, cf. (11.1b). Various categories and lexemes are attested as governors for the dative clitics *mi* and *mu* in (11.2) and (11.3) respectively.<sup>571</sup> As indicated by the superscript figures, clitics exhibit variation with respect to positioning. Table 11.1 summarises the positions’ absolute frequencies. While 2P is the most frequent pattern (11.1b, 11.2c–d, 11.3c, 11.3f), deviations from Wackernagel positioning are attested in the data with 1P (11.2b), 3P (11.1, 11.2a, 11.2c, 11.3a–b, 11.3e), and 4P (11.3d). Clitic Third is particularly well represented. Occurrences of clitic climbing are also attested: the clitic crosses a matrix modal verb and appears in clausal second position in (11.2d). In contrast, the clitic adjoins to the matrix modal verb and occupies 3P with simultaneous verbal circumposition in (11.3e). Bear in mind that contiguity of the clitic to the copula verb *být* ‘(to) be’ is commonly accounted for by mono-clausality. The copula’s function may be attributed to the realisation of tense information or to the establishment of a symmetric relation between entities. However, the finite modal verbs are clearly superordinate to the predicative domain of the clitics’ governors, which is evident from the copulas’ infinitival morphology. Consequently, the modals’ appearance calls for a complex clause analysis, so it is adequate to analyse the occurrence of non-verbally licensed clitics in the modals’ domain as clitic climbing.

<sup>570</sup> Note that the reasoning does not employ Generative Grammar’s theory-dependent notion of Universal Grammar.

<sup>571</sup> Examples (11.2b, 11.2d) label the token *jedno*, which is a numeral, as adverb [ADV], because *jedno* is functionally an adverb and not a numeral proper here.

Table 11.1: Absolute frequencies for attested clausal positions of non-verbally licensed clitics in spoken Czech.

| 1P | 2P | 3P | 4P |
|----|----|----|----|
| 2  | 28 | 15 | 2  |

- (11.1) a. tak<sup>1</sup> mně<sup>2</sup> ho<sup>3</sup> bylo lito [ADV]  
 so me.DAT him.ACC be.PST.3SG.N sorry  
 ‘I felt so bad for him’
- b. no ale zas<sup>1</sup> ho<sup>2</sup> ne-může bejt asi moc no . [NUM]  
 well but again him.GEN NEG-can.PRS.3SG be.INF perhaps a\_lot well  
 ‘well, but I guess that there can’t be certainly a lot of it’ (Cz; ČNK oral\_v1)
- (11.2) a. to<sup>1</sup> je<sup>2</sup> mi<sup>3</sup> jasný (že tam ne-chodiš moc) [ADJ]  
 TO be.PRS.3SG me.DAT clear.NOM that there NEG-walk.PRS.2SG a\_lot  
 ‘it’s obvious to me (that you don’t go there a lot)’
- b. no mi<sup>1</sup> bylo jedno [ADV]  
 well me.DAT be.PST.SG.N one.NOM.SG.N  
 ‘well, I didn’t care’
- c. hlavně<sup>1</sup> je<sup>2</sup> mi<sup>3</sup> [pp s nim] dobře [ADV]  
 mainly be.PRS.3SG me.DAT with him.INS well  
 ‘essentially, I feel comfortable with him’
- d. že<sup>1</sup> mi<sup>3</sup><sup>2</sup> to vlastně může<sub>1</sub> bejt<sub>2</sub> úplně jedno<sub>3</sub> [ADV]  
 that me.DAT this.NOM.SG.N actually can.PRS.3SG be.INF completely one.NOM.SG.N  
 ‘that I actually don’t need to care’
- e. to<sup>1</sup> mi<sup>2</sup> bude kolik? [WH]  
 TO me.DAT be.FUT.3SG how\_much  
 ‘how much will it cost me?’ (Cz; ČNK oral\_v1)
- (11.3) a. prže<sup>1</sup> je<sup>2</sup> mu<sup>3</sup> to prostě trapný [ADJ]  
 because be.PRS.3SG him.DAT this simply embarrassing  
 ‘because it’s plain embarrassing for him’
- b. (dyž odjedou) tak<sup>1</sup> je<sup>2</sup> mu<sup>3</sup> tam smutno [ADV]  
 when leave\_by\_vehicle.PRS.3PL so be.PRS.3SG him.DAT there sad.ADV  
 ‘(when they leave,) he is sad there’

- c. *kolik<sup>1</sup>      mu<sup>2</sup>      bylo      let?* [NN]  
 how\_much him.DAT be.PST.3SG.N summer.GEN.PL  
 ‘how old was he?’ [lit.: ‘how many years did he have?’]
- d. *ale vono<sup>1</sup> už<sup>2</sup>      je<sup>3</sup>      mu<sup>4</sup>      tolik      let* [NN]  
 but it already be.PRS.3SG him.DAT so\_much year.GEN.PL  
 (že už není zajímavěj pro ně)  
 that already NEG.be.PRS.3SG interesting for them  
 ‘but it is already so old (that it isn’t interesting for them anymore)’
- e. *(no myslim) že<sup>1</sup> může<sup>2</sup>      mu<sup>3</sup>      bejt<sup>2</sup> tak čtyřicet<sup>3</sup>* [NUM]  
 well think.PRS.1SG that can.PRS.3SG him.DAT be.INF so forty  
 (něco málo čtyřicet pět třeba nebo tak)  
 something little forty five perhaps or so  
 ‘(well, I think) that he could be forty [years old] (perhaps somewhat close to forty five or so)’
- f. *no a [k čemu]<sup>1</sup> mu<sup>2</sup> to bude* [WH]  
 well and towards what.DAT him.DAT this be.FUT.3SG  
 ‘well and what will that do for him?’ (Cz; ČNK oral\_v1)

## 11.3 Polish

I encountered 64 occurrences of non-verbally licensed short pronoun tokens, which represents 2.1% of the Polish data sample. The evidence involves the types *go*, *mi*, *ci*, *mu*, *się*. There is a strong asymmetry in attestations. Dative clitics make the majority, whereas only three tokens of the accusative/genitive forms have been recorded. To begin with the latter, there are two instances of pronominal *go*, both of which occur with a noun and represent the NP-internal cliticisation discussed by Franks & King (2000: 160–161, 277–278). Utterance (11.4a) is a textbook example for Polish NP-internal clitic placement. The accusative argument clitic *go* is subcategorised for by the nominal head. Note that the NP itself is the complement of the prepositional head *do* ‘to’, which governs the NP’s genitive case. In (11.4b), the clitic has scope over two asyndetically coordinate NPs and is placed in the second conjunct. Again, the noun phrases are headed by prepositions, i.e. a topical preposition *o* ‘about’ governing accusative case. Note that the clitic does not have scope over the conjuncts, which are PPs, but about their complements. A third occurrence of a non-verbally licensed accusative clitic presents the only reflexive form found in the data sample. The clitic *się* in (11.5) is reciprocal in fact and a true argument of the adjective *lizących* ‘licking’. If structures of the type

presented in (11.4) are treated as NP-internal, then structures of the kind in (11.5) may be regarded as AdjP-internal. Although no such data have been brought to light in the Czech sample, Junghanns (2002c: 70–71) reports comparable structures for Czech, while Kolaković et al. (2022: 241–242) provide BCMS evidence.

- (11.4)    a.    bo                masz                cały                rok                [PP do  
                  because have.PRS.2SG whole.ACC.SG year.ACC.SG      to  
                  [NP przygotowania go ]]      tak?  
                  preparing.GEN.SG him.ACC yes  
                  ‘because you have the whole year for preparing it, right?’



- b. nawet jeśli chodzi [PP o [NP przenoszenie ]] [PP o [NN]  
 even if walk.PRS.3SG about carrying.LOC.SG about  
 [NP **dźwiganie go** na plecach ]] (czy w ręce no to.  
 lifting.LOC.SG him.ACC on back.LOC.PL or in hand.ACC.PL well TO  
 szesnaście kilogramów a dwanaście to jest różnica  
 sixteen kilogram.GEN.PL and twelve TO be.PRS.3SG difference.NOM.SG  
 duża)  
 huge.NOM.SG  
 ‘even, if it’s about carrying, about lifting it on the back (or by hand, well, then sixteen  
 kilogrammes versus twelve kilogrammes, that makes a significant difference)’  
 (Pol; NKJP pelcra)

- (11.5) a. [...] ponieważ też patrzę [PP na [NP dwóch facetów [ADJ]  
 because also look.PRS.1SG at two.GEN.PL guy.GEN.PL  
 [AdjP **liżących się** ]]] [...] licking.GEN.PL REFL.ACC  
 ‘... because I also see two guys making out ...’ (Pol; NKJP pelcra)
- b. \*ponieważ patrzę [PP na (\***się**) dwóch (\***się**) facetów [AdjP **się liżących (się)** ]]  
 c. \*ponieważ patrzę [PP na (\***się**) dwóch [AdjP (**się**) **liżących (się)** ] facetów ]  
 d. \*ponieważ patrzę [PP na [AdjP (**się**) **liżących (się)** ] dwóch facetów ]  
 (Pol; constructed from 115a)

Turning to the core of the data, governing categories cover noun [NN], adjective [ADJ], adverb [ADV], and interrogative expressions [WH], but ADV is the most-frequent by far ( $n = 40$ , 62.5%). In one case, (11.8g), the clitic *mu* is licensed by the syntactically complex expression of a PP (*nie*) *na rękę* ‘(not) suit [someone]’, but I attribute it to the nominal category for the sake of convenience. Clitics predominantly occur in 2P, but positions from 1P to 5P are attested, although peripheral positions are rather infrequent except Clitic Third, which makes a good fifth of the cases, cf. Table 11.2. The column for 2P gives two figures, one being in parenthesis, which includes the tokens in (11.4)–(11.5). They are treated separately, because clitics do not enjoy positional flexibility in those island environments.

Table 11.2: Absolute frequencies for attested clausal positions of non-verbally licensed clitics in spoken Polish.

| 1P | 2P      | 3P | 4P | 5P |
|----|---------|----|----|----|
| 2  | 41 (44) | 14 | 3  | 1  |

Example (11.6e) shows *mi* in a mixed cluster with the embedded pronominal object *ich*, which is licensed by the infinitive *wyrzucać* ‘(to) throw away’ subordinate to the adverbial head. The utterance does not exhibit a copula or any other matrix verb. Constructions of that kind motivate the analysis of non-verbally licensed clitics. The dative clitic is unequivocally dependent on the adverb. In examples (11.6c) and (11.8e) the adverbs inflect and appear in the comparative.

- (11.6) a. no      boże      **mi**<sup>1</sup>      to      nie      jest      akurat **potrzebne**      [ADJ]  
well      God.VOC me.DAT TO NEG be.PRS.3SG just necessary.NOM.SG.N  
na przykład pe es pe  
on example PSP  
‘oh God, I just don’t need a PSP [= PlayStation Portable], for instance’
- b. [AdvP bardzo **miło**]<sup>1</sup> **mi**<sup>2</sup>      było      z      tobą      rozmawiać      [ADV]  
very nice me.DAT be.PST.SG with you.INS talk.INF  
‘It was nice talking to you.’
- c. powiem      ci      że<sup>1</sup> tak<sup>2</sup> **mi**<sup>3</sup>      jest      **łatwiej**      [ADV]  
tell.PRS.1SG you.DAT that so me.DAT be.PRS.3SG easier.ADV  
‘I tell you that it is easier for me this way’
- d. ja      ciągle      jak<sup>1</sup>      nieraz<sup>2</sup>      jest<sup>3</sup>      **mi**<sup>4</sup>      **smutno**      [ADV]  
I.NOM constantly when time\_and\_again be.PRS.3SG me.DAT sad.ADV  
(to. y to biorę      właśnie taką      książkę      do ręki [...])  
TO uh TO take.PRS.1SG exactly such.ACC.SG.F book.ACC.SG.F to hand.GEN.SG  
‘[as far as] I [am concerned], when I’m sad time and again, (I always pick up exactly such a [kind of] book ...)’
- e. **szkoda**<sub>1</sub><sup>1</sup>      **mi**<sub>1</sub><sup>2</sup>      **ich**<sub>2</sub><sup>2</sup>      wyrzucać<sub>2</sub>      [NN]  
damage.NOM.SG me.DAT them.ACC throw\_away.INF  
‘for me, they [are] too good to be thrown away’ (Pol; NKJP pelcra)

- (11.7) a. [ nie **zimno** ]<sup>1</sup> **ci**<sup>2</sup> ? [ADV]  
 NEG cold you.DAT  
 ‘aren’t you cold?’
- b. (jeśli już będziesz zdecydowany na to) że<sup>1</sup> jest<sup>2</sup> [ADV]  
 if already be.FUT.2SG certain.NOM.SG.M on this.ACC.SG.N that be.PRS.3SG  
**ci**<sup>3</sup> tak **źle** że że że rezygnujesz z tej roboty [...] [ADV]  
 you.DAT so bad that that that resign.PRS.2SG from this.GEN.SG.F work.GEN.SG  
 ‘(if you finally will be certain about [the fact]) that you feel bad that you will resign from the job ...’
- c. a **szkoda**<sup>1</sup> **ci**<sup>2</sup> czasu? [NN]  
 and damage.NOM.SG you.DAT time.GEN.SG  
 ‘but [is it] a waste of time for you?’ (Pol; NKJP pelcra)
- (11.8) a. no musiała-by-m pomyśleć . co<sup>1</sup> **mu**<sup>2</sup> tak [ADJ]  
 well must.PST.SG.F-COND-AUX.1SG think.INF what him.DAT so  
**potrzebne** nie  
 necessary NEG  
 ‘well, I would need to think about what he actually needs, right’
- b. to za to co<sup>1</sup> jestem<sup>2</sup> **mu**<sup>3</sup> **wdzięczna** [ADJ]  
 TO for this.ACC.SG.N what be.PRS.1SG him.DAT thankful.ACC.SG.F  
 naprawdę mieli-śmy tiko kupić  
 really have.PST.PL.VIR-AUX.1SG Tico.ACC.SG buy.INF  
 ‘so, for that, what I am thankful to him, we should have actually bought a [Daewoo] Tico for him’
- c. ponoć<sup>1</sup> [NP jakieś te robaki ]<sup>2</sup> są<sup>3</sup> [ADJ]  
 supposedly some.NOM.PL.NVIR this.NOM.PL.NVIR worm.NOM.PL be.PRS.3PL  
**potrzebne**<sup>4</sup> **mu**<sup>5</sup> do do systemu trawiennego  
 necessary.NOM.PL.NVIR him.DAT to to system.GEN.SG digestive.GEN.SG.M  
 ‘supposedly, some [sort] of these bugs are necessary for his digestive system’
- d. Mietek nie bo bo<sup>1</sup> **mu**<sup>2</sup> **duszo** [ADV]  
 Mietek NEG because because him.DAT stuffy.ADV  
 ‘Mietek, no, because he [feels] stuffy’

- e. no    bo<sup>1</sup>        tam<sup>2</sup> **mu**<sup>3</sup>        była        **łatwiej**        walizkę        [ADV]  
 well because there him.DAT be.PST.SG.F easier.ADV suitcase.ACC.SG  
 wtachać  
 heave.INF  
 ‘well, because there it was easier for him to heave the suitcase’
- f. wiesz        **mu**<sup>1</sup>        **żał**        jest        trochę        [NN]  
 know.PRS.2SG him.DAT regret.NOM.SG be.PRS.3SG a\_bit  
 ‘you know, he feels some kind of sorry’
- g. więc<sup>1</sup>        chyba<sup>2</sup> [nie [PP na **rękę** ]]<sup>3</sup>        **mu**<sup>4</sup>        pisać        tak        [NN]  
 therefore perhaps NEG on hand.ACC.SG him.DAT write.INF so  
 długie        mejle.  
 long.ACC.PL.NVIR email.ACC.PL  
 ‘so, perhaps [it is] inconvenient for him to write such long mails’
- h. [ **po co** ]<sup>1</sup> **mu**<sup>2</sup>        to        [WH]  
 for what him.DAT this.NOM.SG.N  
 ‘what [does] he [need] it for?’
- i. a [ **na co** ]<sup>1</sup> **mu**<sup>2</sup>        ta        torba?        [WH]  
 and on what him.DAT this.ACC.SG.F bag.ACC.SG  
 ‘and what [does] he [need] this bag for?’ (Pol; NKJP pelcra)

## 11.4 Goralish

Goralish provides us with comparatively rich evidence for non-verbally licensed clitics. Six out of seven clitic types (*cię, go, mi, ci, mu, się*) are attested at least once in a corresponding environment. As before, dative clitics make the vast majority of data. Out of 87 utterances representing 3.2% of the overall Goralish data, 83 tokens are datival clitics. The range of attested positions covers 2P to 5P, but 2P is the most frequent, as can be read from Table 11.3.

Table 11.3: Absolute frequencies for attested clausal positions of non-verbally licensed clitics in Spisz Goralish.

| 2P | 3P | 4P | 5P |
|----|----|----|----|
| 66 | 17 | 2  | 2  |

The accusative/genitive forms in (11.9) show two instances of adverbially licensed clitics (11.9a–b), a syntagm with the clitic being governed by an interjection [INT] (11.9c), and NP-internal reflexive clitic (11.9d).

- (11.9) a. <[...] dało się tak zakraść żeby<sup>1</sup> cię<sup>2</sup> [ADV]  
 give.PST.SG.N REFL.ACC so steal.INF that you.ACC/GEN  
 nie **widno** było>  
 NEG visible be.PST.SG.N  
 ‘...it was possible to steal [it] in such a way that no one saw you’
- b. <(no taki baciarski był nie?) że<sup>1</sup> wszędzie<sup>2</sup> go<sup>3</sup> [ADV]  
 well so lively.NOM.SG.M be.PST.SG.N NEG that everywhere him.GEN  
 było **pełno**>  
 be.PST.SG.N full.ADV  
 ‘(well, he was a rascal, right?) [such] that he couldn’t sit still’ [lit.: ‘everywhere is full of him’]
- c. <(a tu na na dupę się musiało siednąć) [INT]  
 and here on on bottom.ACC.SG REFL must.PST.SG.N sit\_down.INF  
 i **buch**<sup>1</sup> go<sup>2</sup> tymi nogami>  
 and boing him.ACC this.INS.PL leg.INS.SG  
 ‘(and one had to sit here on one’s bottom) and kick him’ [lit.: ‘boing him with the legs’]
- d. <jedna ino scena [NP **wybieranie**<sup>1</sup> się<sup>2</sup> do [NN]  
 one.NOM.SG.F only scene.NOM.SG leaving.NOM/ACC.SG REFL.ACC to  
 kościoła ] zostało nagrane od początku do końca>  
 church.GEN.SG become.PST.SG.F recorded from beginning.GEN.SG to end.GEN.SG  
 ‘one scene only – leaving for church – was recorded from the beginning to the end’  
 (Gor; KoSpi)

Examples (11.10)–(11.12) present selected attestations of dative clitics originating from adjectives [ADJ], adverbs [ADV], nouns [NN], and interrogative expressions [WH] across diverse clausal positions. Adverbs represent the highest portion of non-verbal predicates ( $n = 53$ , 60.9%). Although several examples involve an inflected copula verb, its realisation is not required as long as tense is not specified as non-present, cf. e.g. (11.10g) and (11.12b). Furthermore, discourse conditions may force the pronunciation of the copula, i.e. Contrastive or Verum Focus as in (11.10a). Otherwise, the realisation of a present tense copula may be regarded as optional in principle, cf. (11.10f) with an

overt copula vs. (11.10d) without copula verb. Utterances (11.11a) and (11.12a) exhibit inflected adverbs in the comparative, while (11.10d) and (11.10f) realise the superlative.

- (11.10) a. <no to<sup>1</sup> [PP na to]<sup>2</sup> jest<sup>3</sup> mi<sup>4</sup> jacy [ADJ]  
 well TO on this.ACC.SG.N be.PRS.3SG me.DAT some  
**potrzebny** nie?>  
 necessary.NOM.SG.M NEG  
 ‘well, so, I need one [= a horse] for that, right?’
- b. <mamo jej cosi<sup>1</sup> mi<sup>2</sup> już **niedobrze** [ADV]  
 mom.VOC oh\_dear something me.DAT already unwell  
 (kto wie czy ja już nie będę rodzi...)>  
 who know.PRS.3SG if I.NOM already NEG be.FUT.1SG [give\_birth]  
 ‘mom, oh dear, I already feel somehow bad (who knows, if I won’t [give birth] yet)’
- c. <**trudno**<sup>1</sup> mi<sup>2</sup> jest powiedzieć (no teraz są na [ADV]  
 difficult me.DAT be.PRS.3SG tell.INF well now be.PRS.3PL on  
 pewno inne)>  
 sure different  
 ‘it’s difficult for me to tell (well, they [= relations between speaker and mother-in-law] are different now for sure)’
- d. <(nie nie ruszała m się nigdzie z [ADV]  
 NEG NEG move.PST.SG.F AUX.1SG REFL.ACC nowhere from  
 domu) bo<sup>1</sup> tu<sup>2</sup> mi<sup>3</sup> **najlepiej**>  
 home.GEN.SG because here me.DAT best.ADV  
 ‘(no, I haven’t left [my] home,) ’cause here I [feel] best’
- e. <(to tak ja rozumieć rozumiem) ale tak gadać [ADV]  
 TO yes I.NOM understand.INF understand.PRS.1SG but so speak.INF  
 to<sup>1</sup> raczej<sup>2</sup> jest<sup>3</sup> mi<sup>4</sup> **ciężko**>  
 TO rather be.PRS.3SG me.DAT difficult.ADV  
 ‘(well, yes, I understand, in fact) but talking like that, this is rather difficult for me’
- f. <[...] bo<sup>1</sup> naprawdę<sup>2</sup> [PP na tej wsi]<sup>3</sup> jest<sup>4</sup> [ADV]  
 because really on this.LOC.SG.F village.LOC.SG be.PRS.3SG  
 mi<sup>5</sup> **najlepiej**>  
 me.DAT best.ADV  
 ‘... ’cause I truly feel best [living] in the countryside’

g. <noji tak<sup>1</sup> **mi**<sup>2</sup> trochę tego swetra było [NN]  
 well\_and so me.DAT somewhat this.GEN.SG.N sweater.GEN.SG be.PST.SG.N  
**szkoda**>

damage.NOM.SG

‘I felt some pity about that sweater’

h. <(jeszcze dobrze się do szkoły nie zaczęło [NN]  
 still good.ADV REFL.ACC to school.GEN.SG NEG start.PST.SG.N  
 chodzić już trza było krowy iść paść)  
 walk.INF already necessary be.PST.SG.N cow.ACC.PL go.INF feed.INF  
 marny<sup>1</sup> **mi**<sup>2</sup> **los**>

miserable.NOM.SG.N me.DAT fate.NOM.SG

‘(one hadn’t yet truly started to go to school and already had to go and feed cows) sad fate of mine’

i. <neji potem<sup>1</sup> **mi**<sup>2</sup> był **stop** (potem pięć roków [INT]  
 well\_and then me.DAT be.PST.SG.M stop then five year.GEN.PL  
 nie było potem zaś był zaś był chłopiec)>  
 NEG be.PST.SG.N then again be.PST.SG.M again be.PST.SG boy.NOM.SG

‘well, and then I had a break (then there were five years without [a birth], then there was [a birth] again, it was a boy again)’ (Gor; KoSpi)

(11.11) a. <jak<sup>1</sup> **ci**<sup>2</sup> **lepiej** (to ci zazdrozczą) jak<sup>1</sup> [ADV]  
 when you.DAT better.ADV TO you.DAT be\_envious.PRS.3PL when  
**ci**<sup>2</sup> **ploniej** (to się śmieją z ciebie)>  
 you.DAT more\_hopeless.ADV TO REFL.ACC laugh.PRS.3PL from you.GEN.SG  
 ‘when you’re doing fine, (then everybody’s envious of you) if you’re doing badly (then everybody’s laughing at you)’

b. <czy<sup>1</sup> to<sup>2</sup> **fajnie**<sup>3</sup> **ci**<sup>4</sup> czy<sup>1</sup> **nie-fajnie**<sup>2</sup> **ci**<sup>3</sup>> [ADV]  
 if this.NOM.SG.N fine.ADV you.DAT if NEG-fine.ADV you.DAT  
 ‘if you’re fine or if you’re not fine’

- c. <(pamiętam)            że<sup>1</sup> nie    cz... nie    **wolno**<sup>2</sup>        **ci**<sup>3</sup>            było            [ADV]  
          remember.PRS.1SG   that NEG [...] NEG   allowed.ADV   you.DAT   be.PST.SG.N  
 oglądać    tego            "Teleranka"        (bo    trza            było            rano  
 watch.INF   this.GEN.SG.M   Teleranek.GEN.SG   because necessary   be.PST.SG.N   morning  
 iść    do   sko... d...   paść    te                            krowy)>  
 go.INF to   [...]    [...]   feed.INF this.ACC.PL.NVIR   cow.NOM.PL  
 ‘I remember that one should not watch the Teleranek [= children’s TV shows] (‘cause  
 one had to go in the morning and feed those cows)’
- d. <czy<sup>1</sup>   **ci**<sup>2</sup>            nie    **żał**                            swadziebna   dziewczeczko>                            [NN]  
          if    you.DAT   NEG   regret.NOM.SG   bridal.VOC   girl.DIM.VOC  
 ‘don’t you feel sorry, young bridal girl?’
- e. <jak<sup>1</sup>   **ci**<sup>2</sup>            **miano?**>                            [NN]  
          how   you.DAT   name.NOM/ACC.SG  
 ‘what’s your name?’
- f. <no    to<sup>1</sup>   **chwała**<sup>2</sup>        **ci**<sup>3</sup>            Panie    Boże    za    to                            że            [NN]  
          well   TO   glory.NOM.SG   you.DAT   Lord.VOC   God.VOC   for   this.ACC.SG.N   that  
 że   ktosi            o    nas    myśli            o    tym                            Spiszu>  
 that someone.NOM about   us.LOC   think.PRS.3SG about   this.LOC.SG.N   Spisz.LOC.SG  
 ‘well, so praise be to You, oh Lord, that someone thinks of us, of Spisz’
- g. <**co**<sup>1</sup>   **ci**<sup>2</sup>            z    tej                            szmaty?>                            [WH]  
          what   you.DAT   from   this.GEN.SG.F   cloth.GEN.SG  
 ‘what’s the good of the cloth for you?’
- h. <[ **na co** ]<sup>1</sup>   **ci**<sup>2</sup>            to                            teraz?>                            [WH]  
          on    what   you.DAT   this.NOM.SG.N   now  
 ‘what [do] you [need] it [for] now?’
- i. <to<sup>1</sup>   [ **po co** ]<sup>2</sup>   **ci**<sup>3</sup>            dom?>                            [WH]  
          TO    for    what   you.DAT   house.NOM.SG  
 ‘what [do] you [need] a house [for]?’                            (Gor; KoSpi)

- (11.12) a. <([...] bo            stwierdzi]            że    że<sup>1</sup>   **mu**<sup>2</sup>            tam    **lepiej**                            [ADV]  
                          because ascertain.PRS.3SG   that that him.DAT   there better.ADV  
 (lepiej    tam    zarobi)>  
 better.ADV   there   earn.PRS.3SG  
 ‘(... ‘cause he will find) that it’s better for him there (there he earns better [money])’



- b. <to<sup>1</sup> już<sup>2</sup>      **mu**<sup>3</sup>      **ciężko**      będzie      się      wrócić> [ADV]  
 TO already him.DAT difficult.ADV be.FUT.3SG REFL.ACC return.INF  
 ‘then it will be difficult for him to return’
- c. <ne    to<sup>1</sup>    tak<sup>2</sup>    **ciężko**<sup>3</sup>    jest<sup>4</sup>      **mu**<sup>5</sup>      po prostu> [ADV]  
 well TO so difficult be.PRS.3SG him.DAT simply  
 ‘well, it’s simply somehow difficult for him’
- d. <(ani    nie    wiem)      jak<sup>1</sup>    **mu**<sup>2</sup>      [PP na **miano** ]> [NN]  
 neither NEG know.PRS.1SG how him.DAT on name.ACC.SG  
 ‘(I don’t even know) what his name is’
- e. <[ czterdzieści    cztery    **roki** ]<sup>1</sup>      **mu**<sup>2</sup>      było [NN]  
 forty four year.NOM.PL him.DAT be.PST.SG.N  
 ‘he was forty four years old’ (Gor; KoSpi)

## 11.5 Slovak

Provided the comparatively small sample size, there are few examples for clitics subcategorised for by non-verbs. In sum, 13 sentence tokens have been detected, which corresponds to 1.6% of the Slovak data sample. Revealingly, the attestations cover solely the four dative clitics. Table 11.4 provides that Clitic First (1P) to Clitic Third (3P) positioning is attested for these constructions, but clitics mostly occur in 2P. The Slovak evidence is exhaustively cited in (11.13)–(11.16).

The small sample contains only adjectival and adverbial predicates. There are 8 adjectives and 5 adverbs. The ratio of head adverbs is rather small in comparison to Czech, Polish, and Goralish but could be an artefact of the limited database. Note that there are passive participles among the adjectival predicates, cf. (11.14e–f) and (11.15c–d). Here, the logic of assigning adjective status to the predicates is the same as in the case of the present active participle in Polish (11.5). Of course, they differ from simplex adjectives like those in (11.13a–b), (11.15b), and (11.16) in that they are derived from a verb proper by productive morphology. Still, they exhibit properties typical of adjectives, which is why they are discussed together with other non-verbal predicates.

Table 11.4: Absolute frequencies for attested clausal positions of non-verbally licensed clitics in spoken Slovak.

| 1P | 2P | 3P |
|----|----|----|
| 1  | 8  | 4  |

- (11.13) a. Ale [<sub>NP</sub> všetci upecečkári ]<sup>1</sup> **ti**<sup>2</sup> budú **vd'ační** [ADJ]  
but all UPC-member.NOM.PL you.DAT be.FUT.3PL thankful.NOM.SG.VIR  
'But all UPC-members will be thankful to you'
- b. [...] ktorá<sup>1</sup> **ti**<sup>2</sup> bola taká ne-viem, [ADJ]  
which.NOM.SG.F you.DAT be.PST.SG.F so.NOM.SG.F NEG-know.PRS.1SG  
**či ne-pochopiteľ'ná** [...] [ADJ]  
if NEG-intelligible.NOM.SG.F  
'... which was so – I don't know – if [it was] unintelligible ...' (Slk; SNK s-hovor-7.0)
- (11.14) a. **Smiešne**<sup>1</sup> **mi**<sup>2</sup> bolo [ADV]  
ridiculous.NOM.SG.N mi.DAT be.PST.SG.N  
'I felt ridiculous'
- b. takže<sup>1</sup> je<sup>2</sup> **mi**<sup>3</sup> za nimi **smutno** [ADV]  
such\_that be.PRS.3SG mi.DAT behind them.INS sad.ADV  
'such that I am melancholic about them'
- c. že<sup>1</sup> je<sup>2</sup> **mi**<sup>3</sup> to **Púto** [ADV]  
that be.PRS.3SG me.DAT this.NOM.SG.N sorry  
'that I feel sorry for that'
- d. Lebo<sup>1</sup> bolo<sup>2</sup> **by**<sup>3</sup> **mi**<sup>3</sup> to **Púto** [ADV]  
because be.PST.SG.N COND me.DAT this.NOM.SG.N sorry  
'because I would feel sorry'
- e. bol<sup>1</sup> **mi**<sup>3</sup> zbytok trestu ods [ADJ]  
be.PST.SG.M me.DAT rest.NOM.SG.M penalty.GEN.SG [...] [ADJ]  
**odpustený** amnestiou  
forgive.PASS.PTCP.NOM.SG.M amnesty.INS.SG  
'I have been released from the rest of the penalty by amnesty'
- f. že<sup>1</sup> vlastne<sup>2</sup> on<sup>3</sup> **mi**<sup>4</sup> je **predstavený** [ADJ]  
that actually he.NOM me.DAT be.PRS.3SG superordinate.PASS.PTCP.NOM.SG.M  
'that he is actually my boss' (Slk; SNK s-hovor-7.0)
- (11.15) a. (No ale stále spomínal) ako<sup>1</sup> **mu**<sup>2</sup> bolo **dobre** [ADV]  
well but always mention.PST.SG.M how him.DAT be.PST.SG.N good.ADV  
'(Well, but he always brought up) how well he was doing.'

- b. a som<sup>1</sup> mu<sup>2</sup> vd'ačný s touto cestou, [ADJ]  
 and be.PRS.1SG him.DAT thankful with this.INS.SG.F way.INS.SG  
 (že zachránil môjho otca)  
 that save.PST.SG.M my.ACC.SG.M.ANI father.ACC.SG  
 'and I am thankful to him in this way (that he saved my father)'
- c. aj mu<sup>1</sup> to bolo zaplatené [ADJ]  
 and him.DAT this.NOM.SG.N be.PST.SG.N pay.PASS.PTCP.NOM.SG.N  
 'and he was paid for that'
- d. (komu odpustíte hriechy,) budú<sup>1</sup> mu<sup>2</sup> [ADJ]  
 whom.DAT forgive.PRS.2PL sin.ACC.PL be.FUT.3SG him.DAT  
 odpustené  
 forgive.PASS.PTCP.NOM.PL.NVIR  
 '[if] you forgive anyone's sins, their sins are forgiven' (Slk; SNK s-hovor-7.0)

- (11.16) [...] a ne-bol<sup>1</sup> som<sup>2</sup> si<sup>2</sup> jistý. [ADJ]  
 and NEG-be.PST.SG.M AUX.1SG REFL.DAT sure.NOM.SG.M  
 '... and I didn't feel certain.' (Slk; SNK s-hovor-7.0)

## 11.6 Silesian

While verbs occur most frequently as licensing heads for clitics, other categories than V are also capable of governing pronominal and reflexive clitics in Silesian. However, only two such attestations have been encountered (= 0.5% of the data sample). The evidence is cited in (11.17). Utterance (11.17a) involves the (synchronically) underived noun *žol* 'sorrow; regret' as predicative head of the two clustering clitics *mi cie*. This token represents rhythmic language, as it is the line of a poem or rhyme. Generally, utterances of that type and genre were removed from the data set, as rhythm and rhyme can significantly affect clitic position. However, since the present chapter is not about clitic positioning in the first place but about non-verbal categories licensing clitics, the consideration of rhythmic language does not pose a problem. On the other hand, we find the adverb *ciy'žko* 'difficult' licensing the dative clitic *mu* in (11.17b).

- (11.17) a. Skuli            tyj                szkarupy      na   puklu                      żół                      [NN]  
because\_of this.GEN.SG.F shell.GEN.SG on hunchback.LOC.SG regret.NOM.SG  
**mi          cie**  
me.DAT you.ACC  
'I feel sorry for you because of the crust on your back'
- b. no     ni-jedyn   słaby **ciy"żko mu**            było                do dom                      [ADV]  
well NEG-one weak difficult him.DAT be.PST.3SG.N to house.GEN.SG  
trefić  
reach.INF  
'well, more than a few [were] weak [= because of alcohol], it was difficult for them to get  
[back] home' (Sil; DiG)

## 11.7 Interim conclusion

Chapter 11 has cited data from five of the object languages (Cz, Pol, Gor, Slk, Sil) attesting clitics as dependents of non-verbal heads. Overall, such constructions occur with very low frequency (< 5%) in the data samples. Non-verbally licensed clitics cannot be deemed a core phenomenon even in spoken non-standard varieties hence. On the other hand, the repeated attestations across speakers and lects confront us with an empirical fact. While the sections have a documentary character, some descriptive generalisations are possible:

- There is a predominance of pronominal dative clitics, compared to a relatively small number of accusative/genitive and reflexive clitics as dependents of non-verbal governors.
- Various categories may function as heads, i.e. nouns, adjectives, adverbs, numerals and other quantifiers, interrogative expressions, and even interjections.
- Among those categories, (synchronic) adverbs occur most often.
- The positioning of non-verbally licensed clitics fits what was found for verbally licensed ones: 2P is preferred followed by 3P, and stronger deviations become increasingly rare.

## 12 Wackernagel vs. deviate positioning in Czech

### 12.1 Introductory remarks

Chapter 10 showed that, even though second position placement is dominant in Czech, it is not exclusive, as it amounts to some 80% of occurrences. Positional deviations of a wide range up to 7P are attested. However, the most frequent deviations are 3P and 1P representing around 12% and 4% respectively of the entire Czech data sample. These quantities are quite different from the figures discussed in connection with Hana's (2007: 109) small statistics in Section 4.4.2 on Delayed Placement. Recall that Hana (2007: 109) found that the reflexive clitic *se* occurs with a ratio of 62.5% in 2P, while 3P amounts to 37.5%. Consider also the replication data from SYN v8 that was discussed in this connection. For example, pronominal clitic *ho* exhibited a proportion of  $\frac{2}{3} : \frac{1}{3}$  for 2P and 3P respectively under *protože*, whereas it was the opposite for reflexive *se* with  $\frac{1}{3} : \frac{2}{3}$  for 2P vs. 3P. How do such discrepancies in quantitative data come about and how can we cope with them?

First, bear in mind that we face different data sets representing two distinct linguistic varieties of Czech. The SYN corpora represent edited written texts including fiction (e.g. novels, short stories), non-fiction (e.g. scientific, popular scientific), and journalism (i.e. newspapers, magazines). Journalism makes the vast majority of the SYN v8 corpus, which is unbalanced in this respect.<sup>573</sup> Recall that, in contrast, ORAL is a corpus of spontaneous speech between familiar interlocutors.<sup>574</sup> Second, the data elicitation methods vary between the present thesis's case study in Chapter 10 and the quantitative data from Section 4.4.2. The former is based on random samples of individual clitic types, whereby the underlying corpus query disregarded any surrounding context. The latter elicited specific linguistic structures, i.e. 2P and 3P in embedded clauses under selected subordinators. Due to these disparities, the data samples are not comparable, in fact.

However, the question naturally arises as to what extent the different parameters affect clitic placement. The brief discussion of Hana's (2007: 109) statistics in Section 4.4.2 suggests that the lexical choice of the subordinator is a parameter that influences the occurrence of 2P and 3P. Furthermore, Hana (2007: 109) only queried 3P instantiated by a single noun intervening between subordinator and clitic. One might wonder which results one obtains, e.g., with an intervening verb. We may put the question more generally: Does the syntactic category of the intervening element have an effect on cliticisation? While the choice of a subordinator and syntactic category of the intervening

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<sup>573</sup> Cf. ČNK wiki, Eng.: <https://wiki.korpus.cz/doku.php/en:cnk:syn:verze8>,  
Cz.: <https://wiki.korpus.cz/doku.php/cnk:syn:verze8> (accessed 9 Jan 2025).

<sup>574</sup> Cf. ČNK wiki, Eng.: [https://wiki.korpus.cz/doku.php/en:cnk:oral?s\[\]=oral](https://wiki.korpus.cz/doku.php/en:cnk:oral?s[]=oral),  
Cz.: <https://wiki.korpus.cz/doku.php/cnk:oral> (accessed 9 Jan 2025).

element are at the heart of grammar research, linguistic contrasts between different varieties or instantiations of a language like edited writing and spontaneous speech are usually neglected. However, it is a well-known fact that written and spoken language do not merely differ with respect to the medium of transmission (sound vs. sight). Prototypically, they are associated with a bundle of distinct linguistic features, which, however, are orthogonal to the medium (cf. Biber 1995: 236–279 [Ch. 7]; Koch & Oesterreicher 1985/2012). Variation across such instantiations of language is referred to by the notion of diaphasic or register variation, which traditionally is a research focus in variational sociolinguistics. Within such a framework, the task is to unveil to what degree clitic positioning varies between registers. It is, therefore, valuable to explore register variation here and to consider both spoken and written language, although the thesis primarily focuses on spoken language.

Accordingly, we arrive at a research setup that allows examining clitic positioning as a dependent variable with two values: Wackernagel placement vs. deviate positioning. We may further distinguish between two scenarios of the deviation, i.e. Clitic Third and Clitic First constituting the two most frequent positional deviations and representing adjacent transpositions from canonical second position cliticisation. Choice of subordinator, the syntactic category of the intervening element, and choice of register are treated as independent variables. Furthermore, a purposeful corpus query relies on the operationalisation of the notion of the left periphery to obtain the relevant data. While this is in general not a problem when referring to subordinators, it is far from trivial in the present study. First, we are also interested in Clitic First, but this placement pattern is logically impossible in embedded clauses introduced by a subordinator. Second, speech corpora do not allow for targeting clause boundaries by means of, e.g., punctuation. Therefore, I use coordinating conjunctions somewhat liberally as a clause boundary marker and abstract away from parataxis of constituents smaller than clauses or ellipsis. Note that “coordinating conjunction” substitutes “subordinator” as an independent variable in this respect.

In the following, I elaborate on Hana’s (2007: 109) tentative corpus query to examine 2P vs. 3P and 1P vs. 2P in dependence of the three independent variables.<sup>575</sup> I take both single clitics and clitic clusters into account. The queries exclusively cover constant clitics, but additionally take auxiliary and conditional clitics into consideration. The elicited structures for 2P vs. 3P are presented in (12.1)–(12.4) with the a-examples coding 2P and b-examples coding 3P. Structure (12.1) represents the adjacent transposition of a clitic or clitic cluster and an NP based on Hana’s (2007: 109) original query. Note that NP refers both to plain, non-branching nouns and complex, branching noun phrases involving attributive expressions. Structure (12.2) involves clitic(s) and adverb, which is informed by the results of the case study in Section 10.2. As mentioned above, the adjacent transposition of clitic

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<sup>575</sup> I thank Ruprecht von Waldenfels (p.c., 15 Jun 2023) for motivating me to venture this step.

(cluster) and verb represents a relevant structure and is covered in (12.3). Finally, structure (12.4) accounts for the interaction of clitic(s) and subject pronoun in 2P vs. 3P.

(12.1) a. SUB < CL < NP

b. SUB < NP < CL

(12.2) a. SUB < CL < ADV

b. SUB < ADV < CL

(12.3) a. SUB < CL < V

b. SUB < V < CL

(12.4) a. SUB < CL < SBJ<sub>PRN</sub>

b. SUB < SBJ<sub>PRN</sub> < CL

For practical reasons – and to eliminate ambiguities – only a subset of all subordinators was explicitly selected based on the frequency. Out of the 20 most-frequent conjunctions (POS tag = "J . \*") in the ČNK's database (cf. Table 12.1), I chose the 7 unequivocal subordinators listed in (12.5).<sup>576</sup> The parameter of register variation is accounted for by searching the representative SYN2020 subcorpus (Hnátková et al. 2014; Křen et al. 2020) for edited writing and ORAL v1 for spontaneous speech.

Table 12.1: 20 most-frequent conjunctions (tag = J) in the Czech National Corpus (SYN2015) in descending order.

|                     |                             |                              |                                 |
|---------------------|-----------------------------|------------------------------|---------------------------------|
| 1. <i>a</i> ‘and’   | 6. <i>když</i> ‘when’       | 11. <i>však</i> ‘however’    | 16. <i>kdyby</i> ‘if’           |
| 2. <i>že</i> ‘that’ | 7. <i>aby</i> ‘in order to’ | 12. <i>protože</i> ‘because’ | 17. <i>takže</i> ‘therefore’    |
| 3. <i>ale</i> ‘but’ | 8. <i>nebo</i> ‘or’         | 13. <i>či</i> ‘or’           | 18. <i>až</i> ‘when; until’     |
| 4. <i>i</i> ‘and’   | 9. <i>než</i> ‘than’        | 14. <i>pokud</i> ‘as far as’ | 19. <i>jestli</i> ‘if, whether’ |
| 5. <i>jako</i> ‘as’ | 10. <i>ani</i> ‘neither’    | 15. <i>tedy</i> ‘therefore’  | 20. <i>li</i> ‘if, whether’     |

(12.5) a. *že* ‘that’

b. *když* ‘when’

c. *aby* ‘in order to’

d. *protože* ‘because’

e. *kdyby* ‘if’

f. *takže* ‘therefore’

g. *jestli* ‘if, whether’

<sup>576</sup> ČNK's *Frequency list browser* is available under the following URL: <https://www.korpus.cz/lists/> (accessed 25 Oct 2024). By the time this study is conducted, the most recent database for the frequency lists is SYN2015.

The maximal CQL queries for the structures in (12.1) in written Czech are provided in Figure 12.1 and Figure 12.2 for illustrative purposes. As the queries are quite complex, I shall explain them in more detail for the sake of transparency. The query in Figure 12.1 searches for the string SUB < CL < NP from (12.1a), i.e. subordinator followed by clitic(s) followed by noun phrase.<sup>577</sup> The subordinator is explicitly specified by the parameter [word], which searches for the exact orthographic form specified between the quotation marks. For the seemingly inflecting subordinators *aby*, *kdyby* an economic notation is available using so-called *wildcards* in [word="aby.\*"], [word="kdyby.\*"], i.e. symbols which serve as some sort of a variable or logical operator. Multiple elements can be searched for simultaneously, when separated by the vertical bar ( [ ... | ... ] ), which works as an exclusive disjunction (*either ... or ...*). The subordinator is followed by a series of different expressions, whereby the linear order is respected. The query specifies the occurrence of

- the conditional *by* and its literal and colloquial allomorphs,
- the auxiliary clitic *být* in its standard and colloquial orthographic forms in conjunction with the parts of speech (POS) tag for verb [tag="V.\*"] to reduce the probability that a homonymic non-auxiliary occurs (i.e. adverb *sem* ‘here’, *si* ‘REFL.DAT’),
- the reflexive clitics and their allomorphs in conjunction with the POS tag for short reflexives [tag="P7.\*"] to reduce errors (e.g. from homonymy between REFL.DAT *si* and colloquial AUX.2SG *si*),
- the dative constant clitics in conjunction with the POS tag for short personal pronouns [tag="PH.\*"],
- the accusative constant clitics in conjunction with the POS tag [tag="PH.\*"],
- a numeral [tag="C.\*"],
- a demonstrative [tag="PD.\*"],
- a possessive pronoun [tag="PS.\*"] or possessive reflexive [tag="P8.\*"] (frugally coded as [tag="P[S8].\*"]),
- a sequence of at most four adjectives [tag="A.\*"],
- a noun ["N.\*"].

The suffixes in curly brackets {0, 1} specify the minimum (left) and maximum (right) number of occurrences of the respective element and allow to account for optional elements in a query. The

<sup>577</sup> The presentation of the queries in Figure 12.1 and Figure 12.2 mimics the highlighting colours of KonText’s advanced search for the sake of transparency.



optionality also nicely accounts for redundancies in the queries, which do not need to be explicitly circumvented, e.g. the occurrence of COND after *kdyby*- or the occurrence of AUX after “inflected” COND. For adjectives, sequences of up to 4 elements are allowed {0,4}.<sup>578</sup> Accordingly, the minimum output is subordinator < accusative clitic < head noun. The vertical bar closes off the first part of the query from a second one, which repeats the procedure but purposefully omits accusative clitics, while it takes the dative clitic(s) as mandatory. Another vertical bar introduces a third portion, whereby all pronominal clitics are omitted, but the reflexives are set as obligatory. Thus, the query is supposed to avoid duplication and to return query results only once. Furthermore, the third part particularly prohibits the NP-internal element following the reflexive clitics to bear instrumental case. This specification has been added to sort out occurrences of the syllabic preposition *se* (< *s*) ‘with’ erroneously tagged as short reflexive [tag="P7.\*"].

```
[word="že|když|aby.*|protože|kdyby.*|takže|jestli"]
[word="by|bych|bysem|bys|bychom|bysme|byste"]{0,1}
[word="jsem|sem|jsi|si|jsme|sme|jste|ste" & tag="V.*"]{0,1}
[word="se|si|ses|sis" & tag="P7.*"]{0,1} [word="ti|mi|mu" &
tag="PH.*"]{0,1} [word="tě|ho" & tag="PH.*"] [tag="C.*"]{0,1}
[tag="PD.*"]{0,1} [tag="P[S8].*"]{0,1} [tag="A.*"]{0,4} [tag="N.*"] |
[word="že|když|aby.*|protože|kdyby.*|takže|jestli"]
[word="by|bych|bysem|bys|bychom|bysme|byste"]{0,1}
[word="jsem|sem|jsi|si|jsme|sme|jste|ste" & tag="V.*"]{0,1}
[word="se|si|ses|sis" & tag="P7.*"]{0,1} [word="ti|mi|mu" & tag="PH.*"]
[tag="C.*"]{0,1} [tag="PD.*"]{0,1} [tag="P[S8].*"]{0,1}
[tag="A.*"]{0,4} [tag="N.*"] |
[word="že|když|aby.*|protože|kdyby.*|takže|jestli"]
[word="by|bych|bysem|bys|bychom|bysme|byste"]{0,1}
[word="jsem|sem|jsi|si|jsme|sme|jste|ste" & tag="V.*"]{0,1}
[word="se|si|ses|sis" & tag="P7.*"] [tag="C.*" & tag!= "C...7.*"]{0,1}
[tag="PD.*" & tag!="PD...7.*"]{0,1} [tag="P[S8].*" &
tag!="C[S8]...7.*"]{0,1} [tag="A.*" & tag!="A...7.*"]{0,4} [tag="N.*" &
tag!="N...7.*"]
```

Figure 12.1: CQL query for structures of the type (SUB, CL, NP).

<sup>578</sup> The upper bound of four sequential adjectives for a useful query was determined empirically by several test queries and only chosen for the sake of comprehensiveness. Admittedly, the difference between [tag="A.\*"]{0,2} and [tag="A.\*"]{0,4} is insignificant. The difference is approximately 10±5 output tokens, if there is a difference, at all.

Note that the query focuses on pronominal and reflexive clitics only and intentionally treats conditional and auxiliary clitics merely as facultative, which follows the thesis's subject area. Accordingly, there are no further specifications to particularly query conditional or auxiliary clitics.

```
[word="že|když|aby.*|protože|kdyby.*|takže|jestli" [tag="C.*"]{0,1}
[tag="PD.*"]{0,1} [tag="P[S8].*"]{0,1} [tag="A.*"]{0,2} [tag="N.*"]
[word="by|bych|bysem|bys|bychom|bysme|byste"]{0,1}
[word="jsem|sem|jsi|si|jsme|sme|jste|ste" & tag="V.*"]{0,1}
[word="se|si|ses|sis" & tag="P7.*"]{0,1} [word="ti|mi|mu" &
tag="PH.*"]{0,1} [word="tě|ho" & tag="PH.*"]|
[word="že|když|aby.*|protože|kdyby.*|takže|jestli" [tag="C.*"]{0,1}
[tag="PD.*"]{0,1} [tag="P[S8].*"]{0,1} [tag="A.*"]{0,2} [tag="N.*"]
[word="by|bych|bysem|bys|bychom|bysme|byste"]{0,1}
[word="jsem|sem|jsi|si|jsme|sme|jste|ste" & tag="V.*"]{0,1}
[word="se|si|ses|sis" & tag="P7.*"]{0,1} [word="ti|mi|mu" & tag="PH.*"]
[word!="ti|tě|ho|mi|mu"]|
[word="že|když|aby.*|protože|kdyby.*|takže|jestli" [tag="C.*"]{0,1}
[tag="PD.*"]{0,1} [tag="P[S8].*"]{0,1} [tag="A.*"]{0,2} [tag="N.*"]
[word="by|bych|bysem|bys|bychom|bysme|byste"]{0,1}
[word="jsem|sem|jsi|si|jsme|sme|jste|ste" & tag="V.*"]{0,1}
[word="se|si|ses|sis" & tag="P7.*"] [word!="ti|tě|ho|mi|mu" &
tag!="....7.*"]
```

Figure 12.2: CQL query for structures of the type (SUB, NP, CL).

The query in Figure 12.2 comprises the very same building blocks, but reorganises the linear order between the query strings searching for the clitic (cluster) and the NP elements. Note, however, an important modification of the SUB < NP < CL query. The second and third variants are closed off by negatively defined query strings. The second variant disallows dative and accusative constant clitics ([word!="ti|tě|ho|mi|mu"]) to avoid duplication, while the third variant additionally prohibits any element bearing instrumental case. Bear in mind that the CQL query specifies only a linear string of contiguous elements, whereby the left and right context remain underspecified. We therefore need additional specifications for cases, where we successively remove the last element within a given sequence. Otherwise, the underspecified right context might contain any element in the respective query variants with mandatory pronominal dative or REFL, which includes dative and accusative constant clitics. In other words, we need to avoid double occurrences of the results from query variant one and two, which would skew the quantitative results. The additional specification of prohibited constant clitics is necessary for any 3P query.

Similar procedures were taken for the other structures described by (12.2)–(12.4), but with individual adjustments, where necessary. For (12.2) SUB CL Adv ↔ SUB Adv CL, I eliminated the monosyllabic adverbs *tam*, *tu*, *už*, which occur under the clitic label in the literature. They are excluded to not give rise to doubts about the positional value of the clitic cluster in 3P. For (12.3) SUB CL V ↔ SUB V CL, forms of *být* ‘(to) be’ in PRS.1/2SG are prohibited in the 3P configuration to prevent duplications of auxiliary clitics in second position. Finally, (12.4) SUB CL SBJ<sub>PRN</sub> ↔ SUB SBJ<sub>PRN</sub> CL does not need adaptations thanks to the formal uniqueness of the nominative personal pronouns. Furthermore, bear in mind that, while written data is formally stable in terms of orthography, spoken data show increasing formal variation. For example, different kinds of phonological contractions like consonant cluster simplification etc. need to be accounted for through additional working steps.<sup>579</sup>

The queries for eliciting 2P vs. 1P occurrences are directly derivable from the queries for 2P vs. 3P and vice versa. Basically, queries differ with respect to the first element of the query line, such that the subordinators are substituted by coordinating conjunctions. There are four coordinating conjunctions among the top 20 conjunctions in Table 12.1, which I cite in (12.6). They were used in the corpus query to approximate the left clause boundary for the alternation between 1P and 2P.

- (12.6)
- |    |      |       |
|----|------|-------|
| a. | a    | ‘and’ |
| b. | ale  | ‘but’ |
| c. | i    | ‘and’ |
| d. | nebo | ‘or’  |

## 12.2 Clitic Third: 2P vs. 3P after subordinator

Let us first consider the transposition between clitics in 2P and 3P under subordinators. Table 12.2 and Table 12.3 report the elicited frequencies for written Czech from the balanced SYN2020 corpus and spoken Czech from the ORAL v1 corpus respectively. Figure 12.3 and Figure 12.4 visualise the proportions of 2P vs. 3P by permuted category and subordinator respectively. The diagrams show quite clearly how register in the widest sense, i.e. “writtenness” vs. “spokenness”, affects clitic positioning. Generally, there is a good deal of variation in the realisation of 2P and 3P across permuted categories and subordinators within both corpora.

<sup>579</sup> Covering all variants, i.e. allomorphs/allographs, expands the corpus queries and make them increasingly cumbersome. But before one can account for allomorphic variation, one first must elicit all possible variants occurring in the corpus. For instance, the Czech subordinator *protože* ‘because’ has multiple realisations with different degrees of contraction in spoken Czech that occur in the corpus: *prtože*, *protže*, *prože*, *prže*, *prtže*, *prče*, *pže*.

Beginning with written Czech, 2P is the most frequent placement pattern whereby clitic positioning under *aby-*, *kdyby-*, *jestli* is the most consistent. One might speculate that the predominance of 2P with *aby-* and *kdyby-* is motivated by the underlying morphological structure of these subordinators. Both subordinators are morphologically complex and contain the conditional morpheme {by}. A similar ad hoc assumption could be made for *jestli*, which originally is morphologically complex, too, including the clitic interrogative particle *-li*. The pattern of *když* comes close to those three. All these patterns fluctuate when overt subject pronouns come in, cf. the column {SBJ<sub>PRN</sub>, CL}.

Table 12.2: Frequencies of elicited 2P and 3P occurrences in written Czech from SYN2020.

| Pairs:                     | {NP, CL}          |                   | {ADV, CL}         |                  | {V, CL}           |                 | {SBJ <sub>PRN</sub> , CL} |                | $\Sigma_{(\text{tokens})}$ |
|----------------------------|-------------------|-------------------|-------------------|------------------|-------------------|-----------------|---------------------------|----------------|----------------------------|
| SUB                        | 2P                | 3P                | 2P                | 3P               | 2P                | 3P              | 2P                        | 3P             |                            |
| <i>že</i>                  | 27,753<br>(70.7%) | 11,490<br>(29.3%) | 14,350<br>(90.0%) | 1,601<br>(10.0%) | 58,000<br>(97.5%) | 1,475<br>(2.5%) | 125<br>(20.0%)            | 501<br>(80.0%) | 115,295                    |
| <i>když</i>                | 9,130<br>(97.4%)  | 241<br>(2.6%)     | 4,653<br>(91.0%)  | 460<br>(9.0%)    | 25,010<br>(99.1%) | 218<br>(0.9%)   | 68<br>(33.7%)             | 134<br>(66.3%) | 39,914                     |
| <i>aby-</i>                | 10,559<br>(99.7%) | 27<br>(0.3%)      | 5,241<br>(99.9%)  | 4<br>(0.1%)      | 38,888<br>(99.9%) | 27<br>(0.1%)    | 93<br>(93.0%)             | 7<br>(7.0%)    | 54,846                     |
| <i>protože</i>             | 1,037<br>(37.4%)  | 1,733<br>(62.6%)  | 1,448<br>(72.8%)  | 542<br>(27.2%)   | 5,837<br>(97.9%)  | 125<br>(2.1%)   | 2<br>(1.0%)               | 203<br>(99.0%) | 10,927                     |
| <i>kdyby-</i>              | 1,499<br>(99.7%)  | 5<br>(0.3%)       | 1,051<br>(99.8%)  | 2<br>(0.2%)      | 4,216<br>(99.98%) | 1<br>(0.02%)    | 45<br>(88.2%)             | 6<br>(11.8%)   | 6,825                      |
| <i>takže</i>               | 992<br>(70.3%)    | 419<br>(29.7%)    | 893<br>(85.8%)    | 148<br>(14.2%)   | 2,918<br>(97.2%)  | 84<br>(2.8%)    | 0<br>(—)                  | 32<br>(100%)   | 5,486                      |
| <i>jestli</i>              | 1,171<br>(99.9%)  | 1<br>(0.1%)       | 828<br>(99.8%)    | 2<br>(0.2%)      | 3,912<br>(99.2%)  | 31<br>(0.8%)    | 23<br>(74.2%)             | 8<br>(25.8%)   | 5,976                      |
| $\Sigma_{(\text{tokens})}$ | 52,141            | 13,916            | 28,464            | 2,759            | 138,781           | 1,961           | 356                       | 891            | 239,269                    |

On the other hand, the subordinators *že*, *protože*, *takže* exhibit variation across all transpositions and pattern quite differently from one another: *že* consistently occurs with 2P as the dominant position except for overt subject pronoun, where 3P prevails; *takže* patterns essentially with *že*, but {SBJ<sub>PRN</sub>, CL} is attested exclusively with 3P; *protože* even prefers 3P for {NP, CL} and almost exclusively for {SBJ<sub>PRN</sub>, CL}. Again, one might speculate that the flexibility those subordinators show might have arisen from a common source, which is diachronically the morph *že*. The pair {V, CL} is the only convergent transposition across subordinators almost exclusively preferring 2P. One might

conjecture that the linear contiguity of the verb has an impact on the predominance of 2P. Recall Tables 10.6, 10.8, and 10.10 from Section 10.2.1 which showed that clitic postposition occurs rather infrequently with 2P in the data sample of spoken Czech, whereby this very syntagm is mainly instantiated by clause-initial verbs.

Table 12.3: Frequencies of elicited 2P and 3P occurrences in spoken Czech from ORAL v1.

| Pairs:                     | {NP, CL}       |                | {ADV, CL}        |                | {V, CL}          |               | {SBJ <sub>PRN</sub> , CL} |                | $\Sigma_{(\text{tokens})}$ |
|----------------------------|----------------|----------------|------------------|----------------|------------------|---------------|---------------------------|----------------|----------------------------|
| SUB                        | 2P             | 3P             | 2P               | 3P             | 2P               | 3P            | 2P                        | 3P             |                            |
| <i>že</i>                  | 401<br>(57.2%) | 300<br>(42.8%) | 1,010<br>(73.8%) | 358<br>(26.2%) | 4,747<br>(97.2%) | 136<br>(2.8%) | 16<br>(3.3%)              | 472<br>(96.7%) | 7,440                      |
| <i>když</i>                | 238<br>(85.3%) | 41<br>(14.7%)  | 334<br>(78.6%)   | 91<br>(21.4%)  | 2,577<br>(98.7%) | 33<br>(1.3%)  | 8<br>(5.6%)               | 135<br>(94.4%) | 3,457                      |
| <i>aby-</i>                | 109<br>(80.1%) | 27<br>(19.9%)  | 191<br>(92.3%)   | 16<br>(7.7%)   | 1,103<br>(98.8%) | 13<br>(1.2%)  | 3<br>(13.6%)              | 19<br>(86.4%)  | 1,481                      |
| <i>protože</i>             | 30<br>(21.4%)  | 110<br>(78.6%) | 101<br>(43.3%)   | 132<br>(56.7%) | 493<br>(93.4%)   | 35<br>(6.6%)  | 1<br>(0.3%)               | 360<br>(99.7%) | 1,262                      |
| <i>kdyby-</i>              | 41<br>(80.4%)  | 10<br>(19.6%)  | 75<br>(76.5%)    | 23<br>(23.5%)  | 499<br>(99.2%)   | 4<br>(0.8%)   | 10<br>(37.0%)             | 17<br>(63.0%)  | 679                        |
| <i>takže</i>               | 37<br>(31.4%)  | 81<br>(68.6%)  | 101<br>(48.1%)   | 109<br>(51.9%) | 457<br>(92.0%)   | 40<br>(8.0%)  | 0<br>(—)                  | 246<br>(100%)  | 1,071                      |
| <i>jestli</i>              | 45<br>(73.8%)  | 16<br>(26.2%)  | 92<br>(84.4%)    | 17<br>(15.6%)  | 652<br>(98.3%)   | 11<br>(1.7%)  | 3<br>(10.7%)              | 25<br>(89.3%)  | 861                        |
| $\Sigma_{(\text{tokens})}$ | 901            | 585            | 1,904            | 746            | 10,528           | 272           | 41                        | 1,274          | 16,251                     |

Turning to spoken Czech, variation obviously increases in the data sample, as can be seen in Table 12.3. The overall tendency is still the predominance of 2P. However, 3P occurs more liberally. Observe that 3P is attested under *aby-* and *kdyby-* in spoken Czech, which is intriguing insofar as this configuration implies diacsis, which represents only a small portion, however. A similar trend can be observed for *jestli*. There are two transpositions converging across subordinators but with opposite tendencies. While 2P is clearly the dominant pattern for {V, CL}, 3P is preferably realised for {SBJ<sub>PRN</sub>, CL}. The cases of *aby-* and *kdyby-* are again particularly interesting, for they display the predominance of diacsis, which is in stark contrast to the prevalence of clitic cluster formation elsewhere.

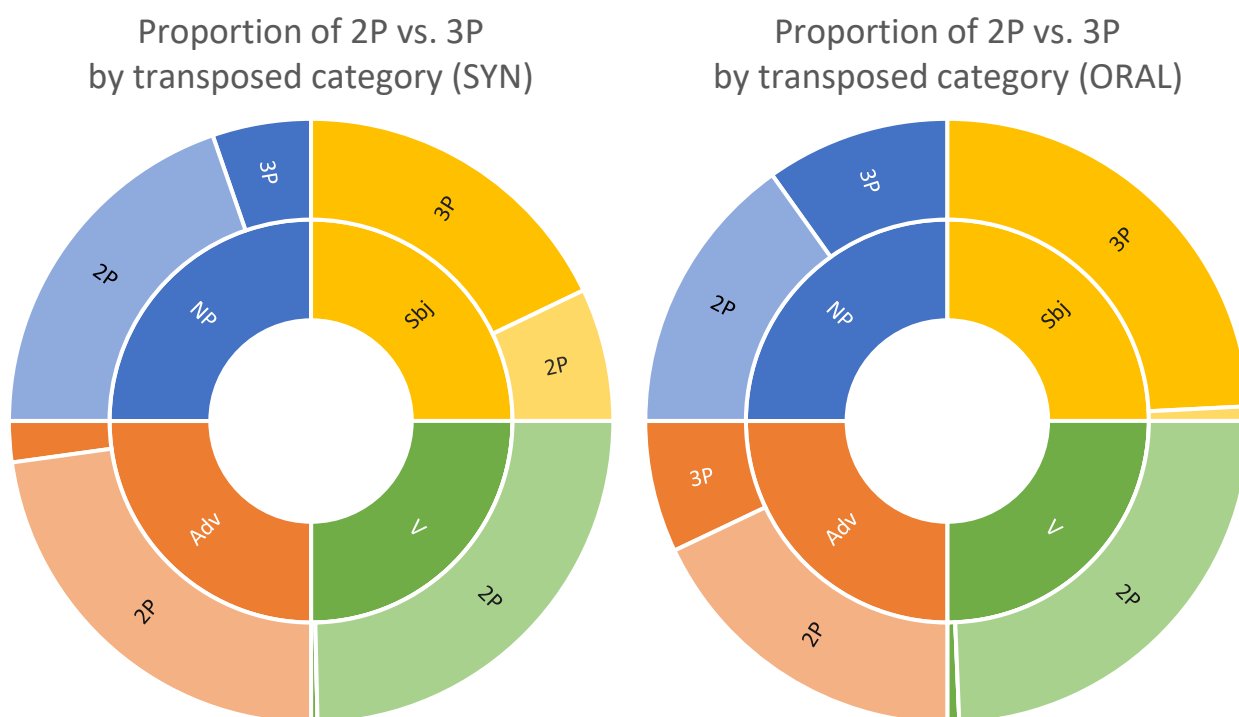


Figure 12.3: Proportions of 2P vs. 3P by transposed category in written vis-à-vis spoken Czech

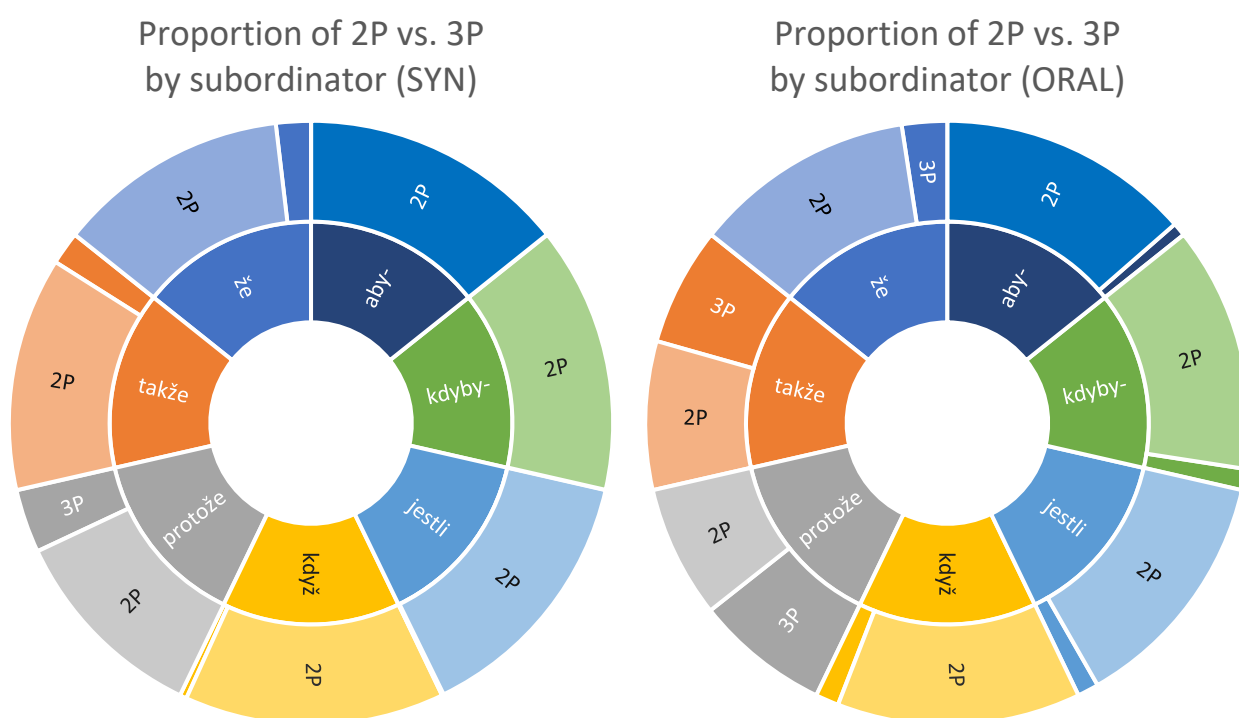


Figure 12.4: Proportions of 2P vs. 3P by subordinator in written vis-à-vis spoken Czech

Comparing the relative frequencies of 2P vs. 3P across corpora and what I will continue to refer to somewhat liberally as “register”, we find, first, that written and spoken Czech share the overall tendency of preferring 2P over 3P. However, we observe a shift in ratios when moving from written

to spoken Czech. As a rule, 3P occurs more often in spoken than in written Czech. The relative frequencies for 3P are always higher in the spoken than in the written corpus with only one exception: we find equal shares for the adjacent transpositions of {SBJ<sub>PRN</sub>, CL} under *takže*. Nevertheless, there are relative frequencies, which are comparable between written and spoken Czech, and thus may indicate converging trends. This is true for the entire column {V, CL} and for {SBJ<sub>PRN</sub>, CL} under *protože*. In contrast, we encounter instances of a reversed pattern with 3P > 2P in spoken Czech, where written Czech exhibits 2P > 3P. This is the case for {NP, CL} under *takže*, for {ADV, CL} under *protože*, and for {SBJ<sub>PRN</sub>, CL} under *aby-*, *kdyby-*, and *jestli*.

## 12.3 Clitic First: 1P vs. 2P after coordinating conjunction

Now, we will examine the transposition of 2P and 1P after coordinating conjunctions. Considering written Czech first, 2P is the most frequent linearisation pattern after coordinating conjunctions, cf. Table 12.4. Clitic First is very rare and does not even reach an occurrence frequency of 1% with some noteworthy exceptions. For the conjunction *i*, there is more than one-fifth ( $\approx 1/5$ ) of occurrences showing first position placement with {V, CL}. We observe a good deal of variation with considerable amounts of 1P cliticisation for the conjunction *nebo*: 1P is the near-to-exclusive pattern for {V, CL} and represents the majority of cases for {ADV, CL}. For transpositions including NPs {NP, CL} and subject pronouns {SBJ<sub>PRN</sub>, CL} the occurrence frequency of 1P oscillates around one-third ( $1/3$ ). Notably, clitics are not attested in 1P with overtly realised subject pronouns, except for *nebo*.

Table 12.4: Frequencies of elicited 1P and 2P occurrences in written Czech from SYN2020.

| Pairs:                     | {NP, CL}       |                    | {ADV, CL}        |                   | {V, CL}          |                     | {SBJ <sub>PRN</sub> , CL} |                  | $\Sigma_{(\text{tokens})}$ |
|----------------------------|----------------|--------------------|------------------|-------------------|------------------|---------------------|---------------------------|------------------|----------------------------|
| CONJ                       | 1P             | 2P                 | 1P               | 2P                | 1P               | 2P                  | 1P                        | 2P               |                            |
| <i>a</i>                   | 5<br>(0.01%)   | 50,465<br>(99.99%) | 40<br>(0.1%)     | 50,785<br>(99.9%) | 59<br>(0.03%)    | 174,393<br>(99.97%) | 0<br>(—)                  | 12,749<br>(100%) | 288,496                    |
| <i>ale</i>                 | 4<br>(0.1%)    | 6,714<br>(99.9%)   | 32<br>(0.3%)     | 11,622<br>(99.7%) | 14<br>(0.1%)     | 19,049<br>(99.9%)   | 0<br>(—)                  | 2,248<br>(100%)  | 39,683                     |
| <i>i</i>                   | 7<br>(0.2%)    | 3,651<br>(99.8%)   | 10<br>(0.9%)     | 1,131<br>(99.1%)  | 37<br>(22.8%)    | 125<br>(77.2%)      | 0<br>(—)                  | 478<br>(100%)    | 5,439                      |
| <i>nebo</i>                | 424<br>(28.4%) | 1,070<br>(71.6%)   | 1,098<br>(66.2%) | 560<br>(33.8%)    | 5,393<br>(97.9%) | 116<br>(2.1%)       | 8<br>(36.4%)              | 14<br>(63.6%)    | 8,683                      |
| $\Sigma_{(\text{tokens})}$ | 440            | 61,900             | 1,180            | 64,098            | 5,503            | 193,683             | 8                         | 15,489           | 342,301                    |

From Table 12.5 we can read a clear preference of 2P after coordinating conjunctions in spoken Czech, too. Considerable amounts of 1P are attested for *nebo* again. However, the tendency to position clitics in 1P is lower than in written Czech, i.e. we observe an increase of 2P for *nebo*. In contrast, the conjunction *i* shows the reverse trend with the pair {V, CL}: While 2P occurs most often in written Czech, 1P is dominant in spoken Czech, where over two-thirds of occurrences are attested in 1P. Perhaps, there might be also a slight tendency for *i* to increase 1P with NP and ADV, but this is rather uncertain due to the low token frequencies. Note that no 1P occurrences are attested with overtly realised subject pronouns as in written Czech – again, except for *nebo*, but with a significantly decreased ratio (< 5%) when compared to written Czech.

Table 12.5: Frequencies of elicited 1P and 2P occurrences in spoken Czech from ORAL v1.

| Pairs:                     | {NP, CL}     |                | {ADV, CL}     |                  | {V, CL}        |                  | {SBJ <sub>PRN</sub> , CL} |                 | $\Sigma_{(\text{tokens})}$ |
|----------------------------|--------------|----------------|---------------|------------------|----------------|------------------|---------------------------|-----------------|----------------------------|
| CONJ                       | 1P           | 2P             | 1P            | 2P               | 1P             | 2P               | 1P                        | 2P              |                            |
| <i>a</i>                   | 2<br>(0.2%)  | 940<br>(99.8%) | 11<br>(0.3%)  | 4,052<br>(99.7%) | 63<br>(1.1%)   | 5,478<br>(98.9%) | 0<br>(—)                  | 2,299<br>(100%) | 12,845                     |
| <i>ale</i>                 | 3<br>(1.3%)  | 227<br>(98.7%) | 10<br>(1.2%)  | 817<br>(98.8%)   | 37<br>(2.7%)   | 1,327<br>(97.3%) | 0<br>(—)                  | 709<br>(100%)   | 3,130                      |
| <i>i</i>                   | 3<br>(4.9%)  | 58<br>(95.1%)  | 1<br>(2.8%)   | 35<br>(97.2%)    | 23<br>(69.7%)  | 10<br>(30.3%)    | 0<br>(—)                  | 5<br>(100%)     | 135                        |
| <i>nebo</i>                | 7<br>(15.2%) | 39<br>(84.8%)  | 52<br>(13.3%) | 339<br>(86.7%)   | 257<br>(74.7%) | 87<br>(25.3%)    | 2<br>(4.3%)               | 44<br>(95.7%)   | 827                        |
| $\Sigma_{(\text{tokens})}$ | 15           | 1,264          | 74            | 5,243            | 380            | 6,902            | 2                         | 3,057           | 16,937                     |

In summary, written and spoken Czech share the overall tendency for 2P after coordinating conjunctions; 1P is rather rare. There is no clear overall trend to increase 1P in spoken Czech, except for *i* with verbs. The conjunction *nebo* shows considerable flexibility of clitic placement in both varieties, but the occurrence frequency for 1P is even lower in spoken than in written Czech. The quantitative data is visualised in Figure 12.5 focusing on transposed categories and in Figure 12.6 representing the conjunctions.



Proportion of 1P vs. 2P  
by transposed category (SYN)



Proportion of 1P vs. 2P  
by transposed category (ORAL)



Figure 12.5: Proportions of 1P vs. 2P by transposed category in written vis-à-vis spoken Czech

Proportion of 1P vs. 2P  
by conjunction (SYN)



Proportion of 1P vs. 2P  
by conjunction (ORAL)

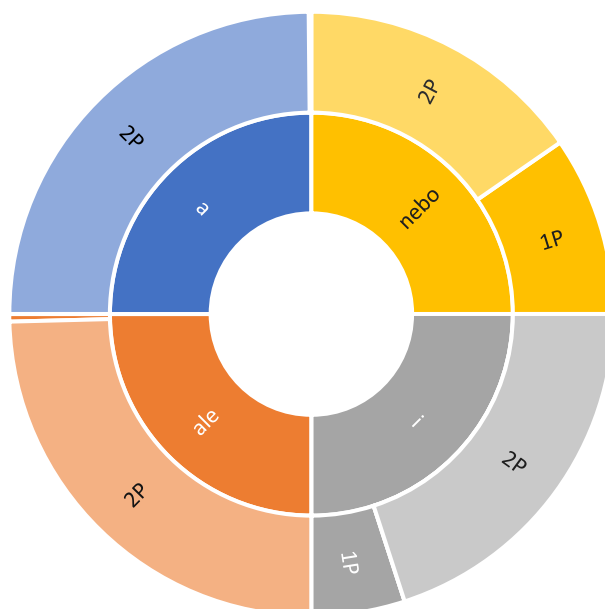


Figure 12.6: Proportions of 1P vs. 2P by conjunction in written vis-à-vis spoken Czech

## 12.4 Delayed Placement after subordinator

The data in this section is purely supplemental and by no means exhaustive. Based on the queries for Clitic Third from Section 12.2, the SYN and ORAL corpora were searched for Delayed Placement with clitics or clitic clusters in fourth and fifth position. The queries are limited, because they only utilise elements of the query strings for 2P vs. 3P and do not exhaust all logically possible combinations. Especially, I did not conduct queries containing the syntagms (ADV, V) in order to get uncontroversial results of 4P and 5P respectively. Depending on one's stance towards constituency, critiques might object that  $ADV < V$  could merely be a complex VP, such that the adverb might not receive constituent status. Moreover, frequencies are not differentiated by subordinator but subsumed under the grammatical label SUB due to the small amount of data. Still, the data is suitable for allowing a glimpse at Delayed Placement in both written and spoken Czech with the caveat that future research is needed for a more comprehensive picture. Table 12.6 summarises absolute frequencies for 4P in written and spoken Czech.<sup>580</sup> Table 12.7 gives the respective quantities for 5P.

Table 12.6: Absolute frequencies of 4P in written and spoken Czech.

| SYN2020 | Syntagm                       | Abs. freq. | ORAL v1 | Syntagm                       | Abs. freq. |
|---------|-------------------------------|------------|---------|-------------------------------|------------|
|         | SUB NP V CL                   | 229        |         | SUB NP V CL                   | 18         |
|         | SUB SBJ <sub>PRN</sub> V CL   | 2          |         | SUB SBJ <sub>PRN</sub> V CL   | 7          |
|         | SUB ADV NP CL                 | 327        |         | SUB ADV NP CL                 | 41         |
|         | SUB SBJ <sub>PRN</sub> NP CL  | —          |         | SUB SBJ <sub>PRN</sub> NP CL  | 2          |
|         | SUB NP SBJ <sub>PRN</sub> CL  | —          |         | SUB NP SBJ <sub>PRN</sub> CL  | 8          |
|         | SUB ADV SBJ <sub>PRN</sub> CL | 53         |         | SUB ADV SBJ <sub>PRN</sub> CL | 28         |
|         | Sum                           | 611        |         | Sum                           | 104        |

Delayed Placement with 4P and 5P is attested for both spoken and written Czech, but with comparatively low frequency. Distribution patterns differ across varieties, but they share tendencies, in that the 4P linearisations in (12.7) are most prominent within both populations.

<sup>580</sup> Frequency data in Table 12.6 and Table 12.7 are tentatively cleared. While some cells still provide raw data directly extracted from the corpus results, some others provide final results. Particularly, false positives were removed from {4P, SUB SBJ<sub>PRN</sub> NP CL} in both written and spoken, {5P, SUB SBJ<sub>PRN</sub> NP V CL} (SYN2020), and {5P, SUB NP SBJ<sub>PRN</sub> V CL} (ORAL v1).

Table 12.7: Absolute frequencies of 5P in written and spoken Czech.

| SYN2020 | Syntagm                         | Abs. freq. | ORAL v1 | Syntagm                         | Abs. freq. |
|---------|---------------------------------|------------|---------|---------------------------------|------------|
|         | SUB ADV NP V CL                 | 12         |         | SUB ADV NP V CL                 | —          |
|         | SUB SBJ <sub>PRN</sub> NP V CL  | —          |         | SUB SBJ <sub>PRN</sub> NP V CL  | 1          |
|         | SUB NP SBJ <sub>PRN</sub> V CL  | —          |         | SUB NP SBJ <sub>PRN</sub> V CL  | —          |
|         | SUB ADV SBJ <sub>PRN</sub> V CL | —          |         | SUB ADV SBJ <sub>PRN</sub> V CL | —          |
|         | Sum                             | 12         |         | Sum                             | 1          |

- (12.7) a. SUB NP V CL  
 b. SUB ADV NP CL  
 c. SUB ADV SBJ<sub>PRN</sub> CL

The figures fit in the general tendency that Delayed Placement becomes rarer with increasing distance to the Wackernagel position. Note that only 1 occurrence of 5P was elicited in ORAL v1 in contrast to  $14 \times$  5P reported in Section 10.2.1 on clitic placement in Czech (cf. Table 10.1, Table 10.10, Table 10.11). Obviously, the queries need some refinement to cover more data. One may speculate that the amount of data could be increased for written Czech, too, provided improved CQL queries. On the other hand, comparing the results from CQL querying for Section 12 and random sampling by individual clitic form for Section 10 indicates that random sampling is a good means of data acquisition. This is true even in case one is interested in rare phenomena.

## 12.5 Interim conclusion

The examination of purposefully elicited instances of 2P vs. 3P and 2P vs. 1P reveals several interesting facts about deviate clitic placement in edited written and spontaneously spoken Czech. The frequency data show a register effect in that written and spoken Czech behave differently. This finding corroborates the arguments put forward in Section 6.3, which endorse the examination of actual speech for genuine grammar research. Generally, while the ratio of 3P is higher in spoken than in written Czech, the trend is reversed for 1P, for which the ratio is smaller in spoken Czech when compared to the written language. Furthermore, spoken Czech cannot be merely characterised by a set of phonological isoglosses and a list of lexical idiosyncrasies. More dimensions need to be taken into consideration. However, we observe some interesting and quite surprising properties in written Czech, too. One of the most significant findings is that the likelihood of occurrence for 3P varies across subordinators and intervening morphosyntactic categories. This implicates that 3P

environments cannot be lumped together and that it is insufficient to refer to a single surrogate category or item, like *že*. Grammatical judgements and theoretical considerations need to take a broader and more diverse data set into account.

The data show that a number of hypotheses argued for in the literature are false. Particularly, the data falsify the claim that DP is a rare phenomenon in Czech (cf. Veselovská 1995: §4.6) and that verbal placement between subordinator and clitic is ungrammatical (cf. Veselovská 1995: §4.6; cf. ex. [28b]).<sup>581</sup> Furthermore, the data demonstrate that 3P is not blocked under subordinators other than *že* ‘that’ (cf. Veselovská 1995: §9.3.5; cf. ex. [39]) and that more than one constituent may intervene between subordinator and clitic (cf. Veselovská 1995: §4.6; cf. ex. [27]). Generally, these results apply to both spoken and written Czech and are in line with Hana’s (2007: 107–111) and Lenertová’s (2004: 136–137) discussion of 3P in Czech. On a related note, the data show significant differences between actual linguistic usage and the prescriptive standard norm followed in edited writing. The SYN data quite consistently show the absence of 3P under *kdyby*, *aby*, and *jestli*, while the subordinator *když* allows it only marginally. In contrast, the ORAL data exhibit a remarkable increase of 3P in these environments, even though 2P remains the dominant pattern. We obtained several thousand tokens with 3P in SYN 2020 ( $n > 19,000$ ) and over 2,500 in ORAL v1 implying that the construction under discussion recurs across writers/speakers, time, locations, genres, source types, and topics. Clitic positioning is not subject to categorical dichotomies, but we are dealing with a more nuanced phenomenon. Multiple factors influence the likelihood of a particular pattern like the occurrence of certain subordinators and grammatical categories, which are associated with a preference of Wackernagel over Delayed Placement or – surprisingly enough – vice versa.

It is more difficult to make statements on 1P due to low occurrence frequencies and an unknown degree of uncertainty as to which domains are coordinated by the conjunctions. We are certainly dealing with a rare phenomenon here. However, it is possible to make some tentative generalisations at this point. Note that, while *nebo* behaves decidedly different than the other coordinating conjunctions, there seems to be some restriction at work regarding the transposition with subject pronouns {SBJ<sub>PRN</sub>, CL} across conjunctions. This is even clearer for spoken Czech, where *nebo* converges with the other conjunctions.<sup>582</sup>

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<sup>581</sup> Cf. also Veselovská (1995: §9.3.5, ex. [40b–c]) for the exemplification of the ungrammatical  $C^0 > V^0 > CL$  order.

<sup>582</sup> One might hypothesise that the distinct behaviour of {SBJ<sub>PRN</sub>, CL} across transpositions (1P/2P, 2P/3P) is grounded in the special status of overt subject pronouns in Czech (and Slavic in general). Note that both cases share the general tendency of placing the subject preferably before the clitic (cluster), which favours 2P over 1P with coordinating conjunctions, but 3P over 2P with subordinators. One relevant issue could be information structure, e.g. topicality of the subject pronoun. Accordingly, the subject occupies the left-most position available within the clausal domain, which is clause-initially in independent clauses but the second constituent following the subordinator in embedded clauses. Another point worth considering could be a syntactic or syntagmatic requirement on subject pronouns to precede clitics. Recall the discussion about precedence patterns in agreement drop in Section 10.8.4 (cf. also Section 10.4.4 on Goralish). I cannot go further here and leave these issues for future research.

The observations explicated here have not been made before, even though certain aspects implicitly underlay the scientific discourse (cf. the cited passages from Veselovská 1995 on 3P/DP). The present approach has proven valuable and may serve for further elaboration on the question of clitic deviate positioning. For instance, further quantitative studies in 2P vs. 3P cliticisation could utilise more, fine-grained register distinctions, include more subordinators, and develop a more detailed account of the syntactic elements intervening between subordinator and clitic, e.g. by differentiating between branching and non-branching units. A refined analysis of Clitic First might aim at eliminating non-clausal tokens from the data sample. A particular aspect sketched in Section 4.4.2 but not addressed here is the variation across types of clitics (e.g. REFL vs PRN). The findings presented here have repercussions for linguistic methodology as discussed in Chapter 6. They demonstrate the capabilities of empirical research in general and corpus linguistics in particular suggesting that it is not sufficient to develop arguments upon a handful of introspectively constructed examples. Such an approach is not only at the risk of leading to incorrect generalisations and false conclusions, but also to miss observations and overlook patterns.

## 13 Information structure and delayed clitic placement in Polish

### 13.1 Introductory remarks

As a first approximation, *information structure* refers to the organisation of elements in an utterance with respect to what kind of information they represent in a discourse. Thereby notions usually occur such as “what is talked about”, “new information”, “old information”, “being important”, or “introducing a contrast”. Two dimensions of organisation are recognised: linear syntax (word and constituency order) and prosodic phonology. Particular positions in a syntactic string are salient locations for certain discourse functions. For instance, elements in clause-final position usually bear the discourse function of FOCUS in Slavic. However, there is no strict one-to-one relation between discourse functions and syntactic positions. On the contrary, discourse functions can be expressed by intonation irrespective of their linear position in a syntactic string.<sup>583</sup> The term *information structure* goes back to Halliday (1967) and the phenomena subsumed under this “convenient cover expression”, as Vallduví (2014) puts it, are referred to by several other labels, e.g. *theme* and *rheme*, *functional sentence perspective* (Prague linguistic school), or *information packaging* (coined by Chafe 1976 and adopted in HPSG). The study of information structure has a long tradition, and it has been conducted in diverse linguistic schools. I cannot do justice to all of these and the vast corpus of literature. As I need to restrict myself, I will focus on the so-called Leipzig model, which was particularly developed for approaching information structure in Slavic (cf. Junghanns 2002b; Junghanns & Lenertová 2008; Junghanns & Zybatow 1997, 2009; Zybatow 1999; Zybatow & Junghanns 1998). Referring to this thread of research benefits from the fact that clitics have been explicitly taken into consideration. Furthermore, the comprehensive approach of the Leipzig model covers all major patterns of information structuring in Slavic. Other bodies of literature are sometimes dedicated only to selected phenomena, e.g. either TOPIC or FOCUS, or prosody. First, I will highlight some previous considerations of the relation between clitics and information structure that can be found in the literature on clitics in Slavic in Section 13.2. Then, I will briefly describe the Leipzig model in Section 13.3. Section 13.4 presents selected data points from Polish showing delayed clitic placement with different information structure configurations. Based on the observations from Section 13.4, I will propose in Section 13.5 that clitics in Polish can be analysed as Wackernagel clitics that are placed within certain information structure domains. Section 13.6 concludes Chapter 13.

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<sup>583</sup> Depending on one’s view of syntax and morphology and the distinction between them, one could add morphological exponents or morpho-syntactic marking as a third dimension, e.g. the Japanese Topic-marking postposition *-wa*.

## 13.2 Previous considerations of information structure and clitics

Two types of clitic constructions have been reflected on with respect to information structure: Delayed Placement and clitic climbing. It is remarkable that information structure is repeatedly mentioned in the context of cliticisation yet does not receive much attention. In other words, notions of information structure are explicitly brought up in the literature on Slavic clitics, but only few contributions take it into consideration seriously, i.e. as a linguistic domain in its own right. The major threads of clitic research in Slavic focus on syntax, phonology, or a combination thereof (cf. also Diesing 2010: 121). I agree with Čavar & Seiss (2011: 4), who state that “information structure has not been the focus of 2PC placement analyses so far, although some work has hinted at relevant implications (e.g. Diesing et al. 2009).”<sup>584</sup> Similarly, Diesing et al. (2009: 72) assert: “The least studied facet of clitic positioning is the impact of information structure, yet intuitively we feel that this is where the action is.” Already Franks & King (2000: 137) noted that “[t]he complex discourse factors interacting with the positioning of the clitic cluster in subordinate clauses require much further research”. I will look at Delayed Placement first, before I go on to discuss clitic climbing. Considerations about the interaction of cliticisation and information structure in Slavic are usually cursory or fragmentary, such that I will present the reader with several pieces of a puzzle that still needs to be put together.

Discussing 3P in Slovak, Franks & King (2000: 137) note that the contrast between 2P and 3P in subordinate clauses is linked to discourse factors. Example (13.1a) presents the “unmarked” order with the clitic *ho* ‘him.ACC’ occupying the second position of the subordinate clause. On the other hand, the AdvP *zajtra ráno* ‘tomorrow morning’ intervenes between the subordinator *že* and the clitic in (13.1b), such that *ho* occurs in Delayed Placement. The AdvP is reported to be stressed and discourse-prominent in this configuration. In a similar vein, Franks (2010: 59–60) states that delayed clitic placement in Czech (13.2) is linked with the intervening element being focused (marked by capital letters indicating stress). This is corroborated by Lenertová (2004: 136), who affirms that the respective constituent in Czech Clitic Third constructions is either topicalised, focused, or emphasised in some other way (recall also the discussion of delayed clitic placement in Section 4.4.2).

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<sup>584</sup> It seems that Čavar & Seiss (2011) are not aware of Junghanns’s (2002c) proposal for clitic climbing in Czech.

- (13.1) a. Myslím, že<sup>1</sup> ho<sup>2</sup> [AdvP zajtra ráno ] u-myje.  
 think.PRS.1SG that him.ACC tomorrow morning PFV-wash.PRS.3SG  
 b. Myslím, že<sup>1</sup> [AdvP zajtra ráno ]<sup>2</sup> ho<sup>3</sup> umyje.  
 ‘I think that he will wash it/him tomorrow morning.’ (Slk; Franks & King 2000: 137)
- (13.2) a. Helena říkala, že<sup>1</sup> se<sup>2</sup> Petr odstěhoval.  
 Helena.NOM say.PST.SG.F that REFL.ACC Petr.NOM move\_out.PST.SG.M  
 ‘Helena said, Peter had moved.’ (Cz; Franks 2010: 59)  
 b. Helena říkala, že<sup>1</sup> PETR<sup>2</sup> se<sup>3</sup> odstěhoval.  
 ‘Peter, Helena said, had moved.’  
 c. %Jistě namítnete, že<sup>1</sup> to<sup>2</sup> VÁM<sup>3</sup> se<sup>4</sup> zítra  
 surely object.PRS.2PL that this.NOM.SG.N you.PL.DAT REFL.ACC tomorrow  
 ne-stane.  
 NEG-happen.PRS.3SG  
 ‘You will surely object that to you it will not happen tomorrow.’  
 d. %Jistě namítnete, že<sup>1</sup> to<sup>2</sup> VÁM<sup>3</sup> ZÍTRA<sup>4</sup> se<sup>5</sup> nestane.  
 ‘You will surely object that to you, tomorrow, it will not happen.’ (Cz; Franks 2010: 60)

Franks & King (2000: 229) also notice a relationship between Clitic Third and information structure in BCMS. Their main argument accounting for 3P is prosodic (cf. also Bošković 2001): An utterance’s initial constituent “is separated by a pause from the rest of the clause”. (Franks & King 2000: 229) The clitic then shifts to the position right-adjacent to the second clausal constituent but occupying second position of the prosodic constituent following the pause. They continue “that the initial phrase is often topicalised or otherwise receives special discourse status.” (Franks & King 2000: 229) Recall example (4.76) repeated in (13.3a), where the time adverbial *svaki dan* ‘every day’ is topicalised and marked off by a pause (#). Franks (2010: 44) cites example (13.3b) saying that “[c]ontrastive focus (when set off prosodically) and parentheticals have a similar delaying effect on clitic placement.”

- (13.3) a. [ Svaki dan ]<sup>1</sup>TOP # [v dogodi ]<sup>2</sup> se<sup>3</sup> najmanje jedna  
 every day happen.PRS.3SG REFL.ACC at\_least one.NOM.SG.F  
 prometna nezgoda.  
 traffic.ADJ.NOM.SG.F accident.NOM.SG  
 ‘Every day at least one traffic accident took place.’ (BCMS; Franks & King 2000: 229)



b. Javili                      su              nam      da<sup>1</sup> # [ prije nekoliko dana ]<sup>2</sup><sub>FOC</sub> # [ na  
 announce.PST.PL.M    AUX.3PL    us.DAT    that      ago    several    day.GEN.PL      on  
 toj                      liniji ]<sup>3</sup><sub>FOC</sub>    #    voz<sup>4</sup>                      je<sup>5</sup>              kasnio              tri  
 that.LOC.SG.F    line.LOC.SG      train.NOM.SG    AUX.3SG    be\_late.PST.SG.M    three  
 sata.  
 hour.GEN.SG

‘They announced that, several days ago, on that line, the train was 3 hours late.’

(BCMS; Franks 2010: 44)

More recently, Franks (2016: 118–119) comments on variable clitic order like (13.4) and diacsis like (13.6) in Slovene with recourse to information structure. Example (13.4a) shows the canonical order DAT < ACC in the cluster, while the order in (13.4b) is reversed. Referring to a suggestion by Adrian Stegovec, Franks remarks that (13.4a) is a neutral reply to (13.5a), while (13.4b) is a contextually appropriate answer to question (13.5b) with the direct object *lopova* ‘thief.ACC.SG’ being fronted. Thus, the word order difference between (13.4a) and (13.4b) reflects different information structures. There is still more research needed on diacsis. However, following Franks’s (2016) considerations, it seems that the discourse function of Focus plays a role in (13.6), which is especially clear from (13.6b). Note that the clitics *ga* and *jo* are contrastively focused here.<sup>585</sup>

(13.4) a. Gospa              **mu**              **ga**              **je**              opisala.  
 lady.NOM.SG    him.DAT    him.ACC    AUX.3SG    describe.PST.SG.F  
 ‘The lady described him to him.’

b. Gospa              **ga**              **mu**              **je**              opisala.  
 lady.NOM.SG    him.ACC    him.DAT    AUX.3SG    describe.PST.SG.F  
 ‘The lady described him to him.’ (Slv; Franks 2016: 117)

(13.5) a. Kdo              **je**              lopova              opisal              Petru?  
 who.NOM    AUX.3SG    thief.ACC.SG    describe.PST.SG.M    Peter.DAT  
 ‘Who described the thief to Peter?’ (Slv; Franks 2016: 118)

b. Lopova              kdo              **je**              opisal              Petru?  
 thief.ACC.SG    who.NOM    AUX.3SG    describe.PST.SG.M    Peter.DAT  
 ‘The thief, who described him to Peter?’ (Slv; constructed from Franks 2016: 118)

<sup>585</sup> Franks (2016: 118) takes the word order facts as evidence for Slovene clitics losing their cliticity and becoming weak pronouns as it has been argued for the case of Polish, even though he emphasises that the process is more advanced in Polish. Jung & Migdalski (2022), too, argue for an ongoing “strengthening” of short pronouns in Slovene, but also Macedonian.

- (13.6) a. ? **So** [AdvP včera] **ga** pretepli?  
 AUX.3PL yesterday him.ACC beat.PST.SG.M  
 ‘Did they beat him up yesterday.’ (Slv; Franks 2016: 118)
- b. ? **A si** [v vidu] **ga** al **jo**?  
 so AUX.2SG see.PST.SG.M him.ACC or her.ACC  
 ‘So did you see HIM or HER?’ (Slv; Franks 2016: 118, n. 46)

Avgustinova & Oliva (1997) represent one of the few accounts that propose a more elaborated picture of the interaction between clitic placement and information structure. They examine clitic positioning in Czech with particular reference to Delayed Placement. In their approach, Wackernagel positioning serves as a means of marking the first clausal position as communicatively significant. Avgustinova & Oliva (1997) adopt a modified version of the Prague school’s approach to Functional Sentence Perspective, where the notion of *communicative dynamism* plays an important role. Roughly speaking, different constituents are necessarily assigned different levels of importance. However, their account allows for multiple elements to share the same *degree of communicative dynamism*, i.e. the same level of importance (cf. Avgustinova & Oliva 1997: 36–37). Consequently, they analyse multiple constituents preceding the clitic cluster as sharing the same communicative weight. In this regard, “several constituents creat[e] together a single ‘semantic unit’ [...] in front of the sentential clitics” (Avgustinova & Oliva 1997: 41). Avgustinova & Oliva (1997: 46) conclude that the “positioning of the sentential clitic in Czech” is not “based on the syntactic constituency of the Czech clause, but rather on its communicative structure.” They continue that

“the word order regularities concerning Czech clitics do not differ from the general word order regularities of Czech (and other free-word-order languages) – i.e. from word order based on communicative rather than syntactic constraints.” (Avgustinova & Oliva 1997: 46)

Note that Avgustinova & Oliva (1997) argue that the syntactic status of the constituents preceding the clitics is important for their ability to co-occur in the communicative unit occupying the first clausal position. Only elements with a similar syntactic status may constitute a unit. They approach the syntactic status in a valency-related manner distinguishing between several kinds of elements: Actants, circonstants, complements, and adjuncts differ with respect to syntactic category (argument NP/clause vs. adverbial) and lexical status in relation to a verbal head (being specified in the predicate’s subcategorisation frame or not).<sup>586</sup> For one thing, they attribute the contrast between

<sup>586</sup> Avgustinova & Oliva (1997: 34) inform that the opposition actant vs. circonstant is a general one, while the distinction of complements and adjuncts are predicate-specific. Actants are subjects and objects, and as such, they must be subcategorised for. Circonstants are adverbial expressions and are purely optional elements that can occur with any

(13.7) and (13.8) to the fact that the relevant constituents (i.e. both PPs) are circonstants in (13.7), whereas the NP *Marii* is an actant in (13.8) (judgements as in the original).

- (13.7) a. (V našem pražském býtě jsme příbuzné  
in our.LOC.SG.M Prague.ADJ.LOC.SG.M flat.LOC.SG AUX.1PL relative.ACC.PL  
ze Saarbrücken o vánocích ještě nějak snesli,) ale<sup>0</sup> [PP na  
from Saarbrücken during Christmas.LOC still somehow bear.PST.PL.VIR but to  
chatu ]<sup>1</sup> [PP v létě ]<sup>2</sup> **jsem**<sup>3</sup> **je**<sup>3</sup> raději  
weekend\_house.ACC.SG in summer.LOC.SG AUX.1PL them.ACC better  
ne-pozvali.  
NEG-invite.PST.PL.VIR  
‘In our Prague flat we bore the relative from Saaerbrücken during Christmas time  
somehow, but we better dit not invite them to the weekend house in summer.’
- b. ?? [PP Na chatu ]<sup>1</sup> [PP v létě ]<sup>2</sup> **jsem**<sup>3</sup> jel  
to weekend\_house.ACC.SG in summer.LOC.SG AUX.1SG go.PST.SG.M  
s rodinou.  
with family.INS.SG
- c. ?? [PP V létě ]<sup>1</sup> [PPna chatu ]<sup>2</sup> **jsem**<sup>3</sup> jel s rodinou.  
Intended: ‘I went to the weekend house with my family in summer.’  
(Cz; Avgustinova & Oliva 1997: 45)

- (13.8) a. ?\* [NP Marii ]<sup>1</sup> [PP v létě ]<sup>2</sup> **jsem**<sup>3</sup> v Praze  
Mary.ACC in summer.LOC.SG AUX.1SG in Prague.LOC  
ne-potkal.  
NEG-meet.PST.SG.M
- b. ?\* [PP V létě ]<sup>1</sup> [NP Marii ]<sup>2</sup> **jsem**<sup>3</sup> v Praze nepotkal.  
Intended: ‘I did not meet Mary in Prague in summer.’  
(Cz; Avgustinova & Oliva 1997: 45)

Hana (2007: 104–105, 111) argues that the assumed constraints on multiple constituents are incorrect and provides counterexamples, cf. (13.9). In (13.9a), there is an NP and a PP preceding the clitics. The NP *Petrovi* is an actant, because it is the verb’s indirect object. The NP is followed by the directive PP *do Francie* ‘to France’, which corresponds to a circonstant. Example (13.9b) involves

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verb. Complements are a subset of actants, but it is not a strict monotonic relation, because some complements can belong to the class of circonstants. Adjuncts are always circonstants.

the direct object NP *všechny sny* ‘all dreams’ and the adverb *najednou* ‘suddenly’. Again, the NP is an actant, whereas the adverb is a circonstant. Hana (2007: 119) refers to these and similar examples as “contrastive theme”, which may be relabelled either as a Contrastive Topic consisting of two arguments or the juxtaposition of a Contrastive Topic and a Contrastive Focus. Hana’s examples show that the putative communicative unit, which precedes the clitic cluster, does not need to be composed of structurally similar elements. But why is (13.8) degraded and (13.9) is not, despite a similar syntactic structure? What seems to be at issue here is the fact that the pre-clitic units need an appropriate communicative embedding. Thus, one may hypothesise that Avgustinova & Oliva’s (1997) example (13.8) is odd, because the construction presupposes a contrast, but lacks overt syntactic material to realise it. Hana’s (2007) example (13.9a), in turn, includes all the syntactic ingredients to verbalise the contrast between the pair (‘Peter’, ‘to France’) and (x, y), whereby x = ‘Martin’ and y = ‘to Hungary’.

- (13.9) a. [NP Petrovi ]<sup>1</sup><sub>TOP</sub> [PP do Francie ]<sup>2</sup><sub>FOC</sub> **bych**<sup>3</sup> to ještě poslal,  
Peter.DAT to France.GEN COND.1SG this.ACC.SG.N still send.PST.SG.M  
(ale Martinovi do Maďarska ani náhodou.)  
but Martin.DAT to Hungary.GEN not\_even chance.INS.SG  
‘I would send Peter to France, but never Martin to Hungary.’
- b. [NP Všechny sny ]<sup>1</sup><sub>TOP</sub> [AdvP najednou ]<sup>2</sup><sub>FOC</sub> **se**<sup>3</sup> **mu**<sup>3</sup>  
all.NOM.PL.NVIR dream.NOM.PL simultaneously REFL.ACC him.DAT  
určitě ne-splní.  
surely NEG-fulfil.PRS.3SG  
‘There is no way that all his dreams will come true at the same time.’

(Cz; Hana 2007: 105)

Turning to clitic climbing (CC), Junghanns (2002c: 83–84) notes that CC is not obligatory in Czech. He argues that a feature-driven movement account cannot capture this seemingly optional phenomenon and suggests that word order rearrangements without semantic correlates are driven by information structure (cf. Junghanns 2002c: 83–86). Junghanns’s (2002c) proposal is based upon the fact that pronominal clitics are regarded as Background material par excellence, which we shall address in a moment. Junghanns’s (2002c) idea is that a clitic escapes the infinitival domain and climbs to the matrix clause, if it belongs to the Background of the complex sentence. Clitic in situ placement is thus a consequence of the infinitive clause being a Topic or Focus domain, cf. (13.10). This is especially clear in the case of topicalised infinitival clauses, cf. (13.11). Obviously, clitics

cannot leave a fronted constituent, because it occupies already the highest possible structure position of the clause. Climbing is thus not available anymore.

- (13.10) a. [ Co ]<sub>FOC</sub> **se**<sub>1</sub> s ním Jirka rozhodl<sub>1</sub> udělat<sub>2</sub>?  
 what REFL.ACC with him.INS Jirka.NOM decide.PST.SG.M do.INF  
 ‘What did Jirka decide to do with it?’  
 b. Jirka **se**<sub>1</sub> rozhodl<sub>1</sub> [ vyhodit<sub>2</sub> **ho**<sub>2</sub> ]<sub>FOC</sub>.  
 Jirka.NOM REFL.ACC decide.PST.SG.M throw\_away.INF him.ACC  
 ‘Jirka decided to throw it away.’ (Cz; Dotlačil 2004: 98)

- (13.11) a. [ Chovat<sub>2</sub> **se**<sub>2</sub> v souladu se svým svědomím ]<sub>TOP</sub>  
 behave.INF REFL.ACC in agreement.LOC.SG with one’s.INS conscience.INS.SG  
 ne-má<sub>1</sub> prý žádnou cenu [...]   
 NEG-have.PRS.3SG allegedly no.ACC.SG.F price.ACC.SG  
 ‘To act in compliance with one’s own conscience, reputedly does not have any value ...’  
 (Cz; Junghanns 2002c: 78)  
 b. [ Sresti<sub>2</sub> **ga**<sub>2</sub> u Kanadi ]<sub>TOP</sub>, Dragan je želio<sub>1</sub>.  
 meet.INF him.ACC in Canada.LOC.SG Dragan.NOM AUX.3SG want.PST.SG.M  
 i. ‘To meet him in Canada is, what Dragan wanted.’  
 ii. ‘Dragan wanted to meet him in Canada.’ (BCMS; Stjepanović 2004: 183)

Stjepanović (2004: 206–207), too, considers the idea of clitic climbing being linked with information structure. She compares clitic climbing in BCMS with object shift in Germanic languages like Dutch, which has been reported to be sensitive to information structure. She says that

“there are no better candidates for object shift than pronominal clitics, because they are always background information. Given that they cannot be new information focus, they have to appear in the shifted position.” (Stjepanović 2004: 206)

However, Stjepanović (2004) does not pursue this idea, such that her suggestion is left without any explication of “what drives ‘clitic climbing’” and “why clitic climbing is optional” (Stjepanović 2004: 206). Only in a footnote, Stjepanović (2004: 207, n. 24) suggests that “information properties [...] play a filtering role” choosing one structural position from a multitude of potential landing sites provided by syntax. This line of thinking is essentially compatible with Junghanns’s (2002c) analysis of clitic climbing in Czech.

In a different perspective, Dotlačil (2004: 93–99, 2007a) ponders the question of why clitic climbing is banned across CP in Czech. He shows that phrasal categories may escape the domain of a full clause, cf. (13.12a). The dislocated constituent has a Focus or Contrastive Topic reading. Dotlačil concludes that a constituent may escape CP, if it is capable of bearing an appropriate discourse function. Clitics, as Dotlačil (2007) emphasises, cannot be focused or topicalised, which is why CP is an island for clitics. It is noteworthy that clitics may be realised outside their licensing domain as in (13.12b), if they are elements of a focused or topicalised constituent (cf. Dotlačil 2004: 98–99). More precisely, clitics cannot be dislocated themselves from the finite clausal domain, but they may leave the clause “aboard” a discourse prominent constituent, which in turn can. Metaphorically speaking, while clitics cannot swim, they can use a discourse-marked element as a raft and leave the island (CP) with a topicalised or focused vehicle.

- (13.12) a. [ Číst<sub>2</sub> ten román ]<sub>TOP/j</sub> jsem říkal, že  
 read.INF this.ACC.SG.M.INA novel.ACC.SG AUX.1SG say.PST.SG.M that  
 ne-budu<sub>1</sub> e<sub>j</sub>.  
 NEG-be.FUT.1SG  
 ‘I said that I would not read the novel.’
- b. [ Číst<sub>2</sub> ho<sub>2</sub> ]<sub>TOP/j</sub> jsem říkal, že ne-budu<sub>1</sub> e<sub>j</sub>.  
 read.INF him.ACC AUX.1SG say.PST.SG.M that NEG-be.FUT.1SG  
 ‘I said that I would not read it.’ (Cz; Dotlačil 2004: 99)

The matter of clitics being Background expressions is repeatedly brought up in research on clitics in Slavic (cf. Bennett 2011: 65 among others). This means that clitics are expressions referring to old, given, or contextually present information. For instance, Avgustinova & Oliva (1997: 36) tell that “(semi)clitics [...] are informationally insignificant, since the reference is fully recoverable from the context, and hence these (semi)clitics do not bear any communicative importance whatsoever [...]”. They continue that the

“clitic cluster is a purely functional (synsemantic) component, i.e. it can be viewed as an auxiliary communicative segment with no informational significance and, hence, with no communicative importance for the particular utterance. The main function of this cluster is to occupy the designated ‘second position’, and thus to define the preceding lexical material as a single substantial communicative segment.” (Avgustinova & Oliva 1997: 38)

Avgustinova & Oliva's (1997) notion of the *informational insignificance* of clitics is thus a paraphrase of the more common notion of *Background*. Working on Bulgarian, Werkmann (2003: 216) specifies object clitics as [–focal]. The occurrence of clitics is linked to the notion of Background-hood, which is inherently represented by clitics (cf. Werkmann 2003: 220). In other words, the feature specification [–focal] forces the clitic to leave the base position inside VP, where it is assigned its  $\theta$ -role while being inserted in the derivation. The VP is assigned Focus by default due to its bottom-/right-most location in the clausal skeleton. The positive Focus feature of the VP conflicts with the clitic's negative Focus value, such that it must leave VP. The clitic thus occupies a position in such a part of the utterance, which is not focused, i.e. Background, in fact (cf. Werkmann 2003: 254–256).

On a related note, Diesing (2010: 112–125, with further references) discusses the contrast between second phrase vs. second word cliticisation in Serbian. She reports that sentences with arguments in initial position highly favour placing clitics second on the level of the syntactic phrase rather than on the level of the word. The situation is reversed for sentence-initial predicates, whereby positioning clitics as second word is more likely than as second phrase. For argument-initial clauses, the less preferred second word placement does also occur. Diesing establishes that information structure is a relevant factor in choosing between both configurations. Second word positioning comes with a contrastive reading, whereas second phrase placement is pragmatically neutral bearing a wide Focus reading. In case of sentence-initial predicates, second word represents the pragmatically neutral default case. Diesing argues that this configuration implies a Topic reading, whereas the dispreferred realisation of clitics as second phrase implicates a given-focus reading.

A similar thread is found in Ćavar & Seiss's (2011) LFG treatment of clitics in Croatian. They argue that the contrast between second phrase and second word cliticisation is a matter of information structure. Ćavar & Seiss (2011) do not distinguish between argument- and predicate-initial constructions but rather look at split NP and PP constituents. They argue that only second phrase positioning allows a neutral Focus interpretation, which is a potential response to the underlying question "What happened?", but is impossible for second word cliticisation (Ćavar & Seiss 2011: 139). Second-word positioning thus results in the focalisation of the split-off host. This is in full agreement with Diesing (2010). Ćavar & Seiss (2011: 139–140) further argue that there are differences in quantifier scope between second phrase and second word placement. While the former is ambiguous and permits a distributive as well as a collective reading of (13.13), the latter brings about a disambiguation that implicates the collective reading only, cf. (13.14).

- (13.13) [NP Koliko članaka ] su svi ti studenti  
 how\_many article.GEN.PL AUX.3PL all this.NOM.PL.M student.NOM.PL  
 pročitali?  
 read.PST.PL.M
- i. ‘What is the number, such that all students read that number of papers?’ [collective]  
 ii. ‘How many articles exist, such that all students read them?’ [distributive]  
 (Hrv; Ćavar & Seiss 2011: 140)

- (13.14) a. [NP Koliko su članaka ] svi ti studenti  
 how\_many AUX.3PL article.GEN.PL all this.NOM.PL.M student.NOM.PL  
 pročitali?  
 read.PST.PL.M
- b. Koliko su svi ti studenti pročitali članaka?  
 how\_many AUX.3PL all this.NOM.PL.M student.NOM.PL read.PST.PL.M article.GEN.PL
- i. What is the number, such that all students read that number of papers?’ [collective]  
 ii. \*‘How many articles exist, such that all students read them?’ [distributive]  
 (Hrv; Ćavar & Seiss 2011: 140)

In sum, there is sufficient evidence to put forward the hypothesis that clitic placement and information structure are interconnected. We can pursue this line of thought irrespective of whether clitic placement is affected by information structure directly, or whether it is an epiphenomenon of more general interactions between information structure and word order. It appears that both perspectives apply, in fact. Various scholars concur in that clitics are Background entities. Therefore, they occur in the left periphery of the clause, which is usually least focal.<sup>587</sup> Prototypically, a communicatively salient element occupies the first position of unembedded main clauses, which gives rise to the Wackernagel pattern. Accordingly, clitic climbing is the default configuration in complex clauses with embedded infinitives. Even though second position cliticisation is the most frequent configuration, there can be more than one salient entity in the left periphery, such that multiple syntactic constituents may precede the clitic cluster provoking Delayed Placement. On the other hand, clitics can be located in a discourse-prominent environment like a topicalised or focused constituent. Consequently, the clitic remains in situ and no climbing occurs. However, the Topic or Focus

<sup>587</sup> An interesting question is the relation of 2P and 1P in this respect. If the left-most position is the most backgrounded one, one should expect clitics to usually occur clause-initially. However, this is not what we find. Maybe one can attribute 2P cliticisation to the fact that the clause-initial position is usually occupied by Topics. Since clitics cannot bear this discourse function, they should appear in the next-best position and realise second position placement. Now consider the observation that not every utterance realises a Topic. If the left-most position is not filled, then clitic first position placement follows rather naturally.



constituent as a whole may be dislocated while carrying the clitic and occupy a salient position within or beyond its original clausal domain. The data and arguments that have been cited in the present section call for a more careful consideration of information structure in Slavic clitic research. The remainder of Chapter 13 presents an attempt of conducting such a kind of research.

### 13.3 The Leipzig model of information structure

The present section briefly summarises the approach to information structure known as the *Leipzig model*. The Leipzig model was primarily developed on the basis of Slavic and stands in the tradition of the Prague linguistic circle. It combines a Minimalist syntax with an independent level of information structure. The Leipzig model was first elaborated for Russian by Gerhild Zybatow and Uwe Junghanns (cf. Junghanns 2002b; Junghanns & Lenertová 2008; Junghanns & Zybatow 1997, 2009; Zybatow 1999; Zybatow & Junghanns 1998) and subsequently for Bulgarian (Janewa 2000; Werkmann 2003, 2008) and Czech (Junghanns 2002b; Junghanns & Lenertová 2008).<sup>588</sup> In contrast to the Prague school's traditional Theme-Rheme or Topic-Comment division, the Leipzig model additionally adopts the Focus-Background division (cf. Junghanns & Lenertová 2008, Junghanns & Zybatow 2009). Unlike most minimalist approaches, the Leipzig model starts from the premise that word order variation is not triggered by purely (morpho-)syntactic features only. Information structure is taken to have its own relevance for word order permutations in Slavic and to be independent from the syntactic core. In fact, the realisation of certain constituent orders is regarded as primarily motivated by information structure in Slavic, while most morphosyntactic features are regarded as weak, such that there is little overt movement by syntax proper (cf. Zybatow 1999: 66–69, Werkmann 2003: 104). However, the concept of information structure in the Leipzig model fits most other recent accounts (cf. De Kuthy 2021). The Comment is the predicative portion of an utterance, predicating over the Topic. The Topic is the traditional theme, i.e. the instance about which a statement is made. The Focus represents highlighted information, while the Background represents portions that are less important. These discourse notions are theoretical primitives.<sup>589</sup>

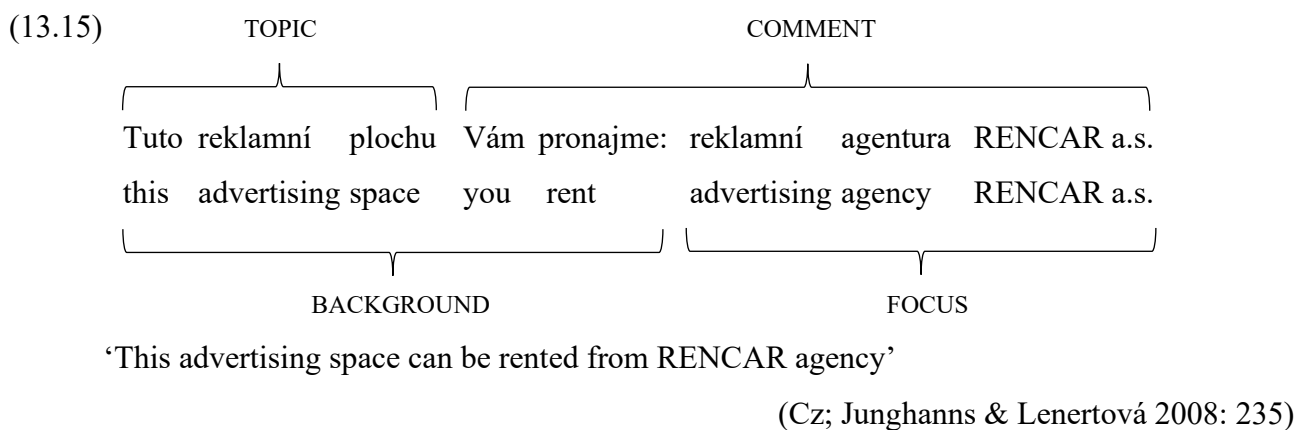
The Topic is conceptualised as “aboutness topic” in the sense of Lambrecht (1994: 118), i.e. “the thing which the proposition expressed by the sentence is *about*.” [Emphasis as in the original.]

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<sup>588</sup> Furthermore, Szucsich (2002) utilises the Leipzig model in his study of the syntax of Russian nominal adverbials. Recently, Däbritz (2021) applied a modified version of the Leipzig model to a diverse sample of five languages of north-western Siberia belonging to three linguistic macro-families (Uralic: Enets, Khanty, Nganasan; Turkic: Dolgan; Tungusic: Evenki).

<sup>589</sup> An alternative approach for the treatment of these discourse functions is presented by Krifka (2008), who does not set Topic and Focus as primitives, but derives them by means of a formal semantic apparatus.

The Comment is a predication about the Topic. However, the Topic is not necessarily a subject, and the Comment does not entail the predicate in any sense of the notion, e.g. a VP. On the other hand, Focus is conceptualised as important information, while the Background is considered less important. Notably, the Leipzig model does not operate with oppositions of “old” vs. “new”, which will be touched upon below. Topic and Focus are the salient entities in their respective structures and receive the features [TOP] and [FOC] respectively. Comment and Background are not assigned a feature, for they are derived subtractively. The disparity of the two structures is illustrated in example (13.15) from Junghanns & Lenertová (2008).



While I follow the Leipzig model in approaching information structure in terms of the Topic-Comment and Focus-Background dichotomies, I do not adopt the Minimalist syntax component due to the meta-theoretical stance of the present thesis.<sup>590</sup> According to the original Leipzig approach, Slavic clauses have an underlying base-generate SVO structure, which means that the so-called Predicate-Internal Subject Hypothesis (PISH; cf. Koopman & Sportiche 1991) is adhered to. Accordingly, a ditransitive VP has the base-generated structure in (13.16).

(13.16) [<sub>VP</sub> SBJ [ V<sup>0</sup> [ IOBJ [ DOBJ ]]]]

Derivational operations (i.e. movement) apply to generate alternative linearisations by displacing VP-internal material to the syntactic superstructure above VP.<sup>591</sup> However, the Leipzig model’s approach to information packaging is not dependent upon a particular syntactic framework and their account may be adopted in other theories, too, including both phrase-structure and dependency grammars.

<sup>590</sup> Cf. Owczarzak (2004) for an orthodox Minimalist approach to word order and information structure in Polish.

<sup>591</sup> Despite the widely held view that Slavic languages have a basic, canonical, or unmarked constituent order, an alternative perspective is proposed by Haider & Szucsich (2022).

Table 13.1 and Table 13.2 summarise the basic patterns of information packaging and salient discourse functions acknowledged by the Leipzig model.

Table 13.1: Topic-Comment structure and its salient discourse functions.

|                   |                                                                                                                                                                                                                                                                     |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Internal Topic    | The Internal Topic occurs in the left periphery. It is an integral part of the clause structure and realises an argument of the verbal predicate, but it can also be an adjunct.                                                                                    |
| Contrastive Topic | Contrastive Topics are a subtype of Internal Topics, which possess a distinctive prosodic contour. A Contrastive Topic is an element of a set of multiple explicit or implicit discourse Topics. Often, Contrastive Topics occur in enumeration-like constructions. |
| External Topic    | An External Topic occurs outside the clause structure. The topicalised element is anaphorically linked with a clause-internal element, which does not need to represent the Topic itself.                                                                           |

While essentially any constituent, which is considered important by the speaker, can bear Focus, the Topic is much more restricted. In terms of linear occurrence, Topics occupy the left-most position in a clause, so they are strictly left-peripheral material. From a semantic point of view, only expressions, which are capable of referring or allow to infer a discourse referent, can be Topics. While an utterance does not need to realise an overt Topic, there is always an overtly Focus-marked unit. Natural Focus occurs per default in the right periphery of the clause, while Contrastive Focus and Verum Focus are mostly positioned in the left periphery or clause-internally. However, the different types of Focus are not in complementary distribution and each Focus type may occur sentence-initially, -internally, or -finally, in fact. This renders the positional distinction redundant, but is not surprising, since Foci are marked prosodically. In other words, we may conclude that, for Slavic, prosodic marking is the primary means of Focus marking, while linear position is only secondary. Topics and Foci appear to represent something like a negative image of one another in this respect (the former can be defined primarily by position even though it is prosodically marked, too).

Table 13.2: Focus-Background structure and its salient discourse functions.

|                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Natural (Neutral) Focus               | Any constituent of arbitrary complexity that is marked as important by the Focus feature. The default position is the right clausal periphery. Focus domains may be determined by the so-called question test, i.e. natural Focus domains always correspond to answers to implicit questions. Focus is prosodically marked by stress on the final element of the Focus domain. Depending on the complexity of the focused phrase, <i>maximal</i> , <i>intermediate</i> , and <i>minimal</i> domains are distinguished. |
| Maximal (Wide) Focus                  | The entire clause is focused.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Minimal (Narrow) Focus                | A single constituent (e.g. argument NP) or head (e.g. verbal head) is focused.                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Intermediate (Medium) Focus           | Neither maximal (wide) nor minimal (narrow) focus; the Focus domain comprises multiple elements (e.g. VP, i.e. verbal head + object NP).                                                                                                                                                                                                                                                                                                                                                                               |
| Contrastive Focus                     | Contrastive Focus serves for contrasting or correcting information and bears a distinctive prosodic contour. It may occur principally in left-peripheral, sentence-medial, and right-peripheral position, although the latter is rarer.                                                                                                                                                                                                                                                                                |
| Verum Focus                           | Verum Focus occurs exclusively on the verb and marks the semantic truth of the denoted event. The Verum Focus may occur sentence-initially, -internally, and -finally, but like Contrastive Focus it is rare in sentence-final position.                                                                                                                                                                                                                                                                               |
| Rightward backgrounding (de-focusing) | Alternatively, natural Focus may be positioned in a non-right-peripheral position, i.e. in left-peripheral (maximal backgrounding/de-focusing) or intermediate position (partial backgrounding/de-focusing).                                                                                                                                                                                                                                                                                                           |
| Thetic clauses                        | Thetic utterances represent maximal Focus, i.e. the entire clause is focused and states a fact. In this respect, I basically follow Ulrich (1986), who considers thetic clauses as monolithic statements of facts, whereas categorical clauses consist of two elements of information structure. Thus, the present study treats maximal or wide Focus simply as thetic clauses.                                                                                                                                        |

Since the Focus-Background structure is grounded on a speaker-oriented notion of “importance”, the question might arise, how the Leipzig model copes with the concept of *old* or *given* vs. *new* information. Such a discourse level is not involved in the original approach, indeed, but it is not excluded and can be implemented in fact. This is exactly what Däbritz (2021) does in his study of information structure in northwest Siberian languages. He modifies the Leipzig model by adding the listener-oriented dimension of *information status* based on previous work by Nissim et al. (2004) and Götze et al. (2007). Following the terminology of Götze et al. (2007), Däbritz (2021: 115) categorises information as *given*, *accessible*, or *new*, which are derived from two binary features, i.e. [ $\pm$ discourse-old] and [ $\pm$ listener-old]. The feature combinations are cited in Table 13.3 adapted from Däbritz (2021: 116). The feature [ $\pm$ discourse-old] specifies whether a discourse referent has been mentioned earlier. On the other hand, the feature [ $\pm$ listener-old] indicates whether the listener is able to determine the respective referent by themselves either by situation, aggregation (i.e. grouping of previously mentioned referents), inference, or common sense. The features are not morphosyntactic features, which are checked against a functional head, but discourse features like [TOP] and [FOC], which are assigned to nodes of a syntactic tree. The same feature bundle can be assigned multiple times.

Table 13.3: Information status as feature combination.

| Discourse-old | Listener-old | Information status       |
|---------------|--------------|--------------------------|
| +             | +            | <i>given</i>             |
| –             | +            | <i>accessible</i>        |
| –             | –            | <i>new</i>               |
| +             | –            | — (logically impossible) |

While Däbritz’s (2021) expansion of the Leipzig model has undeniable merits, I will not utilise it here for practical reasons. First, the Leipzig model serves merely as an analytical toolbox. I will not defend, test, or modify it. Second, the Topic-Comment and Focus-Background structure are sufficient for the purpose of the present study. I leave the implementation of Däbritz’s (2021) proposal for the study of Slavic cliticisation for future research.

## 13.4 Information structure and DP in the corpus

As mentioned earlier, the clitic system of Polish is considered peculiar within Slavic because of what can be paraphrased as “weak restrictions on clitic distribution in the clause” in contrast to the strong

restrictions applying in languages like Czech and BCMS.<sup>592</sup> Recall that the main arguments for the typological classification of Polish are the lack of mandatory Wackernagel placement, of obligatory clitic cluster formation, and of a fixed word order among clitics. Section 10.3 showed that Wackernagel positioning is the most frequent pattern found with Polish pronominal and reflexive clitics, but still we recognise a significant difference in the distribution patterns of Polish and, e.g., Czech, which has a higher ratio of Wackernagel placement than Polish. In other words, Delayed Placement occurs more often in Polish than in Czech. The question arises whether one can still account for Delayed Placement in some principled way. Based on the overview of the interaction of cliticisation and information structure in Section 13.2, the hypothesis may be put forward that delayed clitic placement could be sensitive to notions of information structure in Polish, too. More specifically, we may ask if it is possible to attribute Delayed Placement to certain patterns of information packaging? The goal of the present section is to account for delayed clitic placement in Polish in terms of information structural notions. I will attempt to demonstrate that deviations from Wackernagel positioning in Polish correlate with the occurrence of the salient discourse functions in the relevant environment.

To begin with, consider example (13.17), which is in accord with the assumption of at least two information structure levels, i.e. Topic-Comment and Focus-Background. The utterance consists of two coordinated clauses, itemised as (13.17a) and (13.17b) respectively. In (13.17a), the clause-initial NP *uczelnia* ‘school’ is the grammatical subject as well as the Topic of the clause, while the complex VP *próbuję wyjść z sytuacji z twarzą* is both the grammatical and information-structural predicate. The clause-final PP *z twarzą* ‘with face’ (i.e. ‘without losing face’) is the locus of natural Focus, whereas the linearly preceding material is Background information. The clause (13.17b) introduces a new Topic, i.e. the NP *jej pomysły* ‘her ideas’ with the possessive pronoun *jej*, which is co-referential with the earlier mentioned *uczelnia*. The negated verb *nie przekonują* ‘do not convince’ is the locus of the Focus exponent. The VP represents a minimal Focus domain, for the adjective *młodych* ‘young.ACC/GEN.PL’, which represents a full direct object NP (referring to *studenci* ‘the students’), is placed outside the verbal domain and precedes it. Example (13.17) is well-behaved and exhibits a straightforward pattern. In fact, the example represents written language and is taken from a newspaper article of the *Gazeta Wyborcza*.

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<sup>592</sup> See Section 1.2, Section 3.9, and Section 4.12 for some discussion.

- (13.17) a. TOPIC COMMENT
- [<sub>NP</sub> Uczelnia ]<sub>i</sub> [<sub>VP</sub> próbuje [<sub>VP</sub> wyjść [<sub>PP</sub> z sytuacji ] [<sub>PP</sub> z twarzą ]]],
- school.NOM.SG try.PRS.3SG exit.INF from situation.GEN.SG with face.INS.SG
- BACKGROUND FOCUS
- ‘The school tries to get out of the situation without losing its face,’
- b. TOPIC COMMENT
- ale [<sub>NP</sub> jej pomysły ] [<sub>AdjP</sub> młodych ]<sub>k</sub> [<sub>VP</sub> nie przekonują e<sub>k</sub>].
- but her idea.NOM.PL young.GEN.PL NEG convince.PRS.3PL
- BACKGROUND FOCUS
- ‘but its ideas do not convince the young people.’ (Pol; S19)

After this introduction, let us turn to the spoken data from the NKJP, now. Admittedly, the following discussion is based on a very small data set due to several constraints: 102 data points covering every single clitic type underwent closer inspection (= 3.3% of the entire Polish data sample or 12.1% of the DP subset). I investigated Delayed Placement in declarative clauses only. The reviewed data were selected impressionistically. Discourse functions were identified by means of the context and, whenever possible, auditory data from the corresponding audio recording in the Spokes corpus. However, the Spokes corpus is not simply an alternative version of the NKJP's spoken subcomponent, but it is a distinct corpus, which is developing independently from the NKJP. While it shares a great deal of utterances with NKJP's PELCRA subcorpus, both corpora possess mutually exclusive portions of data. Recall that the data used for the present study have been drawn from the NKJP. Furthermore, although Spokes does provide users with audio recordings in contrast to the NKJP, not all audio files are available. An interesting feature of Spokes is the availability of a recording's respective F0 graph, but it is not aligned with the transcript, unfortunately. So, I cannot make use of it. Finally, I consulted a native speaker in case of doubt, who was especially asked for intuition on prosody in cases where no audio playback was available.<sup>593</sup> Needless to say, the present section cannot provide an exhaustive picture – neither of information structure in Polish, nor of its

<sup>593</sup> A single native speaker represents a far too small number of informants with respect to common standards of quantitative research and statistical soundness. However, as the research question and methodology are not psycholinguistic but corpus linguistic, the native speaker's task was merely to narrow down the pool of useful data points.

relationship with the distribution of clitics. Especially, it cannot provide us with a quantitatively informed picture. However, what the data allow us is to investigate some recurring patterns and get a first impression of what might be going on at the interface between cliticisation and information packaging in Polish. Thus, we can approximate a hypothesis, at least.

We can distinguish several patterns of information structuring co-occurring with delayed clitic placement in the Polish data set. The first pattern I present is Clitic Third (3P) with a focused constituent preceding the clitic immediately, i.e. the clitic is contiguous with the focused expression. Both independent and subordinate clauses exhibit the pattern under discussion. The pattern is exemplified in (13.18), where stressed syllables are marked by capitalisation. Stress is indicated only in cases, when auditory data is available in the Spokes corpus. The translation equivalents of the focused constituents are marked in bold type and a corresponding subscript in the translation line for ease of exposition.

- (13.18) a. no to<sup>1</sup> [NP ja ]<sup>2</sup><sub>FOC</sub> **ci**<sup>3</sup> to mówila-m w  
 well TO I.NOM you.DAT this.ACC.SG.N speak.PST.SG.F-1SG in  
 kaszance  
 kaszanka.LOC.SG  
 ‘well, so, it was **me**<sub>FOC</sub>, who told you – in kaszanka [= lit.: ‘blood sausage’; probably the  
 name of a pub]’
- b. powiedz mi czy<sup>1</sup> [NP ten tata ]<sup>2</sup><sub>FOC</sub> **cię**<sup>3</sup>  
 tell.IMP.2SG me.DAT if this.NOM.SG.M dad.NOM.SG you.ACC  
 odwozi czy<sup>1</sup> [ ja ]<sup>2</sup><sub>FOC</sub> ma[m]<sup>2</sup> **cię**<sup>3</sup> odwieźć  
 drive\_back.PRS.3SG if I.NOM have.PRS.1SG you.ACC drive\_back.INF  
 ‘tell me if **dad**<sub>FOC</sub> is the one who brings you home or if I should bring you [home]’
- c. (nigdy nie odbierała-m telefonu bo nigdy jak  
 never NEG pick\_up.PST.SG.F-1SG telephone.GEN.SG because never when  
 był w torebce) to<sup>1</sup> [AdvP nigdy ]<sup>2</sup><sub>FOC</sub> **go**<sup>3</sup> nie  
 be.PST.SG.M in handbag.LOC.SG TO never him.ACC NEG  
 słyszała-m  
 hear.PST.SG.F-1SG  
 ‘(I never picked up the phone, because, when it was in my handbag,) I **never**<sub>FOC</sub> heard it’



- d. ([ i tak ]<sup>1</sup> [pp u brata ]<sup>2</sup><sub>FOC</sub> **by=m**<sup>3</sup> **się**<sup>3</sup> umył)  
 and so at brother.GEN.SG COND=AUX.1SG REFL.ACC wash.PST.SG.M  
 bo<sup>1</sup> [AdvP tutaj ]<sup>2</sup><sub>FOC</sub> **mi**<sup>3</sup> **się**<sup>3</sup> nie chciało)  
 because here me.DAT REFL.ACC NEG want.PST.SG.N  
 ‘(Either way, I would have washed **at my brother’s place**<sub>FOC</sub>) because I didn’t want [to do it] **here**<sub>FOC</sub>.’
- e. najlepsze że<sup>1</sup> [pp przez SPOdnie ]<sup>2</sup><sub>FOC</sub> **mi**<sup>3</sup> **się**<sup>3</sup> przebił  
 best.ADV that through trouser.ACC.PL me.DAT REFL.ACC break\_through.PST.SG.M  
 ‘the best [thing is] that it stung me **through [my] trousers**<sub>FOC</sub>’
- f. no najem. no to<sup>1</sup> [pp z góry ]<sup>2</sup><sub>FOC</sub> **mu**<sup>3</sup> zapłacili  
 well rent.NOM.SG well TO from mountain.GEN.SG him.DAT pay.PST.PL.VIR  
 ci ludzie  
 this.NOM.PL.VIR people.NOM  
 ‘Well, as far as the rent is concerned, these people paid him **upfront**<sub>FOC</sub>.’
- g. gada że<sup>1</sup> [NP taka dziewczYna ]<sup>2</sup><sub>FOC</sub> **mu**<sup>3</sup> **się**<sup>3</sup>  
 say.PRS.3SG that such.NOM.SG.F girl.NOM.SG him.DAT REFL.ACC  
 podoba tam. u niego nie.  
 like.PRS.3SG there at him.GEN NEG  
 ‘[he] says that there is **this girl**<sub>FOC</sub> he likes – around his place, right’
- h. (daj mi Mirek też papierosa) może<sup>1</sup>  
 give.IMP.2SG me.DAT Mirek.VOC also cigarette.GEN.SG maybe  
 [NumP jednego ]<sup>2</sup><sub>FOC</sub> **se**<sup>3</sup> zapalę  
 one.GEN.SG.M REFL.DAT smoke.PRS.1SG  
 ‘Mirek, give me a cigarette, too, maybe I will smoke **one** [of these]’
- i. no a teraz to<sup>1</sup> [NP [AdvP tylko ] szkołą ]<sup>2</sup><sub>FOC</sub> **się**<sup>3</sup>  
 well and now TO only school.INS.SG REFL.ACC  
 zajęła-m (muszę napisać pracę magisterską)  
 be\_occupied.PST.SG.F-1SG must.PRS.1SG write.INF work.ACC.SG master.ADV.ACC.SG.F  
 ‘Well, recently, I’ve been occupied **only with school**<sub>FOC</sub>. (I must write [my] master thesis.)’  
 (Pol; NKJP pelcra)

Some instances of immediately post-focal Clitic Third are found with Verum Focus (VER.FOC). In (13.19), the clitic follows a focused verb, whose focalisation is neither caused by discourse prominence in terms of importance nor due to contrast. Instead, it is the truth value, which is focused. This means that focalisation specifies whether the event expressed by the predicate is true or not. In

(13.19a), the speaker emphasises the correctness of a previously made statement. The Focus exponent appears on the finite verb entailing that the proposition of the previous utterance is true. The focused verb in (13.19b) also implies the confirmation of the respective truth value (i.e. the parents indeed wanted to have the child). In (13.19c), the speaker is occupied with the loss of her mobile phone. Verum Focus on the verb *miałam* ‘I had’ puts emphasis on the fact that the mobile phone was in the speaker’s possession before the loss. Here, the locative portion of the possessive construction’s semantics becomes immanent. Finally, we encounter a negated verb in (13.19d). In accordance with the account of Verum Focus, the utterance ascertains the truth value FALSE.

- (13.19) a. (chodzi)      no      właśnie<sup>1</sup> [V MÓwię]<sup>2</sup><sub>VER.FOC</sub> **ci**<sup>3</sup>  
 walk.PRS.3SG well actually speak.PRS.1SG you.DAT  
 (że teraz całą                      zimę                      chodził)  
 that now whole.ACC.SG.F winter.ACC.SG walk.PST.SG.M  
 ‘(it works) I’m **telling**<sub>VER.FOC</sub> you (that it worked the entire winter)’
- b. (no to Jola                      mu                      tłumaczy                      mówi                      no)  
 well TO Jola.NOM.SG him.DAT explain.PRS.3SG speak.PRS.3SG well  
 że<sup>1</sup> [V CHCIELi-śmy]<sup>2</sup><sub>VER.FOC</sub> **cię**<sup>3</sup>                      (no urodził-e-ś                      się)<sup>594</sup>  
 that want.PST.PL.VIR-1PL you.ACC well be\_born.PST.SG.M-M-2SG REFL.ACC  
 ‘(Well, so Jola explains [it] to him. She tells [him]) that we [actually] **wanted**<sub>VER.FOC</sub> you.(Well, you were born.)’
- c. (ja      chyba      go                      zgubiła-m                      wiecie ?)  
 I.NOM perhaps him.ACC lose.PST.SG.F-1SG know.PRS.2PL  
 bo<sup>1</sup>      [V miała-m]<sup>2</sup><sub>VER.FOC</sub> **go**<sup>3</sup>  
 because have.PST.SG.F-1SG him.ACC  
 ‘(I lost it maybe, you know?) ’Cause I [surely] **had**<sub>VER.FOC</sub> it.’
- d. Zużka<sup>1</sup>                      [V NIE chce]<sup>2</sup><sub>VER.FOC</sub> **go**<sup>3</sup>  
 Zużka.NOM NEG want.PRS.3SG him.ACC  
 ‘Zużka does **not want**<sub>VER.FOC</sub> him.’ (Pol; NKJP pelcra)

Alternatively, the Focus constituent can be realised non-contiguously to the clitic. The examples in (13.20) show Clitic Third with the focused constituent in first position. Trivially, this pattern is not

<sup>594</sup> The transcription represents the normalisation of an originally non-standard morphological realisation of the finite’s verb form by the speaker. A more accurate transcription is given in (i). Note that the token *my* does not represent the plural subject pronoun but a homophonous non-standard allomorph of *śmy* ‘AUX.1PL’.

(i) [...] że chcieli                      **my**                      **cię**                      [...]  
 that want.PST.PL.VIR AUX.1PL you.SG.ACC

available in subordinate clauses.<sup>595</sup> Note that the clitics occur contiguously to their predicates. Examples (13.20a–c) show verbal circumposition in clitic climbing constructions (or what would be analysed as clitic climbing by default in the absence of counterevidence). The clitics are linearised between the finite matrix verb and an infinitive that licenses them. Utterance (13.20d) exhibits a simple clause with the reflexive dative clitic *se* in postposition, i.e. right-adjoined to its verbal head. I cannot propose any sound generalisation as to whether predicate- or verb-adjacency occurs as a rule in this pattern because of too small a database.

- (13.20) a. nie [NP [AdvP tylko ] głowę ]<sup>1</sup><sub>FOC</sub> muszę<sup>1</sup><sub>2</sub> ci<sup>3</sup><sub>2</sub> zmierzyć<sub>2</sub>  
 NEG only head.ACC.SG must.PRS.1SG you.DAT measure.INF  
 ‘No, I need to measure your **head** **only**<sub>FOC</sub>.’
- b. (taki piękny stół miała-m)  
 such.ACC.SG.M beautiful.ACC.SG.M table.ACC.SG have.PST.SG.F-1SG  
 i [AdvP ciągle ]<sup>1</sup><sub>FOC</sub> musiała-m<sup>1</sup><sub>2</sub> go<sup>3</sup><sub>2</sub> nakrywać<sub>2</sub> ceratą.  
 and always must.PST.SG.F-1SG him.ACC cover.INF tablecloth.INS.SG  
 ‘(I had such a beautiful table) and I **always**<sub>FOC</sub> had to cover it with a tablecloth.’
- c. [NumP każdy ]<sup>1</sup><sub>FOC</sub> może<sup>1</sup><sub>2</sub> się<sup>3</sup><sub>2</sub> ubierać<sub>2</sub> jak się  
 everyone.NOM.SG.M can.PRS.3SG REFL.ACC dress.INF how REFL.ACC  
 chce  
 want.PRS.3SG  
 ‘**everyone**<sub>FOC</sub> may dress howsoever one wants’
- d. [AdvP odWROtnie ]<sup>1</sup><sub>FOC</sub> położyć<sup>2</sup> se<sup>3</sup> (że=by=ś tu  
 inversely lay\_down.IMP.2SG REFL.DAT that=COND=AUX.2SG here  
 miał mordkę)  
 have.PST.SG.M snout.DIM.ACC.SG  
 ‘lay [him] down **the other way** **round**<sub>FOC</sub> (so that you had the snout here)’  
 (Pol; NKJP pelcra)

In a similar fashion, there are attestations for clitic third position placement with a preceding Topic, which may be either contiguous with the clitic like in (13.21) or disjunct from it like in (13.22). The pattern in (13.21) seems to appear consistently in subordinate clauses, which is expected from a purely

<sup>595</sup> Example (13.20a) is structurally ambiguous, compare the bracketing in (i) (with an intonation break additionally marked [#]) and (ii). Ambiguities are a common problem for purely transcription-based corpus research.

(i) nie # [NP [AdvP tylko ] głowę ]<sub>FOC</sub> muszę ci zmierzyć  
 ‘no, it is only your head I need to measure’ = ‘no, I need to measure your head only’  
 (ii) [NP [AdvP nie tylko ] głowę ]<sub>FOC</sub> muszę ci zmierzyć  
 ‘no, it is not only your head I need to measure’

theoretical point of view. If Topics are minimally specified by their linear position as suggested in Section 13.3, then contiguity of a Topic and a clitic in third position can only be achieved in subordinate clauses. However, example (13.21g) seems to be an exception, because it involves an independent main clause with a clause-initial subject pronoun. This is a surprising linearisation in face of the Topic phrase's placement in second position. It is currently not possible to verify the original utterance, since the audio track is unavailable. Still, the English translation line gives the reading relevant here, even though some doubt may remain. Perhaps, the peculiarity of this utterance could be attributed to the unorthodox properties of overt subject pronouns in spoken lects. It is repeatedly noted for West Slavic languages that, despite being *pro*-drop, overt subject pronouns have a higher frequency of use in speech without being pragmatically salient. Under this view, the overt subject pronoun in (13.21g) does not imply any specific information structure, such that the NP *autobusem* were a suitable Topic, indeed. Alternatively, one could assume a pair reading involving subject and instrumental phrase {*ja*, *autobusem*} similar to those examples of Delayed Placement discussed by Hana (2007) for Czech.<sup>596</sup> I cannot resolve this issue here and must leave it for future research.

- (13.21) a. ([...] na przykład. wymogi                      unijne                      jeśli chodzi  
                   for example requirement.NOM.PL    union.ADJ.NOM.PL if    walk.PRS.3SG  
 o    drogi            to jest            tak że.) żeby<sup>1</sup>            [NP unia ]<sup>2</sup><sub>TOP</sub>  
 about road.ACC.PL TO be.PRS.3SG such that in\_order\_to    union.NOM.SG  
 ci<sup>3</sup>            [VP dofinansowała    budowę                      drogi ]<sub>FOC</sub> (to  
 you.DAT            subsidise.PST.SG.F construction.ACC.SG road.GEN.SG TO  
 asfalt    musi                    mieć    jedenaście centymetrów    grubości)  
 asphalt must.PRS.3SG have.INF eleven    centimetre.GEN.PL thickness.GEN.SG  
 ‘(... for example the EU requirements, as far as roads are concerned, it is the case that)  
 for **the EU**<sub>TOP</sub> to subsidise your road constructions (asphalt needs a thickness of eleven  
 centimetres)’
- b. (przypomniało    mi            się)            że<sup>1</sup> [NP teściowa ]<sup>2</sup><sub>TOP</sub>                      cię<sup>3</sup>  
 remind.PST.SG.N me.DAT REFL.ACC that            mother-in-law.NOM.SG you.ACC  
 [V nie    lubi ]<sub>FOC</sub>  
                   NEG like.PRS.3SG  
 ‘(It came to my mind) that [my] **mother-in-law**<sub>TOP</sub> doesn’t like you.’

<sup>596</sup> See Section 4.4.2 and Section 13.2 for relevant examples.

- c. no    bo<sup>1</sup>    [NP ta                      kawa ]<sup>2</sup><sub>TOP</sub>    **mi**<sup>3</sup>    nic                      [v nie  
 well because      this.NOM.SG.F coffee.NOM.SG me.DAT nothing.ACC.SG    NEG  
 daje ]<sub>FOC</sub>  
 give.PRS.3SG  
 ‘**The coffee**<sub>TOP</sub> avails me nothing.’
- d. (jak    [NP ten                      chłopak ]<sub>TOP</sub> powiedział    do mnie    [s że mi  
 when      this.NOM.SG.M boy.NOM.SG say.PST.SG.M to me.GEN    that me.DAT  
 zaraz      przypierdoli]<sub>FOC</sub>.) to<sup>1</sup> [NP ona ]<sup>2</sup><sub>TOP</sub> **mu**<sup>3</sup>      powiedziała [s że  
 immediately slap.PRS.3SG      TO      she.NOM him.DAT tell.PST.SG.F    that  
 zaraz      jak    ja      ci      przerznię      dupę      [...]]<sub>FOC</sub>  
 immediately when I.NOM you.DAT cut\_off.PRS.1SG arse.ACC.SG  
 ‘(When **the guy**<sub>TOP</sub> said to me that he is right to slap me,) **she**<sub>TOP</sub> told him that “I’m right  
 to tear you a new one ...”.’
- e. [...] jak<sup>1</sup>    [NP Monika ]<sup>2</sup><sub>TOP</sub> **se**<sup>3</sup>                      kogoś                      pozna                      z  
                     when      Monika      REFL.DAT someone.ACC get\_to\_know.PFV.PRS.3SG with  
 kim    będzie      wiesz                      chciała                      być      jakoś      dłużej [...] ]  
 whom be.FUT.3SG know.PRS.2SG want.PTCP.SG.F be.INF somehow longer  
 ‘... when **Monika**<sub>TOP</sub> will meet someone, with whom she will, y’know, want to be  
 somehow longer ...’
- f. (tak jest                      tak Ściboreczka) bo<sup>1</sup>    [NP JEJ matka ]<sup>2</sup><sub>TOP</sub>    **się**<sup>3</sup>                      zna  
 so    be.PRS.3SG yes Ściboreczka    because      her mother.NOM.SF REFL.ACC ...  
 nazywała    [ Ściborowska ]<sub>FOC</sub>  
 call.PST.SG.F Ściborowska  
 ‘(That’s right. Yes, Ściboreczka.) ’Cause **her mother**’<sub>STOP</sub> name was Ściborowska.  
 [lit.: ‘Cause **her mother** was called Ściborowska.]’
- g. ja<sup>1</sup>    [NP autobusem ]<sup>2</sup><sub>TOP</sub> **by**<sup>3</sup>=**m**<sup>3</sup>                      **go**<sup>3</sup>                      do domu  
 I.NOM      bus.INS.SG                      COND=AUX.1SG him.ACC to house.GEN.SG  
 [v nie    zabrała ]<sub>FOC</sub>  
                     NEG    take.PST.SG.F  
 ‘**Using the bus**<sub>TOP</sub>, I wouldn’t have taken it [back] home.’ [lit.: ‘**By bus**, I wouldn’t have  
 taken it [back] home.’] (Pol; NKJP pelcra)

The pattern in example block (13.22) with Topics in the first clausal position is excluded from subordinate clauses just as in case of Foci. Again, we observe predicate-adjacency like in the Clitic Third constructions with clause-initial Focus phrases in (13.20).

- (13.22) a. [NP te białka ]<sup>1</sup><sub>TOP</sub> tak<sup>2</sup> ci<sup>3</sup> [V zeSZTYwnią ]<sub>FOC</sub>  
 this.NOM.PL.NVIR egg\_white.NOM.PL such you.DAT become\_stiff.PRS.3PL  
 ‘**The egg whites**<sub>TOP</sub> become this stiff (on your behalf).’
- b. i [NP prasowanie ]<sup>1</sup><sub>TOP</sub> zabiera<sup>2</sup> mi<sup>3</sup> z reguły [NP PÓŁ  
 and ironing.NOM.SG take.PRS.3SG me.DAT from rule.GEN.SG half.ACC.SG  
 godziny ]<sub>FOC</sub> [...] hour.GEN.SG  
 ‘**Ironing**<sub>TOP</sub> takes me half an hour usually ...’
- c. [NP JA ]<sup>1</sup><sub>TOP</sub> dała-m<sup>2</sup> mu [NP piątkę ]<sub>FOC</sub> (bo też więcej nie  
 I.NOM give.PST.SG.F-1SG him.DAT five.ACC.SG because also more NEG  
 miała-m)  
 have.PST.SG.F-1SG  
 ‘**Me**, I<sub>TOP</sub> gave him a fiver (’cause I didn’t have more)’
- d. (wiesz co kurna za Kacpra. to znaczy)  
 know.PRS.2SG what damn for Kacper.ACC.SG this.NOM.SG.N mean.PRS.3SG  
 [PP za dzieci ]<sup>1</sup><sub>TOP</sub> [ na pewno ]<sup>2</sup> się płaci jakiś tam  
 for child.ACC.PL on sure REFL pay.PRS.3SG some.ACC.PL.NVIR there  
 ze zniżkami wiesz ?  
 with concession.INS.PL know.PRS.2SG  
 ‘(you know what, dammit, for Kacper. That is,) **as for children**<sub>TOP</sub>, one surely pays  
 some[thing] with concession, y’know?’ (Pol; NKJP pelcra)

Clitic Third is also possible with two pragmatically loaded constituents preceding the clitic in root clauses. An example is given in (13.23), where a Topic is followed by a contiguous Focus-marked element. Similar constructions are discussed by Hana (2007) for Czech – consider, e.g., example (13.9) in Section 13.2.

- (13.23) [...] [ jutro ]<sup>1</sup><sub>TOP</sub> [ barszcz ]<sup>2</sup><sub>FOC</sub> se<sup>3</sup> zrobię cały  
 tomorrow borscht.ACC.SG REFL.DAT make.PRS.1SG whole.ACC.SG.M.INA  
 gar  
 pot.ACC.SG  
 ‘... **tomorrow**<sub>TOP</sub>, I will make **borscht**<sub>FOC</sub> for myself – a whole pot’ (Pol; NKJP pelcra)

If the linear configuration of two salient discourse functions and a contiguous clitic or clitic cluster encountered in (13.23) is realised in a subordinate clause, then we receive Clitic Fourth positioning as in (13.24)

- (13.24) a. (Ewa mi mówiła ostatnio) że<sup>1</sup> [NP to]<sup>2</sup><sub>TOP</sub>  
 Ewa.NOM me.DAT speak.PST.SG.F lately that this.NOM.SG.N  
 [NumP wszystko]<sup>3</sup><sub>FOC</sub> ci<sup>4</sup> podlicza  
 everything.ACC you.DAT calculate.PRS.3SG  
 ‘(Ewa told me recently) that **it**<sub>TOP</sub> [= an online tool] calculates **everything**<sub>FOC</sub> for you’
- b. (ale oprócz tego że) że<sup>1</sup> [PP na scenie]<sup>2</sup><sub>TOP</sub> [AdvP dobrze]<sup>3</sup><sub>FOC</sub>  
 but beyond this.GEN.SG.N that that on stage.LOC.SG good.ADV  
 się<sup>4</sup> czuję (mnie to bawi)  
 REFL.ACC feel.PRS.1SG me.DAT this.NOM.SG.N amuse.PRS.3SG  
 ‘(but beyond) that, [**on stage**]<sub>TOP</sub>, I feel **well**<sub>FOC</sub> (I enjoy it)’ (Pol; NKJP pelcra)

A slightly different configuration is found in (13.25)–(13.26), which involve a complex Focus constituent consisting of two syntactically independent elements. Example (13.25) shows two syntactic units in a root clause – the restrictive particle *tylko* ‘only’ and the direct object *głowę* ‘head’ – which constitute the Contrastive Focus phrase *tylko głowę* ‘the head only’. Both word tokens can be permuted independently of one another, such that they qualify as constituents, i.e. a particle phrase (PartP) and an NP respectively. Note that the particle *tylko* still would have scope over the NP *głowę*, if the latter were realised within the infinitive VP, as indicated by (13.25b). There is no difference in semantic interpretation and information structure between these two examples and the one in (13.25c). In each case, there is a Contrastive Focus reading. Admittedly, a different interpretation for the linearisation in (13.25b) is available, but the underlying syntactic structure is different. In (13.25b), the restrictive particle *tylko* modifies the NP by narrowing down a set of potential alternatives to a particular entity. In contrast, example (13.25d) presents a sentential adverbial reading of *tylko*, whereby the particle modifies the entire proposition, which results in a contrast with the preceding context. In this case, there is no Contrastive Focus that restricts elements from a set of discourse referents but a natural right-peripheral Focus. A comparable structure is attested in (13.26), which provides an utterance introduced by the particle *to* (glossed ‘TO’). As before, a particle – i.e. *i tak* ‘either way’ – has scope over an adjacent constituent – i.e. the PP *w jednym talerzu* ‘on a single plate’ – in (13.26a). The focused phrase represents a Contrastive Focus. Different linearisations can be realised without changing the discourse properties, cf. (13.26b–c). Finally, the linearisation in (13.26b) is ambiguous and may have different underlying syntactic structures and different

information structures, cf. (13.26d).<sup>597</sup> An alternative interpretation presupposes scope of the particle over the finite verb *robię* ‘I make’, which yields a modification of the entire proposition.<sup>598</sup> The Focus phrase realises plain right-peripheral Focus here.

- (13.25) a. nie [[PartP tylko ]<sup>1</sup> [NP głowę ]<sup>2</sup>]<sub>FOC</sub> muszę<sup>3</sup> ci<sup>4</sup> zmierzyć  
 NEG only head.ACC.SG must.PRS.1SG you.DAT measure.INF  
 (Pol; NKJP pelcra)

b. nie [PartP tylko ] muszę ci [VP zmierzyć [NP głowę ]<sub>FOC</sub>]

c. nie muszę ci [VP zmierzyć [[PartP tylko ] [NP głowę ]<sub>FOC</sub>]]

‘No, I need to measure your **head only**<sub>FOC</sub>.’

d. nie [PartP tylko ] muszę ci [VP zmierzyć [NP głowę ]<sub>FOC</sub>]

‘No, but I need to **measure your head**<sub>FOC</sub>.’

(Pol; constructed from 13.25a)

- (13.26) a. a [NP ja ]<sub>TOP</sub> to<sup>1</sup> [[PartP I TAK ]<sup>2</sup> [PP w jednym talerzu ]<sup>3</sup>]<sub>FOC</sub> se<sup>4</sup>  
 and I.NOM TO and so in one.LOC.SG.M plate.LOC.SG REFL.DAT  
 wszystko robie nie  
 everything.ACC.SG make.PRS.1SG NEG

(Pol; NKJP pelcra)

b. a [NP ja ]<sub>TOP</sub> to [PartP I TAK ] se wszystko robie [PP w jednym talerzu ]<sub>FOC</sub> nie

c. a [NP ja ]<sub>TOP</sub> to se wszystko robie [[PartP I TAK ] [PP w jednym talerzu ]]<sub>FOC</sub> nie

‘And me, I<sub>TOP</sub> prepare everything **on a single plate either way**<sub>FOC</sub>, right?’

d. a [NP ja ]<sub>TOP</sub> to [PartP I TAK ] se wszystko robie [PP w jednym talerzu ]<sub>FOC</sub> nie

‘Anyway, I<sub>TOP</sub> prepare everything on a **(single/the same) plate**<sub>FOC</sub>, right.’

(Pol; constructed from 13.26a)

Two more examples of information-structurally-loaded 4P constructions with two differently discourse marked entities preceding the clitic or clitic cluster are cited in (13.27). Example (13.27a) resembles (13.25a) in that the salient discourse elements precede a verb, which is only followed by

<sup>597</sup> The analysis suggested for (13.23) and (13.24) may be easily put forward in a transformational or phrase-structure-grammatical approach in terms of extraction. The base-generate position of the Focus phrase is VP-internal, which corresponds to the c-examples. The difference between a- and b-examples is then one of full extraction vs. partial extraction. In the b-type constructions, it is only the particles that are dislocated from VP, while in the a-type ones the entire Focus phrase is extracted. This can be modelled using traces or silent copies in the style of Government and Binding Theory and Minimalism respectively, while a HPSG approach employs unsaturated phrases and indexing in this situation and does not utilise a procedural operation of movement.

<sup>598</sup> The attested verb form is *robię* with a missing ogonek, which is a minor transcription error. The orthographically correct form is *robię*. The form *robię* represents the token as it is attested in the Spokes corpus, whereas NKJP’s Pelcra component stores that same data point with the orthographic form *robisz* ‘you<sub>SG</sub> make’.



the clitic. Both example sentences show clitic climbing with verbal circumposition. Note that example (13.27a) is underlyingly ambiguous with repercussions for the interpretation of the clitic position. On the one hand, the string *aneks w kuchni* may represent a simplex NP followed by an independent PP, which I assume here based on the utterance's discourse context in the corpus. On the other hand, the string *aneks w kuchni* could be a single but complex NP, in fact, which would give rise to Clitic Third rather than Clitic Fourth. Without further, especially phonetic evidence, it is impossible to decide which structure is intended by the speaker. Unfortunately, audio data has not been available. The structural assumption pursued here remains with some residual uncertainty, therefore. In contrast, Clitic Fourth is beyond any doubt in example (13.27b), where we encounter the topical subject NP *zachodzik* 'sunset' and a focused adverb *ładnie* 'beautifully'. The clitic follows the particle *też* 'also', which occupies the third position of the clause.

- (13.27) a. [NP *aneks*]<sup>1</sup><sub>TOP</sub> [PP W *kuchni*]<sup>2</sup><sub>FOC</sub> *musieli*<sup>3</sup> *se*<sup>4</sup> *zrobić*  
 kitchenette.ACC.SG in kitchen.LOC.SG must.PST.PL.VIR REFL.DAT make.INF  
 '[The kitchenette]<sub>T</sub>, they had to make it [in the kitchen]<sub>F</sub>.'
- b. o [NP *zachodzik*]<sup>1</sup><sub>TOP</sub> [AdvP *Ładnie*]<sup>2</sup><sub>FOC</sub> *też*<sup>3</sup> *ci*<sup>4</sup> *się*<sup>4</sup>  
 oh sunset.DIM.NOM.SG beautiful.ADV also you.DAT REFL  
*udał*  
 work\_out.PST.SG.M  
 'Oh, the **sunset**<sub>TOP</sub> became **great**<sub>FOC</sub>, too.' [lit.: 'The sunset, you managed [to catch by  
 photograph] beautifully, too.'] (Pol; NKJP pelcra)

Example (13.28a) is the only utterance of the data sample scrutinised for the present section, which exhibits a greater delay than 4P. Particularly, we observe clitic sixth position placement, which we know from Section 10.3.1 is comparatively rare. The example has some interesting features. First, the topicalised phrase is complex and comprises two constituents, i.e. the subject pronoun *ja* and the PP *na przykład* 'for examples' modifying it. The focused NP *szczegóły* 'details' is not contiguous with the Topic, because the finite verb *lubię* 'I like' intervenes. In fact, the utterance exhibits intermediate Focus, i.e. the clitic is surrounded by material of the Focus domain, which is isomorphic to the embedded infinitival VP. Therefore, the bracketing in (13.28b) is more adequate. The NP *szczegóły* 'details' is the Focus exponent bearing the Focus stress, but the Focus domain is isomorphic to the embedded infinitival VP. This state of affairs is reminiscent of Junghanns's (2002c) suggestion that clitic climbing in Czech is sensitive to notions of information structure mentioned in Section 13.2. The clitic does not climb, but stays in situ, since it is an element of a discourse prominent constituent. Note that in contrast to the Leipzig model's standard assumption that the Focus exponent is always

3.28) a. [...] bo<sup>1</sup> [[<sub>NP</sub> ja ]<sup>2</sup> [<sub>PP</sub> na przykład]<sup>3</sup>]<sub>TOP</sub> lubię<sup>4</sup> [<sub>NP</sub> SZCZEgóły]<sup>5</sup><sub>FOC</sub>  
because I.NOM on example.ACC.SG like.PRS.1SG detail.ACC.PL  
**se**<sup>6</sup> oglądać nie  
REFL.DAT watch.INF NEG

b. [...] bo<sup>1</sup> [[<sub>NP</sub> ja ]<sup>2</sup> [<sub>PP</sub> na przykład]<sup>3</sup>]<sub>TOP</sub> lubię<sup>4</sup> [<sub>VP</sub> [<sub>NP</sub> SZCZEgóły<sup>5</sup> ] **se**<sup>6</sup> oglądać ]<sub>FOC</sub> nie  
‘..., ’cause **me for example**, **I**<sub>TOP</sub> like to have a look at **details**<sub>FOC</sub>, right?’

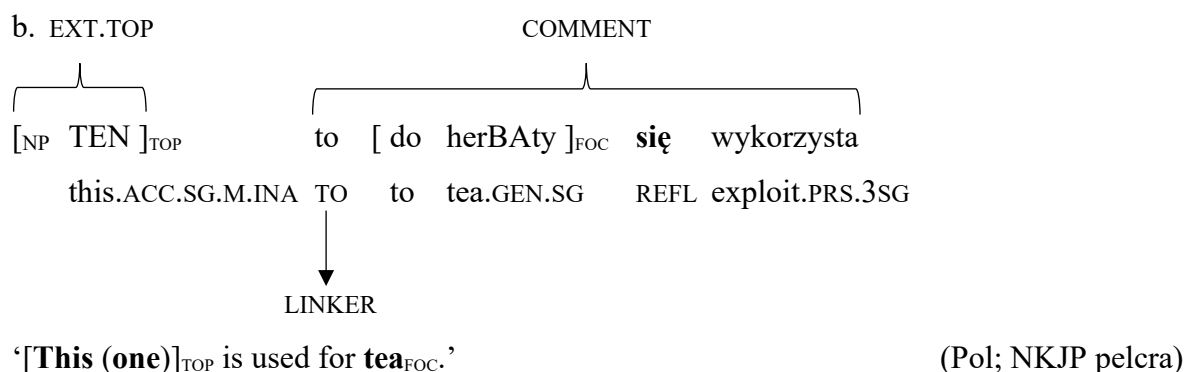
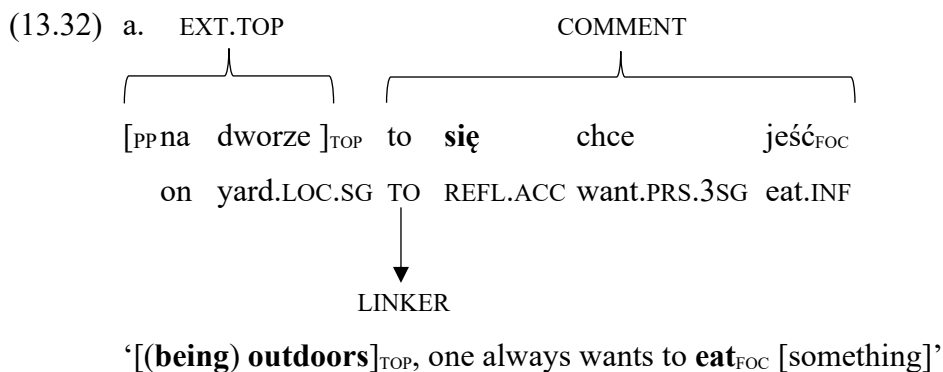
(Pol; NKJP pelc...

- is leads to the more general issue of clitics inside discourse-prominent constituents. Example block (13.30) provides Focus phrases of varying complexity in the canonical right-peripheral clause position. The clitic is placed within the Focus domain, which gives rise to Delayed Placement. However, clitics may surface in the second position of an utterance even though they are inside the Focus domain. Such an instance is found in (13.31) with a focused infinitive VP in clause-initial position. The Focus-bearing verbal head is followed by the clitic, which trivially yields the *ackernagel* pattern.

- (13.30) a. może<sup>1</sup> [ NApisz<sup>2</sup>      **mu**<sup>3</sup>      teraz ]<sub>FOC</sub> (że pod      inny  
 maybe write.IMP.2SG him.DAT now      that under different.ACC.SG.M.INA  
 adres      [...])  
 address.ACC.SG  
 ‘maybe you’ll write him now (that [it should be sent] to another address ...)’
- b. [...] [ zaraz dwa      dni      potem ]<sup>1</sup><sub>TOP</sub> [VP odwołał<sup>2</sup>      **go**<sup>3</sup>      ze  
 soon two day.NOM.PL afterwards      dismiss.PST.SG.M him.ACC from  
 stanowiska      dyrektora]<sub>FOC</sub> [...]  
 position.GEN.SG director.GEN.SG  
 ‘... **only two days later**<sub>TOP</sub>, he **dismissed**<sub>FOC</sub> him from the directorship ...’
- c. ([<sub>NP</sub> ten      nikiel ]<sub>TOP</sub>      to      jest      takie      że      wiesz  
 this.NOM.SG.M nickel.NOM.SG TO be.PRS.3SG such.NOM.SG.N that know.PRS.2SG  
 co) [s jak      patrzysz ]<sub>TOP</sub> to<sup>1</sup> [ aż<sup>2</sup>      **ci**<sup>3</sup>      **się**<sup>3</sup>      w oczach  
 what when look.PRS.2SG TO almost you.DAT REFL.ACC in eye.LOC.PL  
 prze ]<sub>FOC</sub>.  
 press.PRS.3SG  
 ‘(nickel, this is such a thing, you know what,) **when you look [at it]**<sub>TOP</sub>, then **your eyes are going to ache**<sub>FOC</sub>’ [lit.: ‘... then it presses in your eyes’] (Pol; NKJP pelcra)
- (13.31) [<sub>VP</sub> zakręcić<sup>1</sup>      **ci**<sup>2</sup> ]<sub>FOC</sub>      to      trzeba  
 screw\_closed.INF you.DAT this.ACC.SG.N necessary  
 ‘you need to **close**<sub>FOC</sub> it’ [lit.: ‘it’s necessary to **close**<sub>FOC</sub> it (by screwing) for you’]  
 (Pol; NKJP pelcra)

Likewise, clitics may occur in a Topic phrase. Example (13.32) shows a frequently occurring topicalisation strategy in Polish, according to which the left-peripheral Topic phrase is followed by the lexeme *to* ‘this.NOM/ACC.SG.N’, which has a double function here. First, it marks the preceding constituent as Topic. Second, it creates coherence between the Topic and the Comment by syntactically functioning as a linking or subordinator-like element introducing the Comment (cf. Geist & Błaszczak 2000: 131 for an analysis of Polish *to* as conjunction). In fact, the element *to*, which formally is the singular neuter demonstrative pronoun, is a highly polyfunctional element (cf. Zawada 2024 for a recent exploratory corpus study of *to* in Polish). As such, it may be alternatively considered a particle in certain environments, e.g. as mere clause-linker in constructions like *jak ... to ...* ‘when/if

... then ...'.<sup>599</sup> Such occurrences of *to*, which are clearly non-pronominal and non-demonstrative, are simply glossed 'TO' throughout the thesis. Still, there are two important observations for the status of constructions like those in (13.32). First, the diverse functions of *to* are not discrete. Sometimes, several interpretations are available simultaneously and give rise to ambiguities. For example, consider (13.30c) above, where the first *to* following the NP *ten nikiel* is glossed 'TO'. However, it would make perfect sense to gloss it 'this.NOM.SG.N' resulting in an interpretation as demonstrative pronoun. Such an interpretation is eligible, in fact, and can be justified by agreement of the copula verb (*jest*) and the predicative pronoun (*takie*).



Mendoza & Sonnenhauser (2023) discuss such ambiguous instances under the notion of *Oscillation*. Second, the *to*-marked constituent is always an External Topic (EXT.TOP), such that *to* marks the left clause margin like a subordinator.<sup>600</sup>

<sup>599</sup> An example is given in the correlate construction in (i). Here, the linker *to* corresponds to a clause-initial linking adverb like English *then* or German *dann* 'then'.

(i)    no    to    [s    jak    już    możesz    dać    dziewięć    dych ]                                      [s    to    daj  
       well TO    when already can.PRS.2SG    give.INF nine.ACC    ten\_zlotys.GEN.SG    TO give.IMP.2SG  
       stówkę ]  
       hundred\_zlotys.ACC.SG

      'well, so, when you can give [me] nine tenners, then give [me] a ton'                                      (Pol; NKJP pelcra)

<sup>600</sup> Note that in an appropriate environment, the Topic particle *to* can be substituted by an unequivocal demonstrative pronoun, as in the constructed examples (i) and (ii). These structures parallel Junghanns & Zybatow's (2009) Czech example for External Topics in (iii). Finally, *to* may serve as a discourse-functional element in Czech, too. An example of *to* as a Topic marker is cited in (iv) parallel to Polish (13.32b). Furthermore, Šimík (2009) discusses *to* as a Focus

Examples of this construction have been already presented in this section, e.g. (13.18c) and (13.26a), but the Topics did not include any clitic. Topic-internal clitics are cited in (13.33). The Topic phrase in (13.33a) is complex and comprises three different syntactic elements, all of which are dependents of the verb. However, the Topic does not represent a syntactic constituent *sensu stricto* and is a purely communicative unit. Similarly, the Topic in (13.33b) is not an actual constituent, but the subject pronoun accompanied by an object clitic, which has been taken over into what figuratively may be called the “prefield” in reference to the Topic-prominent position before the finite verb in German independent clauses. The *to*-marked External Topic provides an interesting diagnostic for Topic-hood in general as well as for clitics nested in a Topic domain in particular. Although the construction is common in speech, Topic-internal clitics are attested rarely in the examined data sample. Observe that clitics occupy second position, when placed inside a discourse prominent constituent, i.e. within a Topic (13.30) or Focus phrase (13.30, 13.28b).

- (13.33) a. *nie no [ja się z Michaliną]<sub>TOP</sub> to [VP dogaduje*  
 NEG well I.NOM REFL.ACC with Michalina.INS TO get\_along.PRS.1SG  
*w sumie]<sub>FOC</sub>*  
 in sum.LOC.SG  
 ‘No. Well, as far as I and Michalina are concerned, we get along, in fact.’  
 b. *[ja go]<sub>TOP</sub> to piję trochę tak jak wiesz co*  
 I.NOM him.ACC TO drink.PRS.1SG a\_bit such how know.PRS.2SG what.ACC.SG  
*[NP napój w dzień]<sub>FOC</sub>*  
 beverage.ACC.SG in day.ACC.SG  
 ‘Me, I drink it somehow like, you know, a refreshment during the day’  
 (Pol; NKJP *pelcra*)

Three instances of diacsis of co-dependent clitics have been detected in the small data sample examined here. Interestingly, they share an information packaging pattern. The utterances in (13.34)

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marker in Czech. Note that each of Šimík’s (2009) grammatical Czech examples may be perfectly transposed into contemporary Polish.

- (i) *[NP Krzysztof]<sub>TOP</sub> to jest naprawdę śmieszny.*  
 Christopher.NOM TO be.PRS.3SG really funny.NOM.SG.M  
 (ii) *[NP Krzysztof]<sub>TOP</sub> ten jest naprawdę śmieszny.*  
 Christopher.NOM this.NOM.SG.M be.PRS.3SG really funny.NOM.SG.M  
 ‘**Christopher**<sub>TOP</sub>, he’s really funny.’ (Pol; constructed)  
 (iii) *[CP [DP Karel]<sub>TOP</sub>, [CP ten se má dobře ]]*  
 Karel.NOM this.NOM.SG.M REFL.ACC have.PRS.3SG good.ADV  
 ‘**Karel**<sub>TOP</sub>, he has it good.’ (Cz; Trávníček 1931: 135 [cited after Junghanns & Zybatow 2009: 695])  
 (iv) *[NP já]<sub>TOP</sub> to budu objíždět autem (a budu tě hledat)*  
 I.NOM TO be.FUT.1SG drive\_around.INF car.INS.SG and be.FUT.1SG you.ACC search.INF  
 ‘**Me**, I<sub>TOP</sub> will drive around and [I] will look [around] for you.’ (Cz; ČNK oral\_v1)

are parallel in splitting clitics by a focused phrase. This is remarkable insofar as the constructions are rare and one could object that this pattern is a peculiarity of performance. Despite the small database, the similar pattern across different speakers allows putting forward the hypothesis that diaclysis is not merely a plain “error of speech production” but grounded in information packaging. Recall also from Section 4.5 that diaclysis is attested in various Slavic languages, some of which are considered well-behaved Wackernagel clitics languages.<sup>601</sup> Since the pattern is attested in written language (cf. examples in I. Kulik 2023b), it is inadequate to attribute it to “slips of the tongue”.

- (13.34) a. (no zobacz) jak **mi** [NP wiTAmin ]<sub>FOC</sub> **się** chce  
 well look.IMP.2SG how me.DAT vitamin.GEN.PL REFL.ACC want.PRS.3SG  
 cały czas [...] whole.ACC.SG.M time.ACC.SG  
 ‘(well, look) how [much] I want vitamins all the time ...’
- b. i **się** [V NAJszczy ]<sub>FOC</sub> **mi** tu  
 and REFL.ACC piss.PRS.3SG me.DAT here  
 ‘and it [= a dog] will piss here (to my disadvantage)’
- c. (albo może specjalnie) że=**by=m** [NP JA ]<sub>FOC</sub> **mu**  
 or maybe purposefully that=COND=AUX.1SG I.NOM him.DAT  
 na-pisała czy pó-jdziemy po-grać  
 PFV-write.PST.SG.F if PFV-go.PRS.1PL DIM-play.INF  
 ‘(or, perhaps, on purpose) such that I would text him, if we’ll go and play a bit’
- (Pol; NKJP pelcra)

Summing up this section’s findings, there is clear evidence for the interaction of information structure with clitic positioning in Polish. We observe that delayed clitic placement co-occurs with a topicalised or focused constituent preceding the clitic. The salient discourse function may be contiguous with the clitic but does not need to: In independent clauses, it may appear clause-initially, an option logically impossible in embedded clauses introduced by a subordinator. While the occurrence of a single topicalised or focused constituent correlates with Clitic Third, the left-peripheral placement of multiple discourse prominent elements, i.e. a Topic and a Focus, corresponds to an increasing delay, e.g. Clitic Fourth. Increasingly delayed clitic placement on the level of syntactic constituency is also observed, if several syntactic constituents are integrated into a single discourse prominent unit. Besides taking their position in the Background, clitics may be positioned within an information-

<sup>601</sup> Consider also the attestations of diaclysis in the six spoken lects scrutinised in Chapter 10.

structurally prominent constituent; however, this does not need to mirror a syntactic constituent (in case of Topics, at least). As far as the data reviewed here is concerned, Topic- or Focus-internal clitics appear in the second position of the respective discourse function's domain, even though they occupy a delayed position of the entire clause or sentence. Finally, diacrisis manifests itself as the split of clitics by a stressed, Focus-bearing element.

## 13.5 Proposal for a (non-formal) analysis

Let us begin this section by briefly summarising key aspects of the discussion and empirical findings on clitic placement and information structure. Section 10.3 showed that Polish clitics are located in clausal second position to a high extent, while Section 10.8 suggested that Polish primarily differs from other West Slavic 2P languages – and particularly Czech – quantitatively rather than qualitatively. Section 13.4 dealt with Delayed Placement and concluded that deviations from Wackernagel cliticisation can be attributed to certain information structure configurations. Similar observations have been made, for instance, for BCMS and Czech as we learned in Section 13.2. I will therefore propose that information structure is in fact relevant for clitic positioning in Polish (and, perhaps, beyond). Second position clitic placement in Slavic is usually accounted for in terms of syntax and/or phonology. What these accounts have in common is that they refer to a special position within a given domain. It is either the highest available position in the clausal skeleton (sometimes identified as  $C^0$ , sometimes an underspecified head) following a viable host or the left-most position in a prosodic domain, which is non-initial. Corresponding analyses have been proposed for languages like BCMS and Czech, while Polish has defied these kinds of approaches hitherto.

The idea I would like to pursue here is that languages differ as to how they utilise the different linguistic domains for clitic placement. Presumably, some languages might refer to prosody, others to syntax or morphology, and yet other ones to pragmatics. However, languages need not be restricted to only one linguistic level in this respect and may exploit multiple domains. Furthermore, languages might differ with respect to which domain is prioritised over another. My working hypothesis for the remainder of this section is that Polish utilises information structure domains for the purpose of clitic positioning.

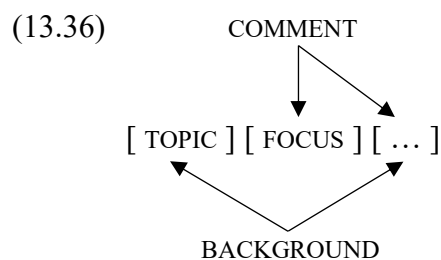
The literature review in Section 13.2 suggests that Wackernagel cliticisation is the placement of a clitic or clitic cluster in second position of the Background by default. The empirical evidence in Section 13.4 suggests that clitics, which are an element of a Topic or Focus domain, occupy second position within this particular domain. Delayed Placement is provoked by the linearisation of one or more discourse-prominent constituents in the left periphery of the Background domain or before it

(cf. Avgustinova & Oliva 1997). The respective configurations are summarised in (13.35), whereby the empirical observations from Section 13.4 are taken into account. Various options of second position cliticisation are schematised in (13.35a–f) followed by Clitic Third in (13.35g–l). The schemes cover both root and subordinate clauses. Scheme (13.35d) represents left-peripheral Focus, while (13.35j) depicts sentence-medial Focus, which is why they seem to be positioned in the Background. They are not Background material, of course, but their position is meant to be derived from the prototypical right-peripheral position. Note that subordinators (SUB) are treated as purely syntactic items without any contribution to information structure in (13.35). This is why the domain introduced by SUB is underspecified and merely labelled as a sentence (S). The relevant information structure domains only follow the subordinator. This is surely a simplified picture, but it is more transparent and facilitates comparison among configurations. It is not entirely clear if the configuration in (13.35i) is valid, for it represents the controversial post-subject placement of a Topic from (13.31g) in Section 13.4. Examples (13.35n–o) illustrate Clitic Fourth, while (13.35p–s) give underspecified DP including arbitrarily late positions.

- (13.35) a. [BACKGROUND X CL ... ] [2P]  
 b. [BACKGROUND SUB CL ... ]  
 c. [BACKGROUND [TOPIC A ] CL ... ]  
 d. [BACKGROUND [FOCUS B ] CL ... ]  
 e. [TOPIC A CL ... ] [COMMENT ... ]  
 f. [FOCUS B CL ... ] [BACKGROUND ... ]  
 g. [TOPIC A ... ] [COMMENT X CL ... ] [3P]  
 h. [FOCUS B ... ] [BACKGROUND X CL ... ]  
 i. ? [BACKGROUND X [TOPIC A ... ] CL ... ]  
 j. [BACKGROUND X [FOCUS B ... ] CL ... ]  
 k. [TOPIC A ... ] [FOCUS B ... ] [COMMENT/BACKGROUND CL ... ]  
 l. [S SUB [TOPIC A ... ] CL ... ]  
 m. [S SUB [FOCUS B ... ] CL ... ]  
 n. [TOPIC A ... ] [FOCUS B ... ] [COMMENT/BACKGROUND X CL ... ] [4P]  
 o. [BACKGROUND SUB [TOPIC A ... ] [FOCUS B ... ] CL ... ]  
 p. [BACKGROUND X ... ] [FOCUS B CL ... ] [DP]  
 q. [TOPIC A ... ] [COMMENT/BACKGROUND X ... ] [FOCUS B CL ... ]  
 r. [S SUB [BACKGROUND X ... ] [FOCUS B CL ... ]]  
 s. [S SUB [TOPIC A ... ] [COMMENT/BACKGROUND X ... ] [FOCUS B CL ... ]]



Note that, usually, certain portions of Comment and Background intersect.<sup>602</sup> That is, elements that are neither Topic nor Focus belong to Comment and Background simultaneously. They are thus non-salient on both levels of information structure assumed in the Leipzig model adopted here. Considering that Comment and Background are essentially not isomorphic, different potential clitic positions manifest, if the clitic is placed with respect to either Background or Comment. In textbook examples like (13.15) in Section 13.3 or (13.17) in Section 13.4, 2P in the Background equates 2P in purely syntactic terms, while 2P in the Comment yields DP. However, Comment and Background may be treated as isomorphic, if Topic and Focus are realised contiguously in the left periphery of the clause. Second position placement in either domain yields the same 4P surface realisation. Note that Comment-Background isomorphism is understood in terms of the Leipzig model's account of negatively defining these notions by means of "subtraction". Consider the scheme in (13.36). While the entire material following the Topic can be still referred to as the Comment, there is no way of appropriately referring to the Background as the Focus's complement in terms of a continuous string of elements. Therefore, it appears to be sensible to refer to the set theoretic notion of *relative complement* to account for the information structurally non-salient "rest", see (13.37). Since the portion of a syntactic string belonging to the non-prominent discourse constituent on either level of information structure is formally definable, it makes sense to make particular reference to this domain for clitic placement. In lack of an appropriate term for this portion of an utterance, I will tentatively refer to this *non-salient "rest"*, as I have phrased it above, as the RESTGROUND.



(13.37)  $\text{COMMENT} \setminus \text{FOCUS} \wedge \text{BACKGROUND} \setminus \text{TOPIC} = \text{COMMENT} \cap \text{BACKGROUND} = \text{RESTGROUND}$

From the previous discussion it follows that clitics can be analysed in terms of information packaging. Accordingly, clitics in Polish are placed in second position within one of five information structure domains distinguished here. Four domains are widely acknowledged, i.e. Topic, Comment, Focus, and Background. The fifth domain has been introduced as a result of the intersection of Comment and Background under exclusion of Topic and Focus. This fifth configuration, which has been

<sup>602</sup> I am abstracting away fromthetic clauses, which are treated here as wide or maximal Foci.

labelled provisionally as *Restground*, arises, if both Topic and Focus occur left-peripherally leaving portions of Comment and Background behind, which become isomorphic then. Consequently, I propose that Wackernagel cliticisation in Polish follows one of the schemes in (13.38).

- (13.38) a. [TOPIC XP CL ]  
           b. [FOCUS XP CL ]  
           c. [COMMENT XP CL ]  
           d. [BACKGROUND XP CL ]  
           e. [RESTGROUND XP CL ]

It is important to bear in mind that this analysis does not abandon syntactic or prosodic constituency. On the contrary, both syntax and phonology are important linguistic levels, which are clearly interrelated with information structure. The schemes in (13.38) merely specify information domains for Wackernagel positioning in Polish. Therefore, the reference to an arbitrary syntactic constituent XP preceding the clitic CL in (13.38) is not a contradiction.

It should be emphasised that the enumeration in (13.38) does not imply any hierarchy but is an unordered list. However, since clitics are regarded as prototypical Background material, it seems natural to regard variant (13.38d), i.e. positioning in the Background domain, as the default case. From this it follows that (13.38b), positioning within the salient Focus domain, is more restricted. By the same token, the Topic-internal placement in (13.38a) may be regarded as being subject to specific conditions. The decision of which domain – Topic or Focus – is the more restricted one for clitic positioning, shall be approached by what may be regarded as the standard textbook examples of the two-layered information structure for natural language. Recall example (13.25) from Section 13.3 repeated as (13.39) in the style of De Kuthy (2021: 1044). The Topic's position overlaps with the Background, such that we may assume a closer relation between Topic-hood and Background-hood based on probability. Since the Background is the Focus's complement, we may hypothesise that Topic-internal clitic placement is more likely to occur than Focus-internal one. This seems to contrast with the occurrence frequency of Focus-internal clitics compared to Topic-internal ones in Section 13.4. Furthermore, it was said that, while not every utterance necessarily realises an overt Topic, each utterance has a focused constituent. Both questions are empirical matters ultimately. In lack of better evidence for the time being, I shall follow the conceptual considerations that favour Topic-internal over Focus-internal clitic placement with the caveat that some corrections may be necessary in the future.

|         |                      |                                             |
|---------|----------------------|---------------------------------------------|
| (13.39) | TOPIC                | COMMENT                                     |
|         | Tuto reklamní plochu | Vám pronajme: reklamní agentura RENCAR a.s. |
|         | BACKGROUND           | FOCUS                                       |

(Cz; Junghanns & Lenertová 2008: 235)

The next question is then, how to deal with the two remaining placement options (13.38c) and (13.38e)? Obviously, both cannot be more prototypical than the Background domain. Variant (13.38c) is dependent only on one dichotomic axis of information packaging, while (13.38e) takes both structures (Topic-Comment, Focus-Background) into account. Since it appears that purely conceptual considerations will hardly provide us with decisive arguments, I will utilise the observation that configuration (13.38e) is the result of constructing an ordered pair of discourse-prominent functions on the level of the syntactic (surface) string. In structuralist terms, (13.38e) is more marked than (13.38c). We may resolve the hierarchy question also by accounting for the resulting surface-syntactic clitic positioning pattern. In an independent clause, configuration (13.38d) should yield 2P. The same expectation comes with (13.38a), whereas the other configurations give rise to delayed clitic placement. Variant (13.38c) is likely to induce 3P, while (13.38e) motivates 4P. Finally, configuration (13.38b) may potentially cause a realisation of the clitic low in the syntactic tree under the Leipzig model's assumption of clause structure. The resulting hierarchy in (13.40) provides us with a scale of unspecified likelihood or prototypicality among domains.

(13.40) *Proposed hierarchy among information structure domains for clitic placement in Polish*

BACKGROUND > TOPIC > COMMENT > RESTGROUND > FOCUS

While assuming that Wackernagel cliticisation in Polish applies within information structure domains might appear surprising and daring at first, there are several points that support such an account. Information structure is an important factor in Slavic word order, while clitic second position placement is a linearisation phenomenon in the first place. The repeatedly mentioned status of clitics as Background material is immediately linked with the notion of information structure. Furthermore, several other phenomena are interrelated with information structure, too, like the distribution of prosodic prominence (e.g. Focus stress). Note that, under this perspective, there is an interesting relation between focused elements and clitics. The former are information-structurally and prosodically salient, whereas the latter are non-salient in both respects.

It is not the present study's primary goal to explicate an analysis in a formal theory of grammar or to implement the proposal in a computer-based language model. These endeavours are left for future research. However, I shall briefly comment on some points. The proposal put forward is

formalisable in any framework that acknowledges information structure as an independent level of linguistic description. Provided the manifold phenomena and linguistic levels interacting, it appears most promising to utilise a framework that allows for the simultaneous evaluation of multiple modules of grammar. Therefore, constraint-based approaches appear to be a means of first choice. On the one hand, one could think of an optimality-theoretic account utilising alignment constraints, which restrict the distribution of syntactic units in information structure domains. The hierarchy in (13.40) (or a revised version thereof) could be utilised for this purpose and be directly implemented in the respective optimality-theoretic constraint hierarchy. Such an approach would also allow for a factorial typology of Wackernagel cliticisation strategies. Languages could be classified as to which strategy or strategies they employ for positioning 2P clitics and what factors are dominant over others. Until recently, prosodic and syntactic notions have played the most prominent role in scientific discourse (e.g. intonation phrase, complementizer position, feature checking etc.), whereas the relevance of information structure has been highlighted in the present study. On the other hand, one could rely on model-theoretic approaches. For instance, there are several accounts within HPSG connecting information packaging with constituent order and prosodic prominence (cf. De Kuthy 2021 for an overview). The asset of such an account lies in the maximally uniform description across linguistic levels, which allows treating syntactic structure, constituent order, prosody, and information structure within the very same feature-value matrices.

On a more general note, while a formalisation is pending, it should be clear that the hypothesis presented in this section needs to be tested against more data. On the one hand, the question is whether an account along the lines presented here harmonises with Wackernagel cliticisation in other languages. On the other hand, Polish itself may serve well for further validation (or falsification). Recall Kolaković et al.'s (2022) triangulation-of-methods approach, which culminates in psycholinguistic experimentation. In this context, the hypothesis put forward for Polish cliticisation on the basis of corpus research needs to be tested experimentally with native speakers.

## 13.6 Interim conclusion

Before concluding Chapter 13, I would like to put forth some critical remarks. First, the database investigated for Section 13.4 is admittedly small and the picture that arose is impressionistic. Both facts are suboptimal from an orthodox empiricist's view. More data is needed to test whether the observations on the one hand, and the generalisations and conclusions drawn from them on the other hand are valid. This is scientists' daily bread. Second, not all data points in the entire data sample seem to provide sound evidence for the approach pursued here. There are cases of delayed clitic

placement, where it is unclear to me if any significant information structure or prosodic pattern can be found. For one thing, consider the co-occurrence of Clitic Third with overtly realised subject pronouns in (13.40). Although it is generally known that the Slavic languages are *pro*-drop languages and the occurrence of a subject pronoun implies some sort of emphasis, the “no-drop” pattern does not necessarily imply prosodic prominence of the subject. On the contrary, both Polish and Czech colloquial speech have been reported to exhibit a high degree of no-drop without affecting the prosodic make-up of an utterance or its discourse properties. This is a clear desideratum and requires further study.<sup>603</sup>

- (13.40) a. *bo<sup>1</sup> ja<sup>2</sup> ci<sup>3</sup> powiem Ewa (bo my mamy)*  
 because I.NOM you.DAT tell.PRS.1SG Ewa.VOC because we.NOM have.PRS.1PL  
 ‘so I tell you, Ewa, because we have [it]’  
 b. *to<sup>1</sup> ja<sup>2</sup> se<sup>3</sup> z-robię tak przy trzydziestce*  
 TO I.NOM REFL.DAT PFV-make.PRS.1SG so by thirty.LOC  
 ‘So I will do like that on my thirtieth [birthday]’ (Pol; NKJP pelcra)

Third, another area still pending further investigation is the information structure of utterances other than declarative clauses. For instance, Junghanns & Zybatow (2009: 704) state that studies of the information structure of interrogative clauses are still a desideratum. This is also true for the present study. I cite the examples in (13.41) for the sake of illustration.

- (13.41) a. *a widać<sup>1</sup> go<sup>2</sup> było zza tej*  
 and see.INF him.ACC be.PST.SG.N from\_behind this.GEN.SG.F  
*kierownicy jak ?*  
*steering\_wheel.GEN.SG how*  
 ‘But could one see him behind the steering wheel[?] How?’  
 b. *no to<sup>1</sup> gdzieś<sup>2</sup> go<sup>3</sup> położyłeś czy co ?*  
 well TO somewhere him.ACC lay\_down.PST.SG.M or what.NOM  
 ‘So, did you put it somewhere or what?’  
 c. *a to<sup>1</sup> [ppw piątek]<sup>2</sup> się<sup>3</sup> umówił-e-ś czy ?*  
 and TO in Friday.ACC REFL.ACC make\_appointment.PST.SG.M-M-AUX.2SG or  
 ‘So did you make the appointment for Friday or ... ?’ (Pol; NKJP pelcra)

<sup>603</sup> A possible research question that might be of interest here is the impact of language contact with German. It is well-established that Czech and Polish have a significant layer of German loans. While the prescriptive norms of the standard written languages are based on a puristic approach, the colloquial varieties retained lexical Germanisms. In case of Polish, contact with Russian – or more generally: East Slavic – is also of considerable interest.

On a completely different note, the quality of a study like the one in Section 13.4 heavily relies on the quality and size of the database. It is a truism that spoken language differs significantly from written language. Therefore, the considerably smaller size of speech corpora as opposed to written corpora imposes serious limitations on researchers. With respect to quality, there are some serious issues that came about while preparing the case study on the relation of information structure and clitic positioning in Polish. An examination of available audio recordings revealed that the transcriptions are not adequate in every case, which has serious repercussions for the analysis of written representations of spoken data. Consider the few illustrative examples in (13.42)–(13.46).

- (13.42) a. czyli nie muszę **go** nigdzie oddawać to dobrze  
 that\_is NEG must.PRS.1SG him.ACC nowhere submit.INF TO good.ADV  
 b. czyli # nie MUszę **go** nigdzie oddawać to dobrze  
 ‘In other words, I don’t have to submit it anywhere. That’s good.’ (Pol; NKJP pelcra)

- (13.43) a. [...] i widzę że Kasia też **go** nakryła  
 and see.PRS.1SG that Kasia.NOM also him.ACC cover.PST.SG.F  
 ceratą [...] tablecloth.INS.SG  
 b. [...] i widzę że Kasia # TEŻ **go** nakryła ceratą [...]  
 ‘... and I see that Kasia covered it with a tablecloth, too ...’ (Pol; NKJP pelcra)

- (13.44) a. [...] to niedobrze **mi się** robi [...] TO unwell me.DAT REFL.ACC make.PRS.3SG  
 b. [...] to # nieDObrze **mi się** robi [...] ‘... then I become sick ...’ (Pol; NKJP pelcra)

- (13.45) a. od razu balejaż **se** zrobisz i już  
 from once.GEN.SG balayage.ACC.SG REFL.DAT make.PRS.2SG and already  
 ‘You make a balayage right away and that’s it.’  
 b. od razu # baLEjaż **se** zrobisz i już  
 ‘Immediately. [You] make a balayage and that’s it.’ (Pol; NKJP pelcra)

- (13.46) a. no    właśnie dopiero **by=ś**                    **se**                    rady                    nie    dał  
                  well   actually barely    COND=AUX.2SG REFL.DAT advice.GEN.SG    NEG    give.PST.SG.M  
                  ‘well, you wouldn’t get along all the more, in fact’
- b. no WŁAŚnie # doPIEro **byś se** rady nie dał  
                  ‘Exactly! All the more, you wouldn’t get along!’ (Pol; NKJP pelcra)

The a-variants represent the actual graphic form of the respective transcriptions. In the b-variant, the reader finds the same utterance, but I have added hash marks (#) to indicate an actual prosodic break, which is audibly in the audio playback. Stress is marked by upper case letters. In all five examples the prosodic break or pause was not transcribed, although prosodic breaks (usually marked by dots “.”) are put in other places. Now, the problem emerges that not only prosodic analyses suffer from inconsistent transcription, but syntactic ones too. Admittedly, the lack of prosodic breaks does not affect the syntactic structures in (13.42)–(13.44). Now, note that the examples involve clitics in a delayed syntactic position, but in second position within their prosodic domain. This is a relevant piece of evidence in view of Bošković’s (2001) prosodic account of clitic placement, which suggests that Wackernagel clitics occupy the second position of their intonation phrase. The unavailability of prosodic information is more severe in (13.45)–(13.46), because it has implications for the syntax. Here, the pause occurs between elements of distinct utterances. It is a commonplace that spoken language abounds in short and truncated utterances. Speech acts do not need to be realised as fully saturated grammatical sentences. So, the hash mark in (13.45) and (13.46) marks the pause between two individual sentences. In other words, the domains for clitic placement are different from what the transcription suggests. We find prosodic second position placement in utterances (13.42)–(13.44), whereas examples (13.45)–(13.46) must be regarded as syntactic second position placement, in fact. Although I am not able to quantify the occurrences of inconsistently transcribed utterances, we must take into account that this finding suggests that the amount of Delayed Placement might potentially decrease, while Wackernagel positioning virtually increase in the Polish data set. Bear in mind that this is a chance discovery. It shows that the availability of audio recordings is of great importance not only for phonology and that corpus developers must meet very high requirements, which they deserve recognition for.

To summarise Chapter 13, the discussion in the previous sections on the relation of information structure and clitics provided that clitics are commonly regarded as Background material. Neither can they bear discourse-prominent functions like Topic or Focus, nor can they be stressed as important or introduce new referents into discourse.<sup>604</sup> On the other hand, the occurrence of discourse-

<sup>604</sup> I am abstracting away from Slovene example (13.6b) in Section 13.2 implying contrastively focused clitics.

prominent elements in the syntactic structure above the clitics has repercussions for cliticisation and may result in Delayed Placement. It has therefore been proposed to account for clitic positioning in terms of information structure. On the basis of the Leipzig model of information structure, the occurrence of delayed clitic placement in Polish has been analysed in a subset of the spoken data sample pulled from the NKJP for the empirical study in Chapter 10. Following the Leipzig model's two-layer distinction of Topic-Comment and Focus-Background structure, it has been shown that DP in Polish may be attributed to certain information structure patterns with salient discourse functions intervening between a clitic or clitic cluster and the left periphery of the clause. It has been furthermore observed that, when a clitic occurs within a Topic or Focus domain, it occupies the second position within that domain. Based on these observations, it has been suggested that information structure domains are relevant for clitic positioning in Polish. Consequently, the daring hypothesis has been put forward that Polish clitics are Wackernagel clitics which are positioned within information structure domains. Consequently, Polish clitics occupy the second position of either Background, Topic, Comment, Focus, or what I tentatively called Restground. This does not necessarily imply that syntactic or prosodic domains were irrelevant for clitic placement in Polish. Keep in mind that syntactic, prosodic, and information structure units can be isomorphic, although this does not need to be the case, of course. Therefore, future research should explore the mapping between these linguistic levels and the consequences for clitic positioning. The hypothesis proposed here might be relevant for languages other than Polish. Having this said, the hypothesis requires a critical assessment by more empirical data.



## 14 Summary and conclusion

### 14.1 Summary of the thesis

Let us recapitulate the thesis first before turning to the discussion and prospects in Section 14.2 and Section 14.3. Chapter 1 opened Part I of the thesis and introduced the overall topic of cliticisation in Slavicist linguistics. It further described the thesis's typological and exploratory-descriptive objectives. Here, I also introduced the meta-theoretical perspective adopted in the thesis to reduce theory-dependent bias, provide a typological comparison, and arrive at a more holistic understanding. Chapter 2 determined the present study's object area, i.e. the distribution of unequivocal pronominal and reflexive clitics (= short pronouns and reflexives, constant clitics) in the clause in six West Slavic spoken lects, i.e. Czech, Polish, (Spisz) Goralish, Slovak, Silesian, and Kashubian. The former three represent the thesis's core objects of study with a larger amount of data, whereas the latter three were used only as supplemental material with successively decreasing sample size. Chapter 3 tackled the problem of the notion of clitic, which has been used in scientific discourse like a self-evident category of linguistic analysis. In fact, there is little agreement on the concept of clitic-hood. Many of the heuristics proposed to identify clitics and distinguish them from independent words and morphological affixes are debatable, as they are language-specific, theory-dependent, or fundamentally flawed. I argued that neither the famous criteria by Zwicky & Pullum (1983) and Zwicky (1985), nor the widely known analysis by Klavans (1995), nor the influential tripartite typology of pronouns by Cardinaletti & Starke (1999a) are suitable for approaching clitics in Slavic. I further showed that the tests suggested by Hana (2007) for Czech are inappropriate and that Franks's (2016) conception of clitic-hood as an instantiation of being a minimal sign is insufficient. Moreover, I contended that the micro-typology of clitic systems in Slavic needs some refinement as contemporary East Slavic is relevant for studying clitics, too. While Rusyn (or southwest Ukrainian) dialects have been considered already by Franks & King (2000: 196–202), these lects are underexplored and deserve more attention. On the other hand, short pronominals in contemporary Russian display properties that are also observable with pronominal clitics in other Slavic languages. Finally, previous critique by Klavans, Bermúdez-Otera & Payne, and Haspelmath regarding notions of clitic-hood were discussed, indicating that the notion itself is controversial to some extent. This is why I refrained from adopting contemporarily wide-spread concepts of clitic-hood, which imply a theoretical, monolithic notion of typological significance, but continued to use the language-particular terminology from traditional Slavic studies and Czech indigenous linguistics mentioned earlier. Chapter 4 reviewed a whole catalogue of phenomena typically associated with cliticisation in

Slavic. While a discussion of prosody at the beginning was rather supplementary, the remainder of the chapter focused on the syntactic properties of clitics in Slavic:

- clitic placement in clausal second position and beyond,
- clitic clustering and cluster-internal ordering,
- clitic climbing across verbal and non-verbal phrase boundaries,
- the restriction on rightwards placement,
- co-occurrence phenomena including haplology of reflexives, agreement drop, and morphonological alterations within the cluster,
- clitic doubling,
- clitics licensed in domains smaller than the clause, i.e. noun phrases.

The data discussed throughout Chapter 4 covered West Slavic, South Slavic, and partly East Slavic and were drawn from the literature and occasionally supplemented by empirical attestations. The literary and data review provided some background for the contentious issue of the micro-typology of clitics in Slavic, which distinguishes special clitic languages covering second position and verb-adjacent placement on the one hand and Polish as a simple clitic language on the other hand. I asserted that the high degree of freedom in distributing clitics in the clause, which is usually regarded as a peculiarity of Polish, is not confined to that language alone. In fact, deviations from prototypical or ideal(ized) Wackernagel cliticisation, as well as related phenomena, can be observed across all relevant cognate languages. These include several degrees of Delayed Placement (e.g. 3P, 4P, 5P), diaclisis (= split clitic constructions), non-canonical orders of clitics, clitic doubling in non-verb-adjacent clitic languages, and the occurrence of pronominal and/or reflexive clitics within noun phrases. Therefore, claims recurring in the literature that such structures are possible only in Polish but absent from special clitic languages are incorrect. I concluded that previous research has missed a significant portion of variation.

This conclusion is the main motivation for the study's empirical agenda, which was elaborated in Part II of the thesis. Its theoretical and methodological key aspects were presented in Chapter 6. An important issue is the tensions arising between introspectively constructed examples commonly employed in formal, especially generative research to substantiate scientific claims, and empirical data elicitation methods like those standardly applied in corpus linguistics and psycholinguistics. While there has been fierce dispute about that subject, I attempted to provide a balanced and unemotional discussion. Nevertheless, I held that several introspectively justified statements and predictions concerning Slavic cliticisation were false as shown in the literary review in Part I. Native speaker linguists are insufficient sources of information in certain respects. Moreover, I addressed

(adherence to) prescriptive normativism as a potential source of bias in introspective data construction and evaluation. Guided by the quality criteria of representativeness, intersubjectivity, and replicability, I made the case for an empirical agenda as the overall mode of scientific practice across disciplines. From this it follows that the thesis is built upon corpus linguistic methodology. I then elaborated on the discrepancies between codified standard languages and native speakers' linguistic usage as the output of their acquired grammar. Native speaker linguists hardly ever address this issue when drawing on introspective data, and it remains often unclear how to draw the line between a learned standard grammar and an acquired native language grammar. This is why I employed spontaneous speech data. Spoken language deserves more careful study, as it is the input for native language acquisition. It is, furthermore, rich in variation and exhibits phenomena that might not be detectable elsewhere, i.e. in large corpora of standardised or standard-like edited writing. Chapter 7 presented the corpora and data sources I used, while Chapter 8 described the workflow of data acquisition and preparation, the latter of which involved the manual annotation of the data samples. An issue which arose in the course of data preparation and analysis is that of positional freedom. Slavic languages exhibit a high degree of word order freedom, while clitics themselves may occupy a range of positions, in fact. I proposed the notion of *permutation space* ( $\pi$ -space), which is an operationalisation of an utterance's syntactic or syntagmatic complexity, and which I devised to measure a clitic's actual clausal position against all hypothetically available positions.

Part III was described as the thesis's empirical core, whereby Chapter 10 was the most comprehensive part. Based on a total of 9,819 manually annotated and semi-automatically analysed clause tokens unevenly distributed across object languages (Czech:  $n = 2,933$ ; Polish:  $n = 3,022$ ; Goralish:  $n = 2,666$ ; Slovak:  $n = 810$ ; Silesian:  $n = 339$ ; Kashubian:  $n = 49$ ), I surveyed clitic placement, clitic clustering, and clitic climbing in each lect. Placement covered clitic positioning in the clause, laterality with respect to the predicate, and the distance from it. Scrutinising clustering, I analysed the size and internal structure of the clitic cluster as well as diacsis in simple and complex clauses. I furthermore recorded which categories of clitics and clitic clusters occupy the absolute clausal-initial position. The main features examined for clitic climbing were target position and verb-adjacency. Beyond that, additional phenomena were detected in individual languages, namely agreement drop in Goralish, Silesian, and – with only one token – Czech, as well as the possessive resultative construction, which was observed only in Czech. After considering each lect individually, a comparative analysis of the quantitative and qualitative measures was offered in Section 10.8. Section 10.9 presented an analysis of the previously established quantitative data within the framework of the permutation space. Chapter 11 documented attestations of pronominal clitics licensed by non-verbal predicates, which is rare, but recurs across speakers and lects. Inspired by Hana's (2007: 109) informal statistics on Wackernagel cliticisation vs. Clitic Third, Chapter 12

contrasted the frequencies of second position cliticisation with Clitic Third and Clitic First in edited written Czech and spontaneously spoken Czech. This case study thus did not only tackle register variation in clitic positioning, but also the discrepancies between written standard norm and actual linguistic usage discussed in Section 6.3. Finally, Chapter 13 presented a survey of delayed clitic placement in Polish based on a selected subset of the Polish data sample under reference to notions of information structure.

The thesis's contributions are twofold. On the one hand, the study unearthed new scientific insights about cliticisation in West Slavic, particularly spoken lects of the Lechitic and Czech-Slovak branch. On the other hand, I proposed and employed some methodological refinements in the study of clitics in Slavic. In sum:

1. I found that Polish short pronouns and reflexives are second position clitics, similar to those in Czech and Slovak. In fact, second position cliticisation – including Wackernagel placement, clitic clustering, and cluster-internal word-order regularity – is the dominant pattern in all the six West Slavic lects under scrutiny. Deviant positions become increasingly rarer with growing delay. Micro-typological differences in pronominal clitic placement between lects are purely quantitative (statistical) but not qualitative (phenomenological).
2. I propose that second position cliticisation in West Slavic is not a grammatical requirement but a statistical preference pattern. Individual lects merely differ with respect to the likelihood of occurrence of second position placement.
3. I discovered that delayed clitic placement represented by Clitic Third in Czech subordinate clauses is a register-sensitive phenomenon. Clitic Third is more eligible both quantitatively (higher frequency of occurrence) and qualitatively (available with more subordinators) in spontaneous speech than in edited writing. Furthermore, the availability of Clitic Third with subordinators containing the conditional clitic morpheme {by} provides that there are significant differences between linguistic usage as output of the native speaker's acquired linguistic knowledge and the prescriptive standard written norm. However, there was no clear-cut contrast for second vs. first position placement. Instead, there seems to be a tendency in speech to accommodate patterns of written language.
4. I demonstrated that delayed clitic placement in Polish can be attributed to the presence of topicalised and/or focused elements in the syntactic context preceding the clitic, which is in line with observations from cognate second position clitic languages like Czech.

Simultaneously, clitics in Polish are positioned second in domains that are determined by notions of information structure (i.e. Topic, Focus, Comment, Background, Restground) rather than in terms of syntactic constituency or prosodic domains.

5. In contrast to previous approaches, my annotation scheme for the analysis of the distribution of clitics in the clause employs elementary parameters (position, laterality, distance). It is thus more precise and more flexible than previous annotations, whose various placement types can be derived by different parameter combinations.
6. I introduced the permutation space ( $\pi$ -space) to operationalise an utterance's complexity, which I developed to analyse a clitic's attested position against all hypothetically available positions as a function of that complexity. In principle, all lects utilise a range of positions offered by complex utterances, but they mainly differ in statistical terms. Importantly, undisputed second position clitic languages are not faithfully limited to second position when their permutation space increases.

## 14.2 Discussion

Section 14.1 gave a plain summary of the thesis and merely enumerated its findings. In the following, I will provide some brief discussion of the thesis's epistemic and methodological contributions to the study of clitics in Slavic and put them into the wider context of this very research tradition.

The case study on the distribution of pronominal and reflexive clitics in Chapter 10 established that, contrary to common opinion, Polish is a Wackernagel clitic language. In fact, Wackernagel cliticisation prevails in every lect that was examined. However, as suggested in finding no. 2 in the previous section, Wackernagel placement is not a strict grammatical requirement prohibiting deviant positionings but a statistical preference pattern in West Slavic. Second position clitic languages allow variation in clitic placement, but they may differ in how much variation they tolerate. From a purely qualitative perspective, the core languages Czech, Polish, Goralish plus supplemental Slovak and Silesian are all attested with the whole variational space of Clitic First (1P) up to Clitic Sixth (6P) or Clitic Seventh (7P) placement, while Kashubian exhibits up to Clitic Fifth (5P). This is an intriguing result, since it shows that these lects permit the same range of variation in distributing clitics in the clause by and large. However, the variation is limited. Recall that, for instance, Polish and Goralish showed a ratio of approximately 66% for second position placement, whereas Czech clitics occupied

second position in slightly more than 80% of cases. Accordingly, one cannot distinguish West Slavic lects as to their grammatical possibilities but merely to the likelihood of occurrence of certain clitic placement patterns. Therefore, the conclusion that both Polish and Czech are typologically similar in being Wackernagel clitic languages is supported qualitatively as well as quantitatively, but they could instantiate different subtypes, perhaps. Polish is more liberal in that it permits positional deviations more often than Czech which is more conservative in this sense. This is not meant to necessarily propose discrete linguistic types, rather we might rather think of a continuum with some languages tending to be liberal more often while others tend to be conservative more often.

One would standardly assume that deviations from the rule do not come about by mere chance but are motivated or conditioned by one or more definable factors. This is quite clear for Czech or BCMS, where Delayed Placement was argued to be a function of information structure as discussed in Section 13.2. The positional variation of clitics in Polish has never been accounted for in terms of conditioning factors. Instead, distributional freedom has been usually treated as an intrinsic property of the clitic system itself, and I am not aware of any serious attempt to explain why a clitic in Polish appears once here and once there (but cf. Bošković's 2001 intonational phrase account).<sup>605</sup> In contrast, Section 13.4 established that Delayed Placement in Polish correlates with the occurrence of a Topic or Focus constituent in the environment preceding the clitic. Thus, we can narrow down Delayed Placement in Polish to the same conditions as in Czech and BCMS.<sup>606</sup>

Note that, when proposing that Polish is a second position clitic language, this does not imply an antisymmetric relation of the kind "Polish is more like Czech". Rather, we should acknowledge corrections applying symmetrically from both sides, such that one then also has "Czech is more like Polish". This holds, because Czech allows for deviations of the very same kind as Polish, even if we submit that the former deviates less often than the latter. This result challenges common opinion, and it calls for new and refined analyses. In fact, it is not news that a Slavic special clitic language exhibits behaviour that for a long time has been thought to be a peculiarity of Polish. Jung & Migdalski (2022: 7–8) draw a parallel between weak pronouns (in Cardinaletti & Starke's 1999a sense) in Polish and clitic pronouns in Macedonian and Slovene, which they argue are in a stage of becoming strengthened to weak pronouns. Jung & Migdalski's (2022) discussion is grounded on the occurrence of the Clitic First configuration in both languages. As we have seen in Section 4.4.1, Clitic First occurs in various Slavic languages. Furthermore, they refer to Stegovec's (2020) observation that Slovene allows

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<sup>605</sup> Ultimately, clitic positioning in Polish is determined by syntactic feature-checking against at least one functional head in Bošković's account, since he asserts that the prosodic condition of being placed initially within an intonation phrase is not effective in Polish.

<sup>606</sup> Furthermore, recall from Section 13.6 that some Polish data points were erroneously annotated as clitic third position placement due to transcription errors, which was recognised only post hoc thanks to auditory data from the Spokes corpus. As a result, the ratio of Wackernagel cliticisation in Polish potentially raises an uncertain amount. This provides some further, even though mild corroboration for the conclusion that Polish is a Wackernagel clitic language, indeed.

ACC < DAT order next to canonical DAT < ACC. Clitic word order variation, too, is attested in several Slavic languages, which was discussed in Section 4.6. So, the present study's empirical findings fit a more general picture which is not confined to Polish or West Slavic alone. We can therefore hypothesise that what linguists originally encountered in Polish cliticisation was perhaps not a peculiarity of an individual titular language (i.e. historical language in Coseriu's terms, E-language in Chomsky's terms) but the patterns of language use unregulated by prescriptive norms. It is noteworthy that Polish indigenous grammaticography has not dedicated special attention to the position of short pronouns as did Czech or – even more – BCMS grammaticography. Bear also in mind that the present study's empirical core is grounded on spontaneous speech, which is less prone to prescriptive normativism than edited writing, as unequivocally evidenced by the case study on clitic positioning in Czech speech and writing in Chapter 12.

The thesis also contributes to the methodology of research in Slavic clitics in several respects. The distributional analysis is based on an annotation scheme that employs elementary parameters, i.e. the numerical value of a clitic's phrasal position, laterality as the clitic's side relative to its predicate, and distance between the clitic and its predicate as number of intervening constituents. In contrast, previous approaches like Reinkowski (2001) and Kosek (2009, 2011, 2015, 2017) employed non-elementary placement categories, while Kolaković et al. (2022) merely distinguished between 2P and DP without further differentiation within the DP category.<sup>607</sup> The system used here is more precise, because it considers cardinalities, and more flexible, since it is able to derive those aforementioned categories. Another novelty of the present study is the proposal of the permutation space, which introduced an operationalisation of an utterance's complexity, against which clitics' attested positions are plotted. This plot was presented in the form of a heat map, and it enabled us to visualise the degree of variation within a data sample against the space available for that variation. The study also exemplifies how quantitative approaches can enrich and complement qualitative research questions. For instance, the results not only allow us to quantify Wackernagel cliticisation and deviations therefrom in West Slavic but also deepen our understanding of cliticisation more generally. The case study on Czech in Chapter 12 is particularly insightful in this respect, as it confronts linguistic structures in different subsystems of a titular language based on "big data". We learnt from this case study quite clearly that the choice of a particular subordinator has an impact on the occurrence of second vs. third position cliticisation. Needless to say, register distinctions or the difference between linguistic usage and linguistic standard norm are relevant for the occurrence and frequency of those cliticisation patterns.

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<sup>607</sup> Admittedly, both Reinkowski (2001) and Kolaković et al. (2022) consider phrase splitting, which is missing in the present thesis but can be easily incorporated.

The corpus study pioneered the exclusive use of spontaneous speech data, which has never been done in Slavic clitic research before, and which is generally underrepresented in empirical linguistic research. Spoken data is valuable for multiple reasons and deserves more careful examination as has been argued in Section 6.4. The key features favouring spontaneous speech over other modalities of language are its naturalness in being unedited and least prone to prescriptive normativism, its ecological validity, prevalence in everyday communication, and relevance for native language acquisition. The last point is of particular interest towards the background of psychologically or cognitively oriented research paradigms in linguistics, as children build their native lexicon and grammar from speech input. It is a plausible assumption that features present in speech are internalised by the child when acquiring its mental grammar and later may become externalised again in linguistic performance (or parole). While this does not imply that there is a one-to-one correspondence between linguistic competence (or langue) and performance (parole), the approach assumes a more direct relation between the acquired linguistic knowledge and patterns of speech than, e.g., between the same knowledge and edited writing, where extra-linguistic factors like normativism and stylistics interfere.

The conclusion that Polish has second position clitics and that West Slavic lects align as to the phenomenology observed with them implies that former typological classifications of Polish clitics as behaving peculiarly in comparison to clitics in other Slavic languages were not well-grounded. On the one hand, this was caused by an incomplete understanding of clitic-hood and the properties associated with it. Wackernagel or second position cliticisation has been conceptualised on the grounds of strict dichotomies, which turned out to be purely theoretical constructs rather than models of the real world. On the other hand, the typology arose from the methodological limitations that accompanied research in Slavic clitics. Most data samples used in Slavic clitic research have been introspectively constructed. The approaches have been rather atomistic, focusing on few constructions while leaving a relevant degree of variation aside. Either consciously or unconsciously, examples were chosen selectively. The issue of researcher bias has been hardly considered seriously in Slavic clitic research (cf. Kolaković et al. 2022: 51–53 for discussion). For instance, a researcher's adherence to the prescriptive norm of a standard language or their neglect of sociolinguistic variation may affect the introspective evaluation process as mentioned earlier. Another issue is judging single constructed utterances without proper context. A lack of ecological validity can distort the assessment of an utterance, which in an out-of-the-blue situation might be considered awkward by a native speaker, but provided with appropriate linguistic environment or discourse context is perfectly fine (cf. Hana 2007: 108–109). We may subsume those issues figuratively as a sort of tunnel-type vision by the researcher.



The present study suggests that clitics in Slavic are not grammatical monoliths but linguistic units with intricate, nuanced properties that deserve careful examination. A number of contradicting or incorrect claims, which appear in the literature, were the thesis's point of departure: Be it Czech's strict second position requirement in the main clause, the impossibility of diverse Clitic Third configurations in the subordinate clause including those with a verb intervening between subordinator and clitic, or the impenetrability of the clitic cluster, among others. I conducted an exploratory corpus investigation to reveal the range of possibilities in clitic distribution and to minimise researcher bias. The result is a mainly quantitative study, which provides us with frequency distributions of various placement patterns. While this approach leaves many structural details aside and does not propose any formal analysis of cliticisation in Slavic, it provides us with new information that future research can rely on. Notably, we have become aware of the actual degree of positional variation as in Section 10 or Section 12 only thanks to a corpus linguistic approach and the utilisation of sufficiently large data samples. Introspective data has not provided us with corresponding information. For instance, while Toman (1996) and Veselovská (1995) explicitly refer to Clitic First and Clitic Third, they do not share a great deal of information on these occurrences.

Corpus linguists and their methodology have been repeatedly criticised by proponents of introspective acceptability/grammaticality judgements and vice versa. While every researcher is – or ideally should be – aware of the advantages and disadvantages of the methods and tools they use, the present study unequivocally shows that corpus research is not inferior to native speaker linguists' introspectively constructed data collections (contra Broekhuis 2010 and Tajsner 2017). Why is this? The present thesis falsified dismissive arguments against empirical methodology. Both the literary review in Part I and the empirical survey in Part III showed that introspective judgements are not always reliable. Admittedly, corpora still do not contain negative evidence, they are incomplete in that they represent only a limited set of linguistic behaviour, and they contain distortion in the form of errors, stuttering or typos, interruptions and many other disfluency phenomena (cf. Broekhuis 2010: 190–192). But this is not crucial and cannot seriously be an argument against the recourse to empirical methodology. It is a truism that every method suffers from some limitation, which applies to introspection, too, as has been explicated throughout the thesis. There is no all-in-one solution in science. This is where complementary methods come into play.

Nevertheless, the present study shows that corpus linguistic research – or any other empirical approach, e.g. experimentation – provides new and relevant data as well as meaningful insights. There is no guarantee that research driven by introspection will always reveal deciding patterns, just as there is no guarantee that a corpus will provide us with the structures we are looking for. Recall the serendipity of agreement drop, which was especially well represented in the Goralish data sample. Until now, there has not been any serious synchronic research on West Slavic clitics concerned with

agreement drop, even though traditional dialectology is aware of the phenomenon for around a century now. Particularly, I am not aware of any introspectively informed investigation of agreement drop in West Slavic and the construction is consistently absent in every overview of clitics in Slavic, e.g. Dimitrova-Vulchanova (1999), Franks (1998/2010, 2009), Franks & King (2000), or Kosta & Zimmerling (2014). The empirically obtained Goralish and Silesian data provided enough evidence for stating interesting hypotheses of the kind in (14.1).

- (14.1)
- a. The construction of agreement drop is realised in a contiguous string: SBJ < CL < V.
  - b. If the string is not contiguous, then SBJ + CL are adjacent leaving V behind.
  - c. Agreement drop increases the likelihood of Delayed Placement, which is a mandatory configuration in subordinate clauses blocking second position cliticisation here.

Recall furthermore from Section 6.2 that empirical research has a corrective function in filtering out unwarranted hypotheses. We can name several corrections like those in (14.2) based on the discussion in Part I, the analysis in Part III, or referring to external empirically oriented publications.

- (14.2)
- a. Strong pronouns are not restricted to [+human] and can refer to [−human] and even [−animate] entities (contra Cardinaletti & Starke 1999a; Kupść 2000b; Witkoś 1998; pro Veselovská 1995; see Section 3.6.5).
  - b. Diaclisis of co-dependent clitics is rare but possible in West Slavic (contra common opinion; pro Lenertová 2004 on Czech; Kolaković et al. 2022 on BCMS; see Section 10.8.2).
  - c. Diaclisis of co-dependent clitics resulting from clitic climbing is rare but possible out of subjunctive *da*-complements in BCMS (contra Aljović 2004; pro Stjepanović 1998, 1999, 2004; cf. Kolaković 2022).
  - d. Clitic Third in Czech subordinate clauses is possible with a verb intervening between subordinator and clitic, even though it rarely occurs (contra Veselovská 1995; pro Hana 2007; see Section 4.4.2, Section 12.2).
  - e. Clitic Third in Czech is possible with a variety of subordinators beyond *že* ‘that’ (contra Veselovská 1995; pro Hana 2007; see Section 12.2).
  - f. Clitic climbing in Czech (and Polish) is not dependent upon the dichotomy between restructuring and non-restructuring environments (contra Rezac 2005; pro Dotlačil 2004; cf. I. Kulik 2023b)

The corrective findings in (14.2) provide enough corroboration for the usefulness and relevance of an empirical approach in the study of language. While clitics in Slavic have experienced decades of research that has been introspectively underpinned or prescriptively-normatively biased, a wide range of tools is at linguists' disposal, including annotated corpora, experimental setups, and diverse statistical procedures, which significantly expand the possibilities of linguistic inquiry. Kolaković et al.'s (2022) examination of BCMS cliticisation takes exactly such a diversified empirical approach. This does not mean that each linguist needs to query corpora, conduct experiments, and run statistical models – perhaps, simultaneously. There seems to be already an established (and necessary) division of labour between several subdisciplines, which future endeavours can rely on. Research in Slavic clitics will have to adapt to a continuously developing and diversifying field in order to receive a comprehensive picture of the regularities and oddities in Slavic cliticisation. Dismissing empirical research in the quest for knowledge, while simultaneously putting forth the strong claim of adhering to the methodological framework of the natural sciences is a severe contradiction. Finally, there is still a place for researchers' intuition, as it guides them in the quest for deepening our knowledge of the world. However, intuition cannot replace method.

### 14.3 Future prospects

Perhaps, the thesis raised more questions than it was able to answer. Open questions point to desiderata and there are numerous threads in the present study that are to be picked up in future research. Broadening the database is an obvious one. On the one hand, there is a great need to expand the data samples of the supplemental lects Slovak, Silesian, and Kashubian. There is no difficulty in obtaining more data for Slovak, for which a complete corpus infrastructure exists, i.e. the SNK. The situation is worse for Silesian and especially Kashubian, whose numbers of speakers are continuously declining. On the other hand, a survey of Sorbian clitics based on authentic, especially spoken data is still pending. This could be accomplished using the GENIE corpus of Lower Sorbian (Marti et al. 2012) and the corpus of Colloquial Upper Sorbian (Breu & Adamou 2011, Breu et al. 2023). Even the investigation of the thesis' core languages Czech, Polish, and Goralish could be refined in subsequent research. Recall that the data samples are based upon arbitrarily chosen quantities of data. As a result, pronominal clitics outnumber reflexive ones, which does not mirror the proportions of the clitic types in usage. Correspondingly, the number of ACC(/GEN) reflexive clitics needs to be significantly increased, because it is the most frequent constant clitic in both written and spoken language in all lects studied here. Furthermore, while the present study draws upon speech, the data of contemporary written language still require empirical analysis, which could pave the way for future

comparative studies, such as the one on Czech in Chapter 12. Another area that deserves further study is agreement drop. I remarked that the construction occurs across the entire Lechitic and Czech-Slovak branches, but the study provided attestations only for Goralish and Silesian – and a single occurrence in Czech. This indicates that agreement drop data needs to be obtained purposefully from corpora rather than by chance. Research should focus on the structural properties of agreement drop constructions, i.e. the possible linear realisations, to clarify whether the word order restriction proposed in Section 10.8.4 is valid. Furthermore, it still needs to be established what the driving force behind agreement drop is and why it is only the first persons that permit it. This also amounts to the question of the potential link between agreement drop in contemporary West Slavic, the diachronic evolution of the East Slavic past tense formation without auxiliary element, and the occasional auxiliary loss in South Slavic I referred to as *deletion of unlikes*.

Another important aspect is the interaction of information structure and cliticisation in Slavic, which has not yet been sufficiently explored in Slavicist research. The analysis in Chapter 13 suggests that Delayed Placement in Polish is actuated by information structure. However, the database needs to be extended, and the proposal explicated in Section 13.5 requires further validation by empirical testing. The hypothesis that clitics in Polish are positioned second in information structure domains needs further validation as does the assumption of the *Restground* domain. My analysis was built upon the Leipzig model's Topic-Comment and Focus-Background structure, but it did not incorporate Däbritz's (2021) extension in terms of information status. However, we may assume that notions of information status may be useful here, if we recall Boudovskaia's (2017) argument from Footnote 372 in Section 4.9.3 that agreement drop in Carpatho-Rusyn is sensitive to *givenness* and *newness*, which are quintessential in information status. Presumably, Däbritz's (2021) proposal to extend the Leipzig model could thus provide researchers of clitics in Slavic with suitable analytical instruments.

A related topic is the question of formal implementation. While the present thesis sporadically provided simple structural schemes, there has not been any attempt to spell out a formal syntactic analysis. This is partly caused by the typologically broad approach taken here, which focuses on quantities, while it simultaneously must disregard detailed analyses of structural properties. There is a rich offer of syntactic theories of both the phrase structure grammar and the dependency grammar type. Provided the interaction of diverse linguistic levels in cliticisation like prosody, syntax, and information structure, the utilisation of a unification formalism appears most promising, for unification grammars evaluate linguistic structures of different kinds within a single formalism. However, an analysis in the framework of Optimality Theory, which is not a unification grammar, could accomplish this, too. On a related note, it would be interesting to follow an approach along the lines of Müller (2023) and compare diverse linguistic theories as to how they cope with clitic phenomena in (West) Slavic. Such an enterprise would allow to evaluate models from different

frameworks with respect to their empirical accuracy. It would be of particular interest to see how dependency grammars perform next to phrase structure grammars (cf. Müller 2023: 369–417 for discussion), as there are rich indigenous traditions of dependency grammatical research in the Slavic countries, which paid special attention to the flexible word order.

The case study on clitic deviate placement in written vs. spoken Czech in Chapter 12 touched upon the issue of sociolinguistic variation, more precisely on diaphasic or register variation (cf. Pescuma et al. 2023). The dichotomy of spoken and written language is, however, only a small aspect of this intricate topic and confined to the medial distinction of script vs. speech in the first place. This kind of research could be extended by more principled investigations that purposefully differentiate between, e.g., text genres like belles-lettres, newspapers, and blog posts, or scrutinise varying situational parameters such as, e.g., public vs. private communication, interaction between familiar vs. unfamiliar interlocutors. Investigations differentiating text types or edited vs. non-edited writing can be easily conducted using balanced corpora and web corpora, as they allow restricting corpus queries to well-defined subcomponents mirroring such parameters. The task is more difficult to conduct for speech corpora, but depending on a corpus' infrastructure and metadata, the research question could be recalibrated from a diaphasic to a diatopic perspective, which has hardly been done in Slavic clitic research (but cf. Reinkowski 2001; Kolaković et al. 2022 on BCMS). Despite the fact that Goralish, Silesian, and Kashubian are all geographically limited lects roofed by Polish, the present thesis did not incorporate a truly diatopic perspective that would distinguish lects according to established macro-areas of Czech, Slovak, and Polish. Questions of diastratic variation can be addressed with reference to socio-demographic parameters covered by metadata like, e.g., level of education, gender, or age. The latter also allows implementing a diachronic perspective, when searching for significant differences in cliticisation patterns across age groups. For example, it remained unclear, whether the comparatively high frequency of REFL < DAT against DAT < REFL in Goralish reflects free variation in a strict sense, or if it could reflect language change with DAT < REFL spreading among younger generations under the influence of Polish. Except for some rather anecdotal remarks in the literature on Clitic First being colloquial (Hana 2007: 76; Toman 1996: 505), signalling familiarity (Hana 2007: 70), or sounding “sloppy” (Lenertová 2004: 149), I have not come across any research that would address questions targeting at social meaning as indexical property of clitic constructions. This kind of research question is nested in the so-called Third Wave of Sociolinguistics (cf. P. Eckert 2012 for an overview). For example, the experimental setup of the Matched Guise Technique (Lambert et al. 1960) is an established paradigm to investigate the social meaning of grammatical constructions (cf. e.g. Varaschin et al. 2024 on Topic Drop in German).

From a methodological perspective, the annotation scheme developed for the present study could be elaborated by extending it by further parameters. For instance, it was mentioned in

Section 10.4.4 that the relative frequency of agreement drop could not be calculated exactly but only approximately, because neither tense nor person were coded. In a similar fashion, the annotation did not include information about clause type like independent main vs. subordinate clause. This is why the permutation space was calculated upon incomplete information in Section 10.9. Both circumstances arose from the situation that agreement drop and potential positions as a function of syntagmatic complexity had not been anticipated when the annotation scheme was developed. Agreement drop was discovered only during the annotation of the Goralish data sample, in fact. The permutation space, too, is a later development in the work history of the thesis and came into being only after the annotations were completed. Information on clause types could also include more detailed information like the syntactic structure underlying complex clauses (i.e. raising, control, ACI) and their deepness of embedding, which was discussed in connection with clitic climbing in Section 4.7.1. Recall also that discontinuous constituents were not annotated. Finally, as the corpus study was solely form-based, the annotation did not encompass the clitics' functions and, thus, disregarded potential contrasts between argument vs. non-argument clitics. Accordingly, the annotation should be revised for future research to increase precision and cover more constructions. An enriched annotation would furthermore strengthen the qualitative perspective and allow the analysis of linguistic structures in more detail again.

Lastly, the present study is entirely corpus linguistic, such that the conclusions, hypotheses, and suggestions presented here need further validation. It was argued in Section 6.1 that replicability is a key aspect of scientific practice. A simple way of addressing this need would be to replicate the corpus study with another corpus sample, where appropriate and possible. This is basically how the case study in Chapter 12 evolved, which arose out of the discussion of Hana's (2007: 109) informal statistics of clitic second-position vs. third-position placement in subordinate clauses introduced by *že* 'that'. However, according to Kolaković et al.'s (2022) triangulation of methods, the subsequent step to corpus research is psycholinguistic experimentation as a complementary approach (cf. also Gilquin & Gries 2009). As far as I am aware, there have been neither perception nor production experiments tackling the issue of clitic positioning in the clause, or the interaction between information structure and cliticisation in West Slavic. However, there exists already a growing body of experimental research on various aspects of cliticisation in South Slavic (cf. Diesing 2009; Kolaković et al. 2022; Tabachnik et al. 2024), which may serve as a reference point for studies in West Slavic. Clitic phenomena in Slavic have received a great deal of attention since the 1990s, but obviously, there is still a lot more work to do to unearth the mechanisms underlying them.

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## Appendix

Tables A.1–A.4 provide the calculated (observed)  $\chi^2$ -values of the pairwise test for homogeneity of clitic positional distribution in Chapter 10 for Czech (§10.2.1), Polish (§10.3.1), [Spisz] Goralish (§10.4.1), and Slovak (§10.5.1). No tests have been conducted for Silesian and Kashubian due to scarce data. The critical  $\chi^2$ -value (for  $p = 0.5$ ,  $df = 2$ ) is 5.991. Cells containing significant results are highlighted with red/orange shading. Significant results allow to refute the  $H_0$ , which is that there is no difference in the frequency distributions among constant clitics within a particular lect.

Table A.1: Matrix of calculated  $\chi^2$ -values for the homogeneity test of clitic pairs in Czech ( $p = 0.5$ ,  $df = 2$ ) [reproducing Table 10.5].

|           | <i>tě</i> | <i>ho</i> | <i>mi</i> | <i>ti</i> | <i>mu</i> | <i>se</i> | <i>si</i> |
|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|
| <i>tě</i> |           | 9.11      | 12.88     | 24.92     | 15.08     | 29.45     | 24.74     |
| <i>ho</i> |           |           | 0.99      | 5.55      | 1.10      | 6.49      | 4.94      |
| <i>mi</i> |           |           |           | 1.97      | 1.97      | 4.09      | 1.97      |
| <i>ti</i> |           |           |           |           | 4.78      | 1.89      | 0.19      |
| <i>mu</i> |           |           |           |           |           | 3.32      | 3.60      |
| <i>se</i> |           |           |           |           |           |           | 0.88      |
| <i>si</i> |           |           |           |           |           |           |           |

Table A.2: Matrix of calculated  $\chi^2$ -values for the homogeneity test of clitic pairs in Polish ( $p = 0.5$ ,  $df = 2$ ).

|            | <i>cię</i> | <i>go</i> | <i>mi</i> | <i>ci</i> | <i>mu</i> | <i>się</i> | <i>se</i> |
|------------|------------|-----------|-----------|-----------|-----------|------------|-----------|
| <i>cię</i> |            | 10.17     | 25.95     | 0.40      | 15.34     | 21.41      | 27.60     |
| <i>go</i>  |            |           | 23.47     | 6.83      | 0.76      | 2.46       | 9.17      |
| <i>mi</i>  |            |           |           | 25.81     | 29.01     | 21.39      | 10.17     |
| <i>ci</i>  |            |           |           |           | 10.90     | 16.68      | 23.68     |
| <i>mu</i>  |            |           |           |           |           | 2.11       | 10.00     |
| <i>się</i> |            |           |           |           |           |            | 3.65      |
| <i>se</i>  |            |           |           |           |           |            |           |

Table A.3: Matrix of calculated  $\chi^2$ -values for the homogeneity test of clitic pairs in Goralish ( $p = 0.5$ ,  $df = 2$ ).

|            | <i>cię</i> | <i>go</i> | <i>mi</i> | <i>ci</i> | <i>mu</i> | <i>się</i> | <i>se</i> |
|------------|------------|-----------|-----------|-----------|-----------|------------|-----------|
| <i>cię</i> |            | 7.44      | 2.91      | 0.21      | 13.64     | 15.84      | 19.64     |
| <i>go</i>  |            |           | 1.32      | 7.68      | 4.34      | 2.86       | 10.98     |
| <i>mi</i>  |            |           |           | 2.75      | 6.08      | 6.52       | 14.83     |
| <i>ci</i>  |            |           |           |           | 12.24     | 15.50      | 24.85     |
| <i>mu</i>  |            |           |           |           |           | 1.25       | 24.95     |
| <i>się</i> |            |           |           |           |           |            | 17.66     |
| <i>se</i>  |            |           |           |           |           |            |           |

Table A.4: Matrix of calculated  $\chi^2$ -values for the homogeneity test of clitic pairs in Slovak ( $p = 0.5$ ,  $df = 2$ ).

|           | <i>ma</i> | <i>ťa</i> | <i>ho</i> | <i>mi</i> | <i>ti</i> | <i>mu</i> | <i>sa</i> | <i>si</i> |
|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|
| <i>ma</i> |           | 1.30      | 0.25      | 3.00      | 3.74      | 0.5       | 0.12      | 1.52      |
| <i>ťa</i> |           |           | 0.41      | 5.55      | 0.66      | 1.17      | 1.51      | 1.00      |
| <i>ho</i> |           |           |           | 3.71      | 2.08      | 0.45      | 0.40      | 0.93      |
| <i>mi</i> |           |           |           |           | 8.67      | 1.71      | 2.00      | 2.33      |
| <i>ti</i> |           |           |           |           |           | 3.15      | 3.99      | 2.13      |
| <i>mu</i> |           |           |           |           |           |           | 0.20      | 0.36      |
| <i>sa</i> |           |           |           |           |           |           |           | 1.07      |
| <i>si</i> |           |           |           |           |           |           |           |           |

# Erklärung

Ich erkläre,

dass mir die geltende Promotionsordnung der Fakultät bekannt ist;

dass ich die Dissertation selbst angefertigt, keine Textabschnitte eines Dritten oder eigener Prüfungsarbeiten ohne Kennzeichnung übernommen und alle von mir benutzten Hilfsmittel, persönlichen Mitteilungen und Quellen in meiner Arbeit angegeben habe;

dass mich niemand bei der Auswahl und Auswertung des Materials sowie bei der Herstellung des Manuskripts unterstützt hat;

dass die Hilfe einer kommerziellen Promotionsvermittlung nicht in Anspruch genommen wurde und dass Dritte weder unmittelbar noch mittelbar geldwerte Leistungen von mir für die Arbeiten erhalten haben, die im Zusammenhang mit dem Inhalt der vorgelegten Dissertation stehen;

dass ich die Dissertation noch nicht als Prüfungsarbeit für eine staatliche oder andere wissenschaftliche Prüfung eingereicht habe;

dass ich nicht die gleiche, eine in wesentlichen Teilen ähnliche oder eine andere Abhandlung bei einer anderen Hochschule als Dissertation eingereicht habe.

Jena, 16.01.2025

Irenäus Kulik

Ich erkläre,

dass in Textverarbeitungsprogrammen integrierte Assistenten bzw. Editoren sowie große Sprachmodelle (*large language models*, LLMs) während der Publikationsvorbereitung ausschließlich zur sprachlichen Korrektur in grammatischer, orthographischer und formaler Hinsicht eingesetzt wurden;

dass diese Systeme weder an der inhaltlichen Erstellung noch an der inhaltlichen Überarbeitung des wissenschaftlichen Textes beteiligt waren.

Berlin, 01.07.2026

Irenäus Kulik